ESAIM: COCV 30 (2024) 91
https://doi.org/10.1051/cocv/2024081

ESAIM: Control, Optimisation and Calculus of Variations
www.esaim-cocv.org

VISCOSITY SOLUTIONS OF CENTRALIZED CONTROL PROBLEMS IN MEASURE SPACES

Averil Aussedat[1,*], Othmane Jerhaoui[2] and Hasnaa Zidani[1]

Abstract. This work focuses on a control problem in the Wasserstein space of probability measures over $\mathbb{R}^d$. Our aim is to link this control problem to a suitable Hamilton–Jacobi–Bellman (HJB) equation. We explore a notion of viscosity solution using test functions that are locally Lipschitz and locally semiconvex or semiconcave functions. This regularity allows to define a notion of viscosity and a Hamiltonian function relying on directional derivatives. Using a generalization of Ekeland's principle, we show that the corresponding HJB equation admits a comparison principle, and deduce that the value function is the unique solution in this viscosity sense. The PDE tools are developed in the general framework of Measure Differential Equations.

Mathematics Subject Classification. 35F21, 35R06, 49Lxx.

Received March 4, 2024. Accepted November 7, 2024.

1. Introduction

In this paper, we consider a Mayer control problem over the metric space of probability measures equipped with the Wasserstein distance. This class of problems is particularly suited for modelling physical situations where the state variable is known only up to a density of probability [1]. It also provides a convenient formalism for problems involving the motion of populations, encompassing both discrete and continuous formulations ([2, 3]; see also the survey [4]).

These problems have been extensively studied in cases where the state variable lies in some finite-dimensional vector space. An effective approach is to connect the control problem with a Hamilton–Jacobi–Bellman (HJB) partial differential equation. These equations are typically understood in the sense of viscosity solutions, which is a dedicated weak formulation adapted to the nonlinear nature of HJB equations. This approach originated in the work of Crandall, Ishii, and Lions [5, 6]. The numerical methods developed for the HJB equation can then be employed to solve the original problem. Our objective with this work is to contribute to the extension of Hamilton–Jacobi techniques into the space of measures.

Considering probability measures as the state space poses several difficulties. We examine an infinite-dimensional subset of probability measures endowed with the Monge–Kantorovich distance, also known as the *Wasserstein distance*, derived from optimal transport theory. The Wasserstein space is not a Banach space,

Keywords and phrases: Hamilton–Jacobi, Wasserstein, viscosity solutions.

[1] INSA Rouen Normandie, Normandie Univ, LMI UR 3226, 76000 Rouen, France.
[2] Université de Rennes, INSA Rennes, CNRS, IRMAR – UMR 6625, 35000 Rennes, France.

* Corresponding author: averil.aussedat@insa-rouen.fr

© The authors. Published by EDP Sciences, SMAI 2024

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

and defining a partial differential equation (PDE) within it is not straightforward. Additional technical challenges arise from the lack of local compactness and convexity of the distance function. Despite these challenges, measures are rich objects that can be viewed as points in a geodesic space, laws of random variables, or generalizations of densities. Each interpretation brings its own set of techniques from geometry, analysis, or probability theory.

Establishing a differential calculus in the Wasserstein space has been an actively researched problem over the past two decades. The foundational work by Otto introduced a pseudo-Riemannian calculus [7, 8], offering striking reformulations of the porous medium equation as a gradient flow. Simultaneously, gradient flows in the Wasserstein space were investigated using techniques from general metric spaces [9], leading to the construction of a natural tangent cone [10]. The emergence of mean-field games [11] emphasized the necessity for proper generalizations of solutions of partial differential equations with a measure variable, laying the ground for the extension of viscosity solutions to this setting.

The approach by test functions is very classical when working in $\mathbb{R}^d$, but entered only recently in the literature over the Wasserstein space. The reason is that the squared distance does not admit a Wasserstein gradient, or equivalently an L-derivative [12], at all measures. Recent works succeeded in finding smoothing procedures in order to provide sufficiently smooth test functions [13–15]. With this technology, [14] obtains a comparison principle for bounded and continuous semisolutions, and [13] a strong comparison principle in the particular case of a Hamiltonian involving a degenerate divergence term, thus calling for an *ad hoc* notion. A strong comparison principle is also proved with a test function approach in [16] in the presence of noise, which helps to control the regularity of the maxima.

In a close spirit, the works [17–20] consider a modification of the definition using test functions and suitable upper/lower envelopes or approximations of the Hamiltonian. [20] obtains a strong comparison principle by relying on a penalization by an energy with compact sublevels, an assumption that is removed in [17–19] with the help of the gradient flow structure of their problems. The latter series of work does apply in more general spaces than the Wasserstein space, but seems difficult to extend out of variational problems.

On the other hand, the approach by semidifferentials is more common, since it corresponds to allowing non-differentiable test functions. Here one can distinguish two tendencies. First, one can define semidifferentials as elements of $L^2_\mu(\mathbb{R}^\mathrm{d}; \mathrm{T}\,\mathbb{R}^\mathrm{d})$ or Tan_μ, the common feature being that these elements are supported on graphs [12, 21–25]. In all these works, comparison principles are obtained under the assumption that at least one of the semisolutions to be compared is uniformly continuous. Secondly, one could enlarge the definition to allow for measure-valued elements in the semidifferentials, as in [26, 27]. The first of these works obtains a strong comparison principle for Eikonal-type equations by successfully drawing a link with metric viscosity, and the second one a strong comparison principle on the torus for a Hamiltonian arising in mean-field games. Let us point that the latter work seems the closest in spirit to ours, with the conceptual difference that the structure of differentiability is build with the Lions lifting procedure, and does not relate to the geodesic structure, hence preventing generalization to other classes of space.

The present paper provides a notion of viscosity solutions that relies mostly on the geodesic structure of the space, and a strong comparison principle that takes advantage of the allowance of the squared distance in the test function sets. The class of Hamiltonian under investigation allow to treat first-order control problems with sufficiently regular dynamic, although it does not seem to extend to the case where diffusion enters the continuity equations. In the Wasserstein space, this line of investigation was initiated in [28, 29], where the underlying space was taken as a compact manifold.

The contributions of the present work are the following: we allow the dynamics to depend on the measure variable, which may have full support in the non-compact space $\mathbb{R}^d$. We give a formulation of the Hamilton–Jacobi equation that supports a strong comparison principle, and we show that in the case of Hamilton–Jacobi–Bellman equation, the unique solution is the value function of the control problem. In particular, the comparison principle is valid on Hamiltonians that are defined using directional derivatives along elements of the general tangent cone, and could be used in the context of Eikonal equations.

The paper is organized as follows. The setting of the problem is detailed in Section 2. Section 3 gathers the essential elements of the theory of the Wasserstein space needed in the subsequent sections. The control problem

is studied in Section 4, where we introduce the value function and discuss its properties. Section 5 focuses on a general Hamilton–Jacobi equation, providing the definition of a viscosity solution and discussing the comparison principle. Finally, the case of Hamilton–Jacobi–Bellman equations is addressed in Section 6.

2. Setting of the problem

In this section, we establish the notations, outline the problem under investigation, and specify the assumptions of the paper. Consider $\mathcal{B}_{\mathbb{R}^\mathrm{d}}$ the Borel $\sigma-$algebra of $\mathbb{R}^\mathrm{d}$, and let $g : \mathbb{R}^\mathrm{d} \mapsto \mathbb{R}$ be a Borel-measurable function. The notation $\#$ will be used to denote the push-forward operator on measures, defined for any Borel measure μ as

$$g\#\mu(A) \coloneqq \mu\left(g^{-1}(A)\right) \qquad \forall A \in \mathcal{B}_{\mathbb{R}}.$$

Notations. Space of measures. For any Polish space (Ω, d), $\mathscr{P}(\Omega)$ will denote the space of Borel probability measures on Ω. In the sequel, we consider the subset $\mathscr{P}_2(\Omega)$ of measures with finite second moment, *i.e.*, measures such that for some (thus any) $o \in \Omega$, there holds

$$\mathscr{P}_2(\Omega) \coloneqq \left\{ \mu \in \mathscr{P}(\Omega) \;\middle|\; \int_{x\in\mathbb{R}^\mathrm{d}} d^2(o,x)\mathrm{d}\mu(x) < \infty \right\}.$$

For any couple $(\mu,\nu) \in \mathscr{P}_2(\Omega)^2$, we denote by $\Gamma(\mu,\nu)$ the set of probability measures on $\Omega\times\Omega$ with marginals μ and ν, further referred to as the set of *transport plans*. An example of such plans, or couplings, is given by the product measure $\mu\otimes\nu$, showing that the set $\Gamma(\mu,\nu)$ is never empty. We endow $\mathscr{P}_2(\Omega)$ with a metric, here chosen as the Monge–Kantorovich distance with $p = 2$ - also called $2-$*Wasserstein distance* in the literature - and defined as

$$d_{\mathcal{W},\Omega}(\mu,\nu) \coloneqq \sqrt{\inf_{\eta\in\Gamma(\mu,\nu)} \int_{(x,y)\in(\Omega)^2} d^2(x,y)\mathrm{d}\eta(x,y)}.$$

The set of plans realizing the infimum is denoted $\Gamma_o(\mu,\nu)$. According to Theorem 4.1 in [8], this set is always nonempty. In the sequel, we will simply denote $d_{\mathcal{W}}$ the Wasserstein distance when the state space is clear from the context.

Throughout this paper, we consider $\mathbb{R}^\mathrm{d}$ as the underlying space. Its tangent space is defined as $\mathrm{T}\,\mathbb{R}^\mathrm{d} = \bigcup_{x\in\mathbb{R}^\mathrm{d}}\{x\}\times\mathrm{T}_x\,\mathbb{R}^\mathrm{d}$, where $\mathrm{T}_x\,\mathbb{R}^\mathrm{d}$ represents the tangent space to $\mathbb{R}^\mathrm{d}$ at the point x. To maintain a clear distinction between points and velocities, we refrain from identifying $\mathrm{T}_x\,\mathbb{R}^\mathrm{d}$ with $\mathbb{R}^d$. Let $T > 0$ be a fixed final time-horizon. We will use the notation

$$X \coloneqq]0,T[\times\mathscr{P}_2(\mathbb{R}^\mathrm{d}), \qquad d_X^2((t,\mu),(s,\nu)) \coloneqq |t-s|^2 + d_{\mathcal{W}}^2(\mu,\nu).$$

Trajectories in the Wasserstein space. Consider a controlled dynamical system of the form

$$\partial_s\mu_s + \mathrm{div}\,\left(f(\cdot,\mu_s,u(s))\#\mu_s\right) = 0, \quad s\in[t,T], \qquad \mu_t = \nu \tag{2.1}$$

where $\nu \in \mathscr{P}_2(\mathbb{R}^\mathrm{d})$ is an initial configuration of the system at a time $t \in [0,T]$. Here the divergence operator is distribution-valued, and defined as

$$\mathrm{div}\,(\xi)(\varphi) \coloneqq -\int_{(x,v)\in\mathrm{T}\,\mathbb{R}^\mathrm{d}} \langle\nabla_x\varphi, v\rangle\,\mathrm{d}\xi(x,v) \qquad \forall\varphi\in\mathcal{C}_c^\infty(\mathbb{R}^\mathrm{d};\mathbb{R}).$$

We denote $(\mu_s^{t,\nu,u})_{s\in[t,T]}$ the solution of (2.1), which is understood in the sense of distributions. Here, the control input $u(\cdot)$ is supposed to be a measurable function, *i.e.*

$$u(\cdot) \in L^1([t,T];U) \coloneqq \{v(\cdot) : [t,T] \mapsto U \mid v(\cdot) \text{ is Lebesgue-measurable}\},$$

where $U \subset \mathbb{R}^\kappa$ is a set of admissible controls, and $f : \mathbb{R}^\mathrm{d} \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times U \mapsto \mathrm{T}\,\mathbb{R}^\mathrm{d}$ is a given controlled and measure-dependant dynamic. The study of the dynamical system (2.1) will be carried out in Section 4.1.

The control problem and the value function. Now, consider a terminal cost $\mathfrak{J} : \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}$. The control problem we address in this paper is in Mayer form, and it consists of the following minimization problem:

$$\text{Find } u \in L^1([t,T];U) \qquad \text{such that} \quad \mathfrak{J}\left(\mu_T^{t,\nu,u}\right) \leqslant \mathfrak{J}(\mu_T^{t,\nu,v}) \quad \forall v \in L^1([t,T];U). \tag{2.2}$$

The adjustments in the case of a Bolza problem with Lipschitz running cost are discussed in Section 6.3. The *value function* associated to this control problem is defined as

$$V : [0,T] \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}, \qquad V(t,\nu) \coloneqq \inf\left\{\mathfrak{J}\left(\mu_T^{t,\nu,u}\right) \;\middle|\; u \in L^1([t,T];U)\right\}. \tag{2.3}$$

The aim of the paper is to characterize the value function as the unique *viscosity solution* of a suitable Hamilton–Jacobi–Bellman (HJB) equation of the form

$$-\partial_t V(t,\mu) + H\left(\mu, D_\mu V(t,\mu)\right) = 0, \qquad V(T,\mu) = \mathfrak{J}(\mu).$$

The definition of the Hamiltonian H and the meaning of the derivative $D_\mu V(t,\mu)$ will be made precise in Sections 5 & 6. In these sections, we will also develop the notion of viscosity solution in the Wasserstein space.

Running assumptions. Let us precise the main assumptions of the paper. We say that an application $m : \mathbb{R}^+ \to \mathbb{R}^+$ is a modulus of continuity if it is continuous, nondecreasing and if $m(0) = 0$.

Assumption [A1] (Control set). The set $U \subset \mathbb{R}^\kappa$ is compact.

Assumption [A2] (Structure of the dynamic). There exist constants $[f]$, $|f|_{0,\infty}$ such that

- f is Lipschitz-continuous in the space and measure variables with constant $[f]$, in the sense that for all $(x,y,\mu,\nu,u) \in \mathbb{R}^\mathrm{d} \times \mathbb{R}^\mathrm{d} \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times U$,

$$|f(x,\mu,u) - f(y,\nu,u)| \leqslant [f]\left(|x-y| + d_{\mathcal{W}}(\mu,\nu)\right).$$

- there exists a modulus of continuity $m_{f,u} : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $(x,\mu,u,v) \in \mathbb{R}^\mathrm{d} \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times U \times U$,

$$|f(x,\mu,u) - f(x,\mu,v)| \leqslant (1+|x|)m_{f,u}(|u-v|).$$

- For all $(\mu,u) \in \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times U$, there holds $|f(0,\mu,u)| \leqslant |f|_{0,\infty}$.

Assumption [A3] (Regularity of the terminal cost). The terminal cost $\mathfrak{J} : \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}$ is locally uniformly continuous, *i.e.* for each $R > 0$, there exists a modulus of continuity $m_{\mathfrak{J},R} : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$|\mathfrak{J}(\mu) - \mathfrak{J}(\nu)| \leqslant m_{\mathfrak{J},R}(d_{\mathcal{W}}(\mu,\nu)) \qquad \forall (\mu,\nu) \in \left(\overline{\mathscr{B}}(\delta_0,R)\right)^2,$$

where $\mathscr{B}(\delta_0,R)$ stands for the ball centred in the Dirac measure δ_0 and with radius R.

Under these assumptions, our objective is to show that the value function uniquely solves a HJB equation (in an appropriate sense). In the classical theory of HJ equations in Banach spaces, the Hamiltonian is typically defined as a function of gradients or, more generally, over the space of linear functionals. In this work, we introduce the Hamiltonian through a *metric cotangent bundle* $\mathbb{T}$, which serves as an alternative to the conventional mapping $p \mapsto \langle \nabla\phi, p\rangle$. The key idea behind the construction of $\mathbb{T}$ is to encapsulate local approximations of test functions in a manner that accommodates the geometric features of the underlying space. Unlike traditional linear structures, the metric cotangent bundle is composed of continuous and positively homogeneous mappings, which may lack linearity but remain suitable for our framework. In the context of HJB equations, linearity is not a strict requirement for establishing uniqueness and characterizing the value function. Instead, the essential interpretation of HJB equations revolves around enforcing growth conditions along the trajectories of the control problem, which relies fundamentally on one-sided derivatives rather than full gradient information.

The challenges arising in the Wasserstein space are twofold: firstly, the space lacks local compactness. This issue can be addressed by employing adapted Ekeland principles, as previously demonstrated in Hilbert spaces in [30]. Secondly, the Wasserstein space exhibits positive curvature, which proves to be unfavourable for stability, contrasting with the reasoning applicable to negatively curved spaces. The critical aspect here lies in the fact that, broadly speaking, the directional derivative of a convex map enjoys upper semicontinuity, while concave maps have lower semicontinuous directional derivatives. This makes Hypothesis 3.4 in [29], which assumes some upper/lower semicontinuity of the Hamiltonian, unattainable in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$.

3. Preliminaries on the Wasserstein space

The space $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, when endowed with the Wasserstein distance, is a geodesic space. A *constant speed geodesic parameterized over* $[0,1]$, or in short a *geodesic*, is a curve $(\mu_t)_{t\in[0,1]} \subset \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ satisfying $d_{\mathcal{W}}(\mu_t,\mu_s) \leqslant |t-s|\, d(\mu_0,\mu_1)$ for all $(s,t) \in [0,1]^2$. We first recall some results on geodesics, and then define directional derivatives along them.

3.1. Representation of geodesics of $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$

Let $\mu,\nu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, and $\pi_x, \pi_y : (\mathbb{R}^{\mathrm{d}})^2 \mapsto \mathbb{R}^{\mathrm{d}}$ be the canonical projections $\pi_x((x,y)) = x$ and $\pi_y((x,y)) = y$. In $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, it is known (see [9], Thm. 7.2.2) that constant speed geodesics coincide with trajectories of the form

$$\mu_t = ((1-t)\pi_x + t\pi_y)\#\eta \qquad \forall t \in [0,1], \quad \eta = \eta(x,y) \in \Gamma_o(\mu,\nu). \tag{3.1}$$

The uniqueness of geodesics in the space $\mathbb{R}^{\mathrm{d}}$ also allows for another equivalent representation, by means of probability measures over the tangent space $\mathrm{T}\,\mathbb{R}^{\mathrm{d}} = \bigcup_{x\in\mathbb{R}^{\mathrm{d}}} \{x\} \times \mathrm{T}_x\,\mathbb{R}^{\mathrm{d}}$. We will denote $(x,v) \in \mathrm{T}\,\mathbb{R}^{\mathrm{d}}$ a generic tangent element, and π_x, π_v the canonical projections. Let $\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$ be the set of *initial velocities*

$$\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu = \left\{\gamma \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}}) \mid \pi_x\#\gamma = \mu\right\}.$$

Define the scalar multiplication $\cdot$ of velocities as $t\cdot\gamma = (\pi_x, t\pi_v)\#\gamma$. To each element $\gamma \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$, we associate a curve of measures by the *exponential map*

$$\exp_\mu : \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu \mapsto \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}), \qquad \exp_\mu(t\cdot\gamma) \coloneqq (\pi_x + t\pi_v)\#\gamma.$$

The sets $\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$ and $\Gamma(\mu, \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})) = \bigcup_{\nu\in\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})} \Gamma(\mu,\nu)$ are in bijection through the map $\Psi : \Gamma(\mu, \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})) \mapsto \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$ given by

$$\Psi(\eta) = (\pi_x, \pi_y - \pi_x)\#\eta, \qquad \Psi^{-1}(\gamma) = (\pi_x, \pi_x + \pi_v)\#\gamma. \tag{3.2}$$

The mapping Ψ is bicontinuous in the respective Wasserstein topologies.

Remark 3.1 (Manifold case)**.** Let us stress that Ψ is a bijection owing to the uniqueness of geodesics in $\mathbb{R}^{\mathrm{d}}$. Indeed, if E is a manifold over which an exponential map $\exp : TE \mapsto E$ is defined, there can be several initial velocities of geodesics in the set $\{(x, v) \in TE \mid \exp_x(v) = y\}$. The corresponding theory is developed in [31] (see in particular Def. 1.4), and in [28] for the associated geodesic viscosity and HJB equations in the compact case.

Any geodesic induced by an optimal transport plan $\eta \in \Gamma_o(\mu, \nu)$ via (3.1) is equivalently represented using (3.2) as

$$\exp_\mu(t \cdot \Psi(\eta)) = (\pi_x + t\pi_v)\#\Psi(\eta) = (\pi_x + t(\pi_y - \pi_x))\#\eta = \mu_t.$$

We define the set of *initial velocities of geodesics* through the identification (3.2):

$$\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_{\mu,o} \coloneqq \left\{\Psi(\eta) \mid \eta \in \Gamma_o(\mu, \nu),\ \nu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})\right\}.$$

Following the notation of [32], we will denote $\exp_\mu^{-1}(\nu)$ the set of *initial velocities of geodesics* linking μ to ν, *i.e.*

$$\exp_\mu^{-1}(\nu) \coloneqq \left\{\gamma \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_{\mu,o} \text{ with } (\pi_x + \pi_v)\#\gamma = \nu\right\}. \tag{3.3}$$

We may equivalently define $\exp_\mu^{-1}$ as

$$\exp_\mu^{-1}(\nu) = \left\{\gamma \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})_\mu \;\middle|\; \exp_\mu(\gamma) = \nu,\ \int_{(x,v)\in \mathrm{T}\,\mathbb{R}^{\mathrm{d}}} |v|^2 \, d\gamma(x, v) = d^2_{\mathcal{W}}(\mu, \nu)\right\}.$$

This set is always nonempty, since $\Gamma_o(\mu, \nu)$ is nonempty (see for instance Thm. 1.7 of [33]). It is compact in $(\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}}), d_{\mathcal{W}})$ as the image of the compact $\Gamma_o(\mu, \nu)$ through the continuous identification (3.2). We refer to it as the set of initial velocities of geodesics issued from μ and reaching ν.

3.2. The tangent cone

In the sequel, we denote

$$\mathrm{T}^2\,\mathbb{R}^{\mathrm{d}} \coloneqq \left\{(x, v_1, v_2) \;\middle|\; x \in \mathbb{R}^{\mathrm{d}},\ v_i \in \mathrm{T}_x\,\mathbb{R}^{\mathrm{d}}\right\},$$

where $|(x, v_1, v_2) - (\overline{x}, \overline{v}_1, \overline{v}_2)|^2 \coloneqq |x - \overline{x}|^2 + \sum_{i=1}^2 |v_i - \overline{v}_i|^2$. Given $\xi_1, \xi_2 \subset \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})_\mu$, we define

$$\Gamma_\mu(\xi_1, \xi_2) \coloneqq \left\{\alpha \in \mathscr{P}(\mathrm{T}^2\,\mathbb{R}^{\mathrm{d}}) \;\middle|\; (\pi_x, \pi_{v_i})\#\alpha = \xi_i,\ i \in \{1, 2\}\right\}.$$

This particular set of transport plans is only allowing transfer of mass between pairs (x, v) and (y, w) such that $x = y$. For each $\mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, let $W_\mu : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})_\mu \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})_\mu \to \mathbb{R}^+$ be defined by

$$W^2_\mu(\xi, \zeta) \coloneqq \inf_{\alpha\in\Gamma_\mu(\xi,\zeta)} \int_{(x,v,w)\in\mathrm{T}^2\,\mathbb{R}^{\mathrm{d}}} |v - w|^2 \,\mathrm{d}\alpha(x, v, w). \tag{3.4}$$

As per [10], Theorem 4.4, W_μ is a metric over $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})_\mu$ and the infimum is always attained. Moreover, disintegrating $\xi = \xi_x \otimes \mu$ and $\zeta = \zeta_x \otimes \mu$ allows to get a representation of W_μ as (see [10], Prop. 4.2)

$$W^2_\mu(\xi, \zeta) \coloneqq \int_{x\in\mathbb{R}^{\mathrm{d}}} d^2_{\mathcal{W}}(\xi_x, \zeta_x) \,\mathrm{d}\mu(x). \tag{3.5}$$

As an useful particular case, we record that

$$W_\mu^2(\xi, 0_\mu) = \int_{(x,v)\in \mathrm{T}^2\,\mathbb{R}^\mathrm{d}} |v|^2 \,\mathrm{d}\xi(x,v) =: \|\xi\|_\mu^2. \tag{3.6}$$

We denote $\Gamma_{\mu,o}(\xi,\zeta)$ the subset of $\Gamma_\mu(\xi,\zeta)$ where the infimum of (3.4) is realized.

Definition 3.2 (Tangent cone [10], Def. 4.1)**.** For each $\mu \in \mathscr{P}_2(\mathbb{R}^\mathrm{d})$, we define

$$\mathbf{Tan}'_\mu\, \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \coloneqq \left\{\alpha\cdot\xi \;\middle|\; \alpha\in\mathbb{R}^+, \xi\in \exp_\mu^{-1}(\sigma) \text{ for some } \sigma\in\mathscr{P}_2(\mathbb{R}^\mathrm{d})\right\},$$

and $\mathbf{Tan}_\mu\, \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \coloneqq \overline{\mathbf{Tan}'_\mu\, \mathscr{P}_2(\mathbb{R}^\mathrm{d})}^{W_\mu}$.

The pre-tangent cone $\mathbf{Tan}'_\mu$ is the set of velocities ξ such that there exists $\varepsilon>0$ with $s\mapsto \exp_\mu(s\cdot\xi)$ being a geodesic between its endpoints over the time interval $[0,\varepsilon]$. Owing to [10], Proposition 4.30, there exists a well-defined projection

$$\pi^\mu : \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^\mathrm{d})_\mu \to \mathbf{Tan}_\mu\, \mathscr{P}_2(\mathbb{R}^\mathrm{d}), \qquad \pi^\mu(\gamma) = \operatorname*{argmin}_{\overline{\gamma}\in \mathbf{Tan}_\mu\, \mathscr{P}_2(\mathbb{R}^\mathrm{d})} W_\mu(\gamma,\overline{\gamma}). \tag{3.7}$$

Note that by picking some $\alpha\in\Gamma_{\mu,o}(\xi,\zeta)$, we directly have

$$d^2_{\mathcal{W},\mathrm{T}\,\mathbb{R}^\mathrm{d}}(\xi,\zeta) \leqslant \int_{(\mathrm{T}\,\mathbb{R}^\mathrm{d})^2} |x-y|^2 + |v-w|^2 \, d[(\pi_x,\pi_v),(\pi_x,\pi_w)]\#\alpha = W_\mu^2(\xi,\zeta). \tag{3.8}$$

Moreover, each transport plan $\alpha\in\Gamma(\xi,\zeta)$ induces a transport plan between $\exp_\mu(t\cdot\xi)$ and $\exp_\mu(t\cdot\zeta)$ by $\beta \coloneqq (\pi_x + t\pi_v, \pi_x + t\pi_w)\#\alpha$. Consequently

$$d^2_{\mathcal{W}}\left(\exp_\mu(t\cdot\xi), \exp_\mu(t\cdot\zeta)\right) \leqslant W_\mu^2(t\cdot\xi, t\cdot\zeta) = t^2 W_\mu^2(\xi,\zeta). \tag{3.9}$$

Let $(\mu,\nu)\in(\mathscr{P}_2(\mathbb{R}^\mathrm{d}))^2$. Following [34], Definition 4.1, we define an application $W_{(\mu,\nu)} : \mathscr{P}(\mathrm{T}\,\mathbb{R}^\mathrm{d})_\mu \times \mathscr{P}(\mathrm{T}\,\mathbb{R}^\mathrm{d})_\nu \to \mathbb{R}^+$ by

$$W^2_{(\mu,\nu)}(\xi,\zeta) \coloneqq \inf\left\{ \int_{(\mathrm{T}\,\mathbb{R}^\mathrm{d})^2} |v-w|^2 \,\mathrm{d}\omega(x,v,y,w) \;\middle|\; \begin{array}{c} \omega\in\Gamma(\xi,\zeta), \\ \pi_{(x,y)}\#\omega\in\Gamma_o(\mu,\nu) \end{array} \right\}. \tag{3.10}$$

The map $W_{(\mu,\nu)}$ computes the difference between ξ and ζ by taking only paths whose projection on the base space is a geodesic. It is coherent with the tangent cone structure, since $W_{(\mu,\mu)}(\xi,\zeta) = W_\mu(\xi,\zeta)$ for all $\xi,\zeta\in\mathscr{P}(\mathrm{T}\,\mathbb{R}^\mathrm{d})_\mu$. However, the application $W_{(\mu,\nu)}$ does not satisfy the triangular inequality (see [34], Rem. 4).

3.3. The metric cotangent bundle

Definition 3.3 (Directionally differentiable map)**.** We say that an application $\varphi : \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}$ is directionally differentiable at $\mu\in\mathscr{P}_2(\mathbb{R}^\mathrm{d})$ if for all $\xi\in\mathbf{Tan}_\mu\,\mathscr{P}_2(\mathbb{R}^\mathrm{d})$, the limit

$$\lim_{t\searrow 0} \frac{\varphi(\exp_\mu(t\cdot\xi)) - \varphi(\mu)}{t} =: D_\mu\varphi(\xi) \tag{3.11}$$

exists. The application $D_\mu\varphi : \mathbf{Tan}_\mu\,\mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}$ is called the differential at μ of φ.

Notice that in (3.11), we do not assume the limit to be uniform in $\xi \in \mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. It is immediate that $D_\mu\varphi$ is positively homogeneous, *i.e.* $D_\mu\varphi(\alpha\cdot\xi) = \alpha D_\mu\varphi(\xi)$ for any $\alpha \geqslant 0$. Moreover, assume that φ is Lipschitz with constant $[\varphi]$ in some ball centred in μ. Then using (3.9),

$$\left|D_\mu\varphi(\xi) - D_\mu\varphi(\bar{\xi})\right| \leqslant [\varphi] \lim_{t\searrow 0} \frac{d_{\mathcal{W}}(\exp_\mu(t\cdot\xi), \exp_\mu(t\cdot\bar{\xi}))}{t} = [\varphi]\, W_\mu(\xi,\bar{\xi}). \tag{3.12}$$

Hence $D_\mu\varphi$ is Lipschitz in $\left(\mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}), W_\mu\right)$. The above leads us to the following definition.

Definition 3.4 (Metric cotangent bundle)**.** Let

$$\mathbb{T}_\mu \coloneqq \left\{ p : \mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R} \;\middle|\; \begin{array}{c} p \text{ Lipschitz w.r.t. } W_\mu, \\ \text{and positively homogeneous} \end{array} \right\}, \tag{3.13}$$

and $\mathbb{T} \coloneqq \bigcup_{\mu\in\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})} \{\mu\} \times \mathbb{T}_\mu$. The sets $\mathbb{T}_\mu$ are stable by the pointwise operations $(p+q)(\xi) \coloneqq p(\xi) + q(\xi)$ and $(\alpha p)(\xi) \coloneqq \alpha p(\xi)$. We endow $\mathbb{T}$ with the application

$$\|\cdot\| : \mathbb{T} \to \mathbb{R}^+, \qquad (\mu, p) \mapsto \|p\|_\mu \coloneqq \sup_{\xi\in\mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}), \|\xi\|_\mu = 1} |p(\xi)|\,. \tag{3.14}$$

The application $\|\cdot\|_\mu$ induces a norm on $\mathbb{T}_\mu$, and we recover $|p(\xi)| \leqslant \|p\|_\mu \|\xi\|_\mu$. The metric cotangent bundle contains all the infinitesimal approximations of "sufficiently smooth maps", generalizing the set of linear applications. A partial differential equation in the space of measures involves elements of $\mathbb{T}$, and we will naturally define the Hamiltonian as a function of $\mathbb{T}$ into $\mathbb{R}$.

4. The control problem

4.1. Trajectories

The celebrated results of [9] indicate that absolutely continuous curves in the Wasserstein space coincide with the solutions of the continuity equation in the sense of distributions. The recent work of [35, 36] raised the theory of continuity equations and continuity inclusions in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ to a level comparable to that of the Caratheodory differential inclusions in $\mathbb{R}^d$. Let us mention that the study of dynamical systems driven by measure-valued, discontinuous dynamics is drawing attention (see the Measure Differential Equations (MDE) of [34, 37]), although it is known that in the Lipschitz setting, solutions of MDEs and continuity equations coincide ([37], Them. 1). In this section, we first reformulate the controlled dynamical system (2.1) in order to apply the results of [36], gather some estimates and properties needed in the sequel, and study the properties of the value function.

4.1.1. Existence, representation and regularity

Denote $\mathscr{P}(U)$ the set of probability measures over the compact U, which is itself a compact set when endowed with the squared Wasserstein distance. Let $L^1([t,T];\mathscr{P}(U))$ be the space of Lebesgue-measurable curves $\omega : [t,T] \to \mathscr{P}(U)$. Define

$$F : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \times \mathscr{P}(U) \to \mathcal{C}(\mathbb{R}^{\mathrm{d}}; \mathrm{T}\,\mathbb{R}^{\mathrm{d}}), \qquad (\mu,\omega) \mapsto F_\omega[\mu] \coloneqq \int_{u\in U} f(\cdot,\mu,u)\mathrm{d}\omega(u).$$

Under [A2], routine computations show that for all $\mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, all $\omega, \varpi \in \mathscr{P}(U)$ and $x \in \mathbb{R}^{\mathrm{d}}$, there holds

$$|F_\omega[\mu](x) - F_\varpi[\mu](x)| \leqslant [f]\,(1+|x|)\sqrt{\inf_{\alpha\in\Gamma(\omega,\varpi)} \int_{(u,v)\in U^2} m_{f,u}^2(|u-v|)\mathrm{d}\alpha(u,v)}.$$

As $m_{f,u}$ is continuous and U compact, the right hand-side of the above inequality is continuous with respect to $d_{\mathcal{W},U}$. Consequently, the application $\omega \mapsto F_\omega[\mu]$ is continuous from $(\mathscr{P}(U), d_{\mathcal{W},U})$ to $\mathcal{C}\left(\mathbb{R}^{\mathrm{d}}; \mathrm{T}\,\mathbb{R}^{\mathrm{d}}\right)$ endowed with the topology of convergence over compact sets. Therefore, $[t,T] \ni s \mapsto F_{\omega(s)}[\mu]$ is Lebesgue-measurable for each $\omega \in L^1([t,T];\mathscr{P}(U))$. For some fixed $\nu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, consider the associated continuity equation

$$\partial_s \mu_s + \mathrm{div}\,\left(F_{\omega(s)}[\mu_s]\#\mu_s\right) = 0 \quad s \in [t,T], \qquad \mu_t = \nu. \tag{4.1}$$

Combining Theorems 2.18, Proposition 2.22 and Theorem 4.2 of [36], we get the following.

Theorem 4.1 (Existence, uniqueness and representation of the solution)**.** *Assume [A1] and [A2]. For each $\omega \in L^1([t,T];\mathscr{P}(U))$, there exists an unique trajectory $(\mu_s^{t,\nu,\omega})_{s\in[t,T]} \in AC\left([t,T];\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})\right)$ solution of (4.1) in the sense of distributions. Moreover $s \mapsto F_{\omega(s)}[\mu_s^{t,\nu,\omega}]$ is Lebesgue-measurable, there exist constants $m = m_{f,T}$ and $M = M_{\nu,f,T}$ such that $\forall t \leqslant s, \tau \leqslant T$,*

$$d_{\mathcal{W}}(\mu_s^{t,\nu,\omega}, \delta_0) \leqslant m\,(1 + d_{\mathcal{W}}(\nu,\delta_0))\,, \qquad d_{\mathcal{W}}(\mu_s^{t,\nu,\omega}, \mu_\tau^{t,\nu,\omega}) \leqslant M\,|\tau - s|\,.$$

and the solution is given by the pushforward $\mu_s^{t,\nu,\omega} = \Phi_s^t\#\nu$, where $\Phi_s^t : \mathbb{R}^{\mathrm{d}} \to \mathbb{R}^{\mathrm{d}}$ is the well-defined flow of

$$\frac{\mathrm{d}}{\mathrm{d}s}\Phi_s^t(x) = F_{\omega(s)}[\mu_s^{t,\nu,\omega}](\Phi_s^t(x)), \qquad \Phi_t^t(x) = x. \tag{4.2}$$

Choosing $\omega(s) = \delta_{u(s)}$ for some $u \in L^1([t,T];U)$, Theorem 4.1 brings well-posedness of the controlled system (2.1). The pushforward representation allows to obtain various estimates directly from the underlying dynamical system. In particular, a Grönwall estimate yields that

$$\left|\Phi_s^{0,x}(x) - x\right| \leqslant s\left([f]\,|x| + |f|_{0,\infty}\right)\mathrm{e}^{[f]s}, \tag{4.3}$$

and

$$d_{\mathcal{W}}(\mu_{t+s}^{t,\nu,\omega}, \nu) \leqslant s\left([f]\,d_{\mathcal{W}}(\nu,\delta_0) + |f|_{0,\infty}\right)\mathrm{e}^{[f]s} \tag{4.4}$$

for all $\omega \in L^1([t,T];\mathscr{P}(U))$, $\nu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, $x \in \mathbb{R}^{\mathrm{d}}$ and $0 \leqslant t, s, t+s \leqslant T$ (see Appendix E).Define the reachable sets from (t,ν) at time T by the flow of (4.1) as

$$\mathcal{R}_T^{t,\nu} := \left\{\mu_T^{t,\nu,\omega} \;\middle|\; \omega \in L^1([t,T];\mathscr{P}(U))\right\}. \tag{4.5}$$

Lemma 4.2 (Lipschitz-continuity of the reachable sets)**.** *Assume [A1] and [A2]. There exists a constant $[\mathcal{R}]$ depending only on f and T such that for all $t \in [0,T]$ and $\nu, \overline{\nu} \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$,*

$$\max\left(\sup_{\mu\in\mathcal{R}_T^{t,\nu}}\inf_{\overline{\mu}\in\mathcal{R}_T^{t,\overline{\nu}}} d_{\mathcal{W}}(\mu,\overline{\mu}), \sup_{\overline{\mu}\in\mathcal{R}_T^{t,\overline{\nu}}}\inf_{\mu\in\mathcal{R}_T^{t,\nu}} d_{\mathcal{W}}(\mu,\overline{\mu})\right) \leqslant [\mathcal{R}]d_{\mathcal{W}}(\nu,\overline{\nu}).$$

Proof. Using chained Grönwall estimates (see Appendix E), we have that two solutions $s \mapsto \mu_s^{t,\nu,\omega}$ and $s \mapsto \mu_s^{t,\overline{\nu},\omega}$ associated to the same control $\omega \in L^1([t,T];\mathscr{P}(U))$ satisfy

$$d_{\mathcal{W}}\left(\mu_T^{t,\nu,\omega}, \mu_T^{t,\overline{\nu},\omega}\right) \leqslant \exp\left([f]\,(T-t)\left(\mathrm{e}^{[f](T-t)}+1\right)\right) d_{\mathcal{W}}(\nu,\overline{\nu}).$$

The claim follows by approximating each $\mu = \mu_T^{t,\nu,\omega} \in \mathcal{R}_T^{t,\nu}$ by $\mu_T^{t,\overline{\nu},\omega}$, and defining $[\mathcal{R}] \coloneqq \mathrm{e}^{[f]T\left(\mathrm{e}^{[f]T}+1\right)}$. □

Lemma 4.3 (Smooth case)**.** *Let $0 \leqslant t \leqslant T$ and $\omega \in L^1([t,T];\mathscr{P}(U))$ be a constant control $\omega(s) \equiv \varpi \in \mathscr{P}(U)$. Then the unique solution $(\mu_s^{t,\nu,\omega})_{s\in[t,T]}$ satisfy*

$$\lim_{h\searrow 0} \frac{d_{\mathcal{W}}\left(\mu_{t+h}^{t,\nu,\omega}, \exp_\mu(h\cdot F_\varpi[\nu])\right)}{h} = 0.$$

The proof relies only on Grönwall estimates, and is deferred to Appendix A.

4.1.2. Convex relaxation of the dynamic

Let $\overline{\mathrm{co}}F : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \rightrightarrows \mathcal{C}(\mathbb{R}^{\mathrm{d}};\mathrm{T}\,\mathbb{R}^{\mathrm{d}})$ be given by

$$\overline{\mathrm{co}}F[\mu] \coloneqq \{F_\omega[\mu] \mid \omega \in \mathscr{P}(U)\} = \left\{ \int_{u\in U} f(\cdot,\mu,u)\mathrm{d}\omega(u) \;\middle|\; \omega \in \mathscr{P}(U) \right\}. \tag{4.6}$$

For each μ, the set $\overline{\mathrm{co}}F[\mu] \subset \mathcal{C}(\mathbb{R}^{\mathrm{d}};\mathrm{T}\,\mathbb{R}^{\mathrm{d}})$ is closed in the topology of uniform convergence on compact sets; indeed, as this topology is metrizable, it suffices to show that $(b_n)_n \subset \overline{\mathrm{co}}F[\mu]$ and $b_n \to_n b$ uniformly over the compacts implies $b \in \overline{\mathrm{co}}F[\mu]$. As $\mathscr{P}(U)$ is compact ([9], Prop. 7.1.5), some subsequence of ω_n converges to a measure $\omega \in \mathscr{P}(U)$, and [A2] yields that for each $R > 0$,

$$\sup_{|x|\leqslant R}\left|\int_{u\in U} f(x,\mu,u)d[\omega_n-\omega]\right| \leqslant (1+R)\inf_{\alpha\in\Gamma(\omega_n,\omega)}\int_{(u,v)\in U^2} m_{f,u}\left(|u-v|\right)\mathrm{d}\alpha \underset{n\to\infty}{\longrightarrow} 0.$$

By uniqueness of the limit, $b(x) = \int_{u\in U} f(x,\mu,u)\mathrm{d}\omega(u)$, and $b \in \overline{\mathrm{co}}F[\mu]$.

Remark 4.4 (Link with the closed convex envelope of [36])**.** In our case, $\overline{\mathrm{co}}F[\mu]$ is equal to the closure of $\mathcal{M}$ in the topology of uniform convergence over compact sets, where

$$\mathcal{M} \coloneqq \bigcup_{N\in\mathbb{N}}\left\{ \sum_{i=1}^N \alpha_i f(\cdot,\mu,u_i) \;\middle|\; \sum_{i=1}^N \alpha_i = 1, \alpha_i \geqslant 0, u_i \in U \right\}.$$

Indeed, we trivially have $\mathcal{M} \subset \overline{\mathrm{co}}F$ through the representation $\omega \coloneqq \sum_{i=1}^N \alpha_i \delta_{u_i}$. On the other hand, let $\bar{b} = F_\omega[\mu] \in \overline{\mathrm{co}}F$ for some $\omega \subset \mathscr{P}(U)$. For each $n \in \mathbb{N}$, cover the compact U by a finite measurable partition $(U_i^n)_{i\in[\![1,N_n]\!]}$ of diameter inferior to $1/n$, and pick $u_i^n \in U_i^n$. Owing to [A2], there holds for each $R > 0$

$$\begin{aligned}
\left\| \int_U f(\cdot,\mu,u)\mathrm{d}\omega(u) - \sum_{i=1}^{N_n}\omega(U_i^n) f(\cdot,\mu,u_i^n) \right\|_{\mathcal{C}} &\leqslant \sup_{|x|\leqslant R}\sum_{i=1}^{N_n}\int_{U_i^n}\left|f(x,\mu,u)-f(x,\mu,u_i^n)\right|\mathrm{d}\omega(u) \\
&\leqslant (1+R)\sum_{i=1}^{N_n}\int_{U_i^n} m_{f,u}\left(|u-u_i^n|\right)\mathrm{d}\omega(u) \\
&\leqslant (1+R)\, m_{f,u}\left(1/n\right),
\end{aligned}$$

and $\overline{b}$ is the uniform limit over the each compact of the sequence $\left(\sum_{i=1}^{N_n}\omega(U_i^n)f(\cdot,\mu,u_i^n)\right)_n \subset \mathcal{M}$. Thus $\overline{b}\in\mathcal{M}$, and equality holds.

Using a selection argument, the set of solutions $\left\{(\mu_s^{t,\nu,\omega})_{s\in[t,T]} \;\middle|\; \omega\in L^1([t,T];\mathscr{P}(U))\right\}$ coincides with the set of solutions of the continuity inclusion

$$\partial_s\mu_s \in -\operatorname{div}\left(\overline{\operatorname{co}}F[\mu_s]\#\mu_s\right) \quad s\in[t,T], \qquad \mu_t=\nu.$$

Consequently, we have the following.

Theorem 4.5 (Relaxation (Thms. 4.5 and 4.6 of [36]))**.** *Assume [A1] and [A2], and let $\nu\in\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. The set*

$$\left\{(\mu_s^{t,\nu,\omega})_{s\in[t,T]} \;\middle|\; \omega\in L^1([t,T];\mathscr{P}(U))\right\} \subset AC([t,T];\mathscr{P}_2(\mathbb{R}^{\mathrm{d}}))$$

is compact in the topology of the uniform convergence, and is the closure in this topology of the set of trajectories of (2.1), namely

$$\left\{(\mu_s^{t,\nu,u})_{s\in[t,T]} \;\middle|\; u\in L^1([t,T];U)\right\}.$$

4.1.3. Linearization of the trajectory

The following technical Lemma allows us to elude the lack of differentiability of a solution of the dynamical system (4.1), by approximating the said curve only along some sequence.

Lemma 4.6 (Right linear approximation)**.** *Assume [A1] and [A2]. Let $\overline{s}>0$, $(\mu_s)_{s\in[0,\overline{s}]}$ be the solution of (4.1) for some control $\omega\in L^1([0,\overline{s}];\mathscr{P}(U))$. Then there exists $b\in\overline{\operatorname{co}}F[\mu_0]$ and a sequence $(s_n)\searrow 0$ such that*

$$\lim_{n\to\infty}\frac{d_{\mathcal{W}}\left(\mu_{s_n},m_{s_n}\#\mu_0\right)}{s_n}=0, \quad \textit{where} \quad m_s:\mathbb{R}^{\mathrm{d}}\ni x\mapsto x+sb(x)\in\mathbb{R}^{\mathrm{d}}.$$

Proof. Let $(s_n)_n\searrow 0$, and define $b_n:\mathbb{R}^{\mathrm{d}}\mapsto T\,\mathbb{R}^{\mathrm{d}}$ by

$$b_n(x)\coloneqq\frac{1}{s_n}\int_{s=0}^{s_n}F_{\omega(s)}[\mu_0](x)\mathrm{d}s=F_{\overline{\omega}_n}[\mu_0](x), \quad \text{where} \quad \overline{\omega}_n\coloneqq\frac{1}{s_n}\int_{s=0}^{s_n}\omega(s)\mathrm{d}s.$$

Since U is compact, so is $\mathscr{P}(U)$ (see [8], Rem. 6.19). Then, along a (non relabeled) subsequence, $\overline{\omega}_n$ converges to some $\overline{\omega}\in\mathscr{P}(U)$ for the Wasserstein distance over $\mathscr{P}(U)$. Let $b\in\overline{\operatorname{co}}F(\mu_0)$ be given by $b=F_{\overline{\omega}}[\mu_0]$. For each $n\in\mathbb{N}$, let $\eta_n\in\Gamma(\omega_n,\omega)$ such that

$$\widetilde{d_{\mathcal{W}}}(\omega_n,\overline{\omega})\coloneqq\inf_{\eta\in\Gamma(\omega_n,\overline{\omega})}\int_{U^2}m_{f,u}(|u-v|)\mathrm{d}\eta(u,v)\geqslant\int_{U^2}m_{f,u}(|u-v|)\mathrm{d}\eta_n(u,v)-\frac{1}{n}.$$

Using [A2], we have that

$$|b_n(x)-b(x)|\leqslant\int_{U^2}|f(x,\mu_0,u)-f(x,\mu_0,v)|\,\mathrm{d}\eta_n(u,v)\leqslant(1+|x|)\left[\widetilde{d_{\mathcal{W}}}(\omega_n,\overline{\omega})+\frac{1}{n}\right],$$

and we conclude to the local uniform convergence of b_n towards b. We know from Theorem 4.1 that $\mu_s=\Phi_s^{0,\omega}\#\mu_0$, where the semigroup $\Phi_s^{0,\omega}$ is defined in (4.2). Let $m_s:\mathbb{R}^{\mathrm{d}}\mapsto\mathbb{R}^{\mathrm{d}}$ be given by $m_s(x)=x+sb(x)$. Along

the sequence $(s_n)_n$, we have that

$$\begin{aligned}
\left|\Phi^{0,\omega}_{s_n}(x) - m_{s_n}(x)\right| &= \left|\int_{s=0}^{s_n} F_{\omega(s)}[\mu_s]\left(\Phi^{0,\omega}_s(x)\right)\mathrm{d}s - s_n b(x)\right| \\
&\leqslant \int_{s=0}^{s_n} \left|F_{\omega(s)}[\mu_s]\left(\Phi^{0,\omega}_s(x)\right) - F_{\omega(s)}[\mu_0]\left(x\right)\right|\mathrm{d}s + s_n\left|F_{\overline{\omega}_n}[\mu_0](x) - F_{\overline{\omega}}[\mu_0](x)\right| \\
&\leqslant [f]\int_{s=0}^{s_n} d_{\mathcal{W}}(\mu_s,\mu_0) + \left|\Phi^{0,\omega}_s(x) - x\right|\mathrm{d}s + s_n\,[f]\,(1+|x|)\left(d_{\mathcal{W},m_{f,u}}(\overline{\omega}_n,\overline{\omega}) + \frac{1}{n}\right).
\end{aligned}$$

By (4.4), we have $d_{\mathcal{W}}(\mu_s,\mu_0) \leqslant s\mathrm{e}^{[f]s}\left([f]\,d_{\mathcal{W}}(\mu_0,\delta_0) + |f|_{0,\infty}\right)$. Plugging this into the above, we get after simplification

$$\frac{d_{\mathcal{W}}\left(\mu_{s_n}, m_{s_n}\#\mu_0\right)}{s_n} \leqslant 2\,[f]\left[s_n\mathrm{e}^{[f]s_n}\left([f]\,d_{\mathcal{W}}(\mu_0,\delta_0) + |f|_{0,\infty}\right) + \widetilde{d_{\mathcal{W}}}(\overline{\omega}_n,\overline{\omega}) + \frac{1}{n}\right],$$

which goes to 0 when n goes to ∞. Hence the result. □

4.2. Properties of the value function

Recall that the value function is defined as

$$V : [0,T]\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}, \qquad V(t,\nu) \coloneqq \inf\left\{\,\mathfrak{J}(\mu_T^{t,\nu,u}) \;\middle|\; u\in L^1([t,T];U)\,\right\}.$$

From Theorem 4.5 and the assumption [A3] of local uniform continuity of $\mathfrak{J}$, we obtain that the set of solutions of the relaxed system (4.1) may be substituted to the set of solutions of the original problem (2.1) without changing the value function, that is,

$$V(t,\nu) = \inf\left\{\,\mathfrak{J}(\mu_T^{t,\nu,\omega}) \;\middle|\; \omega\in L^1([t,T];\mathscr{P}(U))\,\right\} = \inf_{\mu\in\mathcal{R}_T^{t,\nu}}\mathfrak{J}(\mu),$$

where the reachable set $\mathcal{R}_T^{t,\nu}$ is defined in (4.5). Notice that this equality would stand as well with a running cost, since the relaxation result concerns the whole trajectories and not only the reachable sets. In this deterministic setting, we retrieve the classical Dynamic Programming Principle (DPP): for each $0 < h \leqslant T - t$,

$$V(t,\nu) = \inf_{\mu\in\mathcal{R}_{t+h}^{t,\nu}} V(t+h,\mu). \tag{4.7}$$

Lemma 4.7 (Local uniform continuity of the value function)**.** *Under the assumptions [A1], [A2] and [A3], the function V is locally uniformly continuous in time and space, i.e. for all $R>0$, there exists a modulus $m_{V,R}:\mathbb{R}^+\to\mathbb{R}^+$ such that for all $(t,s)\in[0,T]^2$ and $(\mu,\nu)\in\left(\overline{\mathscr{B}}(\delta_0,R)\right)^2$,*

$$|V(s,\nu) - V(t,\mu)| \leqslant m_{V,R}\left(|t-s| + d_{\mathcal{W}}(\mu,\nu)\right).$$

Proof. Let $R>0$, and denote $R_T \coloneqq R + T\exp[f]\,T\left([f]\,R + |f|_{0,\infty}\right)$ a radius large enough so that $\mathcal{R}_T^{0,\nu}\subset\overline{\mathscr{B}}(\delta_0,R_T)$ for all $\nu\in\overline{\mathscr{B}}(\delta_0,R)$. Let $m_{\mathfrak{J},R_T}$ be a local modulus of continuity of $\mathfrak{J}$ in the ball $\overline{\mathscr{B}}(\delta_0,R_T)$. According to the $[\mathcal{R}]-$Lipschitz continuity of the reachable sets given by Lemma 4.2, we have for all $t\in[0,T]$ and $\nu,\overline{\nu}\in$

$\overline{\mathscr{B}}(\delta_0, R)$ that

$$V(t,\nu) - V(t,\overline{\nu}) \leqslant \sup_{\overline{\mu}\in\mathcal{R}_T^{t,\overline{\nu}}} \inf_{\mu\in\mathcal{R}_T^{t,\nu}} \mathfrak{J}(\mu) - \mathfrak{J}(\overline{\mu}) \leqslant \sup_{\overline{\mu}\in\mathcal{R}_T^{t,\overline{\nu}}} \inf_{\mu\in\mathcal{R}_T^{t,\nu}} m_{\mathfrak{J},R_T}\left(d_{\mathcal{W}}(\mu,\overline{\mu})\right) \leqslant m_{\mathfrak{J},R_T}\left([\mathcal{R}]d_{\mathcal{W}}(\nu,\overline{\nu})\right).$$

On the other hand, let $0 \leqslant t \leqslant s \leqslant T$ and $\nu \in \overline{\mathscr{B}}(\delta_0, R)$. The DPP (4.7) and the Grönwall estimate (4.4) give us

$$\begin{aligned} V(t,\nu) - V(s,\nu) &= \inf_{\mu\in\mathcal{R}_s^{t,\nu}} V(s,\mu) - V(s,\nu) \leqslant \inf_{\mu\in\mathcal{R}_s^{t,\nu}} m_{\mathfrak{J},R_T}\left([\mathcal{R}]d_{\mathcal{W}}(\mu,\nu)\right) \\ &\leqslant m_{\mathfrak{J},R_T}\left([\mathcal{R}]\,|s-t|\exp\left([f]\,T\right)\left([f]\,R + |f|_{0,\infty}\right)\right), \end{aligned}$$

and similarly

$$\begin{aligned} V(s,\nu) - V(t,\nu) &= \sup_{\mu\in\mathcal{R}_s^{t,\nu}} V(s,\mu) - V(s,\nu) \leqslant \sup_{\mu\in\mathcal{R}_s^{t,\nu}} m_{\mathfrak{J},R_T}\left([\mathcal{R}]d_{\mathcal{W}}(\mu,\nu)\right) \\ &\leqslant m_{\mathfrak{J},R_T}\left([\mathcal{R}]\,|s-t|\exp\left([f]\,T\right)\left([f]\,R + |f|_{0,\infty}\right)\right). \end{aligned}$$

Hence V is locally uniformly continuous with a modulus depending on $\mathfrak{J}$, f and T. □

A close look at the proof reveals that if $\mathfrak{J}$ is uniformly continuous and f is globally bounded, the value function would also be globally uniformly continuous. However, since we do not assume this for $\mathfrak{J}$ and f, the value function is only locally uniformly continuous.

5. The Hamilton–Jacobi equation

In this section, we are interested into the following generic Hamilton–Jacobi equation

$$-\partial_t u(t,\mu) + H\left(\mu, D_\mu u(t,\mu)\right) = 0, \qquad u(T,\mu) = \mathfrak{J}(\mu). \tag{5.1}$$

Regularity assumptions on the Hamiltonian $H : \mathbb{T} \to \mathbb{R}$ will be made precise further.

5.1. Notion of viscosity solutions

Recall that $X :=]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ and $d_X^2((t,\mu),(s,\nu)) := |s-t|^2 + d_{\mathcal{W}}^2(\mu,\nu)$. In this section, we precise the definition of a viscosity solution of (5.1). To this aim, we will use a class of *test functions*, that will be more regular than the viscosity solution in order to bear the derivatives. The time variable and the measure variable of the test functions do not play symmetric roles, as weaker regularity on the measure dimension will be compensated by stronger assumptions on the time dimension.

5.1.1. Regularity in the measure variable

Definition 5.1 (Locally semiconcave/convex maps)**.** An application $u : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ is locally semiconcave in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ if for all $R > 0$, there exists $\lambda_R \in \mathbb{R}$ such that for all $\mu,\nu \in \overline{\mathscr{B}}(\delta_0, R)$ and $\eta = \eta(x,y) \in \Gamma_o(\mu,\nu)$, there holds for all $h \in [0,1]$

$$u\left(((1-h)\pi_x + h\pi_y)\#\eta\right) \geqslant (1-h)u(\mu) + hu(\nu) - \frac{\lambda_R}{2}h(1-h)d_{\mathcal{W}}^2\left(\mu,\nu\right).$$

An application $u : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ is semiconvex if $-u$ is semiconcave.

Locally semiconcave/convex maps are directionally differentiable at all points. As an important example, the squared Wasserstein distance is directionally differentiable (see [10], Sect. 4.2) and for any $\sigma \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ and $\xi \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$,

$$D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)(\xi) = \inf_{\eta \in \exp_\mu^{-1}(\sigma)} \inf_{\alpha \in \Gamma_\mu(\xi,\eta)} -2 \int_{(x,v,w)\in \mathrm{T}^2\,\mathbb{R}^{\mathrm{d}}} \langle v, w\rangle \,\mathrm{d}\alpha(x,v,w). \tag{5.2}$$

Remark 5.2. Moreover, given $\sigma \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, we may compute $\|D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)\|_\mu$ as defined in (3.14). Indeed,

$$\begin{aligned}\left|D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)(\xi)\right| &= \left|\inf_{\alpha\in\Gamma_\mu(\xi,\exp_\mu^{-1}(\sigma))} \int_{\mathrm{T}\,\mathbb{R}^{\mathrm{d}}} -2\,\langle v,w\rangle \,\mathrm{d}\alpha(x,v,w)\right| \\ &\leqslant 2 \sup_{\alpha\in\Gamma_\mu(\xi,\exp_\mu^{-1}(\sigma))} \int_{\mathrm{T}\,\mathbb{R}^{\mathrm{d}}} |v|\,|w| \,\mathrm{d}\alpha(x,v,w) \leqslant 2\|\xi\|_\mu d_{\mathcal{W}}(\mu,\sigma).\end{aligned}$$

Hence $\|D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)\|_\mu \leqslant 2 d_{\mathcal{W}}(\mu,\sigma)$. On the other hand, if $\sigma \neq \mu$, letting $\xi \in \exp_\mu^{-1}(\sigma)$,

$$\|D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)\|_\mu \geqslant \left|\frac{D_\mu d^2_{\mathcal{W}}(\cdot,\sigma)(\xi)}{\|\xi\|_\mu}\right| = \left|\lim_{h\searrow 0} \frac{(1-h)^2 d^2_{\mathcal{W}}(\mu,\sigma) - d^2_{\mathcal{W}}(\mu,\sigma)}{h\, d_{\mathcal{W}}(\mu,\sigma)}\right| = 2 d_{\mathcal{W}}(\mu,\sigma),$$

showing equality.

Remark 5.3 (Composition rule)**.** Let $\varphi \in \mathcal{C}^2(\mathbb{R}^+;\mathbb{R}^+)$ be nondecreasing, and consider the composition $\psi : \mu \mapsto \varphi(d^2_{\mathcal{W}}(\delta_0,\mu))$. Denote λ_R a local constant of semiconcavity of φ over $[0,3R]$, and $[\varphi_R]$ a local constant of Lipschitz-continuity of φ over the same domain. Then ψ is semiconcave with modulus $R\lambda_R + [\varphi_R]$ (see [38], Prop. 2.1.12).

The squared Wasserstein distance enjoys another interesting property: its directional derivative along ξ is always equal to that along $\pi^\mu\xi$, the projection onto the tangent cone. This follows from the explicit expression (5.2), which can be rewritten as the infimum of $-2\langle\alpha,\xi\rangle$, where $\langle\cdot,\cdot\rangle$ is the metric scalar product from [10], Definition 4.13. Furthermore, Corollary 4.34 of the same reference states that $\langle\alpha,\xi\rangle = \langle\alpha,\pi^\mu\xi\rangle$, since α is the optimal velocity of a geodesic. This additional structure prompts the following definition.

Definition 5.4 (Geometrically consistent applications)**.** An application $u : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ is geometrically consistent if it is directionally differentiable and if

$$\forall \mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}), \quad \forall \xi \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu, \qquad D_\mu u(\xi) = D_\mu u(\pi^\mu \xi).$$

This property will be used in Section 6 in connection with the choice of the Hamiltonian for the control problem; see Remark 6.1 for further details.

5.1.2. Regularity in the time variable

Definition 5.5 (Locally Lipschitz time derivative)**.** We say that $\varphi : X \to \mathbb{R}$ has locally Lipschitz time derivative if for all $\mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, $\varphi(\cdot,\mu) \in \mathcal{C}^1(]0,T[;\mathbb{R})$, and if the application $(t,\mu) \mapsto \partial_t\varphi(t,\mu)$ is locally Lipschitz in (X, d_X).

Lemma 5.6 (Partial derivatives)**.** *Let $\varphi : X \to \mathbb{R}$ satisfy Definition 5.5. Let $(t,\mu) \in X$ such that $\varphi(t,\cdot)$ is directionally differentiable at μ, and $\xi \in \mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. Then*

$$\lim_{h\searrow 0} \frac{\varphi(t+h, \exp_\mu(h\cdot\xi)) - \varphi(t,\mu)}{h} = \partial_t\varphi(t,\mu) + D_\mu\varphi(t,\mu)(\xi).$$

Proof. Let C be a Lipschitz constant for $\partial_s\varphi$ in the ball $\mathscr{B}\left((t,\mu),\sqrt{1+\|\xi\|_\mu^2}\right)$. Then for all $0<h\leqslant 1$,

$$\begin{aligned}
&\frac{\varphi(t+h,\exp_\mu(h\cdot\xi))-\varphi(t,\mu)}{h}\\
&=\frac{\varphi(t+h,\exp_\mu(h\cdot\xi))-\varphi(t,\exp_\mu(h\cdot\xi))}{h}+\frac{\varphi(t,\exp_\mu(h\cdot\xi))-\varphi(t,\mu)}{h}\\
&=\frac{1}{h}\int_{r=t}^{t+h}\partial_r\varphi(r,\exp_\mu(h\cdot\xi))dr+\frac{\varphi(t,\exp_\mu(h\cdot\xi))-\varphi(t,\mu)}{h}\\
&\in\partial_t\varphi(t,\mu)\pm\frac{C}{h}\int_{r=t}^{t+h}\left[|r-t|+d_{\mathcal{W}}(\exp_\mu(h\cdot\xi),\mu)\right]dr+\frac{\varphi(t,\exp_\mu(h\cdot\xi))-\varphi(t,\mu)}{h}\\
&\subset\partial_t\varphi(t,\mu)\pm Ch\left(1+\|\xi\|_\mu\right)+\frac{\varphi(t,\exp_\mu(h\cdot\xi))-\varphi(t,\mu)}{h}.
\end{aligned}$$

Letting $h\searrow 0$ and using the directional differentiability of $\varphi(t,\cdot)$ at μ, we obtain the result. □

5.1.3. Locally uniform upper semicontinuity

Due to the lack of local compactness of $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, we consider a stronger definition than upper semicontinuity.

Definition 5.7 (Locally uniformly upper semicontinuous)**.** Let (Y,d) be a complete metric space. A locally bounded application $u:Y\to\mathbb{R}$ is said to be locally uniformly upper semicontinuous (luusc) if for any decreasing family of closed bounded sets $(B_n)_{n\in\mathbb{N}}$ such that $B\coloneqq\bigcap_{n\in\mathbb{N}}B_n\neq\emptyset$ and $\lim_{n\to\infty}\sup_{x\in B_n}\inf_{y\in B}d_Y(x,y)\to 0$, there holds

$$\lim_{n\to\infty}\sup_{y\in B_n}u(y)\leqslant\sup_{x\in B}u(x). \tag{5.3}$$

Similarly, we say that u is locally uniformly lower semicontinuous (lulsc) if $-u$ is luusc.

Remark 5.8 (Link with other notions of upper semicontinuity)**.** We gather here some properties, whose proofs are postponed to Appendix B.

- Definition 5.7 is strictly weaker than continuity.
- In general, Definition 5.7 is strictly stronger than upper semicontinuity. However both definitions agree whenever the closed balls of Y are compact.
- Let $\mathcal{S}$ be the set of nonempty closed and bounded subsets of Y. Then Definition 5.7 is equivalent to the upper semicontinuity of the set function $U:B\mapsto\sup_{x\in B}u(x)$ in the Hausdorff topology. This definition makes sense in connection with the (min,+) interpretation of Hamilton–Jacobi equations, as it exactly says that the Maslov measure of density u is upper semicontinuous (see [39, 40]).
- The applications that are simultaneously luusc and lulsc are exactly the locally uniformly continuous applications.
- In $Y=\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, there is no comparison with upper semicontinuity in the narrow topology (see counterexamples in Appendix B).

5.1.4. Definition of viscosity solutions

Gathering the above definitions, we arrive at the following.

Definition 5.9 (Test functions)**.** We define $\mathscr{T}_+$ as the set of $\varphi:]0,T[\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})\to\mathbb{R}$ such that

- φ and $\partial_t\varphi$ are locally Lipschitz from $]0,T[\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ to $\mathbb{R}$,
- for all $t\in]0,T[$, $\varphi(t,\cdot)$ is geometrically consistent in the sense of Definition 5.4,

- for all $t \in]0,T[$, $\varphi(t,\cdot)$ is locally semiconcave.

Similarly, we denote $\mathscr{T}_- \coloneqq \{-\varphi \mid \varphi \in \mathscr{T}_+\}$.

Distinguished members of $\mathscr{T}_\pm$ are the applications of the form $\varphi(t,\mu) = \psi(t) \pm \alpha d^2_{\mathcal{W}}(\mu,\sigma)$, where $\psi \in \mathcal{C}^2(]0,T[;\mathbb{R})$, $\alpha \geqslant 0$ and $\sigma \in \mathscr{P}_2(\mathbb{R}^\mathrm{d})$ is fixed. Once given $\mathscr{T}_\pm$, the definition of viscosity solutions is a natural generalization of the finite-dimensional case.

Definition 5.10 (Viscosity solutions)**.** We say that $u : [0,T] \times \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \to \mathbb{R}$ is a viscosity

- subsolution if it is locally uniformly upper semicontinuous, if $u(T,\cdot) \leqslant \mathfrak{J}$, and if for any $\varphi \in \mathscr{T}_+$ such that $u - \varphi$ reaches a maximum at $(t,\mu) \in X =]0,T[\times \mathscr{P}_2(\mathbb{R}^\mathrm{d})$, there holds

$$-\partial_t \varphi(t,\mu) + H\left(\mu, D_\mu \varphi(t,\mu)\right) \leqslant 0. \tag{5.4}$$

- supersolution if it is locally uniformly lower semicontinuous, if $u(T,\cdot) \geqslant \mathfrak{J}$, and if for any $\psi \in \mathscr{T}_-$ such that $u - \psi$ reaches a minimum at $(t,\mu) \in X$, there holds

$$-\partial_t \psi(t,\mu) + H\left(\mu, D_\mu \psi(t,\mu)\right) \geqslant 0. \tag{5.5}$$

- solution if it is both a sub and a supersolution.

Let us put this notion in context with respect to the existing literature. As customary with viscosity solutions, the specificities of each problem require slight adaptations, leading to a great variety of definitions. Here we only mention the main lines, with no claim to be exhaustive.

From our perspective, there are essentially two distinct and non-equivalent approaches to defining viscosity solutions in the Wasserstein space, which depend on the selection of the tangent cone. The option that is most commonly found in the literature is the *regular tangent cone*

$$\mathrm{Tan}_\mu \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \coloneqq \overline{\{\nabla\varphi \mid \varphi \in \mathcal{C}^\infty_c(\mathbb{R}^\mathrm{d};\mathbb{R})\}}^{L^2_\mu(\mathbb{R}^\mathrm{d};\mathrm{T}\,\mathbb{R}^\mathrm{d})} = \left\{f \in L^2_\mu(\mathbb{R}^\mathrm{d};\mathrm{T}\,\mathbb{R}^\mathrm{d}) \;\middle|\; f\#\mu \in \mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^\mathrm{d})\right\}.$$

There are at least two compelling reasons to consider Tan_μ as an appropriate tangent cone. First, it is sufficient for describing the velocities of absolutely continuous curves, albeit only almost everywhere (see [9]). Second, it serves as the natural space in which two distinct theories have defined their gradients, namely the Lions differentiability [41–43] and the Wasserstein gradient [12, 44]. These constructions aim to establish a Hilbert-like structure on the tangent space and are most effectively applied to the dense subset of regular measures, as defined in [32]).

For this reason, the works [14, 24, 25, 44] consider either gradients, or sub- and superdifferentials *valued in this tangent cone*. From a slightly different perspective, [21] explores Hadamard semidifferentials, which coincide with the same definition in sufficiently smooth scenarios. However, it is also possible to define semidifferentials within the general tangent cone $\mathbf{Tan}_\mu$. As noted in [26], in general, these two approaches do not provide equivalent definitions of viscosity solutions. The problem lies in the computation of the metric slope, over which the definition of viscosity solutions for Eikonal-type equations of [26]. If one considers $\varphi : \nu \mapsto d^2_{\mathcal{W}}\left(\nu, \frac{\delta_{-a}+\delta_a}{2}\right)$ for some $a \in \mathbb{R}^\mathrm{d} \setminus \{0\}$, then the metric slope of φ at $\mu = \delta_0$ writes as

$$\overline{\lim_{\nu\to\mu}} \frac{|\varphi(\nu) - \varphi(\mu)|}{d_{\mathcal{W}}(\mu,\nu)} = \sup_{\xi \in \mathbf{Tan}_\mu \mathscr{P}_2(\mathbb{R}^\mathrm{d}), \|\xi\|_\mu = 1} |D_\mu \varphi(\xi)|.$$

In this case, the supremum is reached at $\xi = \frac{\delta_{(0,-a/|a|)}+\delta_{(0,a/|a|)}}{2}$, with a metric slope equal to $2|a|$. However, the directional derivative of φ at μ along any direction of the form $\xi = (id, f)\#\mu$ vanishes. This simple example

shows that relying on Tan_μ may imply additional smoothness assumptions if one wishes to ensure that both theories are equivalent.

5.2. Comparison principle

The comparison principles, or maximum principles in the literature of elliptic equations, are used in the viscosity theory to provide uniqueness of the viscosity solutions. They draw their name from the corresponding results used over the viscous approximations of the PDE, and evolved jointly with the growing scope of HJB equations. When addressing equations in non-locally compact spaces, it is now common to rely on variations over Ekeland's variational principle [45]: see for instance [19, 26, 46, 47]. This is not the only strategy in use in the literature: one could also modify the definition in order to stay over compact sets, as in [20] or [48] in the pathwise setting.

The perturbed optimization principle will bring, as announced, perturbations. To cope with these additional terms, [24, 25] consider an "enlarged" set of semidifferentials, and a strengthened notion of viscosity solutions. We take another point of view by using the Borwein–Preiss principle, also called *smooth Ekeland principle*, that allows to choose the perturbation in a way that they can be embedded into the test functions. To ease the reading, we factorize the application of this theorem in the following Lemma.

Lemma 5.11 (Perturbed optimization)**.** *Denote* $Y = \overline{X}^2 = [0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \times [0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$*, endowed with the distance*

$$d_Y^2((t,\mu,s,\nu),(\overline{t},\overline{\mu},\overline{s},\overline{\nu})) \coloneqq \left|t-\overline{t}\right|^2 + d_{\mathcal{W}}^2(\mu,\overline{\mu}) + |s-\overline{s}|^2 + d_{\mathcal{W}}^2(\nu,\overline{\nu}).$$

Let $\Phi : Y \to \mathbb{R} \cup \{-\infty\}$ *be upper semicontinuous, proper and upper bounded,* $z^0 \in Y$ *be fixed such that* $A \coloneqq \sup \Phi - \Phi(z_0) < \infty$*, and assume that there exists* $R > 0$ *such that*

$$\left\{ z \in Y \;\middle|\; \Phi(z) \geqslant \Phi(z^0) \right\} \subset \overline{\mathscr{B}}(z^0, R). \tag{5.6}$$

Hence for each $n \in \mathbb{N}_*$*, there exists* $z_n \in Y$ *and a perturbation* $p_n : Y \to \mathbb{R}^+$ *such that*

1. *the perturbed map* $\Phi - p_n$ *reaches a global strict maximum in* z_n*,*
2. *The map* $(t,\mu) \mapsto p_n(t,\mu,s_n,\nu_n)$ *belongs to* $\mathscr{T}_+$*, and* $(s,\nu) \mapsto -p_n(t_n,\mu_n,s,\nu)$ *belongs to* $\mathscr{T}_-$*,*
3. *There exists an application* $\omega_{T,R,A} : \mathbb{N} \to \mathbb{R}^+$ *such that*

$$\sum_{r \in \{t,s\}} |\partial_r p_n(z_n)| + \sum_{\sigma \in \{\mu,\nu\}} (1 + d_{\mathcal{W}}(\sigma,\delta_0)) \left\| D_\sigma p_n(z_n) \right\|_\sigma \leqslant \omega_{T,R,A}(n) \underset{n\to\infty}{\longrightarrow} 0,$$

4. *There holds* $\sup \Phi \leqslant \Phi(z_n) + C_{R,A}(n)$*, where* $C_{R,A}$ *is decreasing towards* 0 *when* $n \to \infty$*.*

The proof of Lemma 5.11 is delayed to Appendix C.

Assumption [A4] (Structure of the Hamiltonian). There exists a constant $[H]$ such that for any $\mu,\nu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, any $a > 0$ and $p,q \in \mathbb{T}_\mu$,

$$|H(\mu,p+q) - H(\mu,p)| \leqslant [H](1 + d_{\mathcal{W}}(\mu,\delta_0)) \|q\|_\mu, \tag{5.7}$$

$$H\left(\mu, -aD_\mu d_{\mathcal{W}}^2(\cdot,\nu)\right) - H\left(\nu, aD_\nu d_{\mathcal{W}}^2(\mu,\cdot)\right) \leqslant [H]\, d_{\mathcal{W}}(\mu,\nu)\left(a d_{\mathcal{W}}(\mu,\nu) + 1\right). \tag{5.8}$$

The condition [A4] indicates a locally Lipschitz behavior of the Hamiltonian. In particular, the condition (5.8) can be seen as a one-sided Lipschitz bound on the variations with respect to the first argument only. In the context of the Hamilton–Jacobi–Bellman equation associated with the control problem (2.2), the Bellman

Hamiltonian satisfies the requirement [A4] as long as the standing hypothesis [A2] is satisfied. This assertion will be demonstrated in Lemmata 6.3 and 6.4.

Theorem 5.12 (Comparison principle)**.** *Assume [A4]. Let $u : [0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ be a locally bounded subsolution of (5.1) and $v : [0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ be a locally bounded supersolution of (5.1), which are such that $u(T,\mu) \leqslant v(T,\mu)$ for all $\mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. Then*

$$u(t,\mu) \leqslant v(t,\mu) \qquad \forall (t,\mu) \in]0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}). \tag{5.9}$$

This proof builds on the ideas of [30], Theorem 3.50, p. 206 developed in Hilbert spaces. The structure is the following: assume by contradiction that the inequality (5.9) is not satisfied. Thus we have an information on the sign of the maximum $(t,\mu,s,\nu) \mapsto u(t,\mu) - v(s,\nu)$ on the diagonal $t = s$, $\mu = \nu$. As in the classical proof, this maximum over the diagonal is approximated by the maximum over the doubled space $\overline{X}^2$ of a perturbation of $u \ominus v$ that penalizes the distance to the diagonal. At the point of maximum, freezing two of the four variables of $\overline{X}^2$ allows to apply the definition of viscosity sub and supersolution, giving two inequalities whose combination will eventually lead to a contradiction.

The proof below has several specificities compared to the classical arguments (see [5], Sect. 3). In the argument adapted to $\mathbb{R}^d$, the points of maximum over the doubled space exist, and admit cluster points on the diagonal. In $\overline{X} = [0,T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, balls are not compact, and there is no reason for the maxima to be attained. This is circumvented by the use of variational principle in the vein of Ekeland's principle. Secondly, the so-constructed sequence of "almost maxima" do not necessarily contain converging subsequences. This is where the stronger form of upper semicontinuity of Definition 5.7 replaces the standard extraction, by working with limits over *sets* that decrease towards the diagonal. As opposite to the Hilbertian setting, we are not able to use a linear perturbation into the Ekeland principle, and we have to manipulate series of squared distances in order to get sufficient smoothness to embed the perturbation into test functions.

Proof. Denote again $X =]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. By abuse of notation, let $X^2 :=]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \times]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ and define the distances

$$\begin{aligned} d^2_X((t,\mu),(\overline{t},\overline{\mu})) &:= \left|t - \overline{t}\right|^2 + d^2_{\mathcal{W}}(\mu,\overline{\mu}), \\ d^2_{X^2}((t,\mu,s,\nu),(\overline{t},\overline{\mu},\overline{s},\overline{\nu})) &:= d^2_X((t,\mu),(\overline{t},\overline{\mu})) + d^2_X((s,\nu),(\overline{s},\overline{\nu})). \end{aligned}$$

Assume by contradiction that

$$\Gamma := \min\left(1, \sup\left\{u(t,\mu) - v(t,\mu) \;\middle|\; (t,\mu) \in \overline{X}\right\}\right) > 0. \tag{5.10}$$

Penalizations. As u and $-v$ are locally bounded and locally uniformly upper semicontinuous, we may build an application $g : \mathbb{R}^+ \to \mathbb{R}^+$ controlling the growth of both u and $-v$, *i.e.* such that

$$\max\left(u(t,\mu), -v(t,\mu)\right) \leqslant g\left(d_{\mathcal{W}}(\mu,\delta_0)\right) \qquad \forall \mu \in \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}). \tag{5.11}$$

Up to regularization, we may assume that g is increasing, of class $\mathcal{C}^2$ and with $g'(\cdot) \geqslant 1$. Denote

$$h : \mathbb{R}^2 \to \mathbb{R}, \qquad h(t,r) := g^2\left((1+r)\,\mathrm{e}^{-4[H]t}\right) + \frac{1}{t}.$$

The application h will play the role of a penalization through the composition

$$h_{\circ d} : \mathbb{R} \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}), \qquad h_{\circ d}(t,\mu) := h(t, d^2_{\mathcal{W}}(\delta_0,\mu)).$$

As $h_{\circ d}(t,\cdot)$ grows strictly faster than u and $-v$ by (5.11), for any $\iota>0$,

$$\begin{aligned}&\max\left[u(t,\mu),-v(t,\mu)\right]-\iota h_{\circ d}(t,\mu)\\&\qquad\leqslant\sup_{r\in\mathbb{R}^+}g\left(r\right)-\iota g^2\left((1+r^2)\mathrm{e}^{-4[H]T}\right)=:[\![g-\iota h]\!]<\infty.\end{aligned}\tag{5.12}$$

Since $g\geqslant 1$ and $g'\geqslant 1$, there holds

$$\partial_t h(t,r)=-8\left[H\right](1+r)\mathrm{e}^{-4[H]t}(gg')\left((1+r)\,\mathrm{e}^{-4[H]t}\right)-\frac{1}{t^2}\leqslant-8\left[H\right]\mathrm{e}^{-4[H]T}.\tag{5.13}$$

Moreover, for any $t>0$ and $r\in\mathbb{R}^+$,

$$\begin{aligned}2\left[H\right](1+r)\,\partial_r h(t,r)&=4\left[H\right](1+r)\left(gg'\right)\left((1+r)\,\mathrm{e}^{-4[H]t}\right)\mathrm{e}^{-4[H]t}\\&=-\frac{1}{2}\partial_t h(t,r)-\frac{1}{2t^2}\leqslant-\frac{1}{2}\partial_t h(t,r).\end{aligned}\tag{5.14}$$

As g is $\mathcal{C}^2$, the map h has a locally Lipschitz time derivative. Moreover, owing to Remark 5.3, the application $\mu\mapsto h_{\circ d}(t,\mu)$ is locally semiconcave for all t, and locally Lipschitz. Hence it may enter in the composition of test functions.

Let $\phi_\iota:\overline{X}^2\to\{-\infty\}\cup\mathbb{R}$ be given by

$$\phi_\iota(t,\mu,s,\nu):=u(t,\mu)-v(s,\nu)-\iota\left(h_{\circ d}(t,\mu)+h_{\circ d}(s,\nu)\right).$$

We consider

$$\Gamma_\iota:=\limsup_{r\to 0}\left\{\phi_\iota(t,\mu,s,\nu)\;\middle|\;(t,\mu,s,\nu)\in\overline{X}^2,\;d_X((t,\mu),(s,\nu))\leqslant r\right\},\tag{5.15}$$

$$\Gamma_{\iota\varepsilon}:=\sup\left\{\phi_\iota(t,\mu,s,\nu)-\frac{d_X^2((t,\mu),(s,\nu))}{\varepsilon}\;\middle|\;(t,\mu,s,\nu)\in\overline{X}^2\right\}.\tag{5.16}$$

Then one has

$$\Gamma_{\iota\varepsilon}\underset{\varepsilon\searrow 0}{\searrow}\Gamma_\iota\qquad\text{for all }\iota>0,\qquad\text{and}\qquad\Gamma_\iota\underset{\iota\searrow 0}{\nearrow}\Gamma_0\geqslant\Gamma.\tag{5.17}$$

Here Γ_0 may be equal to $+\infty$. The arguments of (5.17) are easy but tedious, and devolved in Appendix D to lighten the presentation.

The perturbed maximization. Let $\Phi_{\iota\varepsilon}:\overline{X}^2\to\{-\infty\}\cup\mathbb{R}$ be given by

$$\Phi_{\iota\varepsilon}(t,\mu,s,\nu):=\varphi_\iota(t,\mu,s,\nu)-\frac{d_X^2((t,\mu),(s,\nu))}{\varepsilon}.$$

Let $z^0:=(T,\delta_0,T,\delta_0)$. For each fixed ι,ε, the application $\Phi_{\iota\varepsilon}$ is upper semicontinuous, proper and - using (5.12) - upper bounded in the complete metric space $\left(\overline{X}^2,d_{X^2}\right)$. Moreover, if $\Phi_{\iota\varepsilon}(z)\geqslant\Phi_{\iota\varepsilon}(z^0)$, then

$$\frac{\iota}{2}\left(h_{\circ d}\left(t,\mu\right)+h_{\circ d}\left(s,\nu\right)\right)\leqslant 2[\![g-\frac{\iota}{2}h]\!]-\left(u(T,\delta_0)-v(T,\delta_0)-2\iota h(T,0)\right),\tag{5.18}$$

and there exists $R_\iota > 0$ such that $d_{X^2}(z, z^0) \leqslant R_\iota$. Notice that

$$\sup \Phi_{\iota\varepsilon} - \Phi_{\iota\varepsilon}(z^0) \leqslant 2[\![g - \iota h]\!] - (u(T,\delta_0) - v(T,\delta_0) - 2\iota h(T,0)) =: A_\iota$$

does not depend on ε.

Hence we may apply Lemma 5.11: for each $n \in \mathbb{N}_*$, there exist $z_{\iota\varepsilon n} \in \overline{X}^2$ and a perturbation $p_{\iota\varepsilon n} : \overline{X}^2 \to \mathbb{R}^+$ such that $\Phi_{\iota\varepsilon} - p_{\iota\varepsilon n}$ reaches a maximum at $z_{\iota\varepsilon n}$, the partial functions $(t,\mu) \mapsto p_{\iota\varepsilon n}(t,\mu,s_{\iota\varepsilon n},\nu_{\iota\varepsilon n})$ and $(s,\nu) \mapsto -p_{\iota\varepsilon n}(t_{\iota\varepsilon n},\mu_{\iota\varepsilon n},s,\nu)$ belong respectively to $\mathscr{T}_+, \mathscr{T}_-$, and there exist maps $\omega_\iota, C_\iota : \mathbb{N}_* \to \mathbb{R}^+$ such that

$$\sum_{r\in\{s,t\}} |\partial_r p_{\iota\varepsilon n}(z_{\iota\varepsilon n})| + \sum_{\sigma\in\{\mu,\nu\}} (1 + d_{\mathcal{W}}(\sigma_{\iota\varepsilon n},\delta_0))\|D_\sigma p_{\iota\varepsilon n}(z_{\iota\varepsilon n})\|_{\sigma_{\iota\varepsilon n}} \leqslant \omega_\iota(n), \tag{5.19}$$

$$\Gamma_{\iota\varepsilon} - \Phi_{\iota\varepsilon}(z_{\iota\varepsilon n}) = \sup \Phi_{\iota\varepsilon} - \Phi_{\iota\varepsilon}(z_{\iota\varepsilon n}) \leqslant C_\iota(n)$$

and $\omega_\iota(n), C_\iota(n) \underset{n\to\infty}{\longrightarrow} 0$. Notice that

$$\begin{aligned}\Gamma_{\iota\varepsilon} + \frac{d_X^2((t_{\iota\varepsilon n},\mu_{\iota\varepsilon n}),(s_{\iota\varepsilon n},\nu_{\iota\varepsilon n}))}{2\varepsilon} &\leqslant \Phi_{\iota\varepsilon}(z_{\iota\varepsilon n}) + C_\iota(n) + \frac{d_X^2((t_{\iota\varepsilon n},\mu_{\iota\varepsilon n}),(s_{\iota\varepsilon n},\nu_{\iota\varepsilon n}))}{2\varepsilon}\\ &\leqslant \Gamma_{\iota\frac{\varepsilon}{2}} + C_\iota(n)\end{aligned}$$

so that for each fixed ι, there holds by (5.17)

$$\frac{d_X^2((t_{\iota\varepsilon n},\mu_{\iota\varepsilon n}),(s_{\iota\varepsilon n},\nu_{\iota\varepsilon n}))}{\varepsilon} \leqslant 2\left(\Gamma_{\iota\frac{\varepsilon}{2}} - \Gamma_{\iota\varepsilon} + C_\iota(n)\right) \underset{\varepsilon\to 0, n\to\infty}{\longrightarrow} 0. \tag{5.20}$$

To lighten the notation, we shorten $a_{\iota\varepsilon n}$ in a_* in the sequel, for any $a \in \{t, s, \mu, \nu, p, z, \cdots\}$.

Staying away from T. Let us show that for sufficiently small ι and large n, the points t_*, s_* belong to $]0,T[$. Since $h(0,r) = +\infty$ for all $r \geqslant 0$, the construction of z_* implies $t_*, s_* > 0$. On the other hand, recalling (5.17), let $\iota_0 > 0$ be small enough so that $\Gamma_\iota \geqslant \Gamma/2$ for all $0 < \iota \leqslant \iota_0$. Hence, as $\varepsilon \mapsto \Gamma_{\iota\varepsilon}$ decreases when ε decreases,

$$u(t_*,\mu_*) - v(s_*,\nu_*) \geqslant \Phi_{\iota\varepsilon}(z_*) \geqslant \Gamma_{\iota\varepsilon} - C_\iota(n) \geqslant \Gamma_\iota - C_\iota(n) \geqslant \frac{\Gamma}{2} - C_\iota(n)$$

and there exists $n_\iota \in \mathbb{N}_*$ large enough such that $u(t_*,\mu_*) - v(s_*,\nu_*) \geqslant \frac{\Gamma}{4}$ for all $n \geqslant n_\iota$.

Then for each $0 < \iota \leqslant \iota_0$, $\varepsilon > 0$ sufficiently small and n sufficiently large, $t_{\iota\varepsilon n}, s_{\iota\varepsilon n} < T$ simultaneously. Indeed, recall from (5.18) that $\mu_{\iota\varepsilon n}, \nu_{\iota\varepsilon n} \in \overline{\mathscr{B}}(\delta_0, R_\iota)$ for all ε, n. If there exists a sequence $(\varepsilon_m, n_m) \to_m (0,\infty)$ such that $n_m \geqslant n_\iota$ and $T \in \{t_{\iota\varepsilon_m n_m}, s_{\iota\varepsilon_m n_m}\}$ for all m, then using (5.20),

$$\begin{aligned}\frac{\Gamma}{4} &\leqslant \varlimsup_{m\to\infty} u(t_{\iota\varepsilon_m n_m},\mu_{\iota\varepsilon_m n_m}) - v(s_{\iota\varepsilon_m n_m},\nu_{\iota\varepsilon_m n_m})\\ &\leqslant \varlimsup_{r\searrow 0} \sup\left\{u(t,\mu) - v(s,\nu) \;\middle|\; d_X^2((t,\mu),(s,\nu)) \leqslant r,\ \mu,\nu \in \overline{\mathscr{B}}(\delta_0,R_\iota),\ t,s \in [T-r,T]\right\}\\ &\leqslant \sup\left\{u(t,\mu) - v(s,\nu) \;\middle|\; t = s = T,\ \mu = \nu \in \overline{\mathscr{B}}(\delta_0,R_\iota)\right\}.\end{aligned}$$

The last inequality holds since $(t,\mu,s,\nu) \mapsto u(t,\nu) - v(s,\nu)$ is locally uniformly upper semicontinuous (see Def. 5.7). But this contradicts the assumption that $u(T,\cdot) - v(T,\cdot) \leqslant 0$.

Application of the definition of semisolutions. Define $\varphi_*, \psi_* : X =]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ as

$$\varphi_*(t,\mu) \coloneqq v(s_*,\nu_*) + \iota\left(h_{\circ d}(t,\mu) + h_{\circ d}(s_*,\nu_*)\right) + \frac{d_X^2((t,\mu),(s_*,\nu_*))}{\varepsilon} + p_*(t,\mu,s_*,\nu_*)$$

and

$$\psi_*(s,\nu) \coloneqq u(t_*,\mu_*) - \iota\left(h_{\circ d}(t_*,\mu_*) + h_{\circ d}(s,\nu)\right) - \frac{d_X^2((t_*,\mu_*),(s,\nu))}{\varepsilon} - p_*(t_*,\mu_*,s,\nu).$$

By construction, $\varphi_* \in \mathscr{T}_+$ and $\psi_* \in \mathscr{T}_-$. Recalling that $u - \varphi_*$ reaches a maximum at $(t_*,\mu_*) \in X$, we have

$$\begin{aligned} &- \iota\,\partial_t h_{\circ d}(t_*,\mu_*) - \frac{2(t_*-s_*)}{\varepsilon} - \partial_t p_*(z_*) \\ &+ H\left(\mu_*, \iota D_\mu h_{\circ d}(t_*,\mu_*) + \frac{D_\mu d_{\mathcal{W}}^2(\mu_*,\nu_*)}{\varepsilon} + D_\mu p_*(z_*)\right) \leqslant 0. \end{aligned} \tag{5.21}$$

Let us momentaneously denote $c_* \coloneqq [H]\,(1 + d_{\mathcal{W}}(\mu_*,\delta_0))$. Using the Lipschitz assumption (5.7),

$$\begin{aligned} &H\left(\mu_*, \iota D_\mu h_{\circ d}(t_*,\mu_*) + \frac{D_\mu d_{\mathcal{W}}^2(\mu_*,\nu_*)}{\varepsilon} + D_\mu p_*(z_*)\right) \\ &\geqslant H\left(\mu_*, \frac{D_\mu d_{\mathcal{W}}^2(\mu_*,\nu_*)}{\varepsilon}\right) - c_*\left(\iota \|D_\mu h_{\circ d}(t_*,\mu_*)\|_{\mu_*} + \|D_\mu p_*(z_*)\|_{\mu_*}\right). \end{aligned}$$

Using the regularity of h and the estimate of Remark 5.2 on the differential of the squared distance, we have

$$\begin{aligned} \|D_\mu h_{\circ d}(t_*,\mu_*)\|_{\mu_*} &= \left|\partial_r h(t_*, d_{\mathcal{W}}^2(\mu_*,\delta_0))\right| \|D_\mu d_{\mathcal{W}}^2(\mu_*,\delta_0)\|_{\mu_*} \\ &= \partial_r h(t_*, d_{\mathcal{W}}^2(\mu_*,\delta_0)) 2 d_{\mathcal{W}}(\mu_*,\delta_0) \end{aligned}$$

since $\partial_r h \geqslant 0$. Recalling that the partial derivatives of h satisfy (5.14), we get that

$$2\,[H]\,(1 + d_{\mathcal{W}}(\mu_*,\delta_0)) d_{\mathcal{W}}(\mu_*,\delta_0) \partial_r h(t_*, d_{\mathcal{W}}^2(\mu_*,\delta_0)) \leqslant -\frac{1}{2}\partial_t h\left(t_*, d_{\mathcal{W}}^2(\mu_*,\delta_0)\right). \tag{5.22}$$

Combining the four last estimates, we obtain

$$\begin{aligned} &- \iota\,\partial_t h_{\circ d}(t_*,\mu_*) - \frac{2(t_*-s_*)}{\varepsilon} - \partial_t p_*(z_*) \\ &+ H\left(\mu_*, \frac{D_\mu d_{\mathcal{W}}^2(\mu_*,\nu_*)}{\varepsilon}\right) + \frac{\iota}{2}\,\partial_t h_{\circ d}(t_*,\mu_*) - c_* \|D_\mu p_*(z_*)\|_{\mu_*} \leqslant 0. \end{aligned} \tag{5.23}$$

On the other hand, $v - \psi_*$ admits a minimum in $(s_*,\nu_*) \in X$. Applying the definition of supersolution and repeating the argument from (5.21) to (5.23), we get

$$\begin{aligned} &\iota \partial_s h_{\circ d}\,(s_*,\nu_*) + \frac{2(s_*-t_*)}{\varepsilon} + \partial_s p_*(z_*) - \frac{\iota}{2}\partial_s h_{\circ d}\,(s_*,\nu_*) \\ &+ [H]\,(1 + d_{\mathcal{W}}(\nu_*,\delta_0))\,\|D_\nu p_*(z_*)\|_{\nu_*} + H\left(\nu_*, -\frac{D_\nu d_{\mathcal{W}}^2(\mu_*,\nu_*)}{\varepsilon}\right) \geqslant 0. \end{aligned} \tag{5.24}$$

Taking the difference between (5.23) and (5.24) and the inequality resulting from plugging (5.23) and (5.19) into (5.21), we get after simplification

$$0 \leqslant H\left(\nu_*, -\frac{D_\nu d^2_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon}\right) - H\left(\mu_*, \frac{D_\mu d^2_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon}\right) + \frac{\iota}{2}\partial_s h_{\circ d}\left(s_*,\nu_*\right) \tag{5.25}$$

$$+\frac{\iota}{2}\partial_t h_{\circ d}(t_*,\mu_*) + \sum_{\tau\in\{s,t\}} \partial_\tau p_*(z_*) + [H] \sum_{\sigma\in\{\mu,\nu\}} \left(1 + d_{\mathcal{W}}(\sigma_*,\delta_0)\right) \left\|D_\sigma p_*(z_*)\right\|_{\sigma_*}. \tag{5.26}$$

Estimates and conclusion. Recall from (5.13) that $-\frac{1}{2}\partial_t h(t,r) \geqslant 4\,[H]\,\mathrm{e}^{-4[H]T} > 0$ for all $t > 0$ and $r \in \mathbb{R}^+$. Using the assumption (5.8) on H, we get

$$H\left(\nu_*, -\frac{D_\nu d^2_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon}\right) - H\left(\mu_*, \frac{D_\mu d^2_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon}\right) \leqslant [H]\, d_{\mathcal{W}}(\mu_*,\nu_*)\left(\frac{d_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon} + 1\right).$$

Using (5.19) to estimate (5.26), we arrive at

$$4\iota\,[H]\,\mathrm{e}^{-4[H]T} \leqslant [H]\, d_{\mathcal{W}}(\mu_*,\nu_*)\left(\frac{d_{\mathcal{W}}(\mu_*,\nu_*)}{\varepsilon} + 1\right) + \omega_\iota(n). \tag{5.27}$$

Keeping ι fixed, letting $\varepsilon \to 0$ and $n \to \infty$, we get from (5.20) that $4\iota\,[H]\,\mathrm{e}^{-4[H]T} \leqslant 0$, which is absurd. Thus $\Gamma \leqslant 0$. □

6. The case of Hamilton–Jacobi–Bellman equations

We now return to the case of control problems. Recall that $X =]0,T[\times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. Consider the dynamic $\overline{\mathrm{co}}F$ defined in (4.6), and let

$$H : \mathbb{T} \to \mathbb{R}, \qquad H\left(\mu,p\right) \coloneqq \sup_{f\in\overline{\mathrm{co}}F[\mu]} -p\left(\pi^\mu(f\#\mu)\right). \tag{6.1}$$

The Hamilton–Jacobi–Bellman equation associated to (6.1) then writes

$$\begin{cases} -\partial_t u(t,\mu) + \sup_{f\in\overline{\mathrm{co}}F[\mu]} -D_\mu u(t,\mu)\left(\pi^\mu(f\#\mu)\right) = 0 & (t,\mu)\in X, \quad (6.2a) \\ u(T,\mu) = \mathfrak{J}(\mu) & \mu\in\mathscr{P}_2(\mathbb{R}^{\mathrm{d}}). \quad (6.2b) \end{cases}$$

Remark 6.1 (The Hamiltonian and geometric consistency)**.** The construction of the Hamiltonian in (6.1), with a projection onto the tangent cone, is the only reason to impose geometric consistency to the test functions. In particular, geometric consistency does not intervene in the comparison principle. One could remove the projection; in this case, the elements of the metric cotangent bundle should be defined on $\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^{\mathrm{d}})_\mu$. The choice of using π^μ has almost no impact, but follows an emerging consensus in the literature to restrict differential objects to (subsets of) $\mathbf{Tan}_\mu$ or Tan_μ.

In the sequel, we verify that the control Hamiltonian (6.1) satisfies the assumptions of the comparison principle, and we then show that the value function is a solution of (6.2).

6.1. Properties of the control Hamiltonian

Let $G : \mathscr{P}_2(\mathbb{R}^d) \rightrightarrows \mathscr{P}_2(\mathrm{T}\,\mathbb{R}^d)$ the probability vector field (in the spirit of [34], Def. 2.1) given by

$$G[\mu] \coloneqq \{b\#\mu \mid b \in \overline{\mathrm{co}}F[\mu]\}\,.$$

Under [A2], this PVF is Lipschitz-continuous in the Hausdorff sense with respect to the application $W_{(\mu,\nu)}$ defined in (3.10), with constant $2\,[f]$. Indeed, given $\mu_0, \mu_1 \in \mathscr{P}_2(\mathbb{R}^d)$ and $b_0 \in \overline{\mathrm{co}}F[\mu_0]$ defined by $b_0(x) = \int_{u\in U} f(x,\mu_0,u)\mathrm{d}\omega(u)$, define $b_1 \coloneqq \int_{u\in U} f(x,\mu_1,u)\mathrm{d}\omega(u)$. Denote $\eta \in \Gamma_o(\mu,\nu)$. Then

$$\begin{aligned} W^2_{(\mu_0,\mu_1)}(b_0\#\mu_0, b_1\#\mu_1) &\leqslant \int_{(\mathbb{R}^d)^2}\int_{u\in U} |f(x,\mu_0,u) - f(y,\mu_1,u)|^2\,\mathrm{d}\omega(u)\mathrm{d}\eta(x,y) \\ &\leqslant 4\,[f]^2\,d^2_{\mathcal{W}}(\mu_0,\mu_1). \end{aligned}$$

Remark 6.2 (PVF formalism)**.** Our motivation in using a general probability vector field is to recover the connection between the metric slope and the directional derivatives. Indeed, in [26], Section 4.3, it is explicit that the regular tangent cone does not provide enough directions, in the sense that there exists measures μ,ν such that

$$\sup_{f\in \mathrm{Tan}_\mu\,\mathscr{P}_2(\mathbb{R}^d)} D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(f\#\mu) < \sup_{\xi\in \mathbf{Tan}_\mu\,\mathscr{P}_2(\mathbb{R}^d)} D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\xi) = \left|D_\mu d^2_{\mathcal{W}}(\cdot,\nu)\right|.$$

Here the last term is the metric slope. The strict inequality comes from the strong convexity of the squared distance of $\mathbb{R}^d$, which makes it sometimes more optimal to split mass than not to, as in the example $\mu = \delta_0$ and $\nu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$ in dimension one. Hence, although we restrain to Tan_μ by using the theory of continuity equations to formulate the control problem, our HJB tools are formulated in the general case.

Lemma 6.3 (Locally Lipschitz behavior of H)**.** *Assume [A2]. There exists a constant $[H]$ such that*

$$|H(\mu,p+q) - H(\mu,p)| \leqslant [H]\,(1 + d_{\mathcal{W}}(\mu,\delta_0))\,\|q\|_\mu \qquad \forall \mu \in \mathscr{P}_2(\mathbb{R}^d),\ p,q \in \mathbb{T}_\mu\,.$$

Proof. Using that $|\sup_{a\in A} f(a) - \sup_{b\in A} g(b)| \leqslant \sup_{a\in A}|f(a) - g(a)|$, one gets

$$|H(\mu,p+q) - H(\mu,p)| \leqslant \sup_{\xi\in G[\mu]} |-p(\pi^\mu\xi) - q(\pi^\mu\xi) + p(\pi^\mu\xi)| \leqslant \|q\|_\mu \sup_{\xi\in G[\mu]} \|\pi^\mu\xi\|_\mu.$$

Using the non-expansivity of the projection (see [10], Cor. 4.37),

$$\begin{aligned} \sup_{\xi\in G[\mu]} \|\pi^\mu\xi\|_\mu = \sup_{f\in\overline{\mathrm{co}}F[\mu]} \|\pi^\mu(f\#\mu)\|_\mu &\leqslant \sup_{f\in\overline{\mathrm{co}}F[\mu]} \|f\#\mu\|_\mu = \sup_{f\in\overline{\mathrm{co}}F[\mu]} \sqrt{\int_{\mathbb{R}^d} |f(x)|^2\,\mathrm{d}\mu} \\ &\leqslant \sqrt{|f|_{0,\infty} + [f]\,d^2_{\mathcal{W}}(\mu,\delta_0)} \leqslant \sqrt{|f|_{0,\infty} + [f]}\,(1 + d_{\mathcal{W}}(\mu,\delta_0))\,. \end{aligned}$$

We may then take $[H] \coloneqq \sqrt{|f|_{0,\infty} + [f]}$. □

Lemma 6.4 (Behavior on the squared distance)**.** *Assume [A2]. There exists a constant C_H such that for all $a \geqslant 0$ and $\mu,\nu \in \mathscr{P}_2(\mathbb{R}^d)$,*

$$H\left(\mu, -aD_\mu d^2_{\mathcal{W}}(\cdot,\nu)\right) - H\left(\nu, aD_\nu d^2_{\mathcal{W}}(\mu,\cdot)\right) \leqslant 2aC_H d^2_{\mathcal{W}}(\mu,\nu).$$

Proof. As H is positively homogeneous, we may take $a = 1$. Let $\xi \in G[\mu]$ and $\zeta \in G[\nu]$. By [28], Theorem 3.8, there holds $D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\pi^\mu \xi) = D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\xi)$. Moreover, using the bijection between $\exp_\mu^{-1}(\nu)$ and $\Gamma_o(\mu,\nu)$, the directional derivative of the squared Wasserstein distance (5.2) writes

$$D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\xi) = \inf_{\eta\in\Gamma_o(\mu,\nu)} \inf_{\alpha\in\Gamma(\xi,\nu),(\pi_x,\pi_y)\#\alpha=\eta} \int_{(x,v,y)} \langle v, -2(y-x)\rangle \,\mathrm{d}\alpha,$$
$$D_\nu d^2_{\mathcal{W}}(\mu,\cdot)(\zeta) = \inf_{\eta\in\Gamma_o(\mu,\nu)} \inf_{\beta\in\Gamma(\zeta,\mu),(\pi_y,\pi_x)\#\beta=\eta} \int_{(x,v,y)} \langle v, -2(y-x)\rangle \,\mathrm{d}\beta.$$

By disintegration, for each $\eta \in \Gamma_o(\mu,\nu)$ and $(\alpha,\beta) \in \Gamma(\xi,\nu)\times\Gamma(\zeta,\mu)$ such that $(\pi_x,\pi_y)\#\alpha = (\pi_y,\pi_x)\#\beta = \eta$, there exists at least one plan $\omega = \omega(x,v,y,w) \in \Gamma(\xi,\zeta) \subset \mathscr{P}\left((\mathrm{T}\,\mathbb{R}^{\mathrm{d}})^2\right)$ such that $(\pi_x,\pi_v,\pi_y) = \alpha$ and $(\pi_y,\pi_w,\pi_x)\#\omega = \beta$. Then

$$\begin{aligned}
&D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\pi^\mu\xi) + D_\nu d^2_{\mathcal{W}}(\mu,\cdot)(\pi^\nu\zeta)\\
&\quad= \inf_{\eta\in\Gamma_o(\mu,\nu)} \inf_{\omega\in\Gamma(\xi,\zeta),\ (\pi_x,\pi_y)\#\omega=\eta} \int_{(x,v,y,w)} \langle v,-2(y-x)\rangle + \langle w,-2(x-y)\rangle \,\mathrm{d}\omega\\
&\quad= \inf_{\eta\in\Gamma_o(\mu,\nu)} \inf_{\omega\in\Gamma(\xi,\zeta),\ (\pi_x,\pi_y)\#\omega=\eta} 2\int_{(x,v,y,w)} \langle v-w, x-y\rangle \,\mathrm{d}\omega\\
&\quad\leqslant \inf_{\eta\in\Gamma_o(\mu,\nu)} \inf_{\omega\in\Gamma(\xi,\zeta),\ (\pi_x,\pi_y)\#\omega=\eta} 2\sqrt{\int_{(x,v,y,w)} |v-w|^2 \,\mathrm{d}\omega \int_{(x,y)} |x-y|^2 \,\mathrm{d}\eta}\\
&\quad= 2W_{(\mu,\nu)}(\xi,\zeta) d_{\mathcal{W}}(\mu,\nu).
\end{aligned}$$

Hence

$$\begin{aligned}
H(\mu,-D_\mu d^2_{\mathcal{W}}(\cdot,\nu)) - H(\nu, D_\nu d^2_{\mathcal{W}}(\mu,\cdot))&\\
&= \sup_{\xi\in G[\mu]} D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\pi^\mu\xi) - \sup_{\zeta\in G[\nu]} -D_\nu d^2_{\mathcal{W}}(\mu,\cdot)(\pi^\nu\zeta)\\
&= \sup_{\xi\in G[\mu]} \inf_{\zeta\in G[\nu]} D_\mu d^2_{\mathcal{W}}(\cdot,\nu)(\xi) + D_\nu d^2_{\mathcal{W}}(\mu,\cdot)(\zeta)\\
&\leqslant 2d_{\mathcal{W}}(\mu,\nu) \sup_{\xi\in G[\mu]} \inf_{\zeta\in G[\nu]} W_{(\mu,\nu)}(\xi,\zeta) \leqslant 4\,[f]\, d^2_{\mathcal{W}}(\mu,\nu)
\end{aligned}$$

by the Lipschitz-continuity in $W_{(\mu,\nu)}$ of G. Taking $C_H = 2\,[f]$ proves the claim. □

Remark 6.5 (Game Hamiltonians)**.** Using the elementary inequality $\inf_A f - \inf_A g \leqslant \sup_A (f-g)$ if all terms are finite, one can show that the Hamiltonian $H : \mathbb{T} \to \mathbb{R}$ given by

$$H(\mu,p) \coloneqq \inf_{a\in A} \sup_{b\in B} -p\left(\pi^\mu(f(\cdot,a,b)\#\mu)\right) - \ell(a,b,\mu)$$

satisfies the assumption [A4] if A, B are compact sets of controls, f satisfies [A2] and ℓ is Lipschitz with respect to the μ variable, uniformly in a, b. Similarly, one could relax f to consider a probability vector field. We do not insist more on the game setting.

6.2. Characterization of the solution

Theorem 6.6 (Complete characterization of (6.2))**.** *Assume [A1], [A2] and [A3]. The value function is the unique viscosity solution in the sense of Definition 5.10 of the Hamilton–Jacobi-Bellman equation (6.2).*

Proof. Let us show that it is a viscosity solution of (6.2). By Lemma 4.7, V is locally uniformly continuous, hence simultaneously luusc and lulsc. By definition, $V(T,\cdot)=\mathfrak{J}$, so that we only have to verify inequalities (5.4) and (5.5).

Subsolution inequality. Let $\varphi\in\mathscr{T}_+$ such that $V-\varphi$ reaches a maximum at $(t,\mu)\in X$, and $[\varphi]$ a local Lipschitz constant in a ball of radius $\max(MT,[f]\,(|f|_{0,\infty}+d_{\mathcal{W}}(\nu,\delta_0)))$, where M is the constant of Theorem 4.1. By Lemma 4.3, each constant relaxed control $\omega\in L^1([t,T];\mathscr{P}(U))$ given by $\omega(s)\equiv\varpi\in\mathscr{P}(U)$ generates a smooth solution $(\mu_s^{t,\nu,\omega})_{s\in[t,T]}$, in the sense that

$$\lim_{h\searrow 0}\frac{d_{\mathcal{W}}\left(\mu_{t+h}^{t,\nu,\omega},(id+hF_\varpi[\nu])\,\#\nu\right)}{h}=0.$$

Consequently, along the curve $h\mapsto\hat{\mu}_h:=(id+hF_\varpi)\,\#\nu$, the DPP (4.7) gives

$$\begin{aligned}\frac{\varphi(t+h,\hat{\mu}_h)-\varphi(t,\nu)}{h}&\geqslant\frac{\varphi(t+h,\hat{\mu}_h)-\varphi(t+h,\mu_{t+h}^{t,\nu,\omega})}{h}+\frac{V(t+h,\hat{\mu}_h)-V(t,\mu)}{h}\\&\geqslant-[\varphi]\,\frac{d_{\mathcal{W}}\left(\mu_{t+h}^{t,\nu,\omega},\hat{\mu}_h\right)}{h}+0.\end{aligned}$$

Multiplicating by -1 and using the chain rule of Lemma 5.6 to take the limit in $h\searrow 0$, we obtain

$$-\partial_t\varphi(t,\nu)-D_\nu\varphi(t,\nu)(F_\varpi[\nu]\#\nu)\leqslant 0.$$

As $\varphi(t,\cdot)$ is geometrically consistent, we have $D_\nu\varphi(t,\nu)(F_\varpi[\nu]\#\nu)=D_\nu\varphi(t,\nu)(\pi^\nu F_\varpi[\nu]\#\nu)$. Taking the supremum over $b=F_\varpi[\nu]\in\overline{\mathrm{co}}F[\nu]$, one recovers the inequality (5.4), so that V is a subsolution.

Supersolution inequality. Let $\psi\in\mathscr{T}_-$ such that $V-\psi$ reaches a minimum in $(t,\nu)\in X$. Since under [A2], the set of solutions issued from (t,μ) is compact in the topology of uniform convergence (see [36], Thm. 4.5), we may find $\omega\in L^1([t,T];\mathscr{P}(U))$ such that $V(t,\nu)=V(t+h,\mu_{t+h}^{t,\nu,\omega})$ for all $h\in[0,T-t]$. Let $[\psi]$ be a local Lipschitz constant of ψ as above. Applying Lemma 4.6, there exist $(h_n)_n\searrow 0$ and $b\in\overline{\mathrm{co}}F[\nu]$ such that $d_{\mathcal{W}}\left(\mu_{h_n}^{t,\nu,\omega},\hat{\mu}_{h_n}\right)=o(h_n)$, where $\hat{\mu}_{h_n}:=(id+h_nb)\,\#\nu$. Then

$$\begin{aligned}&-[\psi]\,\frac{d_{\mathcal{W}}\left(\mu_{h_n}^{t,\nu,\omega},\hat{\mu}_{h_n}\right)}{h_n}+\frac{\psi\left(t+h_n,\hat{\mu}_{h_n}\right)-\psi(t,\nu)}{h_n}\\&\leqslant\frac{\psi\left(t+h_n,\mu_{h_n}^{t,\nu,\omega}\right)-\psi(t,\nu)}{h_n}\leqslant\frac{V\left(t+h_n,\mu_{h_n}^{t,\nu,\omega}\right)-V(t,\nu)}{h_n}=0.\end{aligned}$$

Taking the limit in $n\to\infty$ and using again the chain rule of Lemma 5.6, we get $\partial_t\psi(t,\nu)+D_\nu\psi\,(b)\leqslant 0$. Multiplicating by -1, taking the supremum over $b\in\overline{\mathrm{co}}F[\nu]$, and using the geometric consistency of φ, we obtain

$$\begin{aligned}-\partial_t\psi(t,\nu)+\sup_{b\in\overline{\mathrm{co}}F[\nu]}-D_\nu\psi(b)&=-\partial_t\psi(t,\nu)+\sup_{b\in\overline{\mathrm{co}}F[\nu]}-D_\nu\psi(\pi^\nu b)\\&=-\partial_t\psi(t,\nu)+H\left(\nu,D_\nu\psi(t,\nu)\right)\geqslant 0.\end{aligned}$$

To conclude, assume that there exists another viscosity solution $W:[0,T]\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})\to\mathbb{R}$ of (6.2). By Lemmata 6.3 and 6.4, the control Hamiltonian defined in (6.1) satisfies the assumption [A4]. Applying Theorem 5.12 to the couples (V,W) and (W,V), we have $V\geqslant W$ and $W\geqslant V$ pointwise over $]0,T]\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, so that they coincide. As both are continuous, the equality extends to $t=0$, and the solution is unique. □

6.3. The case of a Bolza problem

To conclude, let us comment on the adjustments that should be made of one considers a running cost. Assume that the value function is defined as

$$V(t,\mu) = \inf_{u(\cdot)\in L^1(t,T;U)} \int_{s=t}^{T} \ell(\mu_s, u(s))\mathrm{d}s + \mathfrak{J}(\mu_T).$$

We suppose that $\ell : \mathscr{P}_2(\mathbb{R}^\mathrm{d}) \times U \to \mathbb{R}$ is Lipschitz with respect to μ and continuous with respect to u. Here a dependence of ℓ with respect to the space variable $x \in \mathbb{R}^\mathrm{d}$ may be encapsulated, for instance by choosing $\ell(\mu, u) = \int_{x\in\mathbb{R}^\mathrm{d}} \widetilde{\ell}(x, \mu, u)\mathrm{d}\mu(x)$.

Relaxed system. Following the course of the paper, let us consider the same relaxation of the control problem. Unlike in the Mayer problem, the relaxation procedure makes the relaxed control explicitly appear in the cost of the control problem: the value function is equal to

$$V(t,\mu) = \inf_{\omega(\cdot)\in L^1(t,T;\mathscr{P}(U))} \int_{s=t}^{t+h} \int_{u\in U} \ell(\mu_s^{t,\nu,\omega}, u)\mathrm{d}\omega(s)(u)\mathrm{d}s + \mathfrak{J}(\mu_T^{t,\nu,\omega}).$$

This may be seen by introducing an additional variable $z_\tau = \int_{s=t}^{\tau} \ell(y_s, u(s))\mathrm{d}s$ and applying the relaxation theorem to the augmented dynamical system in variables (y, z).

HJB equation. In view of the above, one expects the value function to satisfy the Hamilton–Jacobi equation with

$$H(\mu, p) \coloneqq \sup_{\omega\in\mathscr{P}(U)} \left[-p(\pi^\mu(F_\omega[\mu]\#\mu)) + \int_{u\in U} \ell(\mu, u)\mathrm{d}\omega(u) \right].$$

Since $\ell(\cdot, u)$ is Lipschitz, the first item of Assumption [A4] is satisfied. However, H is no longer positively homogeneous, and the second item of Assumption [A4] has to be modified in

$$H\left(\mu, -aD_\mu d^2_{\mathcal{W}}(\cdot,\nu)\right) - H\left(\nu, ad^2_{\mathcal{W}}(\mu,\cdot)\right) \leqslant 2[H]\left(ad^2_{\mathcal{W}}(\mu,\nu) + [\ell]d_{\mathcal{W}}(\mu,\nu)\right),$$

which is satisfied by H. The last step of the proof of the comparison principle Theorem 5.12 will see a term $2[H]d_{\mathcal{W}}(\mu_*, \nu_*)$ appear in (5.27), which goes to 0 with ε, n, and does not change the argument.

Characterization of the value function. Regarding the satisfaction of the HJB equation by the value function, one should pay attention to limits. In the first part of Theorem 6.6, the inequality

$$\frac{V(t+h,\hat{\mu}_h) - V(t,\mu)}{h} \geqslant \frac{1}{h}\int_{s=t}^{t+h}\int_{u\in U} \ell(\hat{\mu}_s, u)\mathrm{d}\varpi(u)\mathrm{d}s \geqslant \int_{u\in U} \ell(\nu, u)\mathrm{d}\varpi(u) - o(s)$$

follows from the uniform continuity of ℓ, and the fact that the control is taken constant. In the supersolution part, one considers instead

$$\frac{V(t+h,\hat{\mu}_h) - V(t,\mu)}{h} = \frac{1}{h}\int_{s=t}^{t+h}\int_{u\in U} \ell(\mu_s^{t,\nu,\omega}, u)\mathrm{d}\frac{1}{h}\int_{s=t}^{t+h}(u)\mathrm{d}s = \left\langle \ell(\nu,\cdot), \frac{1}{h}\int_{s=t}^{t+h}\omega(s)\mathrm{d}s \right\rangle + o(h),$$

and by a close inspection of the proof of Lemma 4.6, one is allowed to consider that the time-averaged measure converges to some relaxed control along a vanishing subsequence $(h_n)_n$, so that the rest of the argument is unchanged.

Appendix A. Approximation of smooth trajectories

Lemma A.1 (Smooth case)**.** *Let $0 \leqslant t \leqslant T$ and $\omega \in L^1([t,T];\mathscr{P}(U))$ be a constant control $\omega(s) \equiv \varpi \in \mathscr{P}(U)$. Then the unique solution $(\mu_s^{t,\nu,\omega})_{s\in[t,T]}$ satisfy*

$$\lim_{h\searrow 0} \frac{d_{\mathcal{W}}\left(\mu_{t+h}^{t,\nu,\omega}, \exp_\mu(h\cdot F_\varpi[\nu])\right)}{h} = 0.$$

Proof. Let again $\Phi_\cdot^t$ be the flow of the ODE (4.2), such that $\mu_s^{t,\nu,\omega} = \Phi_s^t \# \nu$. There holds

$$\begin{aligned}
I &:= d_{\mathcal{W}}^2\left(\mu_{t+h}^{t,\nu,\omega}, \exp_\mu(h\cdot F_\varpi[\nu])\right)\\
&\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} \left|\Phi_{t+h}^t(x) - (x + hF_\varpi[\nu](x))\right|^2 \mathrm{d}\nu(x)\\
&\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} h\int_{s=t}^{t+h} \left|F_\varpi[\mu_s^{t,\nu,\omega}](\Phi_s^t(x)) - F_\varpi[\nu](x)\right|^2 \mathrm{d}s\mathrm{d}\nu(x)\\
&\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} h\int_{s=t}^{t+h}\int_{u\in U} \left|f(\Phi_s^t(x), \mu_s^{t,\nu,\omega}, u) - f(x,\nu,u)\right|^2 \mathrm{d}\varpi(u)\mathrm{d}s\mathrm{d}\nu(x)\\
&\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} h\int_{s=t}^{t+h} [f]^2\left(d_{\mathcal{W}}(\mu_s^{t,\nu,\omega},\nu) + \left|\Phi_s^t(x) - x\right|\right)^2 \mathrm{d}s\mathrm{d}\nu(x).
\end{aligned}$$

Using (4.3) and (4.4),

$$\begin{aligned}
I &\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} h\int_{s=t}^{t+h} [f]^2\left((s-t)\left([f]\left(|x| + d_{\mathcal{W}}(\nu,\delta_0)\right) + 2\left|f\right|_{0,\infty}\right)\mathrm{e}^{[f](s-t)}\right)^2 \mathrm{d}s\mathrm{d}\nu(x)\\
&\leqslant \int_{x\in\mathbb{R}^{\mathrm{d}}} h^2[f]^2\left(h\left([f]\left(|x| + d_{\mathcal{W}}(\nu,\delta_0)\right) + 2\left|f\right|_{0,\infty}\right)\mathrm{e}^{[f]h}\right)^2 \mathrm{d}\nu(x)\\
&\leqslant \left(2h^2\left[f\right]\left([f]\, d_{\mathcal{W}}(\nu,\delta_0) + \left|f\right|_{0,\infty}\right)\mathrm{e}^{[f]h}\right)^2.
\end{aligned}$$

Hence

$$\lim_{h\searrow 0} \frac{d_{\mathcal{W}}\left(\mu_{t+h}^{t,\nu,\omega}, \exp_\mu(h\cdot F_\varpi[\nu])\right)}{h} \leqslant \lim_{h\searrow 0} 2h\left[f\right]\left([f]\, d_{\mathcal{W}}(\nu,\delta_0) + \left|f\right|_{0,\infty}\right)\mathrm{e}^{[f]h} = 0.$$

□

Appendix B. Details on locally uniform upper semicontinuity

Recall that a locally bounded map $u : Y \to \mathbb{R}$ of a complete metric space (Y, d_Y) is luusc if for any decreasing family of bounded closed sets $(B_n)_n$ such that $B = \cap_n B_n \neq \emptyset$ and $\sup_{x\in B_n} \inf_{y\in B} d_Y(x,y) \to_n 0$, there holds

$$\lim_{n\to\infty} \sup_{y\in B_n} u(y) \leqslant \sup_{x\in B} u(x). \tag{B.1}$$

It turns out that we have the following.

Lemma B.1 (Link with other notions of upper semicontinuity)**.** *The following holds.*

1. *The condition (B.1) is strictly weaker than continuity.*

2. *Let $\mathcal{S}$ be the set of nonempty closed and bounded subsets of Y. Then (B.1) is equivalent to the upper semicontinuity of the set function $U : B \mapsto \sup_{x\in B} u(x)$ in the Hausdorff topology.*
3. *The applications that are luusc and lulsc are exactly the locally uniformly continuous applications.*
4. *In general, the condition (B.1) is strictly stronger than upper semicontinuity. However both definitions coincide whenever closed balls of Y are compact.*
5. *In $Y = \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, there is no comparison with upper semicontinuity in the narrow topology.*

Proof. Point 1 is easily seen with the luusc map $u \coloneqq \mathbb{1}_{\{o\}}$, where $o \in Y$ is some fixed point.

Consider the notations of Point 2. If U is locally upper semicontinuous, then (B.1) directly stands. On the other hand, assume that u is luusc. Consider a sequence of bounded nonempty closed sets $A_n \subset Y$ that converge in the Hausdorff distance, that is, there exists $A \subset \mathcal{S}$ such that

$$d_{H,Y}(A_n, A) \coloneqq \max\left(\sup_{x\in A_n}\inf_{y\in A} d_Y(x,y), \sup_{y\in A}\inf_{x\in A_n} d_Y(x,y)\right) \underset{n\to\infty}{\longrightarrow} 0.$$

As A is bounded, the sequence $(A_n)_n$ is contained in some bounded set. Consider $B_n \coloneqq \overline{\bigcup_{m\geqslant n} A_n}$. Then $(B_n)_n$ is a family of closed nonempty sets, uniformly bounded, with nonempty intersection equal to A and such that

$$d_{H,Y}(B_n, A) = \sup_{x\in B_n}\inf_{y\in A} d_Y(x,y) = \sup_{m\geqslant n}\sup_{x\in A_n}\inf_{y\in A} d_Y(x,y) = \sup_{m\geqslant n} d_{H,Y}(A_n, A) \underset{n\to\infty}{\longrightarrow} 0.$$

Hence

$$\overline{\lim_{n\to\infty}}\, U(A_n) = \lim_{n\to\infty}\sup_{m\geqslant n}\sup_{y\in A_m} u(y) \leqslant \lim_{n\to\infty}\sup_{y\in B_n} u(y) \leqslant \sup_{x\in A} u(x) = U(A),$$

and U is locally upper semicontinuous, proving 2.

We turn to Point 3. If u is locally uniformly continuous, let $m_u : \mathbb{R}^+ \to \mathbb{R}^+$ be a local modulus of continuity in a ball containing all the B_n. If $(y_n)_n$ is a maximizing sequence with $y_n \in B_n$, there exists $x_n \in B$ such that $d_Y(y_n, x_n) \leqslant \varepsilon_n \to_n 0$, and

$$\lim_{n\to\infty}\sup_{y\in B_n} u(y) = \lim_{n\to\infty} u(y_n) \leqslant \overline{\lim_{n\to\infty}}\, u(x_n) + m\left(d_Y(y_n, x_n)\right) \leqslant \sup_{x\in B} u(x) + 0.$$

Thus u satisfies (B.1). On the other hand, if a locally bounded map u is both luusc and lulsc, then $(x,y) \mapsto \pm(u(x) - u(y))$ is luusc. The definition of luusc implies then that for some fixed $o \in Y$ and all $R > 0$,

$$\begin{aligned}&\lim_{r\to 0}\sup\left\{|u(x) - u(y)| \;\middle|\; x, y \in \overline{\mathscr{B}}(o,R),\ d(x,y) \leqslant r\right\}\\ &\leqslant \max_{s\in\{-1,1\}}\lim_{n\to\infty}\sup_{\substack{(x,y)\overline{\mathscr{B}}(o,R)^2\\ d(x,y)\leqslant 1/n}} s(u(x) - u(y)) \leqslant 0.\end{aligned}$$

Any continuous modulus superior to $r \mapsto \sup\left\{|u(x) - u(y)| \;\middle|\; x, y \in \overline{\mathscr{B}}(o,R),\ d(x,y) \leqslant r\right\}$ furnishes a local modulus of continuity.

Points 4 and 5 will use similar counterexamples. Notice first that taking $B_n = \overline{\mathscr{B}}(x, 1/n)$ in (B.1) for each $x \in \mathbb{R}^{\mathrm{d}}$, we see that luusc always imply usc. On the other hand, if the closed balls of Y are compact, let $u : Y \to \mathbb{R}$ be upper semicontinuous. Any maximizing sequence $(y_n)_n$ with $y_n \in B_n$ is in particular contained in a sufficiently large closed ball that is independent of n, hence contains a converging subsequence, whose limit belongs to $\overline{B} = B$ owing to the uniform approximation of B by B_n. By upper semicontinuity of y, (B.1) is satisfied.

Stays to exhibit an usc map that is not luusc. By now, we take $Y = \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, and we define

$$G_r \coloneqq \left\{ \mu_\alpha \coloneqq \left(1 - \frac{1}{\alpha^2}\right)\delta_0 + \frac{1}{\alpha^2}\delta_{(\alpha,0,\cdots,0)} \;\middle|\; \alpha \geqslant r \right\},$$

and

$$H \coloneqq \left\{ \nu_\beta \coloneqq \left(1 - \frac{1}{\beta}\right)\mu_\beta + \frac{1}{\beta}\delta_{(0,1,0,\cdots,0)} \;\middle|\; \beta \geqslant 1 \right\}.$$

We have that for all $r \geqslant 1$, the sets G_r, H are nonempty, disjoint and included in $\overline{\mathscr{B}}(\delta_0, 1)$. Moreover, G_r and H are closed in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ for each $r \geqslant 1$: indeed, let $(\mu_{\alpha_i})_i \subset G_r$ be a Cauchy sequence. If $(\alpha_i)_i$ is unbounded, then $(\mu_{\alpha_i})_i$ should converge to its narrow limit δ_0: but this is absurd since $d_{\mathcal{W}}(\mu_\alpha, \delta_0) = 1$ for all $\alpha > 0$. Thus $(\alpha_i)_i$ is bounded by some constant $C > 0$, and some (non relabeled) subsequence converges towards some $\overline{\alpha} \geqslant r$. Computing

$$d_{\mathcal{W}}\left(\mu_{\alpha_n}, \mu_{\overline{\alpha}}\right) = \sqrt{2\left(1 - \frac{\alpha_n \wedge \overline{\alpha}}{\alpha_n \vee \overline{\alpha}}\right)},$$

we see that $\mu_{\alpha_i} \to_i \mu_\alpha \in G_r$. Let us note that here, we proved that the sets $\{\mu_\alpha \mid \alpha \in [r, C]\}$ are compact in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$. By a similar argument, the set H is closed as well.

Consider $u : \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \to \mathbb{R}$ given by $u(\mu) = \mathbb{1}_{G_1}$. As the indicator of a closed set, u is upper semicontinuous. We claim that it is not locally uniformly upper semicontinuous: indeed, define

$$B_n \coloneqq \left\{ \mu \in \overline{\mathscr{B}}(\delta_0, 1) \;\middle|\; \left|\mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\mu)\right| \coloneqq \left| \int_{x \in \mathbb{R}^{\mathrm{d}}} x \mathrm{d}\mu(x) \right| \leqslant \frac{1}{n} \right\}.$$

It is easily verified that $\mu \mapsto \left|\mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\mu)\right|$ is continuous, so that $(B_n)_n$ is a globally bounded decreasing sequence of closed sets, whose intersection $B = \left|\mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\cdot)\right|^{-1}(\{0\}) \cap \overline{\mathscr{B}}(\delta_0, 1)$ is nonempty, and such that

$$\begin{aligned}
\sup_{\mu \in B_n} \inf_{\nu \in B} d_{\mathcal{W}}(\mu, \nu) &\leqslant \sup_{\mu \in B_n} d_{\mathcal{W}}\left(\mu, \left(\frac{n}{\sqrt{1+n^2}}\left(id - \mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\mu)\right)\right)\#\mu\right) \\
&\leqslant \sup_{\mu \in B_n} \sqrt{\left(1 - \frac{n}{\sqrt{1+n^2}}\right)^2 d^2_{\mathcal{W}}(\mu, \delta_0) + \frac{n^2}{1+n^2}\left|\mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\mu)\right|^2} \\
&\leqslant \sqrt{\left(1 - \frac{n}{\sqrt{1+n^2}}\right)^2 + \frac{1}{1+n^2}} \underset{n \to \infty}{\longrightarrow} 0.
\end{aligned}$$

As $\left|\mathrm{Bary}_{\mathbb{R}^{\mathrm{d}}}(\mu_\alpha)\right| = 1/\alpha$, for any n, the intersection $B_n \cap G_1$ is nonempty. However $B \cap G_1 = \emptyset$, so that

$$\lim_{n \to \infty} \sup_{\mu \in B_n} u(\mu) \geqslant \lim_{n \to \infty} u(\mu_n) = 1, \qquad \sup_{x \in B} u(\mu) = 0.$$

Hence u is not luusc, proving Point 4.

We finally turn to Point 5. Let us build an application that is bounded and narrowly upper semicontinuous, but not locally uniformly upper semicontinuous. The strategy is similar to that of Point 4, with the additional requirement to work with narrowly closed sets. Let us show that the narrow closure of G_1 is $G_1 \cup \{\delta_0\}$: indeed, $(\mu_{\alpha_i})_i \subset G_1$ be a narrowly converging sequence. If $(\alpha_i)_i$ is unbounded, the narrow limit is δ_0. If $(\alpha_i)_i$ is bounded,

then we showed in Point 4 that $\{\mu_\alpha \mid \alpha \in [1, C]\}$ is compact in $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, thus narrowly compact, and the narrow limit stays in G_1.

Consider the bounded and narrowly upper semicontinuous function $u \coloneqq \mathbb{1}_{\{\delta_0\} \cup G_1}$. To show that u is not luusc, we consider the family of sets

$$B_n \coloneqq H \cup G_n.$$

We immediately have that $(B_n)_n$ is a decreasing family of nonempty closed sets, whose intersection is H, and all contained in the Wasserstein unit ball centered in δ_0. Moreover,

$$\begin{aligned}\sup_{\mu \in B_n} \inf_{\nu \in B} d_{\mathcal{W}}(\mu_n, \nu) &= \sup_{\alpha \geqslant n} \inf_{\nu \in H} d_{\mathcal{W}}(\mu_\alpha, \nu) \\ &\leqslant \sup_{\alpha \geqslant n} d_{\mathcal{W}}\left(\mu_\alpha, \left(1 - \frac{1}{\alpha}\right)\mu_\alpha + \frac{1}{\alpha}\delta_{(0,1,0,\cdots,0)}\right) \leqslant \sup_{\alpha \geqslant n} \frac{1}{\alpha} \underset{n \to \infty}{\longrightarrow} 0.\end{aligned}$$

For each n, we have $B_n \cap [\{\delta_0\} \cup G_1] = G_n \neq \emptyset$. However, $H \cap [\delta_0 \cup G_1] = \emptyset$. Then

$$\lim_{n \to \infty} \sup_{\mu \in B_n} u(\mu) = 1 \qquad \text{but} \qquad \sup_{\mu \in H} u(\mu) = 0.$$

Hence u is not locally uniformly upper semicontinuous. The application $\mu \mapsto d^2_{\mathcal{W}}(\mu, \delta_0)$ furnishes an example of map that is luusc, since locally uniformly continuous, but not narrowly upper semicontinuous. Consequently, there is no hierarchy between narrow upper semicontinuity and Definition 5.7. □

Appendix C. Perturbed optimization

Lemma C.1 (Perturbed optimization)**.** *Denote* $Y = \overline{X}^2 = [0, T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}}) \times [0, T] \times \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$*, endowed with the distance*

$$d^2_Y((t, \mu, s, \nu), (\bar{t}, \overline{\mu}, \bar{s}, \overline{\nu})) \coloneqq \left|t - \bar{t}\right|^2 + d^2_{\mathcal{W}}(\mu, \overline{\mu}) + \left|s - \bar{s}\right|^2 + d^2_{\mathcal{W}}(\nu, \overline{\nu}).$$

Let $\Phi : Y \to \mathbb{R} \cup \{-\infty\}$ *be upper semicontinuous, proper and upper bounded,* $z^0 \in Y$ *be fixed such that* $A \coloneqq \sup \Phi - \Phi(z_0) < \infty$*, and assume that there exists* $R > 0$ *such that*

$$\left\{z \in Y \;\middle|\; \Phi(z) \geqslant \Phi(z^0)\right\} \subset \overline{\mathscr{B}}(z^0, R). \tag{C.1}$$

Hence for each $n \in \mathbb{N}_*$*, there exists* $z_n \in Y$ *and a perturbation* $p_n : Y \to \mathbb{R}^+$ *such that*

1. *the perturbed map* $\Phi - p_n$ *reaches a global strict maximum in* z_n*,*
2. *There exists an application* $\omega_{T,R,A} : \mathbb{N} \to \mathbb{R}^+$ *such that*

$$\sum_{r \in \{t,s\}} |\partial_r p_n(z_n)| + \sum_{\sigma \in \{\mu,\nu\}} (1 + d_{\mathcal{W}}(\sigma, \delta_0)) \left\|D_\sigma p_n(z_n)\right\|_\sigma \leqslant \omega_{T,R,A}(n) \underset{n \to \infty}{\longrightarrow} 0,$$

3. *The map* $(t, \mu) \mapsto p_n(t, \mu, s_n, \nu_n)$ *belongs to* $\mathscr{T}_+$*, and* $(s, \nu) \mapsto -p_n(t_n, \mu_n, s, \nu)$ *belongs to* $\mathscr{T}_-$*,*
4. *There holds* $\sup \Phi \leqslant \Phi(z_n) + C_{R,A}(n)$*, where* $C_{R,A}$ *is decreasing towards* 0 *when* $n \to \infty$*.*

Proof. The metric space (Y, d_Y) is complete, and Φ satisfies all the assumptions of the Ekeland–Borwein–Preiss–Zhu theorem [49], Theorem 2.5.2. We consider the gauge-type function d_Y^2, and the choice of ponderation

$$\alpha_{n,m} \coloneqq \frac{1}{n2^{m+1}}, \qquad \text{so that} \qquad \alpha_{n,0} = \frac{1}{2n} \quad \text{and} \quad \sum_{m\in\mathbb{N}} \alpha_{n,m} = \frac{1}{n}.$$

Applying Ekeland–Borwein–Preiss–Zhu, we get the existence of some $z_n \in Y$ and a sequence $(z_{n,m})_{m\in\mathbb{N}} \subset Y$ such that

$$\begin{cases} d_Y^2(z_0, z_n) \leqslant \dfrac{A}{\alpha_{n,0}}, \qquad d_Y^2\left(z_{n,m}, z_n\right) \leqslant \dfrac{A}{2^m \alpha_{n,0}} & \forall m \in \mathbb{N}, \qquad \text{(C.2a)} \\ \Phi(z_n) \geqslant \Phi(z^0) + \displaystyle\sum_{m\in\mathbb{N}} \alpha_{n,m} d_Y^2(z^0, z_{n,m}), & \text{(C.2b)} \\ \Phi(z_n) - \displaystyle\sum_{m\in\mathbb{N}} \alpha_{n,m} d_Y^2(z_n, z_{n,m}) > \Phi(\cdot) - \displaystyle\sum_{m\in\mathbb{N}} \alpha_{n,m} d_Y^2(\cdot, z_{n,m}) \quad \forall z \neq z_n. & \text{(C.2c)} \end{cases}$$

Define $p_n : z \mapsto \sum_{m\in\mathbb{N}} \alpha_{n,m} d_Y^2(z, z_{n,m}) \geqslant 0$. Then using (C.2a),

$$\begin{aligned} p_n(z) &= \sum_{m\in\mathbb{N}} \alpha_{n,m} d_Y^2(z, z_{n,m}) \leqslant 2 \sum_{m\in\mathbb{N}} \alpha_{n,m} \left(d_Y^2(z, z_n) + d_Y^2(z_n, z_{n,m})\right) \\ &\leqslant 2\frac{d_Y^2(z, z_n)}{n} + \frac{A}{n\alpha_0} \sum_{m=0} 4^{-m} = 2\frac{d_Y^2(z, z_n)}{n} + \frac{8A}{3} < \infty. \end{aligned}$$

Hence the application p_n is well-defined from Y to $\mathbb{R}^+$. By (C.2c), $\Phi - p_n$ reaches a global strict maximum in z_n.

We turn to Points 2 and 3. The application $p_n(\cdot, \mu, s, \nu)$ is of the form $c + \sum_{m\in\mathbb{N}} 2^{-m-1} \frac{|\cdot - t_{n,m}|^2}{n}$, over a bounded interval, so uniformly convergent. By direct computation, its derivative is Lipschitz in $[0, T]$ with constant $2/n$, and

$$|\partial_t p_n(t, \mu_n, s_n, \nu_n)| \leqslant \frac{1}{n} \sum_{m\in\mathbb{N}} 2^{-m} |t - t_{n,m}| \leqslant \frac{2T}{n} \qquad \forall t \in [0, T]. \qquad \text{(C.3)}$$

As $(t, \mu) \mapsto p_n(t, \mu, s_n, \nu_n)$ writes as a sum of time and measure contributions, its derivative with respect to t is Lipschitz in the whole domain $\overline{X}$. Moreover, by (C.2b), $z_n \in \left\{\Phi \geqslant \Phi(z^0)\right\}$. As $\mu_{n,m} \to_m \mu_n$, the sequence $(\mu_{n,m})_m$ stays in a bounded set of $\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$, and the partial function $p_n(t, \cdot, s, \nu)$ is uniformly convergent for each n. Using the semiconcavity of $d_{\mathcal{W}}^2(\cdot, \mu_{n,m})$ (see [9], Thm. 7.3.2), there holds for any $(\sigma, \xi) \in \mathbf{Tan}\, \mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ and $h \in [0, 1]$ that

$$\begin{aligned} &p_n\left(t_n, \exp_\sigma(h \cdot \xi), s_n, \nu_n\right) - c \\ &\geqslant \sum_{m\in\mathbb{N}} \alpha_{n,m} \left[(1-h) d_{\mathcal{W}}^2(\sigma, \mu_{n,m}) + h d_{\mathcal{W}}^2(\exp_\sigma(\xi), \mu_{n,m}) - h(1-h) d_{\mathcal{W}}^2(\sigma, \exp_\sigma(\xi))\right] \\ &= (1-h) p_n\left(t_n, \sigma, s_n, \nu_n\right) + h p_n\left(t_n, \exp_\sigma(\xi), s_n, \nu_n\right) - \frac{h(1-h)}{n} d_{\mathcal{W}}^2(\sigma, \exp_\sigma(\xi)). \end{aligned}$$

Thus $p_n(t_n, \cdot, s_n, \nu_n)$ is locally semiconcave. As a combination of squared distance, it is geometrically consistent. To prove that $(t, \mu) \mapsto p_n(t, \mu, s_n, \nu_n)$ belongs to $\mathscr{T}_+$, there only stays to show the local Lipschitzianity in the

measure variable. By direct computation, for any $S > 0$ and $\mu, \sigma \in \overline{\mathscr{B}}(\delta_0, S)$, one has

$$\begin{aligned}\frac{|p_n(t_n, \mu, s_n, \nu_n) - p_n(t_n, \sigma, s_n, \nu_n)|}{d_{\mathcal{W}}(\mu, \sigma)} &\leqslant \sum_{m\in\mathbb{N}} \alpha_{n,m} \left(d_{\mathcal{W}}(\mu, \mu_{n,m}) + d_{\mathcal{W}}(\sigma, \mu_{n,m})\right) \\ &\leqslant \sum_{m\in\mathbb{N}} 2\alpha_{n,m} \left(S + d_{\mathcal{W}}(\delta_0, \mu_n) + d_{\mathcal{W}}(\mu_n, \mu_{n,m})\right) \\ &\leqslant \frac{S + \sqrt{2nA}}{n} \sum_{m\in\mathbb{N}} \frac{1 + 2^{-m/2}}{2^m}.\end{aligned}$$

Here we used (C.2a). Let $R > 0$ be given by the assumption (C.1) such that $d_Y\left(z_n, z^0\right) \leqslant R$ independantly of n. By the above, there holds

$$\left(1 + d_{\mathcal{W}}(\mu_n, \delta_0)\right) \|D_\mu p_n(z_n)\|_{\mu_n} \leqslant (1 + R) \frac{R + \sqrt{2nA}}{n} \sum_{m\in\mathbb{N}} \frac{1 + 2^{-m/2}}{2^m}. \tag{C.4}$$

Gathering (C.3) and (C.4), we obtain the application $\omega_{T,R,A}$ that decreases in $n^{-1/2}$. The reasoning over $p_n(t, \mu, \cdot, \nu)$ and $p_n(t, \mu, s, \cdot)$ is symmetric.

Finally, notice that the supremum of Φ over Y is the same as the supremum of Φ over $\overline{\mathscr{B}}(z^0, R)$. In consequence, (C.2c) gives

$$\begin{aligned}\sup \Phi &\leqslant \Phi(z_n) + \sup_{z\in\overline{\mathscr{B}}(z^0,R)} \sum_{m\in\mathbb{N}} \alpha_{n,m} \left[d_Y^2(z, z_{n,m}) - d_Y^2(z_n, z_{n,m})\right] \\ &\leqslant \Phi(z_n) + \sup_{z\in\overline{\mathscr{B}}(z^0,R)} \sum_{m\in\mathbb{N}} \alpha_{n,m} \left[d_Y(z, z_{n,m}) + d_Y(z_n, z_{n,m})\right] d_Y(z, z_n) \\ &\leqslant \Phi(z_n) + \sum_{m\in\mathbb{N}} \alpha_{n,m} \left[2R + 2d_Y(z_n, z_{n,m})\right] 2R \\ &\leqslant \Phi(z_n) + \sum_{m\in\mathbb{N}} \frac{1}{2^{m+1}n} \left[2R + 2\sqrt{\frac{nA}{2^{m+1}}}\right] 2R.\end{aligned}$$

Hence Point 4 by choosing $C_{R,A} : n \mapsto \frac{4R^2}{n} + \frac{2R\sqrt{A}}{\sqrt{n}\left(2\sqrt{2}-1\right)}$. □

Appendix D. Monotonicities

Lemma D.1 (Monotonicities)**.** *Assume (5.10), and that $u, -v$ are bounded from above. Let Γ, Γ_ι and $\Gamma_{\iota\varepsilon}$ be defined as in (5.10), (5.15) and (5.16). Then*

$$\Gamma_{\iota\varepsilon} \underset{\varepsilon\searrow 0}{\searrow} \Gamma_\iota \qquad \text{for all } \iota > 0, \qquad \text{and} \qquad \Gamma_\iota \underset{\iota\searrow 0}{\nearrow} \Gamma_0 \geqslant \Gamma.$$

Proof. Define the additional variables

$$\begin{aligned}\Gamma_r &\coloneqq \min\left(1, \sup\left\{u(t,\mu) - v(s,\nu) \;\middle|\; (t,\mu,s,\nu) \in \overline{X}^2,\ d_X((t,\mu),(s,\nu)) \leqslant r\right\}\right), \\ \Gamma_{\iota r} &\coloneqq \sup\left\{\phi_\iota(t,\mu,s,\nu) \;\middle|\; (t,\mu,s,\nu) \in \overline{X}^2,\ d_X((t,\mu),(s,\nu)) \leqslant r\right\}\end{aligned}$$

so that $\Gamma = \lim_{r\to 0}\Gamma_r$ and $\Gamma_\iota = \lim_{r\to 0}\Gamma_{\iota r}$. For each fixed $\iota > 0$, using the growth of h, the variables $\Gamma_{\iota r}$ and $\Gamma_{\iota\varepsilon}$ are upper bounded. Restricting ι, ε and r to $]0, 1]$, (5.10) gives us that each term is lower bounded. Moreover, we have the monotonicities

$$\Gamma_r \searrow_r, \qquad \Gamma_\iota \nearrow_\iota, \qquad \Gamma_{\iota r} \searrow_r, \qquad \Gamma_{\iota\varepsilon} \searrow_\varepsilon .$$

In consequence, the limits $\lim_r \Gamma_r$, $\lim_r \Gamma_{\iota r}$ and $\lim_\varepsilon \Gamma_{\iota\varepsilon}$ are finite, and $\lim_\iota \Gamma_\iota$ exists in $\mathbb{R}\cup\{+\infty\}$.

Assume that $\Gamma_0 \coloneqq \lim_{\iota\searrow 0}\Gamma_\iota < \Gamma$. Then there exists r_0 and $\alpha > 0$ sufficiently small so that $\Gamma_\iota \leqslant \Gamma_{\iota r} \leqslant \Gamma - \alpha \leqslant \Gamma_r - \alpha$ for all $0 < r \leqslant r_0$ and ι. Consequently, for some $\frac{\alpha}{2}-$optimal point $z_\alpha = (t_\alpha, \mu_\alpha, s_\alpha, \nu_\alpha)$ for the definition of Γ_r, we have for all ι that

$$u(t_\alpha, \mu_\alpha) - v(s_\alpha, \nu_\alpha) - \iota\left(h_{od}(t_\alpha, \mu_\alpha) + h_{od}(s_\alpha, \nu_\alpha)\right) \leqslant u(t_\alpha, \mu_\alpha) - v(s_\alpha, \nu_\alpha) - \frac{\alpha}{2}.$$

Letting $\iota \searrow 0$, we obtain a contradition. Hence $\Gamma_0 \geqslant \Gamma$.

Assume now that for some fixed ι, $\Gamma_{\iota 0} \coloneqq \lim_{\varepsilon\searrow 0}\Gamma_{\iota\varepsilon} < \Gamma_\iota$. Then for sufficiently small ε, there exists $\alpha > 0$ such that $\Gamma_{\iota\varepsilon} \leqslant \Gamma_\iota - \alpha \leqslant \Gamma_{\iota r} - \alpha$ for all r. For each r, denote $z_{\alpha r}$ some $\frac{\alpha}{2}-$optimal point for the definition of $\Gamma_{\iota r}$, with $d_X(t_{\alpha r}, \mu_{\alpha r}, s_{\alpha r}, \nu_{\alpha r}) \leqslant r$ by construction. There holds

$$\phi_\iota(z_{\alpha r}) - \frac{d_X^2((t_{\alpha r}, \mu_{\alpha r}), (s_{\alpha r}, \nu_{\alpha r}))}{\varepsilon} \leqslant \phi_\iota(z_{\alpha r}) - \frac{\alpha}{2}.$$

Hence

$$\frac{\alpha}{2} \leqslant \frac{d_X^2((t_{\alpha r}, \mu_{\alpha r}), (s_{\alpha r}, \nu_{\alpha r}))}{\varepsilon} \leqslant \frac{r^2}{\varepsilon},$$

and letting $r \searrow 0$, we obtain the desired contradiction.

Finally, assume that for some fixed ι, $\Gamma_{\iota 0} \coloneqq \lim_{\varepsilon\searrow 0}\Gamma_{\iota\varepsilon} > \Gamma_\iota$. Then for sufficiently small r, there exists $\alpha > 0$ such that $\Gamma_{\iota\varepsilon} \geqslant \Gamma_{\iota 0} \geqslant \Gamma_{\iota r} + \alpha \geqslant \Gamma_\iota$ for all $\varepsilon > 0$. For each $\varepsilon > 0$, let $z_{\varepsilon\alpha} \in X^2$ be $\frac{\alpha}{2}-$optimal for the definition of $\Gamma_{\iota\varepsilon}$. Hence

$$\phi_\iota(z_{\varepsilon\alpha}) - \frac{d_X^2((t_{\varepsilon\alpha}, \mu_{\varepsilon\alpha}), (s_{\varepsilon\alpha}, \nu_{\varepsilon\alpha}))}{\varepsilon} \geqslant \Gamma_{\iota\varepsilon} - \frac{\alpha}{2} \geqslant \Gamma_\iota + \frac{\alpha}{2}$$

implies

$$\frac{d_X^2((t_{\varepsilon\alpha}, \mu_{\varepsilon\alpha}), (s_{\varepsilon\alpha}, \nu_{\varepsilon\alpha}))}{\varepsilon} \leqslant [\![u]\!] + [\![-v]\!] + 0 - \Gamma_\iota + \frac{\alpha}{2},$$

and for sufficiently small ε, we have $d_X((t_{\varepsilon\alpha}, \mu_{\varepsilon\alpha}), (s_{\varepsilon\alpha}, \nu_{\varepsilon\alpha})) < r$. For this choice of parameters, we get

$$\phi_\iota(z_{\varepsilon\alpha}) - \frac{d_{X_1}^2((t_{\varepsilon\alpha}, \mu_{\varepsilon\alpha}), (s_{\varepsilon\alpha}, \nu_{\varepsilon\alpha}))}{\varepsilon} \geqslant \phi_\iota(z_{\varepsilon\alpha}) + \frac{\alpha}{2},$$

a flagrant contradiction. □

Appendix E. Grönwall estimates

Lemma E.1 (Grönwall estimates)**.** *Assume [A1] and [A2]. Let $(\mu_s^{t,\nu,\omega})_{s\in[t,T]}$ denote the solution of (4.1) issued from $(t,\nu) \in [0,T]\times\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})$ and driven by the control $\omega \in L^1([t,T];\mathscr{P}(U))$. Let $0 \leqslant t \leqslant \overline{s} \leqslant s \leqslant T$,*

$(\nu,\overline{\nu}) \in \left(\mathscr{P}_2(\mathbb{R}^{\mathrm{d}})\right)^2$ and $(\omega,\overline{\omega}) \in \left(L^1([t,T];\mathscr{P}(U))\right)^2$. Then

$$d_{\mathcal{W}}\left(\mu_s^{t,\nu,\omega},\mu_{\overline{s}}^{t,\overline{\nu},\overline{\omega}}\right) \leqslant (s-\overline{s})\left([f]\,(\overline{s}-t)\mathrm{e}^{(s-t)[f]}+\mathrm{e}^{(s-\overline{s})[f]}\right)\left([f]\,d_{\mathcal{W}}(\delta_0,\nu)+|f|_{0,\infty}\right) + \mathrm{e}^{[f](\overline{s}-t)\left(1+\mathrm{e}^{[f](\overline{s}-t)}\right)}\left(d_{\mathcal{W}}(\nu,\overline{\nu})+E_{t,\overline{s},\overline{\nu},\omega,\overline{\omega}}\right),$$

where

$$E_{t,\overline{s},\overline{\nu},\omega,\overline{\omega}} \coloneqq \left(1+(\overline{s}-t)\left(|f|_{0,\infty}+[f]\,d_{\mathcal{W}}\left(\delta_0,\overline{\nu}\right)\right)\mathrm{e}^{(\overline{s}-t)[f]}\right)\int_{r=t}^{\overline{s}} d_{\mathcal{W},m_{f,u}}(\omega(r),\overline{\omega}(r))dr,$$

and

$$d_{\mathcal{W},m_{f,u}}(\varpi,\overline{\varpi}) \coloneqq \inf_{\alpha\in\Gamma(\varpi,\overline{\varpi})}\int_{(u,v)\in U^2} m_{f,u}(|u-v|)\mathrm{d}\alpha(u,v).$$

As particular cases, we record that

$$d_{\mathcal{W}}(\mu_{t+h}^{t,\nu,\omega},\nu) \leqslant h\left([f]\,d_{\mathcal{W}}(\delta_0,\nu)+|f|_{0,\infty}\right)\mathrm{e}^{h[f]},$$
$$d_{\mathcal{W}}\left(\mu_T^{t,\nu,\omega},\mu_T^{t,\overline{\nu},\omega}\right) \leqslant \mathrm{e}^{[f](T-t)\left(1+\mathrm{e}^{[f](T-t)}\right)}d_{\mathcal{W}}(\nu,\overline{\nu}).$$

Proof. Assume w.l.o.g that $s \geqslant \overline{s}$. Denote $\left(\Phi_\tau^{t,\nu,\omega}\right)_{\tau\in[t,s]}$ and $\left(\Phi_\tau^{t,\overline{\nu},\overline{\omega}}\right)_{\tau\in[t,\overline{s}]}$ the respective fluxes of the ODEs

$$\dot{y}_\tau = F_{\omega(\tau)}[\mu_\tau^{t,\nu,\omega}](y_\tau), \qquad \dot{\overline{y}}_\tau = F_{\overline{\omega}(\tau)}[\mu_\tau^{t,\overline{\nu},\overline{\omega}}](\overline{y}_\tau).$$

On the one hand, for $t \leqslant r \leqslant \tau \leqslant s$,

$$\begin{aligned}\left|\Phi_\tau^{t,\nu,\omega}(x)-\Phi_r^{t,\nu,\omega}(x)\right| &\leqslant \int_{\theta=r}^{\tau}\int_{u\in U}\left|f\left(\Phi_\theta^{t,\nu,\omega}(x),\mu_\theta^{t,\nu,\omega},u\right)\right|\mathrm{d}\omega(\theta)(u)\mathrm{d}\theta \\ &\leqslant \int_{\theta=r}^{\tau}\left[[f]\left|\Phi_\theta^{t,\nu,\omega}(x)-\Phi_r^{t,\nu,\omega}(x)\right|+[f]\left|\Phi_r^{t,\nu,\omega}(x)\right|+|f|_{0,\infty}\right]\mathrm{d}\theta,\end{aligned}$$

so that a Grönwall lemma yields

$$\left|\Phi_\tau^{t,\nu,\omega}(x)-\Phi_r^{t,\nu,\omega}(x)\right| \leqslant (\tau-r)\left([f]\left|\Phi_r^{t,\nu,\omega}(x)\right|+|f|_{0,\infty}\right)\mathrm{e}^{(\tau-r)[f]}. \tag{E.1}$$

In particular,

$$\begin{aligned}\left|\Phi_\tau^{t,\nu,\omega}(x)-\Phi_r^{t,\nu,\omega}(x)\right| &\leqslant (\tau-r)\left([f]\left|\Phi_r^{t,\nu,\omega}(x)-x\right|+[f]\,|x|+|f|_{0,\infty}\right)\mathrm{e}^{(\tau-r)[f]} \\ &\leqslant (\tau-r)\left([f]\,(r-t)\mathrm{e}^{(r-t)[f]}+1\right)\left([f]\,|x|+|f|_{0,\infty}\right)\mathrm{e}^{(\tau-r)[f]}.\end{aligned}$$

Taking the square of each side and integrating with respect to ν, we get

$$\begin{aligned}d_{\mathcal{W}}(\mu_\tau^{t,\nu,\omega},\mu_r^{t,\nu,\omega}) &\leqslant \sqrt{\int_{x\in\mathbb{R}^{\mathrm{d}}}\left|\Phi_\tau^{t,\nu,\omega}(x)-\Phi_r^{t,\nu,\omega}(x)\right|^2\mathrm{d}\nu(x)} \\ &\leqslant (\tau-r)\left([f]\,(r-t)\mathrm{e}^{(r-t)[f]}+1\right)\left([f]\,d_{\mathcal{W}}(\delta_0,\nu)+|f|_{0,\infty}\right)\mathrm{e}^{(\tau-r)[f]}.\end{aligned} \tag{E.2}$$

On the other hand, for $\tau \in [t, \overline{s}]$,

$$\begin{aligned}
&\left|\Phi_\tau^{t,\nu,\omega}(x) - \Phi_\tau^{t,\overline{\nu},\overline{\omega}}(y)\right| \\
&\leqslant |x-y| + [f] \int_{r=t}^{\tau} \left|\Phi_r^{t,\nu,\omega}(x) - \Phi_r^{t,\overline{\nu},\overline{\omega}}(y)\right| + d_{\mathcal{W}}\left(\mu_r^{t,\nu,\omega}, \mu_r^{t,\overline{\nu},\overline{\omega}}\right) dr \\
&\qquad + \left(1 + (\tau - t)\left(|f|_{0,\infty} + [f]\,|y|\right) \mathrm{e}^{(\tau-t)[f]}\right) \int_{r=t}^{\tau} d_{\mathcal{W},m_{f,u}}\left(\omega(r), \overline{\omega}(r)\right) dr.
\end{aligned}$$

Applying a second Grönwall lemma,

$$\left|\Phi_\tau^{t,\nu,\omega}(x) - \Phi_\tau^{t,\overline{\nu},\overline{\omega}}(y)\right| \leqslant \left(|x-y| + [f] \int_t^{\tau} d_{\mathcal{W}}\left(\mu_r^{t,\nu,\omega}, \mu_r^{t,\overline{\nu},\overline{\omega}}\right) dr + E_{t,\tau,y,\omega,\overline{\omega}}\right) \mathrm{e}^{[f](\tau-t)},$$

where $E_{t,\tau,y,\omega,\overline{\omega}} \coloneqq \left(1 + (\tau - t)\left(|f|_{0,\infty} + [f]\,|y|\right) \mathrm{e}^{(\tau-t)[f]}\right) \int_{r=t}^{\tau} d_{\mathcal{W},m_{f,u}}\left(\omega(r), \overline{\omega}(r)\right) dr$. Now, let $\eta \in \Gamma_o(\nu, \overline{\nu})$. The plan

$$\eta_{\overline{s}} \coloneqq \left(\Phi_{\overline{s}}^{t,\nu,\omega}, \Phi_{\overline{s}}^{t,\overline{\nu},\overline{\omega}}\right) \# \eta$$

belongs to $\Gamma(\mu_{\overline{s}}^{t,\nu,\omega}, \mu_{\overline{s}}^{t,\overline{\nu},\overline{\omega}})$, so that

$$\begin{aligned}
&d_{\mathcal{W}}(\mu_{\overline{s}}^{t,\nu,\omega}, \mu_{\overline{s}}^{t,\overline{\nu},\overline{\omega}}) \\
&\leqslant \sqrt{\int_{(\mathbb{R}^{\mathrm{d}})^2} \left|\Phi_{\overline{s}}^{t,\nu,\omega}(x) - \Phi_{\overline{s}}^{t,\overline{\nu},\overline{\omega}}(y)\right|^2 \mathrm{d}\eta(x,y)} \\
&\leqslant \mathrm{e}^{[f](\overline{s}-t)} \sqrt{\int_{(\mathbb{R}^{\mathrm{d}})^2} \left(|x-y| + \int_{r=t}^{\overline{s}} [f]\, d_{\mathcal{W}}\left(\mu_r^{t,\nu,\omega}, \mu_r^{t,\overline{\nu},\overline{\omega}}\right) dr + E_{t,\overline{s},y,\omega,\overline{\omega}}\right)^2 \mathrm{d}\eta(x,y)} \\
&\leqslant \mathrm{e}^{[f](\overline{s}-t)} \left(d_{\mathcal{W}}(\nu,\overline{\nu}) + \int_{r=t}^{\overline{s}} [f]\, d_{\mathcal{W}}\left(\mu_r^{t,\nu,\omega}, \mu_r^{t,\overline{\nu},\overline{\omega}}\right) dr + \sqrt{\int_{y\in\mathbb{R}^{\mathrm{d}}} E_{t,\overline{s},y,\omega,\overline{\omega}}^2 \mathrm{d}\overline{\nu}(y)}\right).
\end{aligned}$$

As

$$\begin{aligned}
&\int_{y\in\mathbb{R}^{\mathrm{d}}} E_{t,\overline{s},y,\omega,\overline{\omega}}^2 \mathrm{d}\overline{\nu}(y) \\
&= \int_{\mathbb{R}^{\mathrm{d}}} \left(1 + (\overline{s} - t)\left(|f|_{0,\infty} + [f]\,|y|\right) \mathrm{e}^{(\overline{s}-t)[f]}\right)^2 \left(\int_t^{\tau} d_{\mathcal{W},m_{f,u}}\left(\omega(r), \overline{\omega}(r)\right) dr\right)^2 \mathrm{d}\overline{\nu}(y) \\
&\leqslant \left(1 + (\overline{s} - t)\left(|f|_{0,\infty} + [f]\, d_{\mathcal{W}}\left(\delta_0, \overline{\nu}\right)\right) \mathrm{e}^{(\overline{s}-t)[f]}\right)^2 \left(\int_{r=t}^{\tau} d_{\mathcal{W},m_{f,u}}\left(\omega(r), \overline{\omega}(r)\right) dr\right)^2 \\
&=: E_{t,\overline{s},\overline{\nu},\omega,\overline{\omega}}^2 < \infty,
\end{aligned}$$

we are ready to apply our third Grönwall lemma to get

$$d_{\mathcal{W}}(\mu_{\overline{s}}^{t,\nu,\omega}, \mu_{\overline{s}}^{t,\overline{\nu},\overline{\omega}}) \leqslant \mathrm{e}^{[f](\overline{s}-t)} \left(d_{\mathcal{W}}(\nu,\overline{\nu}) + E_{t,\overline{s},\overline{\nu},\omega,\overline{\omega}}\right) \mathrm{e}^{[f](\overline{s}-t)\mathrm{e}^{[f](\overline{s}-t)}}. \tag{E.3}$$

Combining (E.2) and (E.3), we get the desired result. □

References

[1] R. Coyaud, *Study of Approximations of Optimal Transport Problems and Application to Physics*. These de doctorat, Paris Est (2021).

[2] A. Corbetta, *Multiscale Crowd Dynamics: Physical Analysis, Modeling and Applications*. PhD thesis, Eindhoven University of Technology (2016).

[3] B. Piccoli and A. Tosin, Pedestrian flows in bounded domains with obstacles. *Continuum Mech. Thermodyn.* **21** (2009) 85–107.

[4] J.A. Carrillo, Y.-P. Choi and M. Hauray, The derivation of swarming models: mean-field limit and Wasserstein distances, in *Collective Dynamics from Bacteria to Crowds: An Excursion Through Modeling, Analysis and Simulation*, edited by A. Muntean and F. Toschi. CISM International Centre for Mechanical Sciences. Springer, Vienna (2014) 1–46.

[5] M.G. Crandall, H. Ishii and P.-L. Lions, User's guide to viscosity solutions of second order partial differential equations. *Bull. Am. Math. Soc.* **27** (1992) 1–67.

[6] H. Ishii, Hamilton–Jacobi equations with discontinuous Hamiltonians on arbitrary open sets. *Bull. Fac. Sci. Eng.* **28** (1985) 33–77.

[7] F. Otto, The geometry of dissipative evolution equations: the porous medium equation. *Commun. Part. Differ. Equ.* **26** (2001) 101–174.

[8] C. Villani, Optimal transport. Vol. 338 of *Grundlehren Der Mathematischen Wissenschaften*. Springer Berlin Heidelberg, Berlin, Heidelberg (2009).

[9] L. Ambrosio, N. Gigli and G. Savaré, Gradient Flows. Lectures in Mathematics ETH Zürich. Birkhäuser-Verlag, Basel (2005).

[10] N. Gigli, *On the Geometry of the Space of Probability Measures Endowed with the Quadratic Optimal Transport Distance*. PhD thesis, Scuola Normale Superiore di Pisa, Pisa (2008).

[11] P.-L. Lions, Jeux à champ moyen, 2006/2007. Conférences au Collège de France.

[12] W. Gangbo and A. Tudorascu, On differentiability in the Wasserstein space and well-posedness for Hamilton–Jacobi equations. *J. Math. Pures Appl.* **125** (2019) 119–174.

[13] C. Bertucci and P.L. Lions, An approximation of the squared Wasserstein distance and an application to Hamilton–Jacobi equations, September 2024. Preprint, available at http://arxiv.org/abs/2409.11793.

[14] A. Cosso, F. Gozzi, I. Kharroubi, H. Pham and M. Rosestolato, Master Bellman equation in the Wasserstein space: uniqueness of viscosity solutions. *Trans. Am. Math. Soc.* **377** (2024) 31–83.

[15] A. Cosso and M. Martini, On smooth approximations in the Wasserstein space. *Electron. Commun. Probab.* **28** (2023) 1–11.

[16] S. Daudin, J. Jackson and B. Seeger, Well-posedness of Hamilton–Jacobi equations in the Wasserstein space: non-convex Hamiltonians and common noise, Dec. 2023. Preprint (arXiv:2312.02324).

[17] G. Conforti, R. Kraaij and D. Tonon, Hamilton–Jacobi equations for controlled gradient flows: Cylindrical test functions, 2023. Preprint, available at https://arxiv.org/abs/2302.06571.

[18] G. Conforti, R.C. Kraaij, L. Tamanini and D. Tonon, Hamilton-Jacobi equations for Wasserstein controlled gradient flows: existence of viscosity solutions (2024).

[19] G. Conforti, R.C. Kraaij and D. Tonon, Hamilton-Jacobi equations for controlled gradient flows: The comparison principle. *J. Funct. Anal.* **284** (2023). 1–53.

[20] J. Feng and M. Katsoulakis, A comparison principle for Hamilton–Jacobi equations related to controlled gradient flows in infinite dimensions. *Arch. Rational Mech. Anal.* **192** (2009) 275–310.

[21] Z. Badreddine and H. Frankowska, Solutions to Hamilton–Jacobi equation on a Wasserstein space. *Calc. Var. Part. Differ. Equ.* **61** (2021) 9.

[22] P. Cardaliaguet and M. Quincampoix. Deterministic differential games under probability knowledge of initial condition. *Int. Game Theory Rev.* **10** (2008) 1–16.

[23] S. Daudin and B. Seeger, A comparison principle for semilinear Hamilton–Jacobi–Bellman equations in the Wasserstein space, Aug. 2023. Preprint (arXiv:2308.15174).

[24] C. Jimenez, A. Marigonda and M. Quincampoix, Optimal control of multiagent systems in the Wasserstein space. *Calc. Var. Part. Differ. Equ.* **59** (2020) 1–45.

[25] A. Marigonda and M. Quincampoix, Mayer control problem with probabilistic uncertainty on initial positions. *J. Differ. Equ.* **264** (2018) 3212–3252.

[26] L. Ambrosio and J. Feng, On a class of first order Hamilton–Jacobi equations in metric spaces. *J. Differ. Equ.* **256** (2014) 2194–2245.
[27] C. Bertucci, Stochastic optimal transport and Hamilton–Jacobi–Bellman equations on the set of probability measures, June 2023. Preprint, available at http://arxiv.org/abs/2306.04283.
[28] F. Jean, O. Jerhaoui and H. Zidani, Deterministic optimal control on Riemannian manifolds under probability knowledge of the initial condition. *SIAM J. Math. Anal.* **56** (2024) 3326–3356.
[29] O. Jerhaoui, *Viscosity Theory of First Order Hamilton Jacobi Equations in Some Metric Spaces*. PhD thesis, Institut Polytechnique de Paris, Paris (2022).
[30] G. Fabbri, F. Gozzi and A. Święch, Stochastic optimal control in infinite dimension. Vol. 82 of *Probability Theory and Stochastic Modelling*. Springer International Publishing, Cham (2017).
[31] N. Gigli, Second order analysis on $(P_2(M),W_2)$. *Mem. Am. Math. Soc.* **216** (2009) 1–147.
[32] N. Gigli, On the inverse implication of Brenier–McCann theorems and the structure of $(P_2(M),W_2)$. *Methods Applic. Anal.* **18** (2011) 127–158.
[33] F. Santambrogio, Optimal transport for applied mathematicians. Vol. 87 of *Progress in Nonlinear Differential Equations and Their Applications*. Springer International Publishing, Cham (2015).
[34] B. Piccoli, Measure differential equations. *Arch. Rational Mech. Anal.* **233** (2019) 1289–1317.
[35] B. Bonnet and H. Frankowska, Differential inclusions in Wasserstein spaces: The Cauchy–Lipschitz framework. *J. Differ. Equ.* **271** (2021) 594–637.
[36] B. Bonnet and H. Frankowska, Caratheodory Theory and A Priori Estimates for Continuity Inclusions in the Space of Probability Measures, May 2023. Preprint (arXiv:2302.00963).
[37] G. Cavagnari, A. Marigonda and B. Piccoli, Superposition Principle for Differential Inclusions, in *Large-Scale Scientific Computing*, Vol. 10665, edited by I. Lirkov and S. Margenov. Springer International Publishing, Cham (2018) 201–209.
[38] P. Cannarsa and C. Sinestrari, *Semiconcave Functions, Hamilton—Jacobi Equations, and Optimal Control*. Birkhäuser, Boston, MA (2004).
[39] P. Del Moral and M. Doisy, Maslov Idempotent Probability Calculus, I. *Theory Probab. Applic.* **43** (1999) 562–576.
[40] V.N. Kolokoltsov and V.P. Maslov, Idempotent Analysis and Its Applications. Springer Netherlands, Dordrecht (1997).
[41] P. Cardaliaguet, Notes on Mean Field Games. Available at https://www.ceremade.dauphine.fr/~cardaliaguet/MFG20130420.pdf, 2013.
[42] P. Cardaliaguet, F. Delarue, J.-M. Lasry and P.-L. Lions, The Master Equation and the Convergence Problem in Mean Field Games. Vol. 201 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ (2019).
[43] R. Carmona and F. Delarue, Probabilistic Theory of Mean Field Games with Applications II. Vol. 84 of *Probability Theory and Stochastic Modelling*. Springer International Publishing (2018).
[44] W. Gangbo, T. Nguyen and A. Tudorascu, Hamilton–Jacobi equations in the Wasserstein space. *Methods Applic. Anal.* **15** (2008) 155–184.
[45] I. Ekeland, On the variational principle. *J. Math. Anal. Applic.* **47** (1974) 324–353.
[46] W. Gangbo and A. Święch, Metric viscosity solutions of Hamilton–Jacobi equations depending on local slopes. *Calc. Var. Part. Differ. Equ.* **54** (2015) 1183–1218.
[47] X. Li and J. Yong, Optimal Control Theory for Infinite Dimensional Systems. Birkhäuser, Boston, MA (1995).
[48] C. Wu and J. Zhang, Viscosity solutions to parabolic master equations and McKean–Vlasov SDEs with closed-loop controls. *Ann. Appl. Probab.* **30** (2020) 936–986.
[49] J.M. Borwein and Q.J. Zhu, Techniques of Variational Analysis. CMS Books in Mathematics. Springer-Verlag, New York (2005).

Please help to maintain this journal in open access!

This journal is currently published in open access under the Subscribe to Open model (S2O). We are thankful to our subscribers and supporters for making it possible to publish this journal in open access in the current year, free of charge for authors and readers.

Check with your library that it subscribes to the journal, or consider making a personal donation to the S2O programme by contacting subscribers@edpsciences.org.

More information, including a list of supporters and financial transparency reports, is available at https://edpsciences.org/en/subscribe-to-open-s2o.

A Cauchy-Lipschitz setting for control problems in complete CAT(0) spaces

Averil Aussedat* and Hasnaa Zidani†

Abstract. We provide a framework for optimal control problems in CAT(0) spaces, in which one can recover the existence, uniqueness and a priori estimates of the classical theory. Our framework allows to extend the Aumann-Filippov theorem through a suitable extension of convex-valued dynamics. This formulation is compatible with the recent theory of viscosity solutions of Hamilton-Jacobi equations in CAT(0) spaces based on semiconvexity.

2020 Mathematics Subject Classification 49J15, 49L25, 37L05, 65J08.

1 Introduction

Optimal control theory has its origins in the study of systems governed by ordinary differential equations (ODEs), where the aim is to determine a control that minimizes a given cost functional. This work is concerned with the extension of this theory to complete geodesic spaces with nonpositive curvature in the sense of Alexandrov – the so-called CAT(0) spaces. We define controlled dynamics in CAT(0) spaces as infinitesimal compositions of gradient flows, and our main result is to identify the closure of the trajectories of a given controlled system as the set of solutions of a relaxed dynamical system, whose dynamics span the convex hull of the original dynamics in a suitable sense. Applied to a Mayer control problem, this allows us to provide conditions for the existence of an optimal control, and to identify the relaxation of the problem when these conditions are not met. Furthermore, the value function is shown to satisfy a Hamilton-Jacobi-Bellman (HJB) equation in the sense of [JZ23].

More precisely, given a state space Ω, a compact control set U, a terminal cost $\mathfrak{J} : \Omega \to \mathbb{R}$ and a dynamic f, we seek to

$$\text{Minimize} \quad \mathfrak{J}(y_T^{0,x,u}) \quad \text{over } u \in L^0(0,T;U), \tag{1}$$

under the constraint that $y^{0,x,u}$ satisfies a suitable controlled dynamical system with initial condition $y_0^{0,x,u} = x$. In the case $\Omega = \mathbb{R}^d$, the value function $V(t,x) = \inf_u \mathfrak{J}(y_T^{t,x,u})$ is the unique viscosity solution of the HJB equation

$$-\partial_t V(t,x) + \sup_{u\in U} -\langle D_x V(t,x), f(x,u)\rangle = 0, \qquad (t,x) \in [0,T) \times \mathbb{R}^d, \tag{2}$$

supplemented with $V(T,\cdot) = \mathfrak{J}$. Viscosity solutions were introduced by [Lio82; CL83], and later generalized to infinite-dimensional settings [FGŚ17]. Solving the HJB equation provides not only the value function, but also the global optimal control law and the corresponding optimal trajectories.

*INSA Rouen Normandie, Normandie Univ, LMI UR 3226, F-76000 Rouen, France. averil.aussedat@insa-rouen.fr

†INSA Rouen Normandie, Normandie Univ, LMI UR 3226, F-76000 Rouen, France. hasnaa.zidani@insa-rouen.fr

CAT(0) spaces are models of particular interest in the study of HJB equations on networks, glued manifolds, ramified spaces, and other piecewise-smooth structures [Ach+11; SC13; Rao13; CFS13; CM13; CSM13; IMZ13; BBC14]. To cope with difficulties created by changes of dimension, it has become classical to weaken the regularity of test functions into partial regularity, i.e. piecewise smooth with consistent transition conditions; we refer to [BC24] for an extensive exposition. At the level of metric spaces, one can also bypass the optimal control interpretation by focusing on isotropic dynamics, leading to Eikonal-type equations studied through the metric slope [AF14; GŚ15], growth constraints along paths [GHN15], or Monge solutions [LSZ25]. However, by nature, these works do not address the non-isotropic case $\dot{y}(s) = f(y(s), u(s))$.

In a general metric space, the most complete theory of ODEs is the abstract framework of mutational analysis introduced by Aubin [Aub99], based on a family of semigroups playing the role of Euclidean translations. We refer to [Lor10] for a comprehensive treatment, and to [FL23a; FL23b] for recent far-reaching extensions including well-posedness and invariance results for mutational inclusions in metric spaces, on which we rely heavily. Optimal control problems at this level of generality have been studied in [BF22a; BF22b], where viability and invariance of sets replace the PDE characterization of the value function. Our work inserts naturally within these references, but the restriction to CAT(0) spaces provides additional geometric structure. In particular, it allows us to introduce an explicit class of dynamics based on gradient flows of concave functions, which ultimately leads to a relaxation theorem and a connection with the viscosity theory of [Jer22; JZ23].

The core difficulty in extending the theory to CAT(0) spaces lies in formulating a suitable notion of Lipschitz dynamics and, in turn, the convexity condition required for relaxation theorems. In mutational analysis, a differential inclusion takes the form $\mathring{y}_t \cap f(y_t) \neq \emptyset$, where the mutation $\mathring{y}_t$ is the set of semigroups approximating the curve $t \mapsto y_t$ at first order. In Euclidean spaces, the Aumann-Filippov theorem [Aum65; Fil63] states that the uniform closure of the solution set of such an inclusion coincides with the solution set of the convexified system $\dot{y}_t \in \overline{\mathrm{conv}}\, f(y_t)$. The theory of mutations does not provide a way to convexify dynamics in this sense, and a generalization to metric spaces is not known in general.

CAT(0) spaces enjoy a rich structure of gradient flows that makes such a generalization possible. In a CAT(0) space, one could attempt to write an ODE by imposing that the velocity of a curve coincides with a given section of the tangent cone at almost every time. However, obtaining Lipschitz estimates requires comparing tangent vectors at different base points, which is not straightforward when tangent cones are not isometric – as is already the case for a simple network with three branches. Replacing the section of tangent cones by an application valued in gradient flows of concave functions resolves this issue: gradient flows are semigroups, hence global objects living in a single metric space, and can be assumed Lipschitz with respect to the corresponding distance. Informally, the ODE $\dot{y}_t = f(y_t)$ is replaced by the condition that $h \mapsto y_{t+h}$ follows the gradient flow of $f(y_t)$, where $f(y_t)$ is now a concave function rather than a vector. This construction recovers classical ODEs when $\Omega = \mathbb{R}^d$, and joins the framework of mutational analysis, which we use as a building block.

The main results of the paper are the following.

Well-posedness. We combine mutational analysis with transitions defined by gradient flows to obtain a Cauchy-Lipschitz theory for ODEs in CAT(0) spaces (Proposition 3.8): for any Lipschitz dynamics and initial condition, solutions exist, are unique, and satisfy explicit a priori estimates. Crucially, these solutions admit a characterization by evolutionary variational inequalities with a time-varying right-hand side (Proposition 3.9), which serves as the main technical tool in the relaxation theorem below.

Relaxation. Building on this variational characterization, we prove an analogue of the Aumann-Filippov theorem: the closure of the trajectory set of a controlled system coincides with the

trajectory set of a relaxed system, whose dynamics span the convex hull of the original ones in a suitable sense. This convexity is understood in terms of the tangent cones of the CAT(0) space, and implies in particular that the relaxed trajectories follow, at almost every time, the barycenters of the original dynamics (Proposition 4.10). As a consequence, an optimal control exists whenever $f(\cdot, U)$ has convex images, and the relaxed problem always admits an optimizer in the form of a Young measure (Proposition 5.3).

Hamilton-Jacobi-Bellman equations. Combining the relaxation result with the a priori estimates, we identify the value function of problem (1) as a viscosity solution of a HJB equation in the sense of [JZ23] (Propositions 5.5 and 5.9). The purpose is not to advance the HJB theory itself, but to show that the control framework introduced here fits into an existing, complete theory – from the definition of dynamics to the characterization of optimal trajectories.

Numerical approximation. Finally, we introduce a semi-Lagrangian discretization adapted to the CAT(0) setting. We establish convergence estimates for the approximation of the value function and for the reconstruction of optimal trajectories, and illustrate the method on a robot navigation problem.

The restriction to CAT(0) spaces is essential for the present work. In the general framework of mutational analysis, the dynamics are described through abstract transitions and the existence of suitable semigroups is typically assumed. Here, the geometry of CAT(0) spaces provides a natural and explicit class of transitions through gradient flows of concave functions. This allows us to define Lipschitz dynamics, introduce a meaningful convexification of the dynamics based on geodesic convexity, and ultimately establish the relaxation theorem proved below. It also connects the resulting control problem with the viscosity theory of [JZ23].

The paper is organised as follows. Section 2 collects necessary background on CAT(0) spaces and mutational analysis, including the well-posedness results from [FL23a] that we rely on. Section 3 introduces the specific definition of ODEs we consider, with Section 3.2 providing the variational characterization. The relaxation theorem is proved in Section 4. The link with HJB equations is studied in Section 5. An Appendix gathers some lengthy computations.

2 Preliminaries

Let (Ω, d) be a metric space. We use 0_Ω as a notation for an arbitrary but fixed point of Ω.

The set of continuous functions over Ω is denoted $\mathcal{C}(\Omega)$, with $\mathcal{C}^\infty$, $\mathcal{C}_b$, $\mathcal{C}_c$ respectively denoting smooth (when defined), bounded and compactly supported functions. The Lebesgue measure is denoted $\mathcal{L}$, with $\mathcal{L}_A$ its restriction to a measurable set A. The closure of a set A is denoted $\overline{A}$, and the notation $\mathscr{B}(x, r)$ refers to the open unit ball of centre x and radius $r > 0$.

Given two subsets $A, B \subset \Omega$, we define the Hausdorff distance $d_H(A, B)$ as

$$d_H(A, B) \coloneqq \max\left(\sup_{a\in A}\inf_{b\in B} d(a, b), \sup_{b\in B}\inf_{a\in A} d(a, b)\right).$$

A curve $\gamma \in \mathcal{C}([0, T]; \Omega)$ is absolutely continuous if there exists $g \in L^1(0, T; \mathbb{R}^+)$ such that

$$d(\gamma_s, \gamma_t) \leqslant \int_{\tau=s}^{t} g(\tau)d\tau \qquad \forall\, 0 \leqslant s \leqslant t \leqslant T.$$

The set of absolutely continuous curves is denoted $\mathrm{AC}([0, T]; \Omega)$. Whenever I is a compact interval and Ω is a metric space, the notation $L^0(I; \Omega)$ refers to the set of measurable applications from I to Ω, and $L^p(I; \Omega)$, $p \geqslant 1$, to the subset of $\gamma \in L^0(I; \Omega)$ for which $t \mapsto d^p(\gamma_t, 0_\Omega)$ is integrable.

2.1 CAT(0) spaces

A curve $\gamma : [0,1] \to \Omega$ is called a *geodesic* if $d(\gamma_s, \gamma_t) = |t-s|\, d(\gamma_0, \gamma_1)$ for any $s, t \in [0,1]$. A CAT(0) space is a metric space in which any two points are connected by at least one geodesic, and where

$$d^2(\gamma_t, z) \leqslant (1-t)d^2(\gamma_0, z) + td^2(\gamma_1, z) - t(1-t)d^2(\gamma_0, \gamma_1) \tag{3}$$

for any $z \in \Omega$, geodesic γ and $t \in [0,1]$. The condition (3) implies that there is exactly one geodesic linking $x, y \in \Omega$, which is denoted by $[xy]$ in the sequel. Following classical notations, the evaluation of $[xy]$ at time $t \in [0,1]$ is denoted $(1-t)x \oplus ty$.

The condition (3) is a 2-convexity property of the squared distance, in the following sense.

Definition 2.1. Let $\lambda \in \mathbb{R}$. A function $f : \Omega \to \mathbb{R}$ is λ-convex if for any $x, y \in \Omega$ and $t \in [0,1]$,

$$f((1-t)x \oplus ty) \leqslant (1-t)f(x) + tf(y) - \frac{\lambda}{2}t(1-t)d^2(x,y).$$

The function f is said semiconvex if it is λ-convex for some $\lambda \in \mathbb{R}$, and convex if it is 0-convex. In addition, f is λ-concave if $-f$ is λ-convex, with semiconcave and concave functions defined accordingly.

A geodesic space is said to be *proper* if every closed and bounded set is compact. By the Hopf-Rinow theorem [AKP23, Theorem 2.15], any complete and locally compact geodesic space is proper.

Assumption [A2.2]. In the sequel, we assume that Ω is a complete and proper CAT(0) space.

Typical examples range from $\mathbb{R}^d$ to locally compact metric trees, and include locally compact hyperbolic manifolds such as the cone of SPD matrices endowed with a so-called geometric distance [Bha07].

2.1.1 Tangent cone

A curve γ is called a reparametrized geodesic if there exists $\lambda > 0$ such that $s \mapsto \gamma_{\lambda s}$ coincides with a geodesic on a nontrivial interval. Given such γ, γ' issued from the same point $x \in \Omega$, let

$$d_x(\gamma, \gamma') \coloneqq \limsup_{h \searrow 0} \frac{d(\gamma_h, \gamma'_h)}{h}. \tag{4}$$

The relation $\gamma \sim_x \gamma'$ defined by $d_x(\gamma, \gamma') = 0$ is an equivalence relation on the set of reparametrized geodesics. We denote by $\overrightarrow{xy}$ the equivalence class of $[xy]$, with the convention $0_x \coloneqq \overrightarrow{xx}$. By (3), the function $h \mapsto d(\gamma_h, \gamma'_h)/h$ is nondecreasing, which yields

$$d_x(\overrightarrow{xy}, \overrightarrow{xz}) \leqslant d(y,z), \qquad \forall x, y, z \in \Omega\,. \tag{5}$$

Definition 2.3. Let $x \in \Omega$. The regular tangent cone $\mathrm{T}'_x\,\Omega$ is defined as the quotient of the class of reparameterized geodesics by the equivalence relation $\sim_x$. The tangent cone $\mathrm{T}_x\,\Omega$ is the abstract completion of $\mathrm{T}'_x\,\Omega$ with respect to the metric d_x.

As shown in [BH99, Theorem 3.19], the tangent cone $(\mathrm{T}_x\,\Omega, d_x)$ is itself a geodesic and CAT(0) space. For any $\lambda > 0$ and $v \in \mathrm{T}'_x\,\Omega$, the notation $\lambda \cdot v$ refers to the equivalence class of the reparametrized curves $s \mapsto \gamma_{\lambda s}$, where γ belongs to the equivalence class v. The scaling operation extends to $\mathrm{T}_x\,\Omega$. For any $v, w \in \mathrm{T}_x\,\Omega$ and $\lambda \in \mathbb{R}^+$, define the

$$\text{norm:} \qquad |v|_x \coloneqq d_x(v, 0_x), \tag{6a}$$

$$\text{scalar product:} \qquad \langle v, w \rangle_x \coloneqq \frac{1}{2}\left[|v|_x^2 + |w|_x^2 - d_x^2(v,w)\right]. \tag{6b}$$

Proposition 2.4 ([GN21, Proposition 2.6]). *For any $u, v, w \in \mathrm{T}_x\,\Omega$, $t \in [0,1]$, and $\lambda > 0$, there holds $|\lambda \cdot u|_x = \lambda\,|u|_x$, as well as*

$$\langle \lambda \cdot u, v\rangle_x = \langle u, \lambda \cdot v\rangle_x = \lambda\,\langle u, v\rangle_x\,,$$
$$\langle u, (1-t)v \oplus tw\rangle_x \geqslant (1-t)\,\langle u, v\rangle_x + t\,\langle u, w\rangle_x\,.$$

Remark 2.5 (Lipschitz regularity of the metric scalar product). For any $x \in \Omega$ and $p, q, v \in \mathrm{T}_x\,\Omega$,

$$\langle p, v\rangle_x - \langle q, v\rangle_x \leqslant |v|_x\, d_x(p,q). \tag{7}$$

Indeed, assume first that $v = \overrightarrow{xy}$ for some $y \in \Omega$. Let $\varepsilon > 0$, and let $p_\varepsilon, q_\varepsilon \in \mathrm{T}'_x\,\Omega$ be associated to reparametrized geodesics γ, τ issued from x, defined on a interval $[0, h_\varepsilon]$, and such that $\max\left(d(p,p_\varepsilon), d(q,q_\varepsilon)\right) \leqslant \varepsilon$. Then, for L a Lipschitz constant of $d^2(\cdot, y)$ in $\mathscr{B}(x, \max(|p|_x, |q|_x))$,

$$\begin{aligned}
\langle p, v\rangle_x - \langle q, v\rangle_x &= \frac{D_x d^2(\cdot,y)(q) - D_x d^2(\cdot,y)(p)}{2} \leqslant \lim_{h \searrow 0, h \leqslant h_\varepsilon} \frac{d^2(\gamma_h, y) - d^2(\tau_h, y))}{2h} + L\varepsilon \\
&= \lim_{h \searrow 0, h \leqslant h_\varepsilon} \left(d(\gamma_h, y) + d(\tau_h, y)\right) \frac{d(\gamma_h, y) - d(\tau_h, y)}{2h} + L\varepsilon \\
&\leqslant d(x,y) d_x(p_\varepsilon, q_\varepsilon) + L\varepsilon \leqslant |v|_x\, d_x(p,q) + (2\,|v|_x + L)\,\varepsilon.
\end{aligned}$$

Letting $\varepsilon \to 0$, we conclude that (7) holds for $v = \overrightarrow{xy}$. Since both sides of (7) are positively homogeneous and continuous in v, the formula extends to any $v \in \mathrm{T}_x\,\Omega$.

2.1.2 Derivatives of curves and differentials of functions

The following definition is taken from [AKP23, Definition 6.9].

Definition 2.6. Let $T > 0$, and let $\gamma : [0,T] \to \Omega$ be a curve starting from $x \in \Omega$. The vector $v \in \mathrm{T}_x\,\Omega$ is called the *right derivative* of γ at 0, denoted by $\gamma_0^+ = v$, if there exists a sequence $(v_n)_n \subset \mathrm{T}'_x\,\Omega$ such that $d_x(v_n, v) \to 0$ as $n \to \infty$, and a corresponding sequence $(\gamma^n)_n$ of reparametrized geodesics satisfying

$$\lim_{n\to\infty} \limsup_{\varepsilon \to 0^+} \frac{d\left(\gamma_\varepsilon, \gamma^n_\varepsilon\right)}{\varepsilon} = 0. \tag{8}$$

The *left derivative* γ_t^- is defined as the derivative of the function $h \mapsto \gamma_{t-h}$ at $h = 0$. Moreover, a curve γ is said to be *differentiable* at $t \in (0,T)$ if it admits both a right and a left derivative at t, and these satisfy $\left|\gamma_t^+\right|_{\alpha_t} = \left|\gamma_t^-\right|_{\alpha_t} = \frac{1}{2} d_x(\gamma_t^+, \gamma_t^-)$.

From (8), it is clear that if γ has a right derivative at 0, and if $\sigma : [0,T] \to \Omega$ is another curve starting from the same point, satisfying $d(\gamma_\varepsilon, \sigma_\varepsilon) = o(\varepsilon)$, then σ_0^+ is well defined and equal to γ_0^+. In addition, if γ and θ are curves with respective right derivatives γ_0^+ and θ_0^+ at $t = 0$, then

$$\lim_{h \searrow 0} \frac{d(\gamma_h, \theta_h)}{h} = d_x(\gamma_0^+, \theta_0^+). \tag{9}$$

Indeed, if γ, θ are reparametrized geodesics, this is the definition of $d_x(\cdot,\cdot)$, and the general case follows by approximation.

Derivatives of curves provide a way to define directional derivatives of maps.

Definition 2.7 (Differential [AKP23, Definition 6.15]). Let $\varphi : \Omega \to \mathbb{R}$ be a locally Lipschitz function. Its differential at $x \in \Omega$, if it exists, is the unique map $D_x\varphi : \mathrm{T}_x\,\Omega \to \mathbb{R}$ such that

$$D_x\varphi(\gamma_0^+) = (\varphi \circ \gamma)_0^+ \tag{10}$$

for any curve $\gamma : [0,T] \to \Omega$ with $T > 0$, $\gamma_0 = x$ and γ_0^+ defined.

When it exists, the differential of an application φ is positively homogeneous, in that $D_x\varphi(\lambda v) = \lambda D_x\varphi(v)$ for any $\lambda \geqslant 0$. Moreover, for any local Lipschitz constant $\mathrm{Lip}(\varphi)$ of φ in a ball containing x, there holds

$$|D_x\varphi(v) - D_x\varphi(w)| \leqslant \mathrm{Lip}(\varphi)\, d_x(v,w) \qquad \forall v, w \in \mathrm{T}_x\,\Omega\,. \tag{11}$$

Any semiconcave φ admits a differential at any point [AKP23, Lemma 13.15]. Moreover, in this case, each $D_x\varphi$ is geodesically concave in $\mathrm{T}_x\,\Omega$ in the sense of Definition 2.1 [AKP23, Propostion 6.16]. We will use repeatedly the fact that whenever $\varphi : \Omega \to \mathbb{R}$ is Lipschitz and admits a differential everywhere, and $\gamma \in \mathrm{AC}([0,T];\Omega)$ is differentiable in the sense of Definition 2.6 for a.e. $t \in [0,T]$, then

$$\varphi(\gamma_T) - \varphi(\gamma_0) = \int_{t=0}^{T} D_{\gamma_t}\varphi(\gamma_t^+)\, dt. \tag{12}$$

This follows from the fact that $t \mapsto \varphi(\gamma_t)$ is absolutely continuous, and the equality (10).

If $v = \overrightarrow{xy}$ for some point $y \in \Omega$, then $D_x\varphi(v)$ is the directional derivative of φ along the geodesic $[xy]$. As useful examples, let us mention that

$$D_x d(\cdot,y)(v) = \begin{cases} |v|_x & \text{if } y = x, \\ -\left\langle \frac{\overrightarrow{xy}}{d(x,y)}, v \right\rangle_x & \text{otherwise,} \end{cases} \quad \text{and} \quad D_x d^2(\cdot,y)(v) = -2\left\langle \overrightarrow{xy}, v \right\rangle_x. \tag{13}$$

2.1.3 Gradient flows

In addition to a differential, a function can admit a *metric gradient*, in the following sense.

Definition 2.8 ([AKP23, Def 13.17])**.** Let $\varphi : \Omega \to \mathbb{R}$ with $D_x\varphi : \mathrm{T}_x\,\Omega \to \mathbb{R}$ defined at $x \in \Omega$. A tangent vector $g \in \mathrm{T}_x\,\Omega$ is called a *gradient of* φ at x if $D_x\varphi(v) \leqslant \langle g, v\rangle_x$ for all $v \in \mathrm{T}_x\,\Omega$, and if $D_x\varphi(g) = \langle g, g\rangle_x$. When unique, it is denoted by $\nabla_x\varphi$.

Any locally Lipschitz and semiconcave function admits a unique gradient at every point [AKP23, Proposition 13.19], which satisfies $\nabla_x(\lambda\varphi) = \lambda\nabla_x\varphi$ for all $\lambda \geqslant 0$.

Proposition 2.9 (Existence and uniqueness of gradient flow [AKP23, Prop. 16.15 and 16.19])**.** *Let $E : \Omega \to \mathbb{R}$ be locally Lipschitz and concave. Then there exists a unique semigroup $\vartheta : \mathbb{R}^+ \times \Omega \to \Omega$ such that*

$$\vartheta(0,x) = x, \qquad \vartheta(h,\vartheta(s,x)) = \vartheta(h+s,x), \qquad \vartheta(\cdot,x)_h^+ = \nabla_{\vartheta(h,x)}E,$$

for all $x \in \Omega$ and $h, s \geqslant 0$.

We refer to ϑ as the gradient flow of E, and denote $\vartheta = \mathcal{G}_{\mathcal{F}}(E)$. The gradient flow of the constant function is denoted 0_Θ.

Remark 2.10. Gradient flows in CAT(0) spaces may be defined by the Crandall-Liggett minimizing scheme, and already provide a semigroup as the limit of the (uniquely defined) solutions of the scheme when the discretization parameter goes to 0 (See Def. 3, Th. 1.3 and Th. 2.5 in [May98]). This semigroup can be characterized by Evolutionary Variational Inequalities (EVI), as done in [AGS05, Theorem 4.0.4], in the sense that $\vartheta : \mathbb{R}^+ \times \Omega \to \Omega$ is the limit of the Crandall-Liggett semigroup generated by a concave and locally Lipschitz $E : \Omega \to \mathbb{R}$ if and only if for almost all $t \geqslant 0$, there holds for all $z \in \Omega$ that

$$\frac{d}{dt}\frac{d^2(\vartheta(t,x),z)}{2} \leqslant (-E)(z) - (-E)(\vartheta(t,x)) = E(\vartheta(t,x)) - E(z). \tag{14}$$

Lemma 3.10 below will show that the gradient flow defined in Proposition 2.9 satisfies the EVI (14). As the solution of the EVI is unique, the two notions coincide under our assumptions.

2.2 The Banach closure of DC functions

In this section, we introduce a Banach space of functions to be used in Section 3.1. Given $E : \Omega \to \mathbb{R}$ a Lipschitz function, denote by $|\cdot|_{1,\infty}$ the supremum of the metric slope, i.e.

$$|E|_{1,\infty} \coloneqq \sup_{x\in\Omega} \limsup_{y\to x} \frac{|E(y) - E(x)|}{d(x,y)}. \tag{15}$$

By definition, $|E|_{1,\infty}$ coincides with the Lipschitz constant of E.

Remark 2.11. If E admits a differential $D_x E$ at any point x in the sense of Definition 2.7, then

$$|E|_{1,\infty} = \sup_{(x,v)\in \mathrm{T}\,\Omega, |v|_x=1} |D_x E(v)|\,.$$

Indeed, $|D_x E(v)| \leqslant \limsup_{y\to x} |E(y) - E(x)|/d(x,y)$ for any x. On the other hand, for any geodesic γ, the Lipschitz curve $t \mapsto E(\gamma_t)$ is differentiable at a.e. $t \in [0,1]$, so that by (12),

$$E(\gamma_1) - E(\gamma_0) = \int_0^1 D_{\gamma_t} E\left(\frac{d(\gamma_0,\gamma_1)}{d(\gamma_t,\gamma_1)} \cdot \overrightarrow{\gamma_t\gamma_1}\right) dt \leqslant d(\gamma_0,\gamma_1) \sup_{(x,v)\in \mathrm{T}\,\Omega, |v|_x=1} |D_x E(v)|.$$

Denote $E \sim E'$ if the two Lipschitz functions E, E' differ by a constant, i.e. $|E - E'|_{1,\infty} = 0$.

Definition 2.12. Let $\mathrm{DC}_{\mathrm{Lip}}$ denote the set of Lipschitz functions that can be written as the difference of two concave functions. We define

$$\mathbb{E} \coloneqq \overline{\mathrm{DC}_{\mathrm{Lip}}/\sim}^{|\cdot|_{1,\infty}}.$$

Lemma 2.13. *$\left(\mathbb{E}, |\cdot|_{1,\infty}\right)$ is a Banach space, and each $E \in \mathbb{E}$ is a Lipschitz function.*

Proof. $\mathbb{E}$ is a vector space, since the operations of addition and scalar multiplication are preserved in the quotient by $\sim$. Moreover, $\mathrm{DC}_{\mathrm{Lip}}/\sim$ is a subset of Lipschitz functions up to a constant, which are closed with respect to $|\cdot|_{1,\infty}$ by Arzelá-Ascoli (applied on balls, which are compact since Ω is assumed proper). Hence $\mathbb{E}$ is a Banach space which is a subset of Lipschitz functions. □

We introduce the class of Fréchet-differentiable functions, defined as follows.

Definition 2.14. A Lipschitz and concave function $E : \Omega \to \mathbb{R}$ is Fréchet-differentiable if $D_x E(v) = \langle \nabla_x E, v\rangle_x$ for all $x \in \Omega$ and $v \in \mathrm{T}_x\,\Omega$.

For instance, $E = -d^2(\cdot, y)$ is Fréchet-differentiable by (13). In Euclidean spaces, Definition 2.14 reduces to classical Fréchet-differentiability. However, it may not be preserved by convex combinations. Part of the interest of these functions comes from the following lemma.

Lemma 2.15. *Let $E, E' : \Omega \to \mathbb{R}$ be Lipschitz, concave, and Fréchet-differentiable. Then*

$$\sup_{x\in\Omega} d_x(\nabla_x E, \nabla_x E') = \left|E - E'\right|_{1,\infty}. \tag{16}$$

Proof. Let $x \in \Omega$. By definition of the metric gradient, there holds

$$D_x E(\nabla_x E') \leqslant \left\langle \nabla_x E, \nabla_x E'\right\rangle_x \qquad \text{and} \qquad D_x E'(\nabla_x E) \leqslant \left\langle \nabla_x E', \nabla_x E\right\rangle_x.$$

Therefore

$$\begin{aligned} d_x^2\left(\nabla_x E, \nabla_x E'\right) &= |\nabla_x E|_x^2 + \left|\nabla_x E'\right|_x^2 - 2\left\langle \nabla_x E, \nabla_x E'\right\rangle_x \\ &\leqslant D_x E(\nabla_x E) + D_x E'(\nabla_x E') - D_x E(\nabla_x E') - D_x E'(\nabla_x E) \end{aligned}$$

$$= D_x\left(E' - E\right)(\nabla_x E) - D_x\left(E' - E\right)(\nabla_x E')$$
$$\leqslant \left|E' - E\right|_{1,\infty} d_x\left(\nabla_x E, \nabla_x E'\right).$$

In the last line, we used that $E - E'$ admits a differential at any point, and the estimate (11) with $\mathrm{Lip}(E - E') = |E - E'|_{1,\infty}$ by definition. This yields the inequality $\leqslant$ in (16). Conversely, as E and E' are Fréchet-differentiable, there holds

$$\left|D_x E(v) - D_x E'(v)\right| = \left|\langle \nabla_x E, v\rangle - \langle \nabla_x E', v\rangle\right| \leqslant |v|_x \times d_x(\nabla_x E, \nabla_x E') \qquad \forall\, (x,v) \in \mathrm{T}\,\Omega\,.$$

Here we used the inequality proved in Remark 2.5. Taking the supremum over all v such that $|v|_x = 1$, then over $x \in \Omega$, we recognize the definition of $|E - E'|_{1,\infty}$. This shows (16). □

2.3 Mutational analysis

Mutational analysis relies on the postulate that one is given a family of semigroups, referred to as *transitions*, satisfying some axiomatic properties gathered in Definition 2.16. These transitions are declared to be the elementary motions of the underlying metric space, and a curve will be "mutable" if it is well-approximated at order one by at least one transition. Building on this, a theory of ordinary differential equation was developed, with applications to control theory in mind: we refer the reader to the original treatise [Aub99], to [Lor10] for developments and applications, and to the recent work [FL23a] for far-reaching extensions.

Definition 2.16 (Transition)**.** A function $\vartheta : [0,\infty) \times \Omega \to \Omega$ is a *transition* on (Ω, d) if

(i) $\vartheta(0,x) = x$ for any $x \in \Omega$,

(ii) $\vartheta(t, \vartheta(s,x)) = \vartheta(t+s,x)$ for any $x \in \Omega$ and $t, s \in \mathbb{R}^+$, i.e. ϑ is a semigroup,

(iii) there exists $\alpha(\vartheta) \geqslant 0$ such that for any $x, y \in \Omega$,

$$\limsup_{h \searrow 0} \frac{d(\vartheta(h,x), \vartheta(h,y)) - d(x,y)}{h} \leqslant \alpha(\vartheta) d(x,y),$$

(iv) there exists $\beta(\vartheta) \geqslant 0$ such that for any $x \in \Omega$, $s \mapsto \vartheta(s,x)$ is $\beta(\vartheta)-$Lipschitz.

In [FL23a, Def 2.1], the definition allows for locally Lipschitz transitions and features an additional restriction, which is implied by (iv) in out case. Let Θ be a set of transitions, taken arbitrary for now. It is a metric space for the distance $\mathbb{D}$ defined by

$$\mathbb{D}(\vartheta, \vartheta') \coloneqq \sup_{x \in \Omega} \limsup_{h \searrow 0} \frac{d(\vartheta(h,x), \vartheta'(h,x))}{h} \qquad \forall \vartheta, \vartheta' \in \Theta. \tag{17}$$

Remark 2.17. The quantity $\mathbb{D}$ defined in (17) is indeed a metric on Θ, not merely a pseudo-metric. While $\mathbb{D}$ compares transitions only through their infinitesimal behaviour near $h = 0$, the semigroup property (ii) of Definition 2.16 and a Grönwall argument show that $\mathbb{D}(\vartheta, \vartheta') = 0$ implies $\vartheta = \vartheta'$.

We denote 0_Θ the constant mutation $\vartheta(h,x) = x$. Note that $\mathbb{D}(\vartheta, 0_\Theta) \leqslant \beta(\vartheta)$ for any transition ϑ. A curve $(y_t)_{t \in \mathbb{R}^+}$ is *mutable with respect to* Θ *at time* t, or just mutable at t, it its mutation $\mathring{y}_t$ is nonempty:

Definition 2.18. The mutation of $y \in \mathrm{AC}(\mathbb{R}^+; \Omega)$ at time $t \in \mathbb{R}^+$ is

$$\mathring{y}_t \coloneqq \left\{ \vartheta \in \Theta \;\middle|\; \lim_{h \searrow 0} \frac{d(\vartheta(h, y_t), y_{t+h})}{h} = 0 \right\}.$$

We stress that $\mathring{y}$ is a set, that can be very large since it only restricts the behaviour of ϑ near y_t. Therefore, given an application $f : \Omega \rightrightarrows \Theta$, the corresponding mutational inclusion will write $\mathring{y}_t \cap f(y_t) \neq \emptyset$ for a.e. t. We now state the well-posedness theorem for mutational inclusions subtending the sequel.

Assumption [A2.19]. Assume that $f : \Omega \rightrightarrows \Theta$ satisfies the following:

(i) $\alpha(\vartheta) \leqslant 0$ for any $\vartheta \in f(x)$ and any $x \in \Omega$,

(ii) there exists a constant C_f such that $\beta(\vartheta) \leqslant C_f(1 + d(x, 0_\Omega))$ for any $\vartheta \in f(x)$ and $x \in \Omega$,

(iii) the restriction of f to any compact B is Lipschitz with respect to the Hausdorff distance over $(\Theta, \mathbb{D})$, with constant denoted $\mathrm{Lip}(f)_B$.

As a consequence of [FL23a, Theorem 3.4], we obtain the following.

Proposition 2.20. *Assume that Θ is complete and separable with respect to $\mathbb{D}$. Let f satisfy **[A2.19]**, $T > 0$, and $x_0 \in \Omega$. Let also $y \in \mathrm{AC}([0,T];\Omega)$ be mutable almost everywhere, and assume that there exists a measurable selection $\theta : [0,T] \to \Theta$ of $t \mapsto \mathring{y}_t$ such that $\int_{s=0}^{T} \mathbb{D}(\theta(s), \vartheta_0) ds < \infty$ for some $\vartheta_0 \in \Theta$. Then there exists a curve $z \in \mathrm{AC}([0,T];\Omega)$ solving*

$$\mathring{z}_t \cap f(z_t) \neq \emptyset \qquad \text{for a.e. } t \in \mathbb{R}^+, \qquad\qquad z_0 = x_0, \tag{18}$$

and for some measurable selection $\vartheta : [0,T] \to \Theta$ of $t \mapsto \mathring{z}_t \cap f(z_t)$, there holds

$$d(y_t, z_t) \leqslant \left[d(y_0, x_0) + 2 \int_{s=0}^{t} \inf_{\theta \in \mathring{y}_s} \mathbb{D}\left(\theta, \vartheta_s\right) \, ds \right] e^{2T \mathrm{Lip}(f)_{R_0}(1+R_0)}. \tag{19}$$

Here $R_0 \coloneqq 1 + d(x_0, 0_\Omega) + TC_f(1 + d(x_0, 0_\Omega))e^{TC_f}$. Moreover, the curve z satisfies the estimate

$$d(z_t, x_0) \leqslant tC_f(1 + d(x_0, 0_\Omega))e^{tC_f} \qquad \text{for all } t \geqslant 0. \tag{20}$$

Proof. The assumption $(\mathbb{H}\Theta)$ in [FL23a] is satisfied for $\|\!|\cdot|\!\| \coloneqq d(\cdot, 0_\Omega)$ and $D_r = D_{\mathrm{loc}} = \mathbb{D}$. Since balls in Ω are convex, each $x \in \Omega$ admits a unique projection p_x onto $\overline{\mathscr{B}}(0_\Omega, R_0)$, and $x \mapsto p_x$ is 1-Lipschitz. Set $\mathfrak{F}(t,x) \coloneqq f(p_x)$ and R as above. The assumptions (i)–(vii) of [FL23a, Theorem 3.4] follow directly from **[A2.19]**: items (i)–(iii) use $\alpha(\vartheta) \leqslant 0$ and $\beta(\vartheta) \leqslant C_f(1 + R_0)$; items (iv)–(v) are immediate since $\mathfrak{F}$ has compact values and is time-independent; item (vi) holds for $\lambda_R \equiv \mathrm{Lip}(f)_{R_0}(1 + R_0)$; item (vii) for $\Psi_R(t) \coloneqq C_f(1 + R_0) + \mathbb{D}(\theta(t), \vartheta_0)$. Applying the theorem with $\kappa \coloneqq 2$ gives z and ϑ satisfying (19).

It remains to show $\mathfrak{F}(t, z_t) = f(z_t)$, i.e. that z stays in $\overline{\mathscr{B}}(0_\Omega, R_0)$. Since z is continuous and x_0 is in the interior of the ball, there exists $\tau > 0$ such that $\mathfrak{F}(t, z_t) = f(z_t)$ on $[0, \tau)$. By [Aub99, Proposition 1.3.2],

$$d(z_t, x_0) \leqslant \int_0^t C_f\big(1 + d(z_s, x_0) + d(x_0, 0_\Omega)\big) \, ds,$$

and a Grönwall lemma gives (20). Hence $\tau \geqslant T$, and z solves (18). □

3 Ordinary differential equations in a CAT(0) space

In this short section, we use mutations to define differential equations and differential inclusions in CAT(0) spaces. Instead of working at the level of transitions, we construct an isometry with the set of energies whose gradient flows will be transitions. The advantage is that energies live in a Banach space, and allow to characterize the solutions of differential inclusions by evolutionary variational inequalities (see Proposition 3.9).

3.1 Definition of ODE

3.1.1 Energies and transition semigroups

The functions used to define transitions need not to be very general, and we restrict to some "admissible sets" $\mathcal{E}$ with good regularity properties. The set $\mathcal{E}$ is chosen as a subset of the Banach space $(\mathbb{E}, |\cdot|_{1,\infty})$ given by Definition 2.12, itself a subset of Lipschitz functions.

Definition 3.1. A subset $\mathcal{E} \subset \mathbb{E}$ is an admissible set of energies if

(i) it is complete and separable (for the topology induced by $|\cdot|_{1,\infty}$),

(ii) each $E \in \mathbb{E}$ is concave and Fréchet-differentiable in the sense of Definition 2.14.

In particular, any $E \in \mathcal{E}$ admits a gradient flow in the sense of Proposition 2.9.

Lemma 3.2. *Let $\mathcal{E}$ be an admissible set of energies. For each $E \in \mathcal{E}$, the gradient flow semigroup $\theta \coloneqq \mathcal{G}_{\mathcal{F}}(E)$ is a transition in the sense of Definition 2.16, with $\alpha(\vartheta) = 0$ and $\beta(\vartheta) = |E|_{1,\infty}$.*

Proof. By definition of gradient flow, $\vartheta \in \Theta$ is a semigroup such that $\vartheta(0, x) = x$ for any $x \in \Omega$. Moreover, since E is concave, its gradient flow is contractant [AKP23, Proposition 16.20], i.e.

$$d(\vartheta(h, x), \vartheta(h, y)) \leqslant d(x, y) \qquad \forall x, y \in \Omega \text{ and } h \geqslant 0.$$

This implies that $\alpha(\vartheta)$ can be taken equal to 0 in Definition 2.16.(iii). By [AKP23, Proposition 16.20] again, each curve $h \mapsto \vartheta(h, x)$ is locally Lipschitz. Therefore (12) and Lemma 2.15 yield

$$d(\vartheta(h, x), x) \leqslant \int_{s=0}^{h} \left|\vartheta(\cdot, x)_s^+\right|_{\vartheta(s,x)} ds = \int_{s=0}^{h} \left|\nabla_{\vartheta(h,x)} E\right|_{\vartheta(s,x)} ds \leqslant h\, |E|_{1,\infty}\,.$$

Hence the constant $\beta(\vartheta)$ in Definition 2.16.(iv) can be taken equal to $|E|_{1,\infty}$, and we conclude. □

The following lemma allows to work directly with energies instead of transitions. The advantage is that $\mathbb{E}$ is a Banach space, and we might use the results of Banach-valued L^p spaces.

Lemma 3.3. *Let $\mathcal{E}$ be an admissible set of energies. Then*

$$\mathbb{D}(\mathcal{G}_{\mathcal{F}}(E), \mathcal{G}_{\mathcal{F}}(E')) = \left|E - E'\right|_{1,\infty} \qquad \forall E, E' \in \mathcal{E},$$

i.e. $\mathcal{G}_{\mathcal{F}}$ is an isometry from $\left(\mathcal{E}, |\cdot|_{1,\infty}\right)$ to $(\Theta, \mathbb{D})$.

In particular, $\mathcal{G}_{\mathcal{F}}$ is injective on its image, so that ${\mathcal{G}_{\mathcal{F}}}^{-1} : \Theta \to \mathcal{E}$ is well-defined. Here we recall that $\mathcal{E}$ is a subset of equivalence classes of functions up to a constant.

Proof. Let $\vartheta \coloneqq \mathcal{G}_{\mathcal{F}}(E)$ and $\vartheta' \coloneqq \mathcal{G}_{\mathcal{F}}(E')$. Fix $x \in \Omega$. By definition of a gradient flow, the curves $h \mapsto \vartheta(h, x)$ and $h \mapsto \vartheta'(h, x)$ admit respectively $\nabla_x E$ and $\nabla_x E'$ as right derivative at $h = 0$. Hence, by (9), there holds

$$\limsup_{h \searrow 0} \frac{d\left(\vartheta(h, x), \vartheta'(h, x)\right)}{h} = d_x\left(\nabla_x E, \nabla_x E'\right).$$

Since E and E' are Fréchet-differentiable, the isometry follows from Lemma 2.15. □

The precise choice of $\mathcal{E}$ does not matter in the sequel, but one could doubt that such a set exists. Here is a general construction in an arbitrary complete and proper CAT(0) space.

Definition 3.4. Let $\kappa : \mathbb{R}^+ \to \mathbb{R}^+$ be given by $\kappa(r) = r^2/2$ if $r \in [0, 1]$, and $\kappa(r) = r - 1/2$ otherwise. Define

$$\mathcal{E} \coloneqq \left\{x \mapsto -\alpha\kappa(d(x, x_0)) \;\middle|\; \alpha \in \mathbb{R}^+ \text{ and } x_0 \in \Omega\right\}. \tag{21}$$

We check that $\mathcal{E}$ satisfies Definition 3.1. Since the distance is convex in CAT(0) spaces [BH99, Proposition 2.2], and κ is convex and non-decreasing, the elements of $\mathcal{E}$ are concave and Lipschitz-continuous. The differential of $E = -\alpha\kappa(d(\cdot, x_0))$ can be explicitly computed as

$$D_x E(v) = \left\langle \frac{\alpha\kappa'(d(x,x_0))}{d(x,x_0)} \cdot \overrightarrow{xx_0}, v \right\rangle_x \qquad \forall v \in \mathrm{T}_x\,\Omega,$$

with $D_{x_0}E(v) = 0$ for any $v \in \mathrm{T}_{x_0}\,\Omega$ since $\kappa'(0) = 0$; thus each $E \in \mathcal{E}$ is Fréchet-differentiable. The proof of the following lemma is postponed to the Appendix.

Lemma 3.5. *The set $\mathcal{E}$ defined in (21) is complete and separable in $\left(\mathbb{E}, |\cdot|_{1,\infty}\right)$.*

3.1.2 Well-posedness of dynamical systems

By now and for the rest of the paper, $\mathcal{E}$ is an admissible set of energies as defined in Definition 3.1, and Θ its associated set of transitions given by $\Theta = \mathcal{G}_{\mathcal{F}}(\mathcal{E})$. We now translate the well-posedness results of mutational analysis to this particular choice of transitions. Accordingly, a curve $y \in \mathrm{AC}(\mathbb{R}^+;\Omega)$ is mutable at $t \geqslant 0$ if there exists $E \in \mathcal{E}$ such that $h \mapsto y_{t+h}$ is approximated at order one by the gradient flow $\vartheta := \mathcal{G}_{\mathcal{F}}(E)$ issued from y_t, i.e.

$$\limsup_{h \searrow 0} \frac{d\left(\vartheta(,y_t), y_{t+h}\right)}{h} = 0.$$

Proposition 3.6. *Let $(E_s)_{s\in[0,T]} \in L^1(0,T;\mathcal{E})$. Then for any $x \in \Omega$, there exists a unique curve $y \in \mathrm{AC}([0,T];\Omega)$ such that $y_0 = x$ and $\mathcal{G}_{\mathcal{F}}(E_s) \in \mathring{y}_s$ for a.e. $s \in [0,T]$.*

Proof. We check that [FL23a, Proposition 2.22] applies. The parameter denoted $\gamma(\vartheta)$ corresponds to a bound on $\liminf_{h\searrow 0} \frac{d(\vartheta(h,x),0_\Omega)-d(x,0_\Omega)}{h(1+d(x,0_\Omega))}$, which is given by $|E|_{1,\infty}$ in our setting. Hence the function Γ of the reference can be taken as $s \mapsto |E_s|_{1,\infty}$, which is L^1 by assumption. Item (i) of the assumptions is satisfied with $\vartheta_0 = 0_\Theta$. Item (ii) is satisfied with $A_R \equiv 0$ and $B_R(s) := |E_s|_{1,\infty}$. The additional assumption to obtain uniqueness is trivially satisfied with $A_\rho \equiv 0$. □

We will consider differential inclusions with dynamics satisfying the following assumptions.

Assumption [A3.7]. Consider a set-valued map $F : \Omega \rightrightarrows \mathcal{E}$ such that

(a) for any $x \in \Omega$, the set $F(x)$ is nonempty and compact with respect to $|\cdot|_{1,\infty}$,

(b) F has sublinear growth, i.e. there exists a constant $C_f \geqslant 0$ such that

$$\sup_{E\in F(x)} |E|_{1,\infty} \leqslant C_f\,(1 + d(x,0_\Omega)) \qquad \forall x \in \Omega\,.$$

(c) F is locally Lipschitz in the Hausdorff distance, i.e. for any compact $B \subset \Omega$, there exists a constant $\mathrm{Lip}(F)_B \geqslant 0$ such that $d_H(F(x),F(y)) \leqslant \mathrm{Lip}(F)\,d(x,y)$ for any $x, y \in B$.

Under the assumption **[A3.7]**, we get the following corollary of Proposition 2.20.

Proposition 3.8. *Assume that F satisfies **[A3.7]**. Then for any $T > 0$ and $x \in \Omega$, there exists a solution $y \in \mathrm{AC}(\mathbb{R}^+;\Omega)$ of the mutational inclusion*

$$\mathring{y}_t \cap \mathcal{G}_{\mathcal{F}}(F(y_t)) \neq \emptyset \text{ for a.e. } t \geqslant 0, \text{ and} \qquad y_0 = x. \tag{22}$$

and a measurable selection $(E_s)_{s\in[0,T]} \in L^1(0,T;\mathcal{E})$ such that $E_s \in F(y_s)$ and $\mathcal{G}_{\mathcal{F}}(E_s) \in \mathring{y}_s$ for a.e. $s \in [0,T]$. Moreover,

$$d(y_t,x) \leqslant tC_f\,(1 + d(x,0_\Omega))\,e^{tC_f} \qquad \forall t \geqslant 0.$$

If F is single-valued, the solution is unique.

Proof. By Lemma 3.3, Θ is isometric to $\mathcal{E}$, which is assumed complete and separable. Choosing transitions as gradient flows of concave functions yields that $\alpha(\vartheta) \leqslant 0$ for any $\vartheta \in \Theta$. The constant $\beta(\vartheta)$ in Definition 2.16 is bounded from above by $|E|_{1,\infty}$ if $\vartheta = \mathcal{G}_{\mathcal{F}}(E)$, so that the assumption **[A3.7]** directly implies **[A2.19]** for $f(x) \coloneqq \mathcal{G}_{\mathcal{F}}(F(x))$. Applying Proposition 2.20, the existence follows. If F has single values, the uniqueness follows from the Cauchy-Lipschitz estimate [Aub99, Proposition 1.3.2] and a Grönwall lemma. □

3.2 Variational characterization

The main result of this section is to characterize the solutions of the dynamical system (22) by evolutionary variational inequalities. This will be used to pass to the limit in the relaxation result of the next section.

Proposition 3.9. *Let $F : \Omega \to \mathcal{E}$ be a dynamic satisfying **[A3.7]**. Then a curve $y \in \mathrm{AC}([0,T];\Omega)$ is a solution of the inclusion (22) if and only if there exists a measurable selection $(E_s)_{s\in[0,T]}$ of $s \mapsto F(y_s)$ such that for almost any $s \in [0,T)$, there holds*

$$\frac{d}{ds}\frac{d^2(y_s,z)}{2} \leqslant E_s(y_s) - E_s(z) \qquad \forall z \in \Omega\,. \tag{23}$$

An important particularity of (23) is that the energy E_s varies with the parameter s, which implies that the flow governed by these "EVI" does not generally possess a gradient structure.

The proof of Proposition 3.9 relies on the following two Lemmata.

Lemma 3.10. *Let $(y_s)_{s\in[0,T]}$ be a solution of (22), and $(E_s)_{s\in[0,T]} \in L^1(0,T;\mathcal{E})$ be such that $E_s \in F(y_s)$ and $\mathcal{G}_{\mathcal{F}}(E_s) \in \mathring{y}_s$ for a.e. s. Then (23) is satisfied.*

Proof. For convenience, denote $\vartheta_s \coloneqq \mathcal{G}_{\mathcal{F}}(E_s)$. Let $s \in [0,T)$ such that $\vartheta_s \in \mathring{y}_s$, i.e. such that $d(y_{s+h}, \vartheta(h,y_s)) = o(h)$. Then, for any $z \in \Omega$, the formula (13) yields

$$\lim_{h\to 0}\frac{d^2(y_{s+h},z) - d^2(y_s,z)}{2h} = \lim_{h\to 0}\frac{d^2(\vartheta_s(h,y_s),z) - d^2(y_s,z)}{2h} = -\left\langle \vartheta_s(\cdot,y_s)_0^+, \overrightarrow{y_s z}\right\rangle_{y_s}.$$

By definition of gradient flow, $\vartheta_s(\cdot,y_s)_0^+ = \nabla_{y_s} E_s$. Since $\langle \nabla_{y_s} E_s, v\rangle_{y_s} \geqslant D_{y_s}E_s(v)$ for any $v \in \mathrm{T}_{y_s}\Omega$ by Definition 2.8 of the metric gradient, and E_s is concave, we get

$$\frac{d}{ds}\frac{d^2(y_s,z)}{2} = -\left\langle \nabla_{y_s}E_s, \overrightarrow{y_s z}\right\rangle_{y_s} \leqslant -D_{y_s}E_s(\overrightarrow{y_s z}) \leqslant E_s(y_s) - E_s(z).$$

Since z is arbitrary and $(y_s)_s$ mutable almost everywhere, the claim holds. □

Lemma 3.11. *For $i \in \{1,2\}$, let $E^i \in L^1(0,T;\mathcal{E})$, and $(y^i_s)_{s\in[0,T]} \in \mathrm{AC}([0,T];\Omega)$ be such that for almost any $s \in [0,T)$,*

$$\frac{d}{ds}\frac{d^2(y^i_s,z)}{2} \leqslant E^i_s(y^i_s) - E^i_s(z) \qquad \forall z \in \Omega\,.$$

Then for all $t \in [0,T]$,

$$\frac{d^2(y^1_t,y^2_t)}{2} \leqslant \frac{d^2(y^1_0,y^2_0)}{2} + \int_{s=0}^{t}\left[D_{y^2_s}E^1_s(\overrightarrow{y^2_s y^1_s}) - D_{y^2_s}E^2_s(\overrightarrow{y^2_s y^1_s})\right] ds. \tag{24}$$

Proof. The argument is adapted from [AGS05, Corollary 4.3.3], in which $s \mapsto E^i_s$ are constant in time. Since $E^i \in L^1(0,T;\mathcal{E})$ are Lebesgue-measurable curves valued in the Banach space $\mathbb{E}$, by [Hyt+16, Theorem 2.3.4], there exists a set $A_i \subset [0,T]$ of full Lebesgue measure such that any $t \in A_i$ is a Lebesgue point for E^i. In particular,

$$\lim_{h\searrow 0}\frac{1}{h}\int_{s=t-\alpha h}^{t+(1-\alpha)h}\left|E^i_t - E^i_s\right|_{1,\infty} ds = 0 \qquad \forall \alpha \in [0,1] \text{ and } t \in A_i.$$

Moreover, by the one-dimensional Rademacher theorem [AKP23, Proposition 13.9], both curves y^i are differentiable in the sense of Definition 2.6 at all points of some sets $A_{2+i} \subset [0,1]$ of full Lebesgue measure. Consider then $A_0 \subset [0,T]$ a conegligible subset such that $t \mapsto d^2(y_t^1, y_t^2)$ is differentiable for all $t \in A_0$, and let $t \in (0,T) \cap \bigcap_{i\in\{0,\cdots,4\}} A_i$. Integrating the evolutionary variational inequality (23) over $[t-h,t]$ and $[t,t+h]$ for a sufficiently small $h>0$, there holds

$$\begin{aligned}
\frac{d^2(y_t^1,z) - d^2(y_{t-h}^1,z)}{2} &\leqslant \int_{s=t-h}^{t} \big(E_s^1(y_s^1) - E_s^1(z)\big)\, ds \qquad &\forall z \in \Omega,\\
\frac{d^2(z,y_{t+h}^2) - d^2(z,y_t^2)}{2} &\leqslant \int_{s=t}^{t+h} \big(E_s^2(y_s^2) - E_s^2(z)\big)\, ds \qquad &\forall z \in \Omega\,.
\end{aligned}$$

Recall that by definition of $\mathcal{E}$, E_s^1 is Lipschitz with constant $\left|E_s^1\right|_{1,\infty}$. Hence

$$\begin{aligned}
&\int_{s=t-h}^{t} \big(E_s^1(y_s^1) - E_s^1(z)\big)\, ds \leqslant \int_{s=t-h}^{t} \Big(E_s^1(y_t^1) - E_s^1(z) + \left|E_s^1\right|_{1,\infty} d(y_s,y_t)\Big)\, ds\\
&\leqslant h\left(E_t^1(y_t^1) - E_t^1(z)\right) + \int_{s=t-h}^{t} \Big(\left|E_s^1\right|_{1,\infty} d(y_s,y_t) + \left|E_s^1 - E_t^1\right|_{1,\infty} d(z,y_t^1)\Big)\, ds.
\end{aligned}$$

Dividing by $h>0$ and taking the limit sup in $h \searrow 0$, the second term vanishes. Following the same reasoning with $i=2$, we obtain that for all $z \in \Omega$,

$$\limsup_{h\searrow 0} \frac{d^2(y_t^1,z) - d^2(y_{t-h}^1,z)}{2} \leqslant E_t^1(y_t^1) - E_t^1(z), \tag{25}$$

$$\limsup_{h\searrow 0} \frac{d^2(z,y_{t+h}^2) - d^2(z,y_t^2)}{2} \leqslant E_t^2(y_t^2) - E_t^2(z). \tag{26}$$

As $t \in A_3$, the curve y^1 is differentiable in the sense of Definition 2.6 at t, and there holds

$$\limsup_{h\searrow 0} \frac{d^1(y_t^1,z) - d^1(y_{t-h}^1,z)}{2h} = D_{y_t^1} \frac{d^2(\cdot,z)}{2}((y^1)_t^-) = -\left\langle (y^1)_t^-, \overrightarrow{y_t^1 z} \right\rangle_{y_t^1}.$$

We may choose $z = (1-\alpha)y_t^1 \oplus \alpha y_t^2$ for $\alpha \in (0,1]$, which implies $\frac{1}{d(y_t^1,z)} \cdot \overrightarrow{y_t^1 z} = \frac{\alpha}{d(y_t^1,y_t^2)} \cdot \overrightarrow{y_t^1 y_t^2}$. Using the positive homogeneity of the metric scalar product, we deduce

$$\limsup_{h\searrow 0} \frac{d^2(y_t^1,z) - d^2(y_{t-h}^1,z)}{2h} = \alpha \limsup_{h\searrow 0} \frac{d^2(y_t^1,y_t^2) - d^2(y_{t-h}^1,y_t^2)}{2h}.$$

On the other hand, E^1 is directionally differentiable, so that $E_t^1((1-\alpha)y_t^1 \oplus \alpha y_t^2) - E_t^1(y_t^1) = \alpha D_{y_t^1} E^1(\overrightarrow{y_t^1 y_t^2}) + o(\alpha)$. Dividing by $\alpha > 0$ and letting $\alpha \searrow 0$, we obtain from (25) that

$$\limsup_{h\searrow 0} \frac{d^2(y_t^1,y_t^2) - d^2(y_{t-h}^1,y_t^2)}{2h} \leqslant -D_{y_t^1} E_t^1(\overrightarrow{y_t^1 y_t^2}),$$

and symmetrically

$$\limsup_{h\searrow 0} \frac{d^2(y_{t+h}^2,y_t^1) - d^2(y_t^2,y_t^1)}{2h} \leqslant -D_{y_t^2} E_t^2(\overrightarrow{y_t^2 y_t^1}).$$

By [AGS05, Lemma 4.3.4], there holds that

$$\begin{aligned}
\frac{d}{dt}\frac{d^2(y_t^1,y_t^2)}{2} &\leqslant \limsup_{h\searrow 0} \frac{d^2(y_t^1,y_t^2) - d^2(y_{t-h}^1,y_t^2)}{2h} + \limsup_{h\searrow 0} \frac{d^2(y_t^1,y_{t+h}^2) - d^2(y_t^1,y_t^2)}{2h}\\
&\leqslant \underbrace{-D_{y_t^1} E_t^1(\overrightarrow{y_t^1 y_t^2}) - D_{y_t^2} E_t^1(\overrightarrow{y_t^2 y_t^1})}_{\mathcal{A}} + D_{y_t^2} E_t^1(\overrightarrow{y_t^2 y_t^1}) - D_{y_t^2} E_t^2(\overrightarrow{y_t^2 y_t^1}).
\end{aligned}$$

As E_t^1 is concave, the term $\mathcal{A}$ is nonpositive. Integrating over time yields (24). □

The proof of Proposition 3.9 can now be completed as follows.

Proof of Proposition 3.9. On the one hand, Lemma 3.10 shows that any solution in the sense of mutations satisfies the EVI characterization. On the other hand, assume that y^1 satisfies the EVI (23) for some $(E_s)_{s\in[0,T]} \in L^1(0,T;\mathcal{E})$ with $E_s \in F(y^1_s)$ a.e.. Applying Lemma 3.11 with the constant curve $[0,T] \ni s \mapsto x$, one sees that

$$\frac{d^2(y^1_t,x)}{2} \leqslant 0 + \int_0^t D_x E_s(\overrightarrow{xy^1_s})ds \leqslant \int_0^t |E_s|_{1,\infty}\, d(x,y^1_s)ds \leqslant \int_0^t C_f\left(1+d(0_\Omega,y^1_s)\right) d(y^1_s,x)ds,$$

hence by a Grönwall Lemma, y^1 lies in a compact subset B of Ω on the time interval $[0,T]$. Up to redefining $(E_s)_{s\in[0,T]}$ on a negligible subset, it holds that $E_s \in \bigcup_{z\in B} F(z)$, with the latter set being compact by [AB06, Lemma 17.8].

By Proposition 3.6, there exists an absolutely continuous curve y^2 issued from y^1_0, that satisfies (22) almost everywhere and such that $\mathcal{G}_{\mathcal{F}}(E_s) \in \mathring{y}^2_s$ for a.e. $s \in [0,T]$. Applying Lemma 3.10, y^2 also satisfies (23) for the curve $(E_s)_s$. Then, by Lemma 3.11,

$$\frac{d^2(y^1_t,y^2_t)}{2} \leqslant \frac{d^2(y^1_0,y^2_0)}{2} + \int_{s=0}^t \left[D_{y^2_s}E_s(\overrightarrow{y^2_s y^1_s}) - D_{y^2_s}E_s(\overrightarrow{y^2_s y^1_s})\right] ds = 0,$$

so that $y^1 \equiv y^2$, hence y^1 is a solution of the inclusion (22). □

4 Relaxation theorem

This section contains the main result of the paper. We consider a differential inclusion as defined in Section 3.1, and aim to characterize the closure of the set of trajectories issued from a single point. We first introduce a suitable convexification of dynamics, then study its trajectories, and conclude by an interpretation of the "convexification" in terms of right derivatives in the tangent cones. It will be convenient to use a distance between probability measures, which will be taken as the Kantorovich-Rubinstein distance.

4.1 Kantorovich-Rubinstein distance

In order to be self-contained, we briefly introduce some definitions, and refer the reader to [AGS05, Chap. 5-7], [Vil09, Chap. 6] or [San15] for further details.

Let (X,d_X) be a Polish space. Denote by $\mathcal{M}_+(X)$ the set of Borel nonnegative measures over X, and by $\mathscr{P}(X)$ the subset of probability measures. If $m : (X,d_X) \to (Y,d_Y)$ is a measurable map between Polish spaces, the pushforward $m_\#\mu$ of $\mu \in \mathcal{M}_+(X)$ is the element of $\mathcal{M}_+(Y)$ given by $(m_\#\mu)(A) \coloneqq \mu\left(m^{-1}(A)\right)$ for all measurable $A \subset Y$. The canonical projections from $(X\times X, d_{X\times X})$ are denoted by π_1,π_2, i.e. $\pi_1(x_1,x_2)=x_1$ and $\pi_2(x_1,x_2)=x_2$.

Let $\mathcal{M}_{+,1}(X)$ be the subset of $\mu \in \mathcal{M}_+(X)$ such that $\int_X d_X(0_\Omega,x)d\mu(x) < \infty$ for some $0_\Omega \in X$, and $\mathcal{M}_1(X)$ the set of differences of elements of $\mathcal{M}_{+,1}$. The space $\mathscr{P}_1(X) \coloneqq \mathscr{P}(X) \cap \mathcal{M}_{+,1}(X)$ is Polish when endowed with the Kantorovich-Rubinstein norm

$$|\mu|_{\mathrm{KR}} \coloneqq \sup_{\varphi \text{ 1-Lipschitz}} \int_X \varphi(x)d\mu(x). \tag{27}$$

We will very often use the equivalent formulation of $|\cdot|_{\mathrm{KR}}$ as a transport distance between the positive and negative parts of μ. Namely, assume that $\mu = \mu^+ - \mu^-$ for two nonnegative measures $\mu^\pm$ of equal mass. Any measure $\eta = \eta(dx,dy) \in \mathcal{M}_+(X^2)$ that has first marginal $\pi_{x\#}\eta = \mu^-$, and second marginal $\pi_{y\#}\eta = \mu^+$, furnishes a bound on $|\mu|_{\mathrm{KR}}$ by

$$|\mu|_{\mathrm{KR}} = \sup_{\varphi \text{ 1-Lipschitz}} \int_{X^2} \left[\varphi(y)-\varphi(x)\right] d\eta(x,y) \leqslant \int_{X^2} d(x,y)d\eta(x,y).$$

Such measures are called *transport plans*. In fact, the Kantorovich-Rubinstein distance is obtained by passing to the infimum over such transport plans. As X is separable, one shows that finite combinations of Dirac masses are dense in $\mathscr{P}_1(X)$ with respect to $|\cdot|_{\mathrm{KR}}$.

4.2 Definition of the convexified system

Recall that $\mathcal{E}$ is an admissible set of energies, built as a subset of the Banach space $\mathbb{E}$ of Definition 2.12. If $A \subset \mathbb{E}$, let $\overline{\mathrm{conv}}\, A$ be the smallest convex closed subset of $\mathbb{E}$ containing A.

Definition 4.1. Let $F : \Omega \rightrightarrows \mathcal{E}$. The relaxed dynamics $\overline{\mathrm{co}}\, F$ is defined as

$$\overline{\mathrm{co}}\, F : \Omega \rightrightarrows \overline{\mathrm{conv}}\, \mathcal{E}, \qquad \overline{\mathrm{co}}\, F(x) \coloneqq \overline{\mathrm{conv}}\, F(x) \qquad \forall x \in \Omega\,. \tag{28}$$

We first check that the relaxed differential system driven by the dynamics $\overline{\mathrm{co}}\, F$ admits solutions. To this aim, we show that $\overline{\mathrm{conv}}\, \mathcal{E}$ is itself an admissible set of energies, and apply Proposition 3.8. To ease the argument, we parametrize the convex hull by probability measures on the Banach set $\mathbb{E}$. If $\mu \in \mathscr{P}_1(\mathbb{E})$, we define $\mathrm{Bary}_{\mathbb{E}}(\mu) : \Omega \to \mathbb{R}$ as the function

$$\mathrm{Bary}_{\mathbb{E}}(\mu)(x) \coloneqq \int_{\mathbb{E}} E(x)\, d\mu(E).$$

Lemma 4.2. *The set $\overline{\mathrm{conv}}\, \mathcal{E}$ is an admissible set of energies in the sense of Definition 3.1. The relaxed dynamics can be parametrized as*

$$\overline{co}\, F(x) = \mathrm{Bary}_{\mathbb{E}}(\mathscr{P}_1(F(x))) = \{\mathrm{Bary}_{\mathbb{E}}(\mu) \mid \mu \in \mathscr{P}_1(F(x))\}\,.$$

*Moreover, if F satisfies **[A3.7]**, then so does $\overline{co}\, F$.*

Proof. Let $\mu \in \mathscr{P}_1(\mathbb{E})$. The function $\mathrm{Bary}_{\mathbb{E}}(\mu)$ is real-valued and Lipschitz-continuous; indeed, for $x, y \in \Omega$, there holds

$$|\mathrm{Bary}_{\mathbb{E}}(\mu)(y) - \mathrm{Bary}_{\mathbb{E}}(\mu)(x)| = \left| \int_{\mathcal{E}} (E(y) - E(x)) d\mu(E) \right| \leqslant d(x,y) \int_{\mathcal{E}} |E|_{1,\infty}\, d\mu(E),$$

where we used that $\mathrm{Lip}(E) = |E|_{1,\infty}$ for any $E \in \mathcal{E}$. The integral on the right hand-side is finite since $\mu \in \mathscr{P}_1(\mathbb{E})$. If μ is concentrated on concave functions, then $\mathrm{Bary}_{\mathbb{E}}(\mu)$ is concave, hence belongs to $\mathbb{E}$. Moreover, $\mathrm{Bary}_{\mathbb{E}}(\cdot)$ is 1-Lipschitz from $(\mathscr{P}_1(\mathbb{E}), |\cdot|_{\mathrm{KR}})$ into $(\mathbb{E}, |\cdot|_{1,\infty})$. Indeed, pick $\mu, \mu' \in \mathscr{P}_1(\mathbb{E})$, and $\eta \in \mathscr{P}(\mathbb{E}^2)$ satisfying the marginal constraints $\pi_{1\#}\eta = \mu$ and $\pi_{2\#}\eta = \mu'$. By Jensen's inequality, there holds

$$\left|\mathrm{Bary}_{\mathbb{E}}(\mu) - \mathrm{Bary}_{\mathbb{E}}(\mu')\right|_{1,\infty} = \left| \int_{\mathbb{E}^2} (E - E') d\eta(E, E') \right|_{1,\infty} \leqslant \int_{\mathbb{E}^2} \left|E - E'\right|_{1,\infty} d\eta(E, E').$$

Taking the infimum over η, we get that $|\mathrm{Bary}_{\mathbb{E}}(\mu) - \mathrm{Bary}_{\mathbb{E}}(\mu')|_{1,\infty} \leqslant |\mu - \mu'|_{\mathrm{KR}}$. In addition, there holds $\overline{\mathrm{conv}}\, \mathcal{E} = \overline{\mathrm{Bary}_{\mathbb{E}}(\mathscr{P}_1(\mathcal{E}))}$, where

$$\mathrm{Bary}_{\mathbb{E}}(\mathscr{P}_1(\mathcal{E})) \coloneqq \{\mathrm{Bary}_{\mathbb{E}}(\mu) \mid \mu \in \mathscr{P}_1(\mathbb{E}) \text{ is supported on } \mathcal{E}\}\,.$$

Indeed, $\mathrm{Bary}_{\mathbb{E}}(\mathscr{P}_1(\mathcal{E}))$ is a convex subset of $\mathbb{E}$, and contains $\mathcal{E}$ as the barycenters of Dirac masses. Hence $\overline{\mathrm{conv}}\, \mathcal{E} \subset \overline{\mathrm{Bary}_{\mathbb{E}}(\mathscr{P}_1(\mathcal{E}))}$. On the other hand, using the density of finite combinations of Dirac masses, each $\mathrm{Bary}_{\mathbb{E}}(\mu)$ is a convex combination of elements of $\mathcal{E}$, hence belongs to conv $\mathcal{E}$ (see also Proposition 1.2.12 in [Hyt+16]).

The set $\overline{\mathrm{conv}}\, \mathcal{E}$ is complete by definition. It is separable since $\mathscr{P}_1(\mathcal{E})$ is separable in the topology induced by $|\cdot|_{\mathrm{KR}}$ [Vil09, Theorem 6.18], and $\mathrm{Bary}_{\mathbb{E}}(\cdot)$ is continuous. Hence it satisfies all points of the definition of an admissible set of energies.

We turn to the properties of the relaxed dynamics. By [Vil09, Remark 6.19], if $K \subset \mathbb{E}$ is compact, then $\mathscr{P}_1(K)$ is itself compact. Hence, for all $x \in \Omega$,

$$\overline{\mathrm{co}}\, F(x) = \mathrm{Bary}_{\mathbb{E}}\left(\mathscr{P}_1(F(x))\right) \tag{29}$$

is a compact subset of $\mathbb{E}$. Additionally, the mapping $K \mapsto \mathscr{P}_1(K)$ is 1-Lipschitz from nonempty compact sets endowed with the Hausdorff distance to $(\mathscr{P}_1(\mathbb{E}), |\cdot|_{\mathrm{KR}})$. Indeed, let K and K' be nonempty compacts, and $\mu \in \mathscr{P}_1(K)$. By a measurable selection theorem, for instance [AB06], one can build a measurable function $g : K \to K'$ such that $d(k, g(k)) \leqslant d_H(K, K')$ for any $k \in K$. Denote then $\mu' \coloneqq g_{\#}\mu \in \mathscr{P}(K')$. There holds

$$\left|\mu - \mu'\right|_{\mathrm{KR}} \leqslant \int_{x\in K} d(x, g(x)) d\mu(x) \leqslant d_H(K, K').$$

Switching K and K', we get that $K \mapsto \mathscr{P}_1(K)$ is 1-Lipschitz. As the barycenter is 1-Lipschitz, the composition (29) satisfies the compactness of images, growth and local Lipschitz behaviour required by **[A3.7]**. □

Remark 4.3 (Measures as control). Let $(y_s)_s \in \mathrm{AC}([0,T];\Omega)$ be a solution of the relaxed dynamical system. By Proposition 3.8, there exists $(E_s)_{s\in[0,T]} \in L^1(0,T;\overline{\mathrm{conv}}\,\mathcal{E})$ such that $\mathcal{G}_{\mathcal{F}}(E_s) \in \mathring{y}_s \cap \mathcal{G}_{\mathcal{F}}(\overline{\mathrm{co}}\, F(y_s))$ for a.e. $s \in [0,T]$. Since $\overline{\mathrm{co}}\, F$ has a linear growth, the curve y is bounded. As $F(\cdot)$ is upper semi-continuous with compact values, the set $A \coloneqq \bigcup_{t\in[0,T]} F(y_s)$ is again a compact subset of $\mathcal{E}$. The set $\mathscr{P}_1(A)$ is itself compact with respect to $|\cdot|_{\mathrm{KR}}$, and the restriction of $\mathrm{Bary}_{\mathbb{E}}(\cdot)$ to this set is continuous. Hence the correspondence $E \mapsto \{\mu \in \mathscr{P}_1(A) \mid \mathrm{Bary}_{\mathbb{E}}(\mu) = E\}$ is upper-semicontinuous with nonempty compact values, and one can apply a measurable selection theorem as [AB06, p. 18.13] to obtain a curve $\mu \in L^1(0,T;\mathscr{P}_1(\mathcal{E}))$ such that $\mathrm{Bary}_{\mathbb{E}}(\mu_s) = E_s$ for almost every $s \in [0,T]$.

4.3 The relaxation theorem

This section aims at proving the following result.

Theorem 4.4. *Consider a set $\mathcal{E}$ as in Definition 3.1, and let $F : \Omega \rightrightarrows \mathcal{E}$ satisfy the assumption **[A3.7]**. For any $x \in \Omega$ and $T > 0$, the closure in $\mathrm{AC}([0,T];\Omega)$ of the set of solutions of*

$$\mathring{y}_s \cap \mathcal{G}_{\mathcal{F}}(F)(y_s) \neq \emptyset, \qquad y_0 = x \tag{30a}$$

is given by the set of solutions of

$$\mathring{y}_s \cap \mathcal{G}_{\mathcal{F}}(\overline{\mathrm{co}}\, F)(y_s) \neq \emptyset, \qquad y_0 = x. \tag{30b}$$

Moreover, the latter set is compact in $\mathrm{AC}([0,T];\Omega)$, endowed with $d_{\mathrm{AC}}(y,y') = \sup_{t\in[0,T]} d(y_t, y'_t)$.

The proof of Theorem 4.4 goes by double inclusion. Lemma 4.5 below shows that any Cauchy sequence of solutions of the original inclusion (30a) converges to a solution of the relaxed system (30b). Conversely, Proposition 4.8 constructs approximations of a given trajectory of the relaxed system by solutions of the original one. Once closedness is established, compactness of the set of trajectories follows easily from the Arzelà-Ascoli theorem.

Lemma 4.5. *Consider the assumptions of Theorem 4.4. Let $(y^n)_n \subset \mathrm{AC}([0,T];\Omega)$ be a sequence of solutions of the differential inclusion (22). Assume that $(y^n)_n$ converges uniformly towards some $y \in \mathrm{AC}([0,T];\Omega)$. Then y is a solution of the relaxed dynamical system (30b).*

Proof. The argument begins as in [BF24]. Let $B \subset \Omega$ be a closed ball containing the graph of all solutions of the relaxed system (30b) issued from some y^n_0 or y^0. Denote $\mathrm{Lip}(F)_B$ a local Lipschitz constant of F on B. For each $n \in \mathbb{N}$, let $\varphi^n \in L^1(0,T;\mathcal{E})$ be a measurable

selection of $s \rightrightarrows F(y_s^n)$ such that $\varphi_s^n \in \mathring{y}_s^n$ for almost every $s \in [0,T]$. As F has sublinear growth, $\|\varphi^n\|_{L^1(0,T;\mathcal{E})} \leqslant C$ for some C that is independent of n. Moreover, each φ^n is valued a.e. in $X := \bigcup_{z\in B} F(z) \subset \mathbb{E}$. The compact union of the weakly compact sets $\overline{\mathrm{co}}\, F(y)$ is again weakly compact, so by [DU77, Proposition 7], the set $L^1(0,T;X)$ is itself weakly compact. In consequence, up to a non relabelled subsequence, φ^n converges weakly in the Banach space $L^1(0,T;\mathcal{E})$ towards some φ that is valued in $X \subset \overline{\mathrm{conv}}\,\mathcal{E}$ for a.e. $s \in [0,T]$.

We now show that $\varphi_s \in \overline{\mathrm{co}}\, F(y_s)$ a.e.. Let L be a linear continuous form over $\mathbb{E}$, and $p \in \mathcal{C}^\infty([0,T];\mathbb{R}^+)$. For each $n \in \mathbb{N}$, using the local Lipschitz character of the dynamic,

$$\begin{aligned}\int_0^T p_s L\left(\varphi_s^n\right) ds &\leqslant \int_0^T p_s \max_{b\in F(y_s^n)} L\left(b\right) ds \\ &\leqslant \int_0^T p_s \max_{b\in F(y_s)} L\left(b\right) ds + \sup_{t\in[0,T]} |p_t| \times \|L\| \mathrm{Lip}(F)_B \times \sup_{s\in[0,T]} d(y_s, y_s^n).\end{aligned}$$

As $\psi \mapsto \int_{s=0}^T p_s L\left(\psi_s\right) ds$ lies in the dual of $L^1(0,T;\mathbb{E})$, we may pass to the limit in $n \to \infty$ in the above inequality. Since $p \in \mathcal{C}^\infty([0,T];\mathbb{R}^+)$ and L are arbitrary, there exists a conegligible set $I \subset [0,T]$ such that $\varphi_s \in \overline{\mathrm{conv}}\, F(y_s) = \overline{\mathrm{co}}\, F(y_s)$ for every $s \in I$ [AF90, Th. 2.4.2].

To conclude, we only have to show that $\varphi_s \in \mathring{y}_s$ for a.e. $s \in [0,T]$. By Proposition 3.9, this is equivalent to the evolutionary variational inequalities (23). Let $z \in \Omega$ be fixed and $0 \leqslant s < t \leqslant T$. For any n,

$$\begin{aligned}\frac{d^2(y_t^n, z) - d^2(y_s^n, z)}{2} &= \int_{\tau=s}^t \frac{d}{d\tau}\frac{d^2(y_\tau^n, z)}{2} d\tau \leqslant \int_{\tau=s}^t \left[\varphi_\tau^n(y_\tau^n) - \varphi_\tau^n(z)\right] d\tau \\ &\leqslant \int_{\tau=s}^t \left[\varphi_\tau^n(y_\tau) - \varphi_\tau^n(z)\right] d\tau + \|\varphi^n\|_{L^1(s,t;\mathbb{E})} \sup_{\tau\in[s,t]} d(y_\tau^n, y_\tau).\end{aligned}$$

As the functional $\psi \mapsto \int_{\tau=s}^t \left(\psi_\tau(y_\tau) - \psi_\tau(z)\right) d\tau$ is linear and continuous from $L^1(0,T;\mathbb{E})$ to $\mathbb{R}$, z and $(y_\tau)_\tau$ being fixed and bounded, it is weakly continuous. Consequently, we may pass to the limit in $n \to \infty$ in the previous inequality to get that for all $0 \leqslant s < t \leqslant T$ and $z \in \Omega$,

$$\frac{d^2(y_t, z) - d^2(y_s, z)}{2} \leqslant \int_{\tau=s}^t \left[\varphi_\tau(y_\tau) - \varphi_\tau(z)\right] d\tau. \tag{31}$$

We now have to get back to a differential version in time, while paying attention to the fact that for a.e. time, the EVI must be satisfied for all z. Let $I^0 \subset [0,T]$ be the set of Lebesgue points of $\tau \mapsto \varphi_\tau \in \mathbb{E}$, which is of full measure by [Hyt+16, Theorem 2.3.4]. Fix $s \in I^0$ and $z \in \Omega$. Recall that φ_τ is an equivalence class of functions up to a constant: for the following computation, we consider the element of the class such that $\varphi_\tau(0_\Omega) = 0$. Then $|\varphi_\tau(x)| \leqslant |\varphi_\tau|_{1,\infty}\, d(x, 0_\Omega)$ for any τ, since $\mathrm{Lip}(\varphi_\tau) = |\varphi_\tau|_{1,\infty}$. Consequently,

$$\begin{aligned}\int_{\tau=s}^t |\varphi_\tau(y_\tau) - \varphi_s(y_s)|\, d\tau &\leqslant \int_{\tau=s}^t |\varphi_\tau(y_\tau) - \varphi_\tau(y_s)|\, d\tau + \int_{\tau=s}^t |\varphi_\tau(y_s) - \varphi_s(y_s)|\, d\tau \\ &\leqslant \mathrm{Lip}(y)(t-s) \int_{\tau=s}^t |\varphi_\tau|_{1,\infty}\, d\tau + \int_{\tau=s}^t |\varphi_\tau - \varphi_s|_{1,\infty}\, d(y_s, 0_\Omega) d\tau.\end{aligned}$$

As s is also a Lebesgue point of $\tau \mapsto |\varphi_\tau|_{1,\infty}$, dividing by $t - s > 0$ and sending t to s, we get

$$\lim_{t\to s} \frac{1}{t-s} \int_{\tau=s}^t |\varphi_\tau(y_\tau) - \varphi_s(y_s)|\, d\tau = 0.$$

The same computation shows that $\lim_{t\to s} \frac{1}{t-s} \int_{\tau=s}^t |\varphi_\tau(z) - \varphi_s(z)|\, d\tau = 0$. On the other hand, as $y(\cdot)$ is absolutely continuous, the function $s \mapsto d^2(y_s, z)$ is differentiable at any point of a set

$I^1 \subset [0,T]$ of full Lebesgue measure. Dividing by $t-s$ and passing to the limit in $t \searrow s$ for $s \in I^0 \cap I^1$ in (31), we obtain that

$$\frac{d}{ds}\frac{d^2(y_s,z)}{2} \leqslant \lim_{t\searrow s} \frac{1}{t-s}\int_{\tau=s}^{t} [\varphi_\tau(y_\tau) - \varphi_\tau(z)]\, d\tau = \varphi_s(y_s) - \varphi_s(z).$$

This is the EVI (23), so that by Proposition 3.9, $\varphi_s \in \mathring{y}_s$, and we conclude. □

Let $\mu \in L^1(0,T;\mathscr{P}_1(\mathcal{E}))$ and $I \subset [0,T]$ be a nontrivial interval. For brevity, we denote by $\mathcal{L}_I \otimes \mu$ the measure on $[0,T]\times\mathcal{E}$ given by $\int_I \delta_s \otimes \mu_s ds$, characterized by

$$\int_{[0,T]\times\mathcal{E}} \varphi(s,E) d[\mathcal{L}_I \otimes \mu](s,E) = \int_I \int_{E\in\mathcal{E}} \varphi(s,E) d\mu_s(E) ds$$

for all $\varphi \in \mathcal{C}_b([0,T]\times\mathcal{E};\mathbb{R})$. The measure $\mathcal{L}_I \otimes \mu$ has mass $\mathcal{L}(I)$. This notation is consistent with the case where $\mu_s \equiv \mu'$ is a constant measure, in which case $\mathcal{L}_I \otimes \mu$ is given by the product measure $\mathcal{L}_I \otimes \mu'$. Using a measurable selection of optimal plans [AGS05, Lemma 12.4.7] , one has

$$\left|\mathcal{L}_I \otimes \mu - \mathcal{L}_I \otimes \mu'\right|_{\mathrm{KR}} \leqslant \int_{s\in I} \left|\mu_s - \mu'_s\right|_{\mathrm{KR}} ds.$$

Lemma 4.6 (Uniform weak limit)**.** *Let $I \subset [0,T]$ be a nontrivial closed interval, $K \subset \mathcal{E}$ and $B \subset \Omega$ be compact, and $y \in \mathrm{AC}(I;B)$. Let μ and $(\mu^n)_n$ be elements of $L^1(I;\mathscr{P}_1(K))$ such that*

$$\lim_{n\to\infty} |\mathcal{L}_I \otimes \mu - \mathcal{L}_I \otimes \mu^n|_{\mathrm{KR}} \underset{n\to\infty}{\longrightarrow} 0.$$

Then for any constant $C \geqslant 0$,

$$\sup_{\substack{y'\in\mathrm{AC}(I;B)\\ \mathrm{Lip}(y')\leqslant C}} \left| \int_I \left[D_{y_s}\mathrm{Bary}_{\mathbb{E}}(\mu_s)\left(\overrightarrow{y_s y'_s}\right) - D_{y_s}\mathrm{Bary}_{\mathbb{E}}(\mu^n_s)\left(\overrightarrow{y_s y'_s}\right)\right]\right| \underset{n\to\infty}{\longrightarrow} 0. \tag{32}$$

Proof. We change variables to apply the results of integration of Banach-valued functions. The space $\mathcal{C}(K\times B;\mathbb{R})$ is a Banach space when endowed with the sup norm. For each $s \in I$, let $\rho_s : K\times B \to \mathbb{R}$ be defined by

$$\rho_s(E,z) \coloneqq D_{y_s}E\left(\overrightarrow{y_s z}\right).$$

The function ρ_s is continuous in both variables, with in addition

$$\begin{aligned}\left|\rho_s(E,z) - \rho_s(E',z')\right| &\leqslant \left|D_{y_s}E\left(\overrightarrow{y_s z}\right) - D_{y_s}E'\left(\overrightarrow{y_s z}\right)\right| + \mathrm{Lip}(D_{y_s}E')d_{y_s}\left(\overrightarrow{y_s z}, \overrightarrow{y_s z'}\right)\\ &\leqslant \left|E-E'\right|_{1,\infty} d(y_s,z) + \left|E'\right|_{1,\infty} d(z,z')\end{aligned}$$

by definition of $|\cdot|_{1,\infty}$ and the estimate (5). It is in addition measurable from I to $\mathcal{C}(K\times B;\mathbb{R})$, as the pointwise limit of the continuous maps $s \mapsto (E((1-h)y_s \oplus hz) - E(y_s))/h$ when h goes to 0. Moreover,

$$\sup_{E\in K, z\in B} |D_{y_s}E\left(\overrightarrow{y_s z}\right)| \leqslant \sup_{E\in K, z\in B} |E|_{1,\infty}\, d(y_s,z) \leqslant \sup_{E\in K} |E|_{1,\infty}\, \mathrm{diam}\ B,$$

so that ρ is in $L^1(I;\mathcal{C}(K\times B;\mathbb{R}))$. Let η^n be an optimal transport plan between $\mathcal{L}_I\otimes\mu$ and $\mathcal{L}_I\otimes\mu^n$. Then

$$\int_{I\times\mathcal{E}} \rho_s(E,y'_s) d[\mathcal{L}_I\otimes\mu](E,s) - \int_{I\times\mathcal{E}} \rho_s(E,y'_s) d[\mathcal{L}_I\otimes\mu^n](E,s)$$

$$
\begin{aligned}
&= \int_{(I\times\mathcal{E})^2} \left[\rho_s(E, y'_s) - \rho_{s'}(E', y'_{s'})\right] d\eta^n((s,E),(s',E')) \\
&\leqslant \int_{(I\times\mathcal{E})^2} \left[\rho_s(E, y'_s) - \rho_{s'}(E, y'_s) + \left|E - E'\right|_{1,\infty} d(y_{s'}, y'_{s'}) + \left|E'\right|_{1,\infty} d(y'_{s'}, y'_s)\right] d\eta^n \\
&\leqslant \int_{I^2} \|\rho_s - \rho_{s'}\|_{\mathcal{C}(K\times B;\mathbb{R})}\, d(\pi_s, \pi_{s'})_\# \eta^n(s,s') + \widetilde{C}\, |\mathcal{L}_I \otimes \mu - \mathcal{L}_I \otimes \mu^n|_{\mathrm{KR}}\,, \qquad (33)
\end{aligned}
$$

with $\widetilde{C} \coloneqq \max(\operatorname{diam} B, |K|_{1,\infty} C)$. The last term in (33) goes to 0 by assumption, and we only have to show that the first term vanishes.

Each $\beta^n \coloneqq (\pi_s, \pi_{s'})_\# \eta^n(s,s')$ has both marginals equal to $\mathcal{L}_I$ and converges to $(id, id)_\# \mathcal{L}_I$ [Vil09, Theorem 5.20]. By [Hyt+16, Lemma 1.2.31], there exists $(\rho^m)_{m\in\mathbb{N}} \subset \mathcal{C}_c(I; \mathcal{C}(K\times B;\mathbb{R}))$ converging to ρ in the $L^1(I;\mathcal{C}(K\times B;\mathbb{R}))$ norm. Using the triangular inequality and the marginal conditions on β^n,

$$
\int_{I^2} \|\rho_s - \rho_{s'}\|_{\mathcal{C}(K\times B;\mathbb{R})} d\beta^n(s,s') \leqslant \int_{I^2} \|\rho^m_s - \rho^m_{s'}\|_{\mathcal{C}(K\times B;\mathbb{R})} d\beta^n(s,s') + 2\|\rho^m - \rho\|_{L^1}.
$$

Taking m, then n large enough, we conclude that (33) vanishes when $n \to \infty$. □

Lemma 4.7 (Density of chattering dynamics)**.** *Let $A \subset \mathcal{E}$ be compact, $I \subset [0,T]$ be a nontrivial closed interval, and $\mu \in L^1(I; \mathscr{P}_1(A))$. For any $\varepsilon > 0$, there exists $E^\varepsilon \in L^1(I;A)$ such that*

$$
|\mathcal{L}_I \otimes \mu - (id, E^\varepsilon)_\# \mathcal{L}_I|_{\mathrm{KR}} \leqslant \varepsilon. \qquad (34)
$$

Proof. Any $\mu \in L^1(0,T;\mathscr{P}_1(A))$ can be approximated in the L^1 distance by simple functions of the form $\mu^{(0)} \coloneqq \sum_{j=1}^M \mu_j^{(0)} \mathbb{1}_{J_j}$, where $(J_j)_{j\in\{1,\cdots,M\}}$ are disjoint nontrivial intervals covering $[0,T]$, and $\mu_j^{(0)} \in \mathscr{P}_1(A)$. Each $\mu_j^{(0)}$ can in turn be approximated with respect to $|\cdot|_{\mathrm{KR}}$ by a totally atomic probability measure $\mu_j^{(1)} = \sum_{k=1}^P m_{jk} \delta_{E_{jk}}$, with $E_{jk} \in A$. Partition each J_j in subintervals J_{jk} such that the relative length $|J_{jk}|/|J_j|$ coincides with the mass m_{jk} of the kth atom. Define E^ε to be identically equal to E_{jk} on J_{jk}. With this choice, $E^\varepsilon_\# \mathcal{L}_{J_j} = \mathcal{L}(J_j)\mu_j^{(1)}$, and the plan $\eta \coloneqq \frac{1}{\mathcal{L}(J_j)} \int_{J_j} \delta_s \otimes (E^\varepsilon, id, E^\varepsilon)_\# \mathcal{L}_{J_j} ds$ is a transport plan between $\mathcal{L}_{J_j} \otimes \mu_j^{(0)}$ and $(id, E^\varepsilon)_\# \mathcal{L}_{J_j}$. Hence

$$
\left|\mathcal{L}_{J_j} \otimes \mu_j^{(0)} - (id, E^\varepsilon)_\# \mathcal{L}_{J_j}\right|_{\mathrm{KR}} \leqslant \frac{1}{\mathcal{L}(J_j)} \int_{J_j^2} \left[|s-\tau| + |E^\varepsilon_\tau - E^\varepsilon_\tau|\right] d\tau ds \leqslant \operatorname{diam}(J_j)\mathcal{L}(J_j).
$$

Using the triangular inequality, and the convexity of the Kantorovich-Rubinstein norm with respect to sums in the Banach space of measures [Vil09, Th. 4.8] ,

$$
\begin{aligned}
&\left|\mathcal{L}_{[0,T]} \otimes \mu - (id, E^\varepsilon)_\# \mathcal{L}_{[0,T]}\right|_{\mathrm{KR}} \\
&\leqslant \left|\mathcal{L}_{[0,T]} \otimes \mu - \mathcal{L}_{[0,T]} \otimes \mu^{(0)}\right|_{\mathrm{KR}} + \left|\mathcal{L}_{[0,T]} \otimes \mu^{(0)} - (id, E^\varepsilon)_\# \mathcal{L}_{[0,T]}\right|_{\mathrm{KR}} \\
&\leqslant \int_0^T \left|\mu_s - \mu_s^{(0)}\right|_{\mathrm{KR}} ds + \sum_{j=1}^M \left|\mathcal{L}_{J_j} \otimes \mu_j^{(0)} - (id, E^\varepsilon)_\# \mathcal{L}_{J_j}\right|_{\mathrm{KR}} \\
&\leqslant \int_0^T \left|\mu_s - \mu_s^{(0)}\right|_{\mathrm{KR}} ds + T \sup_{j\in\{1,\cdots,M\}} \operatorname{diam}(J_j).
\end{aligned}
$$

Choosing first $\mu^{(0)}$ close enough to μ, then the partitions $(J_j)_j$ fine enough to have both terms inferior to $\varepsilon/2$, we conclude. □

We can now approximate solutions of the relaxed system by solutions of the original system.

Proposition 4.8. *Consider the assumptions of Theorem 4.4. Let $y \in \mathrm{AC}([0,T];\Omega)$ be a solution of the relaxed dynamical system (30b). For any $\varepsilon > 0$, there exists a solution $z^\varepsilon \in \mathrm{AC}([0,T];\Omega)$ of the original system (22) issued from y_0 and such that $\sup_{t\in[0,T]} d(y_t, z_t^\varepsilon) \leqslant \varepsilon$.*

Proof. Let $N \in \mathbb{N}^*$ be large enough so that $T/N \leqslant \varepsilon$, and $t_n \coloneqq nT/N$ for $n \in \{0,\cdots,N\}$. Let B be a compact containing all solutions of the relaxed system (30b) issued from y_0 up to time T. By [AB06, Lemma 17.8], the union $A \coloneqq \bigcup_{x\in B} F(x)$ is a compact subset of $\mathcal{E}$. Denote $\mathrm{Lip}(F)_B$ a local Lipschitz constant of F on B, and C_B a bound on F on this set, that provides a common Lipschitz constant for all solutions. The rest of the proof is split in two parts.

Step 1: Approximation by a chattering dynamic. By Remark 4.3, there exists a relaxed control $\mu \in L^1(0,T;\mathscr{P}_1(\mathcal{E}))$ such that $\mathcal{G}_\mathcal{F}(\mathrm{Bary}_\mathbb{E}(\mu_s)) \in \mathring{y}_s \cap \mathcal{G}_\mathcal{F}(\overline{\mathrm{co}}\, F(y_s))$ for almost all $s \in [0,T]$. Up to a redefinition on a negligible set, μ_s is concentrated on A. Let $\alpha > 0$. Applying Lemma 4.7 on each interval $[t_n, t_{n+1}]$, there exists $E^\alpha \in L^1(0,T;A)$ such that

$$\sup_{n\in\{0,\cdots,N-1\}} \left|\mathcal{L}_{[t_n,t_{n+1}]} \otimes \mu - (id, E^\alpha)_\# \mathcal{L}_{[t_n,t_{n+1}]}\right|_{\mathrm{KR}} \leqslant \alpha. \tag{35}$$

By the well-posedness result of Proposition 3.6, there exists a unique absolutely continuous trajectory $(y_s^\alpha)_{s\in[0,T]}$ such that $\mathcal{G}_\mathcal{F}(E_s^\alpha) \in \mathring{y}_s^\alpha$ for a.e. $s \in [0,T]$, and $y_0^\alpha = x$. This trajectory is contained in the compact B and Lipschitz with constant C_B. Moreover, by Lemma 3.11,

$$\frac{d^2(y_{t_{n+1}}^\alpha, y_{t_{n+1}}) - d^2(y_{t_n}^\alpha, y_{t_n})}{2} \leqslant \int_{s=t_n}^{t_{n+1}} \left[D_{y_s} E_s^\alpha\left(\overrightarrow{y_s y_s^\alpha}\right) - D_{y_s}\mathrm{Bary}_\mathbb{E}(\mu_s)\left(\overrightarrow{y_s y_s^\alpha}\right)\right] ds$$

for all $n \in \{0,\cdots,N-1\}$. Summing on n, one deduces that

$$d^2\left(y_{t_n}^\alpha, y_{t_n}\right) \leqslant 2N \sup_{\substack{n\in\{0,\cdots,N-1\}\\ y'\in\mathrm{AC}([t_n,t_{n+1}];B)\\ \mathrm{Lip}(y')\leqslant C_B}} \left|\int_{s=t_n}^{t_{n+1}} \left[D_{y_s} E_s^\alpha\left(\overrightarrow{y_s y_s'}\right) - D_{y_s}\mathrm{Bary}_\mathbb{E}(\mu_s)\left(\overrightarrow{y_s y_s'}\right)\right] ds\right|,$$

which goes to 0 when α does by Lemma 4.6. We can therefore choose $\alpha = \alpha_\varepsilon > 0$ small enough so that $N\alpha \leqslant \varepsilon$ and $d(y_{t_n}^\alpha, y_{t_n}) \leqslant \varepsilon$ for all $n \in \{1,\cdots,N\}$. Set $E^\varepsilon \coloneqq E^{\alpha_\varepsilon}$, and similarly for y^ε.

Step 2: Approximation by a trajectory of the original system. At this stage, y^ε may not solve the original system (22), since E_s^ε may not belong to $F(y_s^\varepsilon)$. To recover a solution of (22), we apply the results of [FL23a] as recalled in Proposition 2.20. Namely, by the latter statement, there exists a curve $z^\varepsilon \in \mathrm{AC}([0,T];\Omega)$ such that

$$\mathring{z}_s^\varepsilon \cap \mathcal{G}_\mathcal{F}(F(z_s^\varepsilon)) \neq \emptyset \qquad \text{for a.e. } s \in [0,T], \qquad\qquad z_0^\varepsilon = y_0 = y_0^\varepsilon,$$

and such that for some $C_0 < \infty$ depending on T, B and F, there holds

$$d(y_{t_n}^\varepsilon, z_{t_n}^\varepsilon) \leqslant C_0 \int_{s=0}^{t_n} \inf_{E''\in F(y_s^\varepsilon)} \left|E_s^\varepsilon - E''\right|_{1,\infty} ds \qquad \forall n \in \{1,\cdots,N\}.$$

Let η^ε be an optimal transport plan between $(id, E^\varepsilon)_\#\mathcal{L}_{[0,t_n]}$ and $\mathcal{L}_{[0,t_n]} \otimes \mu$. Recalling that F is Lipschitz with constant $\mathrm{Lip}(F)_B$ over B, there holds

$$\begin{aligned}
&\int_0^{t_n} \inf_{E''\in F(y_s^\varepsilon)} \left|E_s^\varepsilon - E''\right|_{1,\infty} ds = \int_{([0,t_n]\times\mathcal{E})^2} \inf_{E''\in F(y_s^\varepsilon)} \left|E - E''\right|_{1,\infty} d\eta^\varepsilon((s,E),(s',E'))\\
&\leqslant \int_{([0,t_n]\times\mathcal{E})^2} \left[\left|E - E'\right|_{1,\infty} + \inf_{E''\in F(y_{s'})} \left|E' - E''\right|_{1,\infty} + \mathrm{Lip}(F)_B d(y_s^\varepsilon, y_{s'})\right] d\eta^\varepsilon((s,E),(s',E')).
\end{aligned}$$

Here we used that $\inf_{F(x)} g \leqslant \inf_{F(x')} g + \mathrm{Lip}(g)\mathrm{Lip}(F)_B d(x,x')$ for any Lipschitz $g : \mathcal{E} \to \mathbb{R}$ and $x, x' \in B$. The term

$$\int \inf_{E'' \in F(y_{s'})} \left|E' - E''\right|_{1,\infty} d\eta^\varepsilon((s,E),(s',E')) = \int_0^{t_n} \int_{\mathcal{E}} \inf_{E'' \in F(y_{s'})} \left|E' - E''\right|_{1,\infty} d\mu_{s'}(E') ds'$$

vanishes, since $\mu_{s'}$ is supported on $F(y_{s'})$ for a.e. $s' \in [0,t_n]$ by definition. On the other hand, as y is Lipschitz with constant C_B, we can use the triangular inequality to get

$$\begin{aligned}
&\int_{([0,t_n]\times\mathcal{E})^2} \left[\left|E - E'\right|_{1,\infty} + \mathrm{Lip}(F)_B d(y^\varepsilon_s, y_{s'})\right] d\eta^\varepsilon \\
&\leqslant \int_{([0,t_n]\times\mathcal{E})^2} \left[\left|E - E'\right|_{1,\infty} + \mathrm{Lip}(F)_B \left(d(y^\varepsilon_s, y_s) + C_B\left|s - s'\right|\right)\right] d\eta^\varepsilon \\
&\leqslant \widetilde{C}\left|\mathcal{L}_{[0,t_n]} \otimes \mu - (id, E^\varepsilon)_\# \mathcal{L}_{[0,t_n]}\right|_{\mathrm{KR}} + T\mathrm{Lip}(F)_B \sup_{s\in[0,T]} d(y^\varepsilon_s, y_s),
\end{aligned}$$

with $\widetilde{C} \coloneqq \max\left(1, \mathrm{Lip}(F)_B C_B\right)$. By convexity of the Kantorovich-Rubinstein norm and (35), the first term in inferior to $\widetilde{C} n\alpha \leqslant \widetilde{C}\varepsilon$. Consequently,

$$d\left(y_{t_n}, z^\varepsilon_{t_n}\right) \leqslant d\left(y_{t_n}, y^\varepsilon_{t_n}\right) + C_0\left(\widetilde{C}\varepsilon + T\mathrm{Lip}(F)_B \sup_{s\in[0,T]} d(y^\varepsilon_s, y_s)\right).$$

Since y, y^ε and z are Lipschitz with constant C_B, the estimates at times $(t_n)_{n\in\{0,\cdots,N\}}$ imply uniform estimates on $[0,T]$ with an additional error of order εC_B. Sending ε to 0, we conclude to the desired density. □

4.4 Right derivatives of the relaxed flow in the tangent cone

We show that under some assumptions, the right derivative of the solution of the convexified equation is a barycenter of the original dynamics. This reflects the usual convexity of Euclidean spaces. By barycenter, we mean here the following extension to CAT(0) spaces.

Definition 4.9 (Barycenter [Stu03, Prop. 4.3 and 4.4])**.** Let (X, d_X) be a complete CAT(0) space, and μ be in $\mathscr{P}_1(X)$. There exists an unique point $b \in X$, called the barycenter of μ, such that for all $y, z \in X$,

$$\int_X \left[d^2_X(z,x) - d^2_X(y,x)\right] d\mu(x) \geqslant \int_{x\in X} \left[d^2_X(b,x) - d^2_X(y,x)\right] d\mu(x). \tag{36}$$

Moreover, this point satisfies the variance inequality

$$\int_X \left[d^2_X(x,z) - d^2_X(x,b)\right] d\mu(x) \geqslant d^2(z,b) \qquad \forall z \in X. \tag{37}$$

Since this notion coincides with the classical one in Banach spaces, we still denote $\mathrm{Bary}_X(\mu)$ the barycenter of μ.

Recall that if E is concave, then its gradient flow issued from x admits $\nabla_x E$ as right derivative at $t = 0$, with ∇_x the metric gradient defined in Definition 2.8 . Consequently, if $y(\cdot)$ is a solution of the mutational ODE $\mathring{y}_t \ni \mathcal{G}_{\mathcal{F}}(\mathrm{Bary}_{\mathbb{E}}(\mu_t))$, the right derivative coincides with the metric gradient of the function $\mathrm{Bary}_{\mathbb{E}}(\mu_t)$ evaluated at y_t, whenever the former is defined. Using that all energies are Fréchet-differentiable, i.e. $D_x E(v) = \langle \nabla_x E, v\rangle_x$ for all (x,v), we can provide a formula in the relaxed case as well.

Proposition 4.10. *Assume **[A3.7]**. Let $y \in \mathrm{AC}([0,T];\Omega)$ be a solution of the relaxed ODE (30b) driven by $\mu \in L^1(0,T;\mathscr{P}_1(\mathbb{E}))$. Then, for almost any time $t \in [0,T]$, the equality $y^+_t = \mathrm{Bary}_{\mathrm{T}_{y_t}\Omega}(\nabla_{y_t}\#\mu_t)$ holds in the tangent cone $\mathrm{T}_{y_t}\Omega$.*

Proposition 4.10 relies on the following property of Fréchet-differentiable maps.

Lemma 4.11. *Let $A \subset \mathbb{E}$ be a subset of concave Fréchet-differentiable maps, and $\mu \in \mathscr{P}_1(A)$. The function $\mathrm{Bary}_{\mathbb{E}}(\mu)$ is Lipschitz and concave, so that it admits a gradient everywhere in the sense of Definition 2.8. Moreover, for any $x \in \Omega$,*

$$\nabla_x \mathrm{Bary}_{\mathbb{E}}(\mu) = \mathrm{Bary}_{\mathrm{T}_x\Omega}(\nabla_{x\#}\mu). \tag{38}$$

Proof. Let $x \in \Omega$ be fixed. The application $E \to \nabla_x E$ is 1-Lipschitz from $\mathbb{E}$ to $\mathrm{T}_x\Omega$ by Lemma 2.15, so that $\nabla_{x\#}\mu$ belongs to $\mathscr{P}_1(\mathrm{T}_x\Omega)$. Since the tangent cone $\mathrm{T}_x\Omega$ is itself a CAT(0) space [BH99, Theorem 3.19], both sides of (38) are defined. Let $p \coloneqq \mathrm{Bary}_{\mathrm{T}_x\Omega}(\nabla_{x\#}\mu) \in \mathrm{T}_x\Omega$. On the one hand, by definition of Fréchet-differentiable maps, there holds for any $v \in \mathrm{T}_x\Omega$ that

$$D_x\mathrm{Bary}_{\mathbb{E}}(\mu)(v) = \int_{\mathbb{E}} D_x E(v) d\mu(E) = \int_{\mathbb{E}} \langle \nabla_x E, v\rangle_x \, d\mu(E) \leqslant \langle p, v\rangle_x . \tag{39}$$

The last inequality stems from the concavity of the metric scalar product, and the Jensen inequality of [Stu03, Theorem 6.2]. On the other hand,

$$D_x\mathrm{Bary}_{\mathbb{E}}(\mu)(p) = \int_{\mathbb{E}} \langle \nabla_x E, p\rangle_x \, d\mu(E) = \int_{\mathbb{E}} \frac{1}{2}\left[|\nabla_x E|_x^2 + |p|_x^2 - d_x^2(\nabla_x E, p)\right] d\mu(E).$$

Using the variance inequality (37) in the CAT(0) space $\mathrm{T}_x\Omega$,

$$\int_{\mathbb{E}} \left[|\nabla_x E|_x^2 - d_x^2(\nabla_x E, p)\right] d\mu(E) = \int_{\mathrm{T}_x\Omega} \left[d_x^2(0_x, v) - d_x^2(v, p)\right] d\left[\nabla_{x\#}\mu\right](v) \geqslant d_x^2(0_x, p) = |p|_x^2 .$$

Hence $D_x\mathrm{Bary}_{\mathbb{E}}(\mu)(p) \geqslant |p|_x^2$, and equality holds. Combined with (39), this shows that p is a metric gradient of $\mathrm{Bary}_{\mathbb{E}}(\mu)$ in the sense of Definition 2.8. By uniqueness of the gradient for concave maps, $\nabla_x \mathrm{Bary}_{\mathbb{E}}(\mu) = p$. □

We can now conclude to the announced formula.

Proof of Proposition 4.10. By **[A3.7]**, the curve y is locally Lipschitz, so there exists a set $I^0 \subset [0,T]$ of full measure on which y is differentiable [AKP23, Proposition 13.9]. On the other hand, the definition of mutational ODE provides another set $I^1 \subset [0,1]$ of full measure such that for any $t \in I^1$, $h \to y_{t+h}$ is approximated at first order by the gradient flow of $E_t \coloneqq \mathrm{Bary}_{\mathbb{E}}(\mu_t)$. For any $t \in I^0 \cap I^1$, the right derivative of y coincides with that of the gradient flow of E_t issued from y_t at $h = 0$. By Proposition 2.9, the latter is equal to the metric gradient of E_t. Invoking Lemma 4.11, we obtain that for any $t \in I^0 \cap I^1$,

$$y_t^+ = \nabla_{y_t} E_t = \nabla_{y_t} \mathrm{Bary}_{\mathbb{E}}(\mu_t) = \mathrm{Bary}_{\mathrm{T}_{y_t}\Omega}(\nabla_{y_t\#}\mu_t),$$

which is the desired equality. □

5 Hamilton-Jacobi-Bellman approach for optimal control problems

We now apply the previous results to the study of Mayer optimal control problems on Ω. Consider $\mathfrak{J} : \Omega \to \mathbb{R}$ a terminal cost, U a compact set of controls, and $f : \Omega \times U \to \mathcal{E}$ a controlled dynamic. The Mayer problem writes

$$\underset{u \in L^0(t,T;U)}{\text{Minimize}} \quad \mathfrak{J}(y_T^{t,x,u}), \qquad \text{where } y_t^{t,x,u} = x \text{ and } \mathring{y}_s \ni \mathcal{G}_{\mathcal{F}}(f(y_s, u(s))). \tag{40}$$

To study the relaxation of (40), we build auxiliary dynamics as in Definition 4.1. Since

$$\overline{\operatorname{conv}} f(x,U) = \left\{ \operatorname{Bary}_{\mathbb{E}}\left(f(x,\cdot)_{\#}\mu\right) \mid \mu \in \mathscr{P}_1(U) \right\},$$

it is convenient to define *relaxed controls* as elements of $\mathscr{P}_1(U)$, and the *relaxed controlled dynamic* $\overline{\operatorname{co}} f : \Omega \times \mathscr{P}_1(U) \to \overline{\operatorname{conv}} \mathcal{E}$ by

$$\overline{\operatorname{co}} f(x,\mu) \coloneqq \operatorname{Bary}_{\mathbb{E}}\left(f(x,\cdot)_{\#}\mu\right) = \int_U f(x,u) d\mu(u).$$

We point that $\overline{\operatorname{co}} f(x,\mu)$ is a function from Ω to $\mathbb{R}$, with $\overline{\operatorname{co}} f(x,\mu)(y) = \int_U f(x,u)(y) d\mu(u)$. In the context of optimal control, it is customary to refer to time-dependent relaxed control as Young measures, in the following sense.

Definition 5.1. The set of Young measures over a given set of controls U is defined as

$$\mathcal{Y}(t,T;U) \coloneqq L^0(t,T;\mathscr{P}_1(U)).$$

Any $\mu \in \mathcal{Y}(t,T;U)$ generates a unique solution of the mutational inclusion with dynamics $(s,z) \mapsto \overline{\operatorname{co}} f(z,\mu_s)$. The associated control problem writes

$$\underset{\mu \in \mathcal{Y}(t,T;U)}{\text{Minimize}} \quad \mathfrak{J}(y_T^{t,x,\mu}), \qquad \text{where } y_t^{t,x,\mu} = x \text{ and } \mathring{y}_s \ni \mathcal{G}_{\mathcal{F}}\left(\overline{\operatorname{co}} f(y_s,\mu_s)\right). \tag{41}$$

We first obtain the existence of an optimal control by applying the results of the previous section, then turn to the link with Hamilton-Jacobi-Bellman equations.

5.1 Relaxation

We consider the following assumption on f.

Assumption [A5.2]. Let (U,d_U) be a compact metric space and $f : \Omega \times U \to \mathcal{E}$. Assume that

(a) for each $x \in \Omega$, the application $u \mapsto f(x,u)$ is continuous from (U,d_U) to $(\mathcal{E},|\cdot|_{1,\infty})$,

(b) for each compact $B \subset \Omega$, there exists $\operatorname{Lip}(f)_B \geqslant 0$ such that

$$|f(x,u) - f(y,u)|_{1,\infty} \leqslant \operatorname{Lip}(f)_B \, d(x,y) \qquad \forall (x,y) \in B^2 \text{ and } u \in U,$$

(c) there exists $C_f \geqslant 0$ such that $\sup_{u \in U} |f(x,u)|_{1,\infty} \leqslant C_f(1 + d(x,0_\Omega))$ for all $x \in \Omega$.

Under **[A5.2]**, the set-valued dynamic $F(x) \coloneqq f(x,U)$ satisfies the assumption **[A3.7]**. Hence we can apply the results of the previous section to get that (41) is the relaxation of (40).

Proposition 5.3. *Assume that $\mathfrak{J} : \Omega \to \mathbb{R}$ is continuous, and that the controlled dynamics f satisfies **[A5.2]**. Then (41) is the relaxation of (40), in the sense that*

$$\inf_{u \in L^0(t,T;U)} \mathfrak{J}(y_T^{t,x,u}) = \min_{\mu \in \mathcal{Y}(t,T;U)} \mathfrak{J}(y_T^{t,x,\mu}). \tag{42}$$

In the case where the sets $f(\cdot,U)$ are convex as subsets of $\mathcal{E}$, the relaxed dynamic coincides with f, and Proposition 5.3 implies the existence of an optimal control. Otherwise, a relaxed control μ exists for the relaxed problem.

Proof. By Theorem 4.4, the set of trajectories of the relaxed control system is the closure in $\operatorname{AC}([0,T];\Omega)$ of the set of trajectories of the original system, and is compact. Consequently, for $(t,x) \in [0,T] \times \Omega$ fixed,

$$\left\{ y_T^{t,x,\mu} \;\middle|\; \mu \in \mathcal{Y}(t,T;U) \right\} = \overline{\left\{ y_T^{t,x,u} \;\middle|\; u \in L^0(t,T;U) \right\}},$$

and the latter set is compact (since $y \mapsto y_T$ is continuous from $\operatorname{AC}([0,T];\Omega)$ to Ω). As $\mathfrak{J}$ is continuous, (42) holds. □

To go on, we introduce the *value function* of the control problem (40), as

$$V : [0,T] \times \Omega \to \mathbb{R}, \qquad V(t,x) \coloneqq \inf_{u \in L^0(t,T;U)} \mathfrak{J}(y_T^{t,x,u}). \tag{43}$$

By Proposition 5.3, there holds

$$V(t,x) = \min_{\mu \in \mathcal{Y}(t,T;U)} \mathfrak{J}(y_T^{t,x,\mu}). \tag{44}$$

The value function has the interesting property to be constant along optimal trajectories, as a consequence of the Dynamic Programming Principle

$$V(t,x) = \inf_{u \in L^0(t,t+h;U)} V(t+h, y_{t+h}^{t,x,u}) = \min_{\mu \in \mathcal{Y}(t,t+h;U)} V(t+h, y_{t+h}^{t,x,\mu}) \tag{45}$$

for any $(t,x) \in [0,T) \times \Omega$ and $h \in (0, T-t]$. (45) holds in our setting, since trajectories on $[t,T]$ are concatenations of trajectories on $[t,t+h]$ and $[t+h,T]$. Moreover, under our assumptions, the estimate (19) of Proposition 2.20 imply that the set-valued map $(t,x) \mapsto \left\{ y_T^{t,x,\mu} \;\middle|\; \mu \in \mathcal{Y}(t,T;U) \right\}$ is locally Lipschitz in the Hausdorff distance over Ω. As a consequence, if $\mathfrak{J}$ is locally Lipschitz, then V is locally Lipschitz as well. The proofs of these assertions follow verbatim the proofs of the Euclidean case, and we refer the reader to [Bar94, Section 3.1.3] or [BC97, Section 2.1].

5.2 The HJB equation

Our aim is this section is to give a meaning to the Hamilton-Jacobi-Bellman equation (2) in CAT(0) spaces. To define a suitable Hamiltonian, let

$$\mathbb{T}_x \coloneqq \{ p : \mathrm{T}_x \Omega \to \mathbb{R} \mid p \text{ is positively homogeneous and Lipschitz w.r.t. } d_x(\cdot,\cdot) \},$$

and $\mathbb{T} \coloneqq \bigcup_{x \in \Omega} \{x\} \times \mathbb{T}_x$ the *metric cotangent bundle*. We define the Hamiltonian of (40) as

$$H : \mathbb{T} \to \mathbb{R}, \qquad H(x,p) \coloneqq \sup_{u \in U} -p\left(\nabla_y f(x,u)|_{y=x} \right). \tag{46}$$

Here we recall that $y \mapsto f(x,u)(y)$ is a Lipschitz concave from Ω to $\mathbb{R}$, and that $\nabla_y f(x,u)$ stands for its metric gradient at a point $y \in \Omega$, defined as an element of $\mathrm{T}_y \Omega$. The notation $\nabla_y f(x,u)|_{y=x}$ insists on the fact that the differentiation is taken with respect to y, then evaluated at $y = x$. With this definition, we can now focus on

$$-\partial_t v(t,x) + H\left(x, D_x v(t,x)\right) = 0 \quad (t,x) \in [0,T) \times \Omega, \qquad v(T,\cdot) = \mathfrak{J}. \tag{47}$$

5.2.1 Viscosity solutions

Following [JZ23], we introduce viscosity solutions using test functions that are respectively semiconvex or semiconcave for sub and supersolutions. More precisely, consider the sets of test functions

$$\mathscr{T}_\pm \coloneqq \left\{ \varphi : (0,T) \times \Omega \to \mathbb{R} \;\middle|\; \begin{array}{c} \varphi(t,x) = \psi(t) \pm g(x), \text{ with } \psi \in \mathcal{C}^2((0,T);\mathbb{R}) \\ \text{and } g \text{ locally Lipschitz and semiconvex} \end{array} \right\}.$$

Any $\varphi \in \mathscr{T}_\pm$ admits a differential $D_x \varphi \in \mathbb{T}_x$ at all point $x \in \Omega$ in the sense of Definition 2.7. This allows to define a viscosity solution to (47) as follows.

Definition 5.4. An application $v : [0,T] \times \Omega \to \mathbb{R}$ is a

– viscosity subsolution of (47) if it is upper semicontinuous, if $v(T,\cdot) \leqslant \mathfrak{J}$, and if for any $\varphi \in \mathscr{T}_+$ such that $v-\varphi$ reaches a maximum in $(t,x) \in (0,T)\times\Omega$, there holds

$$-\partial_t\varphi(t,x) + H(x, D_x\varphi(t,x)) \leqslant 0. \tag{48}$$

– viscosity supersolution of (47) if it is lower semicontinuous, if $v(T,\cdot) \geqslant \mathfrak{J}$, and if for any $\varphi \in \mathscr{T}_-$ such that $v-\varphi$ reaches a minimum in $(t,x) \in (0,T)\times\Omega$, there holds

$$-\partial_t\varphi(t,x) + H(x, D_x\varphi(t,x)) \geqslant 0. \tag{49}$$

– viscosity solution of (47) if it is both a viscosity subsolution and a viscosity supersolution.

In the case $\Omega = \mathbb{R}^d$, Definition 5.4 admits a larger class of test functions than the classical definition (which uses C^2 functions), so any viscosity solution in the classical sense is a viscosity solution in our sense. The converse holds as well, since existence via the Perron method of [JZ23, Theorem 3.22] ensures that both notions share the same solution, which is unique by the comparison principle of [JZ23, Theorem 4.3].

5.2.2 Characterization of the value function

The aim of this section is to prove that the value function V introduced in (43) is a viscosity solution of (47), in the sense of Definition 5.4. We first prove that V is a viscosity subsolution, by using arguments analogous to the Euclidean setting.

Proposition 5.5. *Assume that $\mathfrak{J}:\Omega\to\mathbb{R}$ is locally Lipschitz, and that $f:\Omega\times U\to\mathcal{E}$ satisfies **[A5.2]**. Then, the value function V given by (43) is a viscosity subsolution of (47) in the sense of Definition 5.4.*

Proof. Under the given assumptions, V is locally Lipschitz, and satisfies the terminal condition by definition. Let $\varphi \in \mathscr{T}_+$ such that $V-\varphi$ reaches a maximum at (t,x). Let $\mathrm{Lip}(f)_B$ be a local Lipschitz constant of f in a ball B containing all solutions of the controlled system issued from x up to time $T-t$. By **[A5.2]**, $|f(x,u)|_{1,\infty}$ is bounded by a constant C_B over $B\times U$. For any fixed $u\in U$, the unique solution $y\in \mathrm{AC}([t,T];\Omega)$ issued from x of the ODE $\mathring{y}_s \ni \mathcal{G}_{\mathcal{F}}(f(y_s,u))$ admits $\nabla_y f(x,u)|_{y=x}$ as a right derivative. Indeed, it is the case for the gradient curve $z(\cdot)$ of $\overline{\mathrm{co}}\, f(x,u)$ issued from x, and by Lemma 4.5,

$$\begin{aligned} d^2(y_t,z_t) &\leqslant 2\int_0^t [D_{z_s}f(y_s,u)\,(\overrightarrow{z_s y_s}) - D_{z_s}f(x,u)\,(\overrightarrow{z_s y_s})]ds \leqslant \mathrm{Lip}(f)_B \int_0^t 2d(y_s,z_s)d(y_s,x)ds \\ &\leqslant \mathrm{Lip}(f)_B\left[\int_0^t d^2(y_s,z_s)ds + \int_0^t d^2(y_s,x)ds\right] \leqslant \mathrm{Lip}(f)_B\left[\int_0^t d^2(y_s,z_s)ds + \frac{C_B t^3}{3}\right]. \end{aligned}$$

By a Grönwall Lemma, $d(y_t,z_t) = o(t)$, so that $y(\cdot)$ and $z(\cdot)$ admit the same right derivative at $t=0$. Owing to the DPP (45), for any $h>0$ such that $t+h\leqslant T$, one has $V(t,x)\leqslant V(t+h,y_{t+h})$. Consequently,

$$V(t+h,y_{t+h}) - \varphi(t+h,y_{t+h}) \leqslant V(t,x) - \varphi(t,x),$$

so that

$$-\frac{\varphi(t+h,y_{t+h})-\varphi(t,x)}{h} \leqslant -\frac{V(t+h,y_{t+h})-V(t,x)}{h} \leqslant 0.$$

Recalling that test functions admit a differential in the sense of Definition 2.7 at all points, we may let h go to 0 and find that

$$-\partial_t\varphi(t,x) - D_x\varphi(t,x)\,(\nabla_y f(x,u)|_{y=x}) \leqslant 0.$$

Taking the supremum over $u\in U$ yields the subsolution inequality (48). □

The supersolution side is slightly more involved. Here one should use the dynamical programming principle along an optimal solution, which may not satisfy the dynamical system if the latter does not have convex images. Hence, one argues with $\varepsilon-$optimal curves instead. However, to complete the argument, one needs to take a limit in an expression of the form $(t,x)\mapsto H(x,D_x\varphi(t,x))$, where $\varphi\in\mathscr{T}_-$ is a test function. In $\mathbb{R}^d$, test functions can be taken $\mathcal{C}^1$. In our setting, we need to rely on the semiconcavity to get the required semicontinuity. Hence, we impose the following assumption to turn semiconcavity along geodesics into a "sufficient semiconcavity" along the transitions.

Assumption [A5.6]. $\mathcal{E}\subset\mathbb{E}$ is an admissible set of energies in the sense of Definition 3.1. We assume that for each compact sets $A\subset\mathcal{E}$ and $B\subset\Omega$, there exist reparameterized geodesics $(\gamma^{E,x})_{(E,x)\in A\times B}$ defined over a nontrivial common interval $[0,t]$ satisfying

$$\gamma_0^{E,x}=x\qquad\text{and}\qquad\lim_{h\searrow 0}\sup_{(E,x)\in A\times B}\frac{d\left(\mathcal{G}_{\mathcal{F}}(E)(h,x),\gamma_h^{E,x}\right)}{h}=0.$$

The set $\mathcal{E}$ constructed in Definition 3.4 satisfies **[A5.6]**, as shown in Appendix.

Lemma 5.7. *Assume that $\mathcal{E}$ satisfies **[A5.6]**. Let $(y_i)_{i\in\mathbb{N}}\underset{i\to\infty}{\longrightarrow}y\in\Omega$ and $E_i\overset{|\cdot|_{1,\infty}}{\underset{i\to\infty}{\longrightarrow}}E\in\mathcal{E}$. For any locally Lipschitz $\varphi:\Omega\to\mathbb{R}$ that is $\lambda-$convex for some $\lambda\in\mathbb{R}$, there holds*

$$\limsup_{i\to\infty}D_{y_i}\varphi\left(\nabla_{y_i}E_i\right)\leqslant D_y\varphi(\nabla_y E).$$

Proof. The argument is based on the classical proof of [Roc70, Theorem 24.5]. To lighten the notation, denote $z_s^i:=\mathcal{G}_{\mathcal{F}}(E_i)(s,y_i)$, and $z_s:=\mathcal{G}_{\mathcal{F}}(E)(s,y)$. As the sets $\{y_i\}_{i\in\mathbb{N}}$ and $\{E_i\}_{i\in\mathbb{N}}$ are relatively compact (Ω,d) and $(\mathcal{E},|\cdot|_{1,\infty})$, we may use assumption **[A5.6]** and approximate each curve z^i at order one by some reparametrized geodesic $\gamma^i=\gamma^{E_i,y_i}$ defined on some interval $[0,t]$ with t independent of i. Let $\varepsilon>0$, and consider $0<h^\varepsilon\leqslant\min(\varepsilon,t)$ small enough so that

$$\frac{\varphi\left(z_{h^\varepsilon}\right)-\varphi(y)}{h^\varepsilon}\leqslant D_y\varphi(\nabla_y E)+\varepsilon.$$

By [Aub99, Proposition 1.3.2] applied with $\vartheta(t)\equiv\mathcal{G}_{\mathcal{F}}(E_i)$ and $\tau(t)\equiv\mathcal{G}_{\mathcal{F}}(E)$, noticing that the contraction constant $\alpha=0$ in our case, there holds

$$d(z^i_{h^\varepsilon},z_{h^\varepsilon})\leqslant d(y_i,y)+h^\varepsilon\mathbb{D}(\mathcal{G}_{\mathcal{F}}(E_i),\mathcal{G}_{\mathcal{F}}(E))\leqslant d(y_i,y)+h^\varepsilon\left|E_i-E\right|_{1,\infty},$$

which goes to 0 when i goes to ∞. Denote by $\mathrm{Lip}(\varphi)_{\mathrm{loc}}$ a local Lipschitz constant of φ in a compact containing all $\overline{\mathscr{B}}(y_i,h^\varepsilon\max_i|E_i|_{1,\infty})$. Consider i_ε large enough so that for all $i\geqslant i_\varepsilon$,

$$|\varphi(y_i)-\varphi(y)|\leqslant\varepsilon h^\varepsilon,\qquad\text{and}\qquad\left|\varphi(z^i_{h^\varepsilon})-\varphi\left(z_{h^\varepsilon}\right)\right|\leqslant\varepsilon h^\varepsilon.$$

Then for $i\geqslant i_\varepsilon$,

$$\begin{aligned}D_y\varphi(\nabla_y E)+\varepsilon&\geqslant\frac{\varphi\left(z_{h^\varepsilon}\right)-\varphi(y)}{h^\varepsilon}\geqslant\frac{\varphi\left(z^i_{h^\varepsilon}\right)-\varphi(y_i)}{h^\varepsilon}-2\varepsilon\\&\geqslant\frac{\varphi\left(\gamma^i_{h^\varepsilon}\right)-\varphi(y_i)}{h^\varepsilon}-\mathrm{Lip}(\varphi)_{\mathrm{loc}}\frac{d\left(\gamma^i_{h^\varepsilon},z^i_{h^\varepsilon}\right)}{h^\varepsilon}-2\varepsilon\\&\geqslant D_{y_i}\varphi(\nabla_{y_i}E_i)-\frac{|\lambda|}{2}h^\varepsilon\left|(\gamma^i)_0^+\right|^2_{y_i}-\mathrm{Lip}(\varphi)_{\mathrm{loc}}\frac{d\left(\gamma^i_{h^\varepsilon},z^i_{h^\varepsilon}\right)}{h^\varepsilon}-2\varepsilon.\end{aligned}$$

Here the last inequality stands since φ is semiconvex. As $\left|(\gamma^i)_0^+\right|_x=|\nabla_{y_i}E_i|_x\leqslant|E_i|_{1,\infty}$ converges towards $|E|_{1,\infty}$ when i goes to $+\infty$, we may pass to the limit sup in $i\to\infty$ and obtain that

$$D_y\varphi(\nabla_y E)+\varepsilon\geqslant\limsup_{i\to\infty}D_{y_i}\varphi(\nabla_{y_i}E_i)-\frac{|\lambda|}{2}h^\varepsilon|E|^2_{1,\infty}-\mathrm{Lip}(\varphi)_{\mathrm{loc}}\sup_{i\in\mathbb{N}}\frac{d\left(\gamma^i_{h^\varepsilon},z^i_{h^\varepsilon}\right)}{h^\varepsilon}-2\varepsilon.$$

Since $h^\varepsilon\leqslant\varepsilon$, we can send ε to 0 and use **[A5.6]** to conclude. □

As a consequence, the Hamiltonian is semicontinuous in the following way.

Lemma 5.8. *Assume **[A5.6]** and **[A5.2]** hold. Let $\varphi : \Omega \to \mathbb{R}$ be locally Lipschitz. Then*

- *if φ is semiconvex, $x \mapsto H(x, D_x\varphi(x))$ is lower semicontinuous.*
- *if φ is semiconcave, $x \mapsto H(x, D_x\varphi(x))$ is upper semicontinuous.*

Proof. Assume first that φ is semiconvex. Then the application $(x, E) \mapsto D_x\varphi(\nabla_x E)$ is upper semicontinuous by Lemma 5.7, so that $(x, u) \mapsto -D_x\varphi(\nabla_y f(x, u)|\nabla_y f(x, u)|_{y=x})$ is lower semicontinuous, and so is the supremum over $u \in U$. On the other hand, if φ is semiconcave, let $x_n \to_n x \in \Omega$. Up to extraction, we may assume that

$$\limsup_{n\to\infty} H(x_n, D_{x_n}\varphi) = \lim_{n\to\infty} H(x_n, D_{x_n}\varphi) = \lim_{n\to\infty} \sup_{u\in U} -D_{x_n}\varphi(\nabla_y f(x_n, u)|_{y=x_n}).$$

The composition $u \mapsto D_{x_n}\varphi\left(\nabla_y f(x_n, u)|_{y=x_n}\right)$ being Lipschitz, its supremum over the compact U is reached at some point $u_n \in U$. Up to further extraction, we may assume that $(u_n)_{n\in\mathbb{N}}$ converges to some $\overline{u} \in U$. By continuity, $|f(x_n, u_n) - f(x, \overline{u})|_{1,\infty}$ goes to 0 when n goes to $+\infty$, so that Lemma 5.7 applied to the function $-\varphi$ yields

$$\limsup_{n\to\infty} -D_{x_n}\varphi(\nabla_y f(x_n, u_n)|_{y=x_n}) \leqslant -D_x\varphi(\nabla_y f(x, \overline{u})|_{y=x}).$$

Hence

$$\begin{aligned}\limsup_{n\to\infty} H(x_n, D_{x_n}\varphi) &= \lim_{n\to\infty} -D_{x_n}\varphi(\nabla_y f(x_n, u_n)|_{y=x_n}) \leqslant -D_x\varphi(\nabla_y f(x, \overline{u})|_{y=x}) \\ &\leqslant \sup_{u\in U} -D_x\varphi(\nabla_y f(x, u)|_{y=x}) = H\left(x, D_x\varphi\right),\end{aligned}$$

and the desired semicontinuity holds. □

With these arguments, one can carry on with the proof of the supersolution property.

Proposition 5.9. *Assume that **[A5.2]** and **[A5.6]** hold, and that $\mathfrak{J} : \Omega \to \mathbb{R}$ is locally Lipschitz. The value function V given by (43) is a viscosity supersolution of (47) in the sense of Definition 5.4.*

Proof. It suffices to prove the viscosity inequality (49). Let $\varphi \in \mathscr{T}_-$ such that $V - \varphi$ reaches a minimum at point $(t, x) \in (0, T) \times \Omega$. By Theorem 4.4, the value function V coincides with the value function W of the relaxed problem. Moreover, the latter admits an optimal relaxed control $\mu \in \mathcal{Y}(t, T; U)$ such that for any $t \leqslant s \leqslant T$,

$$V(t, x) = W(t, x) = W(s, y_s^{t,x,\mu}) = \mathfrak{J}(y_T^{t,x,\mu}).$$

Here $y = (y_s^{t,x,\mu})_{s\in[t,T]}$ is the absolutely continuous curve such that $y_t = x$ and $\overline{\mathrm{co}}\, f(y_s, \mu) \in \mathring{y}_s$ for a.e. $s \in [t, T]$. Taking $s = t + h$ for some small $h > 0$, there holds

$$W\left(t + h, y_{t+h}\right) - \varphi\left(t + h, y_{t+h}\right) \geqslant W(t, x) - \varphi(t, x) - \frac{\varphi(t + h, y_{t+h}) - \varphi(t, x)}{h} \geqslant 0.$$

The trajectory y is mutable at almost every $s \in [t, t + h]$, so that writing $\varphi(s, z) = \psi(s) + g(z)$ for some $\psi \in \mathcal{C}^2((0, T); \mathbb{R})$ and locally Lipschitz semiconcave $g : \Omega \to \mathbb{R}$,

$$0 \leqslant -\frac{\psi(t + h) - \psi(t)}{h} - \frac{1}{h}\int_{s=t}^{t+h} D_{y_s} g\left(\nabla_z \overline{\mathrm{co}}\, f(y_s, \mu_s)|_{z=y_s}\right) ds.$$

On the one hand, $\frac{\psi(t+h)-\psi(t)}{h} \leqslant -\partial_t\psi + O(h)$ by the regularity of ψ. On the other hand, let C_f be a bound on $|\overline{\mathrm{co}}\, f|_{1,\infty}$ in a sufficiently large ball around x. As $d(x, y_s) \leqslant hC_f$ for all $s \in [0, h]$,

$$0 \leqslant -\partial_t\psi + O(h) + \frac{1}{h}\int_{s=t}^{t+h} \sup_{\mu\in\mathscr{P}_1(U)} \left[-D_{y_s} g\left(\nabla_z \overline{\mathrm{co}}\, f(y_s,\mu)|_{z=y_s}\right)\right] ds.$$

By Proposition 4.10 applied to the measure $f(y_s,\cdot)_\#\mu$, there holds

$$\nabla_z \overline{\mathrm{co}}\, f(y_s,\mu)|_{z=y_s} = \mathrm{Bary}_{\mathrm{T}_{y_s}\Omega}\left(\nabla_{y_s\#} f(y_s,\cdot)_\#\mu\right).$$

As g is semiconcave, its differential is geodesically concave in the CAT(0) space $\mathrm{T}_{y_s}\Omega$. Hence the supremum of over $\mu \in \mathscr{P}_1(U)$ is reached for Dirac masses in $\mathscr{P}_1(U)$, that is,

$$\sup_{\mu\in\mathscr{P}_1(U)} -D_{y_s} g\left(\nabla_z \overline{\mathrm{co}}\, f(y_s,\mu)|_{z=y_s}\right) = \sup_{u\in U} -D_{y_s} g\left(\nabla_z f(y_s,u)|_{z=y_s}\right) \leqslant \sup_{z\in\overline{\mathscr{B}}(x,hC_f)} H\left(z, D_z g\right).$$

Since g is semiconcave, by Lemma 5.8, the map $z \mapsto H\left(z, D_z g\right)$ is upper semicontinuous. As such, it admits a maximum point z_h over $\overline{\mathscr{B}}(x, hC_f)$, that must converge to x when h goes to 0, and there holds $\limsup_{h\searrow 0} H(z_h, D_{z_h} g) \leqslant H(x, D_x g)$. Hence, taking the limit sup in $h \searrow 0$, we have that

$$0 \leqslant -\partial_t\psi(t) + H\left(x, D_x g\right) = -\partial_t\varphi(t,x) + H\left(x, D_x\varphi(t,x)\right).$$

This shows that V satisfies the supersolution property. □

Remark 5.10 (Uniqueness). The solution can be proved to be unique by a comparison principle, as shown for this notion in [JZ23, Theorem 4.3]. The assumptions of the cited result are slightly more restrictive than ours, and would require f to be globally Lipschitz and globally bounded. However, these assumptions can be removed by introducing penalizations (see [FGŚ17, Th. 3.50] for an extensive display of modern techniques).

6 Numerical approximation

The framework developed above also serves as a foundation for numerical methods. Indeed, for the energies of (21), gradient flows reduce to geodesic interpolations, so that the approximation of reachable sets, the projection onto a mesh and the minimization over the control set all remain explicit and computable. This allows to adapt classical semi-Lagrangian schemes for HJB equations to the present setting. We describe the scheme in Section 6.1 and illustrate it on a robot navigation problem in Section 6.2.

6.1 Semi-Lagrangian numerical scheme

Approximate reachable sets. Let $x \in \Omega$ and $0 \leqslant t \leqslant s \leqslant T$. The reachable set from (t, x) at time s is

$$R_s^{t,x} := \left\{\, y_s^{t,x,u} \mid u \in L^0(t, s; U) \,\right\},$$

where $y^{t,x,u}$ is the unique solution of the controlled system $\mathring{y}_\sigma \ni \mathcal{G}_{\mathcal{F}}\big(f(y_\sigma, u(\sigma))\big)$ issued from $y_t = x$. Computing $R_s^{t,x}$ exactly would require solving the controlled system for every control. We therefore assume that one has access to a family of approximate reachable sets $\hat{R}_s^{t,x}$ such that, for some $\varpi \geqslant 0$,

$$d_H\Big(\hat{R}_s^{t,x},\, R_s^{t,x}\Big) \leqslant \varpi\,|t-s|^2. \tag{50}$$

In $\mathbb{R}^d$, building such an approximation with a quadratic error in the Hausdorff distance is classical: it follows from standard Euler-scheme estimates for Lipschitz dynamics, on which semi-Lagrangian schemes for Hamilton-Jacobi-Bellman equations are based [BC97; FF14]. The same order is available in our setting for the explicit energies used in the example of Section 6.2, for which the gradient flows reduce to geodesic interpolations. The construction of general Euler schemes for mutational ODEs in CAT(0) spaces, and the corresponding convergence rates, are left for future work.

Approximation of the value function. Let $\hat{\Omega}$ be a mesh, that is, a finite subset of Ω such that for some step $\Delta x > 0$ there holds $\Omega \subset \bigcup_{\hat{x}\in\hat{\Omega}} \overline{\mathscr{B}}(\hat{x}, \Delta x)$. For any $N \in \mathbb{N}_*$, let $0 = t_0 < t_1 < \cdots < t_N = T$ be a regular discretization of $[0, T]$ of step Δt. The value function is approximated by a semi-Lagrangian scheme, computing iteratively a sequence $(\hat{V}_n)_{n\in[\![0,N]\!]}$ of functions $\hat{V}_n : \hat{\Omega} \to \mathbb{R}$ by

$$\hat{V}_N \coloneqq \mathfrak{J}_{|\hat{\Omega}}, \qquad \hat{V}_n(\hat{x}) = \min_{\hat{y}\in\hat{P}_n^{\hat{x}}} \hat{V}_{n+1}(\hat{y}) \quad \forall\, \hat{x} \in \hat{\Omega},\ n \in [\![0, N-1]\!]. \tag{51}$$

Here $\hat{P}_n^{\hat{x}}$ is the set of mesh points closest to the approximate reachable set, defined by

$$\hat{P}_n^{\hat{x}} \coloneqq \big\{\hat{y} \in \hat{\Omega} \mid \exists\, y \in \hat{R}_{t_{n+1}}^{t_n,\hat{x}},\ d(\hat{y}, y) \leqslant \Delta x\big\}.$$

In practice, $\hat{R}_{t_{n+1}}^{t_n,\hat{x}}$ is computed from the bound (50) and stored before the propagation step.

Lemma 6.1. *Under the approximation* (50)*, there holds*

$$\max_{n\in[\![0,N]\!]} \sup_{\hat{y}\in\hat{\Omega}} \Big|\hat{V}_n(\hat{y}) - V(t_n, \hat{y})\Big| \leqslant \mathrm{Lip}(V)\, T \left(\frac{\Delta x}{\Delta t} + \varpi \Delta t\right).$$

Proof. Set $e_n \coloneqq \sup_{\hat{y}\in\hat{\Omega}} |\hat{V}_n(\hat{y}) - V(t_n, \hat{y})|$. For any $\hat{x} \in \hat{\Omega}$, using (50) and the Lipschitz continuity of V,

$$|\hat{V}_n(\hat{x}) - V(t_n, \hat{x})| \leqslant e_{n+1} + \mathrm{Lip}(V)\, d_H\Big(\hat{P}_n^{\hat{x}}, R_{t_{n+1}}^{t_n,\hat{x}}\Big) \leqslant e_{n+1} + \mathrm{Lip}(V)(\Delta x + \varpi\, \Delta t^2).$$

Since $e_N = 0$, summing over n gives $\max_n e_n \leqslant \mathrm{Lip}(V)(N\Delta x + T\varpi\, \Delta t) = \mathrm{Lip}(V)\, T(\Delta x/\Delta t + \varpi\, \Delta t)$. □

Remark 6.2. The error bound of Lemma 6.1 involves a balance between the space step Δx and the time step Δt. Optimizing over Δt for fixed Δx, the minimum is achieved at $\Delta t = \sqrt{\Delta x/\varpi}$, giving the error bound

$$\max_n \sup_{\hat{y}} |\hat{V}_n(\hat{y}) - V(t_n, \hat{y})| \leqslant 2\mathrm{Lip}(V) T \sqrt{\varpi\, \Delta x}.$$

This is the *inverse CFL condition* $\Delta x/\Delta t^2 = \varpi$, which yields the optimal convergence rate $O(\Delta x^{1/2})$ with respect to the space step. This order is consistent with the known convergence rate of semi-Lagrangian schemes in Euclidean spaces [BC97; FF14].

We assume by now that for any $\varepsilon > 0$, we have access to a function $\hat{V} = \hat{V}_\varepsilon : [0, T] \times \Omega \to \mathbb{R}$ such that

$$\sup_{(t,x)\in[0,T]\times\Omega} \Big|\hat{V}(t, x) - V(t, x)\Big| \leqslant \varepsilon.$$

Such a $\hat{V}$ is obtained from Lemma 6.1 by nearest-neighbour extension: setting $\hat{V}(t, x) \coloneqq \hat{V}_n(\hat{x})$ where t_n is the closest time node to t and $\hat{x} \in \hat{\Omega}$ the closest mesh point to x, the Lipschitz continuity of V gives $|\hat{V}(t, x) - V(t, x)| \leqslant \mathrm{Lip}(V)(\Delta x + \Delta t) + \max_n \sup_{\hat{y}} |\hat{V}_n(\hat{y}) - V(t_n, \hat{y})|$,

which is bounded by ε once $\Delta x, \Delta t$ are small enough. This allows to reconstruct solutions as in [RV91].

Algorithm 1: Approximation of the optimal control and trajectory

Input: $x \in \Omega$, $M \in \mathbb{N}_*$, $T > 0$, $\hat{V}$. Set $\Delta s \coloneqq T/M$, $s_m \coloneqq m\Delta s$ for $m \in \{0, \dots, M\}$.

Let $\hat{y}_0 \coloneqq x$;

for $m \in \{0, \dots, M-1\}$ **do**

  z

$\hat{u}^m \in \operatorname*{argmin}_{v \in U} \hat{V}(s_{m+1}, \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, v))(\Delta s, \hat{y}_{s_m}))$;

$\hat{y}_s \coloneqq \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, \hat{u}^m))(s - s_m, \hat{y}_{s_m})$ for $s \in (s_m, s_{m+1}]$;

return $(\hat{u}^m)_{m \in \{0, \cdots, M-1\}}$ and $\hat{y}^M \in \mathrm{AC}([0,T]; \Omega)$.

Next, we measure the error through the loss of optimality $|\mathfrak{J}(\hat{y}^M_T) - V(0,x)|$, which is continuous with respect to $\hat{y}^M \in \mathrm{AC}([0,T];\Omega)$ and vanishes if and only if $\hat{y}^M$ is optimal.

Theorem 6.3. *Let* $C_1 \coloneqq \frac{\mathrm{Lip}(V)\mathrm{Lip}(f)\|f\|_\infty}{2}$ *and* $C_2 \coloneqq \mathrm{Lip}(V)\, e^{\mathrm{Lip}(f)T}$, *and*

$$\Delta_{[0,T]}(M) \coloneqq \sup_{v \in L^0(0,T;U)} \inf_{w \in U^M} \int_0^T d_U\big(v(s),\, w_{\lfloor s/\Delta s \rfloor}\big)\, ds, \tag{52}$$

where U^M *denotes the set of* M*-tuples* $w = (w_0, \dots, w_{M-1}) \in U^M$, *identified with the controls that are constant equal to* w_m *on each* $(s_m, s_{m+1}]$. *Then*

$$\mathfrak{J}(\hat{y}_T) - V(0,x) \leqslant TC_1\, \Delta s + C_2\, \Delta_{[0,T]}(M) + (2M+1)\varepsilon. \tag{53}$$

The quantity $\Delta_{[0,T]}(M) \to 0$ *as* $M \to \infty$ *by density of piecewise constant controls in* $L^0(0,T;U)$. *Choosing a sequence of approximations with* $\varepsilon = \varepsilon_M = o(1/M)$, *the right-hand side of* (53) *vanishes as* $M \to \infty$, *so that any cluster point of* $(\hat{y}^M)_M$ *in* $\mathrm{AC}([0,T];\Omega)$ *is an optimal trajectory of (40).*

Proof. Write $\hat{y} = \hat{y}^M$. By construction $\hat{y}_{s_{m+1}} = \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, \hat{u}^m))(\Delta s, \hat{y}_{s_m})$, and $\hat{u}^m$ realizes the minimum over U of $v \mapsto \hat{V}(s_{m+1}, \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, v))(\Delta s, \hat{y}_{s_m}))$. Telescoping and using $|\hat{V} - V| \leqslant \varepsilon$ at $t = T$,

$$\mathfrak{J}(\hat{y}_T) - V(0,x) \leqslant \varepsilon + \sum_{m=0}^{M-1} \Big[\min_{v \in U} \hat{V}\big(s_{m+1}, \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, v))(\Delta s, \hat{y}_{s_m})\big) - \hat{V}(s_m, \hat{y}_{s_m}) \Big].$$

We bound each bracket by comparison with the dynamic programming principle (45) at $(s_m, \hat{y}_{s_m})$. Replacing $\hat{V}$ by V costs 2ε per step. For a fixed control value $v \in U$, let $z_\sigma \coloneqq \mathcal{G}_{\mathcal{F}}(f(\hat{y}_{s_m}, v))(\sigma, \hat{y}_{s_m})$ be the frozen move and y_σ the exact solution issued from $\hat{y}_{s_m}$ under the constant control v. Since $\mathcal{G}_{\mathcal{F}}$ is an isometry (Lemma 2.15), the a priori estimate (20) gives

$$d(y_{\Delta s}, z_{\Delta s}) \leqslant \int_0^{\Delta s} |f(y_\sigma, v) - f(\hat{y}_{s_m}, v)|_{1,\infty}\, d\sigma \leqslant \mathrm{Lip}(f) \int_0^{\Delta s} d(y_\sigma, \hat{y}_{s_m})\, d\sigma \leqslant \frac{\mathrm{Lip}(f)\|f\|_\infty}{2} (\Delta s)^2,$$

hence $|V(s_{m+1}, z_{\Delta s}) - V(s_{m+1}, y_{\Delta s})| \leqslant C_1 (\Delta s)^2$ by the Lipschitz continuity of V. Restricting the infimum in the DPP from arbitrary controls to controls constant on $(s_m, s_{m+1}]$ introduces, by Grönwall and the Lipschitz continuity of f and V, an error bounded by C_2 times the control-approximation modulus on that sub-interval. Summing over m, the ε terms add up to $(2M+1)\varepsilon$, the frozen-dynamics terms to $M\, C_1 (\Delta s)^2 = TC_1 \Delta s$, and the control-approximation terms to $C_2\, \Delta_{[0,T]}(M)$ by definition of (52). □

6.2 Navigation problem on a stratified CAT(0) domain

Problem description. We model a robot navigating a non-standard geometric space composed of several Euclidean subdomains of dimensions 1, 2 and 3, glued together at junctions, as depicted in Figure 1. This domain is a CAT(0) space by construction, and is representative of the stratified and singular geometries that motivated the present work: unlike most numerical examples in the metric space literature, it is not a perturbation of a Euclidean domain, but a genuinely non-smooth space in which the dimension changes across subdomains.

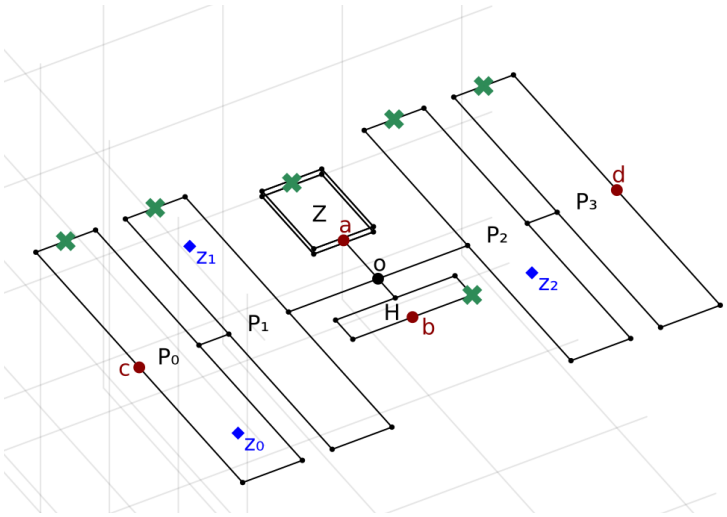


Figure 1: CAT(0) domain in the robot problem.

The domain consists of four Euclidean panes $(P_i)_{i\in\{1,2,3,4\}}$ of dimension 2, identified with $[-\frac{1}{2},\frac{1}{2}]\times[-3,3]$, a 2-dimensional surface H and a 3-dimensional volume Z, linked by linearly parametrized segments. Points $(z_j)_{j\in\{0,1,2\}}$ are the targets, and crosses mark the initial points of the reconstructed optimal trajectories.

Dynamics and optimal control problem. The control set U is *finite*, and the dynamics is *anisotropic*: a constant wind of intensity $w = 2$ blows rightwards. Each control u is described by a target point $x_u(x)$ and a scale $\alpha_u > 0$, and drives the agent through the admissible energy

$$\mathring{y}_s \ni \mathcal{G}_{\mathcal{F}}\big(f(y_s,u_s)\big), \qquad f(x,u) \coloneqq -\,\alpha_u\,\kappa\circ d\big(\cdot,x_u(x)\big) \;\in\; \mathcal{E}, \tag{54}$$

where $\kappa(r) = 2r^2$ if $r \leqslant 1/4$ and $\kappa(r) = r - 1/8$ otherwise. The gradient flow of $f(x,u)$ follows the geodesic $[x\,x_u(x)]$ at speed $\alpha_u\,\kappa'(d(x,x_u(x)))$; since κ is C^1 with $0 \leqslant \kappa' \leqslant 1$ and $x \mapsto d(x,x_u)$ is 1-Lipschitz, **[A5.2]** is satisfied.

On a pane, identified with $[-\frac{1}{2},\frac{1}{2}]\times[-3,3]$, the four controls drive the agent towards the boundary in the directions $\pm e_1, \pm e_2$, with speed w rightwards and 1 otherwise; the agent may also jump towards a neighbouring component, with a scale calibrated so that the speed is consistent across junctions. Away from the target points, where $\kappa' \equiv 1$, the velocities generated by U are

$$\nabla_y f(x,u)\big|_{y=x} \;\in\; \big\{(0,1),\ (0,-1),\ (-1,0),\ (w,0)\big\},$$

which are exactly the extreme points of the unit ball of the distorted norm

$$\|v\|_w \coloneqq \frac{\max(v_1,0)}{w} + \max(-v_1,0) + |v_2|. \tag{55}$$

The convexified dynamics therefore spans *exactly* the set $\{v \in T_x\,\Omega : \|v\|_w \leqslant 1\}$, so that finitely many controls generate a whole convex set of admissible velocities.

Given target points $(z_j)_{j\in\{0,1,2\}}$, indicated in Figure 1, the terminal cost is the distance to the nearest target, $\mathfrak{J}(x) \coloneqq \min_j d(x, z_j)$, which is nonnegative and vanishes exactly at the targets. The optimal control problem is

$$\text{Minimize} \quad \mathfrak{J}\big(y_T^{t,x,u}\big) \quad \text{over } u \in L^0\big(t, T; U\big), \tag{56}$$

subject to $y_t^{t,x,u} = x$ and $\mathring{y}_s^{t,x,u} \ni \mathcal{G}_{\mathcal{F}}\big(f(y_s^{t,x,u}, u_s)\big)$.

HJB equation and interpretation. The Hamiltonian associated to (56) is

$$H(x,p) = \max_{u\in U(x)} \Big(-p\big(\nabla_y f(x,u)\big|_{y=x}\big)\Big),$$

and by Propositions 5.5 and 5.9, the value function $V(t,x) = \inf_u \mathfrak{J}(y_T^{t,x,u})$ is the unique viscosity solution of

$$-\partial_t V(t,x) + H\big(x, D_x V(t,x)\big) = 0, \quad (t,x) \in [0,T) \times \Omega, \tag{57}$$

with terminal condition $V(T,\cdot) = \mathfrak{J}$. Inside a pane and away from the targets, $H(x,\cdot)$ coincides with the support function of the unit ball of (55), that is, with the norm dual to $\|\cdot\|_w$; at the junctions, no interface condition is imposed. The value function has a natural interpretation: $V(t,x)$ is the smallest distance to the targets that can be achieved from x before time T. In particular, $\{V(t,\cdot) = 0\}$ is the set of initial positions from which a target is exactly reached; it grows as t decreases from T to 0.

Simulations and results. Simulations are carried out for $T = 2.5$. For each space step Δx, the time step is chosen so that the inverse CFL condition $\Delta x/\Delta t^2 \simeq 1$ holds (up to the integer rounding of the number of time steps), and the value function is approximated by the semi-Lagrangian scheme (51), yielding $\hat{V}$ on the space-time grid $[\![0,N]\!] \times \hat{\Omega}$. For this test case, the interpretation of V given above allows to compute the exact value function in closed form, which serves as a reference solution. Errors on V are measured with the relative quantity

$$\big|V - \hat{V}\big|_\infty \coloneqq \frac{\max_{n\in[\![0,N]\!]} \max_{\hat{x}\in\hat{\Omega}} \big|V(t_n,\hat{x}) - \hat{V}(t_n,\hat{x})\big|}{\max_{n\in[\![0,N]\!]} \max_{\hat{x}\in\hat{\Omega}} |V(t_n,\hat{x})|}.$$

Numerical trajectories are reconstructed using Algorithm 1 with $\Delta s = \Delta t$, from a fixed set $\mathcal{I}$ of six initial points, indicated by crosses in Figure 1. Their quality is assessed through the loss of optimality

$$\text{Error}(\hat{y}) \coloneqq \big|\mathfrak{J}(\hat{y}_T) - V(0,\hat{y}_0)\big|, \tag{58}$$

where the absolute value accounts for the numerical error on $\hat{V}$: by definition of the value function, $\mathfrak{J}(\hat{y}_T) - V(0,\hat{y}_0)$ is nonnegative and vanishes only for optimal trajectories. Numerical errors are reported in Table 1, and Figure 2 displays $\hat{V}$ for four increasingly refined meshes. Let us stress that the scheme treats junction points and interior points alike: no interface condition is imposed at the junctions. As reported in Table 1, all errors decrease steadily as the mesh is refined, for the value function as well as for the reconstructed trajectories. The observed convergence orders are approximately 0.38 for the value function and 0.46 for the trajectories. The latter is close to the theoretical rate $O(\Delta x^{1/2})$ established in Lemma 6.1 under the inverse CFL condition; the slightly lower rate observed for the value function is consistent with the reduced regularity of V near the junctions, where the pre-asymptotic regime dominates on the range of meshes considered.

Source files are available at https://github.com/averil-aussedat/FLagHada.jl. Simulations were carried on a Dell Inc. Precision 3561 laptop with 11th Gen. Intel I9 processor.

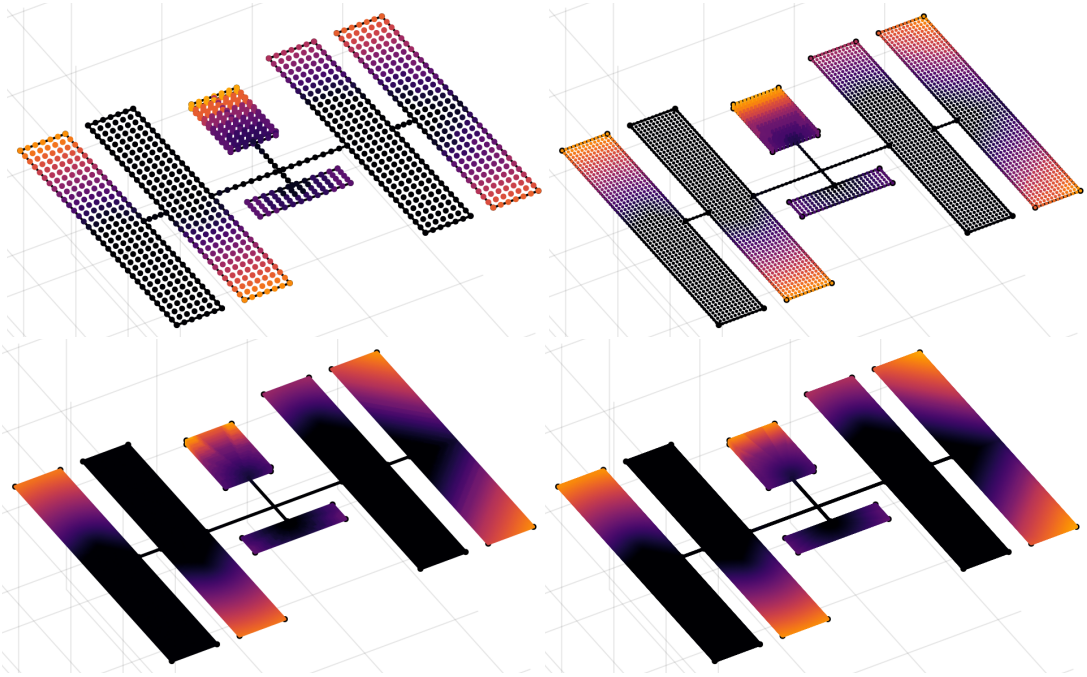


Figure 2: Value function approximation at four increasingly refined space-time meshes. Lighter colors indicate larger values.

Parameters		Errors on V		Loss of optimality		Time (s)
Δt	Δx	global	at $t = 0$	worst	mean	
4.17e-01	1.93e-01	1.36e-01	2.64e-01	2.37e-01	9.94e-02	1.04
2.78e-01	9.36e-02	1.00e-01	1.61e-01	1.83e-01	6.85e-02	1.15
2.08e-01	4.38e-02	6.82e-02	1.16e-01	1.36e-01	4.23e-02	1.70
1.39e-01	2.05e-02	6.15e-02	1.19e-01	8.99e-02	2.90e-02	5.34
9.62e-02	9.61e-03	4.55e-02	8.83e-02	6.52e-02	2.29e-02	38.81
6.58e-02	4.50e-03	2.55e-02	4.95e-02	5.23e-02	1.45e-02	422.74

Table 1: Errors for the robot problem, for decreasing values of $(\Delta t, \Delta x)$ satisfying the inverse CFL condition $\Delta x/\Delta t^2 \simeq 1$. Columns "global" and "at $t = 0$" report the relative error on V over the whole space-time domain and at the initial time, respectively. Columns "worst" and "mean" report the maximum and average of the loss of optimality (58) over the six initial points of $\mathcal{I}$.

Acknowledgments The authors are grateful to the anonymous referee for his/her careful reading of the manuscript, that lead to a significant improvement of the presentation and simplifications in certain arguments.

7 Appendix

This Appendix checks that the set $\mathcal{E}$ is an admissible set of energies satisfying the assumption **[A5.6]**, where

$$\mathcal{E} \coloneqq \{-\alpha\kappa \circ d(\cdot, x_0) \mid \alpha \geqslant 0,\ x_0 \in \Omega\},$$

with $\kappa(r) = r^2/2$ if $r \in [0, 1]$, and $r - 1/2$ otherwise. We first prove separability and completeness with respect to $|\cdot|_{1,\infty}$.

Proof of Lemma 3.5. We begin by separability. Let $\mathcal{A} \subset \mathbb{R}^+, \mathcal{O} \subset \Omega$ be a countable dense subsets. Consider $E \coloneqq -\alpha\kappa \circ d(\cdot, x_0)$ and $\overline{E} \coloneqq -\overline{\alpha}\kappa \circ d(\cdot, \overline{x}_0)$. For any $(x, p) \in \mathrm{T}_x\,\Omega$ such that

$|p|_x = 1$, one has

$$\begin{aligned}\left|d_x E(p) - d_x \overline{E}(p)\right| &= \left|\alpha\kappa'(d(x,x_0))\left\langle \frac{\overrightarrow{xx_0}}{d(x,x_0)}, p\right\rangle_x - \overline{\alpha}\kappa'(d(x,\overline{x}_0))\left\langle \frac{\overrightarrow{x\overline{x}_0}}{d(x,\overline{x}_0)}, p\right\rangle_x\right| \\ &\leqslant |\alpha - \overline{\alpha}| + \overline{\alpha}d(x_0,\overline{x}_0) + \overline{\alpha}\left|\left\langle \frac{\overrightarrow{xx_0}}{d(x,x_0)}, p\right\rangle_x - \left\langle \frac{\overrightarrow{x\overline{x}_0}}{d(x,\overline{x}_0)}, p\right\rangle_x\right|. \end{aligned} \tag{59}$$

Here we used that $\kappa'(r) = \max(0, \min(1, r))$ is nonnegative, bounded by 1 and 1-Lipschitz over $\mathbb{R}^+$. By Remark 2.5, $|\left\langle \frac{\overrightarrow{xx_0}}{d(x,x_0)}, p\right\rangle_x - \left\langle \frac{\overrightarrow{x\overline{x}_0}}{d(x,\overline{x}_0)}, p\right\rangle_x| \leqslant d_x(\frac{\overrightarrow{xx_0}}{d(x,x_0)}, \frac{\overrightarrow{x\overline{x}_0}}{d(x,\overline{x}_0)}) \leqslant d(x_0,\overline{x}_0)$. If now E is fixed, we may choose $\overline{\alpha} \in \mathcal{A}$ and then $\overline{x}_0 \in \mathcal{O}$ rendering (59) arbitrarily small regardless of (x,p), showing that $\mathcal{E}$ is separable in $(\mathcal{E}, |\cdot|_{1,\infty})$.

We turn to completeness. Let $(\overline{E}^n)_n$ be a Cauchy sequence in $(\mathcal{E}, |\cdot|_{1,\infty})$. Denote $E^n = -\alpha^n\kappa \circ d(\cdot, x_0^n) - c_n$ the member of the equivalence class of $\overline{E}^n$ that vanishes in $0_\Omega \in \Omega$. In particular, $\alpha^n = |E^n|_{1,\infty} = \mathrm{Lip}(E^n)$ converges towards a limit $\alpha \in \mathbb{R}^+$. If $\alpha = 0$, then $E^n \to_n 0_{\mathbb{E}}$. Assume by now that $\alpha > 0$, and (up to a shift of indexes) $\alpha^n \geqslant \alpha/2 > 0$ for all $n \in \mathbb{N}$. Let us show that this uniform coercivity forces the sequence $(x_0^n)_n$ to stays in a compact. The sequence $(E^n)_n$ is equiLipschitz by construction, and for each $x \in \Omega$, one has $|E^n(x) - E^m(x)| \leqslant \sup_k \left|E^k\right|_{1,\infty} d(x, 0_\Omega)$. By Arzelà-Ascoli, up to a subsequence, we may assume that E^n converges locally uniformly to some Lipschitz E, that satisfies

$$\begin{aligned}\lim_{d(x,0_\Omega)\to\infty} E(x) &\leqslant \lim_{d(x,0_\Omega)\to\infty} E^n(x) + \sup_k \left|E - E^k\right|_{1,\infty} \\ &\leqslant \lim_{d(x,0_\Omega)\to\infty} -\frac{\alpha}{2}\kappa(d(x,x_0)) - c_n + \sup_k \left|E - E^k\right|_{1,\infty} = -\infty.\end{aligned}$$

In consequence, the level sets $\{E \geqslant c\}$ are bounded in Ω, thus compact. As x_0^n is the maximum of E^n, we have

$$x_0^n \in \{E^n \geqslant 0\} \subset \left\{E \geqslant -\sup_k \left|E - E^k\right|_{1,\infty}\right\},$$

showing that up to a subsequence, x_0^n admits a limit $x_0 \in \Omega$. Invoking (59) again, the associated subsequence $(E^{n_k})_k$ converges towards $E \coloneqq -\alpha\kappa \circ d(\cdot, x_0)$ with respect to $|\cdot|_{1,\infty}$, and the Cauchy sequence $(E^n)_n$ admits $E \in \mathcal{E}$ as a limit. □

We now show that $\mathcal{E}$ satisfies **[A5.6]**, in the sense that for any compact $A \subset \mathcal{E}$ and $B \subset \Omega$, there exist $t > 0$ and reparametrized geodesics $(\gamma^{E,x})_{E\in A, x\in B}$ all defined on $[0,t]$ such that

$$\lim_{h\searrow 0} \sup_{E\in A, x\in B} \frac{d\left(\mathcal{G}_{\mathcal{F}}(E)(h,x), \gamma_h^{E,x}\right)}{h} = 0.$$

We do so by computing explicitly the gradient flows, and deriving a uniform estimate.

Proof. Let $A \subset \mathcal{E}$ be compact for the topology induced by $|\cdot|_{1,\infty}$. Each $E \in A$ writes as $-\alpha^E\kappa \circ d(\cdot, x_0^E)$. First notice that $E \mapsto \alpha^E$ is bounded over A: indeed (assuming that diam $\Omega > 0$) there is some $\underline{\kappa} > 0$ such that for all $x_0^E \in \Omega$, there exists $x \in \Omega$ with $\kappa'(d(x,x_0)) \geqslant \underline{\kappa}$. Hence

$$|E|_{1,\infty} = \sup_{(x,v)\in \mathrm{T}\,\Omega, |v|_x = 1} \left|\alpha^E\kappa'(d(x,x_0^E))\left\langle \frac{\overrightarrow{xx_0^E}}{d(x,x_0^E)}, v\right\rangle_x\right| \geqslant \alpha^E\underline{\kappa},$$

and the boundedness of A implies that of α^E. Since any energy E is concave and Lipschitz, its gradient flow is uniquely defined, and one checks that it is given by

$$\vartheta(h,x) = (1-\tau_h)\, x \oplus \tau_h x_0^E,$$

where $\tau \in \mathcal{C}^1(\mathbb{R}^+;[0,1])$ is identically 0 if $x = x_0^E$, and the unique solution of the Cauchy-Lipschitz ODE

$$\dot{\tau}_h = \frac{\alpha^E \kappa'(d(x,x_0^E)(1-\tau_h))}{d(x,x_0^E)}, \qquad \tau_0 = 0$$

otherwise. This corresponds to following the geodesic linking x to x_0^E with a speed proportional to α^E if x is sufficiently far from x_0^E, and exponentially decreasing in the region where κ becomes quadratic. Let $t_E \coloneqq 1/\dot{\tau}_0$ if $\dot{\tau}_0 > 0$, and 1 if $\dot{\tau}_0 = 0$. The time t_E is lower bounded uniformly on $E \in A$: indeed,

$$\dot{\tau}_0 = \frac{\alpha^E \kappa'(d(x,x_0^E))}{d(x,x_0^E)} = \begin{cases} \frac{\alpha^E}{d(x,x_0^E)} & d(x,x_0^E) \geqslant 1 \\ \alpha^E & d(x,x_0^E) < 1 \end{cases} \leqslant \alpha^E$$

is bounded over the compact A. Hence $t_E \geqslant \max(1, \min_{E\in A} 1/\alpha^E) > 0$.

For any $x \in B$, consider the reparametrized geodesic

$$\gamma^{E,x} : [0,1] \cap [0,t_E] \to \Omega, \qquad \gamma_s^{E,x} \coloneqq \left(1 - \frac{s}{t_E}\right) x \oplus \frac{s}{t_E} x_0^E.$$

The initial velocity of $\gamma^{E,x}$ coincides with $\nabla_x E = \vartheta(\cdot,x)_0^+$ by construction. Moreover,

$$d\left(\vartheta(h,x), \gamma_h^{E,x}\right) \leqslant \left|\tau_h - \frac{h}{t_E}\right| d(x,x_0^E) = |\tau_h - h\dot{\tau}(0)|\, d(x,x_0^E) \leqslant \int_{s=0}^{h}\int_{r=0}^{s} |\ddot{\tau}_r|\, d(x,x_0^E) dr ds.$$

Here $|\ddot{\tau}_r|$ is defined almost everywhere by

$$|\ddot{\tau}_r| = \begin{cases} 0 & \text{if } d(x,x_0^E)(1-\tau_r) > 1, \\ (\alpha^E)^2(1-\tau_r) & \text{if } d(x,x_0^E)(1-\tau_r) < 1. \end{cases}$$

Hence $|\ddot{\tau}_r|\, d(x,x_0^E) \leqslant (\alpha^E)^2$, and the estimate $d\left(\vartheta(h,x), \gamma_h^{E,x}\right) \leqslant \frac{h^2}{2}(\alpha^E)^2$ suffices to show that the approximation is uniform over any compact set. □

References

[Ach+11] Y. Achdou, F. Camilli, A. Cutrì, and N. Tchou. "Hamilton-Jacobi equations on networks". In: *IFAC Proceedings Volumes* 44.1 (Jan. 2011), pp. 2577–2582.

[AKP23] S. Alexander, V. Kapovitch, and A. Petrunin. *Alexandrov geometry: foundations*. Graduate studies in mathematics 236. Providence, Rhode Island: American Mathematical Society, 2023.

[AB06] C. Aliprantis and K. Border. *Infinite Dimensional Analysis*. Berlin/Heidelberg: Springer-Verlag, 2006.

[AF14] L. Ambrosio and J. Feng. "On a class of first order Hamilton–Jacobi equations in metric spaces". In: *Journal of Differential Equations* 256.7 (Apr. 2014), pp. 2194–2245.

[AGS05] L. Ambrosio, N. Gigli, and G. Savaré. *Gradient Flows*. Lectures in Mathematics ETH Zürich. Basel: Birkhäuser-Verlag, 2005.

[Aub99] J. P. Aubin. *Mutational and Morphological Analysis*. Systems & Control: Foundations & Applications. Birkhäuser Boston, MA, 1999.

[AF90] J.-P. Aubin and H. Frankowska. *Set-Valued Analysis*. Boston: Birkhäuser, 1990.

[Aum65] R. J. Aumann. "Integrals of set-valued functions". In: *Journal of Mathematical Analysis and Applications* 12.1 (Aug. 1965), pp. 1–12.

[BF22a] Z. Badreddine and H. Frankowska. "Solutions to Hamilton–Jacobi equation on a Wasserstein space". In: *Calculus of Variations and Partial Differential Equations* 61.1 (Feb. 2022), p. 9.

[BF22b] Z. Badreddine and H. Frankowska. "Viability and invariance of systems on metric spaces". In: *Nonlinear Analysis* 225 (Dec. 2022), p. 113133.

[BC97] M. Bardi and I. Capuzzo-Dolcetta. *Optimal Control and Viscosity Solutions of Hamilton-Jacobi-Bellman Equations*. Boston, MA: Birkhäuser Boston, 1997.

[BBC14] G. Barles, A. Briani, and E. Chasseigne. "A Bellman approach for regional optimal control problems in $\mathbb{R}^N$". In: *SIAM Journal on Control and Optimization* 52.3 (Jan. 2014), pp. 1712–1744.

[Bar94] G. Barles. *Solutions de viscosité des équations de Hamilton-Jacobi*. Mathématiques & applications 17. Springer-Verlag, 1994.

[BC24] G. Barles and E. Chasseigne. *On Modern Approaches of Hamilton-Jacobi Equations and Control Problems with Discontinuities*. Vol. 104. Progress in Nonlinear Differential Equations and Their Applications. Cham: Springer Nature Switzerland, 2024.

[Bha07] R. Bhatia. *Positive definite matrices*. Princeton series in applied mathematics. Princeton (N.J.): Princeton university press, 2007.

[BF24] B. Bonnet-Weill and H. Frankowska. "Carathéodory theory and a priori estimates for continuity inclusions in the space of probability measures". In: *Nonlinear Analysis* 247 (Oct. 2024), p. 113595.

[BH99] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer, 1999.

[CFS13] F. Camilli, A. Festa, and D. Schieborn. "An approximation scheme for a Hamilton–Jacobi equation defined on a network". In: *Applied Numerical Mathematics*. VISCOSITY 2011 73 (Nov. 2013), pp. 33–47.

[CM13] F. Camilli and C. Marchi. "A comparison among various notions of viscosity solution for Hamilton–Jacobi equations on networks". In: *Journal of Mathematical Analysis and Applications* 407.1 (Nov. 2013), pp. 112–118.

[CSM13] F. Camilli, D. Schieborn, and C. Marchi. "Eikonal equations on ramified spaces". In: *Interfaces and Free Boundaries* 15.1 (2013), pp. 121–140.

[CL83] M. G. Crandall and P.-L. Lions. "Viscosity solutions of Hamilton-Jacobi equations". In: *Transactions of the American Mathematical Society* 277.1 (1983), pp. 1–42.

[DU77] J. Diestel and J. J. Uhl. *Vector Measures*. Vol. 15. Mathematical Surveys and Monographs. Providence, Rhode Island: American Mathematical Society, June 1977.

[FGŚ17] G. Fabbri, F. Gozzi, and A. Święch. *Stochastic Optimal Control in Infinite Dimension*. Vol. 82. Probability Theory and Stochastic Modelling. Cham: Springer International Publishing, 2017.

[FF14] M. Falcone and R. Ferretti. *Semi-Lagrangian Approximation Schemes for Linear and Hamilton-Jacobi Equations*. Other Titles in Applied Mathematics 133. Philadelphia: Society for Industrial and Applied Mathematics (SIAM), 2014, p. 319.

[Fil63] A. F. Filippov. "Differential equations with multivalued discontinuous right-hand side". In: *Dokl. Akad. Nauk SSSR* 151.1 (1963). Accessible at https://www.mathnet.ru/links/f213a6e48d91f81f201db1664a2622af/dan28255.pdf, pp. 65–68.

[FL23a] H. Frankowska and T. Lorenz. “Filippov’s Theorem for mutational inclusions in a metric space”. In: *Annali Scuola Normale Superiore - Classe di Scienze* (June 2023), pp. 1053–1094.

[FL23b] H. Frankowska and T. Lorenz. “Invariance of sets under mutational inclusions on metric spaces”. In: *Nonlinear Differential Equations and Applications NoDEA* 30.4 (July 2023), p. 45.

[GŚ15] W. Gangbo and A. Święch. “Existence of a solution to an equation arising from the theory of Mean Field Games”. In: *Journal of Differential Equations* 259.11 (Dec. 2015), pp. 6573–6643.

[GHN15] Y. Giga, N. Hamamuki, and A. Nakayasu. “Eikonal equations in metric spaces”. In: *Transactions of the American Mathematical Society* 367.1 (Jan. 2015), pp. 49–66.

[GN21] N. Gigli and F. Nobili. “A Differential Perspective on Gradient Flows on CAT(kappa)-Spaces and Applications”. In: *The Journal of Geometric Analysis* 31.12 (Dec. 2021), pp. 11780–11818.

[Hyt+16] T. Hytönen, J. van Neerven, M. Veraar, and L. Weis. *Analysis in Banach Spaces*. Cham: Springer International Publishing, 2016.

[IMZ13] C. Imbert, R. Monneau, and H. Zidani. “A Hamilton-Jacobi approach to junction problems and application to traffic flows”. In: *ESAIM: Control, Optimisation and Calculus of Variations* 19.1 (Jan. 2013), pp. 129–166.

[Jer22] O. Jerhaoui. “Viscosity theory of first order Hamilton Jacobi equations in some metric spaces”. PhD thesis. Paris: Institut Polytechnique de Paris, 2022.

[JZ23] O. Jerhaoui and H. Zidani. “Viscosity Solutions of Hamilton-Jacobi Equations in Proper CAT(0) Spaces”. In: *The Journal of Geometric Analysis* 34.2 (Dec. 2023), p. 47.

[Lio82] P.-L. Lions. *Generalized solutions of Hamilton-Jacobi equations*. Research notes in mathematics 69. Boston: Pitman, 1982.

[LSZ25] Q. Liu, N. Shanmugalingam, and X. Zhou. “Discontinuous eikonal equations in metric measure spaces”. In: *Transactions of the American Mathematical Society* 378.01 (Jan. 2025), pp. 695–729.

[Lor10] T. Lorenz. *Mutational Analysis: A Joint Framework for Cauchy Problems in and Beyond Vector Spaces*. Vol. 1996. Lecture Notes in Mathematics. Springer Berlin Heidelberg, 2010.

[May98] U. F. Mayer. “Gradient flows on nonpositively curved metric spaces and harmonic maps”. In: *Communications in Analysis and Geometry* 6.2 (1998), pp. 199–253.

[Rao13] Z. Rao. “Hamilton-Jacobi-Bellman approach for optimal control problems with discontinuous coefficients”. These de doctorat. Palaiseau, Ecole polytechnique, Jan. 2013.

[Roc70] R. T. Rockafellar. *Convex Analysis*. Princeton University Press, Dec. 1970.

[RV91] J. D. L. Rowland and R. B. Vinter. “Construction of optimal feedback controls”. In: *Systems & Control Letters* 16.5 (May 1991), pp. 357–367.

[San15] F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer International Publishing, 2015.

[SC13] D. Schieborn and F. Camilli. “Viscosity solutions of Eikonal equations on topological networks”. In: *Calculus of Variations and Partial Differential Equations* 46.3-4 (Mar. 2013), pp. 671–686.

[Stu03] K.-T. Sturm. “Probability measures on metric spaces of nonpositive curvature”. In: *Contemporary Mathematics*. Vol. 338. American Mathematical Society, 2003, pp. 357–390.

[Vil09] C. Villani. *Optimal Transport*. Vol. 338. Grundlehren der mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009.

Statements and Declarations This work was supported by the National Agency of Research through the grant COSS ANR-22-CE40-0010. The authors have no relevant financial or non-financial interests to disclose. For the purpose of open access, the authors have applied a CC-BY public copyright licence to any Author Accepted Manuscript (AAM) version arising from this submission.

Journal de l'École polytechnique
Mathématiques

Averil Aussedat

Locality of centered tangent cones in the Wasserstein space

Tome 13 (2026), p. 1575-1606.

https://doi.org/10.5802/jep.355

© L'auteur, 2026.

Cet article est mis à disposition selon les termes de la licence Licence internationale d'attribution Creative Commons BY 4.0.
https://creativecommons.org/licenses/by/4.0/

Publié avec le soutien
du Centre National de la Recherche Scientifique

Publication membre du
Centre Mersenne pour l'édition scientifique ouverte
www.centre-mersenne.org
e-ISSN : 2270-518X

Journal de l'École polytechnique
Mathématiques

Tome 13, 2026, p. 1575–1606 DOI: 10.5802/jep.355

LOCALITY OF CENTERED TANGENT CONES IN THE WASSERSTEIN SPACE

by Averil Aussedat

Abstract. — The geometric tangent cone to a probability measure μ is a set of measure-valued applications that are almost geodesics. This is a nonlocal condition, typically lost when conditioning the measure on a given set. We show that if one removes the barycenter of any element of the tangent cone, then the resulting set of centered measure fields is characterized by a local condition. Precisely, centered tangent fields must be concentrated on a family of vector subspaces attached to any point, and these subspaces correspond to the normal spaces to some sets of "dimension k" on which the measure μ is concentrated.

Résumé (Localité des cônes tangents centrés dans l'espace de Wasserstein)

Le cône tangent géométrique en une mesure μ est défini comme un ensemble d'applications à valeurs dans les mesures, chacune décrivant presque une géodésique. Cette propriété n'est pas locale et est généralement perdue quand la mesure est restreinte à un sous-ensemble. Nous montrons qu'il suffit de retirer le barycentre des éléments du cône tangent pour obtenir un ensemble de champs centrés lui-même caractérisé par une condition locale. Plus précisément, les champs tangents centrés sont concentrés sur une famille de sous-espaces vectoriels attachés en chaque point, correspondant aux espaces normaux à certains ensembles de « dimension k » sur lesquels la mesure μ est concentrée.

Contents

Introduction 1576
1. Preliminaries 1578
2. Closed convex cones of centered measure fields 1580
3. Tangent and solenoidal measure fields 1586
4. Properties of the decomposition 1592
Appendix 1603
References 1605

Mathematical subject classification (2020). — 28A15, 51FXX, 35R06.
Keywords. — Wasserstein spaces, tangent cone, DC functions.

This work benefited from the support of the grant ANR-22-CE40-0010 while the author was in Rouen, and from the ERC starting grant ConFine n°101078057 in Pisa.

Introduction

In his inspired article [Lot16], Lott observed that whenever μ is the restriction of a Hausdorff measure to a k-dimensional C^2 manifold, the measures in the geometric tangent cone to μ have a particular structure. It was already known that the barycenter of such a measure must belong to the L^2_μ closure of gradients of smooth compactly supported functions, as proved in full generality in [Gig11]. However, once the barycenter is removed, the remaining *centered part* turns out to be concentrated on the normal directions to the manifold. The purpose of this work is to investigate the general case, using a new line of reasoning that allows us to remove any assumption on the measure $\mu \in \mathscr{P}_2(\mathbb{R}^d)$.

Our main result is a combination of Theorem 3.10, Theorem 4.10 and Proposition 4.12 below, gathered here into a single statement. A set $A \subset \mathbb{R}^d$ is DC_k if, up to a permutation of variables, it is the graph of a map from $\mathbb{R}^k$ to $\mathbb{R}^{d-k}$ whose coordinates are all Differences of Convex (DC) functions. A set that can be covered by countably many DC_k sets is said to be σ-$\mathrm{DC_k}$.

Theorem. — *Let $\mu \in \mathscr{P}_2(\mathbb{R}^d)$. There exists a unique decomposition $\mu = \sum_{k=0}^d m_k \mu^k$ into mutually singular measures such that $m_k\mu^k$ gives 0 mass to DC_{k-1} sets and is concentrated on a σ-DC_k set A_k. Moreover, the centered tangent cone Tan^0_μ splits into $\sum_{k=0}^d m_k \mathsf{Tan}^0_{\mu^k}$, and a centered measure ξ^k belongs to $\mathsf{Tan}^0_{\mu^k}$ if and only if it is concentrated on the normal spaces to A_k, which exist μ^k-almost everywhere.*

For instance, in dimension $d = 2$, let $\mu = \frac{1}{3}\mu^0 + \frac{1}{3}\mu^1 + \frac{1}{3}\mu^2$, where μ^0 is atomic, μ^1 nonatomic and supported on $[0,1] \times \{0\}$, and μ^2 is absolutely continuous. Let ξ be a probability measure on $\mathrm{T}\,\mathbb{R}^d \simeq \mathbb{R}^d \times \mathbb{R}^d$ with finite second moment, first marginal μ and barycenter 0 in each fiber. Since the μ^k are mutually singular, there is a unique way to write $\xi = \frac{1}{3}\sum_{k=0}^2 \xi^k$ with $\pi_{x\#}\xi^k = \mu^k$ for each k. The theorem then states that ξ is tangent to μ if and only if ξ^1 is concentrated on pairs (x, v) with v orthogonal to $[0,1] \times \{0\}$, and ξ^2 is concentrated on (x, v) with $v \in (\mathbb{R}^2)^\perp$, hence $\xi^2 = (\mathrm{id}, 0)_\# \mu^2$.

In dimension one, the splitting $\mu = m_0\mu^0 + m_1\mu^1$ reduces to the atomic/diffuse decomposition. In this case, the statement was already proved in [SW14] (see also [Aus25a]): it says that a centered field $\xi = m_0\xi^0 + m_1\xi^1$ is tangent if and only if $\xi^1 = (\mathrm{id}, 0)_\#\mu^1$. This case is rather special, since σ-DC_0 sets are exactly countable sets, and the only σ-DC_1 set is $\mathbb{R}$ itself. In the general case, one has to pay attention to the regularity of the sets A_k for the intermediate dimensions $0 < k < d$.

The relation between DC_{d-1} sets and optimal transport plans for the 2-Wasserstein distance appeared in the work of [Gig11], which characterizes the measures μ satisfying the conclusion of the Brenier–McCann theorem [Bre91]. Precisely, a measure μ that gives 0 mass to any DC_{d-1} set (and consequently, to any DC_k set for $k \leqslant d-1$) does not admit optimal plans that *split mass*, in the sense that for any other measure ν and optimal transport plan α between μ and ν, there exists a map $f \in L^2_\mu(\mathbb{R}^d; \mathbb{R}^d)$ such that $\alpha = (\mathrm{id}, f)_\#\mu$. The argument runs as follows: optimal plans are known to be concentrated on c-cyclically monotone sets, which, in the case of $c(x, y) = |x - y|^2$,

coincide with subdifferentials of convex functions. If an optimal plan splits mass at some point x, then the convex functions in question must admit several elements in their subdifferential at x, and hence fail to be differentiable there. A remarkable theorem of Zajíček [Zaj79] shows that the non-differentiability set of a convex function can be covered by countably many DC_{d-1} sets, and that any DC_{d-1} set is contained in the non-differentiability set of some convex function. Hence μ allows mass splitting if and only if it charges a DC_{d-1} set.

This work generalizes Gigli's theorem by identifying which part of μ allows optimal plans to split mass in exactly $d-k$ directions, again using Zajíček's theorem. Moreover, and this is the main point, each part is concentrated on a set A_k which admits normal spaces of dimension $d-k$ at μ-almost every point, and the centered tangent measure fields are exactly the measure fields concentrated on these normal spaces. The intuition behind this decomposition is that tangent measures try to get away from μ as fast as possible, and therefore choose normal directions to the concentration sets.

It may seem surprising that no regularity assumption on A_k is needed: this is still a consequence of Zajíček's theorem, since, quite miraculously, singularities of convex functions are described using (differences of) convex functions. Indeed, the regularity of A_k is precisely the one that allows a convex function to fail to be differentiable in $d-k$ independent directions. By Zajíček's theorem, the singular subset of A_k (at which the normal directions are not defined) has the "same size" as non-differentiability sets of convex functions *in at least one more direction*. In the decomposition, these sets are seen by the lower-order measures μ^j for $j<k$, so that each A_k has a tangent plane μ^k-almost everywhere.

Since many properties of Tan^0_μ depend only on the fact that this set is convex in a certain sense, we first focus on a generic horizontally convex closed cone of centered fields. However, the tangent cone is constructed in a canonical way for any μ, and can be compared for different measures. As a by-product, we show that the metric orthogonal Sol_μ of Tan_μ is closed with respect to the Wasserstein distance over $\mathrm{T}\,\mathbb{R}^d$.

We point out that the statement only concerns centered fields. This is a natural assumption, since the opposite case of barycentric fields (induced by a map) cannot be described by a local condition. However, the following can be deduced for any measure field $\xi\in\mathscr{P}_2(\mathrm{T}\,\mathbb{R}^d)_\mu$. First, one can always write $\xi=(\pi_x,\pi_v+b(\pi_x))_\#\xi^0$, where $b\in L^2_\mu$ is the barycenter map, and ξ^0 is centered. It is known that $\xi\in\mathsf{Tan}_\mu$ if and only if both $(\mathrm{id},b)_\#\mu$ and ξ^0 are tangent measure fields: hence the theorem applies to ξ^0. Moreover, anticipating the notation of Section 3, we can further decompose $b=b_D+b_{D^\perp}$, where $b_D(x)$ is the projection of $b(x)$ onto the vector space $D(x)$, and $b_{D^\perp}=b-b_D$. Corollary 3.3 implies that the measure field $\frac{1}{2}\left[(\mathrm{id},-b_{D^\perp})_\#\mu+(\mathrm{id},b_{D^\perp})_\#\mu\right]$ is tangent, regardless of whether b induces a tangent field or not. From this, one recovers that $(\mathrm{id},b_{D^\perp})_\#\mu$ is itself tangent, once again regardless of b (see [Aus25b, Prop. 5.4.4]). Hence, the fact that $(\mathrm{id},b)_\#\mu$ is tangent restricts only its component b_D. This extends the observation of [Lot16] that the barycenter of a tangent map is only constrained through its tangential component.

This article is organized as follows: Section 2 is devoted to closed convex cones of centered measure fields. Section 3 applies the previous results to the centered geometric tangent cone Tan^0_μ and its metric orthogonal Sol^0_μ, providing the candidate decomposition $\mu = \sum m_k \mu^k$. The fact that each μ^k is concentrated on a σ-$\mathrm{DC_k}$ set, and gives 0 mass to DC_{k-1} sets, is proved in Section 4.2. Section 4.3 shows that the normal directions are well-defined μ^k-almost everywhere, and characterize centered tangent measures. In addition, it is shown that the tangent measures in the sense of Preiss are supported on planes associated to Sol^0_μ at μ-almost every point. The appendix collects some lengthy proofs.

Acknowledgements. — The author would like to thank Adolfo Arroyo-Rabasa for pointing out relevant literature, as well as Guy Bouchitté for comments on a preliminary version of this work.

1. Preliminaries

In the sequel, $\Omega = \mathbb{R}^d$ for $d \geqslant 1$. We keep the notation Ω to distinguish points $x \in \Omega$ from vectors $v \in \mathbb{R}^d$. Open balls with center x and radius $r \geqslant 0$ are denoted $\mathscr{B}(x, r)$. The closure of a set $A \subset \Omega$ is denoted $\overline{A}$, and its complement A^c.

Wasserstein spaces. — The set of Borel probability measures on a Polish space X is denoted $\mathscr{P}(X)$. If $X = X_1 \times X_2$ is a product space and $\mu \in \mathscr{P}(X)$, we write $\mu = \mu(dx_1, dx_2)$ to name the variables of X, to be used in the canonical projections $\pi_{x_i} : (x_1, x_2) \to x_i$ for $i \in \{1, 2\}$. The set $\mathscr{P}_2(X) \subset \mathscr{P}(X)$ collects the measures with finite second moment, i.e., with $\int_{x \in X} d^2(x, o)\, d\mu < \infty$ for some $o \in X$.

A measurable function $f : X \to Y$ between two Polish spaces X, Y induces a function $f_\# : \mathscr{P}(X) \to \mathscr{P}(Y)$ by $(f_\#\mu)(A) = \mu(f^{-1}(A))$ for any measurable $A \subset Y$. We refer to $f_\#\mu$ as the *pushforward* of μ by f.

Given $(\mu, \nu) \in \mathscr{P}_2(X) \times \mathscr{P}_2(Y)$, let the set of *transport plans* between μ and ν be

$$\Gamma(\mu, \nu) \coloneqq \big\{\alpha = \alpha(dx, dy) \in \mathscr{P}(X \times Y) \mid \pi_{x\#}\alpha = \mu \text{ and } \pi_{y\#}\alpha = \nu\big\}.$$

The 2-Wasserstein distance $W(\mu, \nu)$ between μ and ν is given by

$$W^2(\mu, \nu) \coloneqq \inf_{\alpha \in \Gamma(\mu,\nu)} \int_{(x,y) \in X \times Y} d^2(x, y) d\alpha(x, y).$$

We refer the reader to [San15] for an introduction to this distance, and only mention that the infimum is attained on a set of *optimal transport plans* denoted $\Gamma_o(\mu, \nu)$.

Measure fields. — Let $\mathrm{T}\,\Omega \coloneqq \{(x, v) \mid x \in \Omega,\ v \in \mathrm{T}_x\,\Omega\}$ be the tangent bundle of Ω, isometric to $\Omega \times \mathbb{R}^d$. When useful, we also denote

$$\mathrm{T}^n\,\Omega \coloneqq \big\{(x, v_1, v_2, \dots, v_n) \mid x \in \Omega,\ v_i \in \mathrm{T}_x\,\Omega \text{ for } i \in [\![1, n]\!]\big\}.$$

For any measure $\mu \in \mathscr{P}_2(\Omega)$, denote by $\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ the set of Borel probability measures $\xi = \xi(dx, dv)$ on $\mathrm{T}\,\Omega$, with finite second moment, and satisfying the marginal condition $\pi_{x\#}\xi = \mu$. These elements can be seen as measure-valued maps in L^2_μ, generalizing vector fields; for this reason, we refer to them as *measure fields*. The

measure fields of the form $\xi = (\pi_x, \pi_y - \pi_x)_\# \eta$ for $\eta \in \Gamma_o(\mu, \nu)$ and $\nu \in \mathscr{P}_2(\Omega)$ are called *velocities of geodesics*. Following [Gig08], we introduce a metric structure on $\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ that takes into account the common marginal μ. Namely, given $\xi, \zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$, define

$$\Gamma_\mu(\xi, \zeta) \coloneqq \big\{\alpha = \alpha(dx, dv, dw) \in \mathscr{P}_2(\mathrm{T}^2\,\Omega) \mid (\pi_x, \pi_v)_\# \alpha = \xi \text{ and } (\pi_x, \pi_w)_\# \alpha = \zeta\big\}.$$

These plans only move mass between pairs (x, v) and (y, w) such that $x = y$. They define a distance W_μ by

$$W_\mu^2(\xi, \zeta) \coloneqq \inf_{\alpha \in \Gamma_\mu(\xi,\zeta)} \int_{(x,v,w) \in \mathrm{T}^2\,\Omega} |v - w|^2 \, d\alpha(x, v, w).$$

The distance to the zero measure field is abbreviated as $\|\xi\|_\mu = \big(\int_{(x,v)} |v|^2 d\xi\big)^{1/2}$. The distance W_μ induces a *metric scalar product* $\langle \cdot, \cdot \rangle_\mu : (\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu)^2 \to \mathbb{R}$ by

$$\langle \xi, \zeta \rangle_\mu \coloneqq \frac{1}{2}\left[\|\xi\|_\mu^2 + \|\zeta\|_\mu^2 - W_\mu^2(\xi, \zeta)\right] = \sup_{\alpha \in \Gamma_\mu(\xi,\zeta)} \int_{(x,v,w) \in \mathrm{T}^2\,\Omega} \langle v, w \rangle d\alpha(x, v, w).$$

Subsets of measure fields. — The *barycenter* of a measure field ξ is the unique element $b \in L^2_\mu(\Omega; \mathbb{R}^d)$ satisfying $\int \varphi(x, v) d\xi = \int \varphi(x, b(x)) d\mu$ for any quadratically growing $\varphi \in \mathcal{C}(\mathrm{T}\,\Omega; \mathbb{R})$ that is linear in v. The measure fields with barycenter 0 are called *centered*, and the set of centered measure fields is denoted $\mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$.

For $\lambda \in \mathbb{R}$ and $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$, define $\lambda \cdot \xi \coloneqq (\pi_x, \lambda \pi_v)_\# \xi$. We say that a subset $\mathsf{Set}_\mu \subset \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ is

– a (positive) cone if $\lambda \cdot \xi \in \mathsf{Set}_\mu$ whenever $\xi \in \mathsf{Set}_\mu$ and $\lambda \geqslant 0$;

– (horizontally) convex if for any $\lambda \in [0, 1]$, $\xi, \zeta \in \mathsf{Set}_\mu$ and $\alpha \in \Gamma_\mu(\xi, \zeta)$, the measure field given by $(\pi_x, (1 - \lambda)\pi_v + \lambda \pi_w)_\# \alpha$ also belongs to Set_μ.

We often omit the adjective "positive" in the sequel. Horizontal convexity is stronger than geodesic convexity, since any transport plan is allowed to produce interpolating curves. It should also be distinguished from convexity in the Banach space of measures, which involves curves of the form $(1 - \lambda)\xi + \lambda\zeta$.

Grassmannian sections. — For brevity, let us agree that a *Grassmannian section* $D : \Omega \rightrightarrows \mathbb{R}^d$ is a measurable multivalued map such that $D(x)$ is a vector subspace, possibly reduced to $\{0\}$, for all $x \in \Omega$. Here measurability is understood in the classical sense, i.e., if for any open set $\mathcal{O} \subset \mathbb{R}^d$, the set $\{x \in \Omega \mid D(x) \cap \mathcal{O} \neq \varnothing\}$ is measurable. The *graph* of D is the set $\operatorname{Graph} D \coloneqq \{(x, v) \mid v \in D(x)\}$, which is a measurable subset of $\mathrm{T}\,\Omega$ [Roc69, Cor. 2.2]. The following lemma will be used.

Lemma 1.1. — *Let $\mu \in \mathscr{P}_2(\Omega)$ and $D : \Omega \rightrightarrows \mathbb{R}^d$ be a Grassmannian section. The set of $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ such that $\xi(\operatorname{Graph} D) = 1$ is closed with respect to the Wasserstein distance $W_{\mathrm{T}\,\Omega}$ on the tangent bundle.*

Proof. — Let $(\xi_n)_n \subset \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ be a sequence such that $\xi_n(\operatorname{Graph} D) = 1$ which converges towards $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ with respect to $W_{\mathrm{T}\,\Omega}(\cdot, \cdot)$. Using a Castaing representation of D [Roc69, Th. 3], one can find d measurable functions $u_1, \ldots, u_d : \Omega \to \mathbb{R}^d$

such that $D(x) = \operatorname{span}\{u_1(x), \dots, u_d(x)\}$ for any $x \in \Omega$. Let $\varepsilon > 0$. By Lusin's theorem [Fed96, 2.3.5], there exists a measurable set $B_\varepsilon \subset \Omega$ such that $\mu(B_\varepsilon) \geqslant 1 - \varepsilon$ and each u_j coincides with a continuous function on B_ε. Since $\mu(B_\varepsilon) = \sup_{\overline{C_\varepsilon} \subset B_\varepsilon} \mu(\overline{C_\varepsilon})$, there is a closed set $C_\varepsilon \subset B_\varepsilon$ closed such that $\mu(C_\varepsilon) \geqslant 1 - 2\varepsilon$. The set

$$G_\varepsilon \coloneqq \bigcup_{x \in C_\varepsilon} \{x\} \times D(x) = \operatorname{Graph} D \cap C_\varepsilon \times \mathbb{R}^d = \bigcup_{x \in C_\varepsilon} \{x\} \times \operatorname{span}\{u_1(x), \dots, u_j(x)\}$$

is measurable and closed by the first and second equalities, respectively. Moreover, $\xi_n(G_\varepsilon) = \mu(C_\varepsilon) \geqslant 1 - 2\varepsilon$ for all $n \in \mathbb{N}$. Since $\zeta \mapsto \int \mathbb{1}_{G_\varepsilon} d\zeta$ is upper semicontinuous with respect to $W_{\mathrm{T}\,\Omega}$ [AGS05, Lem. 5.1.7],

$$\xi(\operatorname{Graph} D) \geqslant \xi(G_\varepsilon) \geqslant \limsup_{n \to \infty} \xi_n(G_\varepsilon) \geqslant 1 - 2\varepsilon.$$

Letting $\varepsilon \searrow 0$, we conclude that $\xi(\operatorname{Graph} D) = 1$. □

2. Closed convex cones of centered measure fields

Throughout this section, we consider a set $\mathsf{Set}^0_\mu \subset \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$ with the following properties.

Assumption 2.1. — The set Set^0_μ is a W_μ-closed, horizontally convex cone of centered measure fields.

Here and below, "cone" means "positive cone". The main motivation is the study of Tan^0_μ and Sol^0_μ carried out in Section 3, which explains our notation.

Proposition 2.2 (Local characterization). — *A set Set^0_μ satisfies Assumption 2.1 if and only if there exists a Grassmannian section $D : \Omega \rightrightarrows \mathbb{R}^d$ such that*

$$\xi \in \mathsf{Set}^0_\mu \iff \left[\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu \text{ and } \xi(\operatorname{Graph} D) = 1\right].$$

As an example, in dimension $d = 2$, the set of centered ξ satisfying $\langle v, (1,0)\rangle = 0$ almost everywhere is W_μ-closed, horizontally convex, stable under multiplication by any scalar, and characterized by the constant map $D \equiv \operatorname{vect}\{(0,1)\}$.

Remark 2.3 (Negative cone). — A direct consequence of Proposition 2.2 is that Set^0_μ is stable under multiplication by any scalar, including negative ones. Such stability was already noticed in the case of the tangent cone in [Gig08], using ad hoc arguments; in this case, the property even holds for map-induced fields. Here it is essential to restrict to centered fields; indeed, already in dimension $d = 1$, the set of measure fields concentrated on (x, v) with $v \geqslant 0$ provides a closed and horizontally convex cone that cannot be characterized by concentration on the graph of a Grassmannian section.

Remark 2.4 (Vertical convexity). — A somewhat surprising implication of Proposition 2.2 is that Set^0_μ is convex as a subset of the Banach space of measures, in the sense that $(1-\lambda)\xi_0 + \lambda\xi_1 \in \mathsf{Set}^0_\mu$ whenever $\xi_0, \xi_1 \in \mathsf{Set}^0_\mu$ and $\lambda \in [0,1]$. Indeed, the superposition $(1-\lambda)\xi_0 + \lambda\xi_1$ remains centered, and concentrated on the graph of D.

The aim of this section is to prove Proposition 2.2. Our strategy is to represent Set^0_μ as the set of centered measure fields that are orthogonal to a family of "simple" measure fields, defined in (3) below. These fields will be used to construct D. The key observation, which justifies our interest in centered fields, is the following.

Lemma 2.5 (Centered is local). — *Let ξ, ζ be measure fields, with ξ centered. Then $\langle \xi, \zeta \rangle_\mu \geqslant 0$, and equality holds if and only if any disintegrations $(\xi_x)_x$, $(\zeta_x)_x$ satisfy $\langle \xi_x, \zeta_x \rangle_{\delta_x} = 0$ for μ-almost every x.*

Proof. — Owing to [Gig08, Prop. 4.2], the metric scalar product can be written as

$$\langle \xi, \zeta \rangle_\mu = \int_{x \in \Omega} \langle \xi_x, \zeta_x \rangle_{\delta_x} d\mu(x). \tag{1}$$

Fix $x \in \Omega$. Identifying ξ_x, ζ_x with measures on $\mathscr{P}_2(\mathrm{T}_x\,\Omega)_{\delta_x} \simeq \mathscr{P}_2(\mathbb{R}^d)$, we may consider the product plan $\alpha_x \coloneqq \xi_x \otimes \zeta_x$ in the definition of the metric scalar product to get

$$\langle \xi_x, \zeta_x \rangle_{\delta_x} \geqslant \int_{v,w \in \mathbb{R}^d} \langle v, w \rangle d\alpha_x = \int_{v \in \mathbb{R}^d} v d\xi_x(v) \int_{w \in \mathbb{R}^d} w d\zeta_x(w).$$

As the first term of the right-hand side vanishes μ-almost everywhere, the integrand in (1) is nonnegative. Hence $\langle \xi, \zeta \rangle_\mu \geqslant 0$, and $\langle \xi, \zeta \rangle_\mu = 0$ implies $\langle \xi_x, \zeta_x \rangle_{\delta_x} = 0$ for μ-a.e. x. The converse follows directly from (1). □

This observation is used as follows. In the sequel, one often has to prove that $\langle \xi, \zeta \rangle_\mu = 0$ for some ξ, ζ lying in cones of interest. By specific arguments, one first obtains the inequality $\langle \xi, \zeta \rangle_\mu \leqslant 0$. In the Hilbertian case, when working with two-sided cones, one would typically take $-1 \cdot \zeta$ in place of ζ, and conclude with the same inequality that $\langle \xi, \zeta \rangle_\mu = 0$. However, we do not a priori assume that Set^0_μ is stable under multiplication by a negative scalar, and the missing inequality is provided by Lemma 2.5.

2.1. The metric orthogonal complement. — Our aim here is to write Set^0_μ as the orthogonal complement of its orthogonal complement. For convenience, we restrict to centered measure fields, since any measure field induced by a map is orthogonal to any centered measure field with respect to $\langle \cdot, \cdot \rangle_\mu$.

Lemma 2.6. — *Let Set^0_μ satisfy Assumption 2.1. Then so does*

$$(\mathsf{Set}^0_\mu)^{\perp 0} \coloneqq \big\{ \zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu \mid \langle \xi, \zeta \rangle_\mu = 0 \text{ for all } \xi \in \mathsf{Set}^0_\mu \big\},$$

i.e., $(\mathsf{Set}^0_\mu)^{\perp 0}$ is a W_μ-closed horizontally convex cone of centered measure fields.

Proof. — By construction, the scalar product is continuous with respect to W_μ, so that $(\mathsf{Set}^0_\mu)^{\perp 0}$ is W_μ-closed. Let $\zeta_0, \zeta_1 \in (\mathsf{Set}^0_\mu)^{\perp 0}$, $\alpha \in \Gamma_\mu(\zeta_0, \zeta_1)$ and $\lambda \in [0,1]$. The measure field $\zeta_\lambda \coloneqq (\pi_x, (1-\lambda)\pi_v + \lambda \pi_w)_\# \alpha$ is centered. To show that $\zeta_\lambda \in (\mathsf{Set}^0_\mu)^{\perp 0}$, let $\xi \in \mathsf{Set}^0_\mu$, and $\beta \in \Gamma_\mu(\zeta_\lambda, \xi)$. We construct a plan $\omega = \omega(dx, dv_0, dv_1, dw) \in \Gamma_\mu(\zeta_0, \zeta_1, \xi)$ such that $(\pi_x, \pi_{v_0}, \pi_{v_1})_\# \omega = \alpha$ and $(\pi_x, (1-\lambda)\pi_{v_0} + \lambda \pi_{v_1}, \pi_w)_\# \omega = \beta$ as follows: first change variables to consider $\widetilde{\alpha} \coloneqq (\pi_x, \pi_v, (1-\lambda)\pi_v + \lambda \pi_w)_\# \alpha \in$

$\Gamma_\mu(\zeta_0,\zeta_\lambda)$, then glue $\widetilde{\alpha}$ and β along their common marginal ζ_λ by [AGS05, Lem. 5.3.2] to produce $\widetilde{\omega} = \widetilde{\omega}(dx,dv,dv_\lambda,dw)$, and return to the original variables with $\omega \coloneqq (\pi_x,\pi_v,\lambda^{-1}(\pi_{v_\lambda}-(1-\lambda)\pi_v),\pi_w)_\#\widetilde{\omega}$. Then

$$\int \langle v,w\rangle d\beta = \int \langle (1-\lambda)v_0+\lambda v_1,w\rangle d\omega \leqslant (1-\lambda)\langle\zeta_0,\xi\rangle_\mu + \lambda\langle\zeta_1,\xi\rangle_\mu = 0.$$

Passing to the supremum over β, we get that $\langle\zeta_\lambda,\xi\rangle_\mu \leqslant 0$. By Lemma 2.5, the inequality $\langle\zeta_\lambda,\xi\rangle_\mu \geqslant 0$ holds since both fields are centered. Hence $\zeta_\lambda \in (\mathsf{Set}^0_\mu)^{\perp 0}$. Lastly, if $\zeta \in (\mathsf{Set}^0_\mu)^{\perp 0}$ and $\lambda \geqslant 0$, we have $\langle\xi,\lambda\cdot\zeta\rangle_\mu = \langle\lambda\cdot\xi,\zeta\rangle_\mu = 0$ for all $\xi \in \mathsf{Set}^0_\mu$, so that $\lambda\cdot\xi \in (\mathsf{Set}^0_\mu)^{\perp 0}$. □

The next result uses elementary tools from the pseudo-Hilbertian structure of $\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$, and is very close to similar statements in [Gig08, Aus25a]. For this reason, we defer the proof to the appendix.

Proposition 2.7 (Projection). — *Let C satisfy Assumption 2.1. Then any $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$ admits a unique metric projection $\pi^\mu_C\xi \in C$, realizing $\inf_{\zeta\in C} W_\mu(\xi,\zeta)$. Moreover, for any $\alpha \in \Gamma_{\mu,o}(\xi,\pi^\mu_C\xi)$, the measure field $(\pi_x,\pi_v-\pi_w)_\#\alpha$ is the metric projection of ξ on $C^\perp \coloneqq \{\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu \mid \langle\zeta,\gamma\rangle_\mu = 0\ \forall\gamma\in C\}$, and we have*

$$\langle\xi,\zeta\rangle_\mu \leqslant \langle\pi^\mu_C\xi,\zeta\rangle_\mu \qquad \forall\zeta\in C, \tag{2}$$

with equality if $\zeta = -1\cdot\zeta = (\pi_x,-\pi_v)_\#\zeta$.

Remark 2.8. — The conclusions of Proposition 2.7 are not sharp; as a consequence of Proposition 2.2, equality will hold in (2) for any $\zeta \in C$. One could additionally prove that $\Gamma_{\mu,o}(\xi,\pi^\mu_C\xi)$ reduces to a singleton that is induced by a map in a certain sense, following the reasoning of [Gig08, Th. 4.33].

Lemma 2.9. — *We have $\mathsf{Set}^0_\mu = ((\mathsf{Set}^0_\mu)^{\perp 0})^{\perp 0}$.*

Proof. — The inclusion $\mathsf{Set}^0_\mu \subset ((\mathsf{Set}^0_\mu)^{\perp 0})^{\perp 0}$ holds by definition. Conversely, let $\xi \in ((\mathsf{Set}^0_\mu)^{\perp 0})^{\perp 0}$, and denote by $\pi^\mu\xi$ its metric projection on Set^0_μ. By Proposition 2.7, for any $\alpha = \alpha(dx,dv,dw) \in \Gamma_{\mu,o}(\xi,\pi^\mu\xi)$, the measure field $(\pi_x,\pi_v-\pi_w)_\#\alpha$ is the projection of ξ on $(\mathsf{Set}^0_\mu)^{\perp 0}$. However, this projection is 0_μ, since $W^2_\mu(\xi,\zeta) = \|\xi\|^2_\mu + \|\zeta\|^2_\mu$ for any $\zeta \in (\mathsf{Set}^0_\mu)^{\perp 0}$. Hence $v = w$ for α-a.e. (x,v,w), and $\xi = \pi^\mu\xi \in \mathsf{Set}^0_\mu$. □

2.2. Reduction to symmetric measure fields. — It will be useful to introduce the following notation: with any $f \in L^2_\mu(\Omega;\mathbb{R}^d)$, we associate

$$\gamma_f \coloneqq \frac{1}{2}\left[(\mathrm{id},-f)_\#\mu + (\mathrm{id},f)_\#\mu\right] \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu. \tag{3}$$

Note that $(\pi_x,-\pi_v)_\#\gamma_f = \gamma_f$. For any $f,g \in L^2_\mu(\Omega;\mathbb{R}^d)$, one can use the transport plan $\frac{1}{2}\left[(\mathrm{id},f,g)_\#\mu + (\mathrm{id},-f,-g)_\#\mu\right]$ to get the estimate $W_\mu(\gamma_f,\gamma_g) \leqslant \|f-g\|_{L^2_\mu}$. Our interest in the measure fields of this form stems from the following lemma.

Lemma 2.10. — *Let $f \in L^2_\mu(\Omega;\mathbb{R}^d)$. A centered measure field ξ is orthogonal to γ_f if and only if for some (thus any) measurable $f^\cdot : \Omega \to \mathbb{R}^d$ in the equivalence class f, we have $\langle v, f^\cdot(x)\rangle = 0$ for ξ-almost every (x,v).*

Proof. — If $\langle v, f^{\cdot}(x)\rangle = 0$ for ξ-almost all (x,v), then any transport plan between ξ and γ is concentrated on (x,v,w) with $w = \pm f^{\cdot}(x)$, and necessarily $\langle v,w\rangle = 0$ almost everywhere. This shows that $\langle \xi,\gamma\rangle_\mu = 0$. Conversely, assume that ξ is centered and $\langle \xi,\gamma\rangle_\mu = 0$. Let $(\xi_x)_{x\in\Omega}$ be a disintegration of ξ, which can be chosen so that $\int_v v d\xi_x = 0$ for all $x\in\Omega$. By Lemma 2.5, we have, for μ-almost every x,

$$\int_{(v,w)\in(\mathbb{R}^d)^2} \langle v,w\rangle d\alpha_x = 0 \qquad \text{for all } \alpha_x \in \Gamma\big(\xi_x, \tfrac{1}{2}\left[\delta_{f^{\cdot}(x)} + \delta_{(-f^{\cdot}(x))}\right]\big). \tag{4}$$

If we show that (4) implies $\langle v, f^{\cdot}(x)\rangle = 0$ for ξ_x-almost every v, then the equality $\int |\langle v, f^{\cdot}(x)\rangle| d\xi = \int_x \int_v |\langle v, f^{\cdot}(x)\rangle| d\xi_x(v) d\mu(x)$ will ensure that $\langle v, f^{\cdot}(x)\rangle = 0$ for ξ-almost every (x,v). In the rest of the proof, we simplify the notation by letting $\zeta := \xi_x \in \mathscr{P}_2(\mathbb{R}^d)^0$ and $\overline{w} := f^{\cdot}(x) \in \mathbb{R}^d$.

Assume by contradiction that there is $\varepsilon_+ > 0$ such that $m_+ := \zeta\{\langle v,\overline{w}\rangle > \varepsilon_+\} > 0$. Then, as $\int_v \langle v,\overline{w}\rangle d\zeta = 0$, there exists $\varepsilon_- > 0$ such that $m_- := \zeta\{\langle v,\overline{w}\rangle < -\varepsilon_-\} > 0$. Let $m := \min(m_-, m_+) > 0$. Construct a transport plan $\alpha \in \Gamma(\zeta, \frac{1}{2}[\delta_{-\overline{w}} + \delta_{\overline{w}}])$ by sending a mass m/m_+ from $\zeta_+ := \zeta \llcorner \{\langle v,\overline{w}\rangle > \varepsilon_+\}$ to $\overline{w}$, a mass m/m_- from $\zeta_- := \zeta \llcorner \{\langle v,\overline{w}\rangle < -\varepsilon_-\}$ to $-\overline{w}$, and splitting the rest evenly; explicitly,

$$\alpha := m\Big[\Big(\frac{1}{m_+}\zeta_+\Big)\otimes\delta_{\overline{w}} + \Big(\frac{1}{m_-}\zeta_-\Big)\otimes\delta_{-\overline{w}}\Big] + \Big(\zeta - \frac{m}{m^+}\zeta_+ - \frac{m}{m_-}\zeta_-\Big)\otimes\Big[\Big(\frac{1}{2}-m\Big)(\delta_{\overline{w}}+\delta_{-\overline{w}})\Big].$$

Then $\pi_{v\#}\alpha = \zeta$, and $\pi_{w\#}\alpha = \frac{1}{2}[\delta_{\overline{w}} + \delta_{-\overline{w}}]$. Moreover,

$$\int_{(v,w)} \langle v,w\rangle d\alpha = \frac{m}{m_+}\int_v \langle v,\overline{w}\rangle d\zeta_+ + \frac{m}{m_-}\int_v \langle v,-\overline{w}\rangle d\zeta_- + 0 \geqslant m\varepsilon_+ + m\varepsilon_- > 0,$$

contradicting (4). Hence $\zeta\{\langle v,\overline{w}\rangle > \varepsilon_+\} = 0$ for all $\varepsilon_+ > 0$. Using that ζ is centered, we conclude that $\zeta = \xi_x$ is concentrated on the orthogonal complement of $\overline{w} = f^{\cdot}(x)$. □

Lemma 2.10 will allow us to reduce a "global" orthogonality to a "local" one. Consider again a W_μ-closed, horizontally convex cone $\mathsf{Set}^0_\mu \subset \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$, and introduce

$$\mathcal{F} := \big\{ f \in L^2_\mu(\Omega;\mathbb{R}^d) \mid \gamma_f \in \mathsf{Set}^0_\mu \big\}.$$

The following result shows that the set $\mathcal{F}$ suffices to characterize $(\mathsf{Set}^0_\mu)^{\perp 0}$.

Lemma 2.11. — *If $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$, then ξ belongs to $(\mathsf{Set}^0_\mu)^{\perp 0}$ if and only if it is orthogonal to all γ_f for $f\in\mathcal{F}$.*

The proof uses a construction that makes it somewhat lengthy, but not complicated.

Proof. — One implication being immediate, we show that if $\langle \xi,\gamma_f\rangle_\mu = 0$ for all $f\in\mathcal{F}$, then $\xi \in (\mathsf{Set}^0_\mu)^{\perp 0}$. This is equivalent to $\pi^\mu\xi = 0_\mu$, where $\pi^\mu\xi$ is the metric projection of ξ on Set^0_μ, given by Proposition 2.7. In particular, as $\gamma_f = (\pi_x, -\pi_v)_\#\gamma_f$ for any $f\in\mathcal{F}$, we have $\langle \xi,\gamma_f\rangle_\mu = \langle \pi^\mu\xi,\gamma_f\rangle_\mu$. Assume by contradiction that $\pi^\mu\xi \neq 0_\mu$: we construct $f\in\mathcal{F}$ such that $\langle \pi^\mu\xi,\gamma_f\rangle_\mu > 0$, contradicting the assumption on ξ.

First consider the constant vector fields $g_i(x) \equiv e_i$, where e_i is the ith element of the canonical basis of $\mathbb{R}^d$. By Lemma 2.10, if $\langle \pi^\mu \xi, \gamma_{g_i} \rangle_\mu = 0$ for all i, then $\langle v, e_i \rangle = 0$ for $\pi^\mu \xi$-almost every (x, v) and each $i \in [\![1, d]\!]$, so that $\pi^\mu \xi = 0_\mu$. Consequently, there must be $i \in [\![1, d]\!]$ such that $\langle \pi^\mu \xi, \gamma_{g_i} \rangle_\mu > 0$.

Pick $\alpha \in \Gamma_{\mu,o}(\pi^\mu \xi, \gamma_{g_i})$. Then $\alpha = \frac{1}{2}(\pi_x, \pi_v, -e_i)_\# \zeta_-^0 + \frac{1}{2}(\pi_x, \pi_v, e_i)_\# \zeta_+^0$ for some $\zeta_\pm^0 \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ such that $\pi^\mu \xi = \frac{1}{2}\zeta_+^0 + \frac{1}{2}\zeta_-^0 \in \mathsf{Set}_\mu^0$. Construct inductively a sequence $(\zeta_+^k, \zeta_-^k)_{k \in \mathbb{N}}$ as follows: assuming that $\frac{1}{2}\zeta_+^k + \frac{1}{2}\zeta_-^k$ belongs to Set_μ^0, consider the transport plan $\beta \coloneqq \frac{1}{2}\zeta_+^k \otimes_\mu \zeta_+^k + \frac{1}{2}\zeta_-^k \otimes_\mu \zeta_-^k$, where for any $\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$, the "pointwise product plan" is defined as

$$\zeta \otimes_\mu \zeta \coloneqq \int_{x \in \Omega} [\delta_x \otimes \zeta_x \otimes \zeta_x]\, d\mu(x) \in \mathscr{P}_2(\mathrm{T}^2\,\Omega)_\mu.$$

Let $\zeta_\pm^{k+1} \coloneqq \big(\pi_x, \frac{1}{2}[\pi_v + \pi_w]\big)_\# \zeta_\pm^k \otimes_\mu \zeta_\pm^k$. Then

$$\frac{1}{2}\zeta_+^{k+1} + \frac{1}{2}\zeta_-^{k+1} = \Big(\pi_x, \frac{\pi_v + \pi_w}{2}\Big)_\# \beta \in \mathsf{Set}_\mu^0$$

by horizontal convexity. The barycenter of each $\zeta_\pm^k$ is preserved along the sequence: indeed, for any $\varphi \in \mathcal{C}(\mathrm{T}\,\Omega; \mathbb{R})$ linear in its second argument and with quadratic growth,

$$\begin{aligned}\int_{(x,v)} \varphi(x,v) d\zeta_+^{k+1} &= \int_{(x,v,w)} \varphi\Big(x, \frac{v+w}{2}\Big) d\left[\zeta_+^k \otimes_\mu \zeta_+^k\right] \\ &= \frac{1}{2}\int_{(x,v)} \varphi(x,v) d\zeta_+^k + \frac{1}{2}\int_{(x,w)} \varphi(x,w) d\zeta_+^k,\end{aligned}$$

hence $\mathrm{Bary}\left(\zeta_\pm^{k+1}\right) = \mathrm{Bary}\left(\zeta_\pm^k\right) =: f_\pm$. In fact

$$\frac{1}{2}f_+ + \frac{1}{2}f_- = \mathrm{Bary}\left(\tfrac{1}{2}\zeta_+^0 + \tfrac{1}{2}\zeta_-^0\right) = \mathrm{Bary}\left(\pi^\mu \xi\right) = 0,$$

so that $f_- = -f_+$. Now, the sequence $(\zeta_+^k)_{k \in \mathbb{N}}$ converges to $(\mathrm{id}, f_+)_\# \mu$: indeed,

$$\begin{aligned}&W_\mu^2\left(\zeta_+^{k+1}, (\mathrm{id}, f_+)_\# \mu\right) = \int_{(x,v,w)} \Big|\frac{v+w}{2} - f_+(x)\Big|^2 d\left[\zeta_+^k \otimes_\mu \zeta_+^k\right] \\ &= 2 \times \frac{1}{4}\int_{(x,v)} |v - f_+(x)|^2 d\zeta_+^k + 2\int_{(x,v,w)} \Big\langle \frac{v - f_+(x)}{2}, \frac{w - f_+(x)}{2} \Big\rangle d\left[\zeta_+^k \otimes_\mu \zeta_+^k\right] \\ &= \frac{1}{2} W_\mu^2\left(\zeta_+^k, (\mathrm{id}, f_+)_\# \mu\right) + 0,\end{aligned}$$

where we used that $\Gamma_\mu(\xi, (\mathrm{id}, g)_\# \mu)$ always reduces to a singleton whenever $g \in L_\mu^2$, and the definition of the pointwise product measure. The same argument implies that $\zeta_-^k \to {}_k(\mathrm{id}, f_-)_\# \mu = (\mathrm{id}, -f_+)_\# \mu$ with respect to W_μ. Consequently, the sequence $\left(\frac{1}{2}\zeta_+^k + \frac{1}{2}\zeta_-^k\right)_{k \in \mathbb{N}} \subset \mathsf{Set}_\mu^0$ converges with respect to W_μ towards the centered field $\frac{1}{2}(\mathrm{id}, f_+)_\# \mu + \frac{1}{2}(\mathrm{id}, f_-)_\# \mu = \gamma_{f_+}$, which must belong to Set_μ^0. Now, recalling the definition of $\zeta_\pm^0$, we have

$$0 < \langle \pi^\mu \xi, \gamma_{g_i} \rangle_\mu = \frac{1}{2}\int_{(x,v)} \langle v, -e_i \rangle d\zeta_-^0 + \frac{1}{2}\int_{(x,v)} \langle v, e_i \rangle d\zeta_+^0 = \int \langle f_+(x), e_i \rangle d\mu,$$

so that $\|f_+\|_\mu > 0$; on the other hand, using that $\langle \zeta, (\mathrm{id}, g)_\# \rangle_\mu = \langle \mathrm{Bary}\,(\zeta), g \rangle_{L^2_\mu}$ for any measure field $\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ and vector field $g \in L^2_\mu(\Omega; \mathbb{R}^d)$,

$$\begin{aligned}\langle \pi^\mu \xi, \gamma_{f_+} \rangle_\mu &\geqslant \frac{1}{2}\langle \zeta^0_+, (\mathrm{id}, f_+)_\# \mu \rangle_\mu + \frac{1}{2}\langle \zeta^0_-, (\mathrm{id}, f_-)_\# \mu \rangle_\mu \\ &= \frac{1}{2}\|f_+\|^2_{L^2_\mu} + \frac{1}{2}\|f_-\|^2_{L^2_\mu} = \|f_+\|^2_{L^2_\mu} > 0.\end{aligned}$$

In conclusion, if $\pi^\mu \xi \neq 0_\mu$, we have constructed $f_+ \in \mathcal{F}$ such that $0 < \langle \pi^\mu \xi, \gamma_{f_+} \rangle_\mu = \langle \xi, \gamma_f \rangle$, contradicting the assumption. Hence $\pi^\mu \xi = 0$, and $\xi \in (\mathsf{Sol}^0_\mu)^{\perp 0}$. □

2.3. Characterization by a Grassmannian section. — We can now turn to the proof of Proposition 2.2. Our aim is to construct a Grassmannian section $D : \Omega \rightrightarrows \mathbb{R}^d$ such that $\xi \in \mathsf{Set}^0_\mu$ if and only if ξ is centered and concentrated on $\mathrm{Graph}\, D$.

Proof of Proposition 2.2. — Assume first that a Grassmannian section $D : \Omega \rightrightarrows \mathbb{R}^d$ is given, and let Set^0_μ be the set of centered measure fields concentrated on $\mathrm{Graph}\, D$. Clearly, Set^0_μ is a convex cone of centered measure fields. Moreover, for each $x \in \Omega$, the set of measures in $\mathscr{P}_2(\mathrm{T}_x\,\Omega) \simeq \mathscr{P}_2(\mathbb{R}^d)$ which are concentrated on the closed set $D(x)$ is closed with respect to the Wasserstein distance. Hence Set^0_μ is closed with respect to $W_\mu \simeq L^2_\mu(\Omega; (\mathscr{P}_2(\mathbb{R}^d); W))$.

Now let Set^0_μ be any W_μ-closed convex cone of centered measure fields. By Lemma 2.9, $\mathsf{Set}^0_\mu = ((\mathsf{Set}^0_\mu)^{\perp 0})^{\perp 0}$. Let $\mathcal{F}_\perp \coloneqq \{ f \in L^2_\mu(\Omega; \mathbb{R}^d) \mid \gamma_f \in (\mathsf{Set}^0_\mu)^{\perp 0} \}$. Using that $W_\mu(\gamma_f, \gamma_g) \leqslant \|f - g\|_{L^2_\mu}$ and that $(\mathsf{Set}^0_\mu)^{\perp 0}$ is W_μ-closed, the set $\mathcal{F}_\perp$ is closed in L^2_μ, hence separable. Consider a countable dense set $(f_n)_{n\in\mathbb{N}} \subset \mathcal{F}_\perp$. From Lemma 2.11 and the continuity of the scalar product, $\xi \in \mathsf{Set}^0_\mu = ((\mathsf{Set}^0_\mu)^{\perp 0})^{\perp 0}$ if and only if $\langle \xi, \gamma_{f_n} \rangle_\mu = 0$ for all $n \in \mathbb{N}$. By Lemma 2.10, the latter condition is equivalent to $\langle v, f_n^\cdot(x) \rangle = 0$ for ξ-almost all (x, v), where $f_n^\cdot : \Omega \to \mathbb{R}^d$ is a measurable map in the L^2_μ-equivalence class f_n. For every $x \in \Omega$, define

$$D(x) \coloneqq \left\{ v \in \mathbb{R}^d \mid \langle v, f_n^\cdot(x) \rangle = 0 \text{ for all } n \in \mathbb{N} \right\}.$$

The map D depends on the precise choice of $(f_n^\cdot)_n$, but only up to a μ-negligible subset. Each $D(x)$ is a vector space, and by [Roc69, Th. 3.(e)], D is measurable as a multivalued map. If ξ is concentrated on the graph of D, then $\langle v, f_n^\cdot(x) \rangle = 0$ for all n ξ-almost everywhere. Conversely, if for any n, there exists B^n such that $\xi(B^n) = 0$ and $\langle v, f_n^\cdot(x) \rangle = 0$ for any $(x, v) \notin B^n$, then $B \coloneqq \bigcup_n B^n$ is still ξ-negligible, and $v \in D(x)$ for any $x \notin B$. Hence $\xi \in \mathsf{Set}^0_\mu$ if and only if $\xi(\mathrm{Graph}\, D) = 1$. □

An interesting corollary of Proposition 2.2 is that Set^0_μ is closed with respect to the (weaker) topology of the Wasserstein distance over the tangent bundle $W_{\mathrm{T}\,\Omega}$, i.e., with cost $c((x, v), (y, w)) \coloneqq \sqrt{|x - y|^2 + |v - w|^2}$. Here, it is necessary to restrict to centered measure fields: the geometric tangent cone Tan_μ is a W_μ-closed, horizontally convex and (two-sided) cone, but is not closed with respect to $W_{\mathrm{T}\,\Omega}$ in general (an example can be found below Proposition 2.10 in [Aus25a]).

Corollary 2.12 ($W_{\mathrm{T}\,\Omega}$-closedness of Set^0_μ). — *Assume that* Set^0_μ *satisfies Assumption* 2.1. *Let* $(\xi_n)_{n\in\mathbb{N}} \subset \mathsf{Set}^0_\mu$ *be a sequence converging to* $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ *with respect to* $W_{\mathrm{T}\,\Omega}$. *Then* $\xi \in \mathsf{Set}^0_\mu$.

Proof. — By [Vil09, Th. 6.9], convergence with respect to $W_{\mathrm{T}\,\Omega}$ is equivalent to the convergence of $\int_{(x,v)} \varphi(x,v)d\xi_n \to_n \int_{(x,v)} \varphi(x,v)d\xi$ for any continuous and quadratically growing $\varphi : \mathrm{T}\,\Omega \to \mathbb{R}$. Consequently, $\int \varphi(x, \mathrm{Bary}\,(\xi)(x))d\mu = 0$ for any φ linear with respect to its second argument, and $\mathrm{Bary}\,(\xi) = 0$. By Lemma 1.1, the set of ξ such that $\xi(\mathrm{Graph}\,D) = 1$ is $W_{\mathrm{T}\,\Omega}$-closed, hence $\xi \in \mathsf{Set}^0_\mu$. □

The characterization of Set^0_μ provides many examples of closed convex cones of centered fields; one just has to choose the map D. However, the following section focuses on two particular subsets of measure fields which are not a priori constructed from such maps, but can be proved to be closed and horizontally convex, yielding an additional structure.

3. Tangent and solenoidal measure fields

We introduce the geometric tangent cone Tan_μ in its classical definition, as well as its metric orthogonal, and immediately restrict our attention to their centered subsets.

Definition 3.1 (Tan^0_μ and Sol^0_μ). — The geometric tangent cone is defined as

$$\mathsf{Tan}_\mu \coloneqq \overline{\{\lambda\cdot\xi \mid \lambda \geqslant 0,\ (\pi_x, \pi_x+\pi_v)_\#\xi \in \Gamma_o(\mu,\nu),\ \nu \in \mathscr{P}_2(\Omega)\}}^{W_\mu} \subset \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu.$$

The solenoidal cone Sol_μ is defined as $(\mathsf{Tan}_\mu)^\perp$, i.e.,

$$\mathsf{Sol}_\mu \coloneqq \big\{\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu \mid \langle \xi, \zeta\rangle_\mu = 0 \text{ for all } \xi \in \mathsf{Tan}_\mu\big\}.$$

The sets Tan^0_μ and Sol^0_μ are defined as the intersections of $\mathsf{Tan}_\mu, \mathsf{Sol}_\mu$ with $\mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$.

Remark 3.2 (Alternative definition). — One can show that for any $\xi, \zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$,

$$\langle \xi, \zeta\rangle_\mu = \langle \xi^0, \zeta^0\rangle_\mu + \langle \mathrm{Bary}\,(\xi), \mathrm{Bary}\,(\zeta)\rangle_{L^2_\mu}, \tag{5}$$

where ξ^0, ζ^0 are the respective centered components of ξ, ζ, i.e.,

$$\xi^0 = (\pi_x, \pi_v - \mathrm{Bary}\,(\xi))_\#\xi.$$

Therefore the sets Tan^0_μ and Sol^0_μ coincide with the sets of metric projections of $\mathsf{Tan}_\mu, \mathsf{Sol}_\mu$ onto the closed, convex cone $\mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$. The orthogonal decomposition (5) also implies that ξ is tangent (resp. ζ solenoidal) if and only if ξ^0 and $(\mathrm{id}, \mathrm{Bary}\,(\xi))_\#\mu$ are tangent (resp. ζ^0 and $(\mathrm{id}, \mathrm{Bary}\,(\zeta))_\#\mu$ solenoidal).

Our first result on Tan^0_μ and Sol^0_μ is an application of Proposition 2.2. If $D : \Omega \rightrightarrows \mathbb{R}^d$ is a Grassmannian section, denote by $D^\perp$ the map such that $D^\perp(x)$ is the orthogonal complement of $D(x)$ in $\mathbb{R}^d$.

Corollary 3.3. — *Let $\mu \in \mathscr{P}_2(\Omega)$. There exists a Grassmannian section $D : \Omega \rightrightarrows \mathbb{R}^d$ such that*

$$\begin{aligned} \zeta \in \mathsf{Sol}^0_\mu &\iff \big[\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu \text{ and } \zeta(\mathrm{Graph}\, D) = 1\big], \\ \xi \in \mathsf{Tan}^0_\mu &\iff \big[\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu \text{ and } \xi(\mathrm{Graph}\, D^\perp) = 1\big]. \end{aligned} \tag{6}$$

In the sequel, D will often be denoted D^{Sol}, and $D^\perp$ will be denoted D^{Tan}.

Proof. — The fact that Tan^0_μ is a W_μ-closed positive cone of centered fields follows directly from the definition. The fact that it is horizontally convex is proved in [Gig08, Prop. 4.25], first by considering optimal plans and using cyclical monotonicity, then by approximation. The decomposition of the metric scalar product in (5) yields that $\mathsf{Sol}^0_\mu = (\mathsf{Tan}^0_\mu)^{\perp 0}$, so that by Lemma 2.6, Sol^0_μ is also a closed convex cone of centered measure fields. Hence the existence of D characterizing Sol^0_μ as in (6) follows from Proposition 2.2. The same result yields a Grassmannian section $\widetilde{D} : \Omega \rightrightarrows \mathbb{R}^d$ characterizing Tan^0_μ; to conclude, it only remains to show that $\widetilde{D} = D^\perp$ μ-almost everywhere. But this is implied by the following equivalences for ξ centered:

$$\begin{aligned} &\xi \text{ concentrated on } (x, v) \text{ such that } v \in \widetilde{D}(x) \\ &\quad\iff \xi \in \mathsf{Tan}^0_\mu \quad \text{(Prop. 2.2)} \\ &\quad\iff \langle \xi, \gamma_f \rangle_\mu = 0 \text{ for any } f \in L^2_\mu \text{ such that } \gamma_f \in \mathsf{Sol}^0_\mu \quad \text{(Lem. 2.9 and 2.11)} \\ &\quad\iff \xi \text{ concentrated on } (x, v) \text{ such that } \langle v, f^\cdot(x) \rangle = 0 \quad \text{(Prop. 2.2)}, \end{aligned}$$

where in the last line, $f^\cdot$ ranges over all measurable functions such that $f^\cdot(x) \in D(x)$ for all $x \in \Omega$, and $\int |f^\cdot|^2 d\mu < \infty$. The fact that sufficiently many of such $f^\cdot$ can be found follows from the Castaing representation of the measurable map D, for instance in [Roc69, Th. 3.(d)]. □

By Corollary 2.12, both Tan^0_μ and Sol^0_μ are closed with respect to the Wasserstein distance $W_{\mathrm{T}\,\Omega}$ on the tangent bundle. In addition, the set of solenoidal measure fields (possibly with nonzero barycenter) is closed in the same topology.

Corollary 3.4 (Sol_μ is $W_{\mathrm{T}\,\Omega}$-closed). — *Let $\mu \in \mathscr{P}_2(\Omega)$. The set Sol_μ is $W_{\mathrm{T}\,\Omega}$-closed.*

Proof. — We first show that any $\zeta \in \mathsf{Sol}_\mu$ is concentrated on Graph D. By Remark 3.2, if $\zeta \in \mathsf{Sol}_\mu$, then $g := \mathrm{Bary}\,(\zeta) \in L^2_\mu(\Omega; \mathbb{R}^d)$ is such that $(\mathrm{id}, g)_\# \mu \in \mathsf{Sol}_\mu$. By [Aus25a, Lem. 2.4 & 2.7], both measure fields $(\mathrm{id}, -g)_\# \mu$ and $\frac{1}{2}\left[(\mathrm{id}, g)_\# \mu + (\mathrm{id}, -g)_\# \mu\right]$ are solenoidal. Applying Corollary 3.3 to the latter, we get that $g(x) \in D^{\mathsf{Sol}}(x)$ for μ-almost every x. As the centered field $\zeta^0 := (\pi_x, \pi_v - g(\pi_x))_\# \zeta$ is solenoidal, it is also concentrated on Graph D, thus so is $\zeta = (\pi_x, \pi_v + g(\pi_x))_\# \zeta^0$.

Now let $(\zeta_n)_{n \in \mathbb{N}} \subset \mathsf{Sol}_\mu$ be a Cauchy sequence with respect to $W_{\mathrm{T}\,\Omega}$, and let $\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ be its limit. By Lemma 1.1, $\zeta(\mathrm{Graph}\, D) = 1$. Consequently, the centered component $\zeta^0 := (\pi_x, \pi_v - \mathrm{Bary}\,(\zeta)(\pi_x))_\# \zeta$ is concentrated on Graph D, hence solenoidal. To show that Bary (ζ) also induces a solenoidal field, it is enough to prove that $\langle f, \mathrm{Bary}\,(\zeta) \rangle_{L^2_\mu} = 0$ for any $f \in L^2_\mu(\Omega; \mathbb{R}^d)$ such that $(\mathrm{id}, f)_\# \mu$ is tangent (by (5)).

In addition, we may take $f = \nabla\varphi$ for $\varphi \in \mathcal{C}^\infty_c(\Omega;\mathbb{R})$ [Gig08, Th. 4.14]. In particular, $(x,v) \mapsto \langle f(x), v\rangle$ is continuous and has quadratic growth, so that

$$\langle f, \mathrm{Bary}\,(\zeta)\rangle_{L^2_\mu} = \int \langle f(x), v\rangle d\zeta = \lim_{n\to\infty} \int \langle f(x), v\rangle d\zeta_n = \lim_{n\to\infty} \langle f, \mathrm{Bary}\,(\zeta_n)\rangle_{L^2_\mu} = 0.$$

In conclusion, both ζ^0 and $(\mathrm{id}, \mathrm{Bary}\,(\zeta))_\#\mu$ belong to Sol_μ, so that ζ is solenoidal. □

One might be tempted to think that Tan_μ is $W_{\mathrm{T}\,\Omega}$-closed by the same argument. This is not correct: it does not hold that $\xi(\mathrm{Graph}\, D^{\mathsf{Tan}}) = 1$ for every tangent ξ. It *does* hold that

– Tan^0_μ is $W_{\mathrm{T}\,\Omega}$-closed (by Corollary 2.12),

– the set of $f \in L^2_\mu(\Omega;\mathbb{R}^d)$ such that $(\mathrm{id}, f)_\#\mu \in \mathsf{Tan}_\mu$ is weakly closed in L^2_μ (by the previous proof),

but the latter set of vector fields is not strongly closed in L^2_μ, and the oscillations captured by the $W_{\mathrm{T}\,\Omega}$-limit may leave D^{Tan}. On the other hand, any map inducing a solenoidal field is already valued in D^{Sol}, so that oscillations can only produce measure fields that are again concentrated on D^{Sol}.

3.1. Stability with respect to restriction. — The aim of this section is to investigate how Tan^0_μ and Sol^0_μ depend on the local properties of the underlying measure μ. We start by showing that the centered solenoidal spaces are stable under restriction of measures. This is valid only for centered measure fields; Remark 3.9 below provides a counterexample in the case of map-induced fields.

Given a measurable set $A \subset \Omega$, denote by $\mu \llcorner A$ the measure given by $(\mu \llcorner A)(B) \coloneqq \mu(A \cap B)$ for any measurable B, and by $\mathrm{T}\,A \subset \mathrm{T}\,\Omega$ the set $\{(x,v) \mid x \in A,\ v \in \mathrm{T}_x\,\Omega\}$.

Proposition 3.5 (Restriction of centered solenoidal measure fields). — *Let $\mu \in \mathscr{P}_2(\Omega)$ and $A \subset \Omega$ be measurable with $\mu(A) \in (0,1)$. Denote $\mu_A \coloneqq (\mu \llcorner A)/\mu(A)$. Then*

$$\mathsf{Sol}^0_{\mu_A} = \left\{(\zeta \llcorner \mathrm{T}\,A)/\mu(A) \mid \zeta \in \mathsf{Sol}^0_\mu\right\}.$$

Proposition 3.5 relies on the following intermediate results, whose proofs are deferred to the appendix. The statements are formulated for Sol_μ instead of Sol^0_μ, since centeredness plays no role there.

Lemma 3.6. — *Let $\mu = (1-\lambda)\mu_1 + \lambda\mu_2 \in \mathscr{P}_2(\Omega)$ for $\mu_i \in \mathscr{P}_2(\Omega)$ and $\lambda \in [0,1]$. Let $\eta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_{\mu_1}$ be the velocity of a geodesic, with $\nu \coloneqq (\pi_x + \pi_v)_\#\eta$ compactly supported. There exists $\gamma \in \mathscr{P}_2(\mathrm{T}\,\Omega)_{\mu_2}$ such that*

$$\xi = (1-\lambda)\eta + \lambda\gamma \tag{7}$$

is the velocity of a geodesic starting from μ.

Lemma 3.6 does not stand without the assumption that ν has compact support. Indeed, consider for instance $\mu_1 = \delta_0$, $\mu_2 = \delta_1$ and ν a Gaussian measure in dimension one. There is only one η such that $(\pi_x, \pi_x + \pi_v)_\#\eta$ is optimal between μ_1 and ν, and its support is not bounded in the velocity variable. To construct a γ satisfying (7), one should be able to ensure that for any $(x,v) \in \mathrm{supp}\,\eta$ and $(y,w) \in \mathrm{supp}\,\gamma$, the

monotonicity condition $|v|^2 + |w|^2 \leqslant |x+v-y|^2 + |y+w-x|^2$ holds. Equivalently, using that $x = 0$ and $y = 1$, one should have $2v - 1 \leqslant 1 + 2w$; since v can be arbitrarily large, no finite w satisfies this. Consequently, in order to use Lemma 3.6, we characterize Sol_μ by orthogonality with respect to velocities of geodesics ending at compactly supported measures.

Lemma 3.7. — *Let $\zeta \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ satisfy $\langle \zeta, (\pi_x, \pi_y - \pi_x)_\# \gamma \rangle_\mu = 0$ for any $\gamma \in \Gamma_o(\mu,\nu)$ with $\nu \in \mathscr{P}_2(\Omega)$ compactly supported. Then $\zeta \in \mathsf{Sol}_\mu$.*

We now turn to our original claim. In the proof, we use the formula given by [Gig08, Prop. 4.2] to deduce a Chasles relation for the metric scalar product, stating that for $\mu \in \mathscr{P}_2(\Omega)$ and $A \subset \Omega$ measurable,

$$\langle \xi, \zeta \rangle_\mu = \int_{x\in\Omega} \langle \xi_x, \zeta_x \rangle_{\delta_x} d\mu(x) = \int_{x\in A} \langle \xi_x, \zeta_x \rangle_{\delta_x} d\mu(x) + \int_{x\in A^c} \langle \xi_x, \zeta_x \rangle_{\delta_x} d\mu(x).$$

Proof of Proposition 3.5. — We proceed by double inclusion. Consider $\zeta \in \mathsf{Sol}^0_\mu$, and let $\zeta_A \coloneqq (\zeta \llcorner \mathrm{T}\,A)/\mu(A)$ and $\zeta_{A^c} \coloneqq (\zeta \llcorner \mathrm{T}\,A^c)/\mu(A^c)$. By Lemma 3.7, it suffices to show that $\langle \zeta_A, \xi^A \rangle_{\mu_A} = 0$ for any $\xi^A \in \mathscr{P}_2(\mathrm{T}\,\Omega)_{\mu_A}$ which induces a geodesic between μ_A and a compactly supported measure. Consider such a ξ^A. Applying Lemma 3.6 with $\mu_1 = \mu_A$, $\mu_2 = \mu_{A^c}$ and $1 - \lambda = \mu(A)$, we obtain a measure field $\xi^{A^c} \in \mathscr{P}_2(\mathrm{T}\,\Omega)_{\mu_{A^c}}$ such that $\xi = \mu(A)\xi^A + \mu(A^c)\xi^{A^c}$ belongs to Tan_μ. As μ_A and μ_{A^c} are mutually singular, we have $\xi^A = (\xi \llcorner \mathrm{T}\,A)/\mu(A)$. Hence

$$0 = \langle \zeta, \xi \rangle_\mu = \mu(A)\langle \zeta_A, \xi^A \rangle_{\mu_A} + \mu(A^c)\langle \zeta_{A^c}, \xi^{A^c} \rangle_{\mu_{A^c}} \geqslant \mu(A)\langle \zeta_A, \xi^A \rangle_{\mu_A} \geqslant 0, \tag{8}$$

where we used that ζ_A and ζ_{A^c} are centered (see Lemma 2.5). Hence $\langle \zeta_A, \xi^A \rangle_{\mu_A} = 0$, and $\zeta_A \in \mathsf{Sol}^0_{\mu_A}$.

Conversely, let $\zeta^A \in \mathsf{Sol}^0_{\mu_A}$, and consider $\zeta \coloneqq \mu(A)\zeta^A + \mu(A^c)0_{\mu_{A^c}}$. In particular, $\zeta^A = (\zeta \llcorner \mathrm{T}\,A)/\mu(A)$. Let $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ be the velocity of a geodesic. By restriction of optimality [Vil09, Th. 4.6], $\xi_A \coloneqq (\xi \llcorner \mathrm{T}\,A)/\mu(A)$ is also the velocity of a geodesic; moreover, by Chasles,

$$\langle \zeta, \xi \rangle_\mu = \mu(A)\langle \zeta^A, \xi_A \rangle_{\mu_A} + \mu(A^c)\langle 0_{\mu_{A^c}}, \xi_{A^c} \rangle_{\mu_{A^c}} = \mu(A)\langle \zeta^A, \xi_A \rangle_{\mu_A} = 0.$$

Hence $\zeta \in \mathsf{Sol}^0_\mu$, and any solenoidal measure field in $\mathsf{Sol}^0_{\mu_A}$ can be written as the restriction of an element of Sol^0_μ. □

As a corollary, we deduce the corresponding statement for Tan^0_μ.

Corollary 3.8 (Restriction of centered tangent measure fields). — *With the same notation as in Proposition* 3.5*,*

$$\mathsf{Tan}^0_{\mu_A} = \left\{ (\xi \llcorner \mathrm{T}\,A)/\mu(A) \mid \xi \in \mathsf{Tan}^0_\mu \right\}.$$

Proof. — First let $\xi \in \mathsf{Tan}^0_\mu$, and denote $\xi_A \coloneqq (\xi \llcorner \mathrm{T}\,A)/\mu(A)$. By Proposition 3.5, any $\zeta^A \in \mathsf{Sol}^0_{\mu_A}$ can be written as $(\zeta \llcorner \mathrm{T}\,A)/\mu(A)$ for some $\zeta \in \mathsf{Sol}^0_\mu$. Hence, using Chasles as in (8), $0 = \langle \xi, \zeta \rangle_\mu \geqslant \mu(A)\langle \xi_A, \zeta^A \rangle_{\mu_A} \geqslant 0$, so that $\xi_A \in \mathsf{Tan}^0_{\mu_A}$. Conversely, if $\xi^A \in \mathsf{Tan}^0_{\mu_A}$, define $\xi \coloneqq \mu(A)\xi^A + \mu(A^c)0_{\mu_{A^c}} \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$. For any $\zeta \in \mathsf{Sol}^0_\mu$,

one has $\langle \xi, \zeta \rangle_\mu = \mu(A) \langle \xi^A, \zeta_A \rangle_{\mu_A}$, where $\zeta_A \coloneqq (\zeta \llcorner \mathrm{T}\, A)/\mu(A)$ belongs to $\mathsf{Sol}^0_{\mu_A}$ by Proposition 3.5. Hence $\langle \xi, \zeta \rangle_\mu = 0$, and $\xi \in \mathsf{Tan}^0_\mu$, completing the proof. □

We emphasize that the argument is not exactly symmetric between Tan^0_μ and Sol^0_μ; the difficulty lies in Lemma 3.6, where an *optimal* plan attached to a measure is "extended" to an optimal plan attached to another measure. Despite many attempts, the author could not find a direct proof of an extension result for *tangent* measure fields: when letting the optimal time decrease to 0, there is no guarantee that the narrow/Wasserstein limit remains tangent. However, optimal plans are sufficient to characterize solenoidal measure fields, so we can first prove the restriction property for Sol^0_μ, and then transfer it to Tan^0_μ.

Remark 3.9 (Necessity of the centered assumption). — Let $\mu \in \mathscr{P}_2(\Omega)$ be the normalized 1-Hausdorff measure restricted to the boundary of the unit square $S \coloneqq \partial[0,1]^2$. Parametrize S by a constant-speed closed curve $\gamma : [0,1] \to \mathbb{R}^2$ rotating clockwise, and let $\zeta \coloneqq (\gamma, \dot\gamma)_\# \mathcal{L}_{[0,1]}$. We claim that ζ is solenoidal; since it is induced by a map, this is equivalent to $\langle \xi, \zeta \rangle_\mu = 0$ for any map-induced tangent ξ. Any such ξ can be approximated arbitrarily well with respect to W_μ by $(\mathrm{id}, \nabla\varphi)_\# \mu$ for some $\varphi \in \mathcal{C}^\infty_c(\Omega; \mathbb{R})$ [Gig11]. As $\langle (\mathrm{id}, \nabla\varphi)_\# \mu, \zeta \rangle_\mu = \int_{[0,1]} \frac{d}{dt} \varphi \circ \gamma \, dt = 0$, the measure field ζ is solenoidal.

However, if $\nu \coloneqq 4\,\mu \llcorner [0,1] \times \{1\}$ is the (normalized) restriction of μ to the top side of the square, then the corresponding restriction $\varsigma \coloneqq (\mathrm{id}, (1,0))_\# \nu$ belongs to Tan_ν, since it induces a geodesic. The reader may check that the centered fields $\frac{1}{2}[\zeta + (\pi_x, -\pi_v)_\# \zeta]$ and $\frac{1}{2}[\varsigma + (\pi_x, -\pi_v)_\# \varsigma]$ are both solenoidal.

3.2. Decomposition according to the dimension of splitting. — With the above material, we can now state our first main result.

Theorem 3.10. — *Let $\mu \in \mathscr{P}_2(\Omega)$. There exists a decomposition $\mu = \sum_{k=0}^d m_k \mu^k$, where $m_k \in [0,1]$ sum to one and $\mu^k \in \mathscr{P}_2(\Omega)$ are mutually singular measures, with the following properties. For each $k \in [\![0, d]\!]$, there exists a Grassmannian section $D_k : \Omega \rightrightarrows \mathbb{R}^d$ such that $\dim D_k \equiv k$, and*

(i) *$\zeta \in \mathsf{Sol}^0_\mu$ if and only if $\zeta = \sum_{k=0}^d m_k \zeta^k$ with $\zeta^k \in \mathsf{Sol}^0_{\mu^k}$ for $k \in [\![0,d]\!]$.*

(ii) *$\xi \in \mathsf{Tan}^0_\mu$ if and only if $\xi = \sum_{k=0}^d m_k \xi^k$ with $\xi^k \in \mathsf{Tan}^0_{\mu^k}$ for $k \in [\![0,d]\!]$.*

(iii) *If $m_k > 0$, $\zeta^k \in \mathsf{Sol}^0_{\mu^k}$ if and only if ζ^k is centered and concentrated on* $\operatorname{Graph} D_k$.

(iv) *If $m_k > 0$, $\xi^k \in \mathsf{Tan}^0_{\mu^k}$ if and only if ξ^k is centered and concentrated on* $\operatorname{Graph} D_k^\perp$.

In addition, the measures $m_k \mu^k$ in the decomposition are unique. This follows directly from the explicit formula given in Theorem 4.10. Figure 1 illustrates Theorem 3.10.

Proof of Theorem 3.10. — By Corollary 3.3, there exists a Grassmannian section $D : \Omega \rightrightarrows \mathbb{R}^d$ such that $\zeta \in \mathsf{Sol}^0_\mu$ if and only if ζ is centered and concentrated on $\operatorname{Graph} D$,

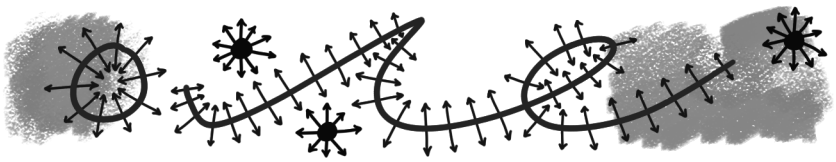


Figure 1. A measure $\mu \in \mathscr{P}_2(\mathbb{R}^2)$ and its section $D^{\mathsf{Tan}} = D^\perp$ characterizing Tan^0_μ.

The measure μ^0 is the sum of the three atoms, and D_0^{Tan} is 2-dimensional. The measure μ^1 is supported on a countable union of DC_1 sets, which, in dimension 2, are graphs of DC functions up to permuting the axes. The direction of the one-dimensional Grassmannian section D_1^{Tan} is represented by the arrows, whose length is irrelevant. The fact that D_1^{Tan} is orthogonal to the set on which μ^1 is concentrated will be proved in Proposition 4.12. The remaining measure μ^2 is transport-regular in the sense that it gives 0 mass to any DC_1 set, and satisfies the conclusion of the Brenier–McCann theorem. Consequently, no optimal plan splits mass, and $D_2^{\mathsf{Tan}} \equiv \{0\}$.

and $\xi \in \mathsf{Tan}^0_\mu$ if and only if $\xi \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$ and ξ is concentrated on $\operatorname{Graph} D^\perp$. For each $k \in [\![0, d]\!]$, define

$$A_k \coloneqq \{x \in \Omega \mid \dim D(x) = k\}\,.$$

Each set A_k is measurable; indeed, by [Roc69, Th. 3.(d)], we have

$$D(x) = \overline{\operatorname{conv}\{g_n(x) \mid n \in \mathbb{N}\}}$$

for some countable family $(g_n)_{n\in\mathbb{N}}$ of measurable maps. Then A_k can be written as the set of x such that any choice of $k+1$ vectors $g_n(x)$ is linearly dependent, and there exist k linearly independent vectors $(g_{n_j}(x))_{j\in[\![1,k]\!]}$, i.e.,

$$\bigcap_{\sigma:[\![1,k+1]\!]\to\mathbb{N}} \left\{\det_{k+1}\left(g_{\sigma_1}, ..., g_{\sigma(k+1)}\right) = 0\right\} \bigcap \bigcup_{\theta:[\![1,k]\!]\to\mathbb{N}} \left\{\det_k\left(g_{\theta(1)}, ..., g_{\theta(k)}\right) \neq 0\right\}.$$

Here $\det_j : (\mathbb{R}^d)^j \to \mathbb{R}$ is the j-determinant, which is continuous. As each g_n is measurable, the set A_k is itself measurable.

If $m_k \coloneqq \mu(A_k) > 0$, define $\mu^k \coloneqq (\mu \llcorner A^k)/\mu(A_k)$. Since $(A_k)_{k=0}^d$ is a partition of Ω, the measures μ^k are mutually singular, and $\mu = \sum_{k=0}^d m_k \mu^k$. Let

$$D_k(x) \coloneqq \begin{cases} D(x) & x \in A_k, \\ \operatorname{vect}\{e_1, \dots, e_k\} & \text{otherwise.} \end{cases}$$

Clearly, D_k is measurable. We now show that μ^k and D_k satisfy the claims.

Let $\zeta \in \mathsf{Sol}^0_\mu$, and write it as $\sum_{k=0}^d m_k \zeta^k$, where $m_k\zeta^k \coloneqq \zeta \llcorner \mathrm{T}\,A_k$. In particular, ζ^k is centered. For each k such that $m_k > 0$, Proposition 3.5 yields that $\zeta^k \in \mathsf{Sol}^0_{\mu^k}$. This proves the first implication of point (i). Conversely, let $(\zeta^k)_{k\in[\![0,d]\!]}$ be a family of centered measure fields such that $\zeta^k \in \mathsf{Sol}^0_{\mu^k}$, and define $\zeta \coloneqq \sum_{k=0}^d m_k \zeta^k$. To show

that $\zeta \in \mathsf{Sol}^0_\mu$, let $\xi \in \mathsf{Tan}^0_\mu$, decomposed as $\sum_{k=0}^d m_k \xi^k$. By Chasles,

$$\langle \zeta, \xi \rangle_\mu = \sum_{k=0}^{d} m_k \langle \zeta^k, \xi^k \rangle_{\mu^k}.$$

By Corollary 3.8, each ξ^k belongs to $\mathsf{Tan}^0_{\mu^k}$, so that every term of the sum vanishes. Hence $\zeta \in (\mathsf{Tan}^0_\mu)^{\perp 0} = \mathsf{Sol}^0_\mu$. This proves (i); the argument is completely symmetric for (ii), with Proposition 3.5 in place of Corollary 3.8.

We turn to point (iii). If $\zeta^k \in \mathsf{Sol}^0_{\mu^k}$, then, by (i), the measure field $\zeta \coloneqq m_k \zeta^k + \sum_{j=0, j\neq k}^d m_j 0_{\mu^j}$ belongs to Sol^0_μ. As such, it is concentrated on $\operatorname{Graph} D$. Therefore so is ζ^k, and since $D = D_k$ for μ^k-almost every x, we get that $\zeta^k(\operatorname{Graph} D_k) = 1$. On the other hand, let $\zeta^k \in \mathscr{P}_2(\mathrm{T}\,\Omega)^0_{\mu^k}$ be concentrated on $\operatorname{Graph} D_k$. As $D_k = D$ μ^k-almost everywhere, ζ^k is concentrated on $\operatorname{Graph} D$. Therefore $\zeta \coloneqq m_k \zeta^k + \sum_{j=0, j\neq k}^d m_j 0_{\mu^j}$ is also concentrated on $\operatorname{Graph} D$, and must belong to Sol^0_μ. By restriction, $\zeta^k = (\zeta \llcorner \mathrm{T}\, A_k)/m_k$ belongs to $\mathsf{Sol}^0_{\mu^k}$. The point (iv) is proved by repeating the argument with $D^\perp$ in place of D. □

4. Properties of the decomposition

Theorem 3.10 relies only on the algebraic properties of tangent and solenoidal cones. We now give a refined description of the measures μ^k in terms of concentration on particular subsets. The argument is based on Zajíček's theorem, recalled below, which characterizes non-differentiability points of convex functions using DC_k sets. We start with some lemmas specific to DC_k and σ-$\mathrm{DC_k}$ sets, and then return to the measures $(\mu^k)_k$.

4.1. Preliminaries on DC_k sets

Definition 4.1 (DC_k and σ-$\mathrm{DC_k}$ sets). — Let $k \in [\![0, d]\!]$. A set $A \subset \mathbb{R}^d$ is a *difference of convex functions of dimension k*, denoted DC_k, if $A = \Phi(\mathbb{R}^k)$ for some function $\Phi : \mathbb{R}^k \to \mathbb{R}^d$ which, up to a permutation of coordinates, can be written as

$$\Phi(X_k) = (X_k, (\varphi_{k+1} - \psi_{k+1})(X_k), \dots, (\varphi_d - \psi_d)(X_k)) \tag{9}$$

where $\varphi_j, \psi_j : \mathbb{R}^k \to \mathbb{R}$ are convex functions for $j \in [\![k+1, d]\!]$. A set that can be covered by countably many DC_k sets will be called σ-$\mathrm{DC_k}$.

DC_k sets are called "c–c hypersurfaces of dimension k" in the original work of Zajíček [Zaj79], and "δ-convex surfaces of dimension k" by [Pav04]. By convention, DC_0 sets are points, and the only DC_d set is $\mathbb{R}^d$. The parametrization yields that DC_k sets are closed; with the notation of (9), a Cauchy sequence in $A = \Phi(\mathbb{R}^k)$ necessarily has its first k components converging in $\mathbb{R}^k$, hence converges to an element of A by continuity of the functions φ_j, ψ_j.

Remark 4.2 (Composition). — Let $0 \leqslant j \leqslant k \leqslant d$. If $A \subset \mathbb{R}^k$ is σ-$\mathrm{DC_j}$, and $\Phi : \mathbb{R}^k \to \mathbb{R}^d$ can be written as a permutation of $(\mathrm{id}_{\mathbb{R}^k}, \varphi_{k+1} - \psi_{k+1}, \dots, \varphi_d - \psi_d)$ for convex functions $\varphi_j, \psi_j : \mathbb{R}^k \to \mathbb{R}$, then $\Phi(A)$ is σ-$\mathrm{DC_j}$ as well. Indeed, this is

trivial if $j = 0$ (since A is countable) or $j = k$ (since $A = \mathbb{R}^k$). Otherwise, let $(A_n)_n$ be a countable family of DC_j sets covering A, each written as $\Phi_n(\mathbb{R}^j)$ for some Φ_n of the form (9). The composition $\Phi \circ \Phi_n$ coincides with the identity on j of its coordinates. Since compositions of DC functions are still DC by [Har59, (I) & (II)], the remaining coordinates of $\Phi \circ \Phi_n$ are DC functions from $\mathbb{R}^j$ to $\mathbb{R}$. Therefore the sets $(\Phi \circ \Phi_n(A_n))_{n\in\mathbb{N}}$ are a countable family of DC_j sets covering A, i.e., A is σ-$\mathrm{DC_j}$.

Theorem (Zajíček's theorem). — *Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a convex function. Then each set*

$$J_k(\varphi) \coloneqq \left\{x \in \mathbb{R}^d \mid \dim \partial_x\varphi \geqslant d-k\right\} \tag{10}$$

is σ-DC_k, i.e., can be covered by countably many DC_k sets. Conversely, if $A \subset \mathbb{R}^d$ is σ-DC_k, there exists $\varphi : \mathbb{R}^d \to \mathbb{R}$ convex such that $A \subset J_k(\varphi)$.

If one is given a single DC_k set A, then the convex function can be chosen "uniformly non-differentiable" over A. This is merely an observation on the construction of [Zaj79], but it will be used in the sequel.

Lemma 4.3. — *Let $A = \Phi(\mathbb{R}^k) \subset \mathbb{R}^d$ be a DC_k set, with Φ as in Definition 4.1. There exist a convex function $\phi : \mathbb{R}^d \to \mathbb{R}$ and $d-k+1$ measurable vector fields $f_k, \dots, f_d$ such that $f_j(x) \in \partial_x\phi$ for each x, and whenever $x \in A$, the vectors $f_k(x), \dots, f_d(x)$ are at distance at least one from each other.*

Proof. — We may permute the coordinates of the space so as to write

$$\Phi = (\mathrm{id}, \varphi_{k+1} - \psi_{k+1}, \dots, \varphi_d - \psi_d)$$

for some convex functions $\varphi_j, \psi_j : \mathbb{R}^k \to \mathbb{R}$. Let $\phi \coloneqq \sum_{j=k+1}^d \phi_j$, where each ϕ_j is given by

$$\phi_j(x_1, \dots, x_d) \coloneqq \max\left(x_j + \psi_j(x_1, \dots, x_k), \varphi_j(x_1, \dots, x_k)\right).$$

Then ϕ_j and ϕ are convex. For convenience, write $x = (X^k, x_{k+1}, \dots, x_d)$ whenever $x \in \mathbb{R}^d$. For $j \in [\![k+1, d]\!]$, let $v_j, w_j : \mathbb{R}^k \to \mathbb{R}^k$ be measurable selections of $\partial\varphi_j$ and $\partial\psi_j$ respectively. Then $\partial_x\phi_j$ contains the vectors $(v_j(X^k), 0) \in \mathbb{R}^d$ and $(w_j(X^k), 0) + e_j$, with $e_j \in \mathbb{R}^d$ the jth vector of the canonical basis. Indeed, since $x_j = \varphi_j(X^k) - \psi_j(X^k)$, we have, for any $y = (Y^k, y_{k+1}, \dots, y_d)$,

$$\begin{aligned}
\phi_j(y) &= \max\left(y_j + \psi_j(Y^k), \varphi_j(Y^k)\right) \\
&\geqslant \max\Big(x_j + \psi_j(X^k) + (y_j - x_j) + \langle w_j(X^k), Y^k - X^k\rangle, \\
&\qquad\qquad \varphi_j(X^k) + \langle v_j(X^k), Y^k - X^k\rangle\Big) \\
&= \phi_j(x) + \max_{z \in \{(w_j(X^k),0)+e_j, (v_j(X^k),0)\}} \langle z, y - x\rangle.
\end{aligned}$$

Recall that $\partial_x\phi = \bigoplus_{j=k+1}^d \partial_x\phi_j$ [Roc70, 10.9 & 7.27]. Hence $\partial_x\phi$ contains the vector $\sum_{j=k+1}^d (v_j(X^k), 0)$ and each $\sum_{j=k+1, j\neq\ell}^d (v_j(X^k), 0) + (w_\ell(X^k), 0) + e_\ell$ for $\ell \in [\![k+1, d]\!]$, which are at pairwise distance at least one. □

The next lemma allows us to pass from the direction of the non-differentiability sets of a convex function to that of the subdifferential. A function φ is semiconvex if $\varphi + \lambda|\cdot|^2/2$ is convex for some $\lambda \in \mathbb{R}$.

Lemma 4.4. — *Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be semiconvex, $A \subset J_k(\varphi)$ be DC_k, and $x \in A$ such that*

– *the dimension of $\partial_x\varphi$ is exactly $d-k$,*
– *the function Φ parametrizing A as in (9) is differentiable at x,*
– *there exist $\varepsilon > 0$ and $r > 0$ such that for any $y \in A \cap \overline{\mathscr{B}}(x,r)$, the set $\partial_y\varphi$ contains $d-k+1$ vectors at pairwise distance at least ε.*

Then $\operatorname{Im}\nabla\Phi(x)$ *is orthogonal to* $\operatorname{span}\{\partial_x\varphi - p\}$ *for any $p \in \partial_x\varphi$.*

Note that the vector space $\operatorname{span}\{\partial_x\varphi - p\}$ is independent of the point $p \in \partial_x\varphi$. Intuitively, Lemma 4.4 generalizes the observation that $\partial_x\varphi$ is orthogonal to any differentiable surface contained in its set of minimum points. The assumption that $\partial_y\varphi$ is sufficiently large for y close to x could be replaced, for instance, by requiring that all $\partial_y\varphi$ near x contain a given (relative) interior point of $\partial_x\varphi$.

Proof. — The function $\Phi : \mathbb{R}^k \to \mathbb{R}^d$ is injective, since it coincides with the identity on k of its coordinates. Let $X \coloneqq \Phi^{-1}(x) \in \mathbb{R}^k$ and $(h_n)_n \subset (0,1)$ be a sequence converging to 0. Fix $e \in \mathbb{R}^k$, and denote $x_n \coloneqq \Phi(X + h_n e) \in A$. By assumption, for sufficiently large n, there exist vectors $(w_n^\ell)_{\ell\in[\![k,d]\!]} \subset \partial_{x_n}\varphi$ at distance at least ε from each other. Since φ is locally Lipschitz, the sequences $(w_n^\ell)_n$ are relatively compact. Extracting subsequences successively, we may assume that $w_n^\ell \to {}_nw^\ell$. By upper semicontinuity, each w^ℓ belongs to $\partial_x\varphi$, and is still at distance at least ε from $w^{\ell'}$ for $\ell' \neq \ell$. Hence the vectors $(w^\ell - w^k)_{\ell\in[\![k+1,d]\!]}$ span a space of dimension $d-k$, which must coincide with $\operatorname{span}\{\partial_x\varphi - w^k\}$ since the latter has dimension $d-k$.

Denoting by λ the semiconvexity constant of φ, we have, for any $\ell, \ell' \in [\![k,d]\!]$,

$$\begin{aligned}\varphi(x_n) &\geqslant \varphi(x) + \langle w^\ell, x_n - x\rangle - \frac{\lambda}{2}|x_n - x|^2\\ &\geqslant \varphi(x_n) + \langle w_n^{\ell'}, x - x_n\rangle + \langle w^\ell, x_n - x\rangle - \lambda|x_n - x|^2.\end{aligned}$$

Dividing by $h_n > 0$ and letting $n \to \infty$, we get

$$\begin{aligned}0 &\geqslant \lim_{n\to\infty}\Big\langle w^\ell - w_n^{\ell'}, \frac{\Phi(X + h_n e) - \Phi(X)}{h_n}\Big\rangle - \frac{\lambda}{h_n}|\Phi(X + h_n e_j) - \Phi(X)|^2\\ &= \langle w^\ell - w^{\ell'}, \nabla\Phi(X)(e)\rangle.\end{aligned}$$

Interchanging ℓ and ℓ', we obtain that $\operatorname{Im}\nabla\Phi(X)$ is orthogonal to $\operatorname{span}\{\partial_x\varphi - w^k\}$. □

We consider the following definition of differentiability, tailored to σ-$\mathrm{DC_k}$ sets.

Definition 4.5 (Tangent plane to a σ-$\mathrm{DC_k}$ set). — Let A be a σ-$\mathrm{DC_k}$ set contained in $\bigcup_n A_n$ for DC_k sets A_n. Let $\mathbb{N}_x \subset \mathbb{N}$ be the subset of $n \in \mathbb{N}$ such that $x \in A_n$, and for $n \in \mathbb{N}_x$, denote by Φ_n a permutation of $(\mathrm{id}_{\mathbb{R}^k}, \varphi_{k+1}^n - \psi_{k+1}^n, \dots, \varphi_d^n - \psi_d^n)$ such

that $A_n = \Phi_n(\mathbb{R}^k)$. Then A admits P as a tangent plane at x if for each $n \in \mathbb{N}_x$ and $j \in [\![k+1, d]\!]$, each φ_j^n, ψ_j^n is differentiable at $\Phi_n^{-1}(x)$, and $\operatorname{Im} \nabla\Phi_n(x) = P$.

Lemma 4.6. — *Let $(A_n)_{n\in\mathbb{N}}$ be a family of DC_k sets, and denote $A \coloneqq \bigcup_{n\in\mathbb{N}} A_n$. Then there exists a σ-$DC_{k\text{-}1}$ set $B \subset A$ such that A admits a tangent plane P at any point $x \in A \setminus B$ in the sense of Definition* 4.5.

Proof. — For each n, denote by $\Phi_n : \mathbb{R}^k \to \mathbb{R}$ a function as in (9) such that $A_n = \Phi_n(\mathbb{R}^k)$. Let $B^{(n)} \subset \mathbb{R}^k$ be the union of the non-differentiability sets of the functions φ_j, ψ_j for $j \in [\![k+1, d]\!]$. Each set $B^{(n)}$ is σ-DC$_{\text{k-1}}$, and by Remark 4.2, the image $\Phi_n(B^{(n)}) \subset \mathbb{R}^d$ is still a σ-DC$_{\text{k-1}}$ subset of $\mathbb{R}^d$. Denote by $P_n(x)$ the image of $\nabla\Phi_n(x)$ at any point $x \in A_n \setminus B^{(n)}$.

We now construct a σ-DC$_{\text{k-1}}$ subset of A outside of which the surfaces A_n cannot intersect transversely. Precisely, for each $n \neq m$, let $B^{(n,m)}$ be the set of $x \in A_n \cap A_m \setminus (B^{(n)} \cup B^{(m)})$ such that $P_n(x) \neq P_m(x)$. We construct a convex function containing $B^{(n,m)}$ in its non-differentiability set J_{k-1}. By Lemma 4.3, there exist convex functions $\phi^{(n)}, \phi^{(m)} : \mathbb{R}^d \to \mathbb{R}$ such that $A_n \subset J_k(\phi^{(n)})$, $A_m \subset J_k(\phi^{(m)})$, and the subdifferentials of $\phi^{(n)}, \phi^{(m)}$ on A_n, A_m contain uniformly separated points. Denote $B_{n,m} \coloneqq J_{k-1}(\phi^{(n)}) \cup J_{k-1}(\phi^{(m)})$, which is σ-DC$_{\text{k-1}}$. By Lemma 4.4, for any $x \in B^{(n,m)} \setminus B_{n,m}$, the affine space containing $\partial_x\phi^{(n)}$ is orthogonal to $P_n(x)$, and similarly for $\partial_x\phi^{(m)}$ and $P_m(x)$. Since $P_n(x) \neq P_m(x)$, we deduce that the affine hulls of $\partial_x\phi^{(n)}$ and $\partial_x\phi^{(m)}$ are not parallel. As a consequence, the convex set

$$\partial_x(\phi^{(n)} + \phi^{(m)}) = \left\{v + w \mid v \in \partial_x\phi^{(n)},\ w \in \partial_x\phi^{(m)}\right\}$$

has dimension at least $d-k+1$. Hence $B^{(n,m)} \setminus B_{n,m}$ is contained in $J_{k-1}(\phi^{(n)}+\phi^{(m)})$, which is σ-DC$_{\text{k-1}}$ since $\phi^{(n)}+\phi^{(m)}$ is convex. Adding back $B_{n,m}$, we obtain that $B^{(n,m)}$ is σ-DC$_{\text{k-1}}$.

Denote $B \coloneqq \bigcup_{n\in\mathbb{N}} B^{(n)} \cup \bigcup_{n,m\in\mathbb{N}^2} B^{(n,m)}$. Then B is σ-DC$_{\text{k-1}}$, and if $x \in A \setminus B$, all sets A_n containing x admit a plane $P_n(x)$ as before. Additionally, $P_n(x) = P_m(x) =: P(x)$ for any $n \neq m$ such that $x \in A_n \cap A_m$, since otherwise x would be in $B^{(n,m)} \subset B$. □

4.2. Characterization of μ^k by concentration on DC_k sets. — We return to measures on $\Omega = \mathbb{R}^d$, with two intermediate results relating the dimension of the map $D^{\mathsf{Tan}} = D^{\perp}$ appearing in Theorem 3.10 to the size of the sets on which μ can be concentrated.

Proposition 4.7 (Large D^{Tan} implies thin concentrations). — *Let $0 \leqslant k \leqslant d-1$, and $\mu \in \mathscr{P}_2(\Omega)$ be a measure such that the Grassmannian section $D^{\mathsf{Tan}} : \Omega \rightrightarrows \mathbb{R}^d$ characterizing Tan^0_μ has dimension at least $d-k$ at μ-almost every point. Then μ is concentrated on a σ-DC_k set.*

For $\varphi : \Omega \to \mathbb{R}$ semiconvex, denote again $J_k(\varphi) \coloneqq \{x \in \Omega \mid \dim \partial_x\varphi \geqslant d-k\}$.

Proof. — We first consider the case of a compactly supported measure, then proceed by exhaustion. Precisely, in the first two steps, we show that whenever μ is compactly

supported and such that $D^{\mathsf{Tan}} = D_\mu^{\mathsf{Tan}}$ has dimension $d-k$, then there exists a measurable σ-$\mathrm{DC_k}$ set B such that $\mu(B) \geqslant 1/2$.

Compact case: construction of B. — Let $(f_j^{\cdot})_{j\in[\![k+1,d]\!]}$ be measurable selections of D^{Tan} such that $(f_{k+1}^{\cdot}(x),\dots,f_d^{\cdot}(x))$ is an orthonormal basis of $D^{\mathsf{Tan}}(x)$ for any $x\in\Omega$. Such maps can be constructed by the Gram–Schmidt orthogonalization algorithm applied to a Castaing representation of D^{Tan}, provided by [Roc69, Th. 3.(d)]. By Proposition 2.2, the measure field

$$\xi \coloneqq \frac{1}{d-k}\sum_{j=k+1}^{d}\frac{1}{2}\left[(\mathrm{id},-f_j^{\cdot})_{\#}\mu + (\mathrm{id},f_j^{\cdot})_{\#}\mu\right]$$

belongs to Tan_μ^0. We show using a Chebyshev-type inequality that if η is close to ξ with respect to W_μ, then it must put mass on balls around each $f_j^{\cdot}(x)$ for any x belonging to a set of large μ-measure.

Let $j\in[\![k+1,d]\!]$ and $s\in\{\pm1\}$. Since $(x,v)\mapsto|v-sf_j^{\cdot}(x)|$ is measurable from $\mathrm{T}\,\Omega$ to $\mathbb{R}$, the sets

$$A_{j,s} \coloneqq \left\{(x,v)\in\mathrm{T}\,\Omega \mid |v-sf_j^{\cdot}(x)|\leqslant 1/2\right\}\subset\mathrm{T}\,\Omega$$

are measurable. Given a set $A\subset\mathrm{T}\,\Omega$, denote $A^x\coloneqq\{v\mid(x,v)\in A\}$. By the disintegration theorem of [Bog07, 10.4.15], for any $\eta\in\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$, there exist measures $(\eta_x)_{x\in\Omega}$ with $\eta_x\in\mathscr{P}(\mathbb{R}^d)$ such that for any $A\subset\mathrm{T}\,\Omega$ measurable, $x\mapsto\eta_x(A^x)$ is measurable, and $\eta(A)=\int_{x\in\Omega}\eta_x(A^x)\,d\mu(x)$. Therefore, the set

$$C\coloneqq\left\{x\in\Omega\;\middle|\;\begin{array}{c}\exists(j,s)\in[\![k+1,d]\!]\times\{\pm1\}\text{ such that}\\ \eta_x(A^x_{j,s})=\eta_x\left(\overline{\mathscr{B}}(sf_j^{\cdot}(x),1/2)\right)\leqslant\frac{1/2}{2(d-k)}\end{array}\right\}$$

is a measurable subset of Ω. Let $\xi_x=\frac{1}{d-k}\sum_{j=k+1}^{d}\frac{1}{2}[\delta_{(x,-f_j^{\cdot}(x))}+\delta_{(x,f_j^{\cdot}(x))}]$ be a particular disintegration of ξ. Whenever $x\in C$, the measure ξ_x places a mass $1/2(d-k)$ on each $\pm f_j^{\cdot}(x)$, whereas η_x places less than $(1/2)/2(d-k)$ on at least one ball of radius $1/2$ centered at one of these points. Consequently, any transport plan α_x between η_x and ξ_x will force a mass of at least $(1/2)/2(d-k)$ to travel a distance greater than $1/2$. In integral form, $\int_{v\in\mathbb{R}^d}|v|^2d\alpha_x\geqslant(1/2)/2(d-k)(1/2)^2$ for any $x\in C$. Taking the infimum over α_x, this yields $W^2(\eta_x,\xi_x)\geqslant 1/16(d-k)$, and integrating over x, we get that

$$\mu(C)=\int_{x\in\Omega}\mathbb{1}_C(x)d\mu(x)\leqslant\int_{x\in\Omega}\frac{W^2(\eta_x,\xi_x)}{1/(16(d-k))}d\mu(x)\leqslant 16(d-k)W^2(\eta,\xi). \tag{11}$$

Here the last inequality holds since $W^2(\xi,\eta)=\int_{x\in\Omega}W^2(\xi_x,\eta_x)d\mu$ [Gig08, Prop. 4.2]. Since ξ is tangent, there exists $\eta\in\mathscr{P}_2(\mathrm{T}\,\Omega)_\mu$ inducing a geodesic on $[0,\tau]$ for some $\tau>0$, and such that $16(d-k)W^2(\eta,\xi)\leqslant\varepsilon$. Let $B\coloneqq C^c$. By (11), $\mu(B)\geqslant 1-\varepsilon$. By definition of C, η_x puts a mass of at least $(1/2)/2(d-k)$ on each $\overline{\mathscr{B}}(\pm f_j^{\cdot}(x),1/2)$ for $x\in B$, so that the restricted measure

$$\gamma\coloneqq\eta\llcorner\bigcup_{j\in[\![k+1,d]\!]}^{s\in\{\pm1\}}A_{j,s}$$

still puts mass on $\mathrm{T}\,B$, i.e., $B \subset \operatorname{supp} \pi_{x\#}\gamma$. Moreover, by restriction of optimality [Vil09, Th. 4.6], γ induces a geodesic on $[0,\tau]$.

Compact case: B is σ-DC_k. — Define φ as the classical explicit Kantorovich potential [Vil09, (5.13)], i.e.,

$$\varphi(x) \coloneqq \frac{1}{2\tau} \sup\Big\{ |\tau v_N|^2 - |x_N + \tau v_N - x|^2 + \textstyle\sum_{i=0}^{N-1} \left(|\tau v_i|^2 - |x_i + \tau v_i - x_{i+1}|^2 \right) \Big\},$$

where the supremum runs over $N \in \mathbb{N}$ and sequences $(x_i, v_i)_{i=1}^N \subset \operatorname{supp}\gamma$. Then $\varphi(x) \geqslant (|\tau v_0|^2 - |x_0 + \tau v_0 - x|^2)/(2\tau)$ by taking $N = 0$. In particular, $\varphi(x_0) \geqslant 0$, and $\varphi(x_0) \leqslant 0$ by cyclical monotonicity. Moreover, $\varphi(x) + |x|^2/(2\tau)$ is a supremum of convex functions, hence φ is semiconvex. Using that $\sup_{a\in A} f(a) - \sup_{a\in A} g(a) \leqslant \sup_{a\in A} f(a) - g(a)$ whenever $\sup_A g < \infty$, we have

$$\begin{aligned}\varphi(x) - \varphi(x_0) &\leqslant \frac{1}{2\tau} \sup_{(x_N,v_N)\in\operatorname{supp}\gamma} |x_N + \tau v_N - x_0|^2 - |x_N + \tau v_N - x|^2 \\ &= \frac{|x_0|^2 - |x|^2}{2\tau} + \frac{|x - x_0|}{\tau} \sup_{(x_N,v_N)\in\operatorname{supp}\gamma} |x_N + \tau v_N|,\end{aligned}$$

with the last supremum bounded by $\operatorname{diam}\operatorname{supp}\mu + \tau(1 + 1/2) < \infty$. Therefore, φ is real-valued. We now show that if $(x, v) \in \operatorname{supp}\gamma$, then $v \in \partial_x\varphi$. For any $\iota > 0$, let $(x_i, v_i)_{i=1}^N \subset \operatorname{supp}\gamma$ be ι-optimal for the definition of $\varphi(x)$. Appending (x, v) to the sequence $(x_i, v_i)_i$, we get that for any $y \in \Omega$,

$$\begin{aligned}\varphi(y) &\geqslant \frac{1}{2\tau}\Big[|\tau v|^2 - |x + \tau v - y|^2 + |\tau v_N|^2 - |x_N + \tau v_N - x|^2 \\ &\qquad\qquad + \sum_{i=0}^{N-1} \left(|\tau v_i|^2 - |x_i + \tau v_i - x_{i+1}|^2 \right) \Big] \\ &\geqslant \langle v, y - x\rangle - \frac{|x-y|^2}{2\tau} + \varphi(x) - \iota.\end{aligned}$$

Letting $\iota \searrow 0$, we obtain that $v \in \partial_x\varphi$. As γ_x puts mass on balls of radius $1/2$ around each $\pm f_j^{\cdot}(x)$ for any $x \in B \subset \operatorname{supp}\mu$, with $|\pm f_j^{\cdot}(x)| = 1$, the convex set $\partial_x\varphi$ has dimension at least $d - k$, so that $B \subset J_k(\varphi)$. By Zajíček's theorem, B is σ-$\mathrm{DC_k}$.

General case: exhaustion. — Consider now $\mu \in \mathscr{P}_2(\Omega)$ arbitrary. Let $R_0 > 0$ be such that $\mu(\overline{\mathscr{B}}(0, R_0)) \geqslant 1/2$. By the restriction formula of Corollary 3.8, the centered tangent cone to $\mu_0 \coloneqq \mu \llcorner \overline{\mathscr{B}}(0, R_0)/\mu(\overline{\mathscr{B}}(0, R_0))$ shares the map D^{Tan} with μ. By the previous steps, there exists a σ-$\mathrm{DC_k}$ set $B_0 \subset \overline{\mathscr{B}}(0, R_0)$ such that $\mu_0(B_0) \geqslant 1/2$, which implies $\mu(B_0) \geqslant 1/4$.

Assume that an increasing sequence of radii R_j and measurable σ-$\mathrm{DC_k}$ sets B_j have been constructed such that $\mu(B_j) \geqslant \mu(B_{j-1}) + (1 - \mu(B_{j-1}))/4$ for $j \in [\![1, n]\!]$. Let R_{n+1} be such that $\mu(B_n^c \cap \mathscr{B}(0, R_{n+1})) \geqslant \mu(B_n^c)/2$, and apply the previous steps to the measure $\mu_{n+1} \coloneqq \mu \llcorner (B_n^c \cap \mathscr{B}(0, R_{n+1}))/\mu(B_n^c \cap \mathscr{B}(0, R_{n+1}))$. This yields a measurable σ-$\mathrm{DC_k}$ set B such that $\mu_{n+1}(B) \geqslant 1/2$. Up to restriction, we may assume

that $B \subset \overline{\mathscr{B}}(0, R_{n+1})$ and is disjoint from B_n. Define $B_{n+1} \coloneqq B_n \cup B$. Then

$$\begin{aligned}\mu(B_{n+1}) &= \mu(B_n) + \mu(B_n^c \cap \mathscr{B}(0, R_{n+1})) \times \mu_{n+1}(B) \geqslant \mu(B_n) + \frac{\mu(B_n^c)}{4} \\ &= \mu(B_n) + \frac{1 - \mu(B_n)}{4}.\end{aligned}$$

The sequence $p_n \coloneqq \mu(B_n)$ satisfies $p_0 \geqslant \frac{1}{4}$ and $p_{n+1} \geqslant \frac{1+3p_n}{4}$, so $\lim_{n\to\infty} p_n = 1$. Therefore, the limit B_∞ of the increasing sequence $(B_n)_n$ satisfies $\mu(B_\infty) = 1$. As a countable union of σ-$\mathrm{DC_k}$ sets, B_∞ is σ-$\mathrm{DC_k}$. □

Remark 4.8 (Support). — The conclusion of Proposition 4.7 cannot be improved to covering the support of μ by DC_k sets. Indeed, consider $\mu = \sum_{n\in\mathbb{N}} \alpha_n \delta_{x_n}$ for a sequence $(x_n)_{n\in\mathbb{N}}$ dense in $[0,1]$. By [Aus25a, Prop. 2.9], Tan^0_μ is equal to $\mathscr{P}_2(\mathrm{T}\,\Omega)^0_\mu$, so that D^{Tan} can be taken identically equal to $\mathbb{R}$. The measure μ is concentrated on the σ-DC_0 (countable) set $\bigcup_{n\in\mathbb{N}}\{x_n\}$, but its support is $[0,1]$.

The following result is a partial converse to Proposition 4.7.

Lemma 4.9 (Thin concentration implies large D^{Tan}). — *Let $0 \leqslant k \leqslant d-1$, and let $\mu \in \mathscr{P}_2(\Omega)$ give positive mass to a measurable DC_k set $A \subset \Omega$. Then the map D^{Tan} characterizing Tan^0_μ in Proposition 2.2 satisfies $\dim D^{\mathsf{Tan}}(x) \geqslant d-k$ for μ-almost every $x \in A$.*

Proof. — By Lemma 4.3, there exist a convex function $\varphi : \Omega \to \mathbb{R}$ and $d-k+1$ measurable vector fields $f_k, \dots, f_d$ such that $f_j(x) \in \partial_x\varphi$ for all x, and for any $x \in A$, $|f_j(x) - f_\ell(x)| \geqslant 1$ for $j \neq \ell \in [\![k, d]\!]$. Let

$$\xi \coloneqq \frac{1}{d-k+1} \sum_{j=k+1}^{d} \frac{(\mathrm{id}, f_j - \mathrm{id})_\#\mu + (\mathrm{id}, f_k - \mathrm{id})_\#\mu}{2}.$$

Then $(\pi_x, \pi_x + \pi_y)_\#\xi$ is concentrated on the graph of the subdifferential of the convex function φ, and ξ is the velocity of a geodesic. By Remark 3.2, the centered measure field $\xi^0 \coloneqq (\pi_x, \pi_v - \mathrm{Bary}\,(\xi)(\pi_x))_\#\,\xi$ is also tangent, and concentrated on $\left(x, \pm\frac{f_j(x)-f_k(x)}{2}\right)$ for $j \in [\![k+1, d]\!]$. Since the latter vectors are independent, ξ^0 splits mass in at least $d-k$ directions, so that $\dim D^{\mathsf{Tan}}(x) \geqslant d-k$ for μ-almost every $x \in A$. □

Combining Proposition 4.7 and Lemma 4.9, we arrive at the following statement.

Theorem 4.10. — *Let $\mu \in \mathscr{P}_2(\Omega)$ be decomposed as $\mu = \sum_{k=0}^d m_k \mu^k$ according to Theorem 3.10. Then, for each $k \in [\![0, d]\!]$ such that $m_k > 0$, the measure μ^k is concentrated on a σ-DC_k set and gives 0 mass to σ-$DC_{k\text{-}1}$ sets. Moreover, for any $k \in [\![0, d]\!]$ and measurable set $A \subset \Omega$,*

$$m_k\,\mu^k(A) = \max_{\substack{B\ \sigma\text{-}DC_k \\ \text{measurable}}} \min_{\substack{C\ \sigma\text{-}DC_{k\text{-}1} \\ \text{measurable}}} \mu(A \cap B \smallsetminus C), \tag{12}$$

and the order of min *and* max *can be reversed.*

Proof. — For each k such that $m_k > 0$, the Grassmannian section D_k characterizing $\mathsf{Tan}^0_{\mu^k}$ has images of dimension $d-k$. Hence, whenever $k < d$, Proposition 4.7 implies that μ^k is concentrated on a σ-$\mathrm{DC_k}$ set. If $k = d$, the only σ-$\mathrm{DC_d}$ set is Ω, and the statement is vacuous. For $k = 0$, we get that μ^0 is purely atomic. Moreover, if $k > 0$, then μ^k must give 0 mass to any DC_{k-1} set, since otherwise Lemma 4.9 would imply that $\dim D_k \geqslant d-(k-1) = d-k+1$ on a set of positive μ^k-measure. For each k such that $m_k > 0$, let A_k be a measurable σ-$\mathrm{DC_k}$ set on which μ^k is concentrated, and define $A_k = \varnothing$ if $m_k = 0$.

Let $k \in [\![0, d]\!]$, and let $A, B, C \subset \mathbb{R}^d$ be measurable such that B is σ-$\mathrm{DC_k}$ and C is σ-$\mathrm{DC_{k\text{-}1}}$. Since μ^j gives 0 mass to any σ-$\mathrm{DC_{j\text{-}1}}$ set, it gives 0 mass to any σ-DC_ℓ set for $\ell \leqslant j-1$. Then

$$\begin{aligned}\mu(A\cap B\smallsetminus C) &= \sum_{j=0}^{d} m_j\mu^j(A\cap B\smallsetminus C)\\ &= \sum_{j=0}^{k-1} m_j\mu^j(A\cap B\smallsetminus C) + m_k\mu^k(A\cap B).\end{aligned} \tag{13}$$

In particular, $\mu(A\cap B\smallsetminus C)\geqslant m_k\mu^k(A\cap B)$ for any such C, with equality if C is chosen as a σ-$\mathrm{DC_{k\text{-}1}}$ set containing $\bigcup_{j\in[\![0,k-1]\!]} A_j$. We get that

$$\min_{\substack{C\ \sigma\text{-}\mathrm{DC_{k\text{-}1}}\\ \text{measurable}}} \mu(A\cap B\smallsetminus C) = m_k\mu^k(A\cap B).$$

Since $m_k\mu^k(A\cap B)\leqslant m_k\mu^k(A)$, with equality if $B = A_k$, we can take the maximum over σ-$\mathrm{DC_k}$ sets B to obtain (12). Using that $\mu^j(A\cap B\smallsetminus C)\leqslant \mu^j(B\smallsetminus C)$, we can also first maximize over A in (13), then minimize over C, to obtain the same result. □

As a consequence of the explicit formula (12), the measures μ^k in Theorem 3.10 are uniquely defined (if $m_k > 0$). For $k = 0$, the measure μ^0 is purely atomic, and collects the atoms of μ; for $k = d$, the measure μ^d gives 0 mass to any DC_{d-1} set, and is the transport-regular part of μ. Let us give some examples.

Example A. — In dimension one, the decomposition reduces to $\mu = m_0\mu^0 + m_1\mu^1$, where μ^0 is concentrated on a σ-$\mathrm{DC_0}$ set and μ^1 gives 0 mass to σ-$\mathrm{DC_0}$ sets. Since σ-$\mathrm{DC_0}$ sets are exactly countable sets, we get that μ^0 is the (normalized) atomic part of μ, while μ^1 is its (normalized) diffuse part. Note that the Cantor part is classified in the same component as the absolutely continuous part of μ.

Example B. — If $\mu = \mathcal{H}^k \llcorner \mathcal{M}$ for $\mathcal{H}^k$ the Hausdorff measure of dimension k, and $\mathcal{M}$ some k-dimensional $\mathcal{C}^2$-manifold with $\mathcal{H}^k(\mathcal{M}) = 1$, then $\mu = \mu^k$. Indeed, covering $\mathcal{M}$ by countably many $\mathcal{C}^2$ charts, and using that $\mathcal{C}^2$ maps are DC [HU85], one sees that $\mathcal{M}$ is σ-$\mathrm{DC_k}$. Moreover, DC_j sets for $j < k$ are negligible for $\mathcal{H}^k$ (as non-differentiability sets of convex functions from $\mathbb{R}^k$ to $\mathbb{R}$). The fact that in this case, the centered elements of the geometric tangent cone are concentrated on the normal directions

to $\mathcal{M}$ was observed by Lott [Lot16], and will be generalized to any $\mu \in \mathscr{P}_2(\Omega)$ in the next section.

Example C. — The regularity in the previous example cannot be weakened to $\mathcal{C}^1$. We sketch a counterexample in dimension $d = 2$, taking inspiration from [Zaj79, Jui11]. Let $f \in \mathcal{C}([0,1];[0,1])$ be continuous and nowhere approximately differentiable, given for instance by [Kha06, Chap. 6]. Define $F : x \mapsto \int_{y=0}^{x} f(y)dy$ and $\mu := (\mathrm{id}, F)_{\#}\mathcal{L}_{[0,1]}$, with $\mathcal{L}$ the Lebesgue measure. We claim that $\mu = \mu^2$, i.e., $D^{\mathsf{Sol}} \equiv \mathbb{R}^2$.

By the above characterization, it suffices to prove that μ gives 0 mass to any DC_1 set. By [AS94], any DC_1 set can be covered by countably many $\mathcal{C}^2$ manifolds up to a set of $\mathcal{H}^1$-measure 0. As μ is absolutely continuous with respect to $\mathcal{H}^1$, it is enough to show that $\mu(A) = 0$ for any $\mathcal{C}^2$ manifold A. Using a parametrization of A by graphs of $\mathcal{C}^2$ functions, this follows from the fact that $\mathcal{L}_{[0,1]}(\{F = \varphi\}) = 0$ for any $\varphi \in \mathcal{C}^2([0,1];\mathbb{R})$. Now, if $\{F = \varphi\}$ has positive Lebesgue measure, one shows that $f(x) = \varphi'(x)$ for any density point $x \in \{F = \varphi\}$. Therefore $\mathcal{L}(\{f = \varphi'\}) > 0$, and since $\varphi' \in \mathcal{C}^1$, the function f admits φ'' as an approximate differential at any density point of $\{f = \varphi'\}$, contradicting the choice of f.

Remark 4.11 (Difference with the decomposability bundle). — In [AM16], the authors introduce the decomposability bundle of a measure μ as a Grassmannian section constructed (roughly) as follows: consider every possible representation of μ as a superposition of 1-Hausdorff measures supported on 1-rectifiable sets, and let D^{AM} be the essential union of the approximate tangent planes to each piece. If $\mu = (\mathrm{id}, F)_{\#}\mathcal{L}_{[0,1]}$ for $F \in \mathcal{C}^1([0,1];\mathbb{R})$, then $D^{\mathrm{AM}}(x, F(x))$ reduces to the classical tangent plane $\mathbb{R} \cdot (1, F'(x))$. This differs from D^{Sol} in example Example C. This raises the following question: can D^{AM} be linked to a Wasserstein tangent cone for another cost?

4.3. Relations with other notions of tangency. — The reader might rightfully complain that so far, no information has been provided concerning the direction of the Grassmannian section D^{Sol} in Corollary 3.3. We now remedy this in two successive steps: first, if μ is concentrated on a σ-$\mathrm{DC_k}$ set A, we show that the tangent planes to A are aligned with D^{Sol} at least μ-almost everywhere. The definition of tangent plane involved is rather strong, and might be useful in applications; however, it uses information coming from a particular set on which μ is concentrated, and lacks an "intrinsic" character. Therefore, in a second step, we show that Preiss tangent measures are concentrated on D^{Sol} at μ-almost every point.

In this section, we consider only $k \in [\![1, d-1]\!]$, since in the extremal cases $k = 0$ and $k = d$, the map D^{Sol} is either $\{0\}$ or the whole space $\mathbb{R}^d$.

Proposition 4.12 (μ^k aligns with D_k^{Sol}). — *Let $k \in [\![1, d-1]\!]$. Assume that $\mu \in \mathscr{P}_2(\Omega)$ is concentrated on a σ-DC_k set A and gives 0 mass to any σ-$DC_{k\text{-}1}$ set. Then for*

μ-almost every point x, A admits a tangent plane $P(x)$ in the sense of Definition 4.5, *and $P(x) = D^{\mathsf{Sol}}(x)$.*

Proof. — Let $(A_n)_n$ be a cover of A by DC_k sets. By Lemma 4.6, there exists a σ-$\mathrm{DC}_{k\text{-}1}$ set $B_0 \subset A$ such that A admits a tangent plane $P = P(x)$ for any $x \in A \setminus B_0$, in the sense of Definition 4.5. Our aim is to show that P is orthogonal to D^{Tan} μ-almost everywhere.

The beginning of the argument follows that of Lemma 4.9. For each A_n, let $\varphi_n : \Omega \to \mathbb{R}$ be a convex function given by Lemma 4.3, i.e., such that $A_n \subset J_k(\varphi_n)$ and $\partial\varphi_n$ admits $d-k+1$ measurable selections $f_k, \dots, f_d$ that are at pairwise distance at least one over A_n. Then, for each $j \in [\![k+1, d]\!]$, the measure field $\frac{1}{2}(\mathrm{id}, f_k - \mathrm{id})_\# \mu + \frac{1}{2}(\mathrm{id}, f_j - \mathrm{id})_\# \mu$ induces a geodesic since φ is convex, so that by Remark 3.2, its projection onto centered measure fields

$$\frac{1}{2}\Big(\mathrm{id}, \frac{f_k - f_j}{2}\Big)_\# \mu + \frac{1}{2}\Big(\mathrm{id}, \frac{f_j - f_k}{2}\Big)_\# \mu$$

is tangent. Since centered tangent measure fields are concentrated on $\operatorname{Graph} D^{\mathsf{Tan}}$ by Theorem 3.10, there exists a μ-negligible set $B_0^{(n)} \subset A$ such that $f_j(x) - f_k(x) \in D^{\mathsf{Tan}}(x)$ for any $x \in A \setminus B_0^{(n)}$. As $D^{\mathsf{Tan}}(x)$ is a subspace of dimension $d-k$, the independent vectors $f_j(x) - f_k(x)$ span it.

We now use the particular choice of φ_n. Recall that for any $x \in A_n \setminus B_0$, the parametrization of A_n is differentiable with differential spanning P. Denote $B_1^{(n)} \coloneqq J_{k-1}(\varphi_n)$ the set of x such that $\dim \partial_x \varphi_n > d-k$, which is σ-$\mathrm{DC}_{k\text{-}1}$. By Lemma 4.4, for any $x \in A_n \setminus (B_0 \cup B_1^{(n)})$,

$$\operatorname{span}\{f_j(x) - f_k(x) \mid j \in [\![k+1, d]\!]\} = P(x)^\perp.$$

Hence, for any $x \in A_n \setminus \big(B_0 \cup B_0^{(n)} \cup B_1^{(n)}\big)$, we have $D^{\mathsf{Tan}}(x) = P(x)^\perp$. Since μ gives 0 mass to σ-$\mathrm{DC}_{k\text{-}1}$ sets, both B_0 and $B_1^{(n)}$ are μ-negligible. Hence $B \coloneqq \bigcup_{n\in\mathbb{N}} B_0 \cup B_0^{(n)} \cup B_1^{(n)}$ is a μ-negligible set such that $P(x) = D^{\mathsf{Tan}}(x)^\perp = D^{\mathsf{Sol}}(x)$ for any $x \in A \setminus B$, as claimed. □

The following definition is taken from [Pre87], where it is shown that any measure μ admits at least one tangent measure ν at μ-almost every point.

Definition 4.13 (Tangent measure in the sense of Preiss). — Let $\mu \in \mathscr{P}(\Omega)$ and $x \in \operatorname{supp}\mu$. A locally finite measure $\nu \in \mathscr{M}_+(\Omega)$ is tangent to μ at x in the sense of Preiss if there exist a vanishing sequence $(h_n)_{n\in\mathbb{N}} \subset (0,1]$, and a sequence $(c_n)_n \in \mathbb{R}_*^+$, such that for any $\varphi \in \mathcal{C}_c(\Omega;\mathbb{R})$,

$$\int_{x\in\Omega} \varphi(x) d\nu(x) = \lim_{n\to\infty} c_n \int_{y\in\Omega} \varphi\Big(\frac{y-x}{h_n}\Big) d\mu(y).$$

Proposition 4.14. — *Let $k \in [\![1, d-1]\!]$. Assume that $\mu \in \mathscr{P}_2(\Omega)$ is concentrated on a σ-DC_k set A and gives 0 mass to any σ-$DC_{k\text{-}1}$ set. Then, for μ-almost every point x, any tangent measure $\nu \in \mathscr{M}_+$ to μ at x in the sense of Definition* 4.13 *is supported on $D^{\mathsf{Sol}}(x)$.*

Proof. — Again, let $(A_n)_n$ be a cover of A by DC_k sets. By Proposition 4.12, there exists a μ-negligible set $B_0 \subset A$ such that for all $x \notin B_0$, A admits $D^{\mathsf{Sol}}(x)$ as a tangent plane in the sense of Definition 4.5. For each n, let $A^{\cup n} \coloneqq \bigcup_{m \leqslant n} A_m$. Then $A = \bigcup_{n\in\mathbb{N}} A^{\cup n}$. By the density theorem [Fed96, 2.9.13], there exists a μ-null set $B^{(n)} \subset A^{\cup n}$ such that every $x \in A^{\cup n} \setminus B^{(n)}$ is a density point of $A^{\cup n}$, i.e.,

$$\lim_{h \searrow 0} \frac{\mu\left(A^{\cup n} \cap \mathscr{B}(x,h)\right)}{\mu(\mathscr{B}(x,h))} = 1. \tag{14}$$

Let $B_1 \coloneqq \bigcup_{n\in\mathbb{N}} B^{(n)}$. Then B_1 is μ-negligible, and for any $x \in A \setminus B_1$, there exists $n_x \in \mathbb{N}$ such that x is a density point of $A^{\cup n_x}$.

Let $x \in A \setminus (B_0 \cup B_1)$ be a point at which the measure μ admits a Preiss tangent measure ν. Following the notations of Definition 4.13, we let $(h_k)_k$ be a vanishing sequence and $(c_k)_k \in \mathbb{R}_*^+$ be such that $\nu = \lim_{k\to\infty} c_k(\frac{\mathrm{id}-x}{h_k})_{\#}\mu$ in C_c'. By the choice of B_0, the set A admits a tangent plan $P \coloneqq D^{\mathsf{Sol}}(x)$ at x in the sense of Definition 4.5. To show that ν is concentrated on P, it is enough to prove that $\nu(\mathscr{B}(0,R_m) \setminus P^\varepsilon) = 0$ for a sequence $(R_m)_m$ going to $+\infty$, and any $\varepsilon > 0$. Here $P^\varepsilon \coloneqq \{y \in \Omega \mid d(y,P) \leqslant \varepsilon\}$. Let $(R_m)_m$ be an increasing sequence going to $+\infty$ and such that $\nu\left(\partial\mathscr{B}(0,R_m)\right) = 0$ for all m, which exists by a classical disintegration argument. For the rest of the proof, $R = R_m$ and $\varepsilon > 0$ are fixed.

Let $I_x \subset [\![0, n_x]\!]$ be the subset of indices n such that $x \in A_n$. In particular, for each $n \in I_x$, there exists $h_n' > 0$ such that $A_n \cap \mathscr{B}(x,Rs) \subset (x+P)^{\varepsilon s}$ for any $s \in (0,h_n')$. On the other hand, for each $n \in [\![0,n_x]\!] \setminus I_x$, the distance $\mathrm{dist}(x,A_n) =: \overline{h}_n$ from the point x to the closed set A_n is positive. Taking the infimum over all $h_n', \overline{h}_n$ for $n \in [\![0,n_x]\!]$, we obtain $\overline{h}_x > 0$ such that $A^{\cup n_x} \cap \mathscr{B}(x,Rs) \subset (x+P)^{\varepsilon s}$, hence

$$\frac{\mu\left((x+P)^{\varepsilon s} \cap A^{\cup n_x} \cap \mathscr{B}(x,Rs)\right)}{\mu\left(A^{\cup n_x} \cap \mathscr{B}(x,Rs)\right)} = 1 \quad \text{for all } s \in (0,\overline{h}_x). \tag{15}$$

For each $\iota \in (0,1)$, consider $\varphi \in \mathcal{C}_c^\infty(\Omega;[0,1])$ such that $\mathbb{1}_{\mathscr{B}(0,(1-\iota)R)} \leqslant \varphi \leqslant \mathbb{1}_{\mathscr{B}(0,R)}$, and $\psi \in \mathcal{C}_c^\infty(\Omega;[0,1])$ such that $\mathbb{1}_{P^\varepsilon \cap \mathscr{B}(0,R)} \leqslant \psi \leqslant \mathbb{1}_{P^\varepsilon \cap \mathscr{B}(0,(1+\iota)R)}$. Then

$$\begin{aligned}\frac{\nu\left(P^\varepsilon \cap \mathscr{B}(0,(1+\iota)R)\right)}{\nu\left(\mathscr{B}(0,(1-\iota)R)\right)} &\geqslant \lim_{k\to\infty} \frac{c_k \int \psi\left(\frac{y-x}{h_k}\right) d\mu}{c_k \int \varphi\left(\frac{y-x}{h_k}\right) d\mu} \geqslant \lim_{k\to\infty} \frac{\mu\left((x+P)^{\varepsilon h_k} \cap \mathscr{B}(x,Rh_k)\right)}{\mu\left(\mathscr{B}(x,Rh_k)\right)} \\ &= \lim_{k\to\infty} \frac{\mu\left(A^{\cup n_x} \cap \mathscr{B}(x,Rh_k)\right)}{\mu\left(\mathscr{B}(x,Rh_k)\right)} \times \frac{\mu\left((x+P)^{\varepsilon h_k} \cap \mathscr{B}(x,Rh_k)\right)}{\mu\left(A^{\cup n_x} \cap \mathscr{B}(x,Rh_k)\right)} = 1.\end{aligned}$$

In the last line, we used respectively (14) and (15). Sending ι to 0, and using that $\nu(\partial\mathscr{B}(0,R)) = 0$, we get

$$\nu\left(P^\varepsilon \cap \overline{\mathscr{B}}(0,R)\right) \geqslant \nu\left(\mathscr{B}(0,R)\right) = \nu\left(P^\varepsilon \cap \overline{\mathscr{B}}(0,R)\right) + \nu\left(\mathscr{B}(0,R) \setminus P^\varepsilon\right).$$

Thus $\nu\left(\mathscr{B}(0,R) \setminus P^\varepsilon\right) = 0$ for all $\varepsilon > 0$ and $R = R_m$, so that ν is supported on P. □

It may happen that the inclusion $\operatorname{supp}\nu \subset D^{\mathsf{Sol}}(x)$ is strict. For instance, in example Example C at the end of Section 4.2, the measure $(\mathrm{id}, F)_{\#}\mu$ with $F \in \mathcal{C}^1$ admits $(\mathrm{id}, F'(x)\cdot\mathrm{id})_{\#}\mathcal{L}_{\mathbb{R}}$ as its unique Preiss tangent measure at $(x,F(x)) \in \mathbb{R}^2$. In particular, ν is supported on a line, but D^{Sol} is equal to $\mathbb{R}^2$ at μ-almost every point.

Appendix

Proof of Proposition 2.7. — The existence and uniqueness of $\pi^\mu_C\xi$ follow verbatim the steps of [Gig08, Prop. 4.30], mimicking the Hilbertian case. We show that for any $\alpha \in \Gamma_{\mu,o}(\xi, \pi^\mu_C\xi)$, the element $\gamma := (\pi_x, \pi_v - \pi_w)_\#\alpha$ belongs to $C^\perp$. Let $\zeta \in C$ be arbitrary, and $\beta = \beta(\pi_x, \pi_u, \pi_v, \pi_w) \in \Gamma_\mu(\xi, \pi^\mu_C\xi, \zeta)$ with $(\pi_x, \pi_u, \pi_v)_\#\beta = \alpha$. By horizontal convexity, $(\pi_x, (1-\varepsilon)\pi_v + \varepsilon\pi_w)_\#\beta \in C$. Then

$$\begin{aligned} W^2_\mu(\xi, \pi^\mu_C\xi) &\leqslant W^2_\mu\left(\xi, (\pi_x, (1-\varepsilon)\pi_v + \varepsilon\pi_w)_\#\beta\right) \leqslant \int_{(x,v,w)} |u - (1-\varepsilon)v - \varepsilon w|^2 \, d\beta \\ &= W^2_\mu(\xi, \pi^\mu_C\xi) + 2\varepsilon \int \langle u - v, v - w\rangle d\beta + \varepsilon^2 \int |v-w|^2 d\beta. \end{aligned}$$

Letting ε tend to 0, we get $0 \leqslant \int\langle u-v, v-w\rangle d\beta$. Choosing in particular $\zeta = r\cdot\pi^\mu_C\xi \in C$ for $r > 0$, and $\beta = (\pi_x, \pi_v, \pi_w, r\pi_w)_\#\alpha$, we get that $\int\langle u - v, v\rangle d\alpha = 0$ for any $\alpha \in \Gamma_{\mu,o}(\xi, \pi^\mu_C\xi)$. Therefore, for a general $\zeta \in C$,

$$\langle \gamma, \zeta\rangle_\mu = \sup_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} \int_{(x,u,v,w)} \langle u - v, w\rangle d\beta \leqslant \int_{(x,u,v)} \langle u - v, v\rangle d\alpha = 0.$$

As the opposite inequality holds by Lemma 2.5, we deduce that $\gamma \in C^\perp$.

Now, for any $\zeta \in C^\perp$, we introduce $\pi^\mu_C\xi$ in the definition of $W^2_\mu(\xi, \zeta)$ to get

$$\begin{aligned} W^2_\mu(\xi,\zeta) &= \inf_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} \int_{(x,u,v,w)} |v + (u - v - w)|^2 \, d\beta \\ &= \|\pi^\mu_C\xi\|^2_\mu + \inf_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} 2\int_{(x,u,v,w)} \langle v, u - v - w\rangle d\beta + |(u-v) - w|^2 d\beta \\ &\geqslant W^2_\mu(\xi,\gamma) + \inf_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} 2\int_{(x,u,v,w)} \langle v, -w\rangle d\beta + W^2_\mu(\gamma,\zeta). \end{aligned}$$

Here we used respectively that $(\pi_x, \pi_v, \pi_v - \pi_w)_\#\alpha$ is a transport plan between ξ and γ, that $\int\langle v, u - v\rangle d\alpha = 0$, and that $(\pi_x, \pi_u - \pi_v, \pi_w)_\#\beta$ is a transport plan between γ and ζ. The middle term is exactly $-2\langle\pi^\mu_C\xi, \zeta\rangle_\mu$, which vanishes since $\pi^\mu_C\xi \in C$ and $\zeta \in C^\perp$. Therefore, for any $\zeta \in C^\perp$, we have $W^2_\mu(\xi,\zeta) \geqslant W^2_\mu(\xi,\gamma) + W^2_\mu(\gamma,\zeta)$, and γ must be the metric projection of ξ onto $C^\perp$.

If now $\zeta \in C$, we have for any $\beta \in \Gamma_\mu(\xi, \pi^\mu_C\xi, \zeta)$ such that $(\pi_x, \pi_u, \pi_v)_\#\beta = \alpha$ that $\int\langle u - v, w\rangle d\beta \leqslant \langle\gamma, \zeta\rangle_\mu = 0$. We deduce that

$$\begin{aligned} \langle\xi, \zeta\rangle_\mu &= \sup_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} \int_{(x,u,v,w)} \langle u - v + v, w\rangle d\beta \\ &\leqslant 0 + \sup_{\substack{\beta\in\Gamma_\mu(\xi,\pi^\mu_C\xi,\zeta)\\(\pi_x,\pi_u,\pi_v)_\#\beta=\alpha}} \int_{(x,u,v,w)} \langle v, w\rangle d\beta = \langle\pi^\mu_C\xi, \zeta\rangle_\mu. \end{aligned}$$

If, in addition, $\zeta = -1 \cdot \zeta$, then

$$\int \langle u - v, w \rangle d\beta = -\int \langle u - v, -w \rangle d\beta \geqslant -\langle \gamma, -1 \cdot \zeta \rangle_\mu = -\langle \gamma, \zeta \rangle_\mu = 0,$$

and equality holds in the penultimate line, so that $\langle \xi, \zeta \rangle_\mu = \langle \pi_C^\mu \xi, \zeta \rangle_\mu$. □

Proof of Lemma 3.6. — Let $M \geqslant 0$ be such that $\operatorname{supp} \nu \subset \overline{\mathscr{B}}(0, M)$, and $(x_0, v_0) \in \operatorname{supp}(\pi_x, \pi_x + \pi_v)_\# \eta$. The formula

$$\varphi(x) \coloneqq \sup \left\{ \textstyle\sum_{i=0}^n |x_i - y_i|^2 - \sum_{i=0}^{n-1} |x_{i+1} - y_i|^2 - |x - y_n|^2 \;\middle|\; \begin{array}{c} n \in \mathbb{N},\ (x_i, y_i)_{i=1}^n \subset \\ \operatorname{supp}(\pi_x, \pi_x + \pi_v)_\# \eta \end{array} \right\}$$

defines a semiconvex function from Ω to $\mathbb{R} \cup \{\infty\}$ with the property that

$$\varphi^c(y) - \varphi(x) = |x - y|^2$$

for $(\pi_x, \pi_x + \pi_v)_\# \eta$-almost every (x, y) [Vil09, Th. 5.10]. The support of $(\pi_x, \pi_x + \pi_v)_\# \eta$ is cyclically monotone, so taking $x = x_0$, we get that $\varphi(x_0) \leqslant 0$. Since each y_i appearing in the supremum is contained in $\operatorname{supp}(\pi_x + \pi_v)_\# \eta = \operatorname{supp} \nu \subset \overline{\mathscr{B}}(0, M)$, one has

$$\begin{aligned} \varphi(x) - \varphi(x_0) &\leqslant \sup \left\{ -|x - y_n|^2 + |x_0 - y_n|^2 \;\middle|\; \begin{array}{c} \exists x_n \in \Omega \text{ such that} \\ (x_n, y_n) \in \operatorname{supp}(\pi_x, \pi_x + \pi_v)_\# \eta \end{array} \right\} \\ &\leqslant |x_0|^2 + 2M|x_0 - x| - |x|^2. \end{aligned}$$

The function φ is bounded from below by $-|x - y_0|^2 + |y_0 - x_0|^2$ by definition, hence locally bounded, and therefore locally Lipschitz since it is semiconvex. Therefore the set-valued subdifferential map $x \mapsto \partial_x \left(\varphi - |\cdot|^2/2\right)$ is compact-valued, and upper semicontinuous in the set-valued sense by [Roc70, Cor. 24.5.1]. By the selection theorem [AB06, 18.13], it admits a measurable selection $f : \Omega \to \mathrm{T}\,\Omega$. Then define $\xi \coloneqq (1 - \lambda)\eta + \lambda(f_\# \mu_2)$. By construction, ξ is still supported on the subdifferential of $\varphi - |\cdot|^2/2$, so $(\pi_x, \pi_x + \pi_v)_\# \xi$ is cyclically monotone, hence optimal. □

Proof of Lemma 3.7

By definition, the set of measure fields of the form $\lambda \cdot (\pi_x, \pi_y - \pi_x)_\# \gamma$, where $\lambda \geqslant 0$ and γ is optimal, is dense in Tan_μ with respect to W_μ. The metric scalar product being continuous and positively homogeneous, one has that $\zeta \in \mathsf{Sol}_\mu$ if and only if $\langle \zeta, (\pi_x, \pi_y - \pi_x)_\# \gamma \rangle_\mu = 0$ for any $\gamma \in \Gamma_o(\mu, \nu)$ and $\nu \in \mathscr{P}_2(\Omega)$. To prove the claim, we have to show that it suffices to consider compactly supported ν. Since we need convergence with respect to W_μ, we cannot approximate the target measure by an arbitrary compactly supported measure and use stability of optimality; hence we construct an explicit approximation. The results on c-transforms that we use may be found in [Vil09, §5], in particular Th. 5.10.

Let $\eta \in \Gamma_o(\mu, \nu)$ for some $\nu \in \mathscr{P}_2(\Omega)$. Let $\varphi : \Omega \to \mathbb{R} \cup \{\infty\}$ be a proper c-convex Kantorovich potential for the pair (μ, ν), and $\varphi^c : y \mapsto \inf_{x \in \Omega} \varphi(x) + |x - y|^2$ its c-transform, which satisfies $\varphi(x) = \sup_{y \in \Omega} \varphi^c(y) - |x - y|^2$. Denote by $\Gamma \subset \Omega^2$ the set of (x, y) such that $\varphi^c(y) - \varphi(x) = |x - y|^2$, which contains $\operatorname{supp} \eta$ and is cyclically monotone. Our strategy is to "truncate" Γ in the y variable, as follows.

For each $R > 0$, define

$$\varphi_R(x) \coloneqq \sup_{y \in \overline{\mathscr{B}}(0,R)} \varphi^c(y) - |x-y|^2 = \sup_{y \in \Omega} \varphi^c(y) - |x-y|^2 - \chi_R(y),$$

where $\chi_R(y) = 0$ if $|y| \leqslant R$, and $+\infty$ otherwise. The function φ_R is c-convex by definition, bounded from above by φ, bounded from below by a quadratic polynomial, and for any $x_0 \in \operatorname{dom} \varphi$,

$$\varphi_R(x) - \varphi_R(x_0) \leqslant \sup_{y \in \overline{\mathscr{B}}(0,R)} |x_0 - y|^2 - |x-y|^2 \leqslant |x_0|^2 + 2R|x_0 - x| - |x|^2.$$

Hence φ_R is locally bounded, and locally Lipschitz since it is semiconvex.

By a classical computation, the set $\Gamma_R \subset \Omega^2$ of pairs (x,y) such that $\varphi_R(x) = \varphi^c(y) - |x-y|^2 - \chi_R(y)$ is cyclically monotone, and contained in $\Omega \times \overline{\mathscr{B}}(0,R)$. Note that $\Gamma \cap (\Omega \times \overline{\mathscr{B}}(0,R)) \subset \Gamma_R$; indeed, whenever $(x,y) \in \Gamma$ with $|y| \leqslant R$, we have $\varphi(x) = \varphi^c(y) - |x-y|^2 \geqslant \varphi^c(z) - |x-z|^2$ for any $z \in \mathbb{R}^d$. In particular, the supremum on the ball of radius R is also attained at y, and $\varphi_R(x) = \varphi(x) = \varphi^c(y) - |x-y|^2 - \chi_R(y)$. We deduce that

$$\operatorname{supp} \eta \cap \big(\Omega \times \overline{\mathscr{B}}(0,R)\big) \subset \Gamma_R. \tag{16}$$

For each $R > 0$, the correspondence $x \mapsto \{y \mid (x,y) \in \Gamma_R\}$ is upper semicontinuous with compact images. By [AB06, 18.13], it admits a measurable selection $f_R : \Omega \to \Omega$, which satisfies $|f_R(x)| \leqslant R$ for all $x \in \Omega$. Define η_R by replacing the part of η that lies outside $\Omega \times \overline{\mathscr{B}}(0,R)$ by the measurable selection f_R; explicitly,

$$\eta_R \coloneqq \eta \llcorner \big(\Omega \times \overline{\mathscr{B}}(0,R)\big) + (\pi_x, f_R(\pi_x))_{\#} \eta \llcorner (\Omega \times \overline{\mathscr{B}}(0,R)^c).$$

Recalling (16), η_R is a probability measure concentrated on the cyclically monotone set Γ_R, hence an optimal transport plan between its marginals. The measure $\pi_{y\#}\eta_R$ is supported on $\overline{\mathscr{B}}(0,R)$ by construction, and since $|f_R(x)| \leqslant R$, one has

$$\begin{aligned} W_\mu^2 \left((\pi_x, \pi_y - \pi_x)_{\#}\eta, (\pi_x, \pi_y - \pi_x)_{\#}\eta_R\right) &\leqslant \int_{x,y \in \Omega, |y|>R} |f_R(x) - y|^2 \, d\eta \\ &\leqslant \int_{y \in \Omega, |y|>R} (R + |y|)^2 d\nu \underset{R\to\infty}{\longrightarrow} 0. \end{aligned}$$

Recalling that $\langle \zeta, (\pi_x, \pi_y - \pi_x)_{\#}\eta_R \rangle_\mu = 0$ for any R by assumption, we deduce that $\langle \zeta, (\pi_x, \pi_y - \pi_x)_{\#}\eta \rangle_\mu = 0$. Since η and ν were arbitrary, ζ is solenoidal. □

Conflict of interest. — For the purpose of open access, the author has applied a CC-BY public copyright license to any Author Accepted Manuscript (AAM) version arising from this submission. Data availability is not applicable to this article as no new data were created or analyzed in this study.

References

[AM16] G. Alberti & A. Marchese – "On the differentiability of Lipschitz functions with respect to measures in the Euclidean space", *Geom. Funct. Anal.* **26** (2016), no. 1, p. 1–66.

[AB06] C. Aliprantis & K. Border – *Infinite dimensional analysis*, Springer-Verlag, Berlin, Heidelberg, 2006.

[AGS05] L. Ambrosio, N. Gigli & G. Savaré – *Gradient flows*, Lectures in Math. ETH Zürich, Birkhäuser-Verlag, Basel, 2005.

[AS94] G. Anzellotti & R. Serapioni – "Ck-rectifiable sets", *J. reine angew. Math.* **453** (1994), p. 1–20.

[Aus25a] A. Aussedat – "On the structure of the geometric tangent cone to the Wasserstein space", *J. Differential Equations* **442** (2025), p. 113520.

[Aus25b] ———, "Optimal control problems and Hamilton-Jacobi-Bellman equations in some curved metric spaces", PhD Thesis, INSA Rouen Normandie, 2025, available at `https://theses.hal.science/tel-05133552/`.

[Bog07] V. I. Bogachev – *Measure theory*, Springer, Berlin, Heidelberg, 2007.

[Bre91] Y. Brenier – "Polar factorization and monotone rearrangement of vector-valued functions", *Comm. Pure Appl. Math.* **44** (1991), no. 4, p. 375–417.

[Fed96] H. Federer – *Geometric measure theory*, Classics in Math., Springer, Berlin, Heidelberg, 1996.

[Gig08] N. Gigli – "On the geometry of the space of probability measures endowed with the quadratic optimal transport distance", PhD Thesis, Scuola Normale Superiore di Pisa, Pisa, 2008, `https://ricerca.sns.it/retrieve/e3aacdfd-ef40-4c98-e053-3705fe0acb7e/Gigli_Nicola.pdf`.

[Gig11] ———, "On the inverse implication of Brenier-McCann theorems and the structure of $(P_2(M), W_2)$", *Methods Appl. Anal.* **18** (2011), no. 2, p. 127–158.

[Har59] P. Hartman – "On functions representable as a difference of convex functions", *Pacific J. Math.* **9** (1959), no. 3, p. 707–713.

[HU85] J.-B. Hiriart-Urruty – "Generalized differentiability/duality and optimization for problems dealing with differences of convex functions", in *Convexity and duality in optimization* (J. Ponstein, ed.), Springer, Berlin, Heidelberg, 1985, p. 37–70.

[Jui11] N. Juillet – "On displacement interpolation of measures involved in Brenier's theorem", *Proc. Amer. Math. Soc.* **139** (2011), no. 10, p. 3623–3623.

[Kha06] A. B. Kharazishvili – *Strange functions in real analysis*, 2nd ed., Pure and Applied Math., no. 272, Chapman & Hall/CRC, Boca Raton, FL, 2006.

[Lot16] J. Lott – "On tangent cones in Wasserstein space", *Proc. Amer. Math. Soc.* **145** (2016), no. 7, p. 3127–3136.

[Pav04] D. Pavlica – "On the points of non-differentiability of convex functions", *Comment. Math. Univ. Carolinae* **45** (2004), no. 4, p. 727–734.

[Pre87] D. Preiss – "Geometry of measures in $\mathbb{R}^n$: Distribution, rectifiability, and densities", *Ann. of Math. (2)* **125** (1987), no. 3, p. 537–643.

[Roc69] R. T. Rockafellar – "Measurable dependence of convex sets and functions on parameters", *J. Math. Anal. Appl.* **28** (1969), no. 1, p. 4–25.

[Roc70] ———, *Convex analysis*, Princeton University Press, Princeton, NJ, 1970.

[San15] F. Santambrogio – *Optimal transport for applied mathematicians*, Progress in nonlinear differential equations and their appl., vol. 87, Springer International Publishing, 2015.

[SW14] M. Sedjro & M. Westdickenberg – "Characterization of the tangent space of monotone transport plans in $\mathbb{R} \times \mathbb{R}$ with prescribed first projection", 2014, `arXiv:1411.3836`.

[Vil09] C. Villani – *Optimal transport*, Grundlehren Math. Wissen., vol. 338, Springer Berlin Heidelberg, Berlin, Heidelberg, 2009.

[Zaj79] L. Zajíček – "On the differentiation of convex functions in finite and infinite dimensional spaces", *Czechoslovak Math. J.* **29** (1979), no. 3, p. 340–348.

Manuscript received 6th February 2026
accepted 2nd October 2026

Averil Aussedat, Dipartimento di Matematica, Università di Pisa,
largo Pontecorvo 5, 56127 Pisa, Italy
E-mail : `averil.aussedat@dm.unipi.it`
Url : `https://averil-aussedat.github.io/`

Characterization of measures on the real line that are critically unstable under small shifts

Averil Aussedat*

Abstract

We study the perturbation of a measure $\mu \in \mathscr{P}(\mathbb{R})$ consisting in superposing two copies of μ, each slightly shifted by a small distance $\pm h$. The difference between μ and its perturbation is measured with a Wasserstein distance. For any μ, this distance is bounded from above by h. We show that measures for which this critical rate is achieved when h goes to 0 are characterized as the ones giving most of their mass to some particular porous sets. This is used to identify which measures μ on the real line have a 2-Wasserstein tangent cone equal to the set of directions inducing curves with maximal initial speed.

Keywords Wasserstein distances; porous sets; tangent cone; Cantor sets.
MSC 49Q22, 28A80, 26A30.

1 Introduction

The set $\mathscr{P}(\mathbb{R})$ of probability measures can be endowed with the family of Wasserstein distances $W_p(\cdot,\cdot)$, measuring the minimal L^p-travel distance for an exponent $p \in [1,\infty)$. All these distances coincide in some particular cases: for instance, if $\mu = \delta_0$ and $\mu_h \coloneqq \frac{1}{2}[\delta_{-h} + \delta_h]$, then $W_p(\mu,\mu_h) = h$ for any $h \geqslant 0$ and $p \in [1,\infty)$. This article focuses on the set of measures satisfying the same inequality at the limit when h goes to zero. Precisely, given $\mu \in \mathscr{P}(\mathbb{R})$, we consider the superposition of its left and right shift

$$\mu_h \coloneqq \frac{1}{2}\left[(id-h)_\#\mu + (id+h)_\#\mu\right],$$

and ask whether it holds that

$$\limsup_{h\searrow 0} \frac{W_p(\mu,\mu_h)}{h} = 1. \tag{1}$$

The inequality $\leqslant$ always holds, and for absolutely continuous measures, the limit sup in is equal to 0 instead of 1 (see [Aus25b, Lemma 5.4.7 and Theorem 5.2.23]). Hence the condition (1) is a critical behavior, and we expect any μ satisfying it to be very concentrated. It turns out that the set of such μ is independent of $p \in [1,\infty)$ (see Remark 3.2 below). The first result of the paper states that (1) is equivalent to the fact that μ gives almost all its mass to an element of a very restrictive class $\mathscr{A}$ of porous set.

Definition 1.1. A measurable set $A \subset \mathbb{R}$ belongs to $\mathscr{A}$ if for some $(s_n)_{n\in\mathbb{N}} \subset (0,1]$ going to 0 when $n \to \infty$, there holds

$$\lim_{n\to\infty} \sup_{x\in A} \inf \left\{\tau \in (0,1) \;\middle|\; \mathscr{B}(x,s_n) \cap A \subset \overline{\mathscr{B}}(x,\tau s_n)\right\} = 0. \tag{2}$$

In words, in any ball of radius s_n around $x \in A$, the set A is contained in the smaller ball of radius τs_n for some τ that goes to 0 with s_n. Sets in $\mathscr{A}$ are porous in the sense that each ball $\mathscr{B}(x,s_n)$ centred at $x \in A$ contains a ball of diameter larger than $(1-\tau)s_n$ that does not intersect A (see for instance [Mer+03]), but Definition 1.1 is stronger in general: the same sequence of scales $(s_n)_n$ is shared by all points $x \in A$, and τ is as small as desired.

Theorem 1.2. *Let $\mu \in \mathscr{P}(\mathbb{R})$. The following conditions are equivalent:*

(a) the equality (1) holds for some (thus any) $p \in [1,\infty)$;

(b) for any $\varepsilon > 0$, there exists $A \in \mathscr{A}$ such that $\mu(A) \geqslant 1-\varepsilon$.

*Dipartimento di Matematica, Università di Pisa, largo Pontecorvo 5, 56127 Pisa, Italy. averil.aussedat@dm.unipi.it

The motivation of Theorem 1.2 is to solve a pending question on the geometry of the 2-Wasserstein space. For $p = 2$, one can use the existing theory of Alexandrov spaces [AKP23] to construct a geometric tangent cone attached to each measure [AGS05; Gig08]. The elements of this cone can be represented by measure-valued applications in $L^2_\mu(\mathbb{R}^d; \mathscr{P}_2(\mathbb{R}^d))$. The subset of elements induced by a map, i.e. of the form $x \mapsto \delta_{f(x)}$ for some $f \in L^2_\mu(\mathbb{R}^d; \mathbb{R}^d)$, has proved to be useful in studying differentiability of functions of measures [BG97; Ott01; AGS05; GNT08; CD18; GT19]. In recent years however, the general case of elements that could take values outside of Dirac masses has gained attention, both to generalize continuity equations with "measure fields" and unify discrete and continuous treatments of numerical schemes (formulated with trajectories that "split mass" at the discrete level) [Pic19; Cam+21; CSS23a; CSS23b], or as a theoretically appealing definition for optimization and differentiability over measures [LBD22; Ber24; Sch25; Ber25; CST26].

In this transition from map-induced to general elements of the tangent cone, some properties are lost. The one that motivates Theorem 1.2 is the following: if $f \in L^2_\mu(\mathbb{R}^d; \mathbb{R}^d)$ is a vector field, then the fact that $x \mapsto \delta_{f(x)}$ belongs to the tangent cone is equivalent to the fact that $W_2(\mu, (id + hf)_\#\mu)/h = \|f\|_{L^2_\mu} + O(h)$ [Aus25b, Proposition 5.3.8]. This equivalence replaces the definition of the tangent cone, which goes by approximation by geodesics in a strong topology, by a simple computation of the initial speed of the curve $h \mapsto (id + hf)_\#\mu$. Unfortunately, it does not hold in the general case. The second result of this paper is to characterize the set of measures μ for which

$$\xi \in \mathbf{Tan}_\mu \qquad \Longleftrightarrow \qquad \limsup_{h \searrow 0} \frac{W_2(\mu, (\pi_x + h\pi_v)_\#\xi)}{h} = \|\xi\|_\mu. \tag{3}$$

Theorem 1.3. *Let $\mu \in \mathscr{P}_2(\mathbb{R})$. The following conditions are equivalent:*

1. *the equivalence (3) holds;*
2. *the diffuse part of μ satisfies $\mu^d(A) = 0$ for any $A \in \mathscr{A}$.*

Here diffuse means nonatomic, i.e. μ^d is equal to μ minus its atomic part. Examples for which (3) fails can be found in [Aus25a] or Section 5. The case of dimension $d > 1$ is open: the argument in dimension $d = 1$ is eased by the fact that we can reduce to a "canonical" case, with a measure field giving the same mass to each unit direction. In higher dimension, one cannot hope for such a reduction, and the description of the class $\mathscr{A}$ might have to be adapted to a form of anisotropic porosity.

In the last part of the paper, we consider the related problem of measures for which the limit sup in (1) is equal to 0, instead of 1. This is still independent of $p \in [1, \infty)$. The motivation comes from the fact that for a map $g \in L^2_\mu(\mathbb{R}^d; \mathbb{R}^d)$, it is equivalent that $W_2(\mu, (id + hg)_\#\mu) = o(h)$, and that $\langle g, f \rangle_{L^2_\mu} = 0$ for any $f \in L^2_\mu$ inducing a tangent element. Such g are actually the L^2_μ vector fields for which $\operatorname{div}(g\mu) = 0$ in the sense of distributions, and the fact that they let μ "infinitesimally invariant" extends the theorem of Liouville on incompressible flows being driven by divergence-free vector fields. Once again, this does not generalize to measure-valued fields, hence the need for some regularity condition on μ under which the equivalence would stand.

We do not have a fully satisfactory characterization of the set of measures μ such that $W_1(\mu, \mu_h) = o(h)$, besides some regularity of the distribution function of μ. However, we can exhibit such a μ that is singular with respect to the Lebesgue measure. This seems to us a curiosity in itself: indeed, from the definition, it holds that for any Lipschitz function $u : \mathbb{R} \to \mathbb{R}$,

$$\left| \int_{x \in \mathbb{R}} \frac{\frac{1}{2}\left[u(x-h) + u(x+h)\right] - u(x)}{h} d\mu \right| \leqslant \operatorname{Lip}(u) \frac{W_1(\mu, \mu_h)}{h} \underset{h \searrow 0}{\longrightarrow} 0.$$

If the absolute values were inside the first integral, we would conclude to a kind of differentiability of any Lipschitz function $\mu-$almost everywhere. However, the measures for which Lipschitz functions are differentiable a.e. are exactly the absolutely continuous measures [Zah46; FP08]. Here, we do not exactly consider "differentiability", but an averaged version on both the directions and on the points; still, it was not so obvious that some singular measures would appear.

The paper is organized as follows. Section 2 precises the notations, definitions and facts in use in the sequel. Theorem 1.2 is proved in Section 3. Theorem 1.3 is proved in Section 4. Section 5 concludes with some explicit computations around the problem mentioned in the end of the introduction.

2 Preliminaries

We use $O(h)$ to denote a quantity that goes to 0 when h does, and $o(h)$ for a quantity such that $o(h)/h$ goes to 0 when h does, independently of the rate of convergence. The open ball of radius $r \geqslant 0$ around a point $x \in \mathbb{R}$ is denoted $\mathscr{B}(x, r)$. The closure of a set $A \subset \mathbb{R}$ is denoted $\overline{A}$, and its complement $A^c \coloneqq \mathbb{R} \setminus A$. A vanishing sequence $(h_n)_{n \in \mathbb{N}} \subset (0, \infty)$ is a nonincreasing sequence that converges to 0. If $\mu \in \mathscr{P}(X)$ is a Borel probability measure over

a Polish space X, and $f : X \to Y$ is a measurable map between Polish spaces, then $f_{\#}\mu$ is the probability measure on Y determined by $(f_{\#}\mu)(A) = \mu(f^{-1}(A))$ for any measurable $A \subset Y$. The set $\mathrm{T}\,\mathbb{R} \coloneqq \{(x, v) \mid x \in \mathbb{R}, v \in \mathrm{T}_x\,\mathbb{R}\}$ is isometric to $\mathbb{R}^2$.

The following material is classical, and can be found in detailed form in [San15] (in particular Proposition 2.17 for the monotone plan). For the dual formulation, we adopt the sign convention of [Vil09, Theorem 5.10].

Wasserstein distances When μ is a measure on a set $X \times Y$, we use the notation $\mu = \mu(dx, dy)$ to give names to the variables of μ, and later use them in the canonical projections π_x, π_y. The set of measures $\xi = \xi(dx, dv) \in \mathscr{P}(\mathrm{T}\,\mathbb{R})$ that satisfy $\pi_{x\#}\xi = \mu$ is denoted $\mathscr{P}(\mathrm{T}\,\mathbb{R})_\mu$. Each element of $\mathscr{P}(\mathrm{T}\,\mathbb{R})_\mu$ induces a curve of measures denoted

$$h \mapsto \exp_\mu(h \cdot \xi) \coloneqq (\pi_x + h\pi_v)_{\#}\xi.$$

The set of transport plans between $\mu \in \mathscr{P}(X)$ and $\nu \in \mathscr{P}(Y)$ is denoted

$$\Gamma(\mu, \nu) \coloneqq \{\ \eta = \eta(dx, dy) \in \mathscr{P}(X \times Y) \mid \pi_{x\#}\eta = \mu, \text{ and } \pi_{y\#}\eta = \nu\ \}.$$

For $p \in [1, \infty)$, let $\mathscr{P}_p(\mathbb{R})$ be the subset of $\mu \in \mathscr{P}(\mathbb{R})$ such that $\int_{x\in\mathbb{R}} |x|^p d\mu < \infty$. The set $\mathscr{P}_p(\mathrm{T}\,\mathbb{R})_\mu$ is defined analogously. Whenever $\xi \in \mathscr{P}_p(\mathrm{T}\,\mathbb{R})_\mu$, there holds $\exp_\mu(h \cdot \xi) \in \mathscr{P}_p(\mathbb{R})$ for all $h \geqslant 0$. The $p-$Wasserstein distance $W_p(\mu, \nu)$ between $\mu, \nu \in \mathscr{P}_p(\mathbb{R})$ is defined as

$$W_p^p(\mu, \nu) \coloneqq \inf_{\eta\in\Gamma(\mu,\nu)} \int_{(x,y)\in\mathbb{R}^2} |x - y|^p d\eta(x, y).$$

In general, minimizers exist but are not unique, and we denote by $\Gamma_{o,p}(\mu, \nu)$ the set of minimizers (dropping the p when it is clear from the context). In the sequel, we might take Wasserstein distances between measures μ, ν with $0 < \mu(\mathbb{R}) = \nu(\mathbb{R}) \neq 1$; in this case, we adopt the convention

$$W_p^p(\mu, \nu) \coloneqq \mu(\mathbb{R}) \times W_p^p\left(\frac{\mu}{\mu(\mathbb{R})}, \frac{\nu}{\nu(\mathbb{R})}\right).$$

Similarly, we extend $W_p(\cdot, \cdot)$ to $\mathscr{P}(\mathbb{R})^2$ by ∞ on pairs (μ, ν) such that no transport plan yields a finite cost. Since we consider mainly the distance between μ and the shifted average μ_h, it always holds that $W_p(\mu, \mu_h) \leqslant \mu(\mathbb{R})\, h < \infty$, even if $\mu \notin \mathscr{P}_p$.

Monotone plan In dimension one, an explicit minimizer is known, which is in addition unique if $p > 1$. This plan is called the "monotone plan", since it is constructed by gathering all the mass that μ puts on $(-\infty, x]$, and spread it monotonically on $(-\infty, y]$ for some y such that $\nu((-\infty, y)) \leqslant \mu((-\infty, x]) \leqslant \nu((-\infty, y])$. In rigorous terms, define the distribution function of $\mu \in \mathscr{P}(\mathbb{R})$ as

$$F_\mu : \mathbb{R} \to [0, 1], \qquad F_\mu(x) \coloneqq \mu((-\infty, x]).$$

The function F_μ is nondecreasing, and admits a pseudo-inverse $F_\mu^{[-1]} : [0, 1] \to \mathbb{R} \cup \{-\infty\}$ characterized by $F_\mu(x) \geqslant r$ if and only if $x \geqslant F_\mu^{[-1]}(r)$. Explicitly, there holds $F_\mu^{[-1]}(r) \coloneqq \inf\{x \in \mathbb{R} \mid F_\mu(x) \geqslant r\}$. The monotone plan between μ and ν is defined as $\eta \coloneqq (F_\mu^{[-1]}, F_\nu^{[-1]})_{\#}\mathcal{L}_{[0,1]}$. By optimality, it holds for any $p \in [1, \infty)$ that

$$W_p^p(\mu, \nu) = \int_{r\in[0,1]} \left|F_\mu^{[-1]}(r) - F_\nu^{[-1]}(r)\right|^p dr.$$

In the case $p = 1$, the expression simplifies further in $W_1(\mu, \nu) = \int_{x\in\mathbb{R}} |F_\mu(x) - F_\nu(x)|\, dx$. Here note that the integral is taken with respect to the Lebesgue measure.

Dual formulation Let $c : \mathbb{R}^2 \to \mathbb{R}^+$ be any cost of the family $c(x, y) = |x - y|^p$. We follow [Vil09] in defining the c–transform of a function $\varphi : \mathbb{R} \to \mathbb{R}$ as

$$\varphi^c : \mathbb{R} \to \mathbb{R} \cup \{-\infty\}, \qquad \varphi^c(y) \coloneqq \inf_{x\in\mathbb{R}} \varphi(x) + c(x, y).$$

The $p-$Wasserstein distance admits a dual reformulation as

$$W_p^p(\mu, \nu) = \sup_{\varphi\in L^1_\mu(\mathbb{R};\mathbb{R})} \int_{y\in\mathbb{R}} \varphi^c(y) d\nu - \int_{x\in\mathbb{R}} \varphi(x) d\mu.$$

Since the cost cannot decrease if φ is replaced by $\psi(x) \coloneqq \sup_{y\in\mathbb{R}} c(x, y) - \varphi^c(y)$, one can assume that φ is of this form. In particular, if $c(x, y) = |x - y|$, one checks that taking twice c–transforms yields that φ is 1-Lipschitz, and $\varphi^c = \varphi$. Hence, denoting Lip_1 the set of 1-Lipschitz functions, there holds

$$W_1(\mu, \nu) = \sup_{\varphi\in\mathrm{Lip}_1} \int_{y\in\mathbb{R}} \varphi(y) d\nu - \int_{x\in\mathbb{R}} \varphi(x) d\mu. \tag{4}$$

Tangent cones in the case $p = 2$ The geometric tangent cone to $\mu \in \mathscr{P}_2$ is defined as

$$\mathbf{Tan}_\mu \coloneqq \overline{\{\, \lambda \cdot \xi \mid (\pi_x, \pi_x + \pi_v)_\# \xi \in \Gamma_o(\mu, (\pi_x + \pi_v)_\# \xi),\ \text{and } \lambda \geqslant 0 \,\}}^{W_\mu} \subset \mathscr{P}_2(\mathrm{T}\,\mathbb{R})_\mu.$$

Here, $\lambda \cdot \xi \coloneqq (\pi_x, \lambda \pi_v)_\# \xi$ is the scalar multiplication of a measure field ξ by a scalar $\lambda \in \mathbb{R}$. The distance W_μ is a generalization of the L^2_μ distance defined by

$$W^2_\mu(\xi, \zeta) \coloneqq \inf_{\eta \in \Gamma_\mu(\xi,\zeta)} \int_{(x,v,w) \in \mathrm{T}^2\,\mathbb{R}} |v - w|^2 d\eta,$$

where we denote $\mathrm{T}^2\,\mathbb{R} \coloneqq \bigcup_{x \in \mathbb{R}} \{x\} \times (\mathrm{T}_x\,\mathbb{R})^2$, and

$$\Gamma_\mu(\xi, \zeta) \coloneqq \left\{ \eta = \eta(dx, dv, dw) \in \mathscr{P}_2\left(\mathrm{T}^2\,\mathbb{R}\right) \;\middle|\; \begin{array}{c} (\pi_x, \pi_v)_\# \eta = \xi, \text{ and} \\ (\pi_x, \pi_w)_\# \eta = \zeta \end{array} \right\}.$$

The subset of $\eta \in \Gamma_\mu(\xi, \zeta)$ such that $\int_{(x,v) \in \mathrm{T}\,\mathbb{R}} |v|^2 d\eta = W^2_\mu(\xi, \zeta)$ is denoted $\Gamma_{\mu,o}(\xi, \zeta)$. We shorten $W_\mu(\xi, (id, 0)_\# \mu)$ as $\|\xi\|_\mu = \sqrt{\int_{(x,v) \in \mathrm{T}\,\mathbb{R}} |v|^2 d\xi}$. The set $\mathbf{Tan}_\mu$ is stable by barycenter: precisely, [Gig08, Chap. 4] shows that if $\xi \in \mathbf{Tan}_\mu$, then its barycenter, defined as the unique $b \in L^2_\mu(\mathbb{R}; \mathbb{R})$ such that $\int_{(x,v) \in \mathrm{T}\,\mathbb{R}} \varphi(x)\psi(v) d\xi = \int_{x \in \mathbb{R}} \varphi(x)\psi(b(x)) d\mu$ for $\varphi \in \mathcal{C}_b(\mathbb{R}; \mathbb{R})$ and ψ linear, induces an element $(id, b)_\# \mu \in \mathbf{Tan}_\mu$.

Besides the distance $W_\mu(\cdot, \cdot)$ adapted to the tangent cone at μ, one can define a metric scalar product

$$\langle \xi, \zeta \rangle_\mu \coloneqq \frac{1}{2}\left[\|\xi\|^2_\mu + \|\zeta\|^2_\mu - W^2_\mu(\xi, \zeta)\right] = \sup_{\eta \in \Gamma_\mu(\xi,\zeta)} \int_{(x,v,w) \in \mathrm{T}^2\,\mathbb{R}} \langle v, w \rangle \, d\eta(x, v, w).$$

The orthogonal $\mathbf{Tan}^\perp_\mu$, referred to as the set of *solenoidal* measure fields, is defined as the set of ζ such that $\langle \xi, \zeta \rangle_\mu = 0$ for any $\xi \in \mathbf{Tan}_\mu$.

Submeasure estimate The following estimate seems to be folklore, and is included for lack of a precise reference.

Lemma 2.1 (Large-mass submeasures of a given measure are close)**.** *Let $\mu \in \mathcal{M}_+(\mathbb{R})$ be a nonnegative Borel measure with finite mass, supported on a compact K of diameter* diam K*. Let $0 \leqslant \varepsilon \leqslant \mu(\mathbb{R})$, and $\alpha, \beta \in \mathcal{M}_+(\mathbb{R})$ satisfy*

$$\alpha \leqslant \mu, \qquad \beta \leqslant \mu, \qquad \alpha(\mathbb{R}) = \beta(\mathbb{R}) \geqslant \mu(\mathbb{R}) - \varepsilon.$$

Then for any $p \in [1, \infty)$, there holds $W^p_p(\alpha, \beta) \leqslant \varepsilon (\mathrm{diam}\ K)^p$.

Proof. Define $\alpha \wedge \beta$ as $(\alpha \wedge \beta)(A) \coloneqq \inf_{(A_n)_n \in \mathfrak{P}(A)} \sum_n \min(\alpha(A_n), \beta(A_n))$, where $\mathfrak{P}(A)$ is the set of measurable countable partitions $(A_n)_n$ of A. Then

$$(\alpha \wedge \beta)(\mathbb{R}) \geqslant \mu(\mathbb{R}) - \sup_{(A_n)_n \in \mathfrak{P}(\mathbb{R})} \sum_n \max((\mu - \alpha)(A_n), (\mu - \beta)(A_n)).$$

Since $\max(a, b) \leqslant a + b$ whenever $a, b \geqslant 0$, and $(\mu - \alpha)(A_n), (\mu - \beta)(A_n) \geqslant 0$ by assumption,

$$(\alpha \wedge \beta)(\mathbb{R}) \geqslant \mu(\mathbb{R}) - [(\mu - \alpha)(\mathbb{R}) + (\mu - \beta)(\mathbb{R})] \geqslant \mu(\mathbb{R}) - 2\varepsilon.$$

As $\alpha \wedge \beta$ is a submeasure of both α and β, we can define a transport plan between α and β sending $\alpha \wedge \beta$ on itself, and the remaining mass of α on that of β. The previous estimate shows that $\alpha(\mathbb{R}) - (\alpha \wedge \beta)(\mathbb{R}) \leqslant \varepsilon$, and since both α and β are supported on K, the mass travels at most diam K. Consequently, $W^p_p(\alpha, \beta) \leqslant \varepsilon\, (\mathrm{diam}\ K)^p$. □

3 Proof of Theorem 1.2

The strategy for Theorem 1.2 is to exploit the dual reformulation of the Wasserstein distance. Informally, when the quotient $W(\mu, \mu_h)/h$ approaches its maximal value 1, the mass of μ must concentrate on points on which Kantorovich pairs (φ, φ^c) satisfy a certain 3-point inequality. For the cost $c(x, y) = |x - y|$, this inequality implies that φ looks like a downward pointing corner with at least a certain angle: this can happen on at most countably many points, that must be at some controlled distance from each other. For $c(x, y) = |x - y|^p$ with $p > 1$, we have no such interpretation, but computation shows that these points must be sufficiently separated, otherwise the condition $\varphi^c \ominus \varphi \leqslant c$ breaks. The "separation" precisely enforces the presence of holes of size comparable to h on which μ puts very few mass, and intersecting over countably many scales, one gets to the porous sets of Definition 1.1. The converse is based on the construction of a particular Kantorovich potential.

We first prove an estimate which is specific to dimension 1.

Lemma 3.1 (Pointwise bound)**.** *Let μ have finite mass and $\zeta \in \mathscr{P}(\mathrm{T}\,\mathbb{R})_\mu$ be such that $|v| \leqslant R$ for some $R > 0$ and $\zeta-$a.e. $(x,v) \in \mathrm{T}\,\mathbb{R}$. Then, the monotone optimal transport plan $\eta \in \Gamma_o(\mu, \exp_\mu(\zeta))$ is concentrated on pairs (x,y) such that $|y-x| \leqslant R$.*

Proof. Let $\nu \coloneqq \exp_\mu(\zeta)$, and denote $F_\mu, F_\nu : \mathbb{R} \to [0,1]$ the respective distribution functions of μ, ν, i.e. $F_\mu(x) = \mu((-\infty, x])$. By the explicit expression of the monotone optimal transport plan in dimension one [San15, Chap. 2], η is concentrated on pairs (x,y) such that for some $r \in \mathbb{R}$,

$$x = \inf\left\{\overline{x} \mid F_\mu(\overline{x}) \geqslant r\right\}, \qquad \text{and} \qquad y = \inf\left\{\overline{y} \mid F_\nu(\overline{y}) \geqslant r\right\}.$$

As F_μ, F_ν are nondecreasing and upper semi-continuous, the r associated to (x,y) also satisfies $F_\mu(x) \geqslant r$ and $F_\nu(y) \geqslant r$ (although the inequality can be strict in case of Dirac mass). By assumption, for any $x \in \mathbb{R}$, one has

$$F_\nu(x+R) = \int_{(y,w)\in \mathrm{T}\,\mathbb{R}} \mathbb{1}_{(-\infty, x+R]}(y+w) d\zeta \geqslant \int_{y\in\mathbb{R}} \mathbb{1}_{(-\infty,x]}(y) d\mu = F_\mu(x) \geqslant r,$$

hence $y \leqslant x+R$. Symmetrically, one has $\mu = \exp_\nu(\gamma)$ for $\gamma \coloneqq (\pi_x + \pi_v, -\pi_v)_\# \zeta$, which also satisfies $|v| \leqslant R$ for $\gamma-$a.e. (x,v); hence $F_\mu(y+R) \geqslant F_\nu(y) \geqslant r$ for any (x,y) in supp η and associated r, and we conclude that $x \leqslant y + R$. □

Remark 3.2. As a consequence, the property (1) is independent of $p \in [1,\infty)$. Indeed, by Hölder's inequality, the Wasserstein distances are ordered as $W_p(\mu,\nu) \leqslant W_q(\mu,\nu)$ for $p \leqslant q$. On the other hand, by Lemma 3.1, the monotone transport plan η_h between μ and μ_h satisfies $|y-x| \leqslant h$ almost everywhere. Since this plan is optimal for any $p \in [1,\infty)$, it follows that

$$\begin{aligned}\frac{W_1(\mu,\mu_h)}{h} \leqslant \frac{W_p(\mu,\mu_h)}{h} = \left(\frac{W_p^p(\mu,\mu_h)}{h^p}\right)^{1/p} &= \left(\frac{\int |y-x|^p d\eta_h}{h^p}\right)^{1/p} \\ \leqslant \left(\frac{h^{p-1}\int |y-x| d\eta_h}{h^p}\right)^{1/p} &= \left(\frac{W_1(\mu,\mu_h)}{h}\right)^{1/p}.\end{aligned}$$

Hence, both the set of measures $\mu \in \mathscr{P}(\mathbb{R})$ such that (1) holds, and such that $W_p(\mu,\mu_h) = o(h)$, are independent of $p \in [1,\infty)$.

The following lemma is the core of the argument of Theorem 1.2. It states that for any φ, the points where $\frac{1}{2}\left[\varphi^c(x-h) + \varphi^c(x+h)\right] - \varphi(x)$ is comparable to h^p must be separated by some distance itself comparable to h.

Lemma 3.3. *Let $p \in [1,\infty)$ and $\kappa_\gamma \coloneqq (2\gamma-1)^{1/p}$ for any $\gamma \in (1/2, 1]$. Given $\varphi : \mathbb{R} \to \mathbb{R}$, $\gamma \in (1/2,1]$, and $h > 0$, let*

$$S \coloneqq \left\{ x \in \mathbb{R} \;\middle|\; \frac{\frac{1}{2}\left[\varphi^c(x-h) + \varphi^c(x+h)\right] - \varphi(x)}{h^p} \geqslant \gamma \right\}.$$

Then for any $x \in S$, there holds $S \cap \mathscr{B}(x, h(1+\kappa_\gamma)) \subset \overline{\mathscr{B}}(x, h(1-\kappa_\gamma))$.

Proof. Assume that $x, y \in S$ with $x \leqslant y$, and let $\delta \coloneqq |y-x|$. By definition of S, there holds

$$2\gamma h^p \leqslant \frac{\varphi^c(x-h) + \varphi^c(x+h)}{2} - \varphi(x) + \frac{\varphi^c(y-h) + \varphi^c(y+h)}{2} - \varphi(y).$$

Recall that $\varphi^c(\cdot) \coloneqq \inf_{z\in\mathbb{R}} \varphi(z) + |z - \cdot|^p$. Therefore, we can bound the first term by $\psi_{x-h}(z) \coloneqq \varphi(z) + |z-(x-h)|^p$ evaluated at $z = x$, the second term by $\psi_{x+h}(y)$, the third term by $\psi_{y-h}(x)$, and the fourth by $\psi_{y+h}(y)$, to get

$$2\gamma h^p \leqslant \frac{\varphi(x)+\varphi(y)}{2} + \frac{h^p + |x+h-y|^p}{2} - \varphi(x) + \frac{\varphi(x)+\varphi(y)}{2} + \frac{|y-h-x|^p + h^p}{2} - \varphi(y).$$

Rearranging, and using that $2\gamma - 1 > 0$ since $\gamma > 1/2$, we find

$$h^p(2\gamma-1) \leqslant |h-\delta|^p \qquad \Longrightarrow \qquad h\kappa_\gamma = h(2\gamma-1)^{1/p} \leqslant |h-\delta|.$$

The above inequality forces $\delta = |y-x|$ to be either in $[0, h(1-\kappa_\gamma)]$, or in $[h(1+\kappa_\gamma), \infty)$, so that all points of S that are at distance strictly less than $h(1+\kappa_\gamma)$ of x must belong to $\overline{\mathscr{B}}(x, h(1-\kappa_\gamma))$. □

Remark 3.4 (Taking $\gamma < 1/2$). Lemma 3.3 can be proved for arbitrarily small $\gamma > 0$, up to taking more than two points x, y: one considers a chain $x_0, x_1, \cdots, x_N$ of points of S, sums the defining inequalities, and bounds each $\varphi^c(x_i \pm h)$ by the values at the left and right neighbours of the chain. The corresponding κ changes but is still positive.

Proof of Theorem 1.2. We start with (a) ⇒ (b). Denote ξ the measure field given by

$$\xi \coloneqq \frac{1}{2}\left[(id,-1)_{\#}\mu + (id,1)_{\#}\mu\right] \in \mathscr{P}(\mathrm{T}\,\mathbb{R})_\mu, \qquad \text{so that} \qquad \mu_h = \exp_\mu(h\cdot\xi).$$

Let $(h_n)_{n\in\mathbb{N}} \subset (0,1]$ be a vanishing sequence along which the limit sup is reached. For each n, pick $\iota_n \geqslant 0$ going to 0 with n, and then $\varphi_n : \mathbb{R} \to \mathbb{R}\cup\{+\infty\}$ be $h_n^p\iota_n-$optimal in the dual formulation of the Wasserstein distance, i.e.

$$\begin{aligned} 1-\iota_n \leqslant \frac{W_p^p(\mu,\exp_\mu(h_n\cdot\xi))}{h_n^p} &\leqslant \frac{1}{h_n^p}\int_{y\in\mathbb{R}} \varphi_n^c(y) d[\exp_\mu(h_n\cdot\xi)] - \frac{1}{h_n^p}\int_{x\in\mathbb{R}}\varphi_n(x)d\mu + \iota_n \\ &= \int_{x\in\mathbb{R}} \frac{\frac{1}{2}\left[\varphi_n^c(x-h_n)+\varphi_n^c(x+h_n)\right]-\varphi_n(x)}{h_n^p} d\mu + \iota_n. \end{aligned} \tag{5}$$

Let $\varepsilon > 0$, and $(n_m)_m$ be a subsequence such that $\iota_{n_m} \leqslant \varepsilon 2^{-2(m+2)}$. For readability, relabel $\iota_{n_m}, h_{n_m}, \varphi_{n_m}$ as ι_m, h_m, φ_m respectively. Let $\gamma_m \coloneqq 1-2^{-(m+2)} \in (1/2,1)$, and for each m, let

$$S_m \coloneqq \left\{ x\in\mathbb{R} \;\middle|\; \frac{\frac{1}{2}\left[\varphi_m^c(x-h_m)+\varphi_m^c(x+h_m)\right]-\varphi_m(x)}{h_m^p} \geqslant \gamma_m \right\}.$$

By definition of the c-transform, $\frac{1}{2}\left[\varphi_m^c(x-h_m)+\varphi_n^c(x+h_m)\right]-\varphi_m(x) \leqslant h_m^p$ always holds, so that (5) gives

$$1-\iota_m \leqslant \int_{S_m} 1 d\mu + \int_{S_m^c} \gamma_m d\mu + \iota_m = 1-\mu(S_m^c)+\gamma_m\mu(S_m^c)+\iota_m.$$

In consequence, $\mu(S_m^c) \leqslant \frac{2\iota_m}{1-\gamma_m} \leqslant \varepsilon 2^{-(m+1)}$. The measurable set $A \coloneqq \bigcap_{m\in\mathbb{N}} S_m$ satisfies $\mu(A) \geqslant 1-\sum_m \mu(S_m^c) \geqslant 1-\varepsilon$, so we just have to prove that it belongs to $\mathscr{A}$. To this aim, consider the vanishing sequence $s_m \coloneqq \min(1, h_m(1+\kappa_m))$ with $\kappa_m = (2\gamma_m-1)^{1/p}$, and $\tau_m \coloneqq (1-\kappa_m)/(1+\kappa_m)$. By Lemma 3.3, for each m, there holds

$$A\cap\mathscr{B}(x,s_m) \subset S_m\cap\mathscr{B}(x,s_m) \subset \overline{\mathscr{B}}(x,s_m\tau_m) \qquad \forall x\in S_m.$$

Passing to the supremum over $x\in A\subset S_m$, we get

$$\sup_{x\in A}\inf\left\{\tau\in(0,1) \;\middle|\; A\cap\mathscr{B}(x,s_m)\subset\overline{\mathscr{B}}(x,s_m\tau)\right\} \leqslant \tau_m = \frac{1-\kappa_m}{1+\kappa_m} = \frac{1-(2\gamma_m-1)^{1/p}}{1+(2\gamma_m-1)^{1/p}}.$$

Since $\gamma_m \to_m 1$, we may pass to the limit in $m\to\infty$ and conclude that $A\in\mathscr{A}$.

Assume now that Point (b) holds. We prove (a) for $p=1$, and deduce it for any $p\in[1,\infty)$ by Remark 3.2. Let $\varepsilon>0$ and $A\in\mathscr{A}$ such that $\mu(A)\geqslant 1-\varepsilon$. By definition, there exist vanishing sequences $(s_n)_n, (\tau_n)_n \subset (0,1)$ such that $A\cap\mathscr{B}(x,s_n)\subset\overline{\mathscr{B}}(x,s_n\tau_n)$ for any $n\in\mathbb{N}$ and $x\in A$. Let $\psi : x\mapsto d(x,A)=\inf_{z\in A} d(x,z)$ be the (1-Lipschitz) distance function to A. Then, by the dual formula (4),

$$\frac{W_1(\mu,\exp_\mu(h\cdot\xi))}{h} \geqslant \int_{x\in\mathbb{R}} \frac{\frac{1}{2}\left[\psi(x+h)+\psi(x-h)\right]-\psi(x)}{h} d\mu.$$

Taking $h_n \coloneqq \frac{s_n+\tau_n s_n}{2}$, we know that for any $x\in A$, the points $x\pm h_n$ are at distance at least $\frac{s_n(1-\tau_n)}{2}$ of any point of A. Hence $\psi(x\pm h_n)\geqslant\frac{s_n(1-\tau_n)}{2}$ on A. On the other hand, one has $\left|\frac{1}{2}\left[\psi(x+h)+\psi(x-h)\right]-\psi(x)\right|/h \leqslant 1$ since ψ is 1-Lipschitz. Hence

$$\int \frac{\frac{1}{2}\left[\psi(x+h_n)+\psi(x-h_n)\right]-\psi(x)}{h_n} d\mu \geqslant \int_A \frac{s_n(1-\tau_n)}{2h_n} d\mu - \mu(A^c) \geqslant \mu(A)\frac{1-\tau_n}{1+\tau_n} - \varepsilon.$$

Taking the limit in $n\to\infty$, we get that

$$\limsup_{h\searrow 0} \frac{W_1(\mu,\exp_\mu(h\cdot\xi))}{h} \geqslant \mu(A)-\varepsilon \geqslant 1-2\varepsilon = \|\xi\|_\mu - 2\varepsilon.$$

Letting $\varepsilon\searrow 0$, we conclude that Point (a) holds. □

4 Link with the geometric tangent cone to $\mathscr{P}_2(\mathbb{R})$

In this section, we fix $p=2$ and denote $W(\cdot,\cdot)$ the corresponding Wasserstein distance. Our aim is to characterize the measures $\mu\in\mathscr{P}_2(\mathbb{R})$ for which $\xi\in\mathbf{Tan}_\mu$ if and only if

$$\limsup_{h\searrow 0}\frac{W(\mu,\exp_\mu(h\cdot\xi))}{h} = \|\xi\|_\mu. \tag{6}$$

The main difficulty is to show that whenever some $\xi\notin\mathbf{Tan}_\mu$ does satisfy (6), then some nonatomic submeasure σ of μ also does with the symmetric unit measure field $\frac{1}{2}\left[(id,-1)_{\#}\sigma+(id,1)_{\#}\sigma\right]$. By [Aus25a, Proposition 2.9], the latter field is not tangent to σ, and is in fact orthogonal to $\mathbf{Tan}_\sigma$ in the sense of the metric scalar product $\langle\cdot,\cdot\rangle_\sigma$.

4.1 Reduction to centred measure fields

We start by proving that whenever $\zeta \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R})_\mu$ satisfies (6), then the centred field $\zeta^0 \coloneqq (\pi_x, \pi_v - \mathrm{Bary}\,(\xi)(\pi_x))_\#\zeta$, obtained by removing the barycenter of ζ, also does. The intuition behind is that map-induced and centred measure fields are orthogonal for the metric scalar product, i.e. for any $b \in L^2_\mu$, there holds

$$\left\langle \zeta^0, (id, b)_\#\mu \right\rangle_\mu = \int_{x\in\mathbb{R}} \int_{v\in \mathrm{T}_x\,\mathbb{R}} \langle v, b(x)\rangle \, d\zeta^0_x(v) d\mu(x) = 0.$$

The same computation shows that $W^2_\mu(\xi,\zeta) = W^2_\mu(\xi^0,\zeta^0) + \|b_\xi - b_\zeta\|^2_{L^2_\mu}$. This "infinitesimal" orthogonality appears in a weaker form before the limit, as seen in the following lemma.

Lemma 4.1. *Let $\zeta = (\pi_x, \pi_v + b(\pi_x))_\#\zeta^0$ for some ζ^0 with null barycenter. Then*

$$\limsup_{h\searrow 0} \frac{W^2(\mu, \exp_\mu(h\cdot\zeta))}{h^2} \leqslant \limsup_{h\searrow 0} \frac{W^2(\mu, \exp_\mu(h\cdot\zeta^0))}{h^2} + \|b\|^2_\mu. \tag{7}$$

Consequently, if the left hand-side of (7) is equal to $\|\zeta\|^2_\mu = \|\zeta^0\|^2_\mu + \|b\|^2_{L^2_\mu}$, then the centred measure field ζ^0 satisfies the limit sup condition (6).

Proof. For any $\xi, \zeta \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R})_\mu$, there holds

$$\left| \limsup_{h\searrow 0} \frac{W(\mu, \exp_\mu(h\cdot\xi))}{h} - \limsup_{h\searrow 0} \frac{W(\mu, \exp_\mu(h\cdot\zeta))}{h} \right| \leqslant \sqrt{W^2_\mu(\xi^0,\zeta^0) + \|b_\xi - b_\zeta\|^2_{L^2_\mu}}.$$

Hence both sides of (7) are continuous with respect to ζ^0 and b, respectively in the $W_\mu-$ and $L^2_\mu-$topologies. Consequently, we may prove (7) in the case where $|v| \leqslant R$ for ζ^0-a.e. (x,v) and some $R>0$, and $b = \sum_{i=1}^N b_i \mathbb{1}_{C_i}$ for a partition $(C_i)_{i=1}^N$ of $\mathbb{R}$ into finitely many intervals such that $\mu(\partial C_i) = 0$ for each i.

Denote $\mu_i \coloneqq \mu \llcorner C_i$, with $\zeta_i \coloneqq \zeta \llcorner \{(x,v)\,|\,x\in C_i\}$, and ζ^0_i defined in the same way. Then $\mu = \sum_i \mu_i$, and by linearity of the pushforward, $\exp_\mu(h\cdot\zeta) = \sum_i \exp_{\mu_i}(h\cdot\zeta_i)$. By the convexity of the squared Wasserstein distance in the Banach space of measures [Vil09, Theorem 4.8],

$$W^2(\mu, \exp_\mu(h\cdot\zeta)) \leqslant \sum_{i=1}^N W^2(\mu_i, \exp_{\mu_i}(h\cdot\zeta_i)).$$

By definition of C_i, $\zeta_i = (\pi_x, \pi_v + b_i)_\#\zeta^0_i$, with $b_i \in \mathbb{R}$ a constant. Let $\eta = \eta(dx,dy) \in \Gamma_o(\mu_i, \exp_{\mu_i}(h\cdot\zeta^0_i))$. Then $(\pi_x, \pi_y + hb_i)_\#\eta$ transports μ_i to $\exp_\mu(h\cdot\zeta_i)$, so that

$$\begin{aligned} W^2(\mu_i, \exp_{\mu_i}(h\cdot\zeta_i)) &\leqslant \int_{(x,y)\in\mathbb{R}^2} |x - (y+hb_i)|^2 d\eta \\ &= W^2(\mu_i, \exp_{\mu_i}(h\cdot\zeta^0_i)) - 2\int_{(x,y)\in\mathbb{R}^2} \langle x-y, hb_i\rangle \, d\eta + \mu(C_i)|hb_i|^2. \end{aligned}$$

Since $\int y d\eta = \int (x+hv) d\zeta^0_i = \int x d\mu_i$, the middle term vanishes. Summing on i, we get

$$W^2(\mu, \exp_\mu(h\cdot\zeta)) \leqslant \sum_{i=1}^N W^2(\mu_i, \exp_{\mu_i}(h\cdot\zeta^0_i)) + h^2\|b\|^2_\mu. \tag{8}$$

Stays to show that $\sum_{i=1}^N W^2(\mu_i, \exp_{\mu_i}(h\cdot\zeta^0_i)) \leqslant W^2(\mu, \exp_\mu(h\cdot\zeta^0)) + h^2 O(h)$. To this aim, let $\eta \in \Gamma_o(\mu, \exp_\mu(h\cdot\zeta^0))$, and decompose it as $\sum_i \eta_i$ for $\eta_i \coloneqq \eta_h \llcorner C_i \times \mathbb{R}$. Let also $\nu_i \coloneqq \pi_{y\#}\eta_i$. Since $\sum_i \nu_i = \exp_\mu(h\cdot\zeta^0) = \sum_i \exp_{\mu_i}(h\cdot\zeta^0_i)$, there holds

$$\sum_{i=1}^N F_{\nu_i} = F_{\exp_\mu(h\cdot\zeta^0)} = \sum_{i=1}^N F_{\exp_{\mu_i}(h\cdot\zeta^0_i)}. \tag{9}$$

Here we recall that $F_\sigma(x) \coloneqq \sigma((-\infty, x])$ for any measure σ. However, by Lemma 3.1, the plan η is supported on pairs (x,y) with $|y-x| \leqslant hR$. Consequently, both ν_i and $\exp_{\mu_i}(h\cdot\zeta^0_i)$ are supported on points that are at distance at most hR from C_i. This implies that $F_{\nu_i}(x) = F_{\exp_{\mu_i}(h\cdot\zeta^0_i)}(x) = 0$ for $x \leqslant \min C_i - hR$, and $\mu(C_i)$ for $x \geqslant \max C_i + hR$. Hence, evaluating (9) at $x \in C_i$ that is at distance at least hR from the boundary ∂C_i, there holds $F_{\nu_i}(x) = F_{\exp_{\mu_i}(h\cdot\zeta^0_i)}(x)$. Using the expression of the monotone optimal plan, we deduce that the optimal transport plan α_i between ν_i and

$\exp_{\mu_i}(h\cdot\zeta_i^0)$ is concentrated on pairs (x,y) such that if $x\in(\min C_i+hR,\max C_i-hR)$, then $y=x$. In addition, $|y-x|\leqslant 2hR$, since mass only moves within balls of radius hR around $\min C_i$ and $\max C_i$. It follows that

$$\begin{aligned}W^2(\nu_i,\exp_{\mu_i}(h\cdot\zeta_i^0)) &= \int_{(x,y)\in\mathbb{R}^2}|y-x|^2d\alpha_i = \int_{x\in\overline{\mathscr{B}}(\min C_i,hR)\cup\overline{\mathscr{B}}(\max C_i,hR),y\in\mathbb{R}}|y-x|^2d\alpha_i\\ &\leqslant (2hR)^2\mu\left(\overline{\mathscr{B}}(\min C_i,hR)\cup\overline{\mathscr{B}}(\max C_i,hR)\right).\end{aligned}$$

Since $\mu(\partial C_i)=0$, the above is of order $h^2O(h)$. Now, by the second triangular inequality,

$$\begin{aligned}&W^2(\mu,\exp_\mu(h\cdot\zeta^0)) = \sum_{i=1}^N W^2(\mu_i,\nu_i)\\ &\geqslant \sum_{i=1}^N W^2(\mu_i,\exp_{\mu_i}(h\cdot\zeta_i^0)) - W(\nu_i,\exp_{\mu_i}(h\cdot\zeta_i^0))\left(W(\nu_i,\exp_{\mu_i}(h\cdot\zeta_i^0))+2W(\mu_i,\nu_i)\right)\\ &\geqslant \sum_{i=1}^N W^2(\mu_i,\exp_{\mu_i}(h\cdot\zeta_i^0)) - W(\nu_i,\exp_{\mu_i}(h\cdot\zeta_i^0))3hR\mu(C_i).\end{aligned}$$

Combining the two last lines provides the desired estimate in (8), and we conclude. □

4.2 Reduction to the unit symmetric measure field

We now prove that whenever *some* centred measure field $\xi\notin\mathbf{Tan}_\mu$ satisfies (6), then μ admits a nonatomic submeasure on which the unit symmetric measure field also satisfies (6). The argument uses various operations to modify a measure field and still satisfy (6); for readability, we introduce the notation

$$\Xi[\mu]\coloneqq\left\{\zeta\in\mathscr{P}_2(\mathrm{T}\,\mathbb{R})_\mu\;\middle|\;\mathrm{Bary}\,(\zeta)=0\quad\text{and}\quad\limsup_{h\searrow 0}\frac{W(\mu,\exp_\mu(h\cdot\zeta))}{h}=\|\zeta\|_\mu\right\}.$$

If μ is a nonnegative Borel measure with finite second moment and mass $m>0$, we still denote $\Xi[\mu]\coloneqq m\,\Xi[\mu/m]$.

Lemma 4.2 (Technical manipulations)**.** *Let $\mu\in\mathscr{P}_2(\mathbb{R})$. Then:*

- *(closure) the set $\Xi[\mu]$ is closed with respect to W_μ.*
- *(nonnegative cone) $\zeta\in\Xi[\mu]$ implies that $(\pi_x,\lambda\pi_v)_\#\zeta\in\Xi[\mu]$ for any $\lambda\geqslant 0$.*
- *(stability by restriction) For any measurable A with $\mu(A)>0$, $\zeta\in\Xi[\mu]$ implies that $\zeta\llcorner\{(x,v)\,|\,x\in A\}\in\Xi[\mu\llcorner A]$.*
- *(stability by truncation) If $f\in L^2_\mu(\mathbb{R};\mathbb{R}^+)$ satisfies $\frac{1}{2}\left[(id,-f)_\#\mu+(id,f)_\#\mu\right]\in\Xi[\mu]$, then for any $R>0$, there holds $\frac{1}{2}\left[(id,-\min(f,R))_\#\mu+(id,\min(f,R))_\#\mu\right]\in\Xi[\mu]$.*

Proof. The set of measure fields with 0 barycenter is itself a $W_\mu-$closed nonnegative cone that is stable by restriction. The first and second point are direct, since the limit sup condition is $W_\mu-$continuous and positively homogeneous. To show the third point, denote $\zeta_A\coloneqq\zeta\llcorner\{(x,v)\,|\,x\in A\}$. Then, using the convexity of the squared Wasserstein distance in the Banach space of measures,

$$\begin{aligned}\int_{(x,v)\in A\times\mathbb{R}}|v|^2d\zeta+\int_{(x,v)\in A^c\times\mathbb{R}}|v|^2d\zeta &= \|\zeta\|_\mu^2=\limsup_{h\searrow 0}\frac{W^2(\mu,(\pi_x+h\pi_v)_\#\zeta)}{h^2}\\ &\leqslant\limsup_{h\searrow 0}\frac{W^2(\mu\llcorner A,(\pi_x+h\pi_v)_\#\zeta_A)+W^2(\mu\llcorner A^c,(\pi_x+h\pi_v)_\#\zeta_{A^c})}{h^2}\\ &\leqslant\limsup_{h\searrow 0}\frac{W^2(\mu\llcorner A,(\pi_x+h\pi_v)_\#\zeta_A)}{h^2}+\int_{(x,v)\in A^c\times\mathbb{R}}|v|^2d\zeta.\end{aligned}$$

As converse inequality always holds, we deduce that $\zeta_A\in\Xi[\mu\llcorner A]$.

The last point is reached by a limit procedure. Let $\varepsilon>0$. Define a sequence $(f_n)_{n\in\mathbb{N}}\subset L^2_\mu$ by $f_0\coloneqq f$, and the following construction. For each n, let $A\coloneqq\{f_n\geqslant R+\varepsilon\}$, and denote $\gamma_n\coloneqq\frac{1}{2}\left[(id,-f_n)_\#\mu+(id,f_n)_\#\mu\right]$. We construct a transport plan $\alpha\in\Gamma_\mu(\gamma_n,\gamma_n)$ that sends v to $-v$ on A, and v to v on A^c, i.e.

$$\alpha\coloneqq\frac{1}{2}\left(id,f_n,f_n\mathbb{1}_{A^c}-f_n\mathbb{1}_A\right)_\#\mu+\frac{1}{2}\left(id,-f_n,f_n\mathbb{1}_A-f_n\mathbb{1}_{A^c}\right)_\#\mu.$$

Let $t \coloneqq R/(R+\varepsilon) \in (0,1)$, and $\gamma_{n+1} \coloneqq (\pi_x, (1-t)\pi_v + t\pi_w)_\# \alpha$. By the displacement semiconcavity of the squared Wasserstein distance [AGS05, Theorem 7.3.2], there holds

$$\begin{aligned} &W^2\left(\mu, \exp_\mu(h\cdot\gamma_{n+1})\right) \\ &\geqslant (1-t)W^2(\mu, \exp_\mu(h\cdot\gamma_n)) + tW^2(\mu, \exp_\mu(h\cdot\gamma_n)) - t(1-t)h^2 \int_{(x,v,w)\in \mathrm{T}^2\,\mathbb{R}} |v-w|^2 d\alpha. \end{aligned}$$

(Precisely, we took $\mu^1 = \mu^2 = \exp_\mu(h\cdot\gamma_n)$ and $\mu^3 = \mu$, along with the plan $\boldsymbol{\mu}^{1\,2} = (\pi_x + h\pi_v, \pi_x + h\pi_w)_\#\alpha$. Then, still using the notation of [AGS05], $W^2_{\boldsymbol{\mu}^{1\,2}}(\mu^1,\mu^2) = \int |(x+hv)-(x+hw)|^2 d\alpha = h^2 \int |v-w|^2 d\alpha$.) Dividing by $h^2 > 0$ and taking the limit sup in $h \searrow 0$, we get that

$$\begin{aligned} \limsup_{h\searrow 0} \frac{W^2\left(\mu, \exp_\mu(h\cdot\gamma_{n+1})\right)}{h^2} &\geqslant \|\gamma_n\|^2_\mu - t(1-t)\int_{(x,v,w)\in \mathrm{T}^2\,\mathbb{R}} |v-w|^2 d\alpha \\ &= \int_{(x,v,w)\in \mathrm{T}^2\,\mathbb{R}} |(1-t)v + tw|^2 d\alpha = \|\gamma_{n+1}\|^2_\mu. \end{aligned}$$

Here we used that $(1-t)|v|^2 + t|w|^2 - t(1-t)|v-w|^2 = |(1-t)v+tw|^2$ for any vectors v, w. Therefore, the measure field γ_{n+1} belongs to $\Xi[\mu]$. As n goes to ∞, the map f_n converges in $\mu-$measure towards some function that coincides with f whenever $f \leqslant R$, and lies in $[R, R+\varepsilon]$ otherwise. Taking $\varepsilon > 0$ small enough, then n large enough, we can approximate $\min(f, R)$ in L^2_μ with arbitrarily small error. Since convergence of maps in L^2_μ implies convergence of their associated symmetric measure field in W_μ, and since $\Xi[\mu]$ is $W_\mu-$closed by the first point, we conclude. □

Lastly, we turn to the main step of the reduction. The strategy is to first restrict to a nonatomic submeasure of μ on which ζ splits mass; then to show that taking averages of the "positive" and "negative" parts does not break the limit sup condition (6); then to normalize the obtained measure field.

Lemma 4.3. *Assume that $\zeta \in \Xi[\mu]$ for some $\zeta \notin \mathbf{Tan}_\mu$. Then there exists a nonatomic measure $\sigma \leqslant \mu$ of positive mass such that $\xi \in \Xi[\sigma]$ for the symmetric unit measure field $\xi \coloneqq \frac{1}{2}\left[(id,-1)_\#\sigma + (id,1)_\#\sigma\right]$.*

Proof. We modify ζ in successive steps to obtain a symmetric unit measure field, and show that each steps preserves the corresponding set Ξ.

Step 1: restriction to a suitable set Decompose $\mu = \mu^a + \mu^d$, with μ^a purely atomic and μ^d atomless. By [SW14, Theorem 3.7] (see also [Aus25a, Proposition 2.9]), the tangent cone to μ is made of measure fields γ of the form $\gamma^a + (id, f)_\#\mu^d$, where $\pi_{x\#}\gamma^a = \mu^a$ is a measure field on $\mathrm{T}\,\mathbb{R}$ with first marginal (hence same mass as) the atomic part μ^a, and $f \in L^2_{\mu^d}(\mathbb{R};\mathbb{R})$. Hence $\zeta \notin \mathbf{Tan}_\mu$ if and only if $\zeta = \zeta^a + \zeta^d$, with $\pi_{x\#}\zeta^a = \mu^a$, $\pi_{x\#}\zeta^d = \mu^d$ and ζ^d not induced by a map. Since ζ is centred by definition of Ξ, this is equivalent to $\|\zeta^d\|_{\mu^d} > 0$. In consequence, there exists a measurable set $A \subset \mathbb{R}$ such that

$$\mu^a(A) = 0, \qquad \mu^d(A) > 0, \qquad \text{and for } \mu-\text{a.e. } x \in A, \quad \int_{v\in\mathbb{R}} |v|^2 d\zeta_x > 0.$$

By Lemma 4.2, the restricted measure field $\zeta_A \coloneqq \zeta \llcorner \{(x,v)\,|\,x\in A\}$ belongs to $\Xi[\mu \llcorner A]$. For the sequel, denote $\nu \coloneqq \mu \llcorner A$.

Step 2: reduction to a symmetric measure field Now, split ζ_A as follows: consider $\alpha = \alpha(dx, dv, dw) \in \Gamma_{\nu,o}(\zeta_A, \xi)$, with again $\xi \coloneqq \frac{1}{2}\left[(id,-1)_\#\nu + (id,1)_\#\nu\right]$ the symmetric unit field. Then $\alpha = \frac{1}{2}\left[\alpha^- + \alpha^+\right]$, where $(\pi_x, \pi_w)_\#\alpha^\pm = (id, \pm 1)_\#\nu$. Using the other marginal condition, $\zeta_A = (\pi_x, \pi_v)_\#\alpha = \frac{1}{2}(\pi_x,\pi_v)_\#\alpha^- + \frac{1}{2}(\pi_x,\pi_v)_\#\alpha^+ =: \frac{1}{2}\left[\zeta_A^- + \zeta_A^+\right]$.

Denote $f^\pm \in L^2_\nu$ the respective barycenters of $\zeta_A^\pm$. Then $\frac{1}{2}\left[f^- + f^+\right]$ is the barycenter of ζ_A, i.e. 0, hence $f^- = -f^+$. We show that $f^+(x) > 0$ for $\nu-$a.e. $x \in \mathbb{R}$. By construction, $\alpha = (\alpha_x)_x \otimes \nu$, with α_x identified with an optimal plan between $(\zeta_A)_x$ and $\xi_x = \frac{1}{2}\left[\delta_{-1} + \delta_1\right]$. Hence

$$\begin{aligned} &\frac{1}{2}\left[\int_{\mathbb{R}} |v|^2 d(\zeta_A)_x + \int_{\mathbb{R}} |w|^2 d\xi_x - W^2((\zeta_A)_x, \xi_x)\right] = \int_{(v,w)\in\mathbb{R}^2} \frac{1}{2}\left[|v|^2 + |w|^2 - |v-w|^2\right] d\alpha_x \\ &= \int_{(v,w)\in\mathbb{R}^2} \langle v, w\rangle\, d\alpha_x = \frac{1}{2}\int_{(v,w)\in\mathbb{R}^2} \langle v, -1\rangle\, d\alpha_x^- + \frac{1}{2}\int_{(v,w)\in\mathbb{R}^2} \langle v, 1\rangle\, d\alpha_x^+ = f^+(x). \end{aligned}$$

By [Aus26, Lemma 2.9], the leftmost term is always nonnegative, and 0 if and only if $(\zeta_A)_x = \delta_0$, which may happen only on a $\nu-$negligible set by assumption. Thus $f^+ > 0$ $\nu-$a.e..

We now replace ζ_A by $\gamma := \frac{1}{2}\left[(id, f^-)_\# \nu + (id, f^+)_\# \nu\right]$ as follows: on the one hand,

$$\begin{aligned}\|\zeta_A\|_\nu^2 &= \frac{1}{2}\left[\|f^-\|_{L^2_\nu}^2 + \|f^+\|_{L^2_\nu}^2 + \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} |v - f^-(x)|^2 d\zeta_A^- + \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} |v - f^+(x)|^2 d\zeta_A^+\right] \\ &= \|\gamma\|_\nu^2 + \frac{1}{2}\sum_{s\in\{-,+\}} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} |v - f^s(x)|^2 d\zeta_A^s. \end{aligned} \tag{10}$$

On the other hand, using the dual reformulation, there holds

$$W^2(\nu, \exp_\nu(h\cdot\zeta_A)) = \sup_{\varphi\in L^1_\mu} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} \varphi^c(x+hv) d\zeta_A - \int_{x\in\mathbb{R}} \varphi(x) d\nu. \tag{11}$$

For each fixed φ and $z\in\mathbb{R}$, the function $y \mapsto \varphi^c(y) - |y-z|^2 = \inf_{w\in\mathbb{R}} \varphi(w) + |w-y|^2 - |y-z|^2$ is concave. Hence, taking $z = x + hf^\pm(x)$ inside the integral, we can write that

$$\begin{aligned}&\int_{(x,v)\in \mathrm{T}\,\mathbb{R}} \varphi^c(x+hv) d\zeta_A \\ &= \frac{1}{2}\sum_{s\in\{-,+\}} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} \left[\varphi^c(x+hv) - |(x+hv) - (x+hf^s(x))|^2\right] d\zeta_A^s \\ &\quad + \frac{1}{2}\sum_{s\in\{-,+\}} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} |(x+hv) - (x+hf^s(x))|^2 d\zeta_A^s \\ &\leqslant \frac{1}{2}\sum_{s\in\{-,+\}} \int_{x\in\mathbb{R}} \left[\varphi^c(x+hf^s(x)) - 0\right] d\nu + \frac{h^2}{2}\sum_{s\in\{-,+\}} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} |v - f^s(x)|^2 d\zeta_A^s. \end{aligned}$$

In the first term of the last line, we recognize $\int_{y\in\mathbb{R}} \varphi^c(y) d\exp_\nu(h\cdot\gamma)$. Using (10) to identify the second term, and plugging this inequality in (11), we obtain

$$\begin{aligned}W^2(\nu, \exp_\nu(h\cdot\zeta_A)) &\leqslant \sup_{\varphi\in L^1_\mu} \int_{y\in\mathbb{R}} \varphi^c(y) d\exp_\nu(h\cdot\gamma) - \int_{x\in\mathbb{R}} \varphi(x) d\mu + h^2\left[\|\zeta_A\|_\nu^2 - \|\gamma\|_\nu^2\right] \\ &= W^2(\nu, \exp_\nu(h\cdot\gamma)) + h^2\left[\|\zeta_A\|_\nu^2 - \|\gamma\|_\nu^2\right].\end{aligned}$$

Dividing by $h^2 > 0$ and taking the limit sup in $h \searrow 0$, we conclude that $\gamma \in \Xi[\nu]$.

Step 3: reduction to a unit symmetric measure field At this point, we know that there exists $f := f^+ \in L^2_\nu(\mathbb{R};\mathbb{R}^+)$ such that $f(x) > 0$ for $\nu-$a.e. point, and the measure field $\gamma := \frac{1}{2}\left[(id, -f)_\# \nu + (id, f)_\# \nu\right]$ belongs to $\Xi[\nu]$. By Lemma 4.2, we might multiply f by some large $\lambda > 0$ so that $\nu\{\lambda f \geqslant 1\} \geqslant \nu(\mathbb{R})/2$, and further restrict to the measurable set $B := \{\lambda f \geqslant 1\} \subset \mathbb{R}$, while still maintaining $\vartheta \in \Xi[\sigma]$ for $\sigma := \nu \llcorner B$ and $\vartheta := \frac{1}{2}\left[(id, -\lambda f)_\# \sigma + (id, \lambda f)_\# \sigma\right]$. Since $\min(\lambda f, 1) = 1$ for $\sigma-$almost every point, we can apply the last point of Lemma 4.2 to obtain that the unit symmetric measure field does belong to $\Xi[\sigma]$. As $\sigma = \mu \llcorner (A\cap B) \leqslant \mu$ has positive mass, this completes the proof. □

Remark 4.4. If we were interested into measure fields for which the *limit* exists, instead of the limit sup, then a shorter proof could be done: one shows that the set Ξ of centred $\xi \in \mathscr{P}_2(\mathrm{T}\,\mathbb{R})_\mu$ such that $\lim_{h\searrow 0} W(\mu, \exp_\mu(h\cdot\xi))/h = \|\xi\|_\mu$ is a $W_\mu-$closed convex cone of centred measure fields, in the sense given to it in [Aus26]. By Proposition 2.1 therein, there exists a measurable application $D : x \mapsto D(x)$, where $D(x)$ is either $\{0\}$ or $\mathbb{R}$, and such that Ξ is exactly given by the set of measure fields concentrated on the graph of D. The assumption of Lemma 4.3 implies that $D(x) = \mathbb{R}$ on some set A of positive μ^d-mass, and one can directly take $\nu := \mu^d \llcorner A$. However, the dependence to a particular subsequence prevents this argument, and the above proof manually goes through the argument of [Aus26].

4.3 Proof of Theorem 1.3

One of the implications of Theorem 1.3 follows from the successive reductions of the previous sections. For the other implication, one needs to show that the mass that μ puts around a set $A \in \mathscr{A}$ does not play a significant role for small $h > 0$.

Proof. Assume that $\zeta \notin \mathbf{Tan}_\mu$ satisfies $\limsup_{h\searrow 0} W(\mu, \exp_\mu(h\cdot\zeta))/h = \|\zeta\|_\mu$. By Lemma 4.1, the centred measure field $\zeta^0 := (\pi_x, \pi_v - \mathrm{Bary}\,(\zeta)(\pi_x))_\#\zeta$ also does. By Lemma 4.3, there exists a submeasure $\sigma \leqslant \mu^d$ such that (6) holds, i.e. $W(\sigma, \sigma_h)/h \to 1$ when $h \searrow 0$. By Theorem 1.2, σ gives mass to an element of $\mathscr{A}$, and so does μ^d.

Conversely, assume that μ^d puts mass on some $B \in \mathscr{A}$. Since any subset $A \subset B$ also belongs to $\mathscr{A}$, we can choose a compact $A \in \mathscr{A}$ such that $\mu^d(A) > 0$. Let

$$\zeta \coloneqq \frac{1}{2}\left[(id,-1)_{\#}\mu \llcorner A + (id,1)_{\#}\mu \llcorner A\right] + (id,0)_{\#}\mu \llcorner A^c.$$

Since μ^d is nonatomic, the measure field ζ is not tangent [Aus25a, Proposition 2.9]. We claim that $\limsup_{h \searrow 0} W(\mu, \exp_\mu(h\cdot\zeta))/h = \|\zeta\|_\mu = \sqrt{\mu(A)}$: the strategy is to show that when passing to the limit, the neighbourhood of the compact A receives few mass, and the quantity $W(\mu, \exp_\mu(h\cdot\zeta))$ is very close to the sum of the contribution of $\mu \llcorner A$ and $\mu \llcorner A^c$.

Denote by η the optimal transport plan between μ and $\exp_\mu(h\cdot\zeta)$, which is concentrated on pairs (x,y) such that $|y-x| \leqslant h$ by Lemma 3.1. Let $\mu_A \coloneqq \mu \llcorner A$ and $\eta_A \coloneqq \eta \llcorner (A \times \mathbb{R})$. Then, by restriction of optimality [Vil09, Theorem 4.6] and the second triangular inequality,

$$\begin{aligned} W^2(\mu, \exp_\mu(h\cdot\zeta)) &= \int_{(x,y)} |y-x|^2 d\eta \geqslant \int_{(x,y)\in A\times\mathbb{R}} |y-x|^2 d\eta = W^2(\mu_A, \pi_{y\#}\eta_A) \\ &\geqslant W^2(\mu_A, (\mu_A)_h) - W(\pi_{y\#}\eta_A, (\mu_A)_h)\left(2W(\mu_A, \pi_{y\#}\eta_A) + W(\pi_{y\#}\eta_A, (\mu_A)_h)\right) \\ &\geqslant W^2(\mu_A, (\mu_A)_h) - W(\pi_{y\#}\eta_A, (\mu_A)_h) \times (2h + (h+h)). \end{aligned} \tag{12}$$

Let $(s_n)_{n\in\mathbb{N}}, (\tau_n)_{n\in\mathbb{N}} \subset (0,1)$ be the vanishing sequences associated to A, i.e. such that for any $x \in A$ and $n \in \mathbb{N}$, the intersection $A \cap \mathscr{B}(x, s_n)$ is contained in $\overline{\mathscr{B}}(x, \tau_n s_n)$. Since A is compact, we can pick successively a finite set of points $\{b_1, \cdots, b_K\} \subset A$ such that $A \subset \bigcup_{k=1}^K \mathscr{B}(b_k, s_n)$, and for any $k \neq \ell$, there holds $b_k \notin \mathscr{B}(b_\ell, s_n)$. Then, the closed balls $B_k \coloneqq \overline{\mathscr{B}}(b_k, \tau_n s_n)$ cover A, and are separated from each other by at least $s_n - 2\tau_n s_n$.

Consider $h_n \coloneqq \frac{s_n}{2} - \tau_n s_n - \varepsilon_n$, where $\varepsilon_n > 0$ and $\varepsilon_n/h_n \to_n 0$. For this choice, the inflated balls $B_k^{h_n} \coloneqq \overline{\mathscr{B}}(b_k, \tau_n s_n + h_n)$ are disjoint. We distinguish the contributions of each $B_k^{h_n}$ by using the convexity of the squared Wasserstein distance in the Banach sense; denoting $\mu_{A,k} \coloneqq \mu \llcorner (A \cap B_k)$, and $\eta_{A,k} \coloneqq \eta \llcorner ((A\cap B_k)\times\mathbb{R})$, there holds

$$W^2(\pi_{y\#}\eta_A, (\mu_A)_{h_n}) \leqslant \sum_{k=1}^K W^2(\pi_{y\#}\eta_{A,k}, (\mu_{A,k})_{h_n}). \tag{13}$$

Now, both measures $\pi_{y\#}\eta_{A,k}$ and $(\mu_{A,k})_{h_n}$ are submeasures of $\exp_\mu(h_n\cdot\zeta) \llcorner B_k^{h_n}$. Indeed, $\pi_{y\#}\eta = \exp_\mu(h_n\cdot\zeta)$, and the $\eta-$a.e. estimate $|y-x| \leqslant h_n$ shows that $\pi_{y\#}\eta_{A,k}$ is supported on $B_k^{h_n}$. On the other hand, ζ coincides with the unit symmetric measure field on A. To apply Lemma 2.1, we only need to estimate the difference between $\pi_{y\#}\eta_{A,k}(\mathbb{R}) = (\mu_{A,k})_{h_n}(\mathbb{R}) = \mu(A\cap B_k)$ and the mass that $\exp_\mu(h_n\cdot\zeta)$ puts on $B_k^{h_n}$. But

$$\begin{aligned} \int_{(x,v)\in \mathrm{T}\,\mathbb{R}} \mathbb{1}_{B_k^{h_n}}(x+h_n v) d\zeta &= \int_{x\in A} \frac{1}{2}\left[\mathbb{1}_{B_k^{h_n}}(x-h_n) + \mathbb{1}_{B_k^{h_n}}(x+h_n)\right] d\mu + \int_{A^c} \mathbb{1}_{B_k^{h_n}} d\mu \\ &= \frac{1}{2}\left[\mu(A\cap(B_k^{h_n}+h_n)) + \mu(A\cap(B_k^{h_n}-h_n))\right] + \mu\left(A^c \cap B_k^{h_n}\right) \\ &= \mu(A\cap B_k) + \frac{1}{2}\mu\left(A\cap(B_k^{2h_n}\setminus B_k)\right) + \mu\left(A^c\cap B_k^{h_n}\right). \end{aligned}$$

Since $A \subset \bigcup_{k=1}^K B_k$, and B_k, B_ℓ are separated by at least $s_n - 2\tau_n s_n = 2h_n + 2\varepsilon_n > 2h_n$, the sets $A\cap(B_k^{2h_n}\setminus B_k)$ are empty. Applying Lemma 2.1 with $\alpha = \pi_{y\#}\eta_{A,k}$ and $\beta = (\mu_{A,k})_{h_n}$, both submeasures of $\exp_\mu(h_n\cdot\zeta)\llcorner B_k^{h_n}$ having the same mass, we get

$$\begin{aligned} W^2(\pi_{y\#}\eta_{A,k}, (\mu_{A,k})_{h_n}) &\leqslant \left(\exp_\mu(h_n\cdot\zeta)(B_k^{h_n}) - \pi_{y\#}\eta_{A,k}(\mathbb{R})\right)\left(\operatorname{diam} B_k^{h_n}\right)^2 \\ &\leqslant \mu\left(A^c\cap B_k^{h_n}\right) \times 4\left(\tau_n s_n + h_n\right)^2. \end{aligned}$$

The sets $A^c \cap B_k^{h_n}$ are disjoint, so we may sum over $k \in [\![1,K]\!]$ and using (13) to deduce

$$W^2(\pi_{y\#}\eta_A, (\mu_A)_{h_n}) \leqslant 4(\tau_n s_n + h_n)^2 \mu\left(A^c \cap \cup_k B_k^{h_n}\right) \leqslant 4(\tau_n s_n + h_n)^2 \mu(A^{\tau_n s_n + h_n}\setminus A).$$

Note that $\tau_n s_n + h_n = \frac{2\tau_n}{1-2\tau_n}(h_n+\varepsilon_n) + h_n \leqslant c\, h_n$ for some $c > 0$. Hence, we obtain from (12) that

$$W^2(\mu, \exp_\mu(h_n\cdot\zeta)) \geqslant W^2(\mu_A, (\mu_A)_{h_n}) - 8\, c\, h_n^2 \sqrt{\mu\left(A^{ch_n}\setminus A\right)}.$$

As A is compact, there holds $\bigcap_{n\in\mathbb{N}} A^{ch_n}\setminus A = \emptyset$, hence $\mu\left(A^{ch_n}\setminus A\right) = O(h_n)$. To conclude, we only have to show that $W^2(\mu_A, (\mu_A)_{h_n})/h_n^2 \to_n \mu(A)$; this is almost the same computation as in Theorem 1.2, up to the choice of $h_n = \frac{s_n}{2} - \tau_n s_n - \varepsilon_n$ instead of $\frac{s_n+\tau_n s_n}{2}$, and the perturbation by $\frac{3}{2}\tau_n s_n + \varepsilon_n = o(h_n)$ does not affect the result. □

5 Measures for which the limit sup vanishes

5.1 Regularity of the distribution function

The situation for measures μ such that $W_1(\mu, \mu_h) = o(h)$ is much less clear. All absolutely continuous measures have this property, and Proposition 5.5 below exhibits some non-absolutely continuous measure that also does. Of course, measures that give mass to some $A \in \mathscr{A}$ do not. In between, we are only able to say the following.

Theorem 5.1. *Let $\mu \in \mathscr{P}(\mathbb{R})$, and denote $F_\mu : \mathbb{R} \to [0,1]$ the function $F_\mu(x) \coloneqq \mu((-\infty, x])$. It is equivalent that*

1. *$W_1(\mu, \mu_h) = o(h)$;*
2. *for any vanishing sequence $(h_n)_{n\in\mathbb{N}} \subset (0,1]$, there exists a subsequence $(h_{n_k})_{k\in\mathbb{N}}$ and $g \in L^1(\mathbb{R}; \mathbb{R}^+)$ such that*

$$\left| \frac{\frac{1}{2}\left[F_\mu(x - h_{n_k}) + F_\mu(x + h_{n_k})\right] - F_\mu(x)}{h_{n_k}} \right| \leqslant g(x) \qquad \text{for a.e. } x \in \mathbb{R},\ k \in \mathbb{N}. \tag{14}$$

By Remark 3.2, Point 1 implies that $W_p(\mu, \mu_h) = o(h)$ for any $p \in [1, \infty)$. Point 2 can be understood as a regularity property in the following sense: the function F_μ is nondecreasing, so differentiable almost everywhere. Therefore, the left hand-side of (14) converges to 0 for $\mathcal{L}-$almost every x. However, (14) additionally imposes that the error is dominated by an L^1 function g.

Proof. By [San15, Proposition 2.17], the 1-dimensional 1-Wasserstein distance is equivalently computed as

$$\frac{W_1(\mu, \mu_h)}{h} = \int_{x\in\mathbb{R}} \frac{1}{h} \left|F_{\mu_h}(x) - F_\mu(x)\right| dx = \int_{x\in\mathbb{R}} \frac{1}{h} \left| \frac{F_\mu(x-h) + F_\mu(x+h)}{2} - F_\mu(x) \right| dx.$$

Assume Point 1 holds, and let $(h_n)_{n\in\mathbb{N}} \subset (0,1]$ be a vanishing sequence. Choose a subsequence $(n_k)_{k\in\mathbb{N}}$ such that $W_1(\mu, \mu_{h_{n_k}}) \leqslant h_{n_k} 2^{-k}$, and define a function $g \in L^1$ as $g \coloneqq \sum_k \left|\frac{1}{2}\left[F_\mu(\cdot - h_{n_k}) + F_\mu(\cdot + h_{n_k})\right] - F_\mu\right| / h_{n_k}$. Then $g \in L^1(\mathbb{R}; \mathbb{R}^+)$, and Point 2 holds.

Conversely, assume Point 2. Since F_μ is nondecreasing and bounded, it has bounded variation, hence is differentiable Lebesgue-almost everywhere [AFP00, Corollary 3.29]. Consequently, for a.e. $x \in \mathbb{R}$, there holds $\psi_h(x) \coloneqq \left|\frac{1}{2}\left[F_\mu(x-h) + F_\mu(x+h)\right] - F_\mu(x)\right| / h \to 0$ as $h \searrow 0$. Let $(h_n)_{n\in\mathbb{N}}$ be a vanishing sequence realizing the limit sup of $W_1(\mu, \mu_h)/h$ when h goes to 0. By assumption, along some subsequence $(n_k)_k$, the functions $\psi_{h_{n_k}}$ are uniformly bounded by an L^1 function. Applying Lebesgue's dominated convergence, we conclude that $\lim_{k\to\infty} \|\psi_{h_{n_k}}\|_{L^1} = \limsup_{h\searrow 0} W_1(\mu, \mu_h)/h = 0$, which proves Point 1. □

5.2 Classification of uniform Cantor measures

Instead of addressing the full characterization of measures μ such that $W(\mu, \mu_h) = o(h)$, we reduce to a subclass of uniform measures on Cantor sets. To avoid irrelevant complications, we only consider nonincreasing sequences $(\alpha_n)_n \subset (0,1)$, which therefore admit a limit in $[0,1]$. If I is a bounded interval of $\mathbb{R}$, the notation $\alpha * I$ refers to the interval with the same center as I, and length equal to α times that of I.

Definition 5.2 (Cantor set $C(\alpha)$)**.** Let $\alpha = (\alpha_n)_{n\in\mathbb{N}} \subset (0,1)$ be a nonincreasing sequence. Denote $C^0 = I_0^0 \coloneqq [0,1]$. Assuming that C^n is defined as the union of 2^n disjoint intervals $(I_\ell^n)_{\ell=1}^{2^n}$, each with the same length δ_n, define C^{n+1} by deleting the proportion α_n of the middle of each I_ℓ^n:

$$C^{n+1} \coloneqq \bigcup_{\ell\in[\![1,2^n]\!]} I_\ell^n \setminus (\alpha_n * I_\ell^n).$$

Then C^{n+1} is again given by 2^{n+1} intervals. The set $C(\alpha)$ is defined as the intersection of all $(C^n)_{n\in\mathbb{N}}$.

Definition 5.3 (Uniform measure on $C(\alpha)$)**.** Let $\alpha = (\alpha_n)_{n\in\mathbb{N}}$, and consider the notations of Definition 5.2. For each $n \in \mathbb{N}$, let $\mu^n \coloneqq (\mathcal{L} \llcorner C^n) / \mathcal{L}(C^n)$. The measure μ_α is defined as the W_1-limit of $(\mu^n)_{n\in\mathbb{N}}$. Equivalently,

$$\mu_\alpha(I_\ell^n) = 2^{-n} \qquad \forall n \in \mathbb{N} \text{ and } \ell \in [\![1, 2^n]\!]. \tag{15}$$

For instance, the classical triadic Cantor set is obtained with the constant sequence $\alpha_n \equiv 1/3$, and μ_α is the derivative of the Cantor staircase. We are interested in the sequences α such that

$$\limsup_{h\searrow 0} \frac{W(\mu_\alpha, \mu_{\alpha,h})}{h} = 0. \tag{16}$$

The sets $C(\alpha)$ were studied in [HP85]. In their terminology, a set C is Tkadlec if there exists a measure concentrated on C that gives mass to some porous set. They show that $C(\alpha)$ is Tkadlec if and only if $\liminf_{n\to\infty} \alpha_n > 0$. What we show in Propositions 5.4 and 5.5 can be reformulated as "μ_α satisfies (16) if and only if $C(\alpha)$ is non-Tkadlec". However, a quick computation yields that $\mathcal{L}(C(\alpha)) > 0$ if and only if $\sum_{n\in\mathbb{N}} \alpha_n < \infty$. Therefore, any vanishing sequence α that is not summable generates a set $C(\alpha)$ of null Lebesgue measure, but such that μ_α still satisfies (16).

Proposition 5.4. *Assume that* $\lim_{n\to\infty}\alpha_n =: \overline{\alpha} > 0$. *Then (16) fails.*

Proof. Denote again δ_n the length of any interval I_ℓ^n. Since each I_ℓ^n contains two intervals I_k^{n+1}, there holds $\delta_{n+1} \leqslant \delta_n/2$. Let $h_n \coloneqq \alpha_{n-1}\delta_{n-1}/2$ and $\psi : x \mapsto \inf_{y\in C^n} d(x,y)$. As $C(\alpha) \subset C^n$, there holds $\psi(x) = 0$ for $\mu_\alpha-$a.e. $x \in \mathbb{R}$. On the other hand, since ψ is 1-Lipschitz, it is admissible in the dual problem for the 1-Wasserstein distance, and there holds

$$\frac{W_1(\mu_\alpha,\mu_{\alpha,h_n})}{h_n} \geqslant \frac{1}{h_n}\left[\int \psi d\mu_{\alpha,h_n} - \int \psi d\mu_\alpha\right] = \int \frac{\psi(x-h_n)+\psi(x+h_n)}{2h_n} d\mu_\alpha$$
$$= \sum_{\ell=1}^{2^{n-1}} \int_{I_\ell^{n-1}} \frac{\psi(x-h_n)+\psi(x+h_n)}{2h_n} d\mu_\alpha = 2^{n-1}\int_{I_1^{n-1}} \frac{\psi(x-h_n)+\psi(x+h_n)}{2h_n} d\mu_\alpha. \tag{17}$$

In the last equality, we used the fact that each I_ℓ^{n-1} is a translation of I_1^{n-1}, that two intervals I_ℓ^{n-1} are separated by at least $\alpha_{n-2}\delta_{n-2} \geqslant \alpha_{n-1}\delta_{n-1} = 2h_n$ and that the measure μ_α is self-similar. The intersection $C^n \cap I_1^{n-1}$ is the union of two intervals I_L and I_R, respectively the left and right connected components of $I_1^{n-1} \setminus (\alpha_{n-1} * I_1^{n-1})$.

For any $x \in [\max I_L - h_n/2, \max I_L]$, the point $x + h_n$ is at distance at least $h_n/2$ from any point of I_L. On the other hand, $\min I_R - \max I_L = 2h_n$, so that $x + h_n$ is at distance at least h_n of I_R. Therefore $\psi(x + h_n) = \inf_{z\in C^n} |x - z| \geqslant h_n/2$ for such x. Repeating the symmetric reasoning on I_R and using that $\psi \geqslant 0$, we get the lower bound

$$\int_{I_1^{n-1}} \frac{\psi(x-h_n)+\psi(x+h_n)}{2h_n} d\mu_\alpha \geqslant \frac{\mu([\max I_L - \frac{h_n}{2}, \max I_L]) + \mu([\min I_R, \min I_R + \frac{h_n}{2}])}{4}.$$

To conclude, we estimate $\mu_\alpha([\max I_L - h_n/2, \max I_L])$ from below. Consider the rightmost interval J_m of $C^m \cap I_L$ as m grows: for $m = n + 1$, the interval J_m is contained in $[\max I_L - \delta_n/2, \max I_L]$. In general, $J_m \subset [\max I_L - 2^{-(m-n)}\delta_n, \max I_L]$. Let M be large enough so that $2^{-M} \leqslant \overline{\alpha}/2$. When $m = n + M$, we obtain

$$2^{-(m-n)}\delta_n = 2^{-M}\delta_n \leqslant \frac{\overline{\alpha}}{2}\delta_n \leqslant \frac{\alpha_{n-1}\delta_{n-1}}{4} = \frac{h_n}{2},$$

which implies that $J_{n+M} \subset [\max I_L - h_n/2, \max I_L]$. Hence $\mu_\alpha([\max I_L - h_n/2, \max I_L]) \geqslant \mu_\alpha(J_{n+M}) = 2^{-(n+M)}$ by (15). Repeating the argument on I_R, we conclude from (17) that $W_1(\mu_\alpha,\mu_{\alpha,h_n})/h_n \geqslant 2^{n-1}(2^{-(n+M)} + 2^{-(n+M)})/4 = 2^{-M-2}$, independently of n. □

Proposition 5.5. *Assume that* $\lim_{n\to\infty}\alpha_n = 0$. *Then (16) holds.*

Proof. By Lemma 3.1, we can discard the first terms of the sequence and assume that $\alpha_n \leqslant 1/2$ for any $n \in \mathbb{N}$. By [San15, Proposition 2.17], the 1-dimensional 1-Wasserstein distance can be computed by

$$W_1(\mu_\alpha,\mu_{\alpha,h}) = \int_{x\in\mathbb{R}} \left|F_{\mu_{\alpha,h}}(x) - F_{\mu_\alpha}(x)\right| dx,$$

with F_{μ_α} and $F_{\mu_{\alpha,h}}$ the respective distribution functions of μ_α and $\mu_{\alpha,h}$. In our case, there holds in addition that $F_{\mu_{\alpha,h}}(x) = \frac{1}{2}\left[F_{\mu_\alpha}(x-h) + F_{\mu_\alpha}(x+h)\right]$. We can further represent F_{μ_α} as a series by mimicking the construction of μ_α as a limit of $(\mu^n)_{n\in\mathbb{N}}$: let $f_{-1}(x) \coloneqq F_{\mu^0}(x) = \max(0, \min(1, x))$, and for each n, let $f_n \coloneqq F_{\mu^{n+1}} - F_{\mu^n}$. Then $f_n = 0$ out of C^n, and f_n repeats the same pattern on each interval of C^n. Precisely, if $[a, a + \delta_n]$ and $[b, b + \delta_n]$ are two intervals of C^n, then $f_n(a + z) = f_n(b + z) = f_n(z)$ for any $z \in [0, \delta_n]$, with

$$f_n(z) = \begin{cases} 2^{-(n+1)}\frac{z}{\delta_{n+1}} - 2^{-n}\frac{z}{\delta_n} & z \in [0, \delta_{n+1}], \\ 2^{-(n+1)} - 2^{-n}\frac{z}{\delta_n} & z \in (\delta_{n+1}, \delta_n - \delta_{n+1}), \\ 2^{-(n+1)} + 2^{-(n+1)}\frac{z-(\delta_n-\delta_{n+1})}{\delta_{n+1}} - 2^{-n}\frac{z}{\delta_n} & z \in [\delta_n - \delta_{n+1}, \delta_n]. \end{cases} \tag{18}$$

Recall that $\delta_{n+1} = \delta_n(1 - \alpha_n)/2$. One checks that $\|f_n\|_\infty = f_n(\delta_{n+1}) = 2^{-(n+1)}\alpha_n$. Hence $\sum_{n=-1}^\infty f_n$ converges uniformly towards F_{μ_α}. We now bound each term of the series.

For each h, let $n \in \mathbb{N}$ be such that $h \in [\sqrt{\alpha_n}\delta_n, \sqrt{\alpha_{n-1}}\delta_{n-1})$. We have to estimate

$$\int \frac{\left|\frac{1}{2}\left[F_{\mu_\alpha}(\cdot - h) + F_{\mu_\alpha}(\cdot + h)\right] - F_{\mu_\alpha}\right|}{h} dx \leqslant \sum_{k=-1}^\infty \int \frac{\left|\frac{1}{2}\left[f_k(\cdot - h) + f_k(\cdot + h)\right] - f_k\right|}{h} dx.$$

We treat differently the indices $k < n$ and $k \geqslant n$. For any $k \geqslant n$, the application $x \mapsto \frac{1}{2}\left[f_k(x - h) + f_k(x + h)\right] - f_k(x)$ vanishes for any point x at distance at least h from C^n. Hence

$$\sum_{k=n}^\infty \int \frac{\left|\frac{1}{2}\left[f_k(x-h) + f_k(x+h)\right] - f_k(x)\right|}{h} dx \leqslant 2\sum_{k=n}^\infty \frac{\|f_k\|_\infty}{h}\mathcal{L}\left(\{x \in \mathbb{R} \mid d(x, C^n) \leqslant h\}\right)$$
$$\leqslant 2\sum_{k=n}^\infty \frac{2^{-(k+1)}\alpha_k}{h} 2^n\left(\mathcal{L}(I_1^n) + 2h\right) = \sum_{k=n}^\infty \alpha_k 2^{n-k}\left(\frac{\delta_n}{h} + 2\right) \leqslant \sum_{k=n}^\infty \alpha_k 2^{n-k}\left(\frac{1}{\sqrt{\alpha_n}} + 2\right).$$

Using that $\alpha_k \leqslant \alpha_n$ for $k \geqslant n$, the last term is bounded from above by $\sqrt{\alpha_n} + 2\alpha_n$. Now, for $k < n$, the application $x \mapsto \frac{1}{2}\left[f_k(x-h) + f_k(x+h)\right] - f_k(x)$ differs from 0 only around the points of discontinuity of f_k'. For $\alpha_n \leqslant 1/2$, the Lipschitz constant of f_k can be seen in (18) to be the slope of the middle segment, so that $\left|\frac{1}{2}\left[f_k(x-h) + f_k(x+h)\right] - f_k(x)\right| \leqslant 2^{-k}h/\delta_k$. Counting 2^k intervals of C^k, each containing 4 points of discontinuity, each influencing a segment of length $2h$, we can estimate that

$$\int \frac{\left|\frac{1}{2}\left[f_k(\cdot - h) + f_k(\cdot + h)\right] - f_k\right|}{h} dx \leqslant \frac{1}{h} \cdot 2^k \cdot 4 \cdot 2h \cdot \frac{2^{-k}h}{\delta_k} \leqslant \frac{8h}{\delta_k} \leqslant \frac{8h}{2^{n-1-k}\delta_{n-1}}.$$

In the last equality, we used that $\delta_{n+1} \leqslant \delta_n/2$, hence $\delta_k \geqslant 2^{n-1-k}\delta_{n-1}$. As we chose $h \leqslant \sqrt{\alpha_{n-1}}\delta_{n-1}$, there holds $h/\delta_{n-1} \leqslant \sqrt{\alpha_{n-1}}$ and

$$\sum_{k=0}^{n-1} \int \frac{\left|\frac{1}{2}\left[f_k(x-h) + f_k(x+h)\right] - f_k(x)\right|}{h} dx \leqslant \sum_{k=0}^{n-1} 2^{k-n+4}\sqrt{\alpha_{n-1}} \leqslant 16\sqrt{\alpha_{n-1}}.$$

There only stays to bound the contribution of $k = -1$, which is easily seen to be of order $4h$. Gathering the above estimates, we conclude that $W_1(\mu_\alpha, \mu_{\alpha,h})/h \to 0$ when $h \searrow 0$, hence that (16) holds. □

Acknowledgments This work benefited from the support of the ERC Starting Grant ConFine 101078057 in Pisa.

References

[AKP23] S. Alexander, V. Kapovitch, and A. Petrunin. *Alexandrov Geometry: Foundations*. Graduate Studies in Mathematics 236. Providence, Rhode Island: American Mathematical Society, 2023.

[AFP00] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Mathematical Monographs. Oxford ; New York: Clarendon Press, 2000. 434 pp.

[AGS05] L. Ambrosio, N. Gigli, and G. Savaré. *Gradient Flows*. Lectures in Mathematics ETH Zürich. Basel: Birkhäuser-Verlag, 2005.

[Aus26] A. Aussedat. "Locality of Centred Tangent Cones in the Wasserstein Space". *Journal de l'École Polytechnique* 13 (2026), pp. 1575–1606.

[Aus25a] A. Aussedat. "On the Structure of the Geometric Tangent Cone to the Wasserstein Space". *Journal of Differential Equations* 442 (2025), p. 113520.

[Aus25b] A. Aussedat. "Optimal Control Problems and Hamilton-Jacobi-Bellman Equations in Some Curved Metric Spaces". 2025.

[Ber24] C. Bertucci. "Stochastic Optimal Transport and Hamilton–Jacobi–Bellman Equations on the Set of Probability Measures". *Annales de l'Institut Henri Poincaré C* (2024).

[Ber25] C. Bertucci. *The Tangent Space to the Wasserstein Space: Parallel Transport and Other Applications*. Preprint, available at `http://arxiv.org/abs/2512.09763`. 2025.

[BG97] A. Bressan and G. Guerra. "Shift-Differentiability of the Flow Generated by a Conservation Law". *Discrete & Continuous Dynamical Systems - A* 3.1 (1997), pp. 35–58.

[Cam+21] F. Camilli, G. Cavagnari, R. De Maio, and B. Piccoli. "Superposition Principle and Schemes for Measure Differential Equations". *Kinetic & Related Models* 14.1 (2021), p. 89.

[CD18] R. Carmona and F. Delarue. *Probabilistic Theory of Mean Field Games with Applications I*. Vol. 83. Probability Theory and Stochastic Modelling. Springer International Publishing, 2018.

[CSS23a] G. Cavagnari, G. Savaré, and G. E. Sodini. *A Lagrangian Approach to Totally Dissipative Evolutions in Wasserstein Spaces*. Preprint (arXiv:2305.05211). 2023.

[CSS23b] G. Cavagnari, G. Savaré, and G. E. Sodini. "Dissipative Probability Vector Fields and Generation of Evolution Semigroups in Wasserstein Spaces". *Probability Theory and Related Fields* 185.3–4 (2023), pp. 1087–1182.

[CST26] G. Ceccherini Silberstein and D. Tonon. *On Viscosity Solutions of Hamilton-Jacobi Equations in the Wasserstein Space and the Vanishing Viscosity Limit*. Preprint, available at `http://arxiv.org/abs/2512.14568`. 2026.

[FP08] T. Fowler and D. Preiss. "A Simple Proof of Zahorski's Description of Non-Differentiability Sets of Lipschitz Functions". *Real Analysis Exchange* 34.1 (2008), pp. 127–138.

[GNT08] W. Gangbo, T. Nguyen, and A. Tudorascu. "Hamilton-Jacobi Equations in the Wasserstein Space". *Methods and Applications of Analysis* 15.2 (2008), pp. 155–184.

[GT19] W. Gangbo and A. Tudorascu. "On Differentiability in the Wasserstein Space and Well-Posedness for Hamilton–Jacobi Equations". *Journal de Mathématiques Pures et Appliquées* 125 (2019), pp. 119–174.

[Gig08] N. Gigli. "On the Geometry of the Space of Probability Measures Endowed with the Quadratic Optimal Transport Distance". Pisa: Scuola Normale Superiore di Pisa, 2008. 159 pp.

[HP85] P. D. Humke and D. Preiss. "Measures for Which σ-Porous Sets Are Null". *Journal of the London Mathematical Society* s2-32.2 (1985), pp. 236–244.

[LBD22] N. Lanzetti, S. Bolognani, and F. Dörfler. *First-Order Conditions for Optimization in the Wasserstein Space*. Preprint (arXiv:2209.12197). 2022.

[Mer+03] M. E. Mera, M. Morán, D. Preiss, and L. Zajíček. "Porosity, σ–Porosity and Measures". *Institute of Physics Publishing*. Nonlinearity 16 (2003), pp. 247–255.

[Ott01] F. Otto. "The Geometry of Dissipative Evolution Equations: The Porous Medium Equation". *Communications in Partial Differential Equations* 26.1–2 (2001), pp. 101–174.

[Pic19] B. Piccoli. "Measure Differential Equations". *Archive for Rational Mechanics and Analysis* 233.3 (2019), pp. 1289–1317.

[San15] F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer International Publishing, 2015.

[Sch25] A. Schichl. *Non-Linear Degenerate Parabolic Flow Equations and a Finer Differential Structure on Wasserstein Spaces*. 2025.

[SW14] M. Sedjro and M. Westdickenberg. *Characterization of the Tangent Space of Monotone Transport Plans in $R \times R$ with Prescribed First Projection*. Preprint (arXiv:1411.3836). 2014.

[Vil09] C. Villani. *Optimal Transport*. Vol. 338. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009.

[Zah46] Z. Zahorski. "Sur l'ensemble des points de non-dérivabilité d'une fonction continue". *Bulletin de la Société Mathématique de France* 74 (1946), pp. 147–178.

arXiv:2609.29401v1 [math.AP] 24 Sep 2026

THE MONGE-KANTOROVICH TANGENT BUNDLE TO A MEASURE

ADOLFO ARROYO-RABASA[†], AVERIL AUSSEDAT[†], GUY BOUCHITTÉ[*]

[†] *Dipartimento di Matematica, Università di Pisa, Largo Pontecorvo 5, 56127 Pisa, Italy*

[*] *Laboratoire IMATH, Université de Toulon, BP 20132, 83957 La Garde Cedex, France*

Abstract. We study the tangent bundle T_μ to a Radon measure μ in Euclidean space, introduced by Bouchitté, Champion and Jimenez (2005). Its construction, inspired by Monge-Kantorovich optimal transport theory, involves the duality between Lipschitz functions and the so-called Arens-Eells space, a Banach subspace of distributions obtained by completing the set of balanced signed measures. Precisely, a velocity field σ is μ-tangent when the divergence of $\sigma\mu$ lies in the Arens-Eells space. The tangent bundle T_μ defined μ-almost everywhere provides a local projection that allows to construct a μ-tangential gradient operator on Lipschitz functions that is weakly continuous and enjoys integration by parts. In this paper, we introduce a new quantitative estimate involving the tangential and normal components of a given velocity vector field $\sigma \in L^1_\mu(\mathbb{R}^d)$. Specifically, we show that the tangential condition $\sigma \in T_\mu$ holding μ a.e. is equivalent to each of the following two conditions: the convergence in the Arens-Eells space of $h^{-1}\left((id + h\sigma)_\#\mu - \mu\right)$ as $h \to 0$, and the differentiability of Lipschitz functions along σ. Moreover, given a field of non-tangent directions, we construct a Lipschitz function that is not differentiable on a large set, confirming that T_μ agrees with the decomposability bundle of Alberti and Marchese. We finally relate T_μ to the tangent measures of Preiss and survey further properties of the tangential differential calculus.

Keywords. Optimal transport, Kantorovich-Rubinstein distance, Arens-Eells, tangent bundle, differentiability, Lipschitz functions, flat chains, Smirnov decomposition

MSC 2020. 49Q15, 49Q22, 26B05, 28A75, 49J45

AI statement. AI tools were used exclusively for minor language assistance. No AI tools were used for the derivation, proof, calculation, or logical reasoning of the mathematical results presented in this paper.

Contents

1. Introduction 2
2. Preliminaries 5
3. The MK tangent bundle to a measure 11
4. Stability of the MK distance 19
5. Relations with other tangent bundles 32

E-mail address: adolfo.rabasa@unipi.it, averil.aussedat@dm.unipi.it, bouchitte@univ-tln.fr.

Date: September 25, 2026.

6. Further properties of MK bundles and tangential calculus 40
References 46

1. Introduction

The study of calculus on singular or nonsmooth structures is a central theme at the intersection of real analysis, geometric measure theory, and partial differential equations. When working with a smooth manifold, the tangent bundle provides a local frame for defining gradient vector fields and integration by parts. However, when the underlying domain is represented by a general Radon measure μ in $\mathbb{R}^d$, establishing an intrinsic differential calculus becomes significantly more delicate. A fundamental difficulty in this setting is to extend classical differential operators, such as the gradient or divergence, to L^p_μ spaces. In standard Sobolev spaces, the underlying measure μ is the Lebesgue measure and integration by parts ensures that smooth derivations are closable, thus enabling the definition of weak derivatives. This is not the case for a singular measure μ, for which closability can only be restored by pointwise projecting the classical gradient onto a suitable tangent bundle associated with μ. This pointwise tangent space that cannot be recovered from Preiss' tangent measures: in general the tangent cone $\mathrm{Tan}(\mu, x)$ may contain non unique tangent planes or none at all, and still contain arbitrary sets of measures (see [45, 44]).

A natural alternative for constructing such a bundle comes from the calculus of variations: it consists in applying the *De Giorgi* relaxation procedure to integral functionals of the form $\int |\nabla u|^p \, d\mu$. In this manner a tangent bundle was introduced in [20] and further developed in a series of papers [19, 23, 22] (see also [33, 52] and the connection with Sobolev spaces in metric measure spaces in [38]). It is defined μ a.e., with a dimension depending on the local geometry of μ as well as on the exponent $p \in [1, \infty)$. Its good behavior with respect to duality and integration by parts has allowed to treat successfully various applications in applied physics, such as the homogenization of thin structures [25], dimension reduction, and optimal design problems in which the measure μ itself is an unknown minimizing a certain energy (see for example [15, 24, 26] and [13]).

Despite their generality and flexibility, these variational constructions have had little impact on geometric measure theory. In this paper, we show that the variant based on optimal transport yields a tangent bundle $T_\mu(x)$ which retains the variational and duality properties mentioned above and, in addition, is intimately connected with the theory of one-dimensional flat chains and with the differentiability of Lipschitz functions. First introduced in 2005 (see [21]), the bundle $T_\mu(x)$ ultimately agrees with the decomposability bundle $V_\mu(x)$ of Alberti and Marchese [2], who proved that V_μ is also the largest *differentiability bundle*: for any $u \in \mathrm{Lip}(\mathbb{R}^d)$, there exists a linear function $d_V u(x) : V_\mu(x) \to \mathbb{R}$ at μ-almost every point x for which

$$u(x+v) = u(x) + d_V u(x) v + o(|v|) \qquad \text{for } v \in V_\mu(x).$$

Construction of the bundle $T_\mu(x)$. Given a nonnegative Radon measure μ on $\mathbb{R}^d$, we begin by defining a subspace of vector fields $\sigma \in (L^1_\mu)^d$ which satisfy a suitable tangency condition. As will become clear, this condition encodes the first-order stability of the Monge-Kantorovich

distance between μ and its infinitesimal pushforward along the velocity field σ:

$$\mu_h^\sigma := (\mathrm{id} + h\sigma)_\sharp \mu.$$

In turn, the tangent velocity fields form a subspace $\mathcal{T}_\mu$ of $(L^1_\mu)^d$ characterized by a divergence condition: σ belongs to $\mathcal{T}_\mu$ if and only if, in the sense of distributions,

$$\mathrm{div}(\sigma\mu) \in \text{Æ}(\mathbb{R}^d). \tag{1.1}$$

Here $\text{Æ}(\mathbb{R}^d)$, the Arens-Eells space, denotes the completion, with respect to the Kantorovich-Rubinstein norm [37] (denoted $\|\cdot\|_{\mathrm{KR}}$), of the space of signed measures with finite first moment and zero average. This Banach space, known as the canonical pre-dual of $\mathrm{Lip}_0(\mathbb{R}^d)$, has been rediscovered as many times as there are viewpoints on it: introduced by Arens and Eells [7] in 1956, it reappears as a distinguished space of first-order distributions in the completion problem for the Kantorovich norm [35, 21, 16] (see its characterization in (2.5)), while in non linear Banach space geometry, it becomes the Lipschitz-free space [34, 50].

The interest of allowing the right hand-side of (1.1) to be a first-order distribution in $\text{Æ}(\mathbb{R}^d)$, instead of a measure, is that the resulting subspace $\mathcal{T}_\mu$ is closed in $(L^1_\mu)^d$ and enjoys the key locality property that, if $\sigma \in \mathcal{T}_\mu$, then $\theta\sigma \in \mathcal{T}_\mu$ for every scalar function $\theta \in L^\infty_\mu$. These two properties imply the existence of a μ-measurable map $T_\mu : \mathbb{R}^d \to \mathcal{G}_d$ (the Grassmannian of linear subspaces of $\mathbb{R}^d$) such that

$$\sigma \in \mathcal{T}_\mu \iff \sigma(x) \in T_\mu(x)\ \mu\text{ a.e.} \tag{1.2}$$

Summarizing the Monge-Kantorovich (MK) tangent bundle $T_\mu(x)$ is defined through the PDE condition (1.1), which allows the local property (1.2). An immediate consequence of this construction is that any Borel vector measure $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ can be decomposed as the sum $\lambda = \lambda_T + \lambda_N$ of its tangential part $\lambda_T = (P_\mu\sigma)\,\mu$ and of its normal part $\lambda_N = (P^\perp_\mu\sigma)\,\mu$, where $\mu = |\lambda|$, $\sigma = d\lambda/d|\lambda|$ and $P_\mu(x), P^\perp_\mu$ denote the orthogonal projections onto $T_\mu(x)$ and $T^\perp_\mu(x)$ respectively.

At this level, we can foresee an important connection with the geometric measure theory: the divergence condition (1.1) is equivalent to the fact that $\sigma\mu$ being a flat chain of finite mass in the classical sense of Whitney and Federer [31, 32]:

$$\mathcal{T}_\mu = \left\{ \sigma \in (L^1_\mu)^d : \sigma\mu \in \mathbf{F_1}(\mathbb{R}^d) \right\},$$

an equivalence which has been used recently to obtain a simple proof of the Ambrosio-Kirchheim Flat Chain Conjecture [6] in the context of one dimensional metric currents [9]. A consequence of the equality (1.2) is that the set of flat chains $\mathbf{F_1}(\mathbb{R}^d)$ forms a complemented subspace of $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$, a fact that we have not found stated in the literature (see Theorem 3.5). More precisely, it holds a direct topological sum

$$\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) = \mathbf{F_1}(\mathbb{R}^d) \oplus \ker \mathcal{P}_\mathbf{F},$$

where the linear projection map $\mathcal{P}_\mathbf{F} : \lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) \to \lambda_T \in \mathbf{F_1}(\mathbb{R}^d)$ is defined such that λ_T uniquely solves the minimal distance problem

$$\mathrm{dist}(\lambda, \mathbf{F_1}(\mathbb{R}^d)) = \inf \left\{ |\lambda - \eta|(\mathbb{R}^d) : \eta \in \mathbf{F_1}(\mathbb{R}^d) \right\}.$$

Two features of this statement go beyond the pointwise characterisation recalled above: it is metric, in the sense that λ_T is the *nearest* flat chain and that the distance is computed explicitly as $|\lambda_N|(\mathbb{R}^d)$; and it is linear, the map $\lambda \mapsto \lambda_T$ being a 1-Lipschitz projection.

Before presenting our main results, we point out that the MK-bundle T_μ defined above is local (see Subsection 3.1) and enjoys all the properties one would expect: decomposition of μ into pieces of constant dimension, chain and product rules, stability under Lipschitz push-forward, and the definition of a generalized varifold and of a mean curvature of μ as an element of $(\mathbf{Æ}(\mathbb{R}^d))^d$. Some of these properties already appear in [21] or are adapted from [17, 18]. They are summarized in Section 6, together with the equivalence $\mu \ll \mathcal{L}^d$ if and only if $T_\mu(x) = \mathbb{R}^d$, μ a.e. which we obtain by extending to flat chains the structure theorem of De Philippis and Rindler [29, 10].

Main results. Our main contributions are presented in Section 4. First, following the closability argument of [21], we recall the construction of the tangential gradient operator $u \in \mathrm{Lip}(\mathbb{R}^d) \to \nabla_\mu u \in (L^\infty_\mu(\mathbb{R}^d))^d$ which is a weak-star continuous operator and satisfies the integration by parts formula:

$$\langle -\operatorname{div} \sigma\, \mu, u\rangle \;=\; \int \langle \sigma, \nabla_\mu u\rangle \, d\mu \quad \forall (\sigma, u) \in \mathcal{T}_\mu \times \mathrm{Lip}(\mathbb{R}^d).$$

Then we show that, for every pair $(\sigma, u) \in \mathcal{T}_\mu \times \mathrm{Lip}(\mathbb{R}^d)$, we have:

(i) $\lim_{h\to 0^+} \|D^\sigma_h \mu + \operatorname{div}(\sigma\,\mu)\|_{\mathrm{KR}} = 0$, where we set $D^\sigma_h \mu := \frac{\mu^\sigma_h - \mu}{h}$;

(ii) $\langle \nabla_\mu u(x), \sigma(x)\rangle = \lim_{h\to 0} \frac{u(x+h\sigma(x)) - u(x)}{h}$ μ a.e.

Assertion (i) is a stability result: for $\sigma \in \mathcal{T}_\mu$, the difference quotients $D^\sigma_h \mu$ converge in $\mathbf{Æ}(\mathbb{R}^d)$, and their limit is the distributional limit $-\operatorname{div}(\sigma\mu)$. The assertion (ii) identifies the weak tangential gradient $\nabla_\mu u$ of any Lipschitz function $u \in \mathrm{Lip}(\mathbb{R}^d)$ as the pointwise directional derivative of u in the direction σ.

The two assertions above, contained in our main result (Theorem 4.1), are proved by starting with measures μ associated with Lipschitz curves, then by extending to pairs (μ, σ) with $\sigma \in \mathcal{T}_\mu$, by using the Smirnov decomposition of normal currents [49] combined with an approximation argument allowing to pass from $\sigma \in \mathcal{T}_\mu$ to $\sigma\mu$ being a normal current. Since this limit is linear in $\sigma(x)$ and $T_\mu(x)$ is already a linear subspace, we can lift (ii) to differentiability with respect to the whole subspace $T_\mu(x)$:

$$u(x+v) \;=\; u(x) + \langle \nabla_\mu u(x),\, v\rangle + o(|v|) \qquad \text{as } T_\mu(x) \ni v \to 0. \tag{1.3}$$

Next we improve upon the pointwise view of differentiability by testing the previous convergence on vector fields σ which belong to $(L^1_\mu)^d$ and do not belong to $\mathcal{T}_\mu$. The convergence stated in (i) extends as follows:

$$\lim_{h\to 0^+} F_h(\mu, \sigma) = 2 \int |\sigma - P_\mu \sigma|\, d\mu, \tag{1.4}$$

where $F_h(\mu, \sigma)$, the defect of uniform differentiability over C^1 1-Lipschitz maps, is given by

$$F_h(\mu, \sigma) \;:=\; \sup_{u \in \mathrm{Lip}_1 \cap C^1} \int \left(\frac{u(x + h\sigma(x)) - u(x)}{h} - \langle \sigma, \nabla u\rangle \right) d\mu(x).$$

Since this limit vanishes exactly when σ has no component in $\mathcal{N}_\mu$, the bundle $\mathcal{T}_\mu$ is precisely the set of velocity fields along which the Kantorovich-Rubinstein norm is stable – equivalently, along which the difference quotients $D^\sigma_{h_n}\mu$ form a Cauchy sequence in $\text{Æ}(\mathbb{R}^d)$ for every $h_n \to 0^+$. Corollary 4.6 quantifies the case when $\sigma \notin \mathcal{T}_\mu$: the difference quotients remain asymptotically at distance exactly $\int |\sigma_N|\, d\mu$ from $-\operatorname{div}(\sigma_T\mu)$. Formula (1.4) moreover quantifies the defect created by the non-tangential part of the velocity.

In Section 5 we compare T_μ with other bundles. We first show that T_μ coincides with the decomposability bundle V_μ of [2] (assertion (i) of Proposition 5.1); this proof is simple and measure-theoretic. The differentiability results described above provide a second, independent route to the same identification, based on optimal transport and duality. Exploiting the dual formulation of the normal bundle $\mathcal{N}_\mu = (\mathcal{T}_\mu)^\perp$ (see (3.8)), we provide a simple construction (Proposition 5.4) of a Lipschitz function whose directional derivatives fail to exist a.e. along any vector field which is not tangent to T_μ, thereby recovering the converse of Rademacher's Theorem obtained in [2]. Combined with (1.3), this identifies T_μ as the maximal differentiability bundle for Lipschitz maps (Theorem 5.5), independently of the results of [2]. We finally relate T_μ to the Preiss tangent measures of μ (Proposition 5.11): at μ-a.e. point x, every tangent measure is invariant under translations along $T_\mu(x)$, so that the MK bundle is always contained in the Preiss tangent plane of μ when the latter exists.

2. Preliminaries

Let (X, d_X) and (Y, d_Y) be metric spaces. The complement of a set $A \subset X$ is denoted A^c, and its indicator by $\mathbb{1}_A$, i.e. $\mathbb{1}_A(x) = 1$ if $x \in A$, and 0 otherwise. $C_0(X;Y)$ is the set of continuous functions from X to Y. The set of Lipschitz functions from X to Y is denoted $\operatorname{Lip}(X;Y)$, with $\operatorname{Lip}(u)$ being the Lipschitz constant of an element $u \in \operatorname{Lip}(X,Y)$. If X is a Banach space, we define the following subsets of $\operatorname{Lip}(X;Y)$:

$$\operatorname{Lip}_0(X;Y) := \{u \,:\, u(0) = 0\}\,, \qquad \operatorname{Lip}_1(X;Y) := \{u \,:\, \operatorname{Lip}(u) \leqslant 1\}\,.$$

The support of a function f is denoted $\operatorname{spt} f$. The set of compactly supported functions of class C^∞ is $\mathcal{D}(\mathbb{R}^d)$, with its dual $\mathcal{D}'$ being the set of distributions.

Let $\mathcal{B}(\mathbb{R}^d)$ be the Borel σ-algebra of $\mathbb{R}^d$. The Lebesgue measure is denoted $\mathcal{L}^d$, the Hausdorff measure of dimension α by $\mathcal{H}^\alpha$, and the Dirac mass at a point $x \in \mathbb{R}^d$ by δ_x. If $A \subset \mathbb{R}^d$ is measurable, the restriction of a measure μ to A is given by $\mu \llcorner A$, with $(\mu \llcorner A)(B) := \mu(A \cap B)$. Two measures are mutually singular, which we write as $\mu \perp \nu$, if there exists $A \in \mathcal{B}(\mathbb{R}^d)$ such that $\mu(A^c) = \nu(A) = 0$. At the opposite, the notation $\mu \ll \nu$ means that μ is absolutely continuous with respect to ν, i.e. $\nu(A) = 0$ implies $\mu(A) = 0$ for $A \in \mathcal{B}$.

The set $\mathcal{M}(\mathbb{R}^d)$ collects all Borel signed measures, with $\mathcal{M}_+(\mathbb{R}^d)$ being the subset of nonnegative measures. We also write $\mathcal{M}(\mathbb{R}^d)^k = \mathcal{M}(\mathbb{R}^d;\mathbb{R}^k)$ to denote the set of $\mathbb{R}^k$-valued measures with $\|\lambda\|_{\mathcal{M}(\mathbb{R}^d)^k} < +\infty$, where the total variation norm is defined as

$$\|\lambda\|_{\mathcal{M}(\mathbb{R}^d)^k} := \sup\Big\{ \langle \lambda, \varphi\rangle \,:\, \varphi \in C_0(\mathbb{R}^d;\mathbb{R}^k),\ |\varphi(x)| \leq 1 \Big\}\,.$$

Any $\lambda \in \mathcal{M}(\mathbb{R}^d)^k$ writes as $\lambda = \sigma|\lambda|$, where $|\lambda| \in \mathcal{M}_+(\mathbb{R}^d)$ is a nonnegative measure with $\|\lambda \llcorner A\|_{\mathcal{M}(\mathbb{R}^d)^k} = |\lambda|(A)$ for any $A \in \mathcal{B}(\mathbb{R}^d)$, and $\sigma \in L^1_{|\lambda|}(\mathbb{R}^d;\mathbb{R}^k)$ valued in the unit ball of $\mathbb{R}^k$. We refer to σ as the polar of λ.

2.1. **Convex analysis.** Let X be a normed space and let $h : X \to \mathbb{R} \cup \{+\infty\}$ be a convex function. Recall that the Moreau-Fenchel conjugate of h is defined on the dual space X^* by

$$h^*(x^*) := \sup_{x\in X} \big\{ \langle x, x^*\rangle - h(x)\big\} \qquad \forall\, x^* \in X^*.$$

Clearly, h^* is convex and lower semi-continuous with respect to the weak-$*$ topology on X^*. The existence of $x_0^* \in X^*$ such that $h^*(x_0^*) < +\infty$ (h^* is said to be *proper*) means that h admits an affine minorant. In that case, the supremum of all affine continuous minorants is given by the biconjugate of h defined on X by

$$h^{**}(x) := \sup_{x^*\in X^*} \big\{ \langle x, x^*\rangle - h^*(x^*)\big\} \qquad \forall\, x \in X.$$

Besides we define the lower semicontinuous envelope of h at $x \in X$ by

$$\overline{h}(x) = \inf_{x_n\to x} \liminf_n h(x_n).$$

We will use two classical results. The first one is a consequence of Hahn-Banach theorem and can be found in [30]; the second one is due to J.J. Moreau [41] in the infinite dimensional case, see also [14].

Proposition 2.1. *Assume that h^* is proper. Then $h^{**} = \overline{h}$.*

Proposition 2.2. *Assume that there exists $r > 0$ such that $\sup\{h(x) \,:\, \|x\| \le r\} < \infty$. Then:*

(i) h is continuous at 0, while h^* is coercive and attains its minimum on X^*;

(ii) we have the equalities: $h(0) = h^{**}(0) = -\min h^*$.

2.2. **Measurable multifunctions.** Fix a Radon measure $\mu \in \mathcal{M}_+(\mathbb{R}^d)$. To every family $\mathcal{F}$ of Borel functions from $\mathbb{R}^d$ to a separable Banach space Y, we may associate a closed valued and μ-measurable multifunction $\Gamma : \mathbb{R}^d \to 2^Y$ such that:

(i) $u(x) \in \Gamma(x)$ μ-a.e. for any $u \in \mathcal{F}$,

(ii) Any μ-measurable multifunction Γ' satisfying (i) is such that $\Gamma(x) \subset \Gamma'(x)$ μ a.e..

The existence of such Γ can be proved in an even more general context (see [27]) and it is unique up to the equality μ-a.e.. The usual notation used for Γ is $\mu - \text{ess } \bigcup\{u, u \in \mathcal{F}\}$. In the case where $\mathcal{F}$ is a subset of $L^1_\mu(\mathbb{R}^d, Y)$ (being Y for instance a finite dimensional normed space), a very simple characterization of $\Gamma = \mu - \text{ess } \bigcup\{u, u \in \mathcal{F}\}$ is obtained by considering a countable dense subset $\{u_n, n \in \mathbb{N}\}$ of $\mathcal{F}$ and then clearly $\Gamma(x) = \overline{\{u_n(x), n \in \mathbb{N}\}}$. Another straightforward observation is that:

– if $\overline{\mathcal{F}}$ denotes the closure of $\mathcal{F}$ in L^1_μ, then

$$\mu - \text{ess } \bigcup\{u;\ u \in \mathcal{F}\} = \mu - \text{ess } \bigcup\{u;\ u \in \overline{\mathcal{F}}\}.$$

– If $\mathcal{F}$ is a vector subspace, then so is $\mu - \text{ess } \bigcup\{u, u \in \mathcal{F}\}$.

We now particularize to $\mathcal{F}$ being a $\mathcal{D}$-stable vector subspace of $L^1_\mu(\mathbb{R}^d; Y)$ in the following sense:

$$u \in \mathcal{F}, \qquad \varphi \in \mathcal{D}(\mathbb{R}^d) \quad \Longrightarrow \quad \varphi\, u \in \mathcal{F}. \tag{2.1}$$

Proposition 2.3. *Let $\mathcal{F}$ be a $\mathcal{D}$-stable vector subspace of $L^1_\mu(\mathbb{R}^d; Y)$ and let $F_\mu := \mu - \text{ess } \bigcup\{u;\ u \in \mathcal{F}\}$ be the associate bundle. Then we have the equality*

$$\overline{\mathcal{F}} = \{u \in L^1_\mu(\mathbb{R}^d; Y) \,:\, u(x) \in F_\mu(x)\ \mu - \textit{a.e.}\}. \tag{2.2}$$

Note that, the inclusion $\subset$ holds true without assuming (2.1).

Proof. By approximation, the condition (2.1) implies that $u\,\mathbb{1}_A \in \overline{\mathcal{F}}$ whenever $u \in \overline{\mathcal{F}}$ and A is μ-measurable. The wished equality then follows from Hiai-Umegaki [36, Th. 3.1 p.158]. For the sake of completeness, we give a short proof of the non trivial inclusion $\mathcal{G} \subset \overline{\mathcal{F}}$ being $\mathcal{G}$ the right hand side of (2.2) assuming that Y is separable. Since $\mathcal{G}$ is closed, by passing to the orthogonal in $(L^\infty_\mu)^d$, the wished inclusion is equivalent to $\mathcal{F}^\perp \subset \mathcal{G}^\perp = \left\{ v \in (L^\infty_\mu)^d : v(x) \in F^\perp_\mu(x)\ \mu - \text{a.e.} \right\}$. Let $v \in \mathcal{F}^\perp$ and $\{u_n, n \in \mathbb{N}\}$ be a countable dense subset of $\mathcal{F}$ so that $F_\mu(x) = \overline{\{u_n(x), n \in \mathbb{N}\}}$. Then, for every n and any Borel subset A, we have that $u_n \mathbb{1}_A$ belongs to $\mathcal{F}$, hence $\int_A \langle v, u_n \rangle\, d\mu = 0$. By localizing, we infer that $\langle v, u_n \rangle = 0$ holds μ a.e, hence $v(x) \in F^\perp_\mu(x)$ as wished. □

2.3. **Monge-Kantorovich distance.** The Monge-Kantorovich distance between two probability measures μ, ν with finite first moment, i.e. $\int |x| d\mu < \infty$, is given by

$$W_1(\mu, \nu) := \inf_\gamma \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|\, d\gamma(x, y), \tag{2.3}$$

where the infimum is taken among all admissible transport plans γ such that $\pi^\sharp_1 \gamma = \mu$ and $\pi^\sharp_2 \gamma = \nu$. Here $\pi^\sharp_1$ and $\pi^\sharp_2$ are the usual push-forward operators associated to the projections π_1 and π_2 from $\mathbb{R}^d \times \mathbb{R}^d$ on the first and second factors respectively. By homogeneity the definition above extends naturally to finite measures $\mu, \nu \in \mathcal{M}_+(\mathbb{R}^d)$ such that $\int \mu = \int \nu$. The core of this theory is the Kantorovich-Rubinstein dual representation

$$W_1(\mu, \nu) = \sup \left\{ \int u\, d\nu - \int u\, d\mu \ :\ u \in \mathrm{Lip}_1(\mathbb{R}^d) \right\},$$

where $\mathrm{Lip}_1(\mathbb{R}^d)$ stands for the set of Lipschitz functions whose Lipschitz constant $\mathrm{Lip}(u)$ is smaller than 1. It is easy to check that the supremum above is achieved provided μ and ν have a finite first order moment and share the same mass. On the other hand, it is clear from the dual formulation that $W_1(\mu, \nu)$ only depends on the difference $f = \nu - \mu$. Accordingly we define the following vector subspace of signed measures on $\mathbb{R}^d$:

$$\mathcal{M}_{0,1}(\mathbb{R}^d)(\mathbb{R}^d) := \left\{ f \in \mathcal{M}(\mathbb{R}^d) \ :\ \int f = 0 \quad , \quad \int |x|\, d|f|(x) < +\infty \right\} \tag{2.4}$$

endowed with the Kantorovich-Rubinstein (KR) norm:

$$\|f\|_{\mathrm{KR}} \ = W_1(f_+, f_-) := \sup \left\{ \int u\, df \ :\ u \in \mathrm{Lip}_1(\mathbb{R}^d) \right\}.$$

In particular, $\delta_x - \delta_0 \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ for any $x \in \mathbb{R}^d$. In turn $\mathcal{M}_{0,1}(\mathbb{R}^d)(\mathbb{R}^d)$ is not complete as a normed space. Indeed there exists Cauchy sequences (f_n) which are not bounded in total variation, as for instance:

$$f_n = \sum_{j=1}^n (\delta_{b_j} - \delta_{a_j}) \quad \text{where} \quad \sum_j |b_j - a_j| < +\infty,$$

which converges in the sense of $\mathcal{D}'$ to a first order distribution f given by the sum of the series.

2.4. **Completion of $\mathcal{M}_{0,1}(\mathbb{R}^d)$.** Owing to its relation with the Arens-Eells (free Lipschitz fields) theory, in which more general metric spaces than $\mathbb{R}^d$ are considered, we will denote by $\text{Æ}(\mathbb{R}^d)$ the completion of $\mathcal{M}_{0,1}(\mathbb{R}^d)$ with respect to the KR norm. In our context, $\text{Æ}(\mathbb{R}^d)$ can be identified as a subset of distributions $f \in \mathcal{D}'_1(\mathbb{R}^d)$ of order ≤ 1 (see for instance [21]).

Definition 2.4. A distribution $f \in \mathcal{D}'(\mathbb{R}^d)$ belongs to $\text{Æ}(\mathbb{R}^d)$ if and only if, for every (φ_n) in $\mathcal{D}(\mathbb{R}^d)$, the following implication holds:

$$\varphi_n \to c \in \mathbb{R}, \quad \sup|\nabla \varphi_n| \leq C \implies \langle f, \varphi_n \rangle \to 0 \,. \tag{2.5}$$

Proposition 2.5. *(i) Every $f \in \text{Æ}(\mathbb{R}^d)$ can be extended in a unique way into a linear form on $\mathrm{Lip}(\mathbb{R}^d)$ (still denoted f) such that*

$$\langle f, u \rangle \leq C \,\mathrm{Lip}(u) \quad , \quad \langle f, 1 \rangle = 0 \,.$$

The best constant C is given by $C = \|f\|_{\mathrm{KR}}$ where we have set

$$\|f\|_{\mathrm{KR}} := \sup \left\{ \langle f, u \rangle \,:\, u \in \mathrm{Lip}_1(\mathbb{R}^d) \right\}.$$

(ii) For every $f \in \text{Æ}(\mathbb{R}^d)$, there exists a potential $u \in \mathrm{Lip}_1(\mathbb{R}^d)$ such that

$$\langle f, u \rangle = \|f\|_{\mathrm{KR}}.$$

(iii) Equipped with the KR-norm, $\text{Æ}(\mathbb{R}^d)$ is a Banach space which contains $\mathcal{M}_{0,1}(\mathbb{R}^d)$ as a dense subspace.

The proof of Proposition 2.5 can be found in [21, 16] where the analysis is restricted to measures compactly supported in a convex compact subset of $\mathbb{R}^d$, further generalized in metric spaces in [9]. Actually, in light of the Arens-Eells theory (see [7, 35, 43, 50]), it is well known that the dual of $\text{Æ}(\mathbb{R}^d)$ can be identified as

$$\text{Æ}(\mathbb{R}^d)^* \sim \mathrm{Lip}_0(\mathbb{R}^d) := \{u \in \mathrm{Lip}(\mathbb{R}^d) : u(0) = 0\}$$

by considering the canonical map $\Psi : u \in \mathrm{Lip}_0(\mathbb{R}^d) \mapsto \langle \cdot, u \rangle \in \text{Æ}(\mathbb{R}^d)^*$. This map is an isometry, i.e. $\|\Psi\|_{\text{Æ}(\mathbb{R}^d)^*} = \mathrm{Lip}(u)$, and its inverse is given by $\Psi^{-1} : f \in \text{Æ}(\mathbb{R}^d) \mapsto u$ where $u(x) = \langle f, \delta_x - \delta_0 \rangle$. As a consequence, the weak-star convergence of an equi-Lipschitz sequence in $\mathrm{Lip}(\mathbb{R}^d)$ is equivalent to its pointwise convergence.

Definition 2.6. A sequence (u_n) in $\mathrm{Lip}(\mathbb{R}^d)$ is said to be weakly star convergent to u if

$$\sup_n \mathrm{Lip}(u_n) < +\infty \quad \text{and} \quad u_n(x) \to u(x) \quad \text{for every } x \in \mathbb{R}^d.$$

In this case, we shall write shortly $u_n \overset{*}{\rightharpoonup} u$.

Note that the convergence above implies the uniform convergence of (u_n) on compact subsets of $\mathbb{R}^d$. Moreover, in virtue to Banach-Steinhaus theorem, it holds:

$$u_n \overset{*}{\rightharpoonup} u \iff \langle f, u_n \rangle \to \langle f, u \rangle \quad \text{for every } f \in \text{Æ}(\mathbb{R}^d). \tag{2.6}$$

Therefore every equi-Lipschitz sequence (u_n) in $\mathrm{Lip}_0(\mathbb{R}^d)$ admits a weak-star converging subsequence. For the reader's convenience, we give a short proof of the dense inclusion $\mathcal{M}_{0,1}(\mathbb{R}^d) \subset \text{Æ}(\mathbb{R}^d)$ stated in assertion (iii) of Proposition 2.5.

Proof. Let $f \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ and let us check that it satisfies (2.5). As f has integral 0, we can assume that $c = u_n(0) = 0$. Therefore $|u_n|(x) \le |x|$ and, since $u_n \to 0$ uniformly on compact subsets, we get that for every $R > 0$,

$$\limsup_{n\to\infty} |\langle f, u_n\rangle| \le \limsup_{n\to\infty} \int |u_n|\, d|f|(x) \le \int_{\{|x|\ge R\}} |x|\, d|f|(x).$$

The conclusion follows by letting $R \to \infty$ in virtue of the integrability condition in (2.4). The density is deduced from $\mathcal{Æ}(\mathbb{R}^d)^* \sim \mathrm{Lip}_0(\mathbb{R}^d)$: indeed, a linear form in $\mathcal{Æ}(\mathbb{R}^d)^*$ vanishing on $\mathcal{M}_{0,1}(\mathbb{R}^d)$ is represented by a Lipschitz function u with $u(0) = 0$, which satisfies in particular $\langle u, \delta_x - \delta_0\rangle = 0$ for any $x \in \mathbb{R}^d$. It follows that $u \equiv 0$, and $\mathcal{M}_{0,1}(\mathbb{R}^d)$ is dense in $\mathcal{Æ}(\mathbb{R}^d)$. □

Remark 2.7. In view of Definition 2.6, by using convolution by smooth kernels and truncations arguments, it is easy to check that $\mathcal{D}(\mathbb{R}^d)$ is weakly-star dense in $\mathrm{Lip}(\mathbb{R}^d)$. Accordingly, we have:

$$\|f\|_{\mathrm{KR}} = \sup\Big\{ \langle f, \varphi\rangle : \varphi \in \mathcal{D} \cap \mathrm{Lip}_1(\mathbb{R}^d) \Big\} \quad \text{for every } f \in \mathcal{Æ}(\mathbb{R}^d). \tag{2.7}$$

It is tempting to use this equality to extend the definition of the KR norm to all elements of the dual of $\mathrm{Lip}(\mathbb{R}^d)$. However, as a linear form $T \in (\mathrm{Lip}(\mathbb{R}^d))'$ is not uniquely determined by its restriction to smooth functions (the closure of $\mathcal{D}(\mathbb{R}^d)$ is a subspace of C^1), to be consistent, we would have to chose specific Hahn-Banach extensions T of elements $f \in \mathcal{Æ}(\mathbb{R}^d)$ (which are not unique).

2.5. **Transport measures versus currents.** A transport measure associated with a source term f is a vector measure $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ such that $-\operatorname{div}\lambda = f$ in the sense of distributions. We are interested in two cases: $f \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ or $f \in \mathcal{Æ}(\mathbb{R}^d)$. By identifying such a λ to a one dimensional currents with finite mass $\|\lambda\|$, we fall into the terminology of the Federer-Flemming theory of currents [31, 42] from which we borrow the notation.

Definition 2.8. We will denote

$$\mathbf{N_1}(\mathbb{R}^d) := \Big\{ \lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) : -\operatorname{div}\lambda \in \mathcal{M}_{0,1}(\mathbb{R}^d) \Big\} \quad \text{(normal 1-currents)}, \tag{2.8}$$

$$\mathbf{F_1}(\mathbb{R}^d) := \Big\{ \lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) : -\operatorname{div}\lambda \in \mathcal{Æ}(\mathbb{R}^d) \Big\} \quad \text{(flat chains of finite mass)}. \tag{2.9}$$

In the language of currents, $-\operatorname{div}\lambda$ represents the boundary ∂T of the 1-current T associated to λ by $\langle T, \varphi\rangle = \int \varphi\, d\lambda$. A key result is the following:

Proposition 2.9. *(i) The linear map $\lambda \in \mathbf{F_1}(\mathbb{R}^d) \subset \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) \mapsto -\operatorname{div}\lambda \in \mathcal{Æ}(\mathbb{R}^d)$ is surjective and satisfies:*

$$\|-\operatorname{div}\lambda\|_{\mathrm{KR}} \le \|\lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \qquad \textit{for every } \lambda \in \mathbf{F_1}(\mathbb{R}^d). \tag{2.10}$$

Moreover, for every $f \in \mathcal{Æ}(\mathbb{R}^d)$, we have the equality

$$\|f\|_{\mathrm{KR}} = \min\Big\{ \|\lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} : -\operatorname{div}\lambda = f,\ \lambda \in \mathbf{F_1}(\mathbb{R}^d) \Big\}. \tag{2.11}$$

(ii) $\mathbf{F_1}(\mathbb{R}^d)$ is the closure of $\mathbf{N_1}(\mathbb{R}^d)$ in the Banach space $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$;

(iii) $\mathbf{F_1}(\mathbb{R}^d)$ has the following local property:

$$\lambda \in \mathbf{F_1}(\mathbb{R}^d) \implies \theta\,\lambda \in \mathbf{F_1}(\mathbb{R}^d) \quad \textit{for any } \theta \in L^1_{|\lambda|}(\mathbb{R}^d). \tag{2.12}$$

The first assertion already appeared in [21]. The density property stated in assertion (ii) plays a central role in the proof of the Flat Chain Conjecture in general 1 metric currents (see for instance [9]).

Proof. (i) By (2.7) applied to $f = -\operatorname{div}\lambda$, recalling that $\langle f, \varphi\rangle = \langle \lambda, \nabla\varphi\rangle$,

$$\| -\operatorname{div}\lambda\|_{\mathrm{KR}} = \sup\Big\{ \langle \lambda, \nabla\varphi\rangle : \varphi \in \mathcal{D}(\mathbb{R}^d), |\nabla\varphi| \le 1 \Big\} \le \|\lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)}.$$

If $f \in \mathcal{M}_{0,1}(\mathbb{R}^d)$, the equality (2.11) is the classical Beckmann dual representation of the Monge-Kantorovich distance $W_1(f_+, f_-)$. The existence of an admissible competitor λ and the fact that the infimum is attained is ensured once $W_1(f_+, f_-) = \|f\|_{\mathrm{KR}} < +\infty$. This is always the case for $f \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ since, by the triangular inequality, $W_1(f_+, f_-) \le \int |x||f|(dx)$. To extend (2.11) to general source terms $f \in \boldsymbol{\mathcal{E}}(\mathbb{R}^d)$, consider an approximating sequence (f_n) in $\mathcal{M}_{0,1}(\mathbb{R}^d)$ such that $f_n \to f$ in $\boldsymbol{\mathcal{E}}(\mathbb{R}^d)$. Then there exists λ_n a minimizer of (2.11) for f_n, such that $\int |\lambda_n| = \|f_n\|_{\mathrm{KR}}$ and $-\operatorname{div}\lambda_n = f_n$. Then the sequence (λ_n) is bounded in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ and admits at least one weak-star cluster point λ, which satisfies:

$$\int |\lambda| \le \|f\|_{\mathrm{KR}} \quad , \quad -\operatorname{div}\lambda = f.$$

Here we used the lower semicontinuity of the total variation in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ with respect to the weak-star convergence (which implies the convergence in the sense of distributions) and also the fact that $\|f_n - f\|_{\mathrm{KR}} \to 0$ implies that $f_n \to f$ in $\mathcal{D}'(\mathbb{R}^d)$. Then, by applying (2.10), we deduce that $\int |\lambda| = \|f\|_{\mathrm{KR}}$, whence the optimality of λ in (2.11).

(ii) We prove that $\mathbf{F_1}(\mathbb{R}^d)$ is closed in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$. Let (λ_n) be a sequence in $\mathbf{F_1}(\mathbb{R}^d)$ and $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ such that $\|\lambda_n - \lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \to 0$. Then, from (2.10), we deduce that $f_n := -\operatorname{div}\lambda_n$ is a Cauchy sequence in $\boldsymbol{\mathcal{E}}(\mathbb{R}^d)$; hence there exists $f \in \boldsymbol{\mathcal{E}}(\mathbb{R}^d)$ such that $\|f_n - f\|_{\mathrm{KR}} \to 0$ and passing to the limit in $\mathcal{D}'(\mathbb{R}^d)$, we get $f = -\operatorname{div}\lambda$ thus $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$.

Next we consider an element $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ and prove the existence of a sequence (λ_n) in $\mathbf{N_1}(\mathbb{R}^d)$ such that $\|\lambda_n - \lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \to 0$. Denoting $f := -\operatorname{div}\lambda$ (then $f \in \boldsymbol{\mathcal{E}}(\mathbb{R}^d)$), we observe that it is enough to find a sequence $(\overline{\lambda}_n)$ in $\mathbf{N_1}(\mathbb{R}^d)$ such that $\|\overline{\lambda}_n - \overline{\lambda}\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)}$ where $\overline{\lambda}$ is any element $\overline{\lambda} \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ such that $-\operatorname{div}\overline{\lambda} = f$. Indeed, if it is the case, then $\lambda - \overline{\lambda}$ is divergence free and therefore belongs to $\mathbf{N_1}(\mathbb{R}^d)$. Then the sequence $\lambda_n = \overline{\lambda_n} + (\lambda - \overline{\lambda})$ will satisfy the desired requirement.

Construction of $\overline{\lambda}_n \to \overline{\lambda}$: Since $f \in \boldsymbol{\mathcal{E}}(\mathbb{R}^d)$, there exists a sequence (f_n) in $\mathcal{M}_{0,1}(\mathbb{R}^d)$ such that $\|f_n - f\|_{\mathrm{KR}} \to 0$. Possibly after extracting a subsequence, we can assume that $\varepsilon_n := \|f_{n+1} - f_n\|_{\mathrm{KR}}$ satisfies $\sum_1^\infty \varepsilon_n < +\infty$. Setting $f_0 = 0$ and $g_k := f_{k+1} - f_k$ we have $f_n = \sum_{k=0}^n g_k$ and we can express f as the sum of the absolutely convergent series:

$$f = \sum_{k=0}^{\infty} g_k \quad , \quad g_k \in \mathcal{M}_{0,1}(\mathbb{R}^d) \quad , \quad \|g_k\|_{\mathrm{KR}} = \varepsilon_k.$$

Then letting $\chi_k \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ be optimal in (2.11) for g_k, the equality $\|\chi_k\| = \varepsilon_k$ allows to construct a series converging absolutely in total variation. We set:

$$\overline{\lambda}_n := \sum_{k=0}^{n-1} \chi_k \quad , \quad \overline{\lambda} := \sum_{k=0}^{\infty} \chi_k.$$

Since the convergences $\overline{\lambda}_n \to \overline{\lambda}$ in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ and $f_n \to f$ in $\text{Æ}(\mathbb{R}^d)$ imply the convergence in the sense of distributions, by passing to the limit in the equation $-\operatorname{div}\overline{\lambda}_n = f_n$, we deduce that $\overline{\lambda}$ satisfies $-\operatorname{div}\overline{\lambda} = f$ in $\mathcal{D}'(\mathbb{R}^d)$.

(iii) Let $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ and consider a smooth scalar function $\theta \in \mathcal{D}(\mathbb{R}^d)$. Then, since for every $u \in \mathcal{D}(\mathbb{R}^d)$, we have $\langle -\operatorname{div}(\theta\,\lambda), u\rangle = \langle \lambda, \nabla(\theta u) - u\nabla\theta\rangle$, the criterion (2.5) under which $\operatorname{div}(\theta\lambda)$ belongs to $\text{Æ}(\mathbb{R}^d)$ is clearly satisfied. Accordingly we have $\theta\,\lambda \in \mathbf{F_1}(\mathbb{R}^d)$. Given now $\theta \in L^1_{|\lambda|}(\mathbb{R}^d)$, we can take a sequence of smooth functions (θ_n) such that $\theta_n \to \theta$ in $L^1_{|\lambda|}(\mathbb{R}^d)$. Then $\lambda_n = \theta_n\,\lambda$ converges to $\theta\,\lambda$ in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$. Hence we can conclude that $\theta\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ since, by the assertion (ii), we know that $\mathbf{F_1}(\mathbb{R}^d)$ is a closed subspace of $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$. □

Remark 2.10. The locality property of assertion (iii) turns out to be crucial, although it a straightforward consequence of (ii). We stress that the implication (2.12) is false if we replace $\mathbf{F_1}(\mathbb{R}^d)$ by $\mathbf{N_1}(\mathbb{R}^d)$. Indeed, $\operatorname{div}(\theta\lambda)$ may not have finite total variation if the scalar function θ is discontinuous (take for instance $\theta = \mathbb{1}_B$ where B has infinite perimeter).

Remark 2.11. In [21], $\mathbf{F_1}(\mathbb{R}^d)$ was introduced as the set of *tangential measures*. The reason behind this name is that in smooth cases, such measures λ are tangential to the geometric support of $|\lambda|$. The forthcoming construction of the MK tangent bundle will result in a precise formulation of this property.

3. The MK tangent bundle to a measure

3.1. **Construction and locality.** Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure. We consider the following subspace of $(L^1_\mu(\mathbb{R}^d))^d$:

$$\mathcal{T}_\mu := \Big\{\sigma \in (L^1_\mu(\mathbb{R}^d))^d \ : \ -\operatorname{div}(\sigma\mu) \in \text{Æ}(\mathbb{R}^d)\Big\} \tag{3.1}$$

By (2.9), $\mathcal{T}_\mu$ coincides with the subspace $\big\{\,\sigma \in (L^1_\mu(\mathbb{R}^d))^d : \sigma\,\mu \in \mathbf{F_1}(\mathbb{R}^d)\,\big\}$. Therefore it inherits the properties of $\mathbf{F_1}(\mathbb{R}^d)$ given in the assertions (ii) and (iii) of Proposition 2.9, namely:

- $\mathcal{T}_\mu$ is a closed subspace of $(L^1_\mu(\mathbb{R}^d))^d$;
- If $\sigma \in \mathcal{T}_\mu$, we have $\sigma\,\theta \in \mathcal{T}_\mu$ for every $\theta \in L^\infty_\mu$.

It follows that $\mathcal{T}_\mu$ is determined by a local multivalued application T_μ.

Proposition 3.1. *There exists a μ-measurable multivalued application $T_\mu : \mathbb{R}^d \rightrightarrows \mathbb{R}^d$ such that $T_\mu(x)$ is a vector subspace of $\mathbb{R}^d$, possibly reduced to $\{0\}$, and*

$$\sigma \in \mathcal{T}_\mu \quad \Longleftrightarrow \quad \sigma(x) \in T_\mu(x) \quad \mu\text{-a.e.} \qquad (\Longleftrightarrow\ \sigma\,\mu \in \mathbf{F_1}(\mathbb{R}^d)). \tag{3.2}$$

Proof. As in Proposition 2.3, the existence of $T_\mu(\cdot)$ follows from [36, Th. 3.1 p.158]. □

Definition 3.2 (MK-tangent bundle)**.** The multivalued application T_μ characterized in (3.1) is the MK-tangent bundle to the measure μ. Its orthogonal bundle is denoted

$$N_\mu(x) \coloneqq (T_\mu(x))^\perp.$$

A main interest of the construction of $\mathcal{T}_\mu$ lies in the following property.

Lemma 3.3 (Locality)**.** *Let μ, ν be two Radon measures and $B \in \mathcal{B}(\mathbb{R}^d)$. Then*

$$\mu \llcorner B = \nu \llcorner B \implies T_\mu(x) = T_\nu(x) \quad \textit{a.e. on } B. \tag{3.3}$$

Proof. Denoting $\mu_B = \mu \llcorner B$, the implication above is clear if we can show that $T_\mu(x) = T_{\mu_B}(x)$ holds μ a.e. in B. Let $\sigma \in \mathcal{T}_{\mu_B}$, then $\sigma\mu_B = \widetilde{\sigma}\mu$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$, where $\widetilde{\sigma}$ is the extension of σ by 0 outside of B. Therefore $\widetilde{\sigma}(x) \in \mathcal{T}_\mu$. Accordingly we have $T_{\mu_B}(x) \subset T_\mu(x)$ for μ-a.e. $x \in B$. In the opposite direction, if $\sigma \in \mathcal{T}_\mu$, then $\sigma\, \mathbb{1}_B\mu = \sigma\, \mu_B$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$ by the locality property (2.3). It follows that $\sigma \in T_{\mu_B}$, whence $\sigma(x) \in T_{\mu_B}(x)$ for μ-a.e. $x \in B$. □

As a corollary of the property (3.3), we get the following.

Lemma 3.4. *The following implications hold:*

$$\mu = \mu_1 + \mu_2 \ \textit{and}\ \mu_1 \perp \mu_2 \implies T_\mu(x) = \begin{cases} T_{\mu_1}(x) & \mu_1 \textit{ a.e.}, \\ T_{\mu_2}(x) & \mu_2 \textit{ a.e.}. \end{cases} \tag{3.4}$$

$$\nu \ll \mu \implies T_\nu(x) = T_\mu(x) \quad \nu \textit{ a.e.} \tag{3.5}$$

Proof. (3.4) follows from (3.3) by considering a Borel subset B such that $\mu \llcorner B = \mu_1$ and $\mu \llcorner B^c = \mu_2$. To show (3.5), we write $\nu = \theta\mu$ where θ is the Radon-Nikodym derivative of ν with respect to μ. Then $\sigma \in \mathcal{T}_\nu$ implies that $-\operatorname{div}(\sigma\theta\mu) \in \text{Æ}(\mathbb{R}^d)$, hence $\sigma(x) \in T_\mu(x)$ holds μ a.e. on $\{\theta > 0\}$. This means that $T_\nu(x) \subset T_\mu(x)$ holds ν almost everywhere. In the opposite direction, assume that σ belongs to $\mathcal{T}_\mu$. Then, since $\sigma\,\mu \in \mathbf{F_1}(\mathbb{R}^d)$, we also have that $\sigma\,\nu = \theta\sigma\mu \in \mathbf{F_1}(\mathbb{R}^d)$ by the locality property (2.12). Therefore $\sigma \in \mathcal{T}_\nu$ and $\sigma(x) \in T_\nu(x)$ holds ν a.e.. □

3.2. **Projection theorem and orthogonal bundle.** The next result extends estimates obtained in [16] allowing us to show that $\mathbf{F_1}(\mathbb{R}^d)$ is a complemented space of $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$. To the best of our knowledge, this seems to be a new result. We refer to [40, Section 3] for a similar idea of decomposition to study purely non-flat currents.

Theorem 3.5. *Let $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$. Then there exists a unique solution $\lambda_T \in \mathbf{F_1}(\mathbb{R}^d)$ to the minimal distance problem*

$$\|\lambda - \lambda_T\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \;=\; \inf\Big\{ \|\eta - \lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} : \eta \in \mathbf{F_1}(\mathbb{R}^d) \Big\} \;=: \operatorname{dist}(\lambda, \mathbf{F_1}(\mathbb{R}^d)).$$

Moreover the projection map $\mathcal{P}_\mathbf{F} : \lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) \mapsto \lambda_T \in \mathbf{F_1}(\mathbb{R}^d)$ is linear and 1-Lipschitz. Accordingly, we have the topological direct sum

$$\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) \;=\; \mathbf{F_1}(\mathbb{R}^d) \oplus \mathbf{F_1^\perp}(\mathbb{R}^d)\,,$$

where $\mathbf{F_1^\perp}(\mathbb{R}^d) := \ker \mathcal{P}_\mathbf{F} = \Big\{ \lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) : \frac{d\lambda}{d|\lambda|} \in N_{|\lambda|} \ \ |\lambda| - a.e. \Big\}.$

Proof. Let $\lambda = \sigma\mu$ be the polar decomposition of λ, with $|\sigma| = 1$ μ a.e.. Denoting P_μ the orthogonal projection on T_μ and $P_\mu^\perp$ the orthogonal projection on $T_\mu^\perp$, we can decompose σ as $\sigma = \sigma_T + \sigma_N$, where

$$\sigma_T(x) := P_\mu(x)(\sigma(x)), \qquad \sigma_N(x) := P_\mu^\perp(x)(\sigma(x)).$$

By (3.2), $\lambda_T := \sigma_T\,\mu$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$ and therefore:

$$d(\lambda, \mathbf{F_1}(\mathbb{R}^d)) \;\le\; \|\lambda - \lambda_T\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \;=\; \int |\sigma_N|\, d\mu.$$

On the other hand, any $\eta \in \mathbf{F_1}(\mathbb{R}^d)$ admits a Lebesgue-Nikodym decomposition $\eta = w\,\mu + \eta_s$ where $|\eta_s| \perp \mu$. Using (3.4), we infer that $w(x) \in T_\mu(x)$ at μ a.e. $x \in \mathbb{R}^d$. It follows that

$$\|\lambda - \eta\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \;=\; \int \sqrt{|\sigma_T - w|^2 + |\sigma_N|^2}\, d\mu + \int |\eta_s| \;\ge\; \int |\sigma_N|\, d\mu.$$

Hence the minimal distance of λ to $\mathbf{F_1}(\mathbb{R}^d)$ is reached at $\lambda_T = \eta$ if and only if $w = \sigma_T$ and $\eta^s = 0$, that is $\lambda_T = P_\mu(\sigma)\mu$.

Next, for every $\eta = \xi\, m$ with $m \in \mathcal{M}_+(\mathbb{R}^d)$ and $\xi \in (L^1_m)^d$, we set

$$\mathcal{P}_{\mathbf{F}}(\eta) := P_m(\xi)\, m. \tag{3.6}$$

Thanks to the homogeneity and local property (see (3.5)) of the MK tangent bundle, it is easy to check that the latter definition does not depend on the decomposition used for η. In particular, using the polar decomposition of a given $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ as before, we obtain the relation $\lambda_T = \mathcal{P}_{\mathbf{F}}(\lambda)$. Now, let us demonstrate that $\mathcal{P}_{\mathbf{F}}$ is a linear operator. The homogeneity follows directly from (3.6). To check the additivity, we consider arbitrary $\lambda_1, \lambda_2 \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ and set $m := |\lambda_1| + |\lambda_2|$. Then λ_1, λ_2 and $\lambda_1 + \lambda_2$ are absolutely continuous with respect to m. Using (3.6) and exploiting the linearity of the local projection P_m, we find that

$$\begin{aligned}\mathcal{P}_{\mathbf{F}}(\lambda_1 + \lambda_2) &= P_m\left(\frac{d\lambda_1}{dm} + \frac{d\lambda_2}{dm}\right) m \\ &= P_m\left(\frac{d\lambda_1}{dm}\right) m + P_m\left(\frac{d\lambda_2}{dm}\right) m = \mathcal{P}_{\mathbf{F}}(\lambda_1) + \mathcal{P}_{\mathbf{F}}(\lambda_2).\end{aligned}$$

Finally, the inequality $\|\mathcal{P}_{\mathbf{F}}(\lambda)\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \le \|\lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)}$ entails that $\mathcal{P}_{\mathbf{F}}$ is 1-Lipschitz. □

Example 3.6. Let $\lambda = \sigma\,\mu$ in $\mathbb{R}^2$, where $\mu = \mathcal{H}^1 \llcorner A$ with $A := [0,1] \times \{0\}$, and $\sigma(x,0) = (h(x), k(x))$. Assuming that h, k are smooth functions on $[0,1]$, it is easy to check that $f = -\operatorname{div}\lambda$ is given by

$$f = h(1)\delta_{(1,0)} - h(0)\delta_{(0,0)} - h'(x)\mathcal{H}^1 \llcorner A - \operatorname{div}\left((0, k(x))\mathcal{H}^1 \llcorner A\right).$$

Accordingly $\lambda \in \mathbf{N_1}(\mathbb{R}^2)$ if and only if $k \equiv 0$. Indeed, by testing with $\varphi_n(x,y) = k(x)\theta(ny)$ where $\theta \in \mathcal{D}(\mathbb{R}, [-1,1])$ is such that $\theta(0) = 0$ and $\theta'(0) = 1$, we infer that the total variation of f is minorized by $n\int_0^1 k^2(x)\,dx$, and therefore blows up to infinity as $n \to \infty$ unless k vanishes. Therefore, the projection of λ onto $\mathbf{F_1}(\mathbb{R}^2)$ is given by $\lambda_T = (h(x), 0)\mu$, achieving the distance $\int |\lambda_N| = \int_0^1 |k(x)|\,dx$. Eventually the reader can check easily that, for general $(h,k) \in (L^1_\mu)^2$, we have $\lambda \in \mathbf{F_1}(\mathbb{R}^2)$ if and only if $k = 0$ holds μ-a.e., while requiring that $\lambda \in \mathbf{N_1}$ implies the additional conditions that h and k belong to $\mathrm{BV}(0,1)$.

A consequence of Theorem 3.5 is that a vector field $\sigma \in (L^1_\mu)^d$ satisfies $\sigma(x) \in T^\perp_\mu(x)$ μ a.e if and only if $\sigma\,\mu \in \mathbf{F}^\perp_{\mathbf{1}}(\mathbb{R}^d)$, or equivalently if $\mathrm{dist}(\sigma\,\mu, \mathbf{F}_{\mathbf{1}}(\mathbb{R}^d)) = \int |\sigma|\, d\mu$. Actually it is possible to give a direct dual characterization of such elements σ that are in $(L^\infty_\mu)^d$, that is of the subspace

$$\mathcal{N}_\mu := \mathcal{T}^\perp_\mu = \Big\{ \sigma \in (L^\infty_\mu)^d : \sigma(x) \in T^\perp_\mu(x)\ \mu \text{ a.e.} \Big\}, \tag{3.7}$$

where the second identity is a direct consequence of Proposition 3.1.

As we will show in Proposition 3.8 below, one has

$$\sigma \in \mathcal{N}_\mu \quad \Longleftrightarrow \quad \exists u_n \in \mathcal{D}(\mathbb{R}^d) \ : \ u_n \overset{*}{\rightharpoonup} 0 \ \text{ in } \mathrm{Lip}(\mathbb{R}^d)\ ,\ \nabla u_n \to \sigma \ \ \mu \text{ a.e..} \tag{3.8}$$

This characterization follows from the computation of the Moreau-Fenchel conjugate of the convex functional $G_\sigma : \mathrm{Lip}(\mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ defined by

$$G_\sigma(u) := \begin{cases} \int \langle \sigma, \nabla u \rangle \, d\mu & \text{if } u \in (\mathcal{D} \cap \mathrm{Lip}_1)(\mathbb{R}^d), \\ +\infty & \text{otherwise.} \end{cases} \tag{3.9}$$

Note that G_σ is invariant by addition of a constant, so that it can be identified with a functional on $\mathrm{Lip}_0(\mathbb{R}^d)$.

Lemma 3.7. *Let $\sigma \in (L^1_\mu)^d$. Then, for every $g \in \text{Æ}(\mathbb{R}^d)$, the conjugate of G_σ in the duality between $\mathrm{Lip}_0(\mathbb{R}^d)$ and $\text{Æ}(\mathbb{R}^d)$ is given by*

$$G^*_\sigma(g) = \min_{\lambda \in \mathcal{M}^d} \left\{ \int_{\mathbb{R}^d} |\lambda - \sigma\,\mu| \ : \ -\mathrm{div}\,\lambda = g \right\}. \tag{3.10}$$

Proof. We consider the perturbation functional $h : (C_0(\mathbb{R}^d))^d \to \mathbb{R}$ defined by:

$$h(p) := \inf \left\{ \int \langle \sigma, \nabla u + p \rangle \, d\mu - \langle g, u \rangle : u \in \mathcal{D}(\mathbb{R}^d),\ |\nabla u + p| \le 1 \right\}.$$

One checks that

(i) $h(0) = -G^*_\sigma(g)$;
(ii) h is convex as it can be written as $h(p) = \inf \big\{ H(u,p) : u \in \mathrm{Lip}(\mathbb{R}^d) \big\}$ where H is convex in both variables on $\mathrm{Lip}(\mathbb{R}^d) \times (C_0(\mathbb{R}^d))^d$ (see [30, Lemma 2.1 p.50]);
(iii) $h(p) \le \int \langle \sigma, p \rangle \, d\mu$ if $|p| \le 1$ (simply taking $u = 0$ as a competitor).

Therefore h is continuous at $p = 0$ and, by Proposition 2.2, its Fenchel conjugate $h^* : \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ is coercive, achieves its minimum and satisfies the equalities:

$$h(0) = h^{**}(0) = -\min \Big\{ h^*(\lambda) : \lambda \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d) \Big\}. \tag{3.11}$$

We deduce that

$$\begin{aligned}
h^*(\lambda) &= \sup\left\{ \langle \lambda, p\rangle - h(p) : p \in (C_0(\mathbb{R}^d))^d \right\} \\
&= \sup\left\{ \langle \lambda, p\rangle + \langle g, u\rangle - \int \langle \sigma, \nabla u + p\rangle \, d\mu : (u,p) \in \mathcal{D} \times (C_0)^d,\ |\nabla u + p| \le 1 \right\} \\
&= \sup\left\{ \langle \lambda - \sigma\,\mu, q\rangle - \langle \lambda, \nabla u\rangle + \langle g, u\rangle : (u,q) \in \mathcal{D} \times (C_0)^d,\ |q| \le 1 \right\} \\
&= \sup\left\{ \langle \lambda - \sigma\,\mu, q\rangle : q \in (C_0)^d, |q| \le 1 \right\} + \sup\left\{ -\langle \lambda, \nabla u\rangle + \langle g, u\rangle : u \in \mathcal{D} \right\} \\
&= \begin{cases} \int |\lambda - \sigma\mu| & \text{if } -\operatorname{div}\lambda = g, \\ +\infty & \text{otherwise.} \end{cases}
\end{aligned}$$

Note that, by Proposition 2.9, the domain of h^* is not empty since the map $\lambda \in \mathbf{F_1}(\mathbb{R}^d) \to -\operatorname{div}\lambda \in \mathcal{Æ}(\mathbb{R}^d)$ is surjective (in particular, we have $h^*(\lambda) \le \|\lambda\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} + \|\sigma\|_{L^1_\mu}$ for every $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$). The equality (3.10) and the existence of a minimal λ then follow from (3.11). □

Proposition 3.8. *Let $\sigma \in (L^1_\mu)^d$ and denote $(\sigma_T, \sigma_N) := (P_\mu\sigma, P^\perp_\mu\sigma)$. Then there exists a sequence $(u_n) \subset \mathcal{D} \cap \operatorname{Lip}_1(\mathbb{R}^d)$ such that:*

$$u_n \overset{*}{\rightharpoonup} 0 \ \text{ in } \operatorname{Lip}(\mathbb{R}^d)\,, \qquad \nabla u_n \to \frac{\sigma_N}{|\sigma_N|}\,\mathbb{1}_{\{\sigma_N \neq 0\}} \quad \mu \text{ a.e.} \tag{3.12}$$

As a consequence, the equivalence (3.8) *holds and also the equality:*

$$\mathcal{N}_\mu = \left\{ \sigma \in (L^\infty_\mu)^d : \exists u_n \in \mathcal{D}(\mathbb{R}^d) : u_n \overset{*}{\rightharpoonup} 0 \text{ in } \operatorname{Lip}(\mathbb{R}^d),\ \nabla u_n \overset{*}{\rightharpoonup} \sigma \text{ in } (L^\infty_\mu)^d \right\} \tag{3.13}$$

Remark 3.9. In [21], the construction of the tangent bundle T_μ was obtained by orthogonality, starting from the condition (3.8) to define the normal space $\mathcal{N}_\mu$ and proving that $\mathcal{N}^\perp_\mu$ satisfies the conditions of Proposition 2.3.

Proof. By Proposition 2.1, since G_σ is convex and proper on $\operatorname{Lip}(\mathbb{R}^d)$, its lower semicontinuous envelope with respect to the weak-star topology given by

$$\overline{G}_\sigma(v) := \inf_{v_n \overset{*}{\rightharpoonup} v} \liminf_n G_\sigma(v_n) \tag{3.14}$$

coincides with the biconjugate G^{**}_σ, namely:

$$\overline{G}_\sigma(v) = G^{**}_\sigma(v) := \sup\left\{ \langle g, v\rangle - G^*_\sigma(g) : g \in \mathcal{Æ}(\mathbb{R}^d) \right\}.$$

In particular, by taking $v = 0$, we obtain

$$-\overline{G}_\sigma(0) = \inf\left\{ G^*_\sigma(g) : g \in \mathcal{Æ}(\mathbb{R}^d) \right\} = d(\sigma\mu, \mathbf{F_1}(\mathbb{R}^d)),$$

where the last equality follows from (3.10). In view of Theorem 3.5, we know that $d(\sigma\mu, \mathbf{F_1}(\mathbb{R}^d)) = \int |\sigma_N|\, d\mu$. Therefore, by applying (3.14) to $v = 0$, we obtain a sequence (v_n) in $\mathcal{D} \cap \operatorname{Lip}_1$ such that $v_n \overset{*}{\rightharpoonup} 0$ and

$$\int \langle \sigma, \nabla v_n\rangle \, d\mu = G_\sigma(v_n) \underset{n\to\infty}{\longrightarrow} \overline{G}_\sigma(0) = -\int |\sigma_N|\, d\mu.$$

Setting $u_n := -v_n$, we claim that the sequence (u_n) satisfies (3.12). Indeed, since $-\operatorname{div}(\sigma_T\mu)$ belongs to $\mathbf{Æ}(\mathbb{R}^d)$ and $u_n \overset{*}{\rightharpoonup} 0$, (2.5) implies that $\int \langle \sigma_T, \nabla u_n\rangle \, d\mu \to 0$. It follows that $|\sigma_N| - \langle \sigma_N, \nabla u_n\rangle$, which is non negative, converges strongly to 0 in L^1_μ. Possibly after extracting a subsequence, we can then assume that $\langle \sigma_N, \nabla u_n\rangle \to |\sigma_N|$ at μ a.e. point. Since $|\nabla u_n| \leq 1$, this entails that $\nabla u_n \to \frac{\sigma_N}{|\sigma_N|}\mathbb{1}_{\{\sigma_N \neq 0\}}$ as claimed in (3.12).

We now show the equality (3.13) and the equivalence (3.8). Let us denote

$$\mathcal{A} := \Big\{ \sigma \in (L^\infty_\mu)^d : \exists u_n \in \mathcal{D}(\mathbb{R}^d) : u_n \overset{*}{\rightharpoonup} 0 \ \text{ in } \mathrm{Lip}(\mathbb{R}^d),\ \nabla u_n \to \sigma \ \ \mu \text{ a.e.} \Big\}$$
$$\widetilde{\mathcal{A}} := \Big\{ \sigma \in (L^\infty_\mu)^d : \exists u_n \in \mathcal{D}(\mathbb{R}^d) : u_n \overset{*}{\rightharpoonup} 0 \ \text{ in } \mathrm{Lip}(\mathbb{R}^d),\ \nabla u_n \overset{*}{\rightharpoonup} \sigma \ \text{ in } (L^\infty_\mu)^d \Big\}.$$

Clearly $\mathcal{A} \subset \widetilde{\mathcal{A}}$. On the other hand, one checks that $\widetilde{\mathcal{A}} \subset \mathcal{N}_\mu$. Indeed let (u_n) be a sequence in $\mathcal{D}(\mathbb{R}^d)$ such that $u_n \overset{*}{\rightharpoonup} 0$ and $\nabla u_n \overset{*}{\rightharpoonup} \sigma$ in $(L^\infty_\mu)^d$. Then, $-\operatorname{div}(\sigma_T\mu)$ belongs to $\mathbf{Æ}(\mathbb{R}^d)$, by applying (2.6), we get $\int \langle \nabla u_n, \sigma_T\rangle \, d\mu = \langle -\operatorname{div}(\sigma_T\mu), u_n\rangle \to 0$. It follows that $\int |\sigma_T|^2 \, d\mu = \int \langle \sigma, \sigma_T\rangle \, d\mu = \lim_{n\to\infty} \int \langle \nabla u_n, \sigma_T\rangle \, d\mu = 0$, hence $\sigma_T = 0$ holds μ a.e. and therefore $\sigma \in \mathcal{N}_\mu$.

The converse inclusion $\mathcal{N}_\mu \subset \mathcal{A}$ needed for (3.8) is more involved. In order to conclude, we are reduced to show the following claims:

$$\mathcal{A}^\perp \subset \mathcal{T}_\mu, \tag{3.15}$$
$$\mathcal{A} = \widetilde{\mathcal{A}} \quad \text{and} \quad \widetilde{\mathcal{A}} \text{ is a weakly-star closed subspace of } (L^\infty_\mu)^d. \tag{3.16}$$

Indeed, by passing to the orthogonal in (3.15), we get the inclusion $\mathcal{N}_\mu = \mathcal{T}_\mu^\perp \subset \mathcal{A}^{\perp\perp}$ while, by (3.16), the identities $\mathcal{A} = \widetilde{\mathcal{A}} = \mathcal{A}^{\perp\perp}$ all hold true.

Proof of (3.15). Let $\zeta \in (L^1_\mu)^d$ such that $\zeta \in \mathcal{A}^\perp$. Denoting $\zeta_N = P^\perp_\mu(\zeta)$, we consider the normal vector function of $(L^\infty_\mu)^d$ defined by

$$\sigma := \frac{\zeta_N}{|\zeta_N|}\,\mathbb{1}_{\zeta_N \neq 0}\,,$$

for which we may retrieve a sequence satisfying (3.12): there exists $(u_n)_n$ in $\mathrm{Lip}_1(\mathbb{R}^d)$ such that $u_n \overset{*}{\rightharpoonup} 0$ and (since $\mathcal{A} \subset \mathcal{A}$) $\nabla u_n \overset{*}{\rightharpoonup} \sigma$ in $(L^\infty_\mu)^d$. Therefore σ belongs to $\mathcal{A}$ and in particular $\int |\zeta_N| \, d\mu = \langle \zeta, \sigma\rangle = 0$. This proves $\zeta \in \mathcal{T}_\mu$, as desired.

Proof of (3.16). Consider the vector space $V := \big\{ (u, \nabla u) : u \in \mathcal{D}(\mathbb{R}^d)\ ,\ u(0) = 0 \big\}$, which we identify isometrically with a subspace of $\mathrm{Lip}_0(\mathbb{R}^d) \times (L^\infty_\mu)^d$, the dual of $\mathbf{Æ}(\mathbb{R}^d) \times (L^1_\mu)^d$. Denoting by $\overline{V}^{*\sigma}$ the weak-star closure of V, we may look at $\widetilde{\mathcal{A}}$ as the intersection of $\overline{V}^{*\sigma}$ with the closed subspace $\{0\} \times (L^\infty_\mu)^d$. It follows that $\widetilde{\mathcal{A}}$ is weakly-star closed. It remains to show the equality $\mathcal{A} = \widetilde{\mathcal{A}}$. The inclusion $\mathcal{A} \subset \widetilde{\mathcal{A}}$ is true since the convergence μ a.e. of a uniformly bounded approximating sequence (∇u_n) implies its weak-star convergence in $(L^\infty_\mu)^d$. The other set contention $\widetilde{\mathcal{A}} \subset \mathcal{A}$ is a consequence of Mazur's Lemma: there exists a convex combination v_j of the elements of (u_n) satisfying $v_j \overset{*}{\rightharpoonup} 0$ in $\mathrm{Lip}(\mathbb{R}^d)$ and $\nabla v_j \to \sigma$ in $(L^1_\mu)^d$. By convexity of the Lipschitz seminorm also $\mathrm{Lip}(v_j) \leq C$ where C is the uniform

Lipschitz constant of $(u_n)_n$. Moreover, up to extracting a subsequence, we may assume that $\nabla v_j \to \sigma$ μ a.e.. This shows that $\sigma \in \mathcal{A}$. □

3.3. **Closability, weak tangential gradient, integration by parts.** In this section, μ is a Radon measure in $\mathcal{M}_+(\mathbb{R}^d)$. Based on the construction of the tangent bundle T_μ, we obtain a notion of tangential derivation which allows to extend classical operations of differential calculus to Lipschitz functions.

Let us consider the operator $A_\mu : \mathrm{Lip}(\mathbb{R}^d) \to (L^\infty_\mu)^d$ with domain $D(A) = \mathcal{D}(\mathbb{R}^d)$ obtained by setting

$$(A_\mu u)(x) \;=\; P_\mu(x)(\nabla u(x)) \quad \text{for every } x \in \mathbb{R}^d. \tag{3.17}$$

Note that $D(A)$ is weakly-star dense in $\mathrm{Lip}(\mathbb{R}^d)$ (see Remark 2.7).

Lemma 3.10. *A_μ is weakly-star closable in the sense that:*

$$u_n \in \mathcal{D}(\mathbb{R}^d), \quad u_n \overset{*}{\rightharpoonup} 0 \text{ in } \mathrm{Lip}(\mathbb{R}^d), \quad P_\mu(\nabla u_n) \overset{*}{\rightharpoonup} \xi \text{ in } (L^\infty_\mu)^d \implies \xi = 0.$$

Proof. For every $\sigma \in (L^1_\mu)^d$, we have $P_\mu\sigma \in \mathcal{T}_\mu$, hence $f := -\operatorname{div}((P_\mu\sigma)\,\mu)$ belongs to $\text{Æ}(\mathbb{R}^d)$. By applying (2.6), it follows that:

$$\int \langle \xi, \sigma\rangle \, d\mu = \lim_n \int \langle P_\mu(\nabla u_n), \sigma\rangle \, d\mu = \lim_n \int \langle \nabla u_n, P_\mu\sigma\rangle \, d\mu = \lim_n \langle u_n, f\rangle = 0.$$

Since σ is arbitrary in $(L^1_\mu)^d$, we conclude that $\xi = 0$ as wished. □

In virtue of Lemma 3.10, we can extend A_μ in a unique way to all $u \in \mathrm{Lip}(\mathbb{R}^d)$, defining a weakly-star linear continuous operator

$$\nabla_\mu : \mathrm{Lip}(\mathbb{R}^d) \mapsto (L^\infty_\mu)^d. \tag{3.18}$$

Section 6.6 below shows that ∇_μ is a derivation in the sense of Weaver. More precisely, we have the following characterization:

Theorem 3.11. *(i) Let $u \in \mathrm{Lip}(\mathbb{R}^d)$ and (u_n) be any sequence in $C^1 \cap \mathrm{Lip}(\mathbb{R}^d)$ such that $u_n \overset{*}{\rightharpoonup} u$. Then the sequence $(P_\mu(\nabla u_n))$ admits a unique weak-star cluster point $\xi \in (L^\infty_\mu)^d$. Moreover ξ is independent of the sequence (u_n) and, by setting $\nabla_\mu u = \xi$ in* (3.18)*, we define a linear weakly-star continuous operator.*

(ii) Conversely, for any pair $(u, \zeta) \in \mathrm{Lip}(\mathbb{R}^d) \times \mathcal{N}_\mu$, there exists a sequence (u_n) in $\mathcal{D}(\mathbb{R}^d)$ such that $u_n \overset{*}{\rightharpoonup} u$ while the convergence $\nabla u_n \to \nabla_\mu u + \zeta$ holds μ a.e. and weakly-star in $(L^\infty_\mu)^d$. In particular, we have the convergences μ a.e.:

$$P_\mu(\nabla u_n) \to \nabla_\mu u \quad , \quad P^\perp_\mu(\nabla u_n) \to \zeta.$$

Proof. (i) The sequence $(P_\mu(\nabla u_n))$ is bounded in $(L^\infty_\mu)^d$ and admits at least one weak-star cluster point ξ. By Lemma 3.10, ξ is unique and does not depend of the approximating sequence (u_n). Note that such a sequence exists since $\mathcal{D}(\mathbb{R}^d)$ is weakly-star dense in $\mathrm{Lip}(\mathbb{R}^d)$ (see Remark 2.7). The fact that ∇_μ is linear, weakly-star continuous and coincides with $P_\mu \circ \nabla$ on smooth functions is straightforward.

(ii) Take any sequence (φ_n) in $\mathcal{D}(\mathbb{R}^d)$ such that $\varphi_n \overset{*}{\rightharpoonup} u$. Then, $(\nabla\varphi_n)$ is bounded in $(L^\infty_\mu)^d$ and, up to extracting a subsequence, converges weakly-star to some η. Observing that the

latter convergence holds also weakly in $(L^2_{\mathrm{loc},\mu})^d$, by Mazur's lemma, we can pick v_n in the convex hull of $\{\varphi_m, m \geq n\}$ such that $P_\mu(\nabla v_n) \to \eta$ strongly in $(L^2_{\mu,\mathrm{loc}})^d$. Clearly the Lipschitz constant of v_n remains smaller than the uniform Lipschitz bound of (φ_n) and $v_n \to u$ pointwise as well. Therefore $v_n \overset{*}{\rightharpoonup} u$ and, possibly after the extraction of a new subsequence, we arrive to the convergence $\nabla v_n \to \eta$ holding μ a.e.. From the assertion (i), we know that $P_\mu(\nabla v_n) \overset{*}{\rightharpoonup} \nabla_\mu u$, hence $P_\mu(\eta) = \nabla_\mu u$ so that $\nabla v_n \to \nabla_\mu u + P^\perp_\mu(\eta)$. Then, observing that $\sigma := \zeta - P^\perp_\mu(\eta)$ belongs to $\mathcal{N}_\mu$, it is enough to consider $u_n := v_n + w_n$, where (w_n) is the sequence in $\mathcal{D}(\mathbb{R}^d)$ obtained by applying (3.8). Indeed this sequence satisfies $w_n \overset{*}{\rightharpoonup} 0$ and $\nabla w_n \overset{*}{\rightharpoonup} \sigma$ pointwise. Accordingly $u_n \overset{*}{\rightharpoonup} u$ and $\nabla u_n \to \nabla_\mu u + \zeta$ pointwise as wished. □

An important feature of the *weak tangential gradient operator* ∇_μ is that it satisfies the following integration by parts formula:

$$\int \langle \sigma, \nabla_\mu u \rangle \, d\mu = \langle -\operatorname{div}(\sigma\mu), u \rangle \quad \text{for every } (u, \sigma) \in \operatorname{Lip}(\mathbb{R}^d) \times \mathcal{T}_\mu \tag{3.19}$$

where the duality bracket is intended in the sense of the duality between $\mathbf{Æ}(\mathbb{R}^d)$ and $\operatorname{Lip}(\mathbb{R}^d)$. This follows from the next lemma where we show that (3.19) actually characterizes $\nabla_\mu u$.

Lemma 3.12. *Let $u \in \operatorname{Lip}(\mathbb{R}^d)$ and $\xi \in (L^\infty_\mu)^d$ be such that $\xi(x) \in T_\mu(x)$, μ a.e.. Then*

$$\xi = \nabla_\mu u \iff \int \langle \sigma, \xi \rangle \, d\mu = \langle -\operatorname{div}(\sigma\mu), u \rangle \quad \textit{for every } \sigma \in \mathcal{T}_\mu. \tag{3.20}$$

Proof. Let $\sigma \in \mathcal{T}_\mu$. Then, by (2.9) and (2.5), the distribution $-\operatorname{div}(\sigma\mu)$ belongs to $\mathbf{Æ}(\mathbb{R}^d)$. By Theorem 3.11, for any smooth sequence in $\mathcal{D}(\mathbb{R}^d)$ such that $u_n \overset{*}{\rightharpoonup} u$, we have $\nabla_\mu u_n \overset{*}{\rightharpoonup} \nabla_\mu u$ in $(L^\infty_\mu)^d$. Therefore:

$$\begin{aligned}\langle -\operatorname{div}(\sigma\mu), u \rangle &= \lim_{n\to\infty} \langle -\operatorname{div}(\sigma\mu), u_n \rangle = \lim_{n\to\infty} \int \langle \sigma, \nabla u_n \rangle \, d\mu \\ &= \lim_{n\to\infty} \int \langle \sigma, \nabla_\mu u_n \rangle \, d\mu = \int \langle \sigma, \nabla_\mu u \rangle \, d\mu \,,\end{aligned}$$

where, in the last line, we used the fact that $\sigma \in \mathcal{T}_\mu$. It follows that the right hand side of (3.20) holds if and only if $\xi - \nabla_\mu u$ belongs to $\mathcal{N}_\mu = \mathcal{T}^\perp_\mu$. As $\xi(x) - \nabla_\mu u(x) \in T_\mu(x)$, this implies that $\xi = \nabla_\mu u$. □

As a consequence of Lemma 3.12, we get a useful localization property of the tangential gradient which supplements the locality property (3.5).

Lemma 3.13 (Locality of the tangential gradient)**.** *Let $u, v \in \operatorname{Lip}(\mathbb{R}^d)$ and μ, ν two Radon measures in $\mathbb{R}^d$. Then*

$$u = v \quad \mu \textit{ a.e.} \qquad \Longrightarrow \qquad \nabla_\mu u = \nabla_\mu v \quad \mu \textit{ a.e.}, \tag{3.21}$$

$$\nu \ll \mu \qquad \Longrightarrow \qquad \nabla_\mu u = \nabla_\nu u \quad \nu \textit{ a.e.}. \tag{3.22}$$

Proof. Since ∇_μ is linear, does not see constants and $u = u^+ - u^-$ with $u^\pm \geq 0$ Lipschitz and vanishing μ-a.e., we may assume $u \geq 0$. The proof of (3.21) uses the pointwise differentiability

along tangent fields established in the next section (assertion (ii) of Theorem 4.1). Let $\sigma \in \mathcal{T}_\mu$. At μ-a.e. x we have $u(x) = 0$, hence

$$\langle \nabla_\mu u(x), \sigma(x) \rangle = \lim_{h \to 0^+} \frac{u(x + h\sigma(x))}{h} \;\geq\; 0.$$

Applying this to $-\sigma \in \mathcal{T}_\mu$ gives the reverse inequality, so that $\langle \nabla_\mu u, \sigma \rangle = 0$ μ-a.e. for every $\sigma \in \mathcal{T}_\mu$, i.e. $\nabla_\mu u \in \mathcal{N}_\mu$. Since $\nabla_\mu u(x) \in T_\mu(x)$, (3.21) follows.

In order to prove (3.22), consider $\sigma \in \mathcal{T}_\nu$. Then $\tilde{\sigma} := \sigma \frac{d\nu}{d\mu}$ belongs to $\mathcal{T}_\mu$ by the local tangency criterion (3.2). Since $\tilde{\sigma}\, \mu = \sigma\, \nu$, by using twice (3.19), we get

$$\int \langle \sigma, \nabla_\mu u \rangle \, d\nu = \int \langle \tilde{\sigma}, \nabla_\mu u \rangle \, d\mu = \langle -\operatorname{div}(\tilde{\sigma}\mu), u \rangle = \langle -\operatorname{div}(\sigma\nu), u \rangle.$$

The equality (3.22) follows by applying (3.20) to the measure ν and $\xi = \nabla_\mu u$. □

Remark 3.14. Identity (3.21) extends the classical locality of the weak derivative for Sobolev functions with $\mu = \mathcal{L}^d$: if $u, v \in W^{1,p}(\mathbb{R}^d)$, then $\nabla u \equiv \nabla v$ $\mathcal{L}^d$ almost everywhere on $\big\{\, x \in \mathbb{R}^d : u(x) = v(x) \,\big\}$. As an example, one can consider μ as the 1-Hausdorff measure on the segment $[0,1] \times \{0\} \subset \mathbb{R}^2$, with $u(x,y) = y$. Then $u = 0$ at μ a.e. point, and $\nabla_\mu u = 0$, even though $\nabla u \neq 0$.

Remark 3.15. The closability result appearing in Lemma 3.10 (see also in [21]) has been recently extended in [1] to more general differential operators and function spaces. In particular, Theorem 1.3 and Section 1.7 therein yield another proof of the fact that $T_\mu = V_\mu$, relying on [2].

4. Stability of the MK distance

In this section, we investigate the first-order stability of the Monge-Kantorovich distance under infinitesimal displacements of a general measure μ along velocity fields $\sigma \in (L^1_\mu)^d$, given by $\mu^\sigma_h = (\mathrm{id} + h\sigma)_\sharp \mu$. Our main result shows that the Monge-Kantorovich tangent bundle $\mathcal{T}_\mu$ characterizes the subset of stable directions. As an application, we provide a proof that Lipschitz functions are differentiable almost everywhere in the directions of the MK tangent bundle, recovering a result previously established in [2].

4.1. **Statements.** Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure and $h > 0$. For any velocity field $\sigma \in (L^1_\mu)^d$, and any $u \in C^1$, we set:

$$\Delta^\sigma_h(u)(x) := \frac{u(x + h\sigma(x)) - u(x)}{h} - \langle \nabla u(x), \sigma(x) \rangle, \tag{4.1}$$

$$F_h(\mu, \sigma) := \sup_{u \in \mathrm{Lip}_1 \cap C^1(\mathbb{R}^d)} \left\{ \int \Delta^\sigma_h(u)(x) \, d\mu \right\}, \tag{4.2}$$

and

$$F_0^+(\mu, \sigma) := \limsup_{h \to 0} F_h(\mu, \sigma), \qquad F_0^-(\mu, \sigma) := \liminf_{h \to 0} F_h(\mu, \sigma). \tag{4.3}$$

If $\sigma \in \mathcal{T}_\mu$, then according to (3.18), the expression (4.1) is extended to all $u \in \mathrm{Lip}(\mathbb{R}^d)$ by

$$\Delta^\sigma_h(u)(x) := \frac{u(x + h\sigma(x)) - u(x)}{h} - \langle \nabla_\mu u(x), \sigma(x) \rangle. \tag{4.4}$$

In this case, by Remark 2.7, the supremum in (4.2) is the KR norm of an element of $\mathbf{Æ}(\mathbb{R}^d)$, which is expressed through the difference quotients $D_h^\sigma \mu = \frac{\mu_h^\sigma - \mu}{h}$ with $\mu_h^\sigma := (\mathrm{id} + h\sigma)_\sharp \mu$:

$$F_h(\mu, \sigma) = \|D_h^\sigma \mu + \mathrm{div}(\sigma\, \mu)\|_{\mathrm{KR}} \qquad \text{for } \sigma \in \mathcal{T}_\mu. \tag{4.5}$$

Note that $D_h^\sigma \mu$ belongs to $\mathbf{Æ}(\mathbb{R}^d)$ for every $\sigma \in (L^1_\mu)^d$ since it is a balanced measure, with $\|D_h^\sigma \mu\|_{\mathrm{KR}} \le \int |\sigma|\, d\mu$. The main result of this section is the following stability result.

Theorem 4.1. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure and let T_μ be its MK-tangent bundle. Then:*

(i) for any vector field $\sigma \in (L^1_\mu(\mathbb{R}^d))^d$, we have the equalities

$$F_0^+(\mu, \sigma) \;=\; F_0^-(\mu, \sigma) \;=\; F(\mu, \sigma) := \; 2 \int |\sigma_N|\, d\mu, \tag{4.6}$$

where $\sigma_N = P_\mu^\perp \sigma$ denotes the normal component of σ. As a consequence, we have the following equivalence:

$$\sigma \in \mathcal{T}_\mu \iff \lim_{h \to 0} F_h(\mu, \sigma) = 0. \tag{4.7}$$

(ii) Let $\sigma \in \mathcal{T}_\mu$. Then $\Delta_h^\sigma(u) \to 0$ in L^1_μ, for every $u \in \mathrm{Lip}(\mathbb{R}^d)$. Consequently, any Lipschitz function u is differentiable in the direction σ, namely:

$$\frac{u(x + h\sigma(x)) - u(x)}{h} \to \langle \nabla_\mu u(x), \sigma(x) \rangle \quad \mu \text{ a.e. and in } L^1_\mu(\mathbb{R}^d).$$

(iii) Every $u \in \mathrm{Lip}(\mathbb{R}^d)$ is differentiable at μ-a.e. x with respect to the whole subspace $T_\mu(x)$, with differential $\langle \nabla_\mu u(x), \cdot \rangle$:

$$u(x + v) \;=\; u(x) + \langle \nabla_\mu u(x),\, v \rangle + o(|v|) \qquad \text{as } T_\mu(x) \ni v \to 0. \tag{4.8}$$

The proof of Theorem 4.1 is contained in the subsection 4.3, after some comments and preliminary results.

Remark 4.2. Since the family of functionals $(F_h(\mu, \cdot))$ is equi-Lipschitz in $(L^1_\mu)^d$ (see Lemma 4.8 below), the pointwise convergence (4.6) implies that $F_h(\mu, \cdot)$ Γ-converges to $F(\mu, \cdot)$ as $h \to 0$ in the Banach space $(L^1_\mu)^d$.

Remark 4.3. The factor 2 appearing in the limit $F(\mu, \sigma)$ might look counterintuitive since we expect the two terms $\int \left(\frac{u(x+h\sigma) - u(x)}{h} \right) d\mu$ and $\int \langle \nabla u, \sigma \rangle\, d\mu$ in (4.2) to share the same sign while both of them have absolute values not larger than $\int |\sigma|\, d\mu$. The simple example below illustrates that it is not the case as far as one considers the supremum over $\mathrm{Lip}_1(\mathbb{R}^d)$.

Example. Let $\mu = \mathcal{H}^1 \llcorner ([0,1] \times \{0\})$ in $\mathbb{R}^2$ and $\sigma = (0, 1)$. Then $\int |\sigma_N|\, d\mu = 1$ and by taking the 1-Lipschitz function $u_h(x_1, x_2) := |x_2 - h^2|$ for $h \le 1$ (which we may mollify near the line $\{x_2 = h^2\}$, at no cost for the values used below, so as to meet the requirement $u \in C^1$ of (4.2)), we obtain the lower bound:

$$\begin{aligned} F_h(\mu, \sigma) &\ge \int_0^1 \left(\frac{u_h(x_1, h) - u_h(x_1, 0)}{h} - \langle \nabla u_h(x_1, 0), (0, 1) \rangle \right) dx_1 \\ &= \int_0^1 \left(\frac{h - h^2 - h^2}{h} + 1 \right) dx_1 = 2 - 2h. \end{aligned}$$

Remark 4.4. The convergence $F_h(\mu,\sigma) \to 0$ obtained when $\sigma \in \mathcal{T}_\mu$ cannot be applied if alternatively, we consider $\widetilde{F_h}$ defined by:

$$\widetilde{F_h}(\mu,\sigma) := \sup_{u\in \mathrm{Lip}_1\cap C^1(\mathbb{R}^d)} \left\{ \int |\Delta_h^\sigma(u)|\, d\mu \right\}.$$

Despite the fact that, for any Lipschitz function u, the integral above vanishes as $h \to 0$ thanks to the μ-a.e. differentiability in the direction σ (see assertion (ii)), we cannot expect this convergence to be uniform with respect to $u \in \mathrm{Lip}_1(\mathbb{R}^d)$. This is confirmed by the example below.

Example. Let $d = 1$ and consider $\mu = \mathcal{L}_{[0,1]}$, and $\sigma(x) = 1$ for any $x \in [0,1]$. Then $\sigma \in L^1_\mu$ belongs to $\mathcal{T}_\mu$. However, we claim that

$$\liminf_{h\searrow 0} \sup_{u\in \mathrm{Lip}_1} \int_{x\in[0,1]} \left| \frac{u(x+h)-u(x)}{h} - u'(x) \right| dx \;\geq \frac{2}{\pi}. \tag{4.9}$$

Indeed, for each $h > 0$, the function $u : x \mapsto \frac{h}{2\pi}\sin(2\pi\frac{x}{h})$ is 1-Lipschitz and vanishes at $x = 0$. There holds

$$\left| \frac{u(x+h)-u(x)}{h} - u'(x) \right| = \left| \frac{\sin(2\pi\frac{x}{h}+2\pi)-\sin(2\pi\frac{x}{h})}{2\pi} - \cos(2\pi x/h) \right| = |\cos(2\pi x/h)|.$$

Since the function $|\cos(2\pi y)|$ is 1-periodic, the integral $\int_{[0,1]} |\cos(2\pi\frac{x}{h})| dx$ converges to the mean value $\int_{[0,1]} |\cos(2\pi y)| dy = \frac{2}{\pi}$, hence the lower bound (4.9).

Remark 4.5. A natural variant of $F_h(\mu,\sigma)$ is obtained by averaging two opposite velocity fields. Setting $\widehat{\mu_h^\sigma} := \frac{1}{2}\big(\mu_h^\sigma + \mu_h^{-\sigma}\big)$, one has

$$\sup_{u\in \mathrm{Lip}_1\cap C^1(\mathbb{R}^d)} \int \frac{\Delta_h^\sigma(u) + \Delta_h^{-\sigma}(u)}{2}\, d\mu \;=\; \left\| \frac{\widehat{\mu_h^\sigma}-\mu}{h} \right\|_{\mathrm{KR}} \;=\; \frac{1}{h}\, W_1\big(\widehat{\mu_h^\sigma},\mu\big),$$

which, in contrast with (4.5), involves no divergence term. As this quantity is bounded by $\frac{1}{2}\big(F_h(\mu,\sigma)+F_h(\mu,-\sigma)\big)$, the criterion (4.7) gives $\mathcal{T}_\mu \subset \mathcal{T}_\mu^{sym}$, where $\mathcal{T}_\mu^{sym}$ denotes the set of those $\sigma \in (L^1_\mu)^d$ for which $h^{-1}W_1(\widehat{\mu_h^\sigma},\mu) \to 0$. The inclusion is strict in general. This is demonstrated in [11], where the analogous W_2 question is studied for $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, provides uniform measures on Cantor subsets of $[0,1]$ with $\mathcal{T}_\mu = \{0\}$ for which $\sigma \equiv 1$ belongs to $\mathcal{T}_\mu^{sym}$. A characterization of $\mathcal{T}_\mu^{sym}$ seems to be still out of reach.

Next we derive a relation involving a given velocity field $\sigma \in (L^1_\mu)^d$ and the asymptotic behavior as $h \to 0_+$ of the sequence $D_h^\sigma\mu$. Although the total variation of this sequence blows up in general, it admits an uniform upper bound in the Banach space $\boldsymbol{Æ}(\mathbb{R}^d)$, namely $\|D_h^\sigma\mu\|_{\mathrm{KR}} \leq \int |\sigma|\, d\mu$. Then, even if $D_h^\sigma\mu$ fails to converge in $\boldsymbol{Æ}(\mathbb{R}^d)$, we can define its *asymptotic radius* $\rho := \inf_{f\in\boldsymbol{Æ}(\mathbb{R}^d)} H(\sigma,f)$ where

$$H(\sigma,g) := \limsup_{h\to 0_+} \|D_h^\sigma\mu - g\|_{\mathrm{KR}}. \tag{4.10}$$

Accordingly the *asymptotic center* of the sequence $\{D_h^\sigma\mu\}$ is the (possibly empty) set of minimizers $\big\{\, g \in \boldsymbol{Æ}(\mathbb{R}^d) : H(\sigma,g) = \rho \,\big\}$. Clearly $\{D_h^\sigma\mu\}$ is a Cauchy sequence in $\boldsymbol{Æ}(\mathbb{R}^d)$ if and only if $\rho = 0$. In this case, the asymptotic center reduces to the singleton $\{f\}$, where f is

the limit of $\{D_h^\sigma \mu\}$, which exists since $\text{Æ}(\mathbb{R}^d)$ is complete. If $\rho > 0$, despite the fact that the functional $H(\sigma, \cdot) : \text{Æ}(\mathbb{R}^d) \to \mathbb{R}_+$ defined in (4.10) is convex Lipschitz and coercive, it is not clear that the asymptotic center is non-empty since $\text{Æ}(\mathbb{R}^d)$ is not reflexive.

Corollary 4.6. *Let $\sigma \in (L^1_\mu)^d$ be a velocity field and denote $\sigma_T = P_\mu \sigma$ and $\sigma_N = P_\mu^\perp \sigma$ its tangential and normal components respectively; then we define*

$$f_T := -\operatorname{div}(\sigma_T \mu) \quad , \; r := \int |\sigma_N| \, d\mu \, . \tag{4.11}$$

(i) $f_T \in \text{Æ}(\mathbb{R}^d)$ and the functional $H(\sigma, \cdot)$ defined in (4.10) *satisfies the inequalities*

$$r \;\le\; H(\sigma, g) \;\le\; r + \|g - f_T\|_{\mathrm{KR}} \qquad \text{for every } g \in \text{Æ}(\mathbb{R}^d). \tag{4.12}$$

(ii) the sequence $\{D_h^\sigma \mu\}$ is a Cauchy sequence in $\text{Æ}(\mathbb{R}^d)$ if and only if $\sigma \in \mathcal{T}_\mu$. In this case, we have

$$\lim_{h \to 0} \|D_h^\sigma \mu + \operatorname{div}(\sigma\, \mu)\|_{\mathrm{KR}} = 0. \tag{4.13}$$

(iii) If $\sigma \notin \mathcal{T}_\mu$, then the sequence $\{D_h^\sigma \mu\}$ has positive asymptotic radius $\rho = r = \int |\sigma_N| \, d\mu$ and asymptotic center reducing to the singleton $\{f_T\}$. In particular we have

$$\lim_{h \to 0} \|D_h^\sigma \mu + \operatorname{div}(\sigma_T\, \mu)\|_{\mathrm{KR}} = \int |\sigma_N| \, d\mu. \tag{4.14}$$

Proof. By (3.1)-(3.2), we already know that f_T belongs to $\text{Æ}(\mathbb{R}^d)$. Since $\sigma_T \in \mathcal{T}_\mu$, it follows from (4.5) and (4.7) that

$$\lim_{h \to 0_+} \|D_h^{\sigma_T} \mu - f_T\|_{\mathrm{KR}} \;=\; 0.$$

On the other hand, we have the upper bound:

$$\|D_h^{\sigma_T} \mu - D_h^\sigma \mu\|_{\mathrm{KR}} \le \int |\sigma - \sigma_T| \, d\mu = \int |\sigma_N| \, d\mu = r.$$

Therefore, for every $g \in \text{Æ}(\mathbb{R}^d)$, we infer that:

$$\begin{aligned} H(\sigma, g) &= \limsup_{h \to 0_+} \|g - D_h^\sigma \mu\|_{\mathrm{KR}} \\ &\le \|g - f_T\|_{\mathrm{KR}} + \limsup_{h \to 0_+} \|f_T - D_h^{\sigma_T} \mu\|_{\mathrm{KR}} + \limsup_{h \to 0_+} \|D_h^{\sigma_T} \mu - D_h^\sigma \mu\|_{\mathrm{KR}} \\ &\le \|g - f_T\|_{\mathrm{KR}} + 0 + r. \end{aligned}$$

Let us now establish the lower bound $H(\sigma, g) \ge r = \int |\sigma_N| \, d\mu$. For every $u \in \mathcal{D} \cap \mathrm{Lip}_1(\mathbb{R}^d)$, one has

$$H(\sigma, g) \ge \liminf_{h \to 0_+} \|D_h^\sigma \mu - g\|_{\mathrm{KR}} \ge \liminf_{h \to 0_+} \langle D_h^\sigma \mu - g, u \rangle = \int \langle \sigma, \nabla u \rangle \, d\mu - \langle g, u \rangle \tag{4.15}$$

where to pass to the last equality, we used the the distributional convergence $D_h^\sigma \mu \;\to\; -\operatorname{div}(\sigma\, \mu)$. Let us apply the inequality above replacing u by the sequence (u_n) in $\mathcal{D} \cap \mathrm{Lip}_1(\mathbb{R}^d)$ given in (3.12) (see Proposition 3.8) (i.e. $u_n \overset{*}{\rightharpoonup} 0$ and $\nabla u_n \to \frac{\sigma_N}{|\sigma_N|} \mathbb{1}_{\{\sigma_N \neq 0\}}$ μ a.e.), we conclude that

$$H(\sigma, g) \ge \liminf_n \int \langle \sigma, \nabla u_n \rangle \, d\mu - \langle g, u_n \rangle \;\ge \int |\sigma_N| \, d\mu,$$

whence (4.12) and the assertion (i).

(ii) Assume that $D_h^\sigma \mu$ is a Cauchy sequence in $\text{Æ}(\mathbb{R}^d)$. Then, since $\text{Æ}(\mathbb{R}^d)$ is complete, we have that $D_h^\sigma \mu \to g$ for a suitable $g \in \text{Æ}(\mathbb{R}^d)$. Since this convergence holds also in the distributional sense, the only possibility is that $g = -\operatorname{div}(\sigma\,\mu)$. Therefore, by (3.1), $\sigma \in \mathcal{T}_\mu$. Conversely, if $\sigma \in \mathcal{T}_\mu$, then $\sigma = \sigma_T$, $\sigma_N = 0$ and by applying (4.12) to $g = -\operatorname{div}(\sigma\,\mu)$, we get $H(\sigma, g) = 0$, hence the strong convergence $D_h^\sigma \mu \to g$ in $\text{Æ}(\mathbb{R}^d)$.

(iii) The equality $\rho = r$ and (4.14) follow by applying (4.12) to $g = f_T$. There remains to prove that no other element of $\text{Æ}(\mathbb{R}^d)$ is an asymptotic center. Assume that $H(\sigma, g) = r$ for some $g \in \text{Æ}(\mathbb{R}^d)$. By taking the supremum with respect to u (or $-u$) in $\mathcal{D} \cap \operatorname{Lip}_1(\mathbb{R}^d)$ in the right hand side of (4.15), we obtain

$$H(\sigma, g) \geq G_\sigma^*(g) = \min_{\lambda \in \mathcal{M}^d} \left\{ \int_{\mathbb{R}^d} |\lambda - \sigma\,\mu| \ : \ -\operatorname{div}\lambda = g \right\},$$

where G_σ is defined in (3.9) while the second equality follows from (3.10). Let $\overline{\lambda} = w\,\mu + \lambda^s$ be optimal in the minimum above, being $w\mu$ the absolutely continuous part and $\lambda^s \perp \mu$. Since $g \in \text{Æ}(\mathbb{R}^d)$ and $T_{|\lambda|} = \mathcal{T}_\mu$ holds μ a.e. (in virtue of (3.5)), we infer that $w \in \mathcal{T}_\mu$. Therefore

$$H(\sigma, g) \geq G_\sigma^*(g) = \int \sqrt{|w - \sigma_T|^2 + |\sigma_N|^2}\, d\mu + \int |\lambda^s|.$$

Then we observe that the equality $H(\sigma, g) = r = \int |\sigma_N|\, d\mu$ implies that

$$\int \left(\sqrt{|w - \sigma_T|^2 + |\sigma_N|^2} - |\sigma_N| \right) d\mu = \int |\lambda^s| = 0,$$

hence $\overline{\lambda} = \sigma_T\,\mu$ and $g = -\operatorname{div}(\sigma_T\,\mu)$, as wished. □

Remark 4.7. The right hand-side of (4.14) has a coefficient 1, instead of the coefficient 2 in (4.6). This comes from the fact that (4.6) compares the differential quotient with $\operatorname{div}(\sigma\mu) = \operatorname{div}(\sigma_N\mu) + \operatorname{div}(\sigma_T\mu)$, while (4.14) only contains the latter term $\operatorname{div}(\sigma_T\mu)$. The term $\operatorname{div}(\sigma_N\mu)$ accounts for the distance between $D_0^\sigma\mu$ and $\text{Æ}(\mathbb{R}^d)$ in the bidual of $\text{Æ}(\mathbb{R}^d)$, measured with a suitable extension of the Kantorovich-Rubinstein norm.

4.2. **Preliminary results.** In the proof of the assertion (i) of Theorem 4.1, we will need some properties of $F_h(\mu, \cdot)$ and $F_0^\pm(\mu, \cdot)$. For convenience, we define counterparts on every vector measure $\lambda \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$ by

$$\Phi_h(\lambda) := F_h(\mu, \sigma), \quad \text{where} \quad \mu = |\lambda|,\ \sigma = \frac{d\lambda}{d|\lambda|}. \tag{4.16}$$

Accordingly we let $\Phi_0^\pm(\lambda) = F_0^\pm(\mu, \sigma)$. Similarly, in the proof of the assertion (ii) of Theorem 4.1, we need to associate the following functional on $\operatorname{Lip}(\mathbb{R}^d)$:

$$\Psi_h(\lambda)(u) := \int \widehat{\Delta_h^\sigma}(u)\, d\mu \qquad \text{where} \quad \mu = |\lambda|,\ \sigma = \frac{d\lambda}{d|\lambda|}, \tag{4.17}$$

where the function $\widehat{\Delta_h^\sigma}(u)$ is defined by

$$\widehat{\Delta_h^\sigma}(u) := \sup_{s \in (0,h]} |\Delta_s^\sigma(u)|. \tag{4.18}$$

Note that the condition $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ implies that $\sigma \in \mathcal{T}_\mu$ which is necessary in order to extend the definition of $\Delta_s^\sigma(u)$ given by (4.4) to Lipschitz functions.

Lemma 4.8. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$. Then:*

(i) For every $\sigma \in (L^1_\mu)^d$ and every Borel subset $B \in \mathcal{B}(\mathbb{R}^d)$, we have

$$F_h(\mu, \sigma \mathbb{1}_B) = F_h(\mu \llcorner B, \sigma).$$

(ii) The map $B \mapsto F_h(\mu, \sigma \mathbb{1}_B)$ is subadditive on disjoint Borel subsets. By taking the limit sup, one deduces that F_0^+ is also subadditive.

(iii) Φ_h and $F_h(\mu, \cdot)$ are positively homogeneous, in that the equalities

$$\Phi_h(t\lambda) = t\Phi_h(\lambda), \quad F_h(t\mu, \sigma) = tF_h(\mu, \sigma)$$

hold for every $t \in \mathbb{R}_+$. Hence $\Phi_0^\pm$ and $F_0^\pm$ are positively homogeneous. Moreover $F_h(\mu, t\sigma) = t\, F_{th}(\mu, \sigma)$, so that $F_0^\pm(\mu, \cdot)$ is positively homogeneous in σ as well.

(iv) The functionals $F_h(\mu, \cdot)$, $F_0^\pm(\mu, \cdot)$ are 2-Lipschitz in $(L^1_\mu)^d$, while Φ_h, $\Phi_0^\pm$ and Ψ_h are 3-Lipschitz in $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$. Specifically, for any $\lambda_1, \lambda_2 \in \mathcal{M}^d$ and $u \in \mathrm{Lip}_1$,

$$|\Phi_h(\lambda_2) - \Phi_h(\lambda_1)| \;\le\; 3\|\lambda_1 - \lambda_2\|, \tag{4.19}$$

$$|\Psi_h(\lambda_2)(u) - \Psi_h(\lambda_1)(u)| \;\le\; 3\|\lambda_1 - \lambda_2\|. \tag{4.20}$$

Proof. The assertion (i) follows from $\Delta_h^{\sigma \mathbb{1}_B} = \mathbb{1}_B \Delta_h^\sigma$. The subadditivity (ii) is direct from the definition as a supremum of integrals. The assertion (iii) is straightforward once one notices that $\frac{d\lambda}{d|\lambda|} = \frac{d(t\lambda)}{d|t\lambda|}$ for any $t > 0$. For the assertion (iv), let $u \in \mathrm{Lip}_1(\mathbb{R}^d)$. For any $\sigma_1, \sigma_2 \in (L^1_\mu)^d$,

$$\left|\Delta_h^{\sigma_2}(u) - \Delta_h^{\sigma_1}(u)\right| \;\le\; \frac{|u(x + h\sigma_2) - u(x + h\sigma_1)|}{h} + |\langle \nabla u, \sigma_2 - \sigma_1 \rangle| \;\le\; 2\,|\sigma_2 - \sigma_1|\,, \tag{4.21}$$

so that, for every μ, $F_h(\mu, \cdot)$ and $F_0^\pm(\mu, \cdot)$ are 2-Lipschitz. For Φ_h and Ψ_h the measure varies as well, and we compare the two integrands on $m := |\lambda_1| + |\lambda_2|$. Set $p_i := \frac{d\lambda_i}{dm}$, $a := |p_1|$, $b := |p_2|$, $\omega_i := p_i/|p_i|$, $\theta := |\omega_1 - \omega_2|$ and $V_i := \Delta_h^{\omega_i}(u)(x)$, so that $\int \Delta_h^{\sigma_i}(u)\, d|\lambda_i| = \int |p_i|\, V_i\, dm$ while $|V_i| \le 2$ and, by (4.21), $|V_1 - V_2| \le 2\theta$. Decomposing $aV_1 - bV_2$ as $a(V_1 - V_2) + (a-b)V_2$ and as $b(V_1 - V_2) + (a-b)V_1$, and using $|p_1 - p_2|^2 = (a-b)^2 + ab\,\theta^2$ together with $ab \ge \min\{a,b\}^2$, we get the pointwise bound

$$|aV_1 - bV_2| \;\le\; 2\big(\min\{a,b\}\,\theta + |a-b|\big) \;\le\; 2\sqrt{2}\,|p_1 - p_2| \;\le\; 3\,|p_1 - p_2|. \tag{4.22}$$

Integrating (4.22) against m and taking the supremum over u gives (4.19), and (4.20) follows from the same computation with $V_i := \widehat{\Delta_h^{\omega_i}}(u)(x)$. The tangential gradients adapt well to the argument, since $|\lambda_i| \ll m$ forces $\nabla_{|\lambda_i|} u = \nabla_m u$ at $|\lambda_i|$-a.e. point by (3.22), and taking absolute values and the supremum over $s \in (0, h]$ are 1-Lipschitz operations. □

Next we consider a measure Q on the Banach space $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$, endowed with the Borel σ-algebra of the weak-$*$ topology, such that $\int \|\lambda\|\, dQ < +\infty$. Then we denote $\lambda_Q = \int \lambda\, dQ(\lambda)$ where the integral is intended in the weak sense as follows:

$$\langle \lambda_Q, \psi \rangle \;=\; \int \langle \lambda, \psi \rangle\, dQ(\lambda) \quad \text{for every } \psi \in (C_0(\mathbb{R}^d))^d.$$

It is not restrictive for us to assume that Q is supported on a closed ball $B_R := \{\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d) : \|\lambda\| \le R\}$ (for R suitably large) on which the weak-$*$ topology is metrizable and compact.

Lemma 4.9. *Let us assume that the measure Q above satisfies the condition:*

$$\|\lambda_Q\| = \int \|\lambda\|\, dQ(\lambda) < +\infty. \tag{4.23}$$

Then:

(i) $|\lambda_Q| = \int |\lambda|\, dQ(\lambda)$ *and, for Q almost all λ, it holds* $\frac{d\lambda_Q}{d|\lambda_Q|} = \frac{d\lambda}{d|\lambda|}$, $|\lambda|$*-a.e.;*

(ii) For every $h > 0$, the Lipschitz functionals Φ_h satisfies the subadditivity property:[1]

$$\Phi_h(\lambda_Q) \;\le\; \int \Phi_h(\lambda)\, dQ(\lambda). \tag{4.24}$$

(iii) Assume in addition that $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ for Q almost all λ. Then $\lambda_Q \in \mathbf{F_1}(\mathbb{R}^d)$ and for every $h > 0$, we have the equality:

$$\Psi_h(\lambda_Q)(u) \;=\; \int \Psi_h(\lambda)(u)\, dQ(\lambda) \quad \textit{for every } u \in \mathrm{Lip}(\mathbb{R}^d). \tag{4.25}$$

Proof. (i) For any open subset $A \subset \mathbb{R}^d$ and any $\varphi \in (C_0(\mathbb{R}^d))^d$ such that $\mathrm{spt}(\varphi) \subset A$ and $|\varphi| \le 1$, it holds

$$\langle \varphi, \lambda_Q\rangle = \int \langle \varphi, \lambda\rangle\, dQ(\lambda) \le \int |\lambda|(A)\, dQ(\lambda).$$

By passing to the supremum over such φ, we get the inequality $|\lambda_Q| \le \int |\lambda|\, dQ(\lambda)$ as measures in $\mathcal{M}_+(\mathbb{R}^d)$. The assumption implies that these two measures have the same total mass, hence coincide. Setting $\sigma_Q := \frac{d\lambda_Q}{d|\lambda_Q|}$ and $\sigma_\lambda := \frac{d\lambda}{d|\lambda|}$, which are unit vectors, we deduce that:

$$|\lambda_Q| = \langle \sigma_Q, \lambda_Q\rangle = \int \langle \sigma_Q, \sigma_\lambda\rangle |\lambda|\, dQ(\lambda) \le \int |\lambda|\, dQ(\lambda) = |\lambda_Q|,$$

whence the equalities $\langle \sigma_Q, \sigma_\lambda\rangle = 1$ and $\sigma_Q = \sigma_\lambda$ holding $|\lambda|$-a.e. and for Q-almost all $\lambda \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$.

(ii) Denote $\mu := |\lambda_Q|$ and $\mu_\lambda := |\lambda|$. Since $\lambda_Q = \sigma_Q\, \mu$ is the polar decomposition of λ_Q, in virtue of (4.2), (4.16) and recalling the notation (4.1), we have:

$$\Phi_h(\lambda_Q) = F_h(\mu, \sigma_Q) = \sup_{u \in \mathrm{Lip}_1 \cap C^1} \int \Delta_h^{\sigma_Q}(u)\, d\mu. \tag{4.26}$$

[1] That (4.24) is well defined deserves an argument, that is, why Φ_h is Q-measurable. The supremum of $\Delta_h^{\sigma_\lambda}(u)$ in $\Phi_h(\lambda)$ runs over $\mathrm{Lip}_1 \cap C^1$, which is separable for the topology of local uniform convergence of the function and of its gradient. Along a sequence $u_n \to u$ in that topology the integrands $\Delta_h^{\sigma_\lambda}$ converge pointwise while staying bounded by 2, so that the supremum may be restricted to a countable dense family. This reduces the question to the Q-measurability of $\lambda \mapsto \Delta_h^{\sigma_\lambda}(u)$. This is where the assertion (i) is needed: it gives $\sigma_\lambda(x) = \sigma_Q(x)$ at $|\lambda|$ a.e. point x, for Q a.e. λ. This yields for each $u \in \mathrm{Lip}(\mathbb{R}^d) \cap C^1$ that

$$\int \Delta_h^{\sigma_\lambda}(u)\, d|\lambda| = \int \Delta_h^{\sigma_Q}(u)\, d|\lambda| \qquad Q \text{ a.e..}$$

The measurability of the right-hand side is then handled by the other half of (i): the identity $|\lambda_Q| = \int |\lambda|\, dQ$ makes $\lambda \mapsto \int f\, d|\lambda|$ a Q-measurable map for every bounded Borel f, which we apply to $f = \Delta_h^{\sigma_Q}(u)$. Thus Φ_h is a countable supremum of Q-measurable maps.

By the assertion (i), we have that $\mu = \int \mu_\lambda \, dQ(\lambda)$ while the unit vectors σ_Q and σ_λ agree μ_λ -a.e. Accordingly, for every $u \in \mathrm{Lip}_1 \cap C^1$, one has:

$$\begin{aligned}\int \Delta_h^{\sigma_Q}(u)\, d\mu &= \int \left(\int \Delta_h^{\sigma_\lambda}(u)\, d\mu_\lambda \right) dQ(\lambda) \\ &\leq \int F_h(\mu_\lambda, \sigma_\lambda)\, dQ(\lambda) = \int \Phi_h(\lambda)\, dQ(\lambda).\end{aligned}$$

In virtue of (4.26), the inequality (4.24) of the assertion (ii) follows by taking the supremum with respect to u in the left hand side above.

(iii) We show first that $\lambda_Q = \int \lambda \, dQ(\lambda)$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$. Let (φ_n) be a sequence in $\mathcal{D}(\mathbb{R}^d)$ such that $\varphi_n \overset{*}{\rightharpoonup} 0$. By testing the former integral against $\nabla\varphi_n$, we are led to

$$\langle -\operatorname{div} \lambda_Q, \varphi_n \rangle = \langle \lambda_Q, \nabla\varphi_n \rangle = \int \langle \lambda, \nabla\varphi_n \rangle \, dQ(\lambda).$$

Since Q-almost every λ belongs to $\mathbf{F_1}(\mathbb{R}^d)$, we have that $\langle \lambda, \nabla\varphi_n \rangle$ converges pointwise to zero while $|\langle \lambda, \nabla\varphi_n \rangle| \leq C\|\lambda\|$, where $C = \sup \mathrm{Lip}(\varphi_n)$. The dominated convergence can be then applied since $\int \|\lambda\| \, dQ(\lambda) < +\infty$ and we conclude that $\langle -\operatorname{div} \lambda_Q, \varphi_n \rangle \to 0$, whence $\lambda_Q \in \mathbf{F_1}(\mathbb{R}^d)$ in virtue of (2.5).

Next, let us fix $u \in \mathrm{Lip}(\mathbb{R}^d)$. We know that, for Q almost all λ, the equality $\sigma_\lambda = \sigma_Q$ is valid μ_λ a.e. while, in virtue of Lemma 3.13 that we apply to $\mu_\lambda \ll \mu$, we also have $\nabla_\mu u = \nabla_{\mu_\lambda} u$. Accordingly the equality $\widehat{\Delta_h^{\sigma_Q}}(u) = \widehat{\Delta_h^{\sigma_\lambda}}(u)$ holds μ_λ a.e. and we conclude that:

$$\Psi_h(\lambda_Q)(u) = \int \widehat{\Delta_h^{\sigma_Q}}(u)\, d\mu = \int \left(\int \widehat{\Delta_h^{\sigma_\lambda}}(u)\, d\mu_\lambda \right) dQ(\lambda) = \int \Psi_h(\lambda)(u)\, dQ(\lambda). \quad \square$$

We will be using Lemma 4.9 when the measure Q is concentrated on vector measures $\lambda \in \mathbf{N_1}(\mathbb{R}^d)$ which are associated to a family of curves. More precisely, being $T > 0$, we associate to any Lipschitz curve $\gamma \in \mathrm{Lip}_1([0,T]; \mathbb{R}^d)$ the vector measure λ_γ defined by

$$\langle \lambda_\gamma, \varphi \rangle := \int_0^T \langle \varphi(\gamma(t)), \dot\gamma(t) \rangle \, dt \tag{4.27}$$

where the derivative $\dot\gamma$ is Lebesgue a.e. defined in $[0,T]$. The length of the curve $L(\gamma) := \int_0^T |\dot\gamma(t)| dt$ is such that: $\|\lambda_\gamma\| \leq L(\gamma) \leq T$ with possibly a strict inequality; an equality prevents $\gamma(t)$ to go back and forth on a part of $\gamma([0,T])$. Besides we note that the distributional equality $-\operatorname{div} \lambda_\gamma = \delta_{\gamma(T)} - \delta_{\gamma(0)}$ ensures that $\lambda_\gamma \in \mathbf{N_1}(\mathbb{R}^d)$.

Next we consider the map $\Lambda : \gamma \in \mathrm{Lip}_1([0,T]; \mathbb{R}^d) \mapsto \lambda_\gamma \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$. It is continuous when $\mathrm{Lip}_1([0,T]; \mathbb{R}^d)$ is equipped with the uniform norm of $(C_0(\mathbb{R}^d))^d$ and $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$ with the weak-$*$ topology. However it is not injective due to the fact that the curve integral (4.27) is invariant under reparametrization. Then we will consider measures Q of the form $Q = \Lambda^\sharp(\nu)$, where ν is a suitable Borel measure on $\mathrm{Lip}_1([0,T]; \mathbb{R}^d)$ (a closed convex subset of the Banach space $C_0([0,T]; \mathbb{R}^d)$).

The possibility of decomposing a general $\lambda \in \mathbf{N_1}(\mathbb{R}^d)$ as $\lambda = \int \lambda_\gamma d\nu(\gamma)$, in which ν fulfils the mass preserving condition (4.23) of Lemma 4.9, is given by the celebrated Smirnov decomposition theorem [49, Thm C]. Here we use the precise formulation of [46]:

Theorem 4.10 (Smirnov decomposition). *Let $T > 0$ and $\lambda \in \mathbf{N_1}(\mathbb{R}^d)$ be given. Then, there exists a Borel measure $\nu \in \mathcal{M}_+(\mathrm{Lip}_1([0,T];\mathbb{R}^d))$ such that:*

(i) Inclusion: $\|\lambda_\gamma\| = L(\gamma)$ and $\gamma([0,T]) \subset \mathrm{spt}(\lambda)$ for ν-almost every γ;
(ii) Decomposition: $\lambda = \int \lambda_\gamma \, d\nu(\gamma)$, where the integral is intended in the weak sense $\langle \lambda, \psi \rangle = \int \langle \lambda_\gamma, \psi \rangle \, d\nu(\gamma), \ \forall \psi \in C_0(\mathbb{R}^d)^d$;
(iii) Mass preserving: $|\lambda| = \int |\lambda_\gamma| \, d\nu(\gamma)$ as elements of $\mathcal{M}_+(\mathbb{R}^d)$.

The following result is true for the above-mentioned subfamily of curves:

Lemma 4.11. *Let $\gamma \in \mathrm{Lip}_1([0,T];\mathbb{R}^d)$ be such that $\|\lambda_\gamma\| = L(\gamma)$. Then we have*

$$\Phi_h(\lambda_\gamma) \leq 2\, L(\gamma) \quad , \quad \Phi_0^+(\lambda_\gamma) = \lim_{h \to 0} \Phi_h(\lambda_\gamma) = 0. \tag{4.28}$$

Furthermore, for every $u \in \mathrm{Lip}_1(\mathbb{R}^d)$, it holds

$$\Psi_h(\lambda_\gamma)(u) \leq 2\, L(\gamma) \quad , \quad \lim_{h \to 0} \Psi_h(\lambda_\gamma)(u) = 0. \tag{4.29}$$

Proof. After reparametrization, we may assume that $|\dot\gamma(t)| = 1$ a.e. $t \in [0, L(\gamma)]$ while $\gamma(t) = \gamma(T)$ for $t \geq L(\gamma)$. The condition $\|\lambda_\gamma\| = L(\gamma)$ implies that $|\lambda_\gamma| = \gamma^\sharp \mathcal{L}_{[0,L(\gamma)]}$ while $\lambda_\gamma = \sigma_\gamma |\lambda_\gamma|$ where $\sigma_\gamma(\gamma(t)) = \dot\gamma(t)$. Accordingly, we have

$$\Phi_h(\lambda_\gamma) = F_h(|\lambda_\gamma|, \sigma_\gamma) = \sup_{u \in \mathrm{Lip}_1 \cap C^1} \int_0^T \Delta_h^\gamma(u)(t)\, dt, \tag{4.30}$$

where

$$\Delta_h^\gamma(u)(t) := \frac{u(\gamma(t) + h\dot\gamma(t)) - u(\gamma(t))}{h} - \langle \nabla u(\gamma(t)), \dot\gamma(t) \rangle. \tag{4.31}$$

The first inequality in (4.28) follows since $|\Delta_h^\gamma(u)(t)| \leq 2$. To prove the convergence of $\Phi_h(\lambda_\gamma)$ to zero, we decompose

$$\Delta_h^\gamma(u)(t) = A_h^\gamma(u)(t) + R_h^\gamma(u)(t)$$

where we have set:

$$A_h^\gamma(u)(t) := \frac{u(\gamma(t+h)) - u(\gamma(t))}{h} - \langle \nabla u(\gamma(t)), \dot\gamma(t) \rangle$$
$$R_h^\gamma(u)(t) := \frac{u(\gamma(t) + h\dot\gamma(t)) - u(\gamma(t+h))}{h}$$

Then, since u is 1-Lipschitz and recalling that $\gamma(t+h) = \gamma(T)$ whenever $t + h > T$, we get:

$$\int_0^T \Delta_h^\gamma(u)\, dt \leq \int_0^T A_h^\gamma(u)\, dt + \int_0^T \left| \frac{\gamma(t+h) - \gamma(t)}{h} - \dot\gamma(t) \right| dt.$$

Since γ is Lipschitz and a.e. differentiable, the second integral vanishes by dominated convergence and we are reduced to show that $\int_0^T A_h^\gamma(u)\, dt \to 0$. Note that this second integral does not depend on u, so that the estimate obtained below is uniform with respect to u. We notice that

$$\int_0^T \langle \nabla u(\gamma(t)), \dot\gamma(t) \rangle \, dt = u(\gamma(T)) - u(\gamma(0)) \, ,$$
$$\int_0^T \frac{u(\gamma(t+h)) - u(\gamma(t))}{h} \, dt = u(\gamma(T)) - \frac{1}{h} \int_0^h u(\gamma(t))\, dt.$$

Thus, since γ is also 1-Lipschitz, we get:

$$\int_0^T A_h^\gamma(u)(t)\,dt = u(\gamma(0)) - \frac{1}{h}\int_0^h u(\gamma(t))\,dt \le \frac{1}{h}\int_0^h t\,dt = \frac{h}{2},$$

whence (4.28) by sending $h \to 0$.

Let us show (4.29), keeping the same parametrization for the curve γ. Then, for every $u \in \mathrm{Lip}_1(\mathbb{R}^d)$, the tangential gradient of u with respect to $\mu_\gamma := |\lambda_\gamma|$ satisfies, for a.e. $t \in [0, L(\gamma)]$,

$$\big\langle \nabla_{\mu_\gamma} u(\gamma(t)), \dot\gamma(t)\big\rangle \;= (u\circ\gamma)'(t). \tag{4.32}$$

Indeed, at a.e. t both γ and $u\circ\gamma$ are differentiable and, u being Lipschitz, $|u(\gamma(t+h)) - u(\gamma(t) + h\dot\gamma(t))| = o(h)$; hence $(u\circ\gamma)'(t) = D_{\sigma_\gamma}u(\gamma(t))$, where $D_{\sigma_\gamma}u(y) := \lim_{h\to 0}\frac{u(y+h\sigma_\gamma(y)) - u(y)}{h}$ depends on the point $y = \gamma(t)$ only and is defined μ_γ-a.e.. On the other hand, for $\theta \in L^\infty_{\mu_\gamma}$, $\theta\sigma_\gamma \in \mathcal{T}_{\mu_\gamma}$ by locality, and (3.19) gives $\int \theta\big\langle \sigma_\gamma, \nabla_{\mu_\gamma}u\big\rangle\, d\mu_\gamma = \langle -\operatorname{div}(\theta\lambda_\gamma), u\rangle$, whose right-hand side equals $\int_0^T \theta(\gamma(t))\,(u\circ\gamma)'(t)\,dt = \int \theta\, D_{\sigma_\gamma}u\, d\mu_\gamma$ for smooth u by (4.27). Since θ is arbitrary, $\big\langle \sigma_\gamma, \nabla_{\mu_\gamma}u\big\rangle = D_{\sigma_\gamma}u\ \mu_\gamma$-a.e., which is (4.32). Accordingly (4.31) can be written as

$$\Delta_h^\gamma(u)(t) = \frac{u(\gamma(t) + h\dot\gamma(t)) - u(\gamma(t))}{h} - (u\circ\gamma)'(t).$$

Next, according to (4.18), we set:

$$\widehat{\Delta_h^\gamma}(u)(t) := \sup_{s\in(0,h]} |\Delta_s^\gamma(u)(t)|.$$

Clearly $\widehat{\Delta_h^\gamma}(u) \le 2$ and, by the a.e. differentiability of the Lipschitz function $u\circ\gamma$, we have the pointwise convergence $\widehat{\Delta_h^\gamma}(u)(t) \to 0$ as $h \to 0$ holding a.e. Accordingly we conclude by dominated convergence that $\Psi_h(\lambda_\gamma)(u) = \int \widehat{\Delta_h^\gamma}(u)(t)\,dt \to 0$. □

4.3. **Proof of Theorem 4.1.**

Proof of the assertion (i) of Theorem 4.1. We proceed in three steps, successively showing that:

(1) $\Phi_0^+(\lambda) = 0$ when $\lambda \in \mathbf{F}_1(\mathbb{R}^d)$;
(2) $F_0^+(\mu,\sigma) \le 2\int |\sigma_N|\,d\mu$ for any pair $(\mu,\sigma) \in \mathcal{M}_+ \times (L^1_\mu)^d$;
(3) $F_0^-(\mu,\sigma) \ge 2\int |\sigma_N|\,d\mu$.

Step 1 We use Smirnov and Lebesgue's dominated convergence. Let λ be an element of $\mathbf{F}_1(\mathbb{R}^d)$. Then we know from Proposition 2.9 (assertion (ii)) that there exists a sequence $\lambda_n \in \mathbf{N}_1(\mathbb{R}^d)$ such that $\|\lambda_n - \lambda\| \to 0$. Using the Lipschitz property of the assertion (iv) of Lemma 4.8, we have that $\Phi_0^+(\lambda) \le \Phi_0^+(\lambda_n) + 3\|\lambda_n - \lambda\|$. Thus we only need to show that $\Phi_0^+(\lambda) = 0$ for every $\lambda \in \mathbf{N}_1(\mathbb{R}^d)$. Since the distributional divergence of such a vector measure λ is a signed measure of bounded total variation, we can use the Smirnov decomposition (Theorem 4.10). We obtain a decomposition of the form $\lambda = \int \lambda_\gamma\, d\nu(\gamma)$ where $\nu \in \mathcal{M}_+(\mathrm{Lip}_1([0,T];\mathbb{R}^d))$. Let $Q = \Lambda^\sharp(\nu)$ where the map $\Lambda : \gamma \in \mathrm{Lip}_1([0,T];\mathbb{R}^d) \mapsto \lambda_\gamma \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ is continuous when $\mathrm{Lip}_1([0,T];\mathbb{R}^d)$ is equipped with the uniform norm of $(C_0(\mathbb{R}^d))^d$ and $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)$ with the weak-$*$ topology.

From the mass preserving condition (iii) of Theorem 4.10, we infer that Q fulfils the condition (4.23) of Lemma 4.9, while $\lambda = \int \zeta \, dQ(\zeta)$ with $\int \|\zeta\| \, dQ(\zeta) = \int L(\gamma) \, d\nu(\gamma) < +\infty$. Rewriting the subadditivity property (4.24) in terms of ν yields

$$\Phi_h(\lambda) \;\leq\; \int \Phi_h(\lambda_\gamma) \, d\nu(\gamma).$$

Since the equality $\|\lambda_\gamma\| = L(\gamma)$ holds for ν-a.e. γ, we can apply Lemma 4.11. Then (4.28) implies the pointwise convergence $\Phi_h(\lambda_\gamma) \to 0$ ν-a.e. whereas $\Phi_h(\lambda_\gamma) \leq 2L(\gamma)$. Recalling that $\int L(\gamma) \, d\nu(\gamma) < +\infty$, we conclude that $\Phi_h(\lambda) \to 0$ by using dominated convergence. Therefore $\Phi_0^+(\lambda) = 0$ for every $\lambda \in \mathbf{F_1}(\mathbb{R}^d)$ and the proof of Step 1 is complete.

Step 2 Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be given and denote again

$$\mathcal{T}_\mu := \{\sigma \in (L^1_\mu)^d \,:\, \sigma(x) \in T_\mu(x) \; \mu \text{ a.e}\}.$$

We are going to show the inclusion

$$\mathcal{T}_\mu \subset \mathcal{F}_\mu := \{\sigma \in (L^1_\mu)^d : F_0^+(\mu, \sigma) = 0\}. \tag{4.33}$$

Since $F_0^+(\mu, \cdot)$ is Lipschitz continuous (see Lemma 4.8.(iv)), $\mathcal{F}_\mu$ is a closed subset of $(L^1_\mu)^d$. From Step 1, $\mathcal{F}_\mu$ contains every $\sigma \in \mathcal{T}_\mu$ such that $|\sigma| \equiv t > 0$ is constant. Indeed $\lambda := \sigma\mu \in \mathbf{F_1}(\mathbb{R}^d)$ has polar decomposition $\lambda = \frac{\sigma}{t}(t\mu)$, so that, by the positive homogeneity of Lemma 4.8.(iii) (in both arguments), $0 = \Phi_0^+(\lambda) = F_0^+(t\mu, \sigma/t) = F_0^+(\mu, \sigma)$. Next, let $\sigma \in \mathcal{T}_\mu$ be such that $|\sigma|$ takes finitely many values $t_1, \ldots, t_m$, and set $B_j := \{|\sigma| = t_j\}$. By the locality property (3.2), $\sigma \mathbb{1}_{B_j} \in \mathcal{T}_\mu$, hence $\sigma \in \mathcal{T}_{\mu \llcorner B_j}$ with $|\sigma| \equiv t_j$ $\mu \llcorner B_j$-a.e. (if $t_j = 0$, then $F_0^+(\mu \llcorner B_j, \sigma) = 0$ trivially). Using the subadditivity and the localization properties (ii) and (i) of Lemma 4.8, we get

$$F_0^+(\mu, \sigma) \leq \sum_j F_0^+(\mu, \sigma \, \mathbb{1}_{B_j}) = \sum_j F_0^+(\mu \llcorner B_j, \sigma) = 0.$$

Finally, an arbitrary $\sigma \in \mathcal{T}_\mu$ is the L^1_μ-limit of such fields: for $n \in \mathbb{N}$, let $\theta_n := \frac{1}{|\sigma|} \min\left\{n, \frac{1}{n}\lfloor n|\sigma|\rfloor\right\} \mathbb{1}_{\{\sigma \neq 0\}}$. Then $0 \leq \theta_n \leq 1$, so that $\sigma_n := \theta_n \sigma \in \mathcal{T}_\mu$ by (3.2), $|\sigma_n|$ takes values in $\{0, \frac{1}{n}, \frac{2}{n}, \ldots, n\}$, and $\sigma_n \to \sigma$ in L^1_μ by dominated convergence. Since $\mathcal{F}_\mu$ is closed, (4.33) follows.

Eventually let us consider a general vector field $\sigma \in (L^1_\mu)^d$ and its decomposition $\sigma = \sigma_T + \sigma_N$ into its tangential part $\sigma_T = P_\mu \sigma$ and its orthogonal part $\sigma_N = P_\mu^\perp \sigma$. Then, recalling that $F_0^+(\mu, \cdot)$ is 2-Lipschitz, we deduce that $F_0^+(\mu, \sigma) \leq F_0^+(\mu, \sigma_T) + 2\int |\sigma_N| \, d\mu$, whence $F_0^+(\mu, \sigma) \leq 2\int |\sigma_N| \, d\mu$, since we already proved that $F_0^+(\mu, \sigma_T) = 0$.

Step 3 We now show that $F_0^-(\mu, \sigma) \geq 2\int |\sigma_N| \, d\mu$. To that aim, we use the variational representation (3.10) of the conjugate of the functional G_σ (defined in (3.9)) that we apply to the measure $D_h^\sigma \mu = \frac{\mu_h^\sigma - \mu}{h}$. Noticing that

$$F_h(\mu, \sigma) = \sup_{u \in C^1 \cap \mathrm{Lip}_1} \int \Delta_h^\sigma(u) \, d\mu \;\geq\; \sup_{u \in \mathcal{D} \cap \mathrm{Lip}_1} \{\langle D_h^\sigma \mu, u\rangle - G_\sigma(u)\} \;=\; G_\sigma^*(D_h^\sigma \mu).$$

We deduce that

$$F_h(\mu,\sigma) \;\geq\; \min\left\{\int_{\mathbb{R}^d} |\lambda - \sigma\,\mu| \;:\; -\operatorname{div}\lambda = D_h^\sigma\mu\right\}$$
$$= \;\min\left\{\int |w-\sigma|\,d\mu + \int |\lambda^s| \;:\; (w,\lambda^s)\in\mathcal{S}_h\right\}, \tag{4.34}$$

where $\mathcal{S}_h := \big\{\,(w,\lambda^s)\in (L^1_\mu)^d\times\mathcal{M}^d \,\boldsymbol{:}\, -\operatorname{div}(w\mu+\lambda^s) = D_h^\sigma\mu\;,\;\lambda^s\perp\mu\,\big\}$. Since the measure $D_h^\sigma\mu$ belongs to $\boldsymbol{Æ}(\mathbb{R}^d)$, the divergence condition in $\mathcal{S}_h$ implies that $w\mu+\lambda^s$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$. By the locality property (2.12), we infer that $w\in\mathcal{T}_\mu$, while λ^s is an element of $\mathbf{F_1}(\mathbb{R}^d)$ which satisfies $-\operatorname{div}\lambda^s = D_h^\sigma\mu + \operatorname{div} w\mu$. As such, (2.11) yields the lower bound

$$\|\lambda^s\|_{\mathcal{M}(\mathbb{R}^d;\mathbb{R}^d)} \;\geq\; \|D_h^\sigma\mu + \operatorname{div} w\mu\|_{\mathrm{KR}}.$$

Accordingly, for any given $u\in \mathrm{Lip}_1(\mathbb{R}^d)\cap\mathcal{D}(\mathbb{R}^d)$, we get the following inequalities:

$$\begin{aligned}
\int|\lambda^s| &\geq \int\left(\frac{u(x+h\sigma)-u(x)}{h} - \langle\nabla u, w\rangle\right)d\mu\\
&= \int\left(\frac{u(x+h\sigma)-u(x)}{h} - \langle\nabla u,\sigma\rangle\right)d\mu + \int\langle\nabla u,\sigma-w\rangle\,d\mu\\
&= \int\left(\int_0^1 \langle\nabla u(x+s\,h\sigma(x)) - \nabla u(x),\sigma(x)\rangle\,ds\right)\mu(dx) + \int\langle\nabla u,\sigma-w\rangle\,d\mu\\
&\geq \int\langle\nabla u,\sigma-w\rangle\,d\mu - \;\omega_u(h), \qquad \omega_u(h) := \int\min\Big\{2,\;\frac{h}{2}\,\mathrm{Lip}(\nabla u)\,|\sigma|\Big\}|\sigma|\,d\mu,
\end{aligned}$$

where we used that $|\nabla u(x+s\,h\,\sigma)-\nabla u(x)|\leq \min\{2,\mathrm{Lip}(\nabla u)\,sh|\sigma|\}$. Note that $\omega_u(h)\to 0$ as $h\to 0$ by dominated convergence, since $\sigma\in L^1_\mu$. Inserting the last inequality in (4.34), we deduce that for every fixed $u\in\mathrm{Lip}_1\cap\mathcal{D}(\mathbb{R}^d)$, we have

$$F_h(\mu,\sigma)\geq \inf_{w\in\mathcal{T}_\mu}\int\left(|w-\sigma| + \langle\nabla u,\sigma-w\rangle\right)d\mu - \omega_u(h),$$

where $\omega_u(h)$ does not depend on w. Thus by passing to the limit inf in $h\to 0$, we are led to the following lower bound:

$$F_0^-(\mu,\sigma)\geq \inf_{w\in\mathcal{T}_\mu}\int\left(|w-\sigma| + \langle\nabla u,\sigma-w\rangle\right)d\mu.$$

Since the constraint $w\in\mathcal{T}_\mu$ is local, namely $w(x)\in T_\mu(x)$ μ a.e., every competitor w satisfies the pointwise inequality $|w(x)-\sigma(x)| + \langle\nabla u(x),\sigma(x)-w(x)\rangle \geq m(\sigma(x),\nabla u(x),T_\mu(x))$, where for every $(\sigma,z,T)\in\mathbb{R}^d\times\mathbb{R}^d\times\mathcal{G}_d$, we denote the infimum value function

$$m(\sigma,z,T) := \inf_{w\in T}\{|w-\sigma| + \langle z,\sigma-w\rangle\}.$$

Integrating and taking the infimum in w we get

$$F_0^-(\mu,\sigma)\geq\int m(\sigma(x),\nabla u(x),T_\mu(x))\,d\mu.$$

Let us decompose $z = z_T + z_N$ where z_T, z_N denote the orthogonal projections of z on the subspace T and its orthogonal $N = T^\perp$ respectively. In the same way, we write $\sigma =$

$\sigma_T + \sigma_N$. It is convenient to rewrite the infimum problem in term of $\zeta := w - \sigma_T$. After easy manipulations we obtain:

$$m(\sigma, z, T) = \inf_{\zeta \in T} \{|\zeta - \sigma_N| - \langle z_T, \zeta \rangle\} + \langle z_N, \sigma_N \rangle.$$

Then, handling this finite dimensional convex program, we conclude that

$$m(\sigma, z, T) = \begin{cases} |\sigma_N| \sqrt{1 - |z_T|^2} + \langle z_N, \sigma_N \rangle & \text{if } |z_T| \leq 1, \\ -\infty & \text{otherwise.} \end{cases}$$

We end up with the following integral inequality holding for every $u \in \mathrm{Lip}_1 \cap \mathcal{D}(\mathbb{R}^d)$:

$$F_0^-(\mu, \sigma) \geq \int \left(|\sigma_N| \sqrt{1 - |\nabla_\mu u|^2} + \langle \nabla u, \sigma_N \rangle \right) d\mu. \tag{4.35}$$

To conclude this final step, we appeal now to (3.12) to get a sequence $(u_n)_n$ in $\mathrm{Lip}_1 \cap \mathcal{D}(\mathbb{R}^d)$ such that

$$u_n \overset{*}{\rightharpoonup} 0, \qquad \nabla u_n \to \frac{\sigma_N}{|\sigma_N|} \qquad \mu \text{ a.e. on } \{\sigma_N \neq 0\}.$$

In particular, $\nabla_\mu u_n = P_\mu(\nabla u_n)$ converges to 0 μ a.e. on $\{\sigma_N \neq 0\}$. By applying (4.35) to $u = u_n$ and after passing to the limit by dominated convergence, we conclude that $F_0^-(\mu, \sigma) \geq 2 \int |\sigma_N| \, d\mu$. The proof of the assertion (i) of Theorem 4.1 is now complete. □

Proof of the assertion (ii) of Theorem 4.1. Let us fix $u \in \mathrm{Lip}(\mathbb{R}^d)$. Without loss of generality, we can assume that $\mathrm{Lip}(u) \leq 1$. We begin by proving that $\Delta_h^\sigma(u) \to 0$ in L^1_μ for every $\sigma \in \mathcal{T}_\mu$ of the form $|\sigma| = \mathbb{1}_B$, where B is a Borel set of finite μ-measure. Then, setting $\overline{\lambda} = \sigma \, \mu$ and recalling (4.17), we are reduced to show that $\limsup_{h \to 0_+} \Psi_h(\overline{\lambda})(u) = 0$. The condition $\sigma \in \mathcal{T}_\mu$ implies that $\overline{\lambda} \in \mathbf{F_1}(\mathbb{R}^d)$. Then, by the density of $\mathbf{N_1}(\mathbb{R}^d)$ in $\mathbf{F_1}(\mathbb{R}^d)$, we can assume that $\overline{\lambda} \in \mathbf{N_1}(\mathbb{R}^d)$. Indeed by (4.20), we have for every $\lambda \in \mathbf{N_1}(\mathbb{R}^d)$

$$\limsup_{h \to 0_+} \Psi_h(\overline{\lambda})(u) \leq \limsup_{h \to 0_+} \Psi_h(\lambda)(u) + 3 \|\lambda - \overline{\lambda}\|.$$

Thus, assuming now that $\overline{\lambda} \in \mathbf{N_1}(\mathbb{R}^d)$, we can apply Smirnov (Theorem 4.10) in the same way as we did in the proof of the assertion (i): we get a decomposition of the form

$$\overline{\lambda} \; = \; \int \lambda_\gamma \, d\nu(\gamma) \; = \int \lambda \, dQ(\lambda) \; ,$$

where $Q = \Lambda^\sharp(\nu)$ is a Borel measure concentrated on $\mathbf{N_1}(\mathbb{R}^d)$ which satisfies the conditions of the assertion (iii) of Lemma 4.9. In view of (4.25), we deduce that:

$$\limsup_{h \to 0} \Psi_h(\overline{\lambda})(u) \; = \limsup_{h \to 0} \int \Psi_h(\lambda_\gamma)(u) \, d\nu(\gamma).$$

By (4.29), we know that $\Psi_h(\lambda_\gamma)(u) \leq 2 \, L(\gamma)$ while $\Psi_h(\lambda_\gamma)(u) \to 0$. Therefore, recalling that $\int L(\gamma) \, d\nu(\gamma) < +\infty$, by dominated convergence, we conclude that

$$\Psi_h(\overline{\lambda})(u) \; = \; \int \widehat{\Delta_h^\sigma}(u) \, d\mu \; \to 0.$$

By applying Fatou's Lemma, it follows that $\liminf_{h\to 0} \widehat{\Delta_h^\sigma}(u) \to 0$ μ a.e. Observing that the dependence in h is monotone, we conclude that $\widehat{\Delta_h^\sigma}(u)(x)$, hence $\Delta_h^\sigma(u)(x)$, do converge pointwise to 0 at μ almost all $x \in \mathbb{R}^d$. It remains to remove the restriction on $|\sigma|$. For $t > 0$ one checks directly on (4.4) that $\Delta_h^{t\sigma}(u) = t\,\Delta_{h|t|}^\sigma(u)$ with, so that, for every μ-measurable $\theta \geq 0$ which is finite μ a.e.,

$$\limsup_{h\to 0} |\Delta_h^{\theta\sigma}(u)|(x) \;=\; \theta(x)\; \limsup_{h\to 0} |\Delta_h^\sigma(u)(x)|\,,$$

whose right-hand side vanishes by the first part. Let now $\sigma \in \mathcal{T}_\mu$ be arbitrary and put $B_k := \{|\sigma| > \frac{1}{k}\}$, so that $\mu(B_k) \leq k \int |\sigma|\, d\mu < +\infty$. The field $\sigma_k := \frac{\sigma}{|\sigma|}\mathbb{1}_{B_k}$ is a bounded μ-measurable selection of T_μ supported on a set of finite measure, hence belongs to $\mathcal{T}_\mu$ by (3.2), and $|\sigma_k| = \mathbb{1}_{B_k}$; moreover $\sigma = \theta_k\,\sigma_k$ on B_k with $\theta_k := |\sigma|$ finite μ a.e. Therefore $\Delta_h^\sigma(u) \to 0$ μ a.e. on B_k for every k, hence μ a.e. on $\{\sigma \neq 0\} = \bigcup_k B_k$, and trivially on $\{\sigma = 0\}$. Finally the convergence also holds in L^1_μ, by dominated convergence, since $|\Delta_h^\sigma(u)| \leq 2|\sigma| \in L^1_\mu$. The assertion (ii) of Theorem 4.1 is proved. □

Proof of the assertion (iii) of Theorem 4.1. We prove that differentiability along countably many tangent fields upgrades to differentiability along the whole subspace by an equicontinuity argument on the unit sphere. Since $\mathcal{T}_\mu$ is a closed subspace of the separable space $(L^1_\mu)^d$, pick a countable dense family $(\sigma_k) \subset \mathcal{T}_\mu$; by the description of the essential union in Subsection 2.2, $T_\mu(x) = \overline{\{\,\sigma_k(x) : k \in \mathbb{N}\,\}}$ holds μ a.e. For $h > 0$ and $v \in T_\mu(x)$ set

$$g_h(v) \;:=\; \frac{u(x+hv) - u(x)}{h} - \langle \nabla_\mu u(x), v\rangle, \qquad \text{so that} \qquad g_h(tv) = t\, g_{th}(v) \quad (t > 0).$$

What precedes says that $g_h(\sigma_k(x)) \to 0$ as $h \to 0^+$ for μ a.e. x. Discarding the countably many exceptional sets, fix x at which this holds for every k and $T_\mu(x) \neq \{0\}$, and let S be the unit sphere of $T_\mu(x)$, which is compact. On S the g_h are Lipschitz with constant $2\mathrm{Lip}(u)$, independent of h, and by the homogeneity above they tend to 0 on the dense subset $\{\,\sigma_k(x)/|\sigma_k(x)| : \sigma_k(x) \neq 0\,\}$ of S. An equicontinuous family on a compact set which converges pointwise on a dense subset converges uniformly, so that $\sup_S |g_h| \to 0$ as $h \to 0^+$. Finally, for every $v \in T_\mu(x) \setminus \{0\}$,

$$\frac{u(x+v) - u(x) - \langle \nabla_\mu u(x), v\rangle}{|v|} \;=\; g_{|v|}\Big(\frac{v}{|v|}\Big) \;\longrightarrow\; 0 \qquad \text{as } v \to 0,$$

which is (4.8). □

5. Relations with other tangent bundles

5.1. **About some variants.** Different tangent bundles to a measure μ can be constructed by choosing suitable subspaces $\mathcal{W} \subset (L^1_\mu(\mathbb{R}^d))^d$ to which one can associate the μ-measurable multi-function $W_\mu(x) := \mu - \operatorname{ess} \bigcup\{\sigma;\ \sigma \in \mathcal{W}\}$ detailed in Section 2.2. By construction we have $\mathcal{W} \subset \big\{\,\sigma \in (L^1_\mu(\mathbb{R}^d))^d : \sigma(x) \in W_\mu(x)\ \mu \text{ a.e.}\,\big\}$. However considering $W_\mu(x)$ is merely useful when the latter inclusion is an equality. In virtue of Lemma 2.3, this is true if and only if $\mathcal{W}$ is closed and satisfies the stability condition (2.3). This is the case when taking $\mathcal{W}$ to be $\mathcal{T}_\mu$ as in (3.1). Other options are possible taking $\mathcal{W}$ to be one of the following subspaces:

(i) $\mathcal{S}_\mu := \big\{\sigma \in (L^1_\mu(\mathbb{R}^d))^d : \sigma\,\mu \in \mathbf{N_1}(\mathbb{R}^d)\big\}$;

(ii) $\mathcal{V}_\mu := \left\{ \frac{d\lambda}{d\mu} : \lambda \in \mathbf{N_1}(\mathbb{R}^d) \right\}$;

(iii) $\mathcal{T}_\mu^{(p)} := \left\{ \sigma \in (L^{p'}_\mu(\mathbb{R}^d))^d : \ -\operatorname{div}(\sigma\,\mu) \in L^{p'}_\mu(\mathbb{R}^d) \right\}$, with $1 < p < \infty$ and $p' = \frac{p}{p-1}$.

Despite the fact that all theses spaces satisfy (2.3), there are not closed subspaces of $(L^1_\mu(\mathbb{R}^d))^d$ as it is the case for $\mathcal{T}_\mu$. However we may apply Proposition 2.3 to their closure to obtain the equality

$$\overline{\mathcal{W}} \ = \ \left\{ \sigma \in (L^1_\mu(\mathbb{R}^d))^d : \sigma(x) \in W_\mu(x) \ \mu \text{ a.e.} \right\}.$$

Accordingly we can consider the associated tangent bundles, in particular:

(i) $S_\mu(x)$ (appearing in [33]) will be called the *Smirnov* bundle;

(ii) $V_\mu(x)$ coincides with the *Alberti-Marchese* decomposability bundle, this as a consequence of [2, Corollary 6.5];

(iii) $T_\mu^{(p)}$ is, for $p \in (1, \infty)$, the tangent bundle introduced in [20], with the motivation of defining the Sobolev space $H^p_\mu(\mathbb{R}^d)$.

These bundles as related as follows.

Proposition 5.1. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure. All statements below are intended for μ almost $x \in \mathbb{R}^d$.*

(i) $V_\mu(x) = T_\mu(x)$;

(ii) $S_\mu(x) \subset T_\mu(x)$ with possibly a strict inclusion;

(iii) If μ is finite with compact support, then $T_\mu^{(p)}(x) \subset S_\mu(x)$ while $T_\mu^{(p)}(x)$ is non decreasing with respect to $p \in (1, +\infty)$.

Proof. By Lemma 2.3, the equality (i) is true if we show the following equality between subspaces of $(L^1_\mu(\mathbb{R}^d))^d$:

$$\overline{\mathcal{V}_\mu} = \mathcal{T}_\mu \, . \tag{5.1}$$

Let $\tau \in \mathcal{V}_\mu$, i.e. there exists $\lambda \in \mathcal{M}^d$ such that $\operatorname{div} \lambda \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ and $\frac{d\lambda}{d\mu} = \tau$. Then we write λ in the form $\lambda = \chi\, m$ where

$$m = \mu + |\lambda^s| \quad , \quad \chi = \begin{cases} \tau & \mu - \text{a.e.} \\ \frac{d\lambda^s}{d|\lambda^s|} & \lambda^s - \text{a.e.} \end{cases},$$

where λ^s denotes the singular part of λ with respect to μ. Since $\operatorname{div}(\chi\, m)$ belongs to $\mathcal{M}_{0,1}(\mathbb{R}^d) \subset \mathcal{Æ}(\mathbb{R}^d)$, we have $\chi(x) \in T_m(x)$ m a.e.. By applying (3.4) to the decomposition $m = \mu + |\lambda^s|$, we infer that $T_m(x) = T_\mu(x)$ holds μ a.e.. Combined with the equality $\chi = \tau$ holding μ a.e., we conclude that $\tau \in \mathcal{T}_\mu$, hence $\overline{\mathcal{V}_\mu} \subset \mathcal{T}_\mu$ since the latter set is closed.

Conversely, let $\tau \in \mathcal{T}_\mu$. As $\lambda := \tau\,\mu$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$, by the assertion (ii) of Proposition 2.9, there exists a sequence (λ_n) in $\mathbf{N_1}(\mathbb{R}^d)$ such that $\|\lambda_n - \lambda\|_{\mathcal{M}^d} \to 0$. This means that the Lebesgue-Nikodym decomposition $\lambda_n = \tau_n\,\mu + \lambda_n^s$ satisfies

$$\tau_n \to \tau \quad \text{in } L^1_\mu \quad , \quad \|\lambda_n^s\|_{\mathcal{M}^d} \to 0. \tag{5.2}$$

In other words, every tangent field is an L^1_μ-limit of densities of normal 1-currents. By the definition of $\mathcal{V}_\mu$, we have $\tau_n \in \mathcal{V}_\mu$ whence $\tau \in \overline{\mathcal{V}_\mu}$ and $T_\mu \subset \overline{V_\mu}$. The desired equality (5.1) follows and the assertion (i) is established.

The assertion (ii) is straightforward from the inclusion $\mathcal{S}_\mu \subset \mathcal{T}_\mu$. The strict inclusion is shown in the example 5.3 below.

For the assertion (iii), note first that $\mathcal{T}_\mu^{(p)} \subset \mathcal{T}_\mu^{(q)}$ whenever $\mu(\mathbb{R}^d) < +\infty$ and $1 < p \leq q < \infty$, since $L_\mu^{p'} \subset L_\mu^{q'}$; this gives the monotonicity in p. Next, let $\sigma \in \mathcal{T}_\mu^{(p)}$, so that $-\operatorname{div}(\sigma\mu) = g\,\mu$ with $g \in L_\mu^{p'} \subset L_\mu^1$. Testing against $\varphi_n := \varphi(\cdot/n)$, where $\varphi \in \mathcal{D}(\mathbb{R}^d)$ equals 1 on B_1, and letting $n \to \infty$ yields $\int g\,d\mu = \lim_n \int \langle \sigma, \nabla\varphi_n\rangle\, d\mu = 0$ by dominated convergence. Hence $g\mu \in \mathcal{M}_{0,1}(\mathbb{R}^d)$ and $\sigma\mu \in \mathbf{N_1}(\mathbb{R}^d)$, i.e. $\sigma \in \mathcal{S}_\mu$. This shows $\mathcal{T}_\mu^{(p)} \subset \mathcal{S}_\mu$, whence $T_\mu^{(p)}(x) \subset S_\mu(x)$ μ-a.e. by Proposition 2.3. □

Remark 5.2. Amongst all tangent bundles appearing in Proposition 5.1, only $T_\mu(x) = V_\mu(x)$ satisfies the crucial locality property (3.3), as checked in the 1d example 5.3 below (see also [9, 33]). This observation makes it suitable for comparison with the Preiss tangent space and ultimately for deriving rectifiability properties.

Example 5.3. Take μ to be the Lebesgue measure on $\mathbb{R}$ and $\nu = \mu \llcorner C$ where C is a fat compact Cantor subset ($\mu(C) > 0$ and C has empty interior). Then, as the property (3.3) is true for T_μ, $T_\nu(x) = \mathbb{R}$ holds ν a.e.. Take now $\sigma \in L_\mu^1$ such that $\operatorname{div}(\sigma\nu) = (\sigma\,\mathbb{1}_C)'$ is a finite measure. Then $u = \sigma\,\mathbb{1}_C$ can be identified to a compactly supported function in $BV(\mathbb{R})$. Since u vanishes on the complement of C, an open dense subset of $\mathbb{R}$, we deduce that $u = 0$, hence $\sigma = 0$ a.e. on C. Therefore, according to the definition of the Smirnov bundle $\mathcal{S}_\nu$ given above, $S_\nu(x) = \{0\}$ holds ν a.e.. The same degeneracy occurs for the bundles $T_\nu^{(p)}$ which are contained in S_ν.

5.2. **The differentiability bundle.** Here we shortly explain why V_μ introduced before is also a differentiability bundle. Let us say that a function $u : \mathbb{R}^d \to \mathbb{R}$ is *differentiable at x with respect to* a linear subspace $W \subset \mathbb{R}^d$ if, for some linear form L on W,

$$u(x+v) - u(x) \;=\; L(v) + o(|v|) \qquad \text{as } W \ni v \to 0. \tag{5.3}$$

In the remarkable paper [2], Alberti and Marchese proved that every Lipschitz function is differentiable μ a.e. with respect to $V_\mu(x)$ and, in addition, that V_μ is the largest such bundle. This means that any bundle $W(x)$ such that (5.3) holds μ-a.e. for every Lipschitz u satisfies the inclusion $W(x) \subset V_\mu(x)$ for μ-a.e. x. We recover the differentiability along T_μ in assertion (iii) of Theorem 4.1, but its maximality so far only follows through the deep theory of [2] and Proposition 5.1. In order to produce a self-contained proof of this maximality, we construct along any prescribed direction field which is not tangent to T_μ, a Lipschitz function which is not differentiable on a set of arbitrarily large μ-mass. The construction is short and uses only the approximation property (3.12) of $\mathcal{N}_\mu$; the result is weaker than [2, Prop. 4.4 and 4.5], which shows that the set of such functions is large in the sense of Baire.

Proposition 5.4. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a finite measure and let $w \in (L_\mu^\infty)^d$ be a unit vector field such that $w(x) \notin T_\mu(x)$ for μ-a.e. x. Then, for every $\varepsilon > 0$, there exist a compact set K with $\mu(\mathbb{R}^d \setminus K) \leq \varepsilon$ and a Lipschitz function $f : \mathbb{R}^d \to \mathbb{R}$ such that the directional derivative $\partial_{w(x)} f(x)$ fails to exist at μ a.e. $x \in K$.*

Proof. We write $w_T := P_\mu w$ and $w_N := P_\mu^\perp w$ for the tangential and normal parts of w and define

(i) the tangential displacement $t_h := \mathrm{id} + hw_T$,
(ii) the full displacement $z_h := \mathrm{id} + hw = t_h + hw_N$,
(iii) the normal increment

$$N_h u := \frac{u(z_h) - u(t_h)}{h}, \qquad \text{so that} \qquad |N_h u| \le \mathrm{Lip}(u)\,|w_N|.$$

Reduction to the normal fields. Since μ is finite and $w \in (L^\infty_\mu)^d$, the field w_T belongs to $\mathcal{T}_\mu$ by (3.2). Splitting the difference quotient of u along w as $\Delta_h^w u - N_h u + \Delta_h^{w_T} u$, we get from (ii) of Theorem 4.1 that for every $u \in \mathrm{Lip}(\mathbb{R}^d)$, one gets $\Delta_h^w u - N_h u \to \langle \nabla_\mu u, w_T\rangle$ in L^1_μ. We conclude that, at μ a.e. x,

$$\partial_{w(x)} u(x) \text{ exists} \qquad \Longleftrightarrow \qquad \lim_{h\to 0^+} N_h u(x) \text{ exists.} \tag{5.4}$$

It is therefore enough to construct $u \in \mathrm{Lip}(\mathbb{R}^d)$ whose normal increment oscillates on a compact set K as in the statement.

Set-up. By assumption $w_N \neq 0$ μ a.e., so the unit normal field $\sigma := w_N/|w_N|$ is well defined. Fix $\delta \in (0,1]$ such that $\mu(\{|w_N| < \delta\}) < \varepsilon/2$, and introduce the smallness constant

$$\varepsilon_0 \;:=\; \frac{\delta}{8} \;\le\; \tfrac{1}{8}.$$

By Proposition 3.8 there is a sequence $(u_k)_k$ of smooth 1-Lipschitz functions converging weakly-$*$ to zero in $\mathrm{Lip}(\mathbb{R}^d)$, hence uniformly to zero on compact sets, and such that $\nabla u_k \to \sigma$ μ a.e.. By Egorov's and Lusin's theorems together with inner regularity there is a compact set $K \subset \{|w_N| \ge \delta\}$ with $\mu(\mathbb{R}^d \setminus K) \le \varepsilon$ on which σ is continuous and $\nabla u_k \to \sigma$ uniformly. By Tietze's theorem, followed by a projection onto the unit ball, we extend σ to a continuous field on $\mathbb{R}^d$ (still denoted σ) with $|\sigma| \le 1$. Writing $K^s := \{\mathrm{dist}(\cdot, K) \le s\}$, we let ω be a modulus of continuity for σ on the compact set K^1. Throughout, $\|\cdot\|_\infty$ denotes the supremum norm on K^1, so that $\|u_k\|_\infty \to 0$.

Construction of a candidate non-differentiable function. Set $s_0 = h_0 = 1$ and define $s_r, h_r > 0$ inductively: given s_{r-1}, h_{r-1}, choose successively

(a) a function $v_r := u_k$ with k so large that

$$\|v_r\|_\infty \le 2^{-r} \min\{s_{r-1}, h_{r-1}\} \qquad \text{and} \qquad \sup_K |\nabla v_r - \sigma| \le \varepsilon_0 2^{-(r+1)}.$$

This makes v_r small and ∇v_r close to σ on K, with an error decreasing geometrically.
(b) A scale $s_r \le s_{r-1}/4$ small enough so that the previous estimate still holds on s_r-enlargement of K up to a factor 2, i.e.

$$|\nabla v_r - \sigma| \le \varepsilon_0 2^{-r} \quad \text{on } K^{s_r}.$$

This is made possible by continuity of $\nabla v_r - \sigma$ and compactness of K.
(c) A length scale h_r, at which the difference quotients will be read, such that

$$h_r \;\le\; \min\left\{ h_{r-1}\,,\; \frac{s_r}{\sqrt{2}} \right\}.$$

We localise each v_r at its own scale. Let χ_r be a smooth cut-off of $K^{s_{r-1}/2}$ supported in $K^{s_{r-1}}$, whose transition layer costs a gradient $\|\nabla\chi_r\|_\infty \le 3/s_{r-1}$. By (a), the amplitude of v_r compensates this cost geometrically:

$$\|v_r\,\nabla\chi_r\|_\infty \;\le\; 3\cdot 2^{-r}. \tag{5.5}$$

Define the candidate u and its associated partial sums

$$u := \sum_{r\ge 1} (-1)^{r+1}\,\chi_r v_r, \qquad u_N := \sum_{r\le N} (-1)^{r+1}\chi_r v_r.$$

Using (a) once more, the series above converge uniformly, and u_N is C^1 for each N.

Lipschitzianity. We first estimate the gradient of u_N. To derive the bound on K, let $x\in K$. In particular $\chi_r\equiv 1$ near x for every $r\le N$, so that

$$\nabla u_N(x) = \sigma(x)\mathbf{1}_{\{N \text{ odd}\}} + \sum_{r\le N}(-1)^{r+1}(\nabla v_r(x)-\sigma(x)).$$

Since $K\subset K^{s_r}$ for every r, property (b) gives $|\nabla u_N(x)|\le |\sigma(x)| + \varepsilon_0\sum_{r=1}^N 2^{-r}\le 2$. Let us now compute the gradient outside of K. For this, let us choose $x\in\mathbb{R}^d\setminus K$ with $\operatorname{dist}(x,K)\le 1$. Let $M\in\mathbb{N}$ be such that $x\in K^{s_{M-1}}\setminus K^{s_M}$. In particular, we have $\chi_r\equiv 1$ near x for $r\le M-1$ and $\chi_r\equiv 0$ near x for $r\ge M+1$. Thus, near x we get

$$u_N \;=\; (-1)^{M+1}\chi_M v_M\,\mathbf{1}_{\{N\ge M\}} \;+\; \sum_{r\,\le\,\min\{M-1,N\}} (-1)^{r+1}v_r. \tag{5.6}$$

As $x\in K^{s_r}$ for $r\leqslant M-1$, the choice (b) yields

$$\Big|\sum_{r\le M-1}(-1)^{r+1}\nabla v_r(x)\Big| \le |\sigma(x)|+\varepsilon_0\le 2.$$

Adding $|\chi_M\nabla v_M(x)|\le 1$ and (5.5), we obtain $|\nabla u_N(x)|\le 5$. Hence $\operatorname{Lip}(u_N)\le 5$ for every N, and since $u_N\to u$ uniformly we conclude $\operatorname{Lip}(u)\le 5$. Finally, (5.6) with $N=\infty$ expresses u near any $x\notin K$ as a finite sum of C^1 functions, so that u is of class C^1 on $\mathbb{R}^d\setminus K$.

Oscillation of the normal slope on K. Fix $x\in K$ and $\ell\ge 1$. Parametrizing the segment $[t_{h_\ell}(x), z_{h_\ell}(x)]$ by $p_\tau := t_{h_\ell}(x)+\tau\,h_\ell\,w_N(x)$ for $\tau\in[0,1]$, property (c) gives

$$|p_\tau - x| \;\le\; h_\ell(|w_T(x)|+|w_N(x)|) \;\le\; \sqrt{2}\,h_\ell \;\le\; s_\ell.$$

Consequently, for every $r\le\ell$ the whole segment lies in $K^{s_r}\subset\{\chi_r\equiv 1\}$, so that $N_h(\chi_r v_r) = N_h v_r$ there. By the fundamental theorem of calculus

$$\sum_{r\le\ell}(-1)^{r+1}N_h v_r(x) \;=\; \int_0^1 \langle G_\ell(p_\tau),\, w_N(x)\rangle\,d\tau, \qquad G_\ell := \sum_{r\le\ell}(-1)^{r+1}\nabla v_r.$$

Once more, by (b) we have $|G_\ell(p_\tau)-\mathbf{1}_{\{\ell \text{ odd}\}}\sigma(p_\tau)|\le\varepsilon_0$, while $|\sigma(p_\tau)-\sigma(x)|\le\omega(\sqrt{2}\,h_\ell)$ and $\langle\sigma(x),w_N(x)\rangle = |w_N(x)|$ by the definition of σ. Therefore, the integral differs from $\mathbf{1}_{\{\ell \text{ odd}\}}|w_N(x)|$ by at most $[\varepsilon_0+\omega(\sqrt{2}\,h_\ell)]|w_N(x)|$. For the remaining indices $r>\ell$ we use

$\|v_r\|_\infty \le 2^{-r} h_{r-1} \le 2^{-r} h_\ell$, which gives the bound $|N_{h_\ell}(\chi_r v_r)| \le 2\|v_r\|_\infty / h_\ell \le 2^{-r+1}$. Altogether, recalling $|w_N| \le 1$ and $\varepsilon_0 = \delta/8$,

$$|\, N_{h_\ell} u(x) - \mathbf{1}_{\{\ell \text{ odd}\}} |w_N(x)|\,| \;\le\; \frac{\delta}{8} + \omega(\sqrt{2}\, h_\ell) + 2^{-\ell+1} \;\le\; \frac{\delta}{4} \qquad \text{for all } x \in K,$$

the last inequality holding as soon as ℓ is large enough, since $h_\ell \to 0$. As $|w_N| \ge \delta$ on K, we conclude that for every such ℓ and every $x \in K$

$$N_{h_\ell} u(x) \begin{cases} \ge \frac{3}{4}\delta & \text{if } \ell \text{ is odd,} \\ \le \frac{1}{4}\delta & \text{if } \ell \text{ is even.} \end{cases}$$

Thus $\lim_{h\to 0^+} N_h u(x)$ exists at no point of K, and by (5.4) the directional derivative $\partial_{w(x)} u(x)$ fails to exist μ a.e. $x \in K$. □

Theorem 5.5 (T_μ is the largest differentiability bundle)**.** *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure and let $W : \mathbb{R}^d \to \mathrm{Gr}(d)$ be a μ-measurable map such that every Lipschitz function $u : \mathbb{R}^d \to \mathbb{R}$ is differentiable at μ-a.e. x with respect to $W(x)$. Then $W(x) \subset T_\mu(x)$ for μ-a.e. x.*

Proof. Let $(e_i)_i$ be the canonical basis of $\mathbb{R}^d$ and $P_W(x)$ the orthogonal projection onto $W(x)$. Since $W(x)$ is spanned by the vectors $P_W(x)e_i$, the set $\{x : W(x) \not\subset T_\mu(x)\}$ is the union of the μ-measurable sets $E_i := \{x : P_W(x)e_i \notin T_\mu(x)\}$. Assume by contradiction that $\mu(E_i) > 0$ for some i and pick a bounded Borel set $E \subset E_i$ with $\mu(E) > 0$. On E, the unit field $w := P_W e_i / |P_W e_i|$ satisfies $w(x) \in W(x) \setminus T_\mu(x)$, and $T_{\mu \llcorner E} = T_\mu$ μ-a.e. on E by (3.5). Proposition 5.4, applied to the finite measure $\mu \llcorner E$ and to w, provides a Lipschitz function f and a compact set K with $\mu(K \cap E) \ge \mu(E) - \varepsilon > 0$ such that $D_{w(x)} f(x)$ does not exist for $x \in K$. Since $w(x) \in W(x)$ on E, f is not differentiable with respect to $W(x)$ at any $x \in K \cap E$, a contradiction. □

In view of assertion (iii) of Theorem 4.1, T_μ is therefore the largest bundle along which every Lipschitz function is μ-a.e. differentiable.

5.3. **Relation with Preiss tangent measures.** In the literature, blow-up techniques were initially developed to define tangent cones for rectifiable structures such as currents and varifolds (see, e.g., H. Federer [31], W. Allard [4], and the recollection by L. Simon [48]). The notion of tangent measures for a general Radon measure was introduced by D. Preiss [45] (see also [33] for a comparison between Preiss' one and the Smirnov bundle):

Definition 5.6. Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure and $x_0 \in \mathbb{R}^d$. The Preiss tangent space to μ at x_0, denoted $\mathrm{Tan}(\mu, x_0)$, consists of all non zero Radon measures $\nu \in \mathcal{M}_+(\mathbb{R}^d)$ such that, for suitable sequences of reals $c_j > 0$ and $r_j \to 0$, we have

$$\lim_{j\to\infty} c_j \int \varphi\left(\frac{x - x_0}{r_j}\right) d\mu = \int_{\mathbb{R}^d} \varphi(y) d\nu(y) \quad \text{for every } \varphi \in C_c^0(\mathbb{R}^d). \tag{5.7}$$

Remark 5.7. By testing the convergence (5.7) against a continuous function $\varphi \in C^0(\mathbb{R}^d; [0,1])$ such that $\varphi = 1$ on B_R and $\mathrm{spt}(\varphi) \subset B_{R+\delta}$, we infer that the reals c_j, r_j involved (they depend on x_0) are such that, for every $R > 0$ and $\delta > 0$:

$$\nu(B_R) \le \liminf_{j\to\infty} c_j\, \mu\left(B_{r_j R}\right) \le \limsup_{j\to\infty} c_j\, \mu\left(B_{r_j R}\right) \le \nu(B_{R+\delta}) < +\infty.$$

Therefore, by sending $\delta \to 0$, it follows that for every R in the complement of a countable subset of $(0, +\infty)$, there exists a constant $\kappa(x_0, R) > 0$ such that

$$c_j \, \mu \left(B_{r_j \, R} \right) \ \to \kappa(x_0, R) \quad \text{as } j \to \infty. \tag{5.8}$$

Note that the situation is much simpler in the case where μ satisfies the doubling condition around x_0, namely:

$$\limsup_{s \to 0_+} \frac{\mu(B(x_0, 2s))}{\mu(B(x_0, s))} < +\infty.$$

In this case, it is not restrictive to impose that $c_j \mu(B_{r_j}) = 1$.

The main feature of $\mathrm{Tan}(\mu, x_0)$ is that it is non empty positive cone of $\mathcal{M}_+(\mathbb{R}^d)$ μ-a.e., closed under vague convergence and dilation invariant (see [45, Thm 2.5]). For those properties, no regularity assumptions on μ are required. In contrast, when some rectifiability property of μ is expected, the more restrictive notion of tangent space to μ can be used (see [48, 39]).

Definition 5.8. The k-dimensional subspace V of $\mathbb{R}^d$ is said to be tangent to μ at x_0 if there exists a constant $\theta > 0$ (called the *multiplicity*) such that

$$\lim_{h \to 0_+} \frac{1}{h^k} \int \varphi \left(\frac{x - x_0}{h} \right) \, d\mu = \theta \int_V \varphi(y) d\mathcal{H}^k(y) \quad \text{ for every } \varphi \in C_c^0(\mathbb{R}^d). \tag{5.9}$$

Remark 5.9. The property (5.9) implies that (5.7) holds with $\nu = \theta \, \mathcal{H}^k \llcorner V$ being $c_j = r_j^{-k}$ and r_j any vanishing sequence. This means that the possibly large set $\mathrm{Tan}(\mu, x_0)$ contains at least one flat measure (in general not unique). As originally established by Preiss [45] (see [39, Thm 14.12] for an accessible exposition), if $\mathrm{Tan}(\mu, x_0)$ consists only of flat measures, then these measures are concentrated on the same subspace V. Thus, if $\dim V = k$, one has $\mathrm{Tan}(\mu, x_0) = \{\theta \, \mathcal{H}^k \llcorner V \ : \ \theta \in \mathbb{R}_+\}$.

A revealing connection between $\mathrm{Tan}(\mu, x_0)$ defined in (5.7) and $T_\mu(x_0)$ is summarized in the following Proposition which extends [33] where the Smirnov tangent bundle $S_\mu(x_0)$ is considered (see also [8]); recall that, by Proposition 5.1, $S_\mu(x_0)$ can be strictly contained in the MK-bundle $T_\mu(x_0)$. Beforehand we give a related result of convergence, whose proof relies on the density of normal currents in flat chains (assertion (ii) of Proposition 2.9).

Lemma 5.10. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure and σ an element of $\mathcal{T}_\mu$. Then, for μ a.e. $x_0 \in \mathbb{R}^d$, we have the following convergence as $h \to 0_+$:*

$$f_h^{x_0} := \frac{h}{\mu(B(x_0, h))} \, T_h^{x_0}(-\operatorname{div}(\sigma \, \mu)) \ \to 0 \quad \textit{in } \mathcal{D}'(B_1) \tag{5.10}$$

where $T_h^{x_0} : x \mapsto \frac{x - x_0}{h}$.

Proof. Since $\sigma \, \mu$ belongs to $\mathbf{F_1}(\mathbb{R}^d)$, there exists $\lambda_n = w_n \, \mu + \lambda_n^s \in \mathbf{N_1}(\mathbb{R}^d)$ such that

$$\int |\sigma - w_n| \, d\mu + \int |\lambda_n^s| \leq \frac{1}{n} \quad , \quad |\lambda_n^s| \perp \mu.$$

Up to extracting a subsequence, we can assume that $w_n \to \sigma$ μ a.e. as $n \to \infty$. Then setting $f_h^{x_0} = g_{h,n}^{x_0} + r_{h,n}^{x_0}$ where $g_{h,n}^{x_0} := \frac{h}{\mu(B(x_0,h))}\, T_h^{x_0}(-\operatorname{div} \lambda_n)$, we have

$$\begin{aligned}\left\langle r_{h,n}^{x_0}, \varphi \right\rangle &= \fint_{B(x_0,h)} \left\langle \sigma - w_n, \nabla\varphi\left(\frac{x-x_0}{h}\right)\right\rangle d\mu + \frac{1}{\mu(B(x_0,h))}\left\langle \lambda_n^s, \nabla\varphi\left(\frac{x-x_0}{h}\right)\right\rangle \\ &\le \operatorname{Lip}(\varphi)\left(\fint_{B(x_0,h)} |\sigma - w_n|\, d\mu + \frac{|\lambda_n^s|(B(x_0,h))}{\mu(B(x_0,h))}\right)\end{aligned}$$

for every $\varphi \in \mathcal{D}(B_1)$. On the other hand, since $\operatorname{div} \lambda_n$ is a measure:

$$\left\langle g_{h,n}^{x_0}, \varphi \right\rangle \le h\, \sup|\varphi|\, \frac{|\operatorname{div} \lambda_n|(B(x_0,h))}{\mu(B(x_0,h))}.$$

Next we choose a set $B \in \mathcal{B}(\mathbb{R}^d)$ with $\mu(B) = 0$ and such that, for every $x_0 \in B^c$ and $n \in \mathbb{N}^*$:

$$\limsup_{h\to 0} \frac{|\operatorname{div} \lambda_n|(B(x_0,h))}{\mu(B(x_0,h))} < +\infty, \qquad \limsup_{h\to 0} \frac{|\lambda_n^s|(B(x_0,h))}{\mu(B(x_0,h))} = 0,$$
$$\text{and} \qquad \lim_{h\to 0} \fint_{B(x_0,h)} |\sigma - w_n|\, d\mu = |\sigma(x_0) - w_n(x_0)|.$$

Then gathering all, we infer that, for every n and $x_0 \in B^c$, it holds

$$\limsup_{h\to 0} \left|\left\langle f_h^{x_0}, \varphi\right\rangle\right| \le \operatorname{Lip}(\varphi)\, |\sigma - w_n|(x_0) \quad \forall \varphi \in \mathcal{D}(B_1).$$

The convergence (5.10) follows by the pointwise convergence $w_n \to \sigma$ as $n \to \infty$. □

Proposition 5.11. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure. Then for μ a.e. $x_0 \in \mathbb{R}^d$, any measure $\nu \in$ Tan(μ, x_0) satisfies the equality*

$$\nu \;\; = \mathcal{H}^k \llcorner T_\mu(x_0) \otimes \eta \qquad \text{in } \mathbb{R}^d, \tag{5.11}$$

where $k = k(x_0)$ denotes the dimension of the $T_\mu(x_0)$ and η is a Radon measure on $T_\mu^\perp(x_0)$. In particular, if μ admits a k-dimensional tangent space V at x_0 in the sense of (5.9)*, then*

$$T_\mu(x_0) \;=\; V, \tag{5.12}$$

while in general (5.11) *only yields $T_\mu(x_0) \subset V$.*

Proof. Replacing μ by $\mu \llcorner B_m$ does not change T_μ nor Tan(μ, x_0) for $x_0 \in B_m$. Hence, we may assume that μ is finite. Let $P_\mu^1, \dots, P_\mu^d \in \mathcal{T}_\mu$ be the rows of the orthogonal projection P_μ. We fix a μ-negligible Borel set N such that, for every $x_0 \in \mathbb{R}^d \setminus N$, the set Tan$(\mu, x_0)$ is non empty, the convergence (5.10) holds at x_0 for $\sigma = P_\mu^l$ and every $l \in \{1, \dots, d\}$, and x_0 is a μ-Lebesgue point of P_μ.

Let $\nu \in$ Tan(μ, x_0). Then there exist positive reals c_j and $r_j \to 0$ such that (5.7) holds for any $\varphi \in \mathcal{D}(\mathbb{R}^d)$. Fix such an element φ and a finite radius $R > 0$ so large that $\operatorname{spt}(\varphi) \subset B_R$. In virtue of (5.8), $\kappa_j := c_j\, \mu\left(B(x_0, r_j R)\right)$ remains bounded and strictly positive. Thus, possibly after extracting a subsequence, there exists a finite constant $\kappa > 0$ (depending on R) such that

$$0 < \kappa := \limsup_j \kappa_j < +\infty.$$

We claim that

$$\partial_z \nu = 0 \quad \text{in the sense of distributions, for all } z \in V := T_\mu(x_0). \tag{5.13}$$

Fix $z \in V$. Applying (5.10) with $\sigma = P^l_\mu$ and $h = R\, r_j$, we get for every $l \in \{1, \dots, d\}$:

$$\begin{aligned} 0 &= \lim_{j\to\infty} \fint_{B(x_0, Rr_j)} \left\langle P^l_\mu(x), \nabla\varphi\left(\frac{x-x_0}{Rr_j}\right)\right\rangle \mu(dx) \\ &= \lim_{j\to\infty} \fint_{B(x_0, Rr_j)} \left\langle P^l_\mu(x_0), \nabla\varphi\left(\frac{x-x_0}{Rr_j}\right)\right\rangle \mu(dx) \\ &= \frac{1}{\kappa(x_0, R)} \left\langle P^l_\mu(x_0), \int \nabla\varphi(y/R)\, \nu(dy)\right\rangle, \end{aligned}$$

where the second equality follows from x_0 being a μ-Lebesgue point of P_μ, and the third one from (5.7) applied to $\varphi_R = \varphi(\frac{\cdot}{R})$, taking into account (5.8). As φ is arbitrary in $\mathcal{D}(\mathbb{R}^d)$, the equality above yields $-\operatorname{div}(P_\mu(x_0)\,\nu) = 0$, which is equivalent to (5.13).

The fact that $\partial_z \nu = 0$ for all $v \in V := T_\mu(x_0)$ implies that ν is invariant under translations by elements of V. Write $\mathbb{R}^d = V \times V^\perp$, fix a half-open unit cube $Q \subset V$ and set $\eta \in \mathcal{M}(V^\perp)$ by $\eta(B) := \nu(Q \times B)$. The 2^{jk} half-open dyadic subcubes of Q of side 2^{-j} are disjoint, tile Q, and are translates of one another; by translation invariance they carry the same mass, so that $\nu(A \times B) = \mathcal{H}^k(A)\,\eta(B)$ whenever A is a translate of such a subcube. Since every open subset of V is a countable disjoint union of half-open dyadic cubes, additivity and regularity extend this to all Borel sets $A \subset V$, which is (5.11). □

Corollary 5.12. *Let μ be a k-rectifiable measure, that is of the form $\mu = \theta\, \mathcal{H}^k \llcorner S$ where S is a countably $\mathcal{H}^k$ rectifiable subset of $\mathbb{R}^d$ and θ a positive $\mathcal{H}^k$ locally integrable function on S. Then, denoting $T_S(x)$ the classical tangent space (defined at $\mathcal{H}^k$ a.e. $x \in S$), the following equalities are true μ a.e.:*

$$T_\mu(x) = T_S(x), \quad \mathit{Tan}(\mu, x) = \{c\mathcal{H}^k \llcorner T_S(x) \ :\ c > 0\}.$$

Proof. The second equality is classical and also the fact that μ admits $T_S(x)$ as a tangent space in the sense of (5.9). The first equality follows from the latter and (5.12). □

Remark 5.13. By applying Proposition 5.11 in the case $k = d$, we deduce that the Preiss $\operatorname{Tan}(\mu, x_0)$ reduces to $\left\{\, \theta\, \mathcal{L}^d : \theta > 0 \,\right\}$ for μ almost all x_0 (the measure η appearing in (5.11) reduces to a scalar). Unfortunately it is not sufficient to recover that $\mu \ll \mathcal{L}^d$ as stated in Proposition 6.1. In fact, in [45], David Preiss constructed a singular measure $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ with $\operatorname{Tan}(\mu, x_0) = \left\{\, \theta\mathcal{L}^d : \theta > 0 \,\right\}$.

6. Further properties of MK bundles and tangential calculus

6.1. **Decomposition into components of constant dimension.** Denote P_μ the tensor-valued function from $\mathbb{R}^d$ to symmetric tensors $\mathcal{S}^{d\times d}$ associated to the orthogonal projection on $T_\mu(x)$. If μ is multi-dimensional (as it is often the case in the analysis of mechanical structures [20]), this integer-valued function allows to split μ into components of constant dimension:

$$\mu = \sum_{k=0}^{d} \mu_k, \qquad \mu_k := \mu \llcorner \left\{\, \dim T_\mu = k \right\}, \tag{6.1}$$

Note that the components μ_k that we obtain here are mutually singular. The top dimension $k = d$ is special since μ_d is absolutely continuous with respect to the Lebesgue measure $\mathcal{L}^d$. Indeed, by adapting to flat chains the structure theorem for $\mathcal{A}$-free measures of Philippis and Rindler [29], we prove the following.

Proposition 6.1. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be a Radon measure. Then*

$$\dim T_\mu(x) = d \quad \mu \text{ a.e.} \qquad \Longleftrightarrow \qquad \mu \ll \mathcal{L}^d.$$

Proof. Assume that $\mu \ll \mathcal{L}^d$. Then, by applying the property (3.5), we infer that $T_\mu(x) = T_{\mathcal{L}^d}(x)$ holds μ a.e.. The latter set is easily seen to be equal to $\mathbb{R}^d$, since any smooth compactly supported vector field σ belongs to $\mathcal{T}_{\mathcal{L}^d}$. Taking $\sigma(x) \equiv \overline{\sigma}$ in an open ball, we deduce that $\overline{\sigma} \in T_\mu(x)$ for any $\overline{\sigma} \in \mathbb{R}^d$ and μ-a.e. $x \in \mathbb{R}^d$.

Conversely, let $\Lambda \subset \mathbb{R}^{d\times d}$ be the cone of all singular matrices. The distance function $g = \operatorname{dist}(\cdot, \Lambda)$ is 1-Lipschitz, positively 1-homogeneous and vanishes exactly on Λ. Define

$$\Phi(\lambda) \coloneqq \int g\Big(\frac{d\lambda}{d|\lambda|}\Big)\, d|\lambda|^s, \qquad \lambda \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^{d\times d}).$$

Here $|\lambda|^s$ is the part of $|\lambda|$ that is singular with respect to $\mathcal{L}^d$. We observe that Φ is 1-Lipschitz: Indeed, for $\lambda_1, \lambda_2 \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^{d\times d})$, let $\pi = |\lambda_1| + |\lambda_2|$ and write $\lambda_1 = f_1\pi$, $\lambda_2 = f_2\pi$. Because Φ is positively 1-homogeneous and g is 1-Lipschitz we get

$$|\Phi(\lambda_1) - \Phi(\lambda_2)| \;\le\; \int |g(f_1) - g(f_2)|\, d\pi \;\le\; \int |f_1 - f_2|\, d\pi^s \;\le\; \|\lambda_1 - \lambda_2\|_{\mathcal{M}}.$$

The result of De Philippis and Rindler [29, Thm 1.1] is equivalent with the functional Φ vanishing on $\mathbf{N_1}(\mathbb{R}^d)^d$. Therefore, it suffices to show that $\Phi(P_\mu\, \mu) = 0$. The density of $\mathbf{N_1}(\mathbb{R}^d)$ in $\mathbf{F_1}(\mathbb{R}^d)$ applied row-wise to $P_\mu\, \mu$ yields a sequence $(\lambda_j)_j \in \mathbf{N_1}(\mathbb{R}^d)^d$ satisfying $\|P_\mu\, \mu - \lambda_j\|_{\mathcal{M}} \to 0$. In particular $\Phi(P_\mu\, \mu) = \lim_{j\to\infty} \Phi(\lambda_j) = 0$. Hence $d\lambda/d|\lambda|$ is singular for any flat chain, and we deduce that T_μ is at most $(d-1)$-dimensional μ-a.e.. □

Corollary 6.2 (De Philippis–Rindler for flat chains)**.** *Let $T_1, \dots, T_d \in \mathbf{F_1}(\mathbb{R}^d)$ and $\mathbf{T} = (T_1, \dots, T_d) \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^{d\times d})$, then*

$$\operatorname{rank}\left(\frac{d\mathbf{T}}{d|\mathbf{T}|}\right) < d \qquad |\mathbf{T}|^s\text{-almost everywhere.}$$

Proof. Let $\mu = |\mathbf{T}|$ and write $T_j = \sigma_j|\mathbf{T}|$ so that $\sigma_j \in \mathcal{T}_\mu$ for all $j = 1, \dots, d$. By the top-dimensional estimate it follows that $\mu_d \ll \mathcal{L}^d$, which is precisely the sought assertion. □

Remark 6.3 (Dimensional estimates)**.** Proposition 6.1 extends to the following relation between tangential dimension and dimensional estimates:

$$\dim T_\mu(x) \ge k \quad \mu \text{ a.e.} \qquad \Longrightarrow \qquad \mu \ll \mathcal{I}^k \ll \mathcal{H}^k,$$

where $\mathcal{I}^k$ is the k-dimensional integral-geometric measure and $k \in \{1, \dots, d\}$. The proof follows verbatim the proof of sufficiency in the proposition above, by replacing [29, Theorem 1.1] with its geometric version [10, Theorem 1.3] and with $g = \operatorname{dist}(\cdot, \Lambda_k)$, where Λ_k is the cone of matrices with rank less than k.

6.2. **Generalized varifold and mean curvature.** Let us identify the Grassmannian manifold $\mathcal{G}_d$ of $\mathbb{R}^d$ with the subset of those symmetric tensors $P \in \mathcal{S}^{d\times d}$ which are projection matrices (i.e. those with eigenvalues in $\{0,1\}$). Following [17, 18], we can associate every Radon measure $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ with a generalized varifold. This measure is on the product $\mathbb{R}^d \times \mathcal{S}^{d\times d}$ whose first marginal is μ. This varifold $\mathbf{V}_\mu$ is defined by

$$\langle \mathbf{V}_\mu, \varphi\rangle \;:=\; \int \varphi(x, P_\mu(x))\, d\mu \qquad \text{for every } \varphi \in C_0(\mathbb{R}^d \times \mathcal{S}^{d\times d}).$$

Then (6.1) induces the decomposition $\mathbf{V}_\mu = \sum_{k=1}^d \mathbf{V}_{\mu_k}$ where $\mathbf{V}_{\mu_k}$ fits with the classical notion of k-varifold in $\mathbb{R}^d$ when μ_k is a k-rectifiable measure (see [4, 48]). Following [17, 18] (see also [28] for related weak notions of curvature), the first variation of $\mathbf{V}_\mu$ allows to define a generalized mean curvature vector $H(\mu)$ by setting:

$$H(\mu) := \operatorname{div}(P_\mu \mu)$$

where the divergence is taken in the distributional sense in $\mathbb{R}^d$. Note that, since every row of the tensor P_μ belongs to $\mathcal{T}_\mu$, divergence above is component-wise in the Arens-Eells space. Therefore $H(\mu)$ can be identified as an element of $(\text{Æ}(\mathbb{R}^d))^d$.

6.3. **Operations on normal and tangent fields.** In this section, we investigate the modification of tangent and normal spaces when the measure μ is pushed forward by a Lipschitz function. This generalizes the deformation of tangent spaces in manifolds. At first, we consider the dual operation. Let $f \in \operatorname{Lip}(\mathbb{R}^d;\mathbb{R}^m)$, μ be a Radon measure on $\mathbb{R}^d$, and $\nu := f_\#\mu$. Define the (tangential) pullback of $\zeta \in L^\infty_\nu(\mathbb{R}^m;\mathbb{R}^m)$ by f as

$$(f^*\zeta)(x) := (\nabla_\mu f(x))^t(\zeta\circ f)(x) \qquad \text{for } \mu\text{-a.e. } x \in \mathbb{R}^d.$$

Here $\nabla_\mu f(x)$ identifies with the matrix whose i^{th} line is given by $\nabla_\mu f_i$ (see (3.18)). Note that f^* is a linear operator from $L^\infty_\nu(\mathbb{R}^m;\mathbb{R}^m)$ to $L^\infty_\mu(\mathbb{R}^d;\mathbb{R}^d)$ with norm bounded by $\operatorname{Lip}(f)$.

Theorem 6.4. *The pullback operator f^* sends $\mathcal{N}_\nu$ on $\mathcal{N}_\mu$. As a consequence,*

$$\nabla_\mu f(x)(T_\mu) \subset T_\nu(f(x)) \qquad \textit{for } \mu\textit{-a.e. } x \in \mathbb{R}^d. \tag{6.2}$$

Proof. Let $\zeta \in \mathcal{N}_\nu$. By (3.8), we can choose a sequence $(u_n)_n \subset \mathcal{D}(\mathbb{R}^m)$ such that $u_n \overset{*}{\rightharpoonup} 0$ in $\operatorname{Lip}(\mathbb{R}^m)$ and $\nabla u_n \to \zeta$ holds ν-a.e. Since $u_n \to 0$ pointwise and $\operatorname{Lip}(u_n\circ f) \le \operatorname{Lip}(f)\operatorname{Lip}(u_n)$ is bounded, the sequence pulls back to $u_n \circ f \overset{*}{\rightharpoonup} 0$ in $\operatorname{Lip}(\mathbb{R}^d)$. By the weak-star continuity of the tangential gradient (Theorem 3.11), this implies $\nabla_\mu(u_n\circ f) \overset{*}{\rightharpoonup} 0$ in $(L^\infty_\mu)^d$.

For an arbitrary tangent field $\sigma \in \mathcal{T}_\mu$, we might use assertion (ii) of Theorem 4.1 on each coordinate of f to write

$$\begin{aligned}
0 &= \lim_{n\to\infty} \int \langle \nabla_\mu(u_n\circ f), \sigma\rangle\, d\mu = \lim_{n\to\infty}\lim_{h\to 0} \int \frac{u_n(f(x+h\sigma(x))) - u_n(f(x))}{h}\, d\mu \\
&= \lim_{n\to\infty}\lim_{h\to 0} \int \left\langle \nabla u_n(f(x)), \frac{f(x+h\sigma(x)) - f(x)}{h}\right\rangle d\mu \\
&= \lim_{n\to\infty} \int \langle \nabla u_n \circ f,\ \nabla_\mu f\ \sigma\rangle\, d\mu \;=\; \int \langle \zeta\circ f, \nabla_\mu f\, \sigma\rangle\, d\mu = \int \langle f^*\zeta, \sigma\rangle\, d\mu,
\end{aligned}$$

The passage to the limit in the last line is justified by the dominated convergence theorem: $\nabla u_n \circ f \to \zeta\circ f$ holds μ-a.e. (since $\nu = f_\#\mu$), $|\nabla u_n|$ is uniformly bounded and the majorant

$|\nabla_\mu f\ \sigma| \le \mathrm{Lip}(f)|\sigma|$ belongs to L^1_μ. Since σ is arbitrary in $\mathcal{T}_\mu$, we infer that $f^*\zeta$ is orthogonal to $\mathcal{T}_\mu$. Hence the claimed inclusion $f^*(\mathcal{N}_\nu) \subset \mathcal{N}_\mu$.

To prove (6.2), we localize. Let $g : \mathbb{R}^d \to \mathbb{R}^d$ be a μ-measurable selection of T_μ and $\zeta = P^\perp_\nu a$ for $a \in \mathbb{R}^m$. For any measurable set $B \subset \mathbb{R}^d$, it holds that $\mathbb{1}_B g \in \mathcal{T}_\mu$ and

$$f^*\zeta = (\nabla_\mu f)^t P^\perp_\nu a \circ f \in \mathcal{N}_\mu.$$

The composition $P^\perp_\nu \circ f$ is measurable. The set A of μ-Lebesgue points of g, $\nabla_\mu f$ and $P^\perp_\nu \circ f$ is of full measure for μ, and for each $x \in A$ and $B = \mathscr{B}(x, r)$ with $r > 0$, there holds

$$\begin{aligned} 0 &= \int \left\langle (\nabla_\mu f)^t P^\perp_\nu a \circ f, \mathbb{1}_B g \right\rangle d\mu = \int_B \left\langle P^\perp_\nu(f(y))a, \nabla_\mu f(y) g(y) \right\rangle d\mu(y) \\ &= \left\langle a, \int_B P^\perp_\nu(f(y)) \nabla_\mu f(y) g(y) d\mu(y) \right\rangle. \end{aligned}$$

As $a \in \mathbb{R}^m$ is arbitrary, the integral vanishes. Dividing by $\mu(B)$ and passing to the limit as $r \to 0$, we get that $P^\perp_\nu(f(x))\nabla_\mu f(x) g(x) = 0$, hence $\nabla_\mu f(x) g(x) \in T_\nu(f(x))$ on A. Since g was arbitrary in $\mathcal{T}_\mu$, we conclude that $\nabla_\mu f(T_\mu) \subset T_\nu \circ f$ at μ-a.e. point. □

The pullback allows to define a pushforward by duality, and to mirror the stability result of Theorem 6.4 on the tangent bundle. Precisely, given $\sigma \in L^1_\mu(\mathbb{R}^d; \mathbb{R}^d)$ and $f \in \mathrm{Lip}(\mathbb{R}^d; \mathbb{R}^m)$ as before, there exists a unique vector field $f_*\sigma \in L^1_\nu(\mathbb{R}^m; \mathbb{R}^m)$ satisfying

$$\int_{\mathbb{R}^m} \langle \zeta, f_*\sigma \rangle \, d\nu = \int_{\mathbb{R}^d} \langle f^*\zeta, \sigma \rangle \, d\mu \qquad \text{for every } \zeta \in L^\infty_\nu(\mathbb{R}^m; \mathbb{R}^m). \tag{6.3}$$

Explicitly, $f_*\sigma$ is the Radon-Nikodym derivative of the vector-valued measure $f_\# (\nabla_\mu f \sigma \mu)$ with respect to $\nu = f_\#\mu$. The duality formulation allows to recover the following result, which is the pendant of the stability of flat chains by (current) pushforward stated in [31, 4.1.14].

Corollary 6.5. *The pushforward operator f_* of $f \in \mathrm{Lip}(\mathbb{R}^d; \mathbb{R}^m)$ sends $\mathcal{T}_\mu$ on $\mathcal{T}_\nu$.*

Proof. Let $\sigma \in \mathcal{T}_\mu$ and $\zeta \in \mathcal{N}_\nu$. By Theorem 6.4, $f^*\zeta \in \mathcal{N}_\mu$, so that

$$\int \langle \zeta, f_*\sigma \rangle d\nu = \int \langle f^*\zeta, \sigma \rangle d\mu = 0.$$

As ζ is arbitrary, we conclude that $f_*\sigma \in \mathcal{T}_\nu$. □

As a corollary, one obtains an inclusion of tangent bundles constructed from tensor products. In fact, in this case, one can even get the following identity.

Proposition 6.6 (Tensor products)**.** *Let $d = p + q$ and $\mu \in \mathcal{M}_+(\mathbb{R}^p)$, $\nu \in \mathcal{M}_+(\mathbb{R}^q)$ be finite Radon measures. The tangent bundle of the product measure is the product of the tangent bundles:*

$$T_{\mu\otimes\nu}(x, y) \ = \ T_\mu(x) \times T_\nu(y) \qquad \mu \otimes \nu\text{-a.e. } (x, y) \in \mathbb{R}^d.$$

Proof. Inclusion $\subset$. Let $\pi_1(x, y) \coloneqq x$ and $\pi_2(x, y) \coloneqq y$ be the coordinate projections, so that $(\pi_1)_\#(\mu \otimes \nu) = \nu(\mathbb{R}^q)\,\mu$ and $(\pi_2)_\#(\mu \otimes \nu) = \mu(\mathbb{R}^p)\,\nu$. Since scaling a measure by a positive constant does not change its bundle ($\mathcal{T}_{c\,\mu} = \mathcal{T}_\mu$ for $c > 0$), these two pushed-forward measures have bundles T_μ and T_ν respectively. Moreover, the tangential gradient of the linear maps π_i

can be computed using (3.17) as $\nabla_\mu \pi_i = P_\mu \nabla \pi_i$. Hence $\nabla_\mu \pi_i(T_\mu) = \nabla \pi_i(T_\mu) = \pi_i(T_\mu)$. Applying Theorem 6.4, we obtain

$$\pi_1\big(T_{\mu\otimes\nu}(x,y)\big) \subset T_\mu(x), \qquad \pi_2\big(T_{\mu\otimes\nu}(x,y)\big) \subset T_\nu(y) \qquad \mu\otimes\nu\text{-a.e.}$$

The inclusion $T_{\mu\otimes\nu}(x,y) \subset T_\mu(x) \times T_\nu(y)$ follows immediately.
Inclusion $\supset$. Conversely, let $\tau \in \mathcal{T}_\mu$ and set $\sigma(x,y) \coloneqq (\tau(x), 0)$. For any smooth test function $\varphi \in \mathcal{D}(\mathbb{R}^d)$, Fubini's theorem gives

$$\langle -\operatorname{div}(\sigma\, \mu\otimes\nu), \varphi\rangle \;=\; \int_{\mathbb{R}^q} \langle -\operatorname{div}(\tau\mu),\; \varphi(\cdot, y)\rangle \, d\nu(y).$$

If $\varphi_n \overset{*}{\rightharpoonup} 0$ in $\mathrm{Lip}(\mathbb{R}^d)$, then for each fixed y, $\varphi_n(\cdot, y) \overset{*}{\rightharpoonup} 0$ in $\mathrm{Lip}(\mathbb{R}^p)$, meaning the inner bracket tends pointwise to 0. It is moreover uniformly bounded by $\|\operatorname{div}(\tau\mu)\|_{\mathrm{KR}} \sup_n \mathrm{Lip}(\varphi_n)$ by Proposition 2.5. As ν is finite, dominated convergence yields $\langle -\operatorname{div}(\sigma\,\mu\otimes\nu), \varphi_n\rangle \to 0$, and the continuity criterion (2.5) implies $\sigma \in \mathcal{T}_{\mu\otimes\nu}$, i.e., $(\tau(x),0) \in T_{\mu\otimes\nu}(x,y)$ almost everywhere. Choosing a countable generating family $(\tau_k) \subset \mathcal{T}_\mu$ such that $T_\mu(x) = \operatorname{span}\{\, \tau_k(x) : k \in \mathbb{N}\,\}$ μ-a.e. (as in Subsection 2.2) and intersecting the exceptional sets, we obtain $T_\mu(x) \times \{0\} \subset T_{\mu\otimes\nu}(x,y)$ a.e.. The symmetric argument yields $\{0\} \times T_\nu(y) \subset T_{\mu\otimes\nu}(x,y)$, and the conclusion follows since $T_{\mu\otimes\nu}(x,y)$ is a vector subspace. □

6.4. **Tangential chain rule.** The computation in the proof of Theorem 6.4 used a chain rule of the form $\langle \nabla_\mu(u_n \circ f), \sigma\rangle = \langle \nabla u_n \circ f,\; \nabla_\mu f\; \sigma\rangle$ for a smooth u and $\sigma \in \mathcal{T}_\mu$. The following result shows that the chain rule holds with tangential gradients.

Theorem 6.7 (Tangential chain rule)**.** *Let $g \in \mathrm{Lip}(\mathbb{R}^m; \mathbb{R}^k)$. Then, for every $\sigma \in \mathcal{T}_\mu$,*

$$\nabla_\mu(g\circ f)(x)\,\sigma(x) \;=\; \nabla_\nu g(f(x))\,\nabla_\mu f(x)\,\sigma(x) \qquad \mu\text{-a.e.}. \tag{6.4}$$

Proof. Arguing componentwise, we may assume g is scalar. By Remark 2.7, we can choose $g_k \in \mathcal{D}(\mathbb{R}^m)$ such that $g_k \overset{*}{\rightharpoonup} g$ in $\mathrm{Lip}(\mathbb{R}^m)$. As $g_k \circ f \overset{*}{\rightharpoonup} g \circ f$ in $\mathrm{Lip}(\mathbb{R}^d)$, the weak-star continuity of the tangential gradient (Theorem 3.11) ensures that

$$\nabla_\nu g_k \overset{*}{\rightharpoonup} \nabla_\nu g \quad \text{in } L^\infty_\nu, \qquad \text{and} \qquad \nabla_\mu(g_k\circ f) \overset{*}{\rightharpoonup} \nabla_\mu(g\circ f) \quad \text{in } L^\infty_\mu. \tag{6.5}$$

Fix $\phi \in L^\infty_\mu$ and define the tangent field $\tau \coloneqq \phi\,\sigma \in \mathcal{T}_\mu$. Then, arguing as in Theorem 6.4,

$$\int \langle \nabla_\mu(g_k\circ f), \tau\rangle \, d\mu \;=\; \int \langle \nabla g_k \circ f, \nabla_\mu f\, \tau\rangle \, d\mu \;=\; \int \langle \nabla g_k, f_*\tau\rangle d\nu. \tag{6.6}$$

We now pass to the limit as $k\to\infty$. On the target space, since $f_*\tau \in \mathcal{T}_\nu \subset (L^1_\nu)^m$ by Corollary 6.5, the weak-star convergence of ∇g_k yields:

$$\lim_{k\to\infty} \int \langle \nabla g_k, f_*\tau\rangle \, d\nu \;=\; \int \langle \nabla_\nu g, f_*\tau\rangle \, d\nu \;=\; \int \langle \nabla_\nu g\circ f, \nabla_\mu f\, \tau\rangle \, d\mu.$$

On the source space, the weak-star convergence of $\nabla_\mu(g_k\circ f)$ implies the left-hand side of (6.6) converges to $\int \langle \nabla_\mu(g\circ f), \tau\rangle d\mu$. Equating the limits and recalling that $\tau = \phi\sigma$, we get

$$\int \langle \nabla_\mu(g\circ f), \sigma\rangle \phi \, d\mu \;=\; \int \langle \nabla_\nu g\circ f, \nabla_\mu f\; \sigma\rangle \phi \, d\mu.$$

Since $\phi \in L^\infty_\mu$ is arbitrary, the identity holds μ-a.e.. □

6.5. **The case of bi-Lipschitz maps.** By the inclusion (6.2), there always holds

$$\dim T_{f_\#\mu}(f(x)) \;\geq\; \dim(\nabla_\mu f(T_\mu(x))) = \operatorname{rank}\big(\nabla_\mu f(x)\big) \qquad \mu\text{-a.e..} \tag{6.7}$$

We show that if f is bi-Lipschitz on its image, then equality holds. The Example 6.9 below illustrates that injectivity is not sufficient.

Proposition 6.8. *Let $\mu \in \mathcal{M}_+(\mathbb{R}^d)$ be finite and let $f \in \operatorname{Lip}(\mathbb{R}^d;\mathbb{R}^m)$ with $m \geqslant d$. Assume that f is injective, and that its left inverse $f^{-1} : f(\mathbb{R}^d) \to \mathbb{R}^d$ is Lipschitz. Then, denoting $\nu \coloneqq f_\#\mu$, we have*

$$\nabla_\mu f(x)(T_\mu(x)) = T_\nu(f(x)) \qquad \textit{for } \mu\textit{-a.e. } x \in \mathbb{R}^d. \tag{6.8}$$

Proof. Let $g \in \operatorname{Lip}(\mathbb{R}^m;\mathbb{R}^d)$ be a Lipschitz extension of f^{-1}, which exists by Kirszbraun's theorem. Then $f \circ g = \mathrm{id}$ at ν almost any point. Passing to the ν-tangential gradient in the latter equality thanks to (3.21), and using the chain rule of Theorem 6.7, we get

$$P_\nu(y) = \nabla_\nu(\mathrm{id})(y) = \nabla_\nu(f\circ g)(y) = \nabla_\mu f(g(y))\nabla_\nu g(y).$$

Since $P_\nu(y)(w) = w$ for any $w \in T_\nu(y)$, we deduce that $\nabla_\mu f(g(y))$ is surjective on $T_\nu(y)$. Hence, using that $\nabla_\mu f(g(y))(N_\mu) = \{0\}$, there holds

$$\dim(\nabla_\mu f(g(y))(\mathbb{R}^d)) = \dim(\nabla_\mu f(g(y))(T_\mu)) \geqslant \dim(T_\nu(y)) \qquad \text{at } \nu \text{ a.e. } y.$$

As $\mu = g_\#\nu$ and $\nu = f_\#\mu$, integrating against ν yields

$$0 \leqslant \int \left[\dim(\nabla_\mu f(g(y))(T_\mu)) - \dim(T_\nu(y))\right] d\nu = \int \left[\dim(\nabla_\mu f(T_\mu)) - \dim(T_\nu(f(x)))\right] d\mu.$$

However, in virtue of Theorem 6.4, we know that $\nabla_\mu f(x)(T_\mu(x)) \subset T_\nu(f(x))$ holds μ almost everywhere. The wished equality follows. □

To conclude, we construct an injective Lipschitz function which is not bi-Lipschitz, and for which the equality (6.8) does not hold.

Example 6.9. Let $E \subset [0,1]$ be a compact set with empty interior and $\mathcal{L}^d(E) > 0$ (a *fat* Cantor set), and define

$$f(t) \;\coloneqq\; \int_0^t \mathbb{1}_{E^c}(s)\, ds, \qquad t \in \mathbb{R}.$$

Then f is 1-Lipschitz and, since E has empty interior, $f(b) - f(a) = \mathcal{L}^d([a,b] \setminus E) > 0$ for $a < b$. Thus, the map f is strictly increasing, hence injective. Take $\mu \coloneqq \mathcal{L}^d \llcorner [0,1]$, so that $T_\mu(x) = \mathbb{R}$ for every x by Proposition 6.1. Denote $\nu \coloneqq f_\#\mu$.

Since $f' = \mathbb{1}_{E^c}$ a.e., the area formula gives $\mathcal{L}^d(f(E)) \leq \int_E |f'| = 0$, while injectivity gives $\nu(f(E)) = \mu(E) = \mathcal{L}^d(E) > 0$. Thus $\nu \llcorner f(E)$ is a non-zero measure singular with respect to $\mathcal{L}^d$. If the set $A \coloneqq \{\, y : \dim T_\nu(y) = 1 \,\}$ met $f(E)$ in positive ν-measure, then by (3.3) and Proposition 6.1 the measure $\nu \llcorner (A \cap f(E))$ would be both absolutely continuous with respect to $\mathcal{L}^d$ and concentrated on the $\mathcal{L}^d$-null set $f(E)$, hence zero. Therefore

$$\dim T_\nu(f(x)) = 0 < 1 = \dim T_\mu(x) \qquad \text{for } \mu\text{-a.e. } x \in E.$$

Consistently with (6.7), the tangential differential degenerates here: $\nabla_\mu f = f' = 0$ holds $\mathcal{L}^d$-a.e. on E.

6.6. **Weaver derivation.** We point a connection with the analysis on metric spaces already noticed in [38] for the tangent space therein. Let Lip_b stand for Lipschitz and bounded functions. The tangential gradient map $\nabla_\mu : \mathrm{Lip}_b(\mathbb{R}^d) \to (L^\infty_\mu)^d$ is a derivation in the sense of Weaver, i.e. satisfies

(i) *Weak-star continuity:* $\nabla_\mu u_n \overset{*}{\rightharpoonup} \nabla_\mu u$ whenever $u_n \overset{*}{\rightharpoonup} u$ in $\mathrm{Lip}_b(\mathbb{R}^d)$;
(ii) *Leibniz rule:* $\nabla_\mu(uv) = u\nabla_\mu v + v\nabla_\mu u$ for all $u, v \in \mathrm{Lip}_b(\mathbb{R}^d)$.

Proof. The weak-star continuity is a direct consequence of Lemma 3.10 and the linearity of ∇_μ. The Leibniz rule follows by approximation with test functions and the smooth Leibniz rule: if $u_n \overset{*}{\rightharpoonup} u, v_n \overset{*}{\rightharpoonup} v$ in $\mathrm{Lip}_b(\mathbb{R}^d)$, then by dominated convergence $u_n\nabla_\mu v_n \overset{*}{\rightharpoonup} u\nabla_\mu v$ and $v_n\nabla_\mu u_n \overset{*}{\rightharpoonup} v\nabla_\mu u$ in $(L^\infty_\mu)^d$. On the other hand, $\nabla_\mu(u_nv_n) \overset{*}{\rightharpoonup} \nabla_\mu(uv)$, and

$$\nabla_\mu(u_nv_n) = P_\mu(v_n\nabla u_n + u_n\nabla v_n) = v_n\nabla_\mu u_n + u_n\nabla_\mu v_n \overset{*}{\rightharpoonup} v\nabla_\mu u + u\nabla_\mu v$$

in $(L^\infty_\mu)^d$. Uniqueness of the weak-star limit yields the assertion. □

References

[1] G. Alberti, D. Bate, A. Marchese: *On the closability of differential operators.* J. Funct. Anal. **289** (2025), no. 7, 111029.
[2] G. Alberti, A. Marchese: *On the differentiability of Lipschitz functions with respect to measures in the Euclidean space.* Geom. Funct. Anal. **26** (2016), no. 1, 1–66.
[3] G. Alberti, A. Marchese: *On the structure of flat chains with finite mass.* Preprint, arXiv:2311.06099 (2023).
[4] W. K. Allard: *On the first variation of a varifold*, Ann. of Math. (2) 95 (1972), no. 3, 417–491.
[5] L. Ambrosio, N. Gigli, G. Savaré: *Gradient flows in metric spaces and in the space of probability measures.* Second edition, Lectures in Mathematics ETH Zürich, Birkhäuser, Basel (2008).
[6] L. Ambrosio, B. Kirchheim: *Currents in metric spaces.* Acta Math., **185** (2000), no. 1, 1–80.
[7] R.F Arens, J.J Eells: *On embedding uniform and topological spaces*, Pacific Journal of Mathematics, **6** (1956), 397–403.
[8] A. Arroyo-Rabasa: *An elementary approach to the dimension of measures satisfying a first-order linear PDE constraint*, Proc. Amer. Math. Soc. **148** (2020), no. 1, 273–282
[9] A. Arroyo-Rabasa, G. Bouchitté: *Structural properties of one-dimensional metric currents: SBV-representations, connectedness and the flat chain conjecture*, arxiv.org/abs/2508.08212 (2025).
[10] A. Arroyo-Rabasa, G. De Philippis, J. Hirsch, F. Rindler: *Dimensional estimates and rectifiability for measures satisfying linear PDE constraints.* Geom. Funct. Anal. **29** (2019), no. 3, 639–658.
[11] A. Aussedat: *Characterization of measures on the real line that are critically unstable under small shifts*, arxiv.org/abs/2603.00775 (2026).
[12] D. Bate: *Purely unrectifiable metric spaces and perturbations of Lipschitz functions.* Acta Math., **224** (2020), no. 1, 1–65.
[13] K. Bołbotowski, G. Bouchitté: Optimal design versus maximal Monge-Kantorovich metrics. *Arch. Ration. Mech. Anal.*, 243:1449–1524, 2022.
[14] G. Bouchitté: Convex analysis and duality, Encyclopedia of Mathematical Physics, Elsevier vol.11, Oxford (2006), 642–653.
[15] G. Bouchitté, G. Buttazzo: *Characterization of optimal shapes and masses through Monge-Kantorovich equation.* J. Eur. Math. Soc. (JEMS), **3** (2001), 139–168.
[16] G. Bouchitté, G. Buttazzo, L. De Pascale: *The Monge-Kantorovich Problem for Distributions and Applications* Journal of Optimization Theory and Applications, Journal of Convex Analysis **17** (2010), 3&4, 925–943.
[17] G. Bouchitté, G. Buttazzo and I. Fragalà: *Mean curvature of a measure and related variational problems*, Ann. Scuola Norm. Sup. Pisa CI: Sci., XXV(4) (1997), 179–196.
[18] G. Bouchitté, G. Buttazzo and I. Fragalà: *Convergence of Sobolev spaces on varying manifolds*, J. Geom. Anal. 11 (3) (2001) 399–422.

[19] G. Bouchitté, G. Buttazzo, P. Seppecher: *Shape optimization solutions via Monge-Kantorovich equation.* C. R. Acad. Sci. Paris, **324-I** (1997), 1185–1191.
[20] G. Bouchitté, G. Buttazzo, P. Seppecher: *Energies with respect to a measure and applications to low dimensional structures.* Calc. Var., **5** (1997), 37–54.
[21] G. Bouchitté, T. Champion, C. Jimenez: *Completion of the space of measures in the Kantorovich norm.* Riv. Mat. Univ. Parma, **7** (2005), 127–139.
[22] G. Bouchitté, I. Fragalà: *Variational theory of weak geometric structures,Variational theory of weak geometric structures.* The measure method and its applications, Progress in Nonlinear Differential Equations and Their Applications, **51**, 19–40, 2002.
[23] G. Bouchitté, I. Fragalà: *Second-order energies on thin structures: variational theory and non-local effects.* J. Funct. Anal. 204 (2003), no. 1, 228–267.
[24] G. Bouchitté, I. Fragalà: Optimality conditions for mass design problems and applications to thin plates. *Arch. Ration. Mech. Anal.* 184:257–284, 2007.
[25] G. Bouchitté, I. Fragalà, Homogenization of thin structures by two-scale method with respect to measures, SIAM J. Math. Anal. **32** (2001), no. 6, 1198–1226; MR1856245
[26] G. Bouchitté, I. Fragalà and P. Seppecher: *Structural optimization of thin elastic plates: the three dimensional approach.*, Arch. Ration. Mech. Anal. **202** (2011), no. 3, 829–874; MR2854671
[27] G. Bouchitté, M. Valadier, *l.s.c. multifunctions and the essential l.s.c. regularized function*, Ann. Inst. H. Poincaré Anal. Non Linéaire 6 (1989), suppl., 123–149.
[28] B. Buet, G. P. Leonardi, S. Masnou, *Weak and approximate curvatures of a measure: A varifold perspective*, Nonlinear Analysis, **222** (2022), 112983, 34 pp.
[29] G. De Philippis, F. Rindler: *On the structure of $\mathcal{A}$-free measures and applications.* Annals of Mathematics **184** (2016), 1017–1039.
[30] I. Ekeland, R. Temam: *Analyse Convexe et Problèmes Variationnels*, Dunod-Gauthier Villars, Paris, 1976.
[31] H. Federer: *Geometric measure theory*, Die Grundlehren der mathematischen Wissenschaften, Band 153, Springer-Verlag, New York (1969).
[32] H. Federer, W.H. Fleming: *Normal and integral currents.* Ann. of Math. (2), **72** (1960), 458–520.
[33] I. Fragalà and C. Mantegazza: *On some notions of tangent space to a measure*, Proc. Roy. Soc. Edinburgh Sect. A **129** (1999), no. 2, 331–342.
[34] G. Godefroy, N.J. Kalton: *Lipschitz-free Banach spaces.* Studia Math., **159** (2003), no. 1, 121–141.
[35] L. G. Hanin: *Duality for general Lipschitz classes and applications.* Proc. London Math. Soc. **75** (1997), 134–146.
[36] F. Hiai, H. Umegaki: *Integrals, conditional expectations and martingales functions*, J. Multivariate An. 7 ((1977), 149–182.
[37] L.V. Kantorovich, G.S. Rubinstein: *On a function space in certain extremal problems.* Dokl. Akad. Nauk. USSR, **115** (1957), 1058–1061.
[38] D. Lučić, E. Pasqualetto, T. Rajala: *Characterisation of upper gradients on the weighted Euclidean space and applications.* Annali di Matematica Pura ed Applicata, **200** (2021), no. 6, 2473–2513.
[39] P. Mattila: *Geometry of sets and measures in Euclidean spaces*, Cambridge University Press (1995)
[40] A. Marchese, A. Merlo: *A simple proof of the 1-dimensional flat chain conjecture.* Adv. Calc. Var. **19** (2026), no. 1, 61–69.
[41] J.-J. Moreau: Fonctionnelles convexes. *Séminaire Jean Leray* 2:1–108, 1966.
[42] F. Morgan: Geometric measure theory. A beginner's guide, Academic Press, Boston (1988).
[43] Y. Olubummo: *On duality for a generalized Monge-Kantorovich problem.* J. Funct. Anal., **207** (2004), 253–263.
[44] T. O'Neil: *A measure with a large set of tangent measures* Proc. Amer. Math. Soc., 123 (1995), 2217–2221.
[45] D. Preiss:*Geometry of measures on $\mathbb{R}^n$: distribution, rectifiability and densities.* Ann. Math., 125 (1987), 573–643.
[46] A. Rodriguez-Arenas and J. Wengenroth: *Smirnov-decompositions of vector fields*, preprint arXiv:2405.13406 (2024).
[47] R. T. Rockafellar: *Measurable dependence of convex sets and functions on parameters*, Journal of Mathematical Analysis and Applications **28.1** (1969), pp. 4–25.
[48] L. Simon: *Lectures on geometric measure theory*, Proc. C.M.A.3, Australian Natl. U. Canberra (1983).

[49] S. K. Smirnov: *Decomposition of solenoidal vector charges into elementary solenoids and the structure of normal one-dimensional currents*, St. Petersbg. Math. J. 5 (1993), no. 4, 206–238.
[50] N. Weaver: *Lipschitz Algebras*. Second edition, World Scientific, Hackensack, NJ (2018).
[51] H. Whitney: *Geometric Integration Theory*. Princeton University Press, Princeton, NJ (1957).
[52] V. V. Zhikov: *On an extension of the method of two-scale convergence and its applications*, Sb. Math., 191:7 (2000), 973–1014